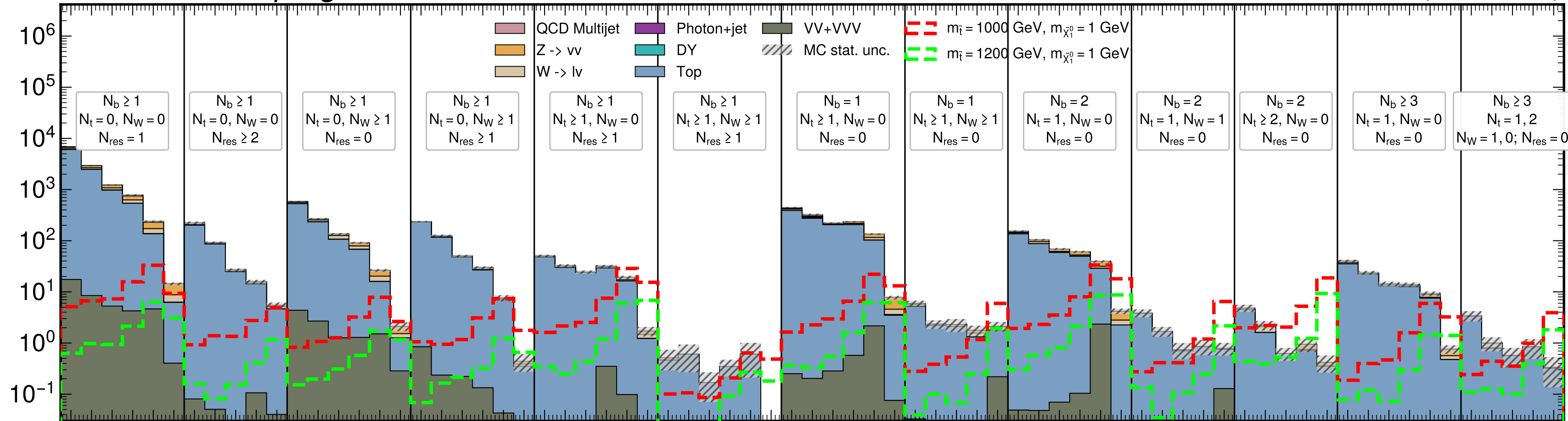
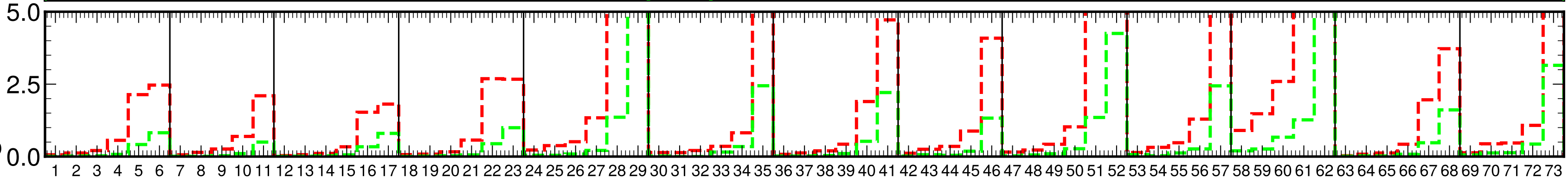


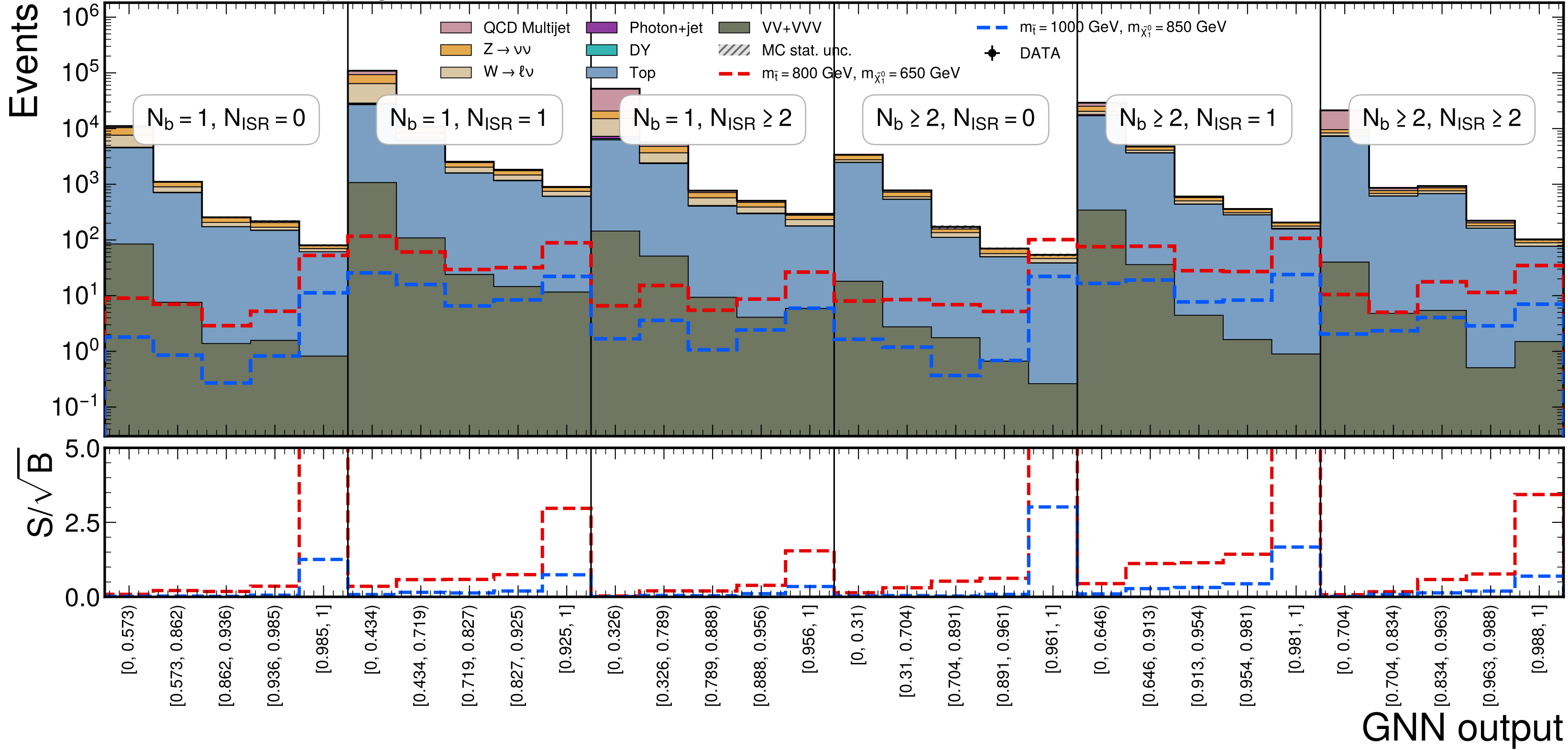
Events



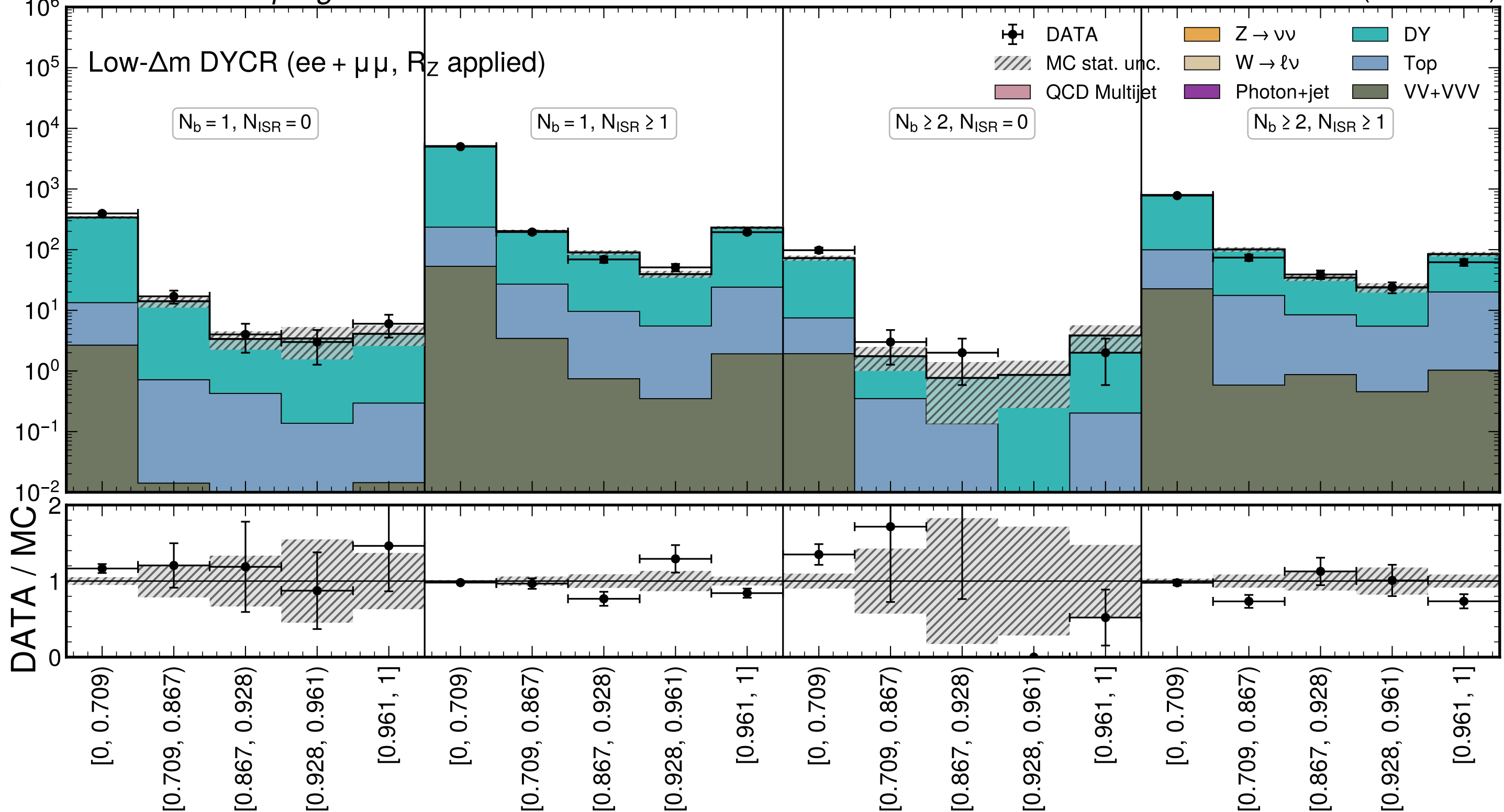
Significance



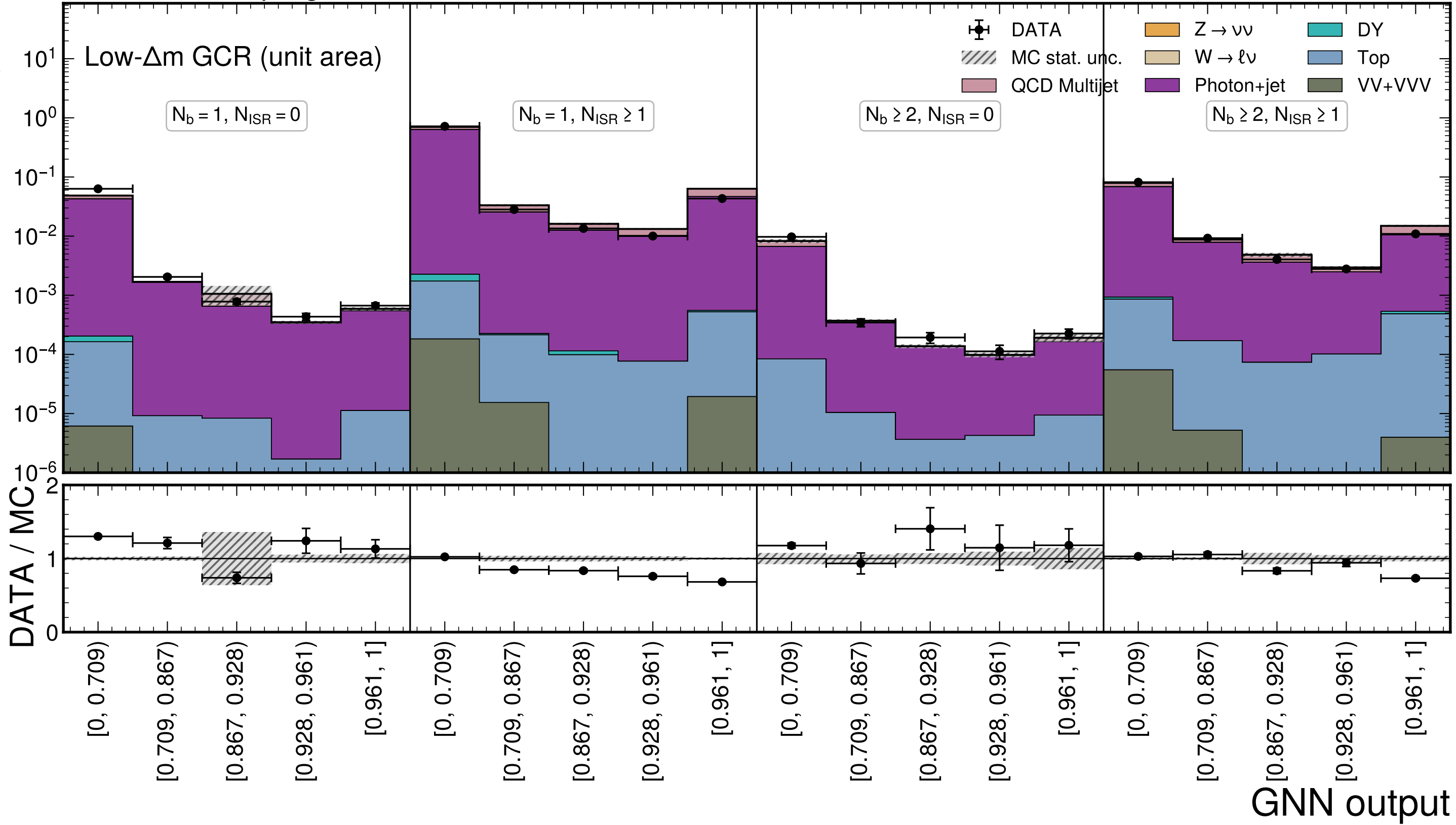
Search bin



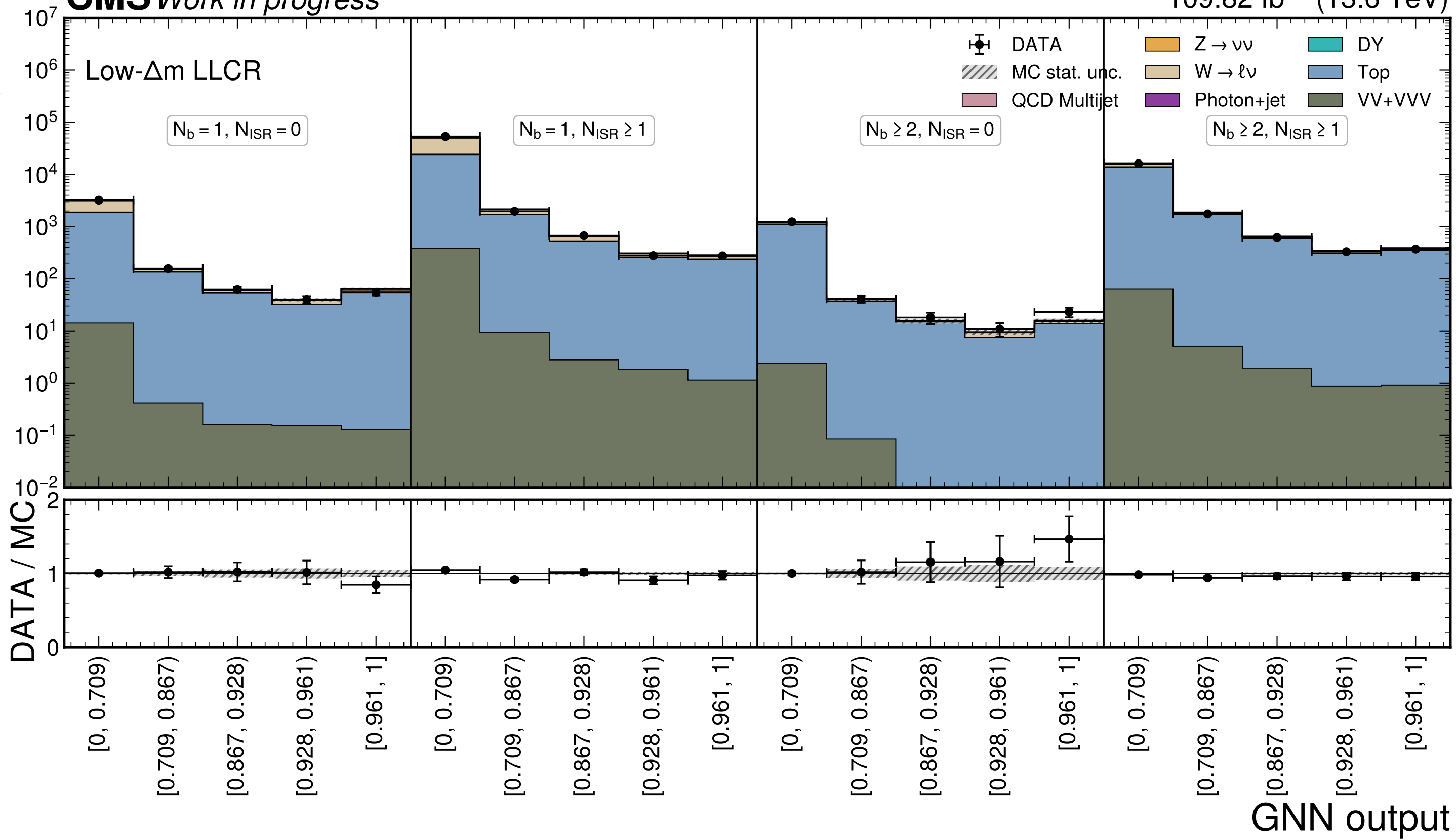
Events / bin



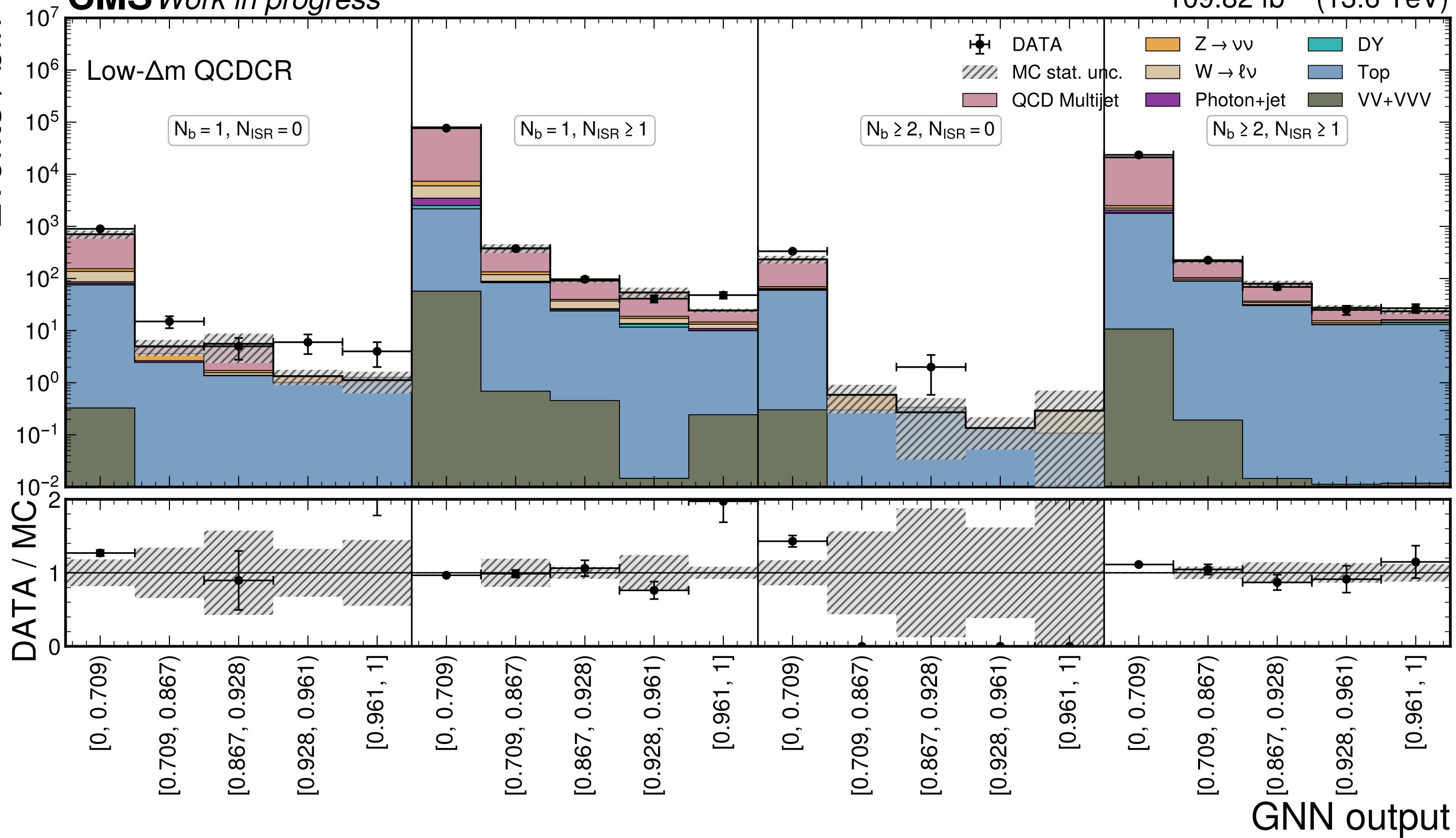
Normalized events / bin



Events / bin



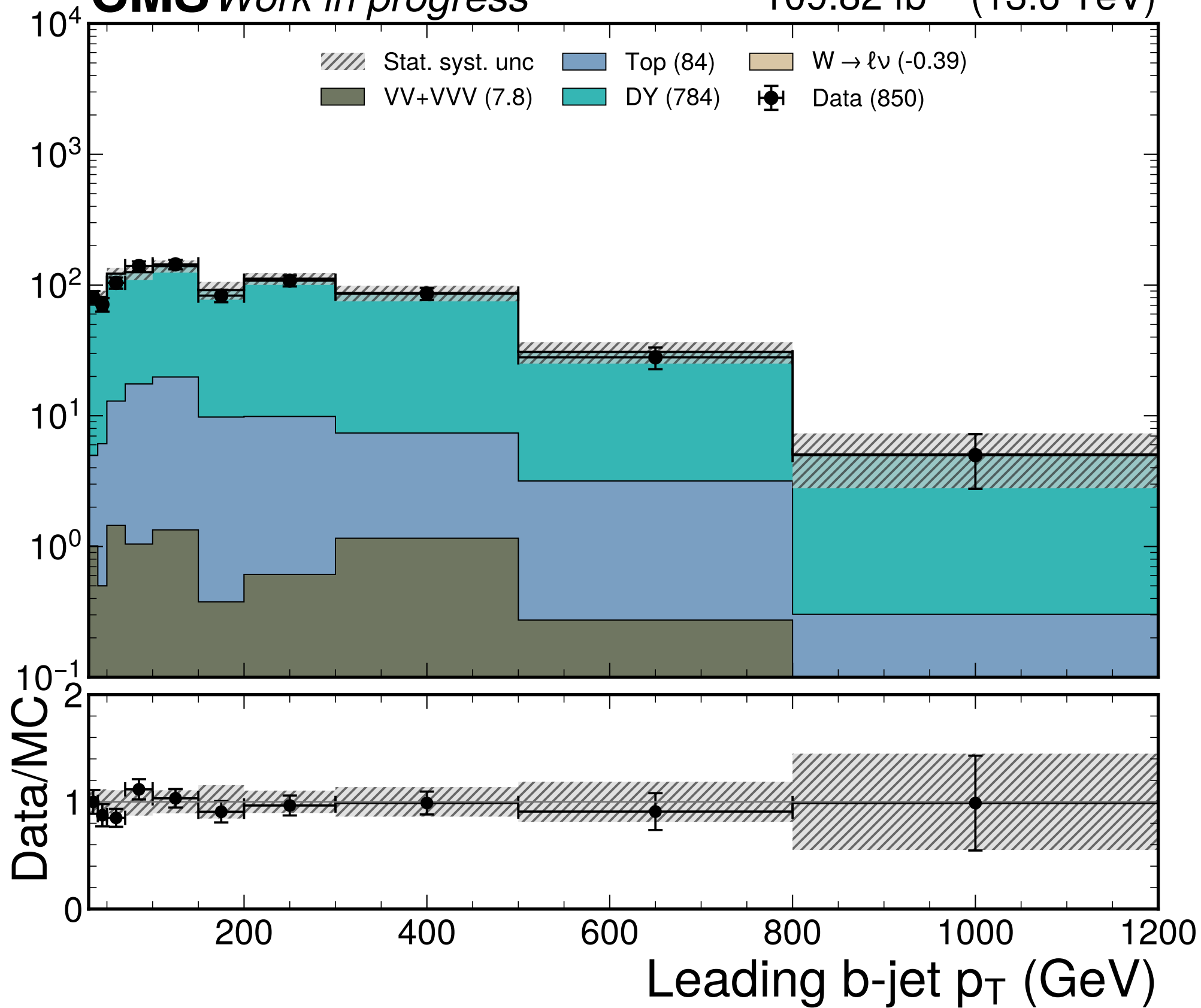
Events / bin



CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)

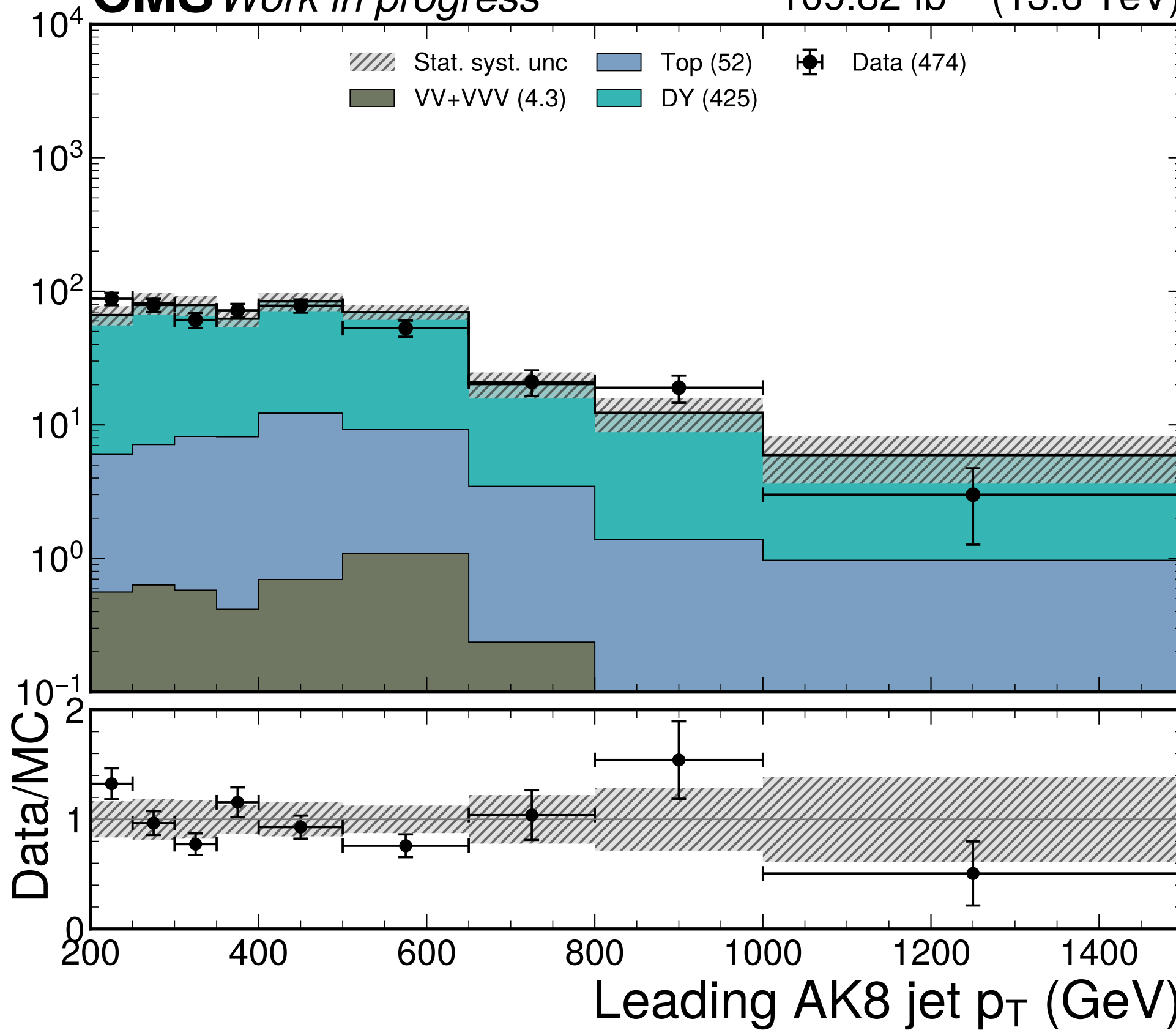
Events



CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)

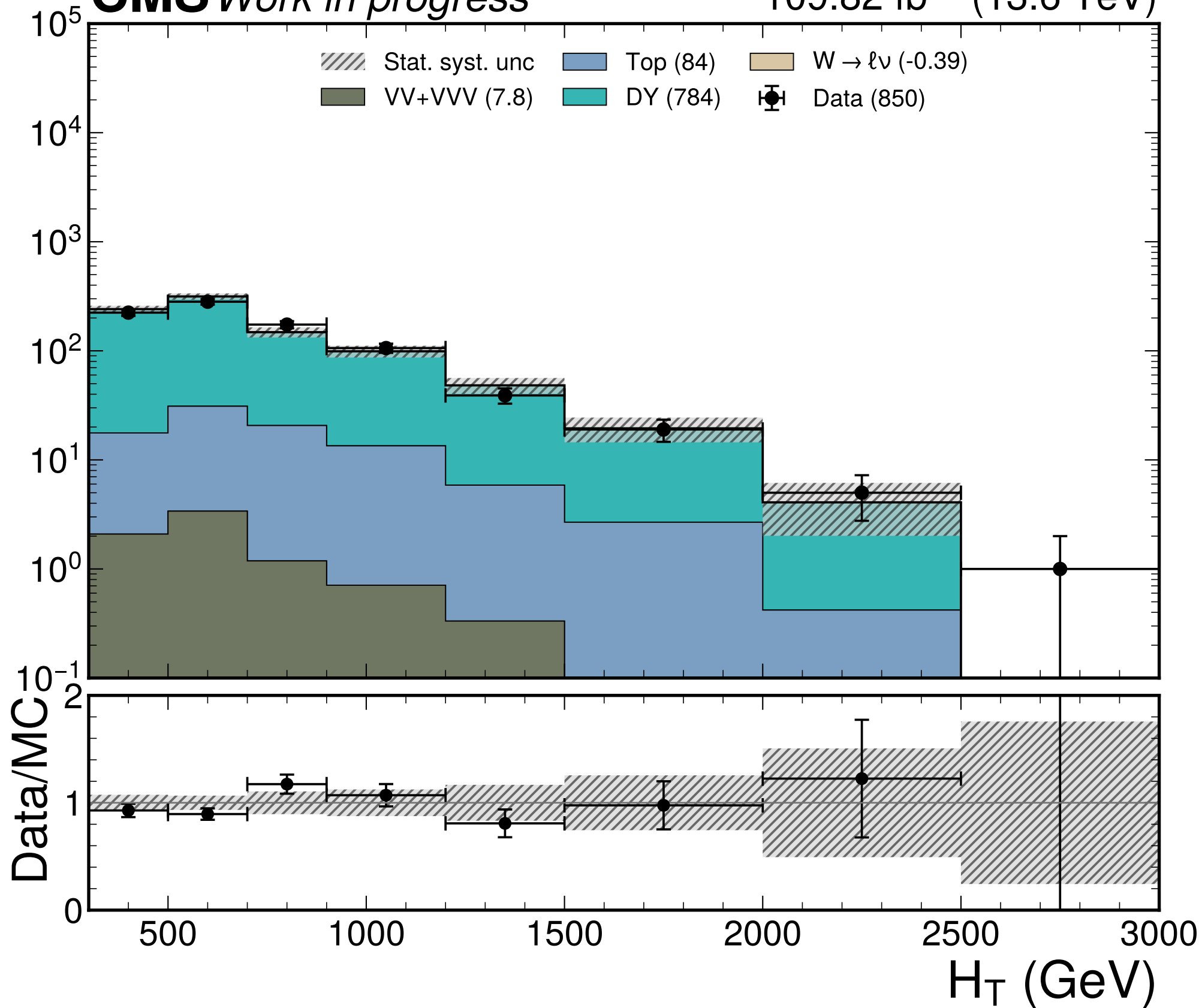
Events



CMS *Work in progress*

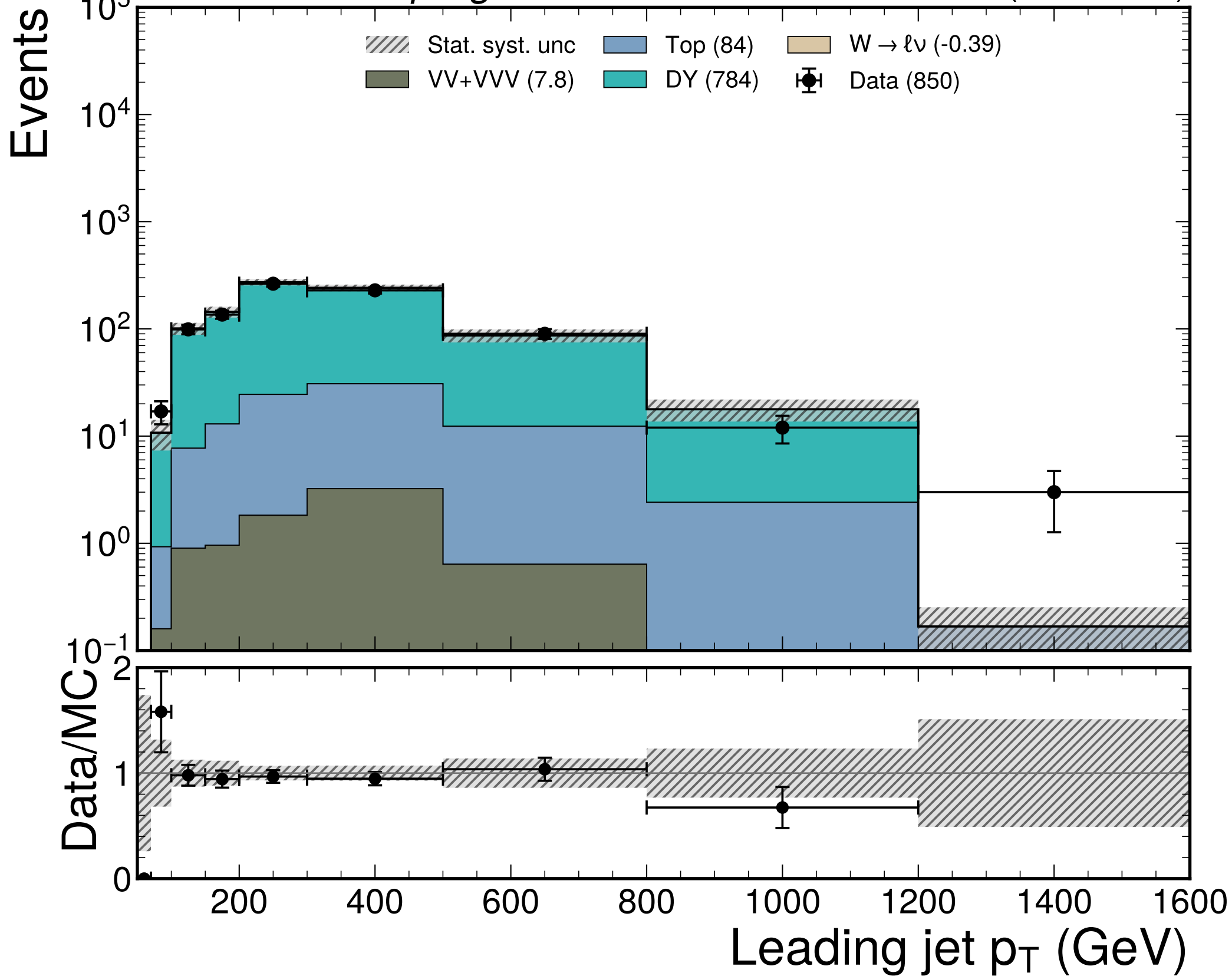
109.82 fb⁻¹ (13.6 TeV)

Events



CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)



CMS *Work in progress*

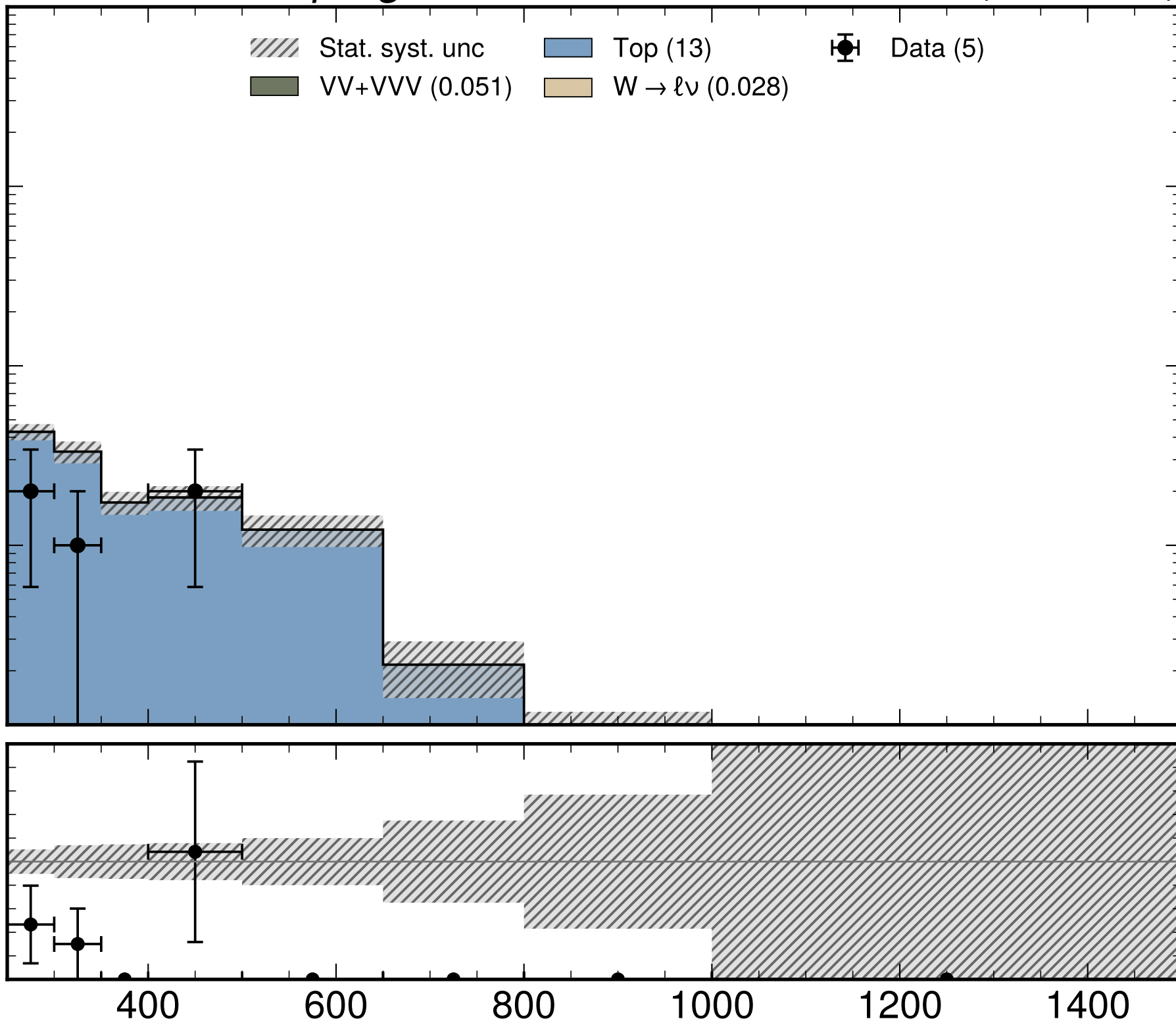
109.82 fb⁻¹ (13.6 TeV)

Events

Stat. syst. unc Top (13) Data (5)
VV+VVV (0.051) W → ℓν (0.028)

Data/MC

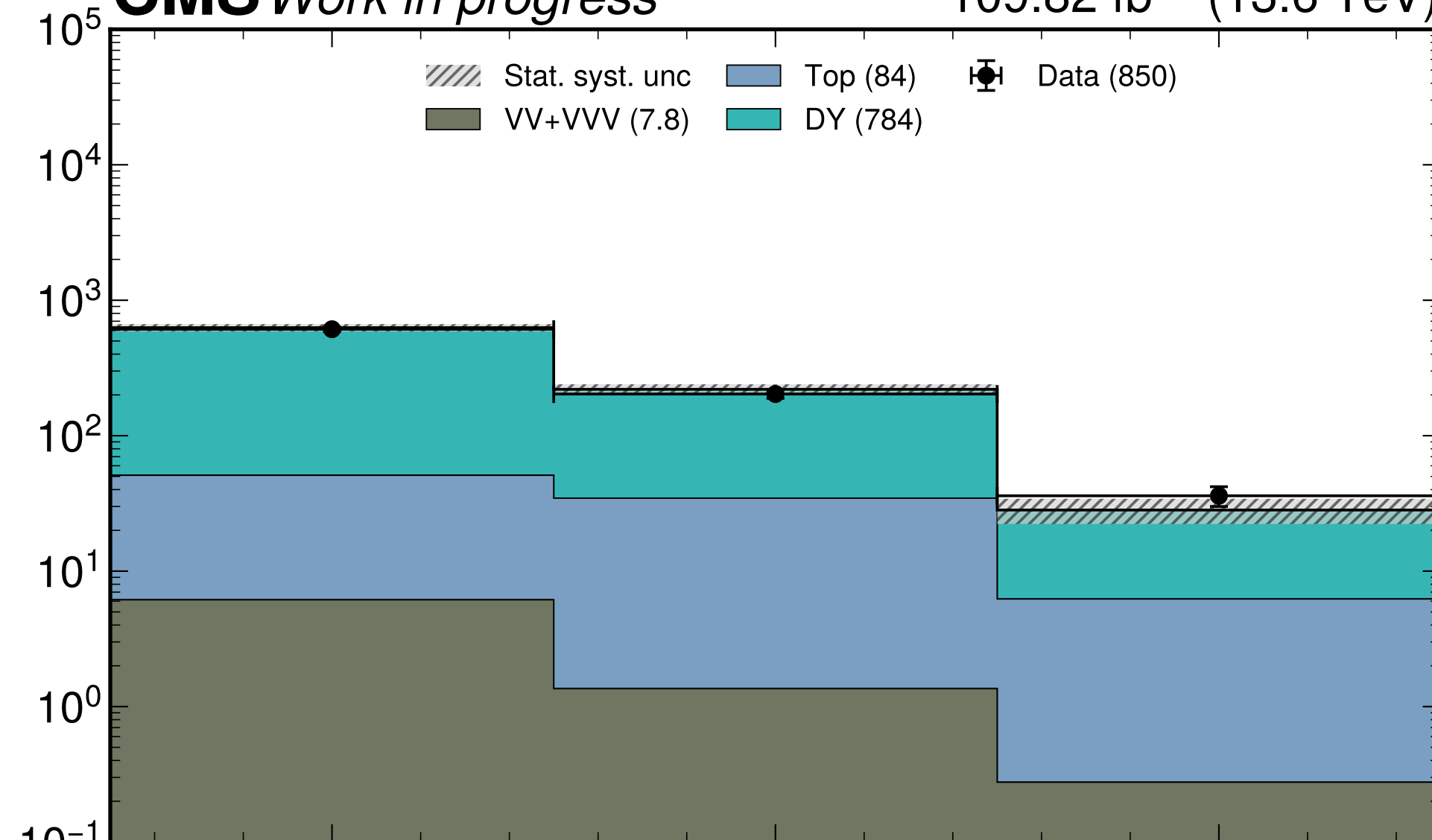
p_T^{miss} (GeV)



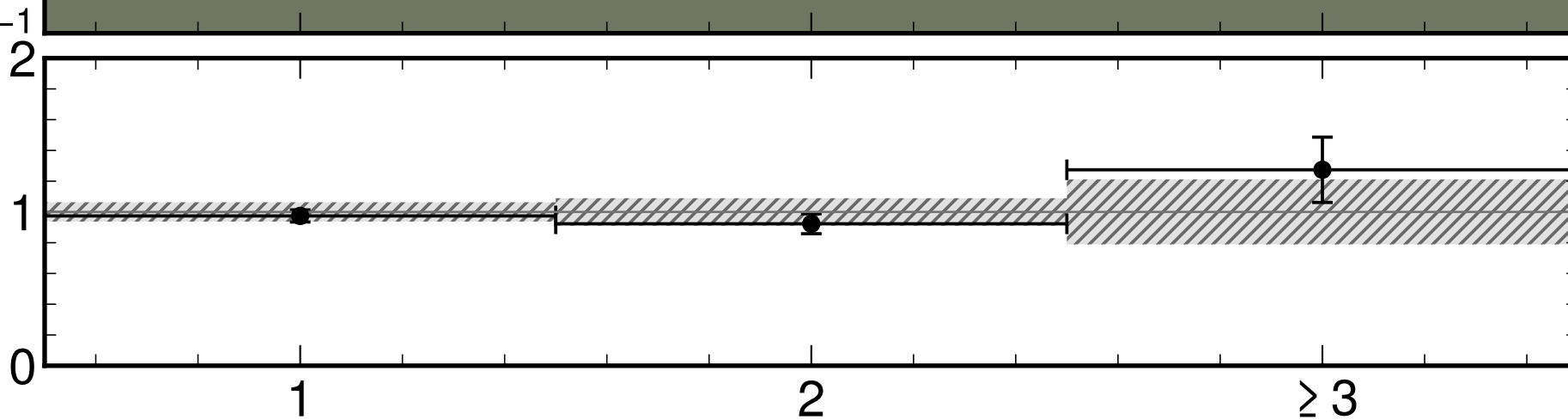
CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)

Events



Data/MC

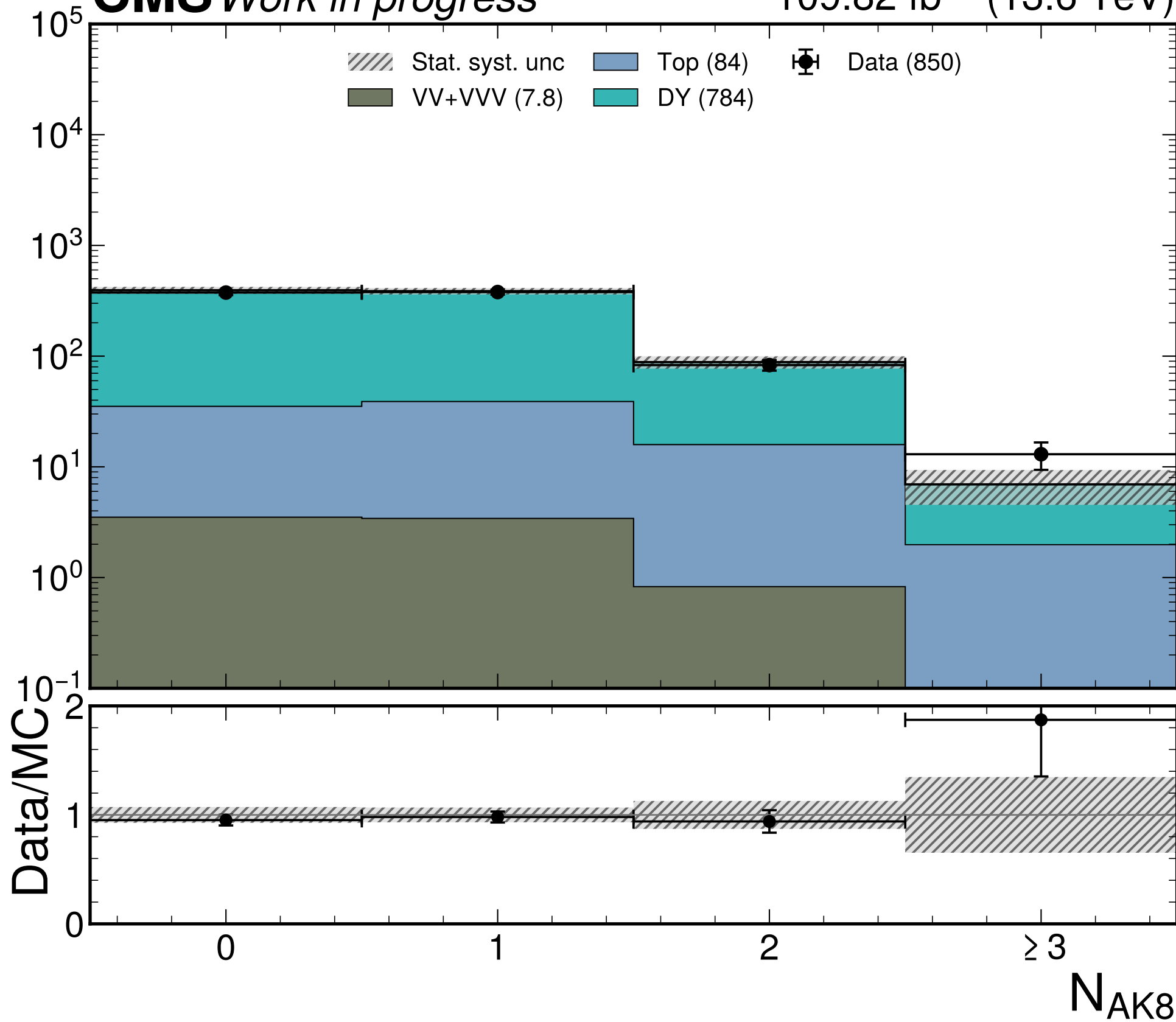


N_b

CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)

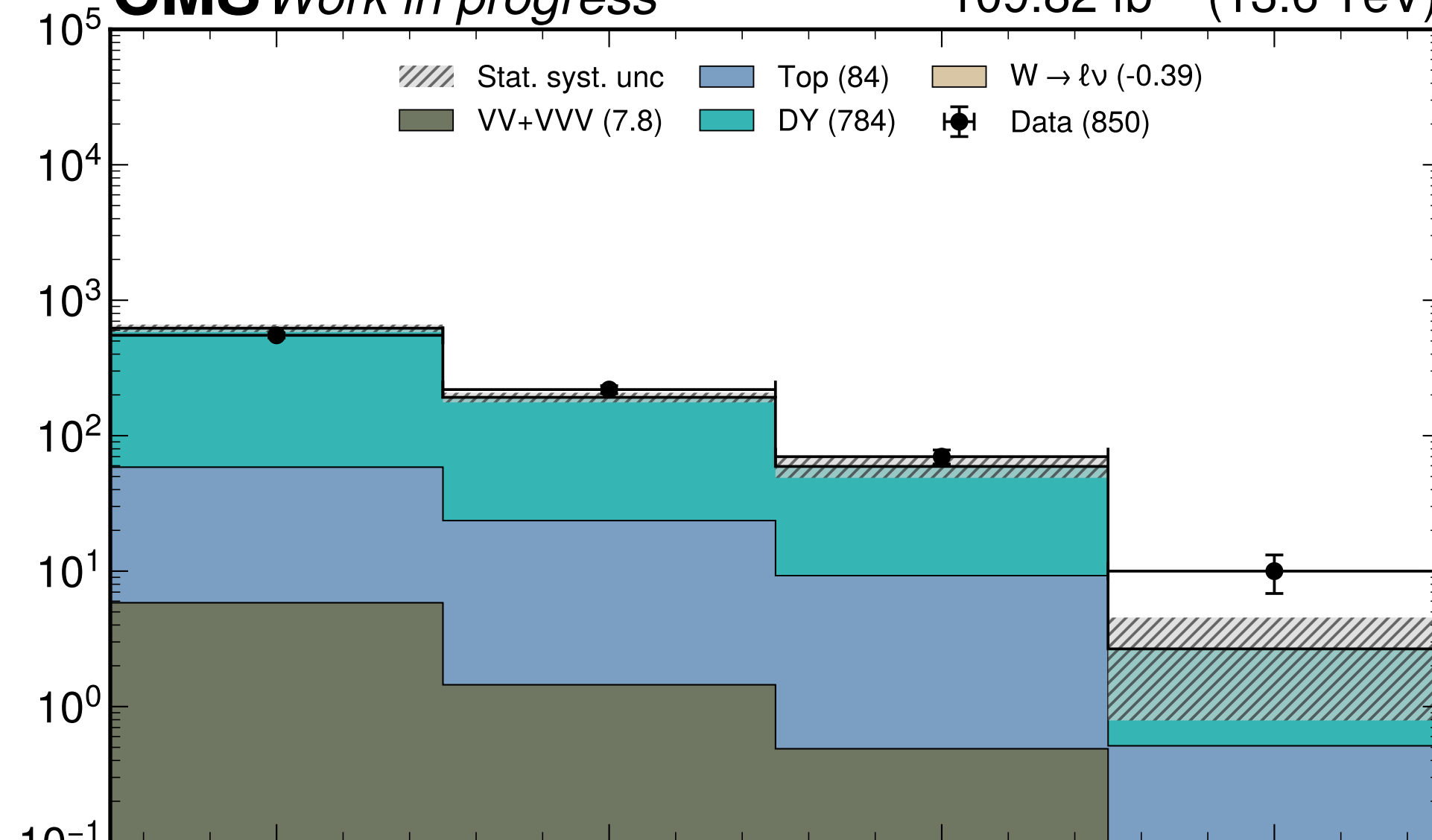
Events



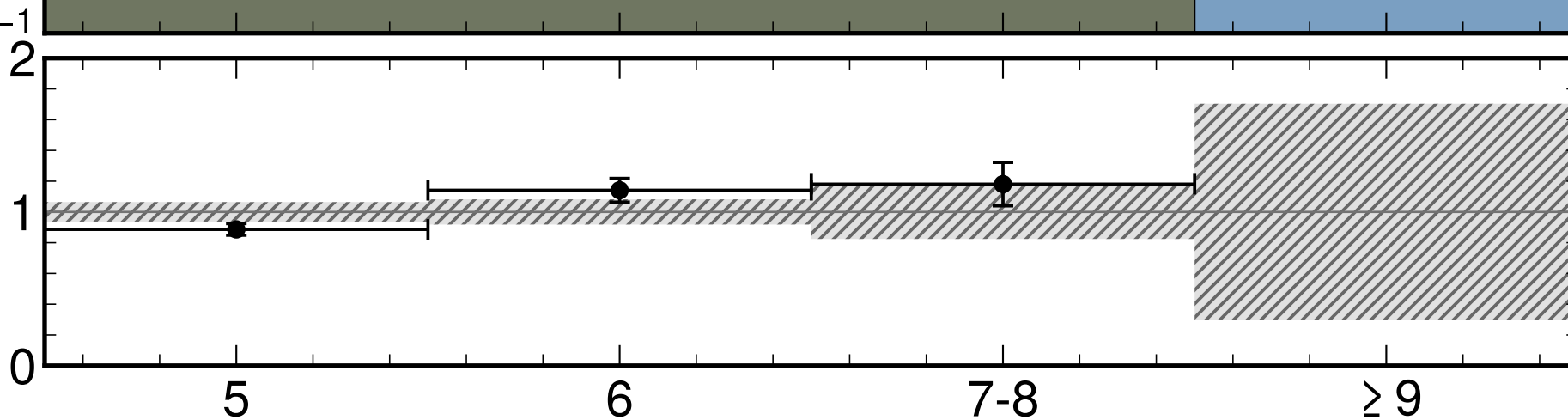
CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)

Events



Data/MC

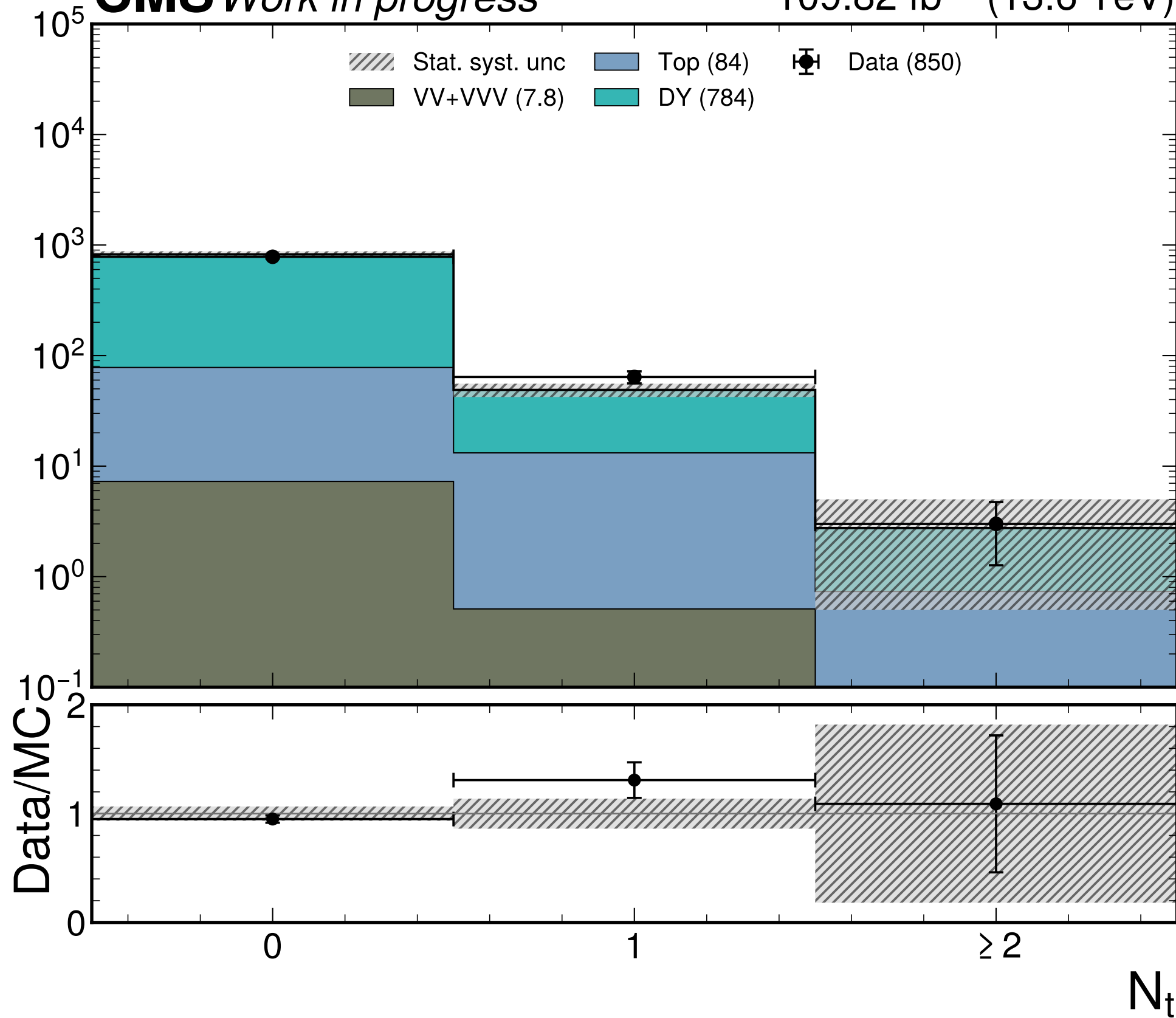


N_j

CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)

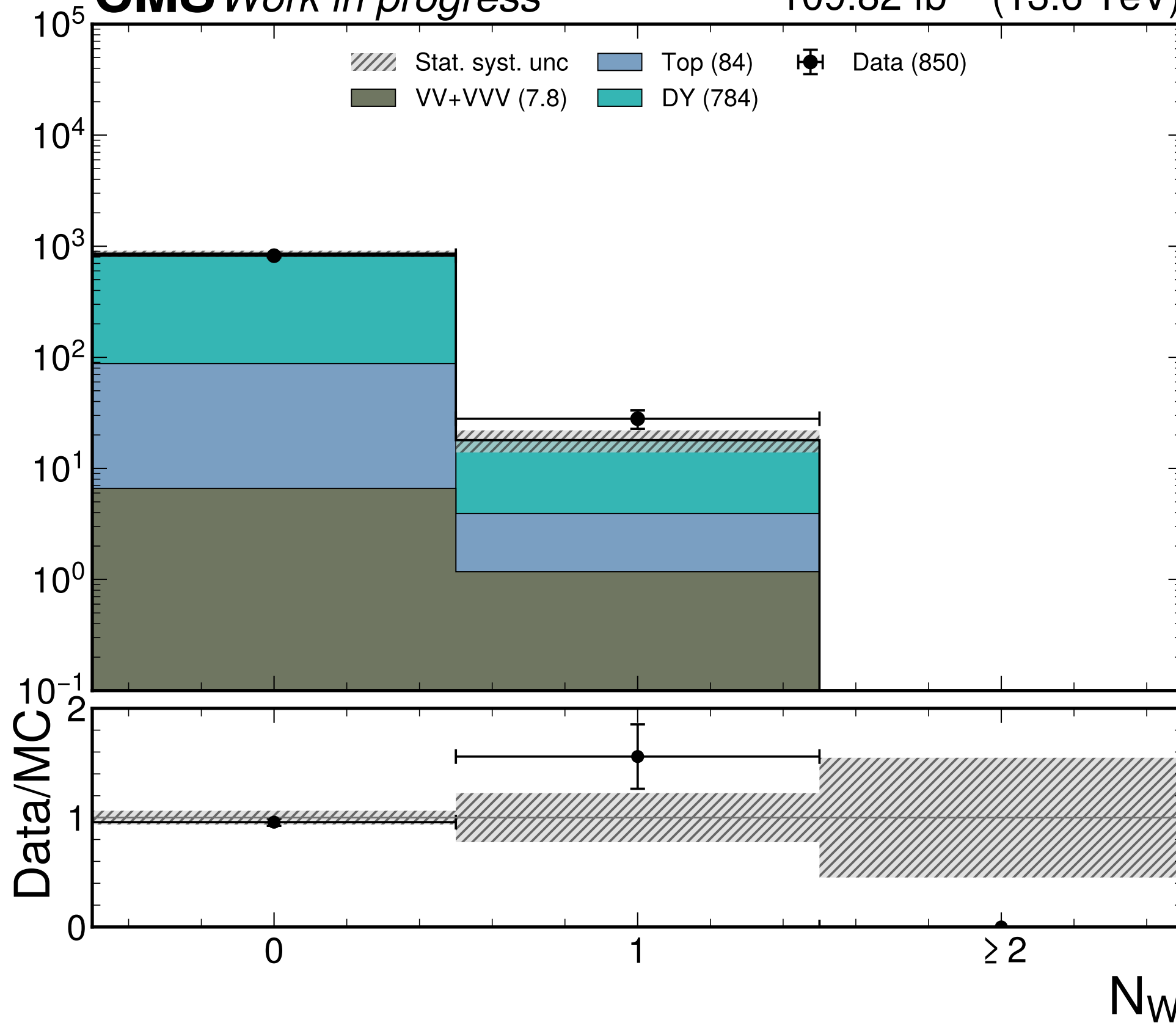
Events



CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)

Events

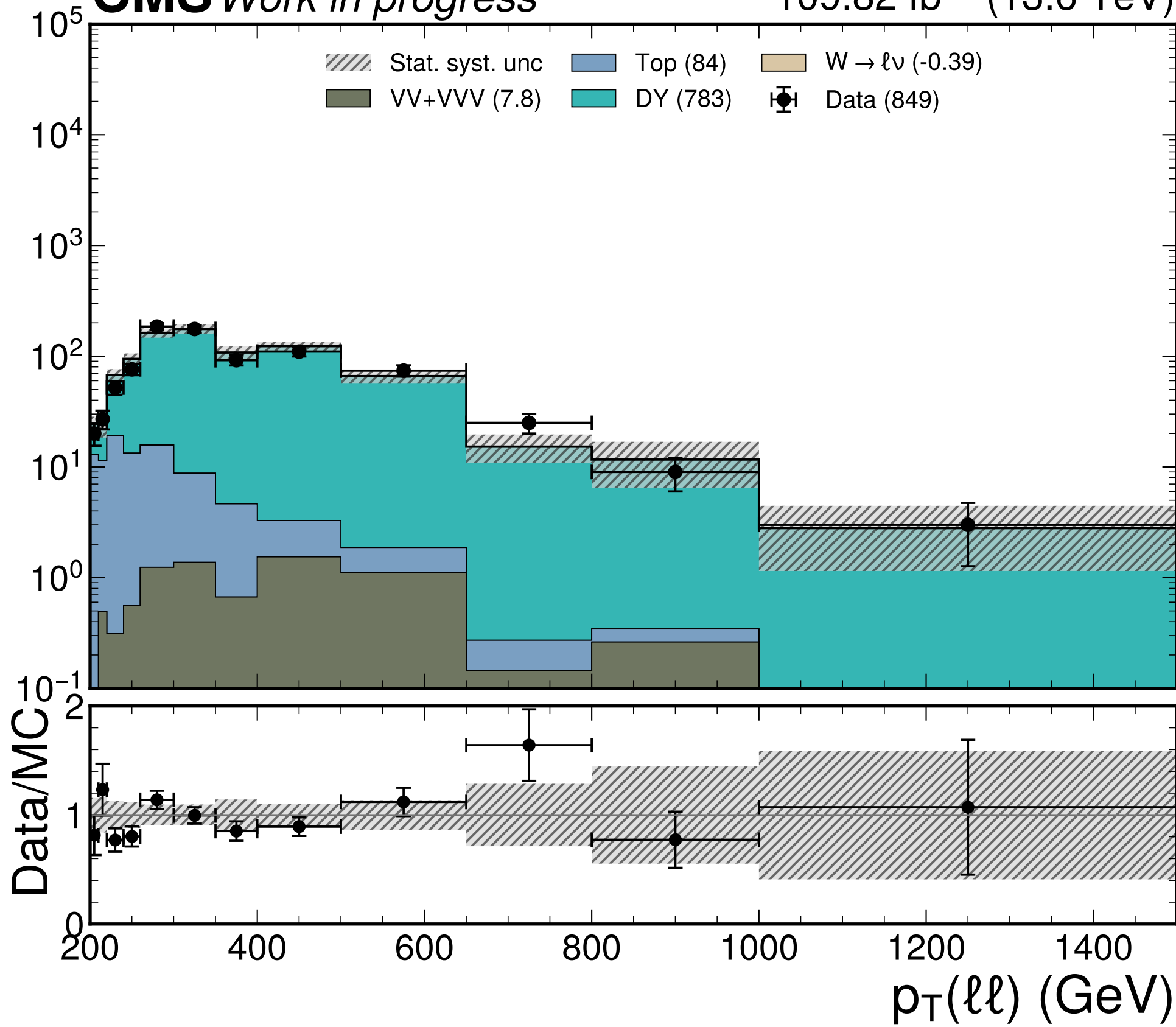


CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)

Events

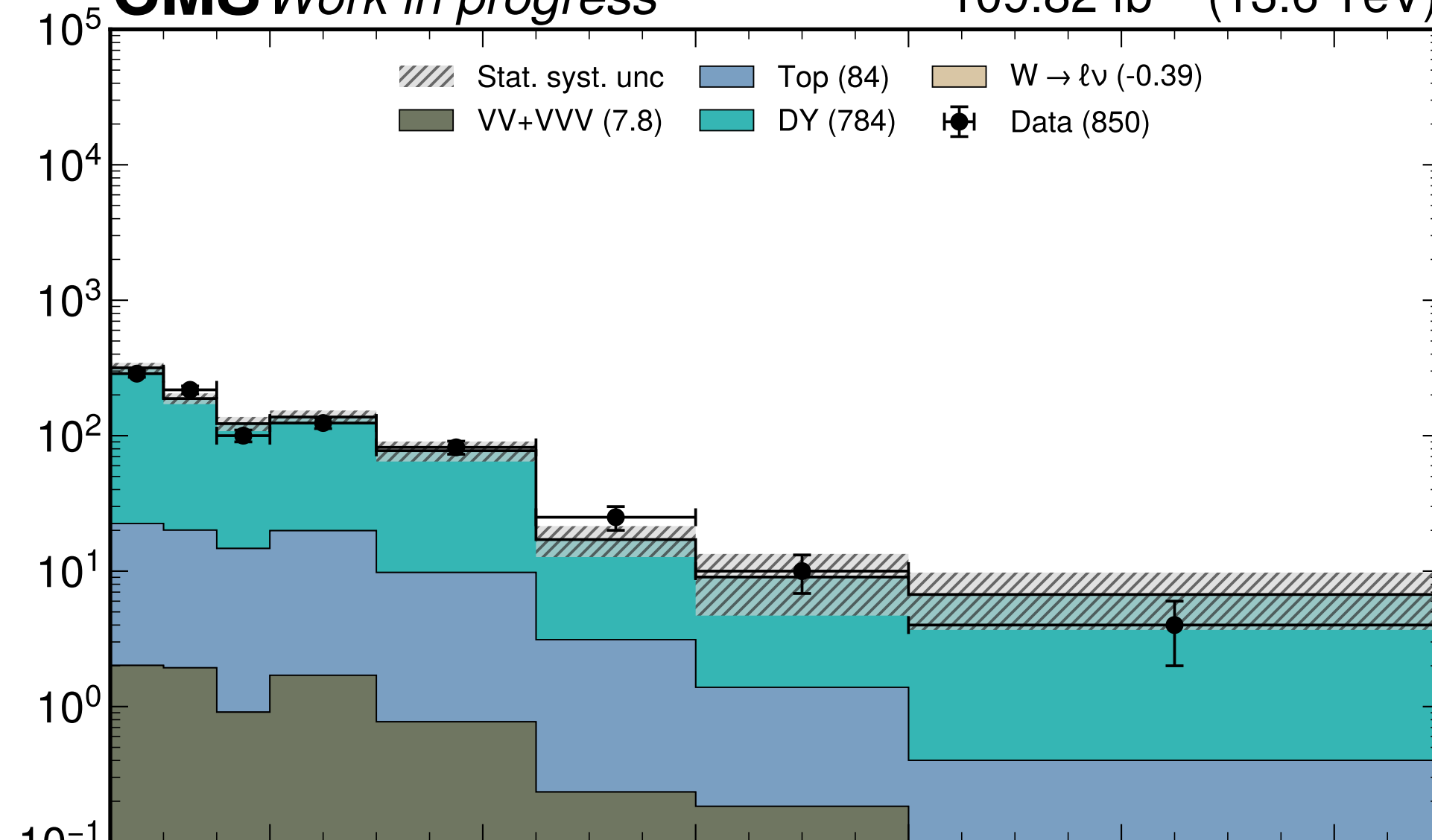
Stat. syst. unc Top (84) W → ℓν (-0.39)
VV+VVV (7.8) DY (783) Data (849)



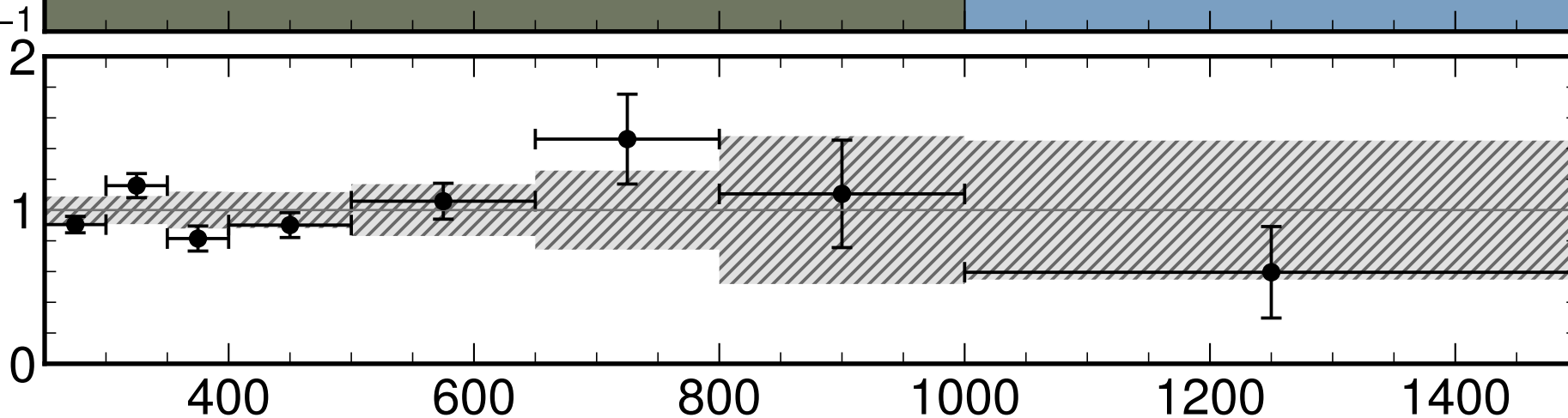
CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)

Events



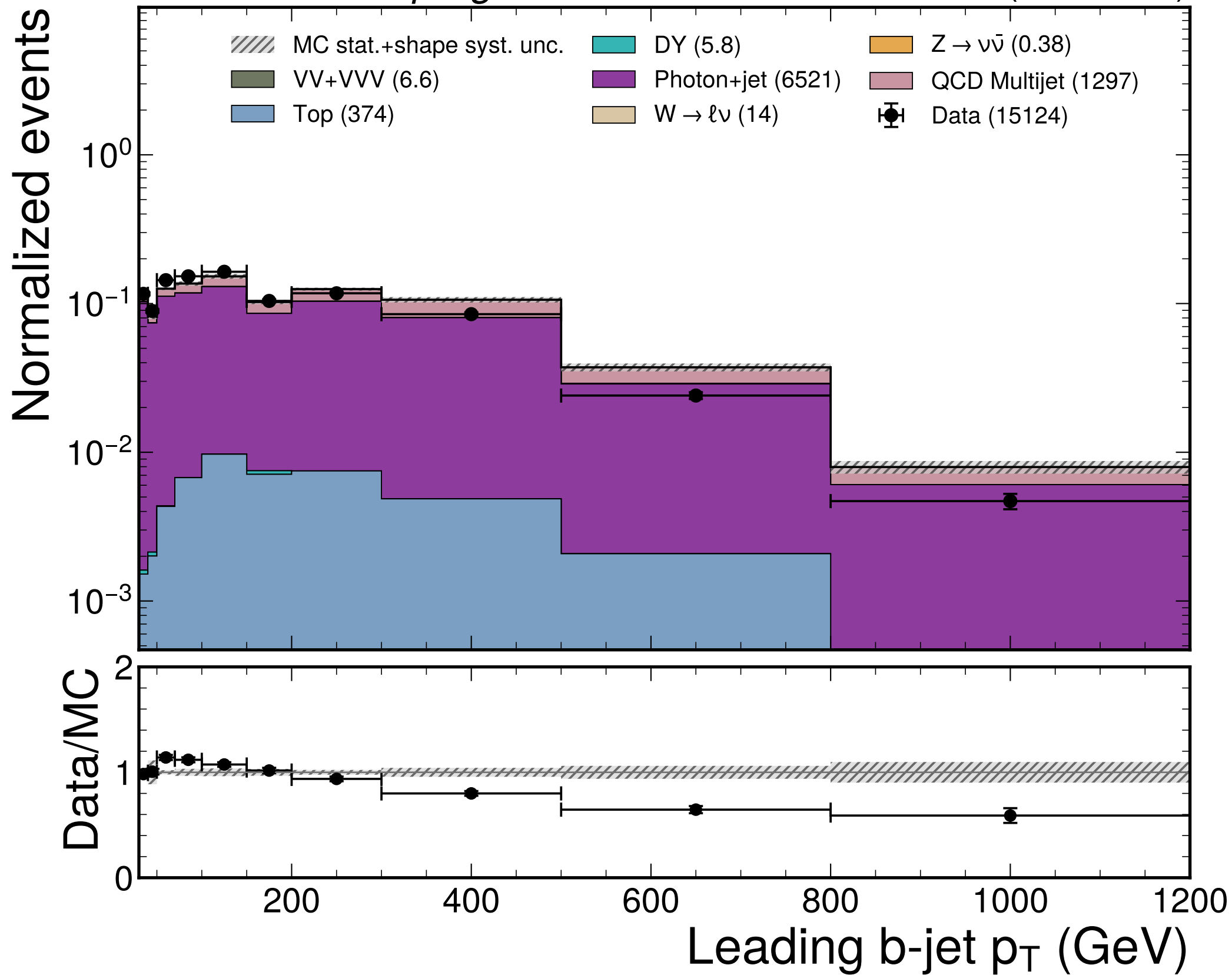
Data/MC



U_T (GeV)

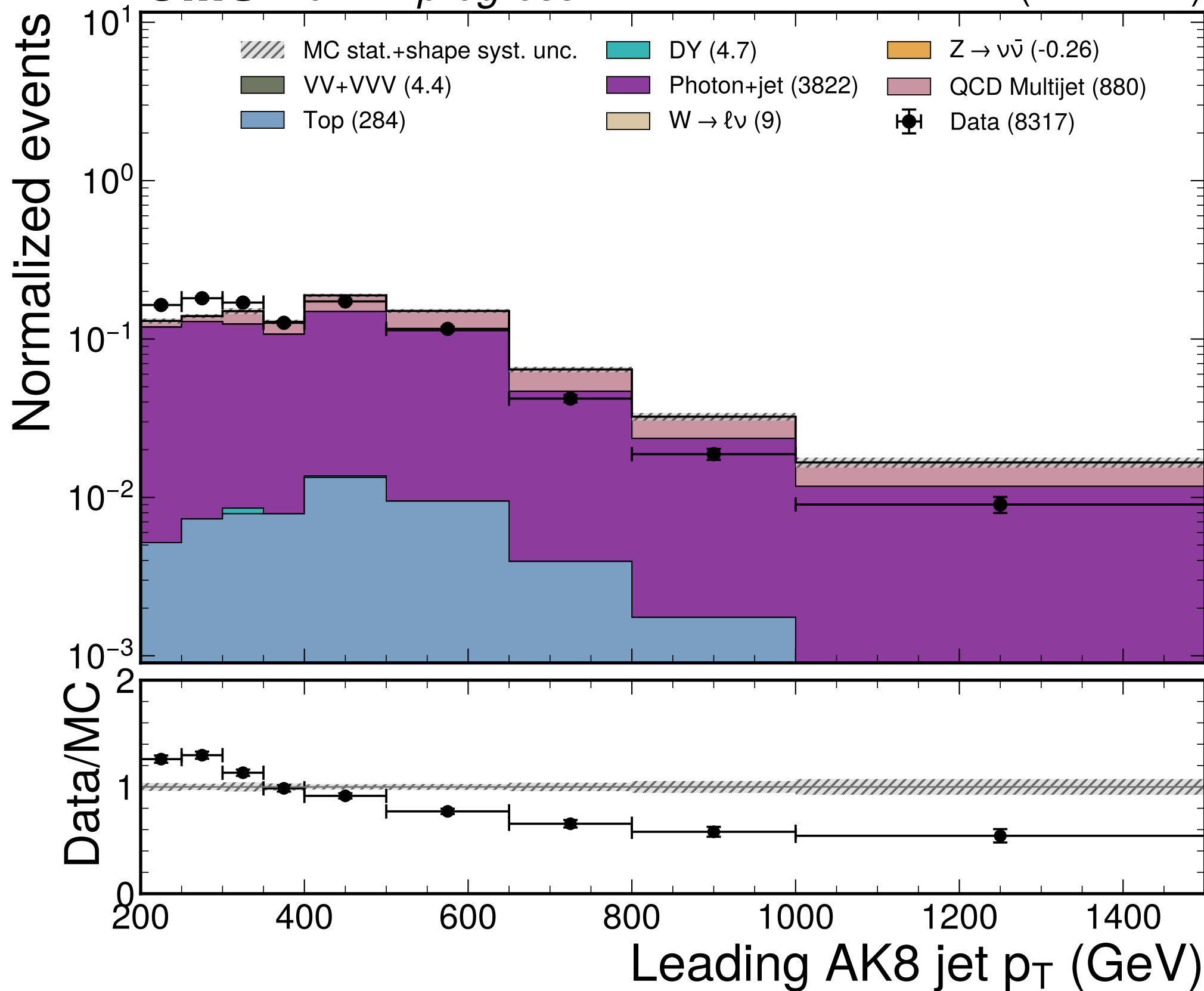
CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)



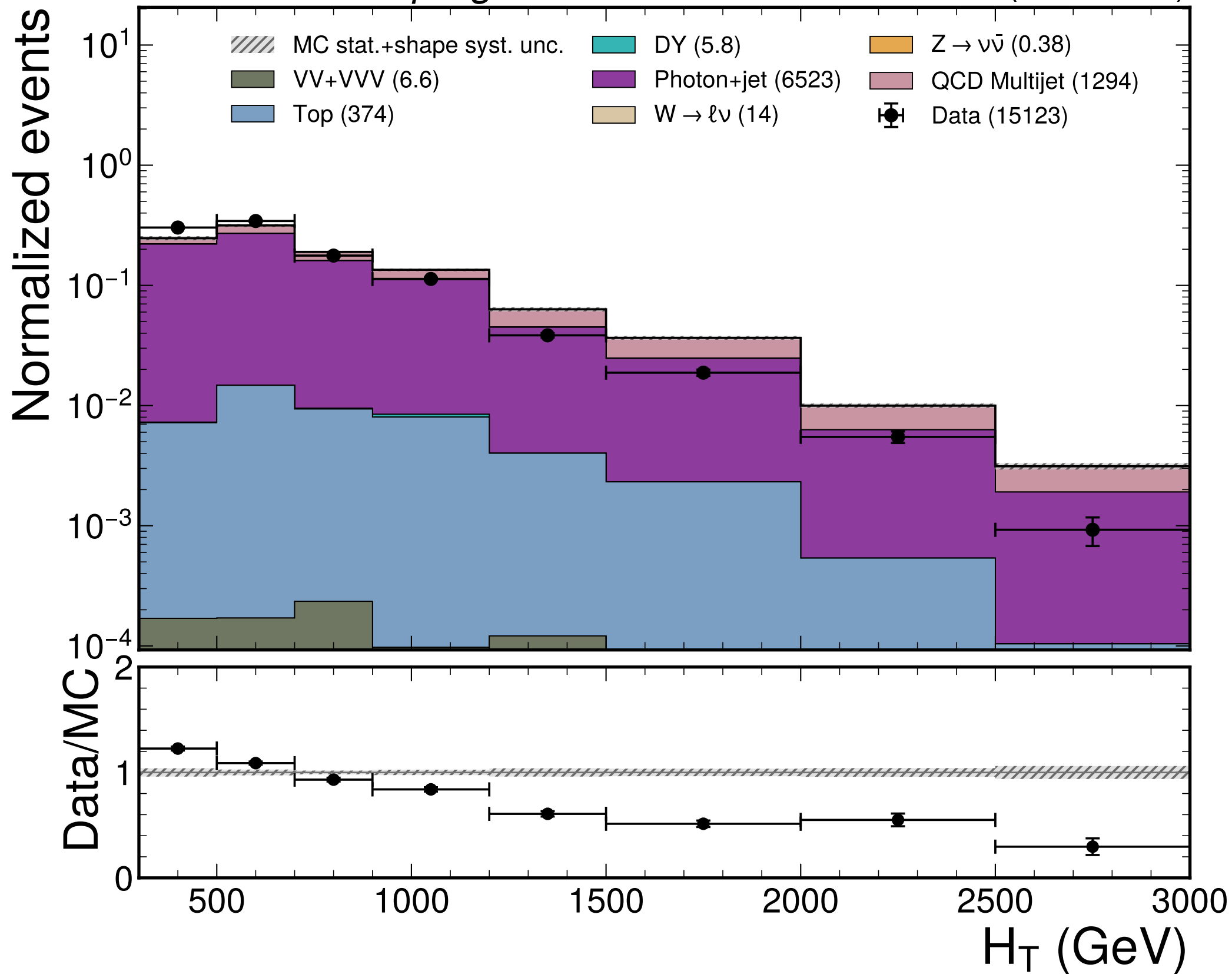
CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)



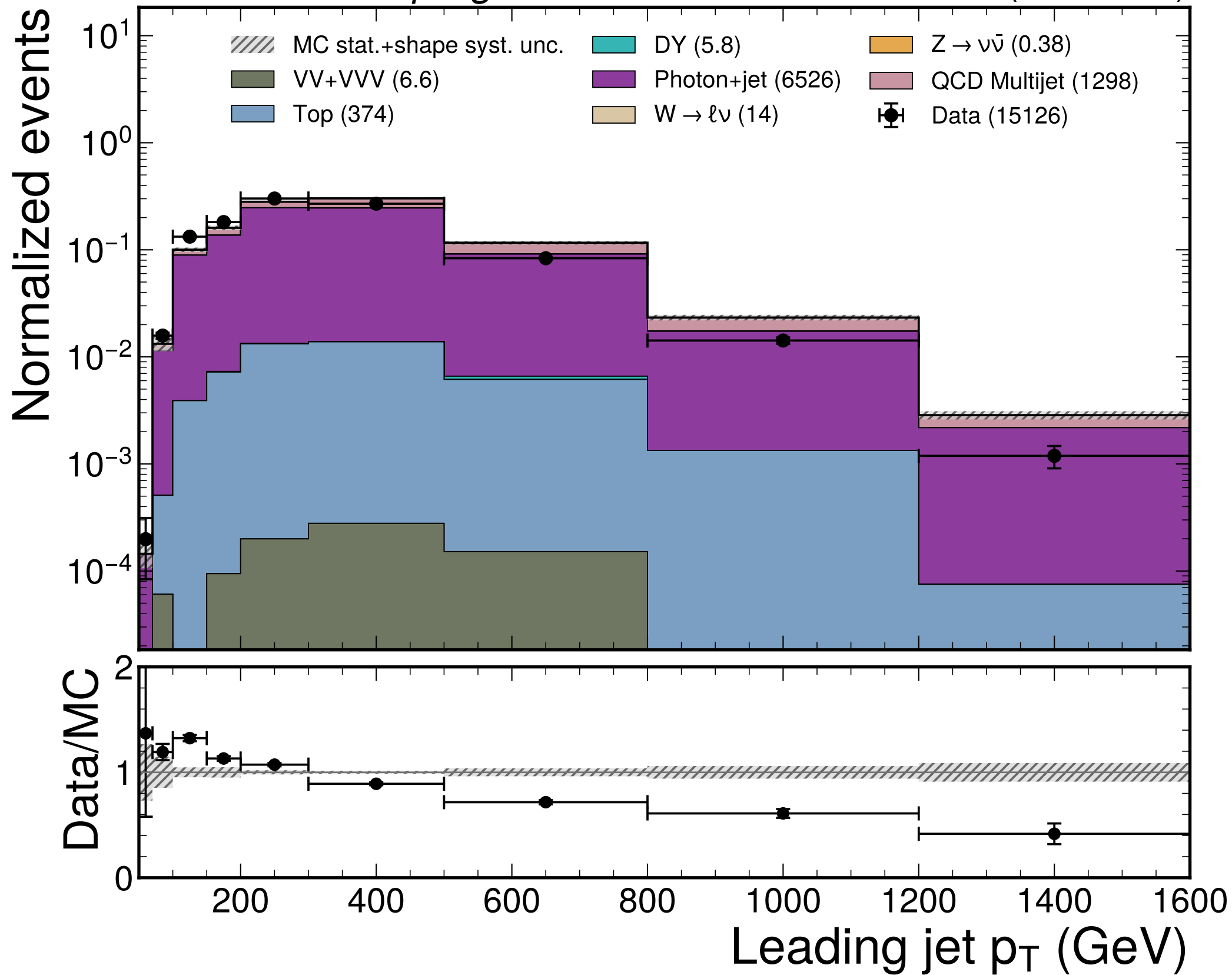
CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)



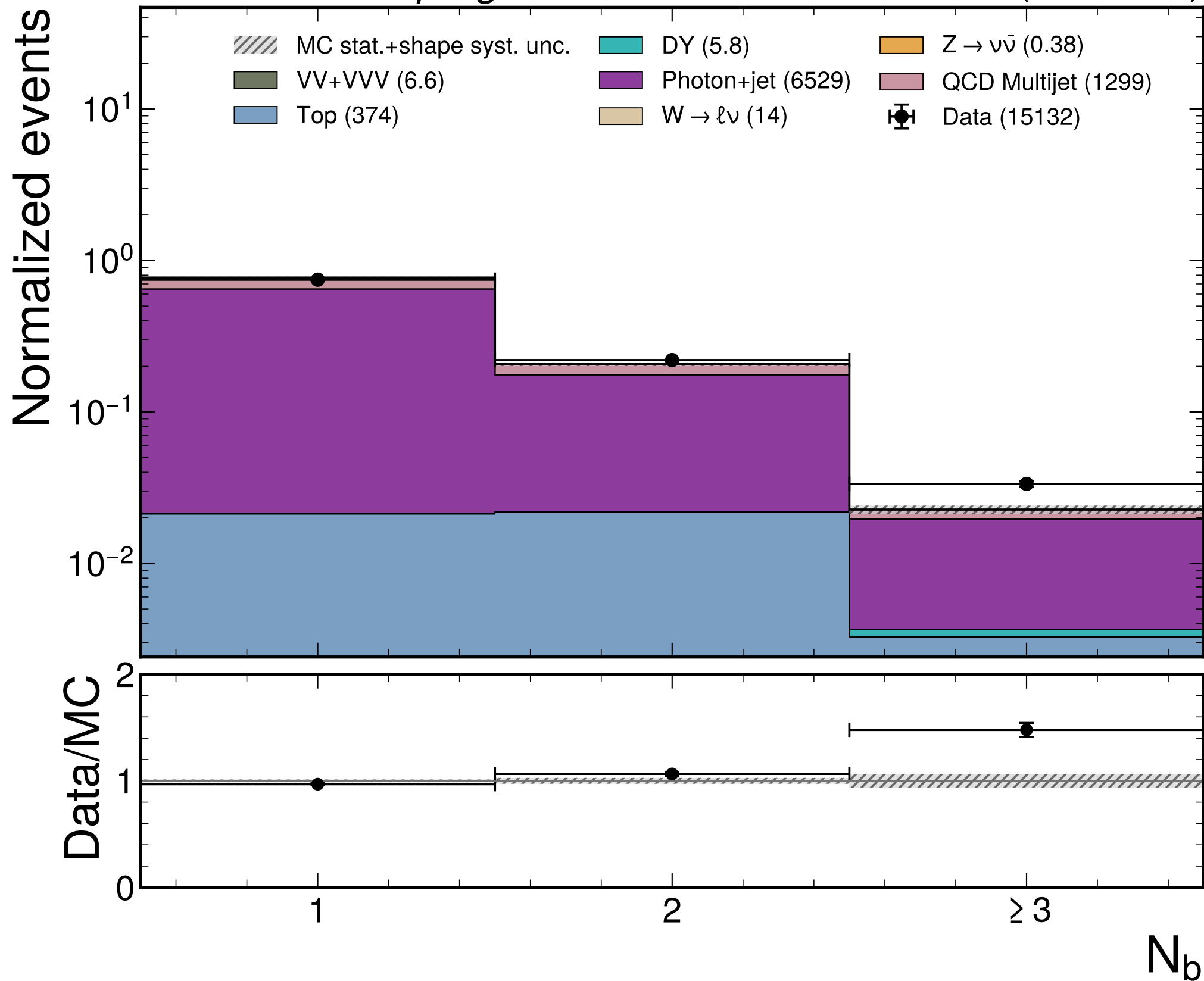
CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)



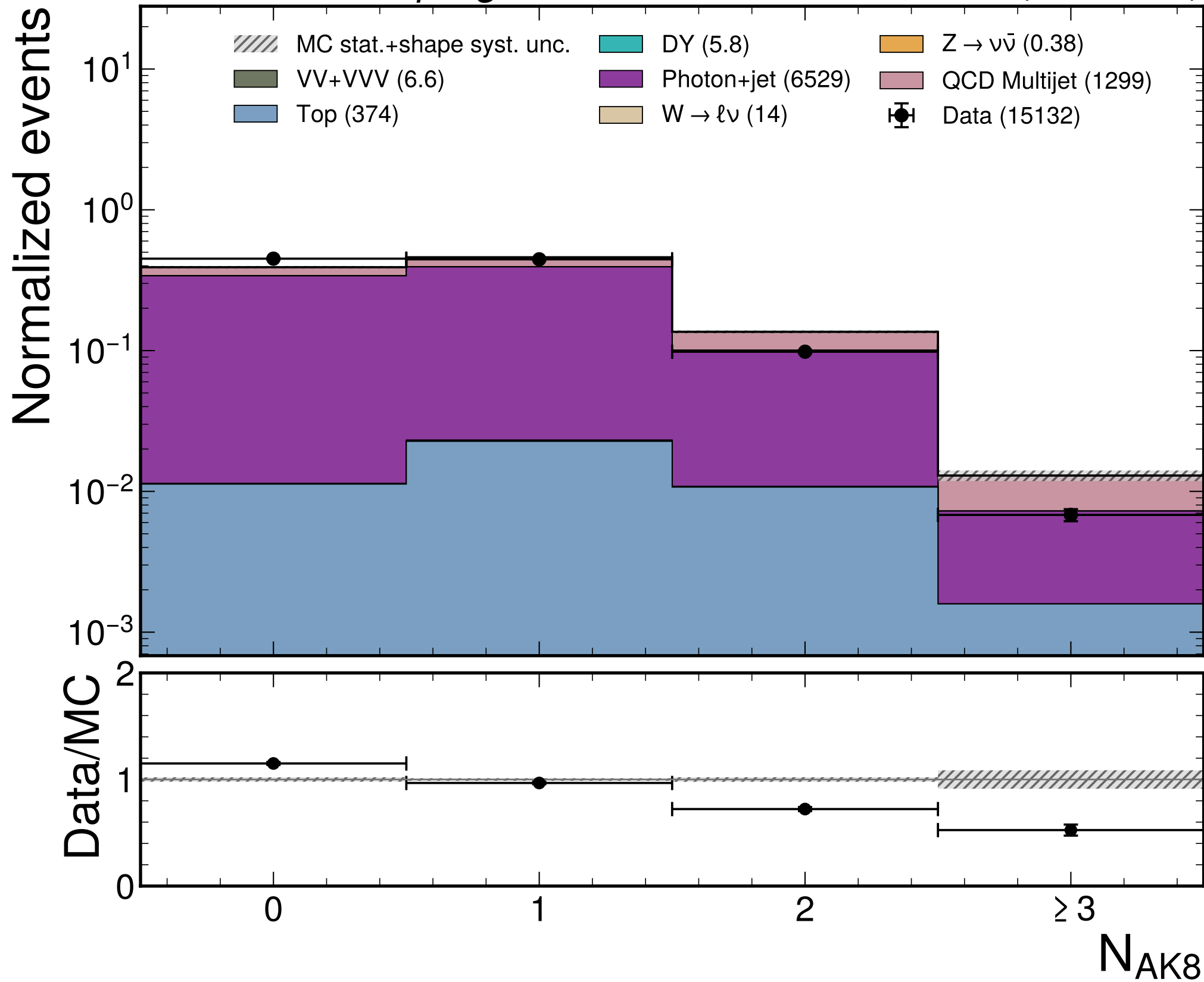
CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)



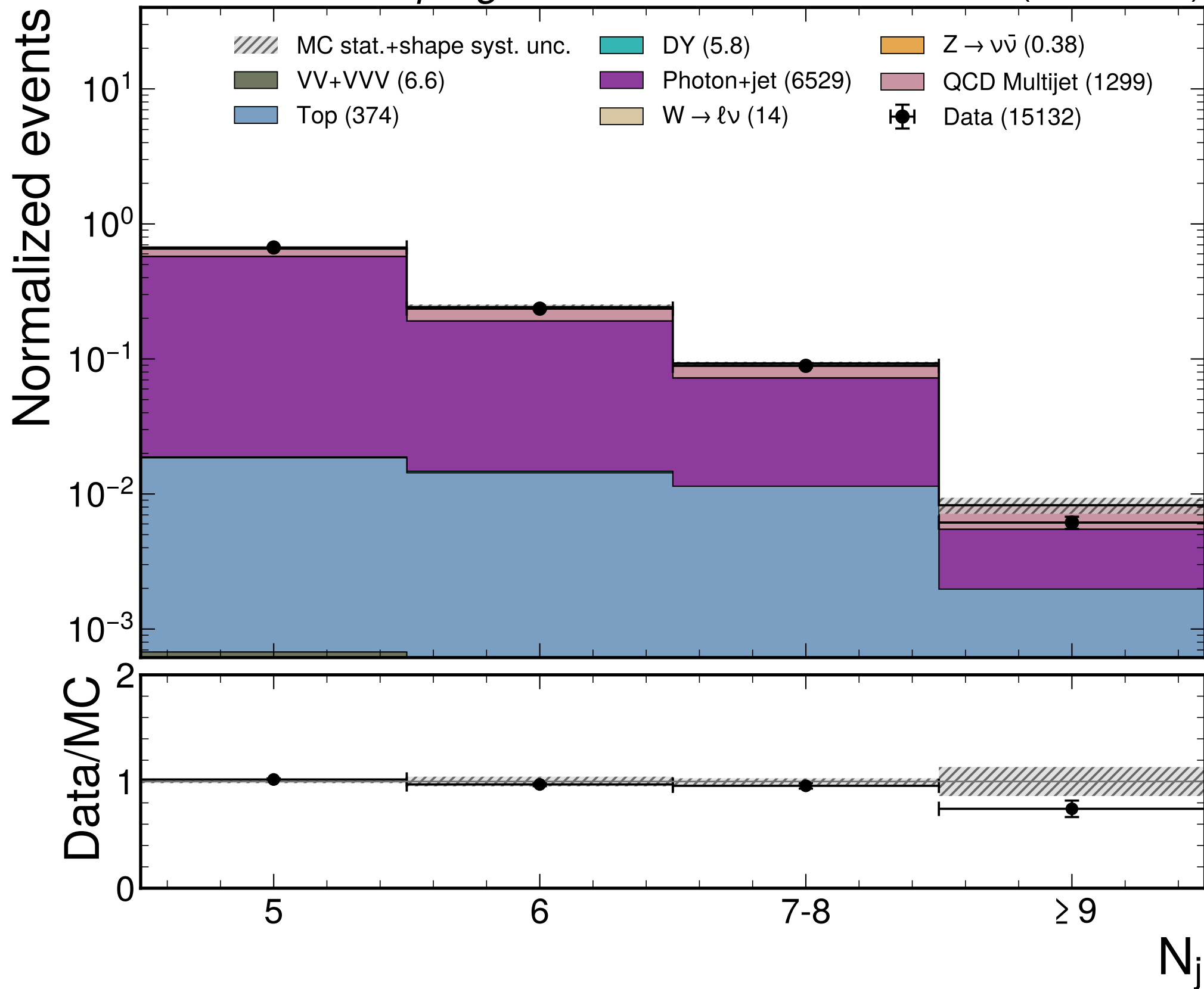
CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)



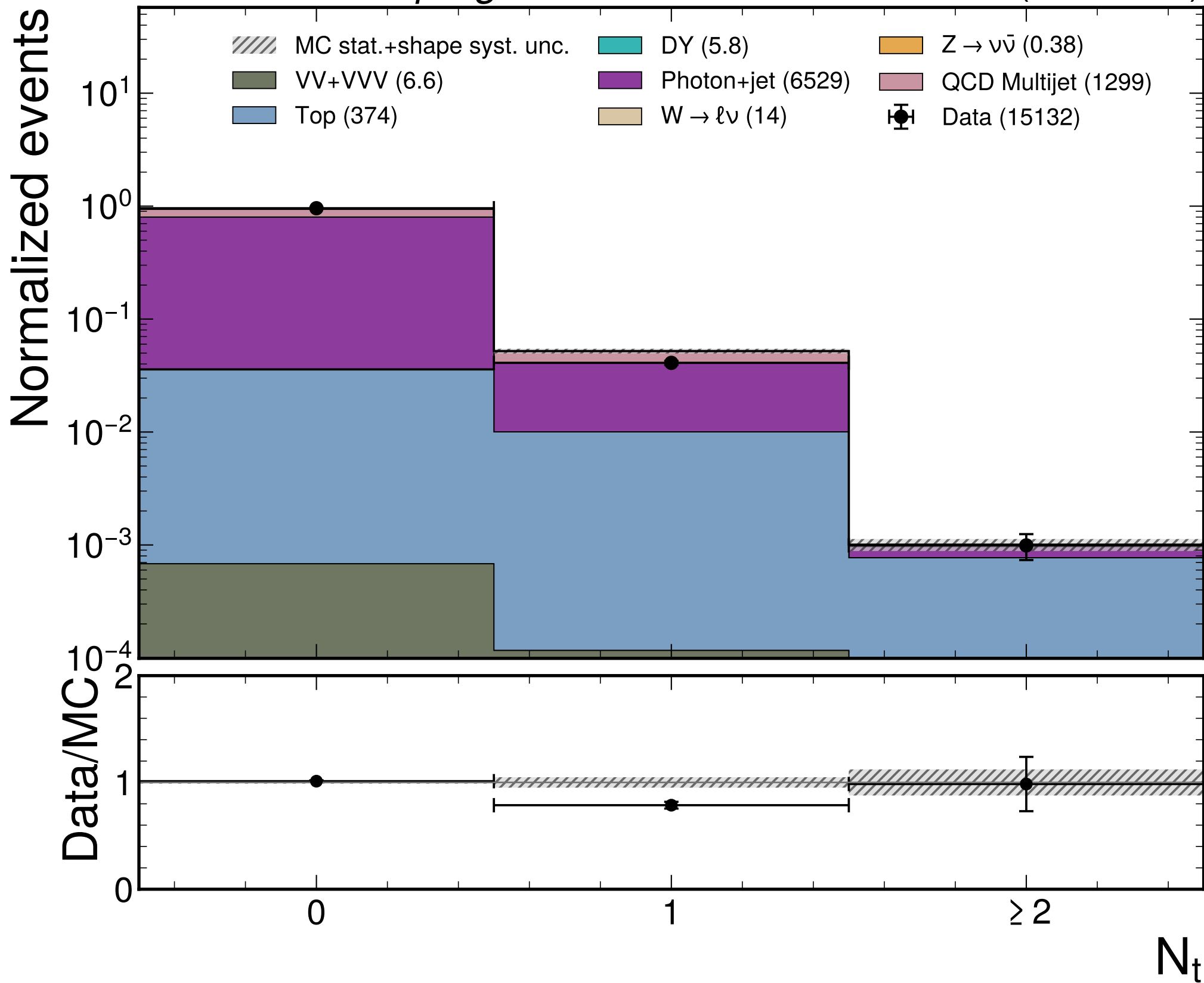
CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)



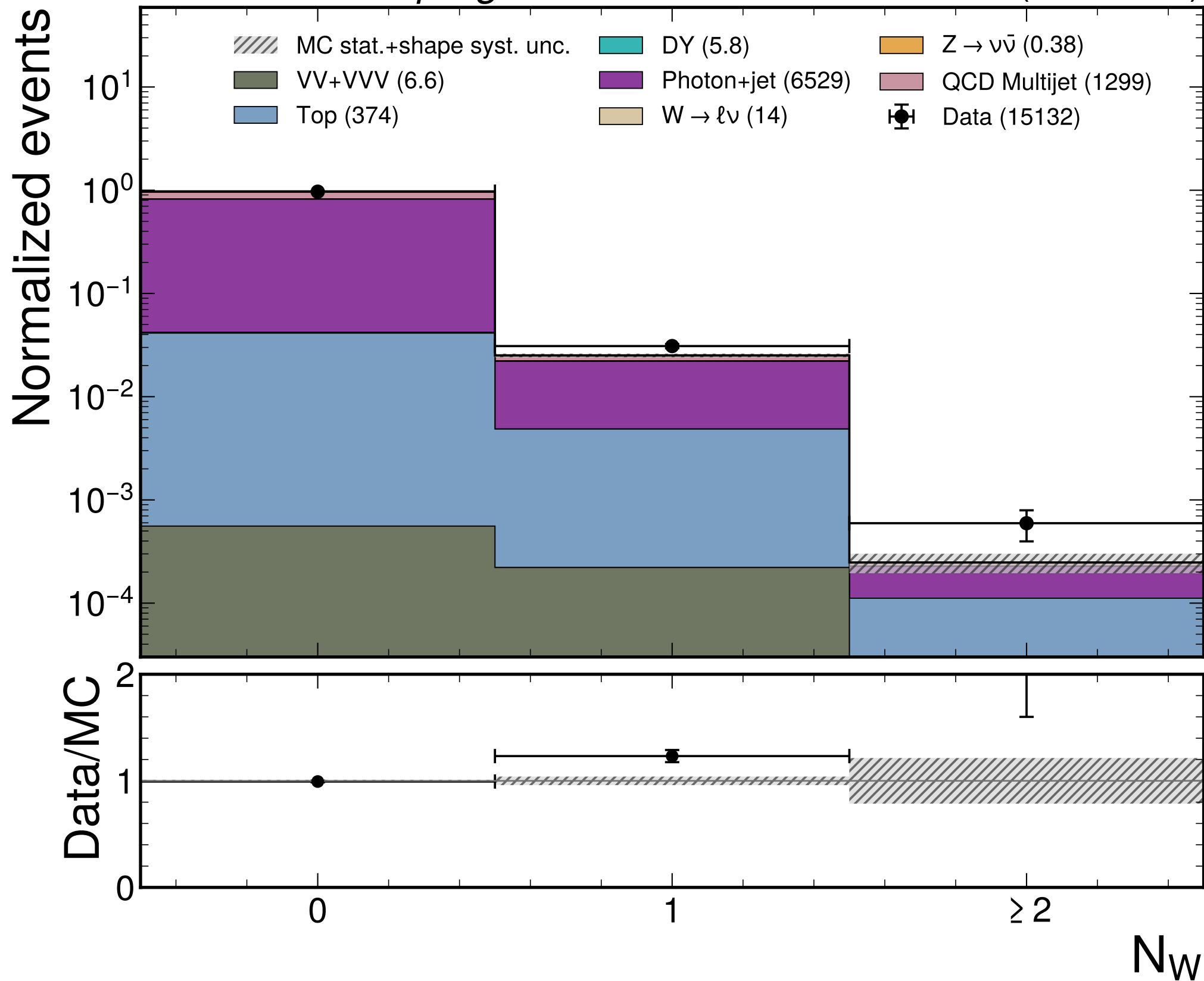
CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)



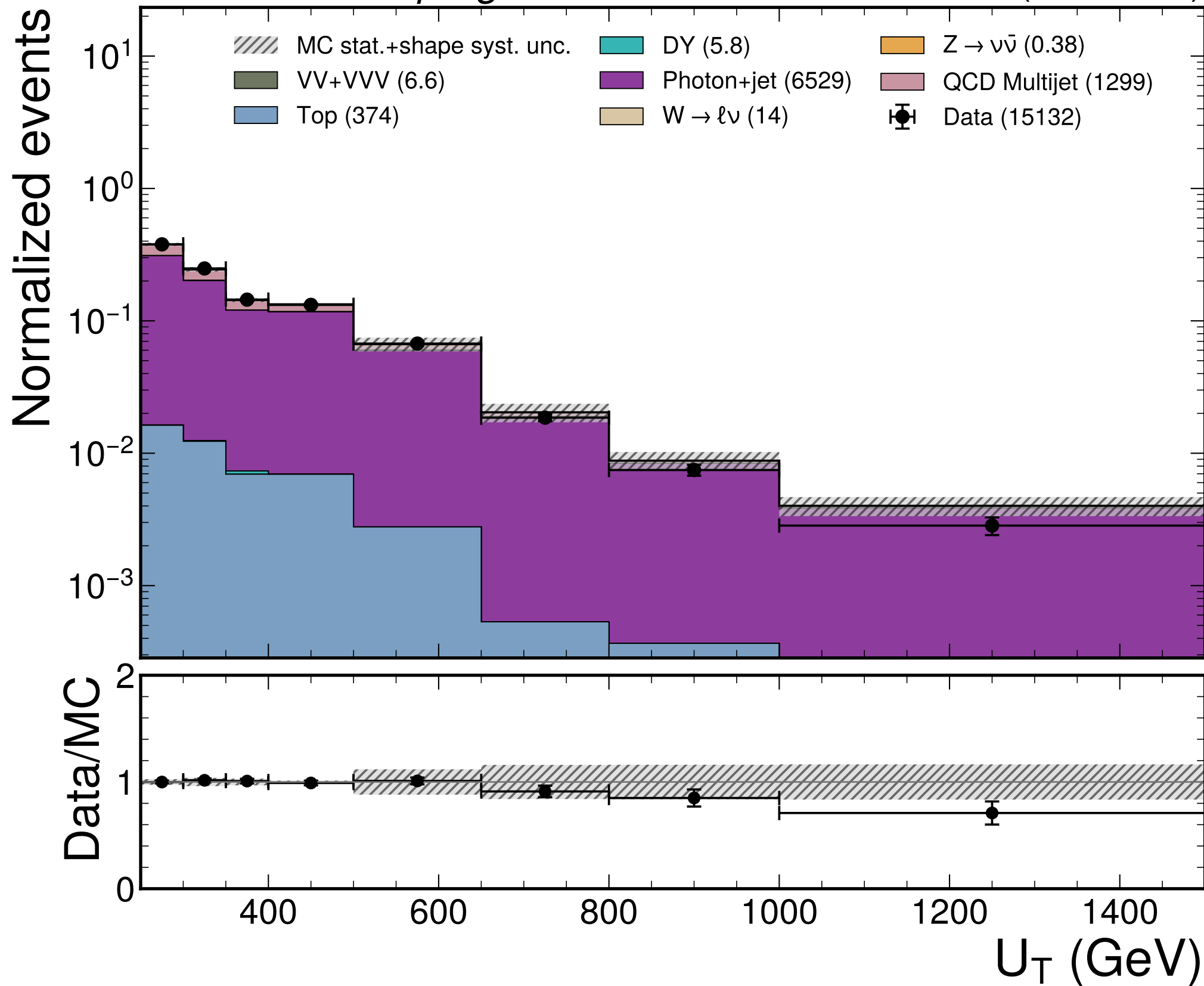
CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)



CMS *Work in progress*

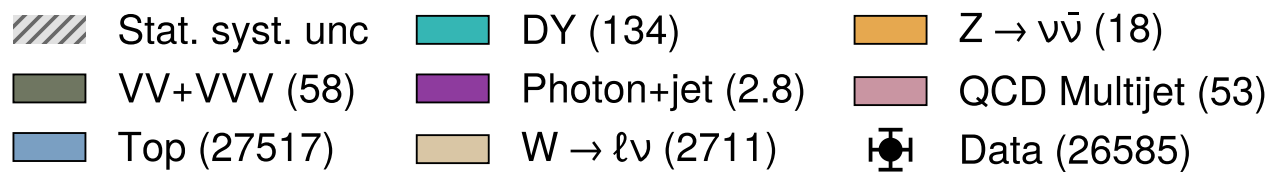
109.82 fb⁻¹ (13.6 TeV)



CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)

Events



Data/MC

10⁻²

2

0

200

400

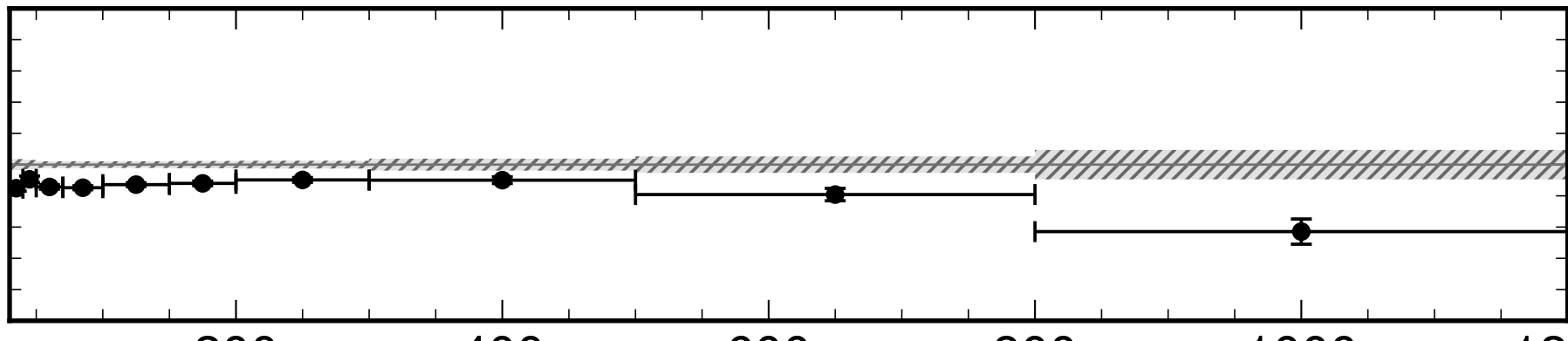
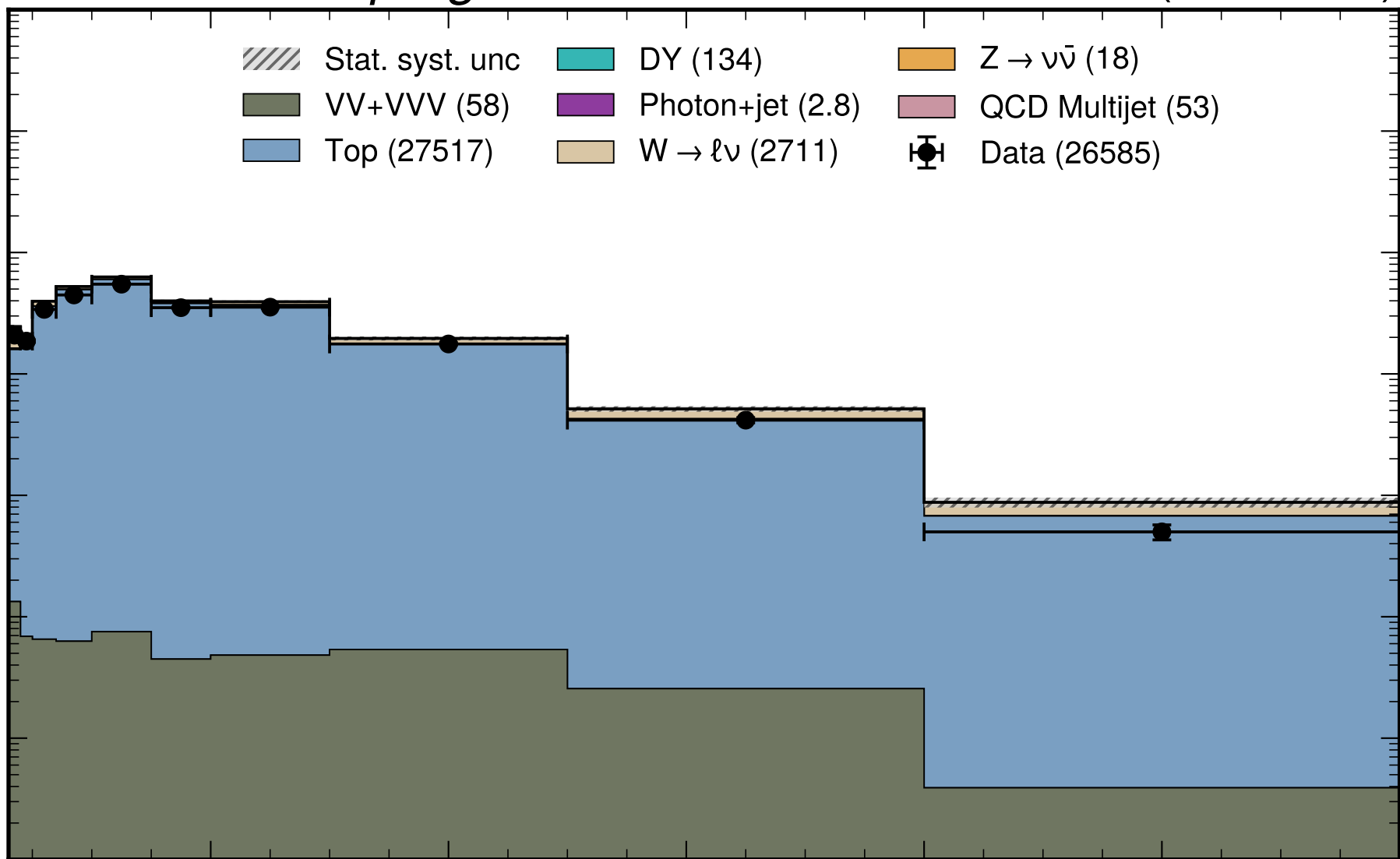
600

800

1000

1200

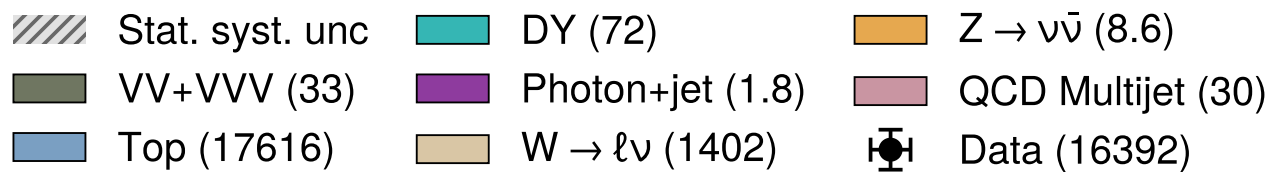
Leading b-jet p_T (GeV)



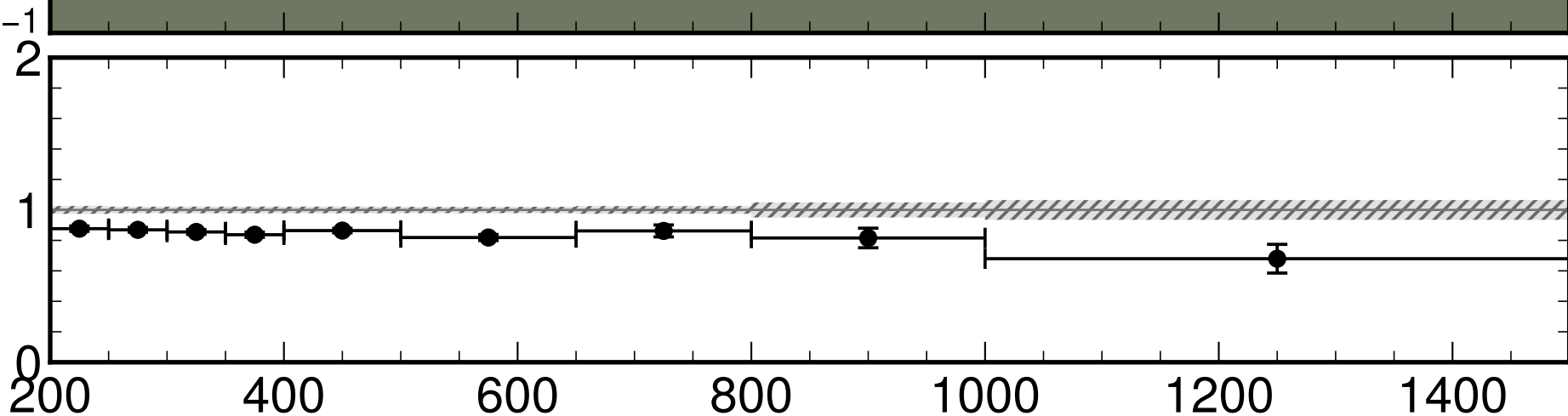
CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)

Events



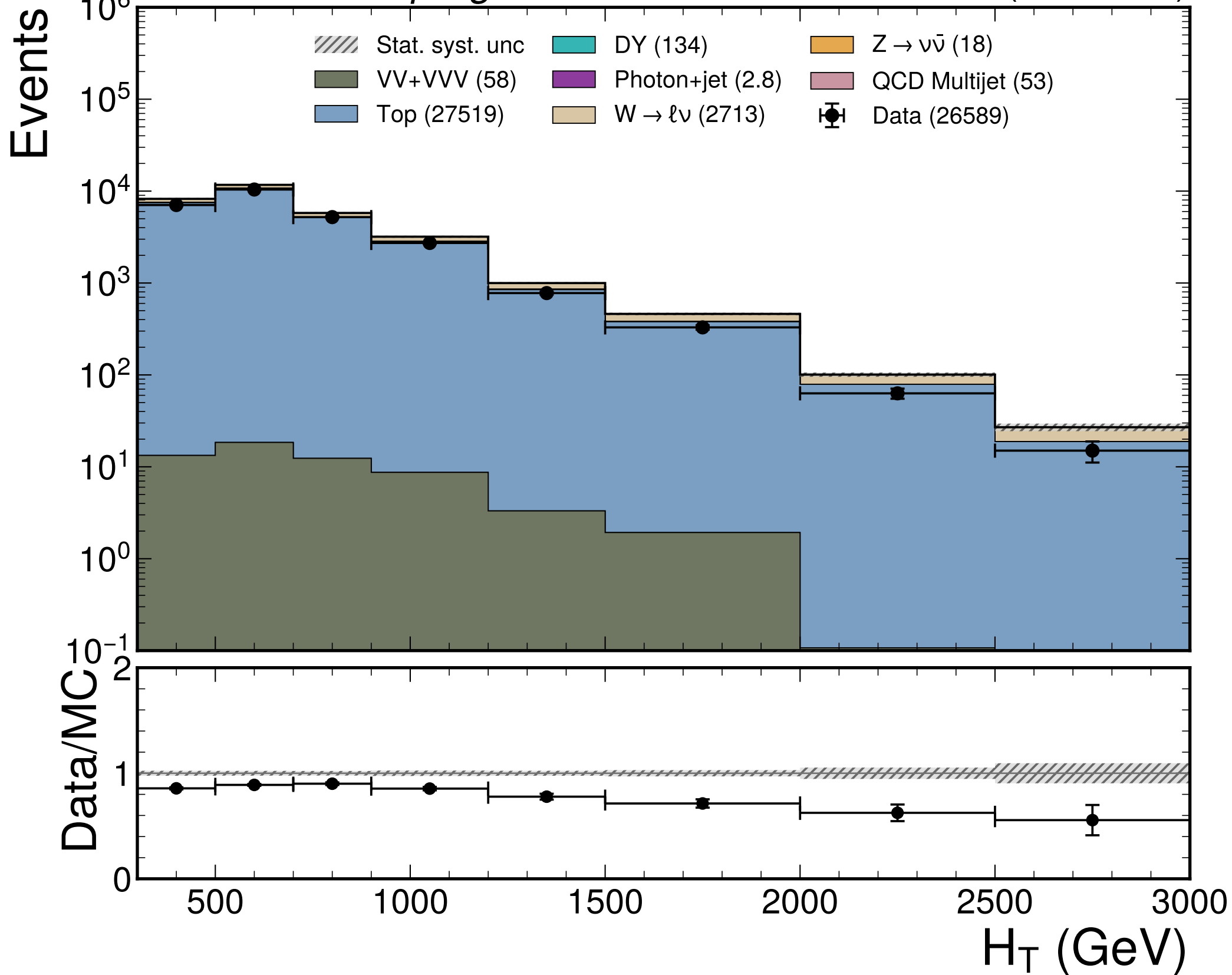
Data/MC



Leading AK8 jet p_T (GeV)

CMS *Work in progress*

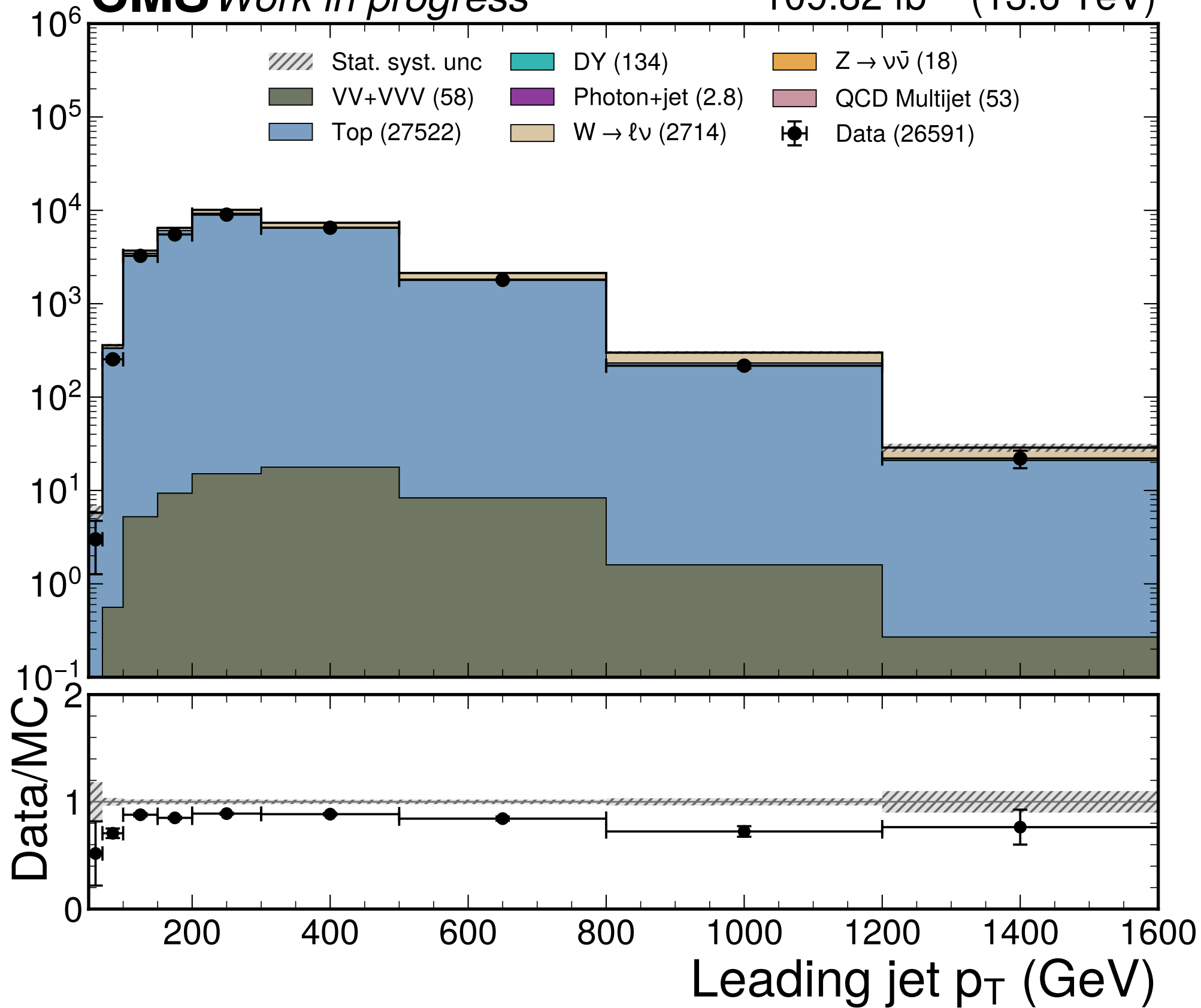
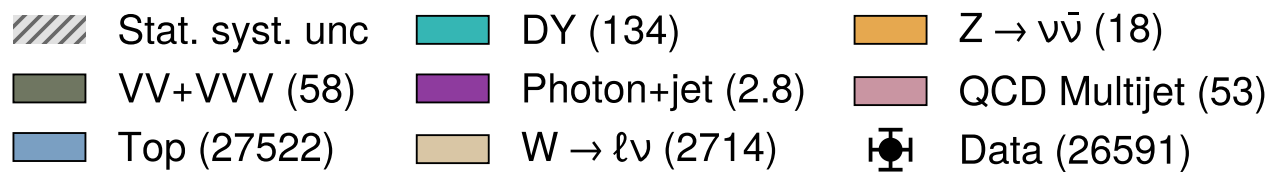
109.82 fb⁻¹ (13.6 TeV)



CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)

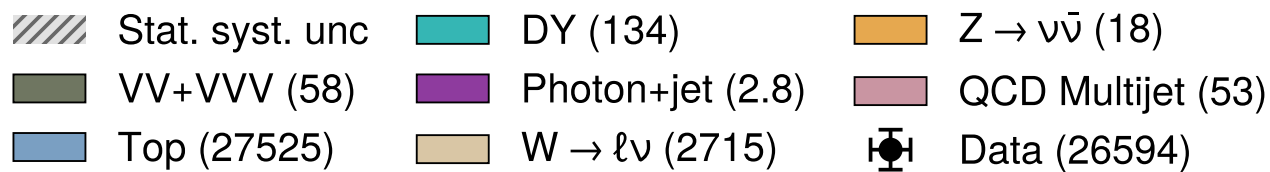
Events



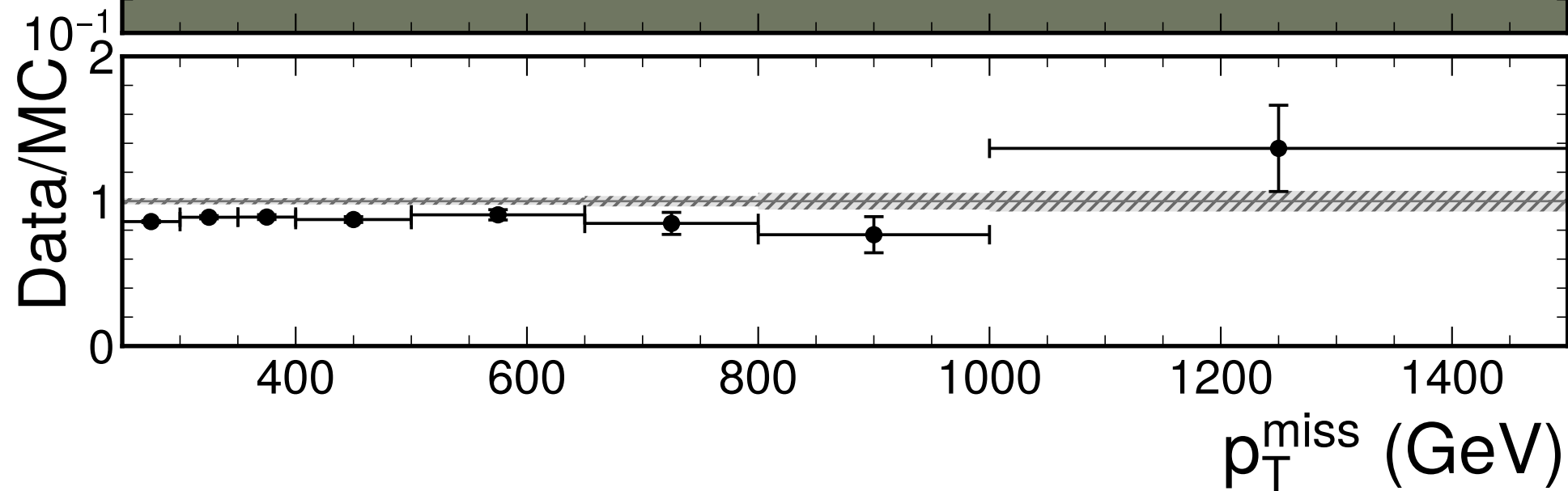
CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)

Events



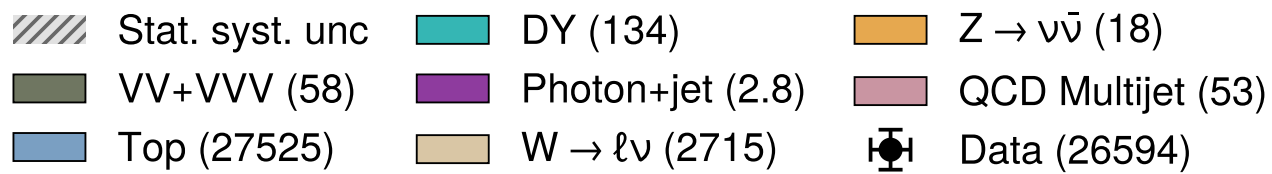
Data/MC



CMS *Work in progress*

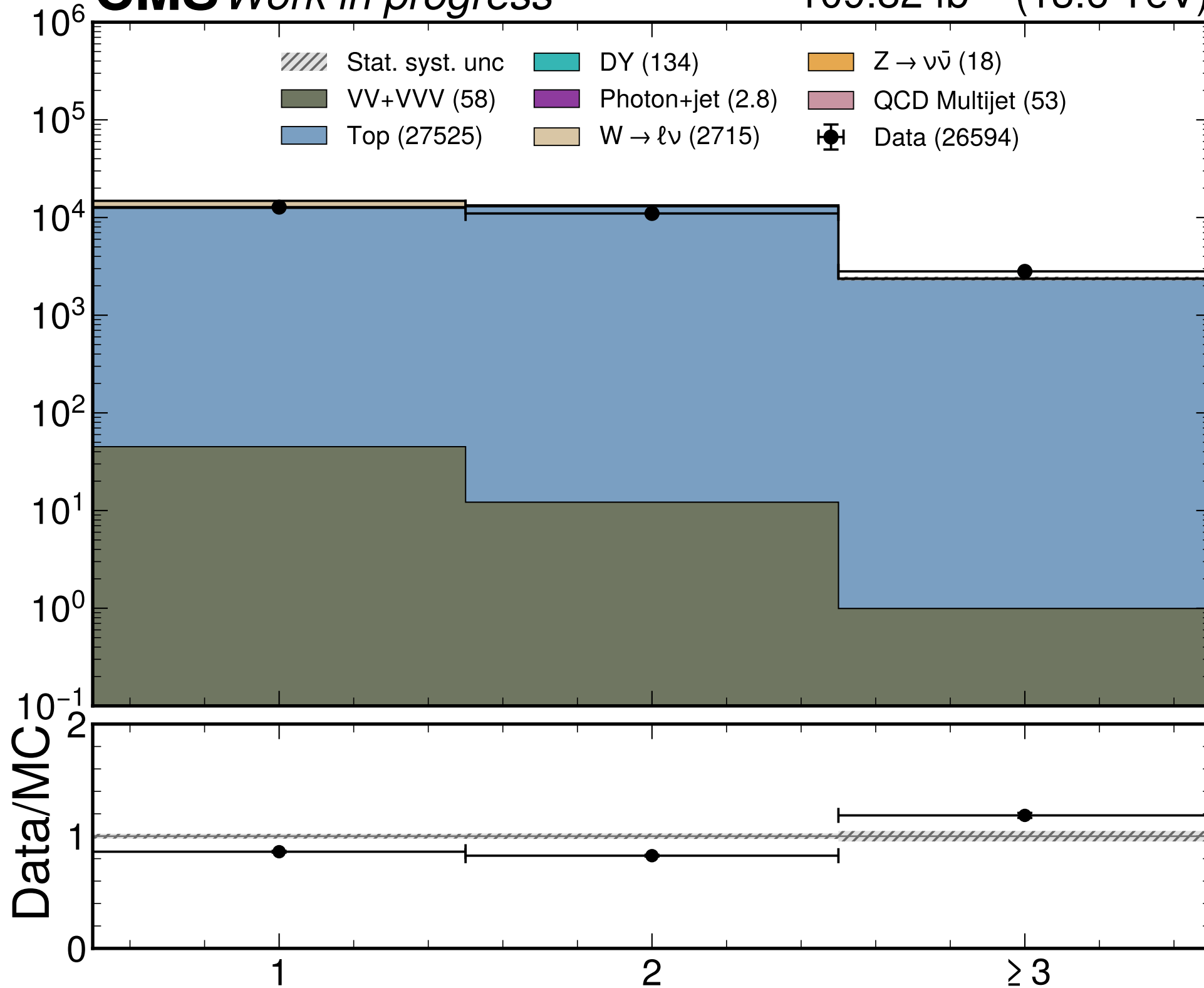
109.82 fb⁻¹ (13.6 TeV)

Events

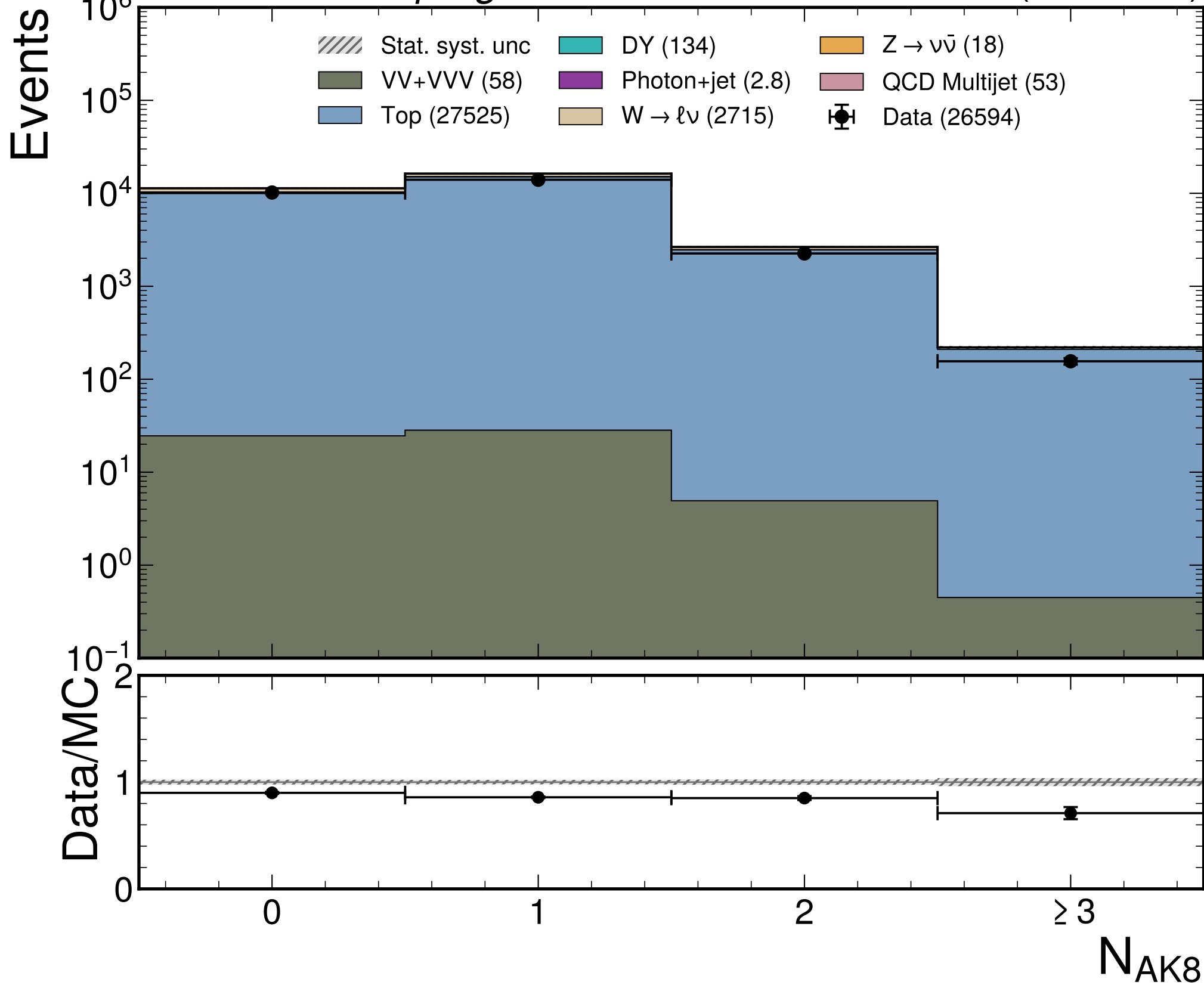


Data/MC

N_b



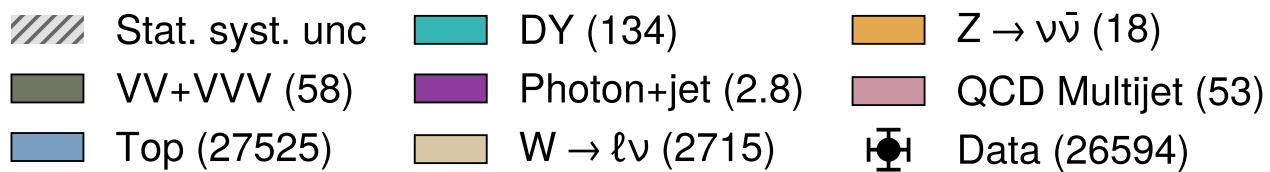
109.82 fb⁻¹ (13.6 TeV)



CMS *Work in progress*

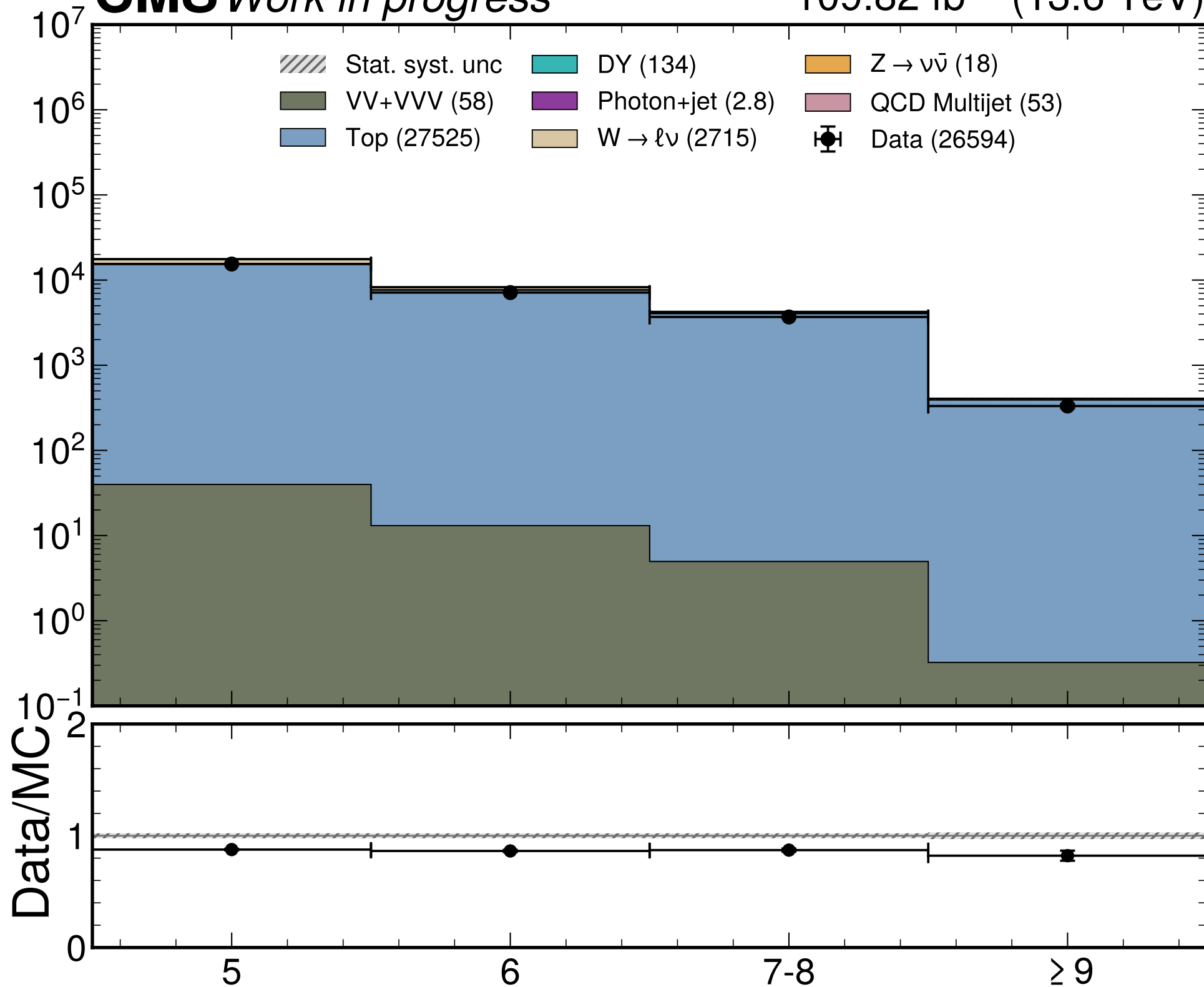
109.82 fb⁻¹ (13.6 TeV)

Events



Data/MC

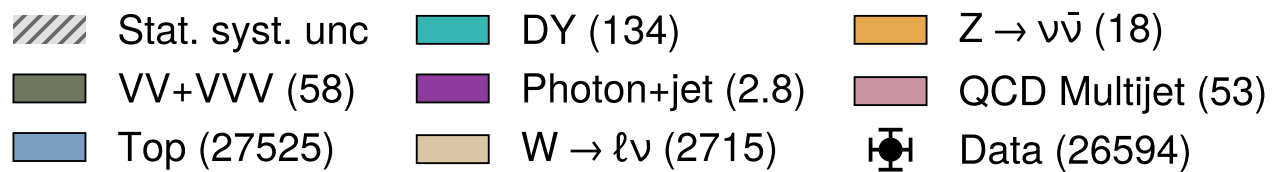
N_j



CMS *Work in progress*

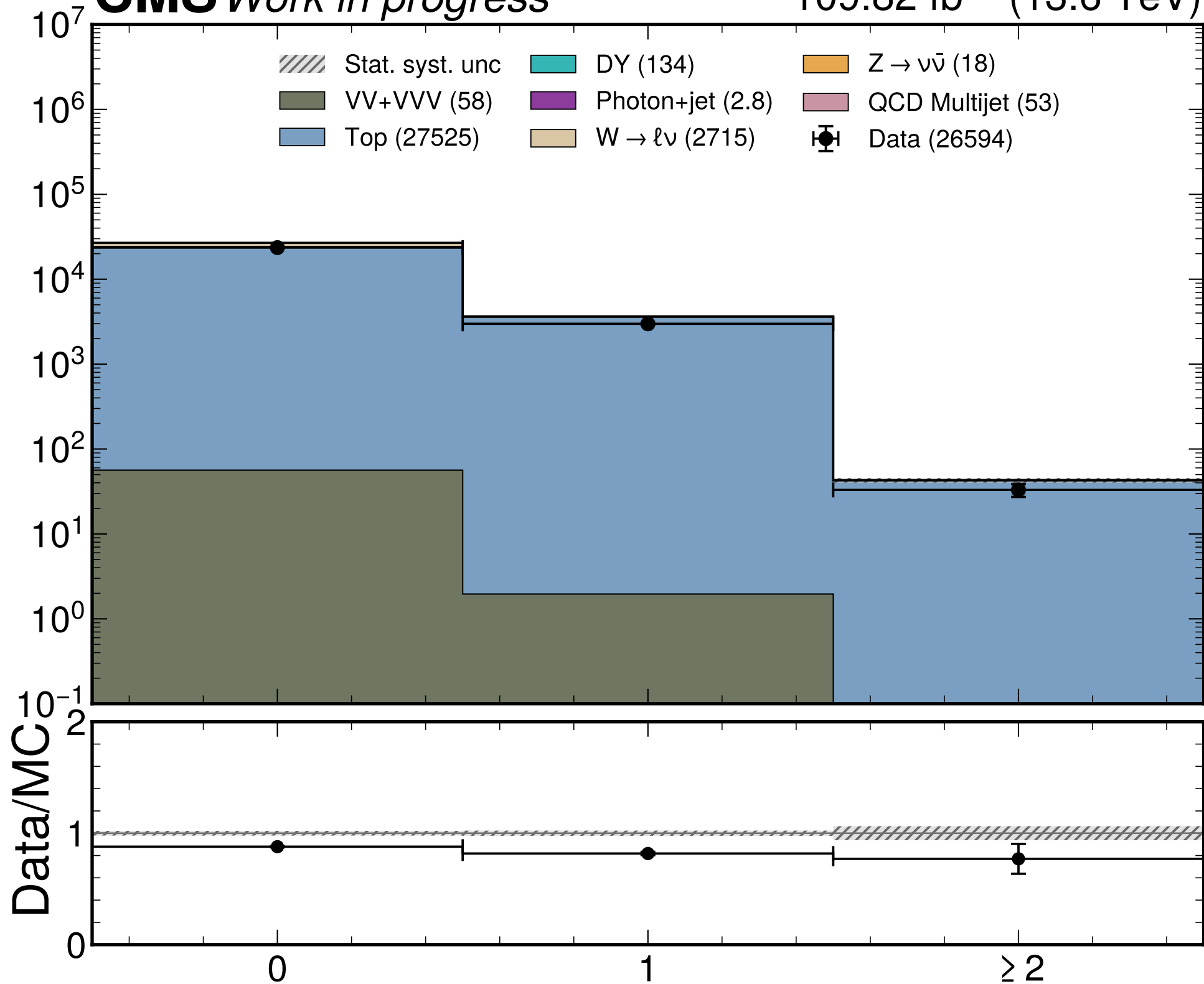
109.82 fb⁻¹ (13.6 TeV)

Events



Data/MC

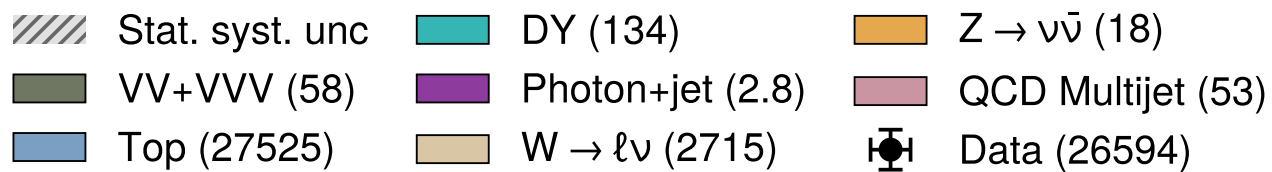
N_t



CMS *Work in progress*

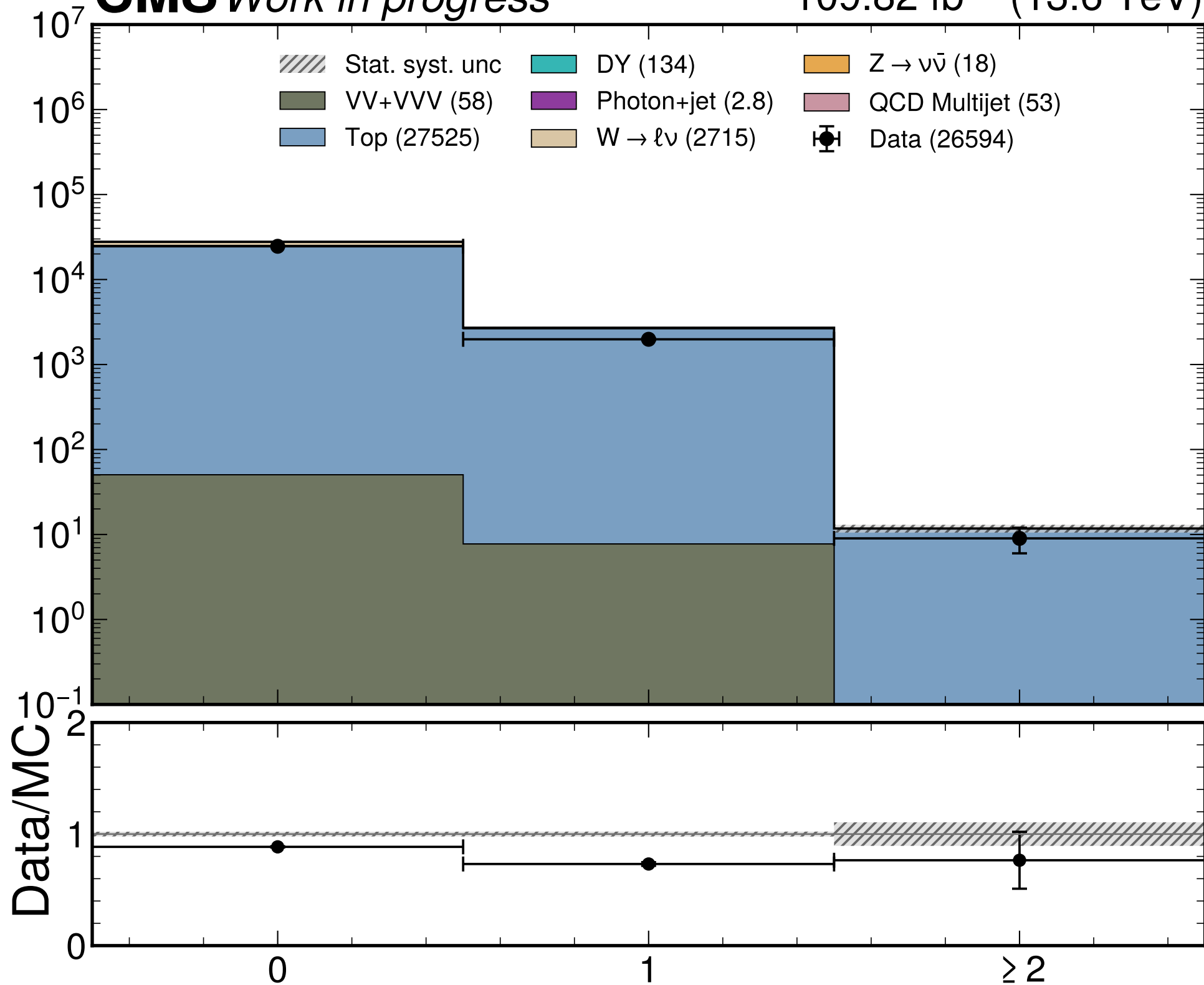
109.82 fb⁻¹ (13.6 TeV)

Events



Data/MC

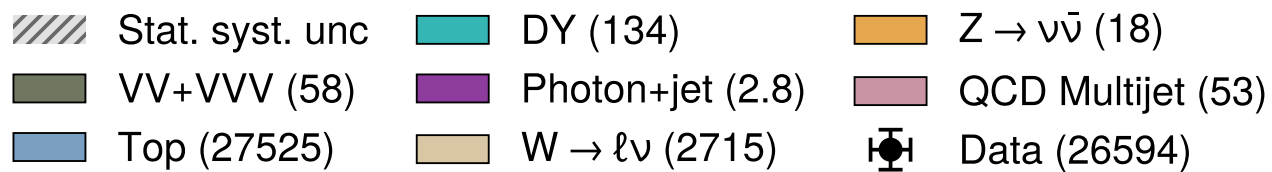
N_W



CMS *Work in progress*

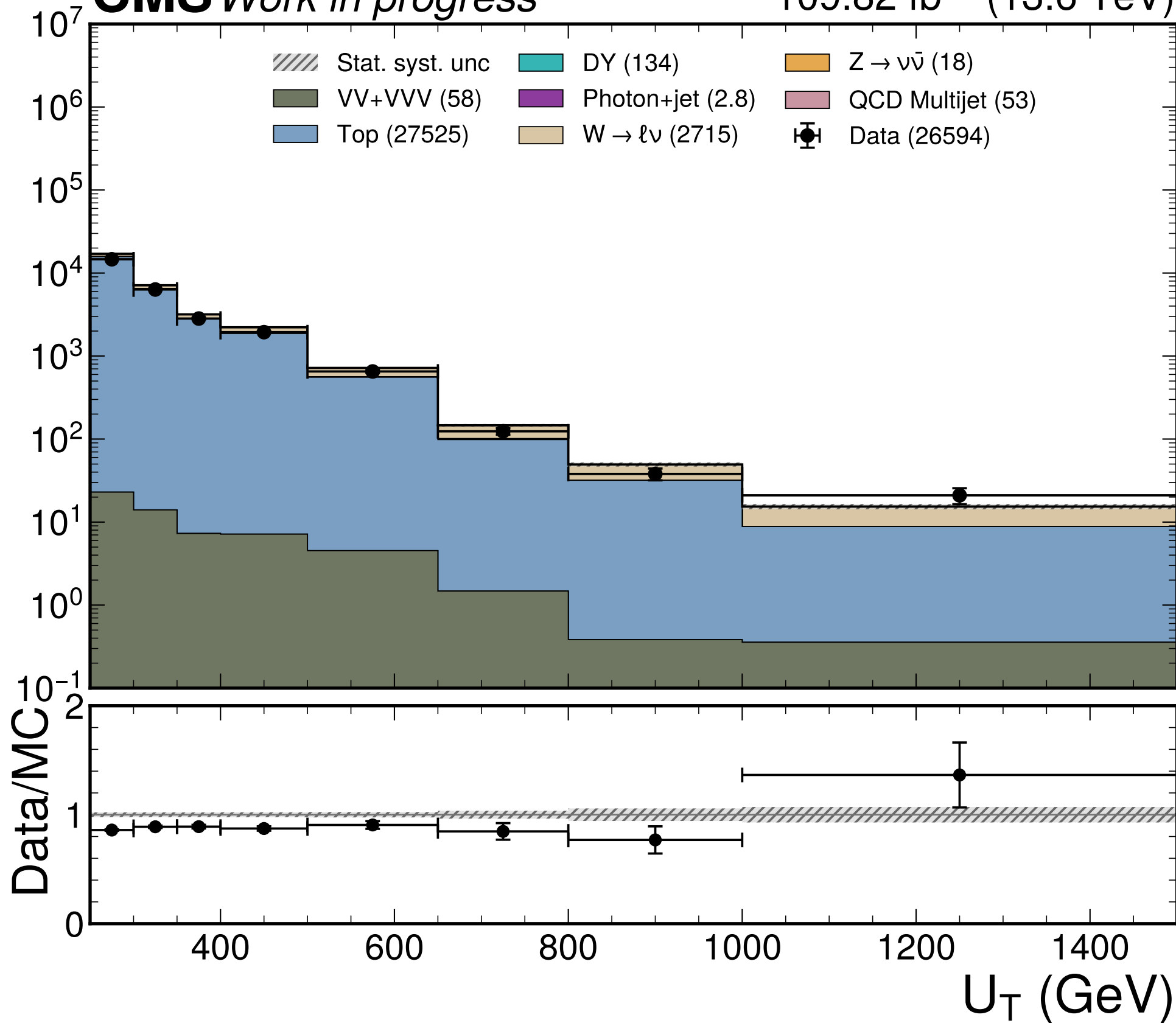
109.82 fb⁻¹ (13.6 TeV)

Events



Data/MC

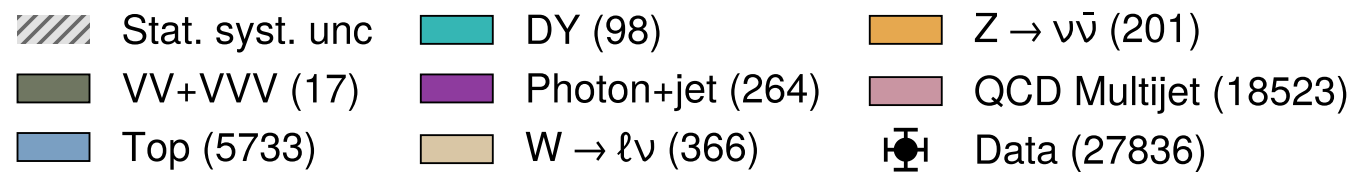
U_T (GeV)



CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)

Events



Data/MC

10⁻¹
2
1
0

200

400

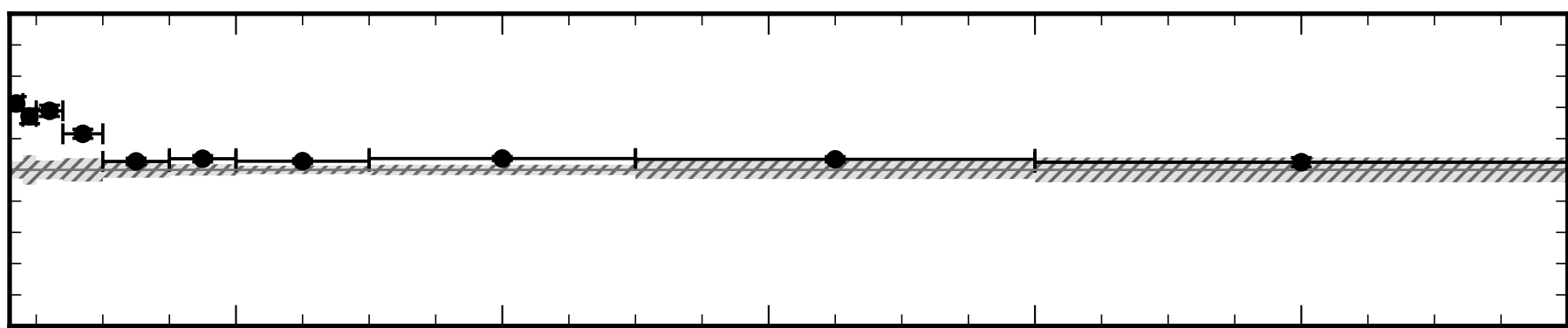
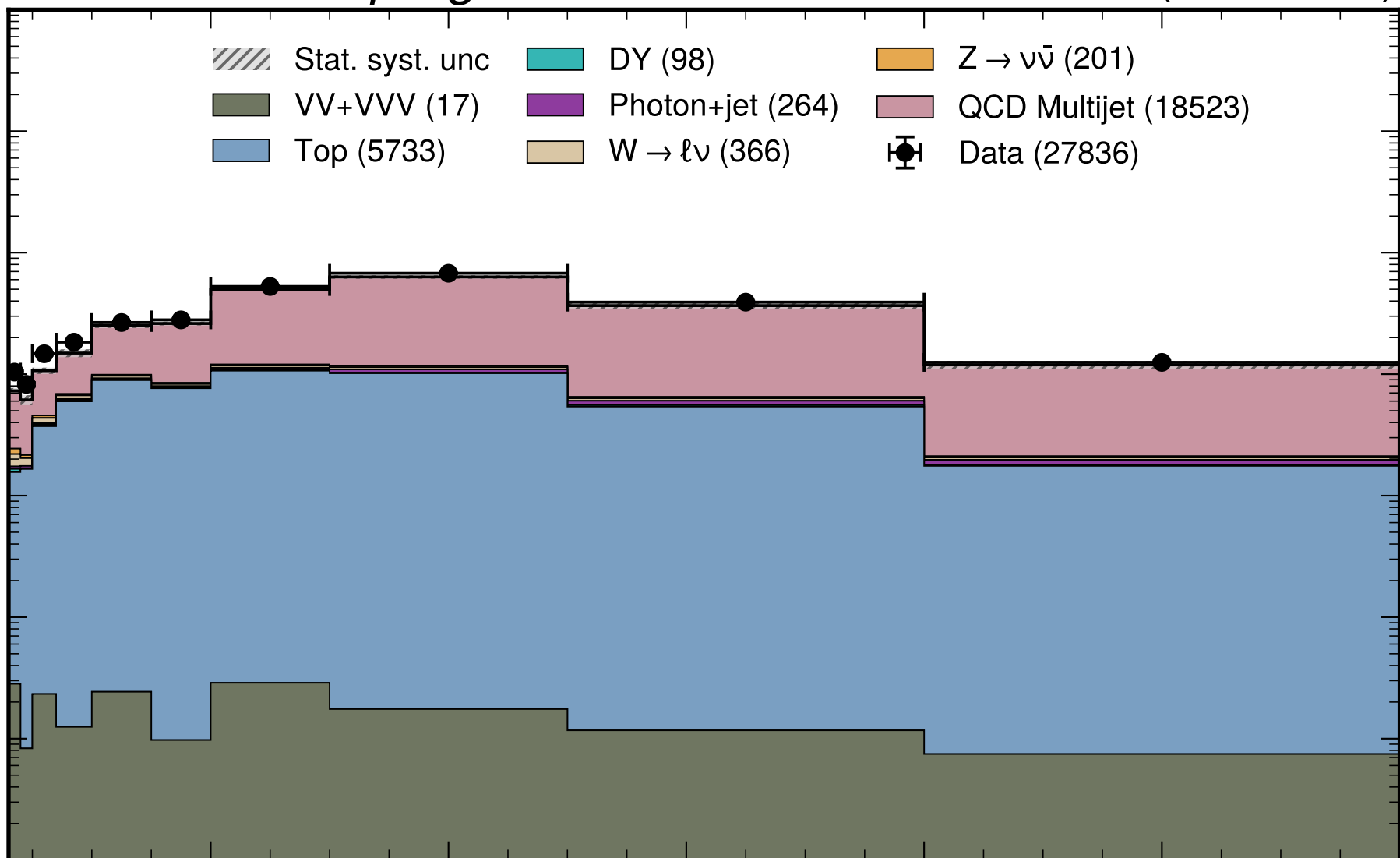
600

800

1000

1200

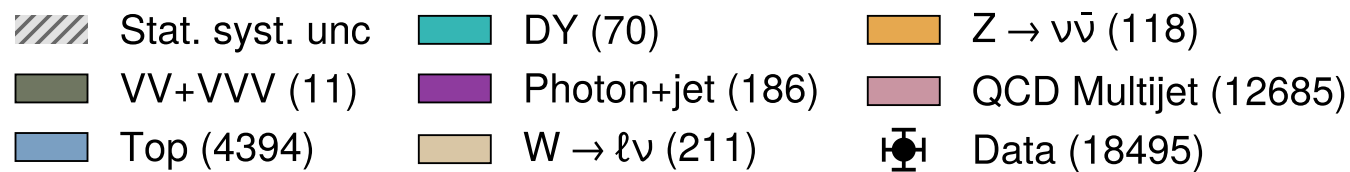
Leading b-jet p_T (GeV)



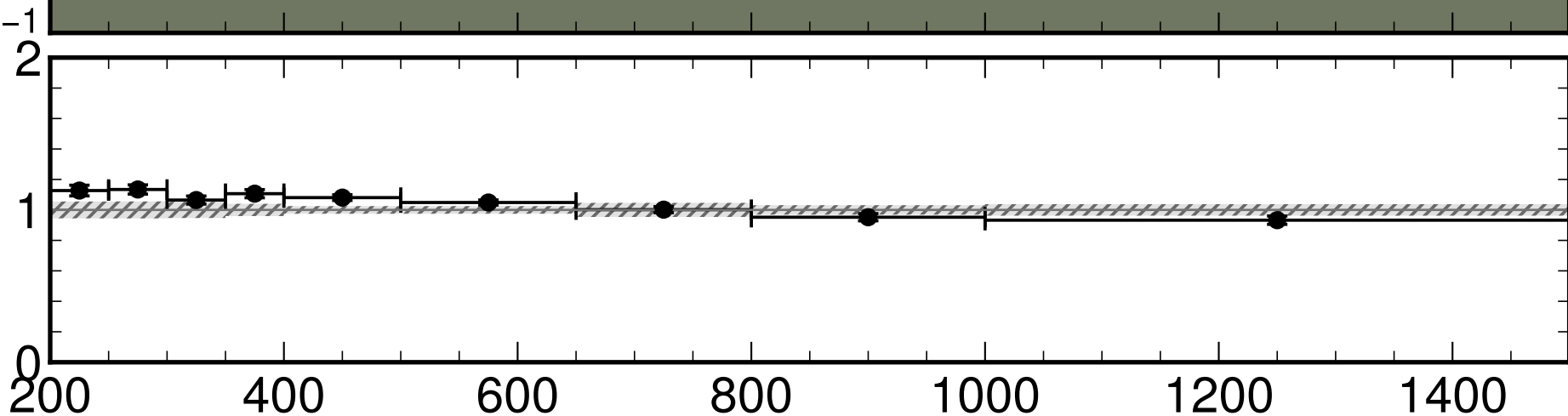
CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)

Events



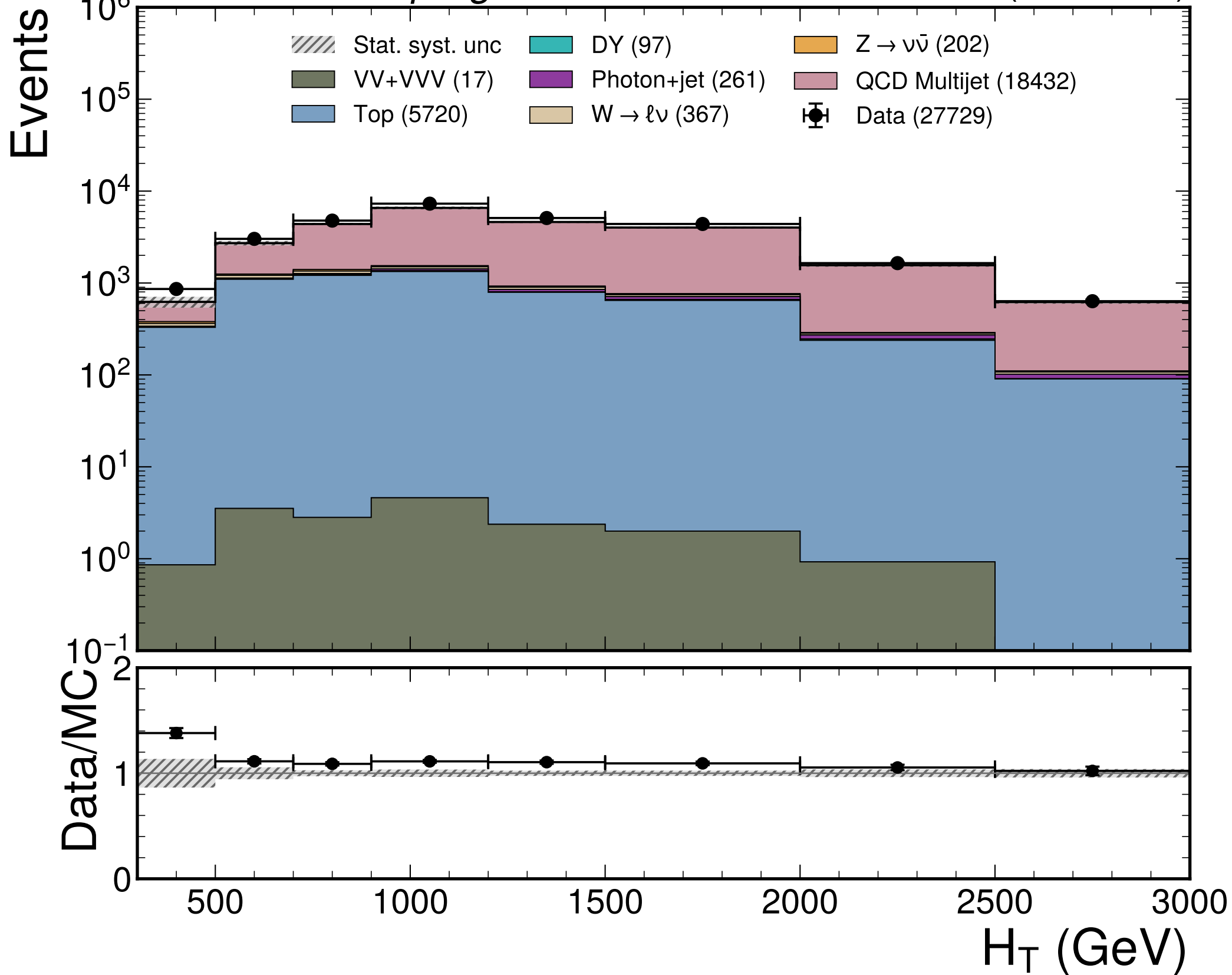
Data/MC



Leading AK8 jet p_T (GeV)

CMS *Work in progress*

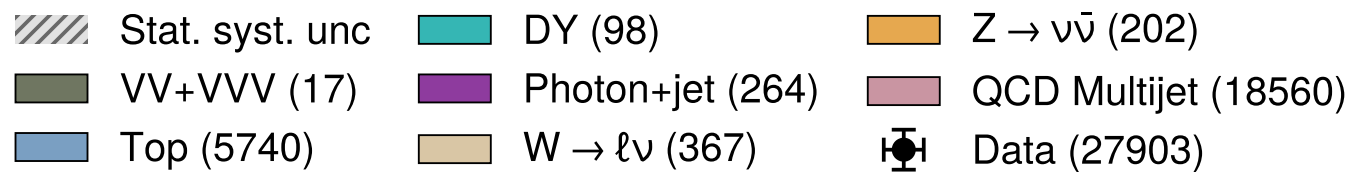
109.82 fb⁻¹ (13.6 TeV)



CMS *Work in progress*

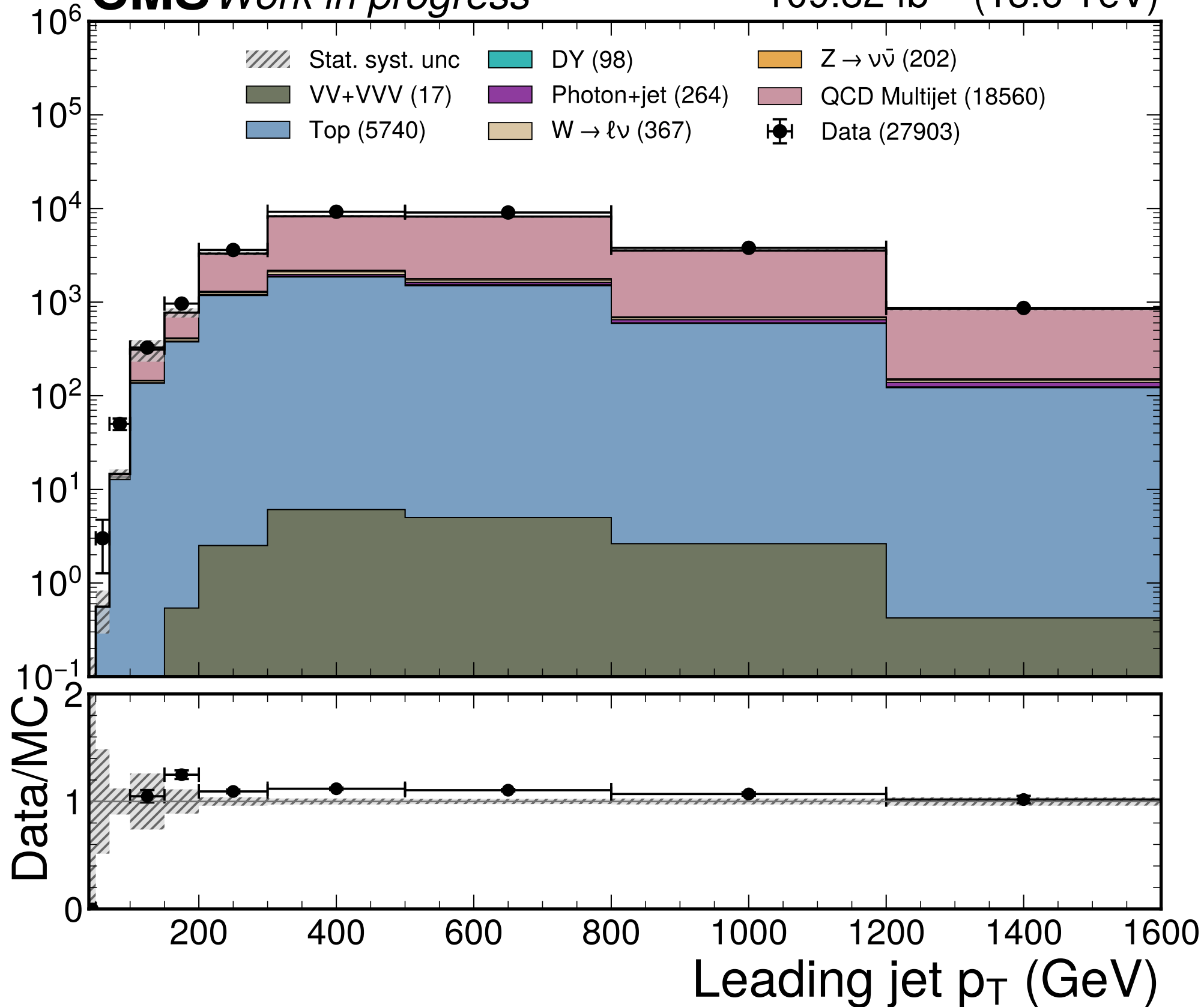
109.82 fb⁻¹ (13.6 TeV)

Events



Data/MC

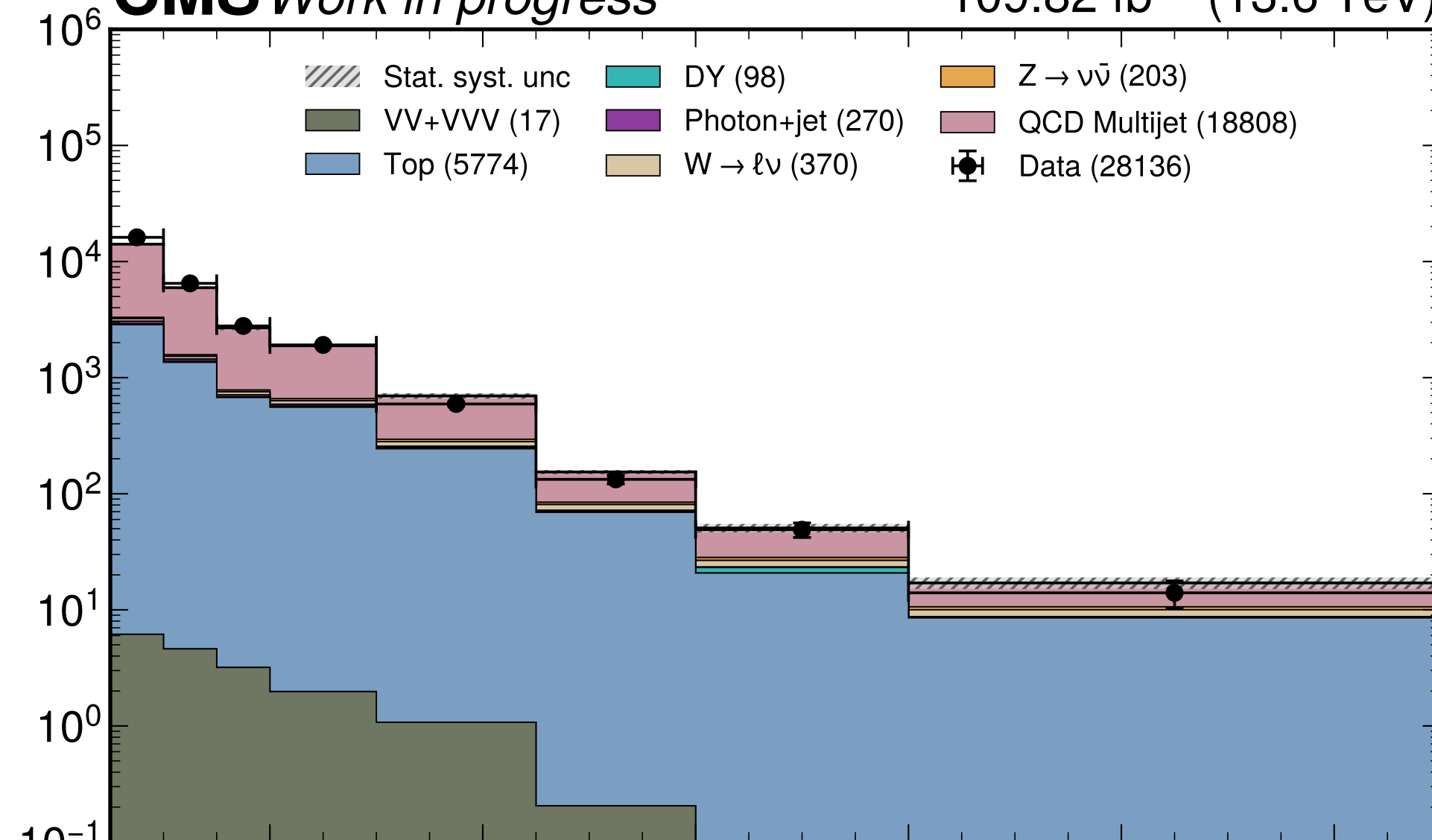
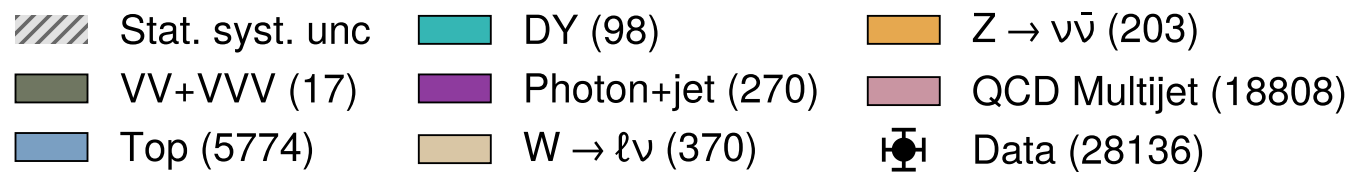
Leading jet p_T (GeV)



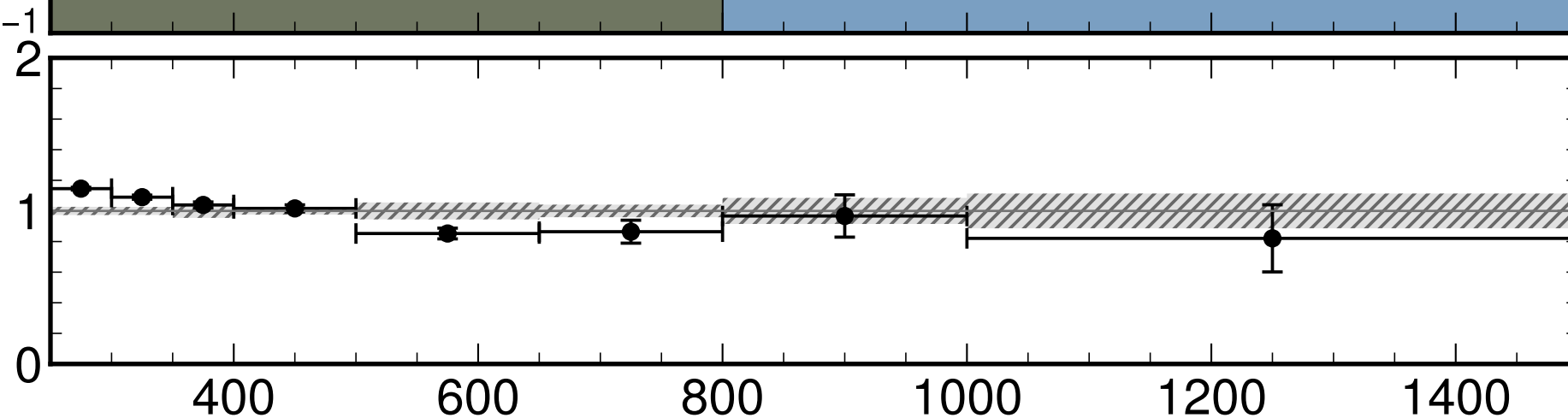
CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)

Events



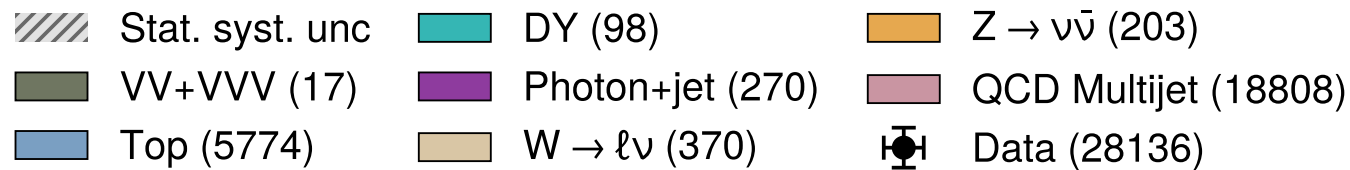
Data/MC



CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)

Events



Data/MC

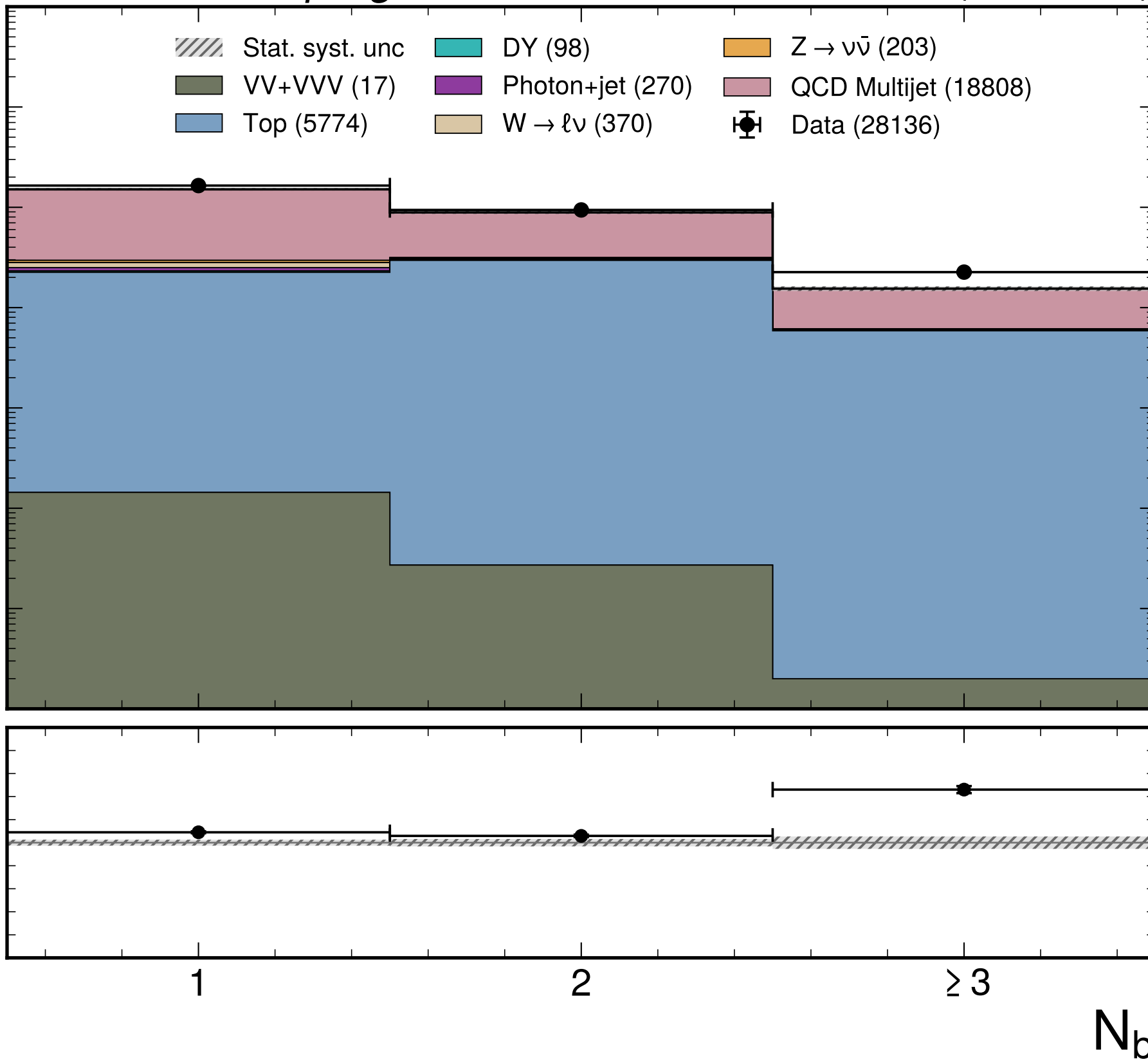
10⁻¹
2
1
0

1

2

≥ 3

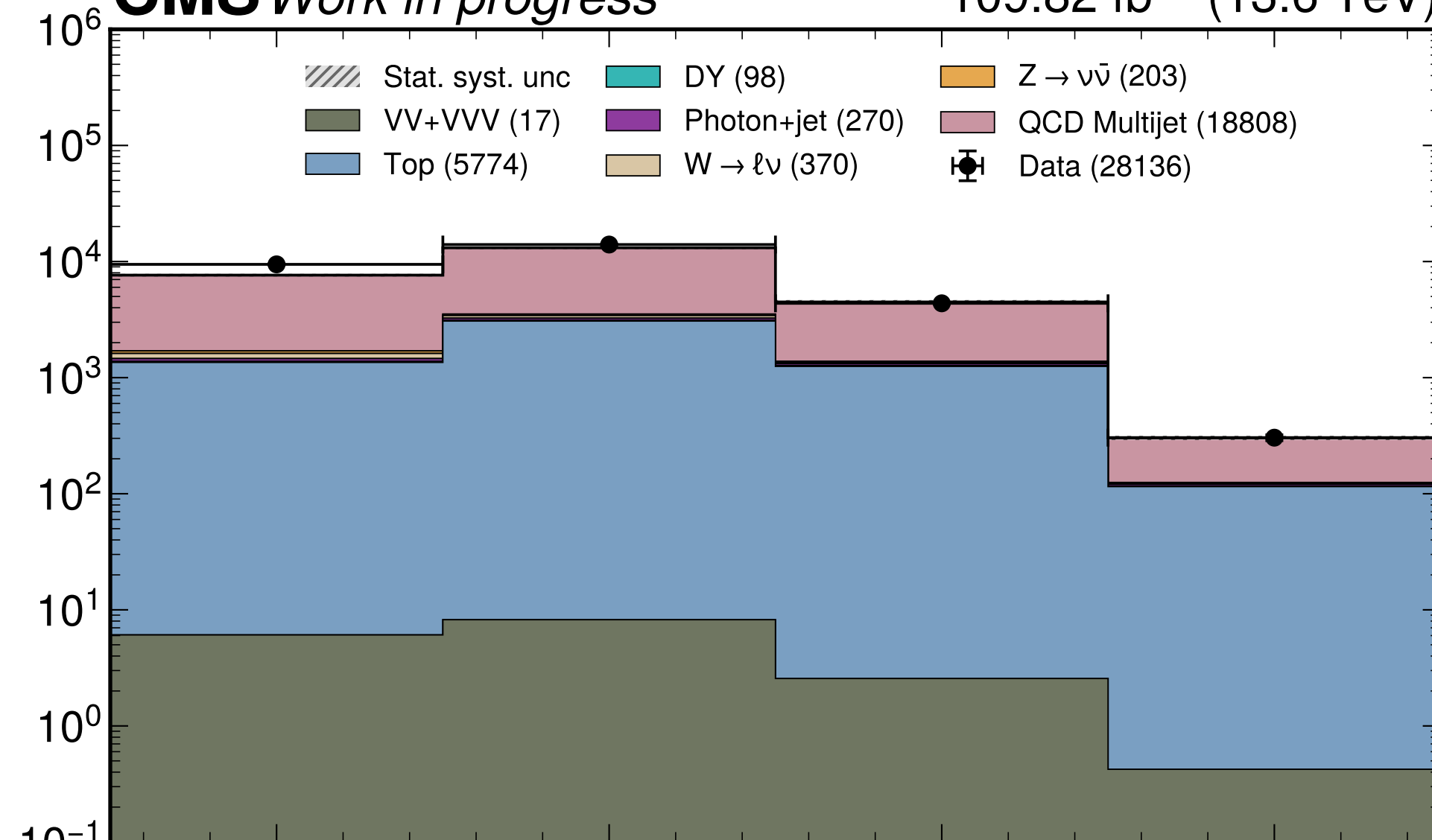
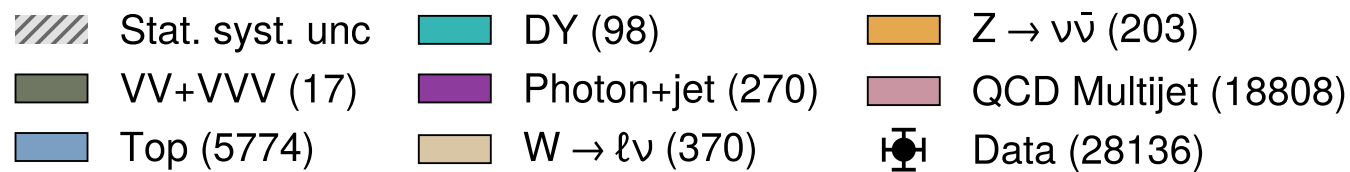
N_b



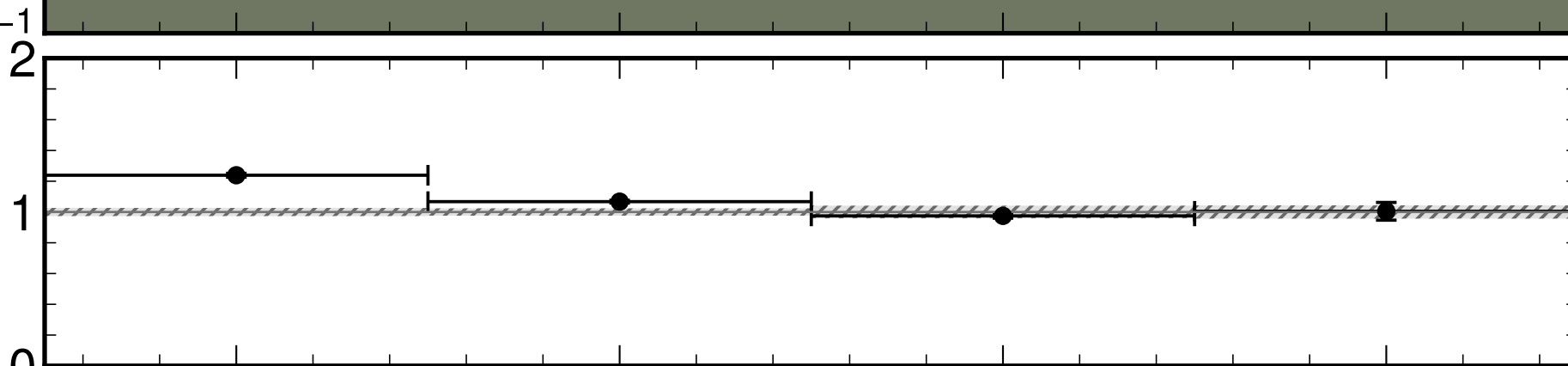
CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)

Events



Data/MC



0

1

2

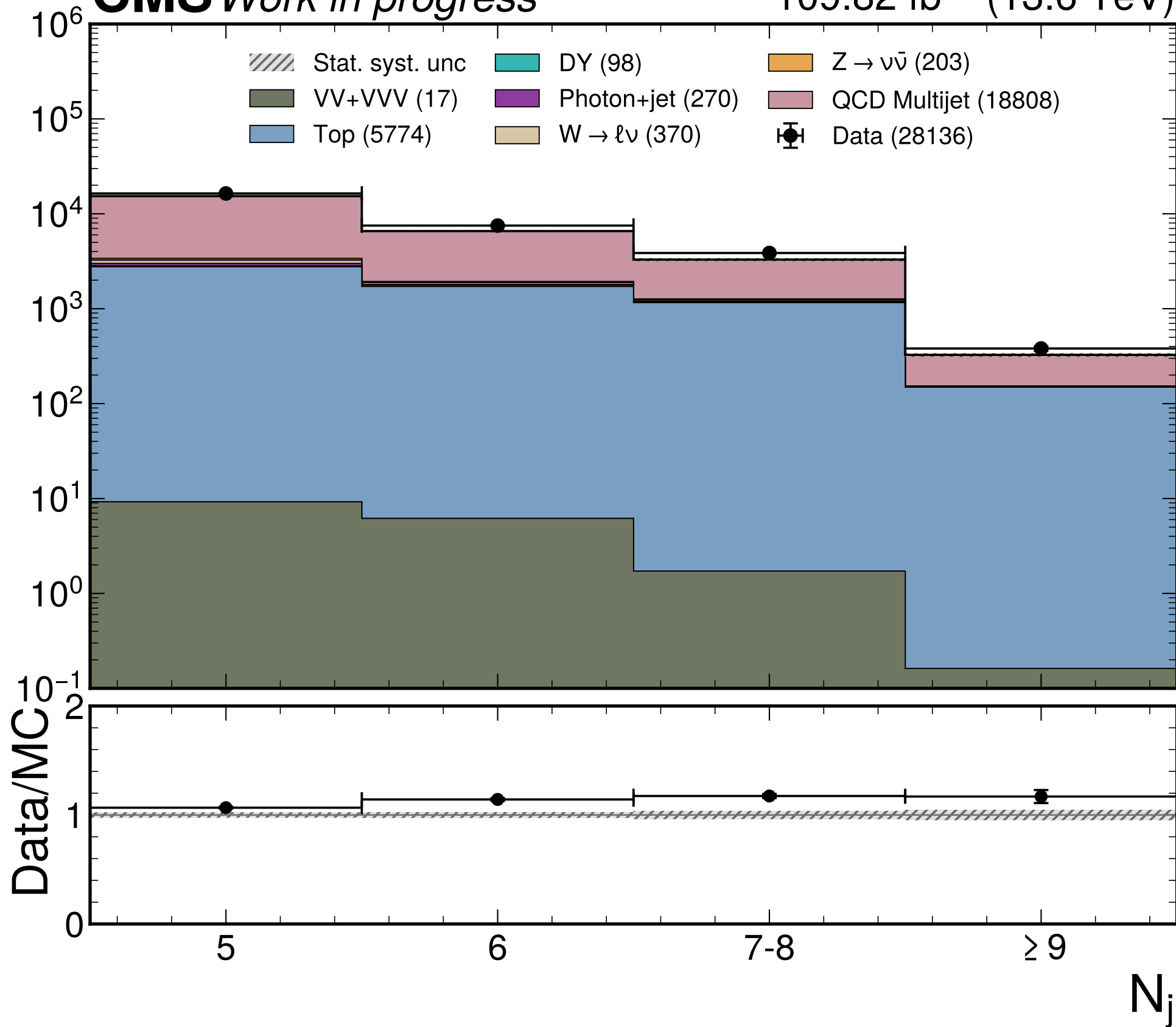
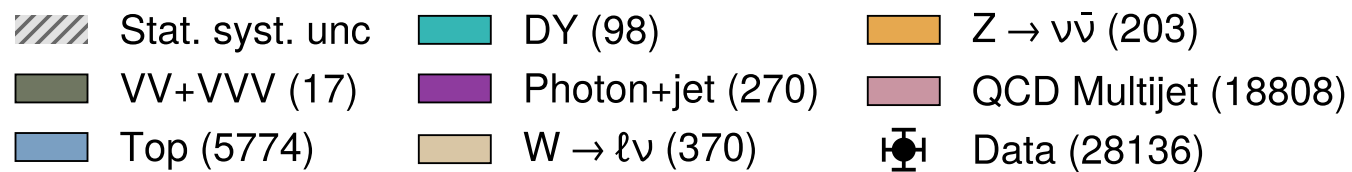
≥ 3

N_{AK8}

CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)

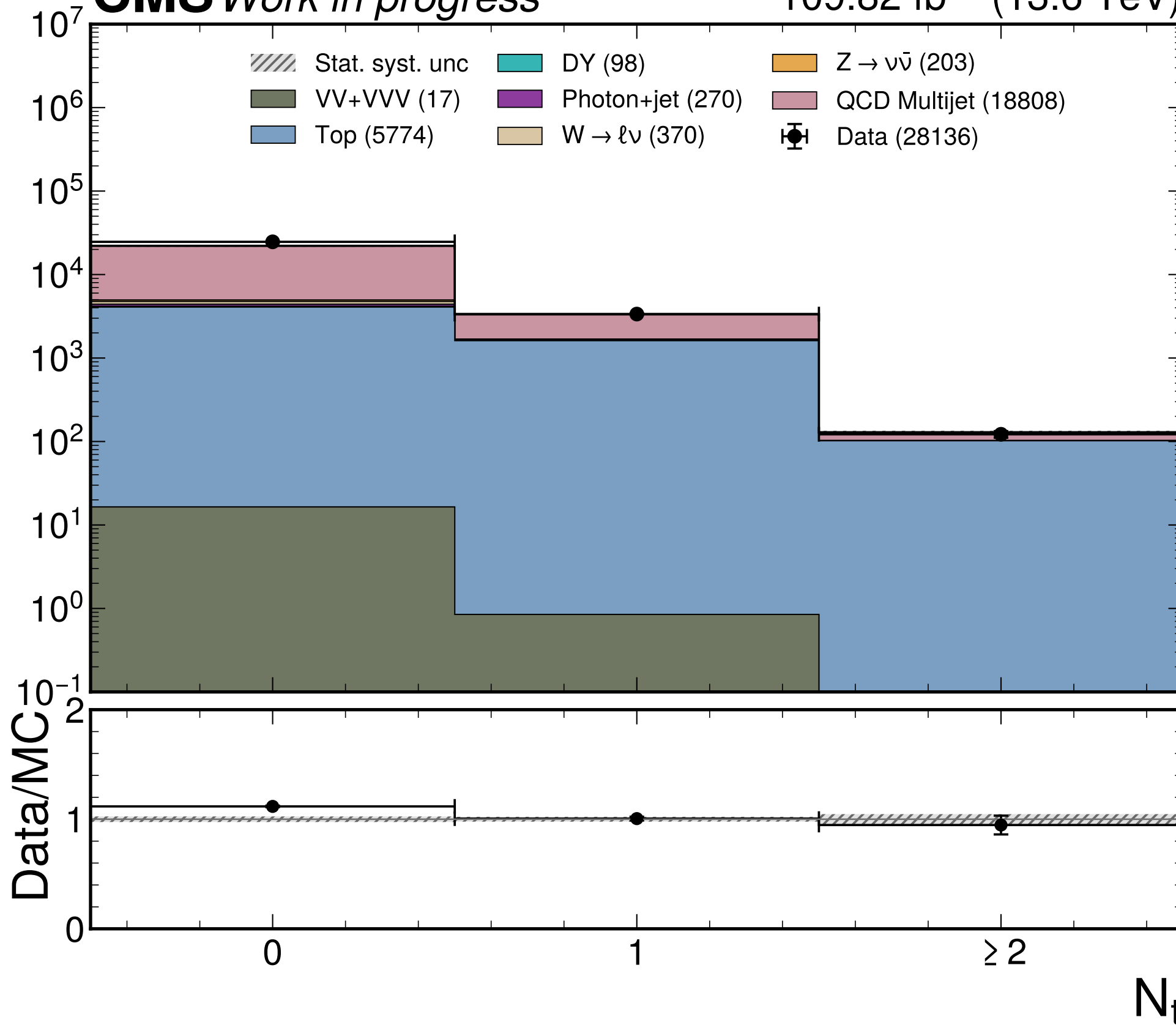
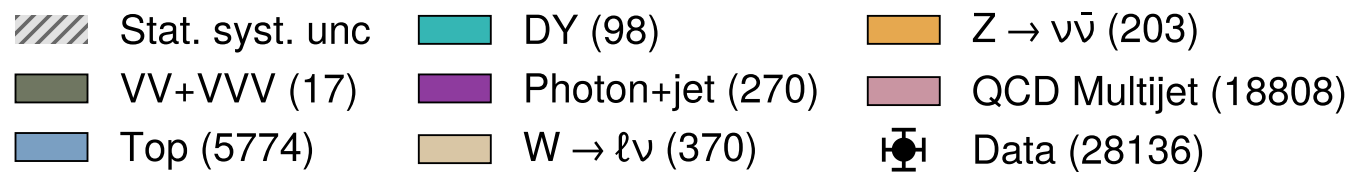
Events



CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)

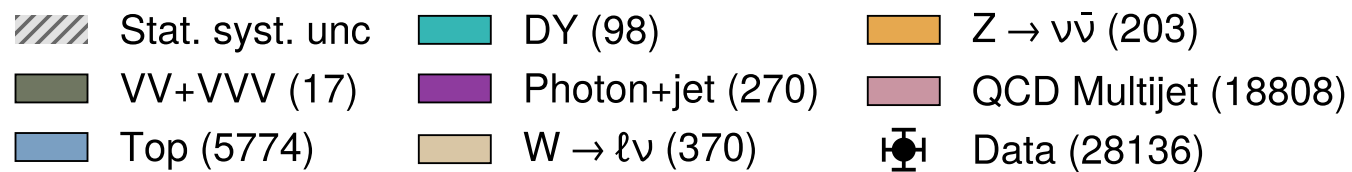
Events



CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)

Events



Data/MC

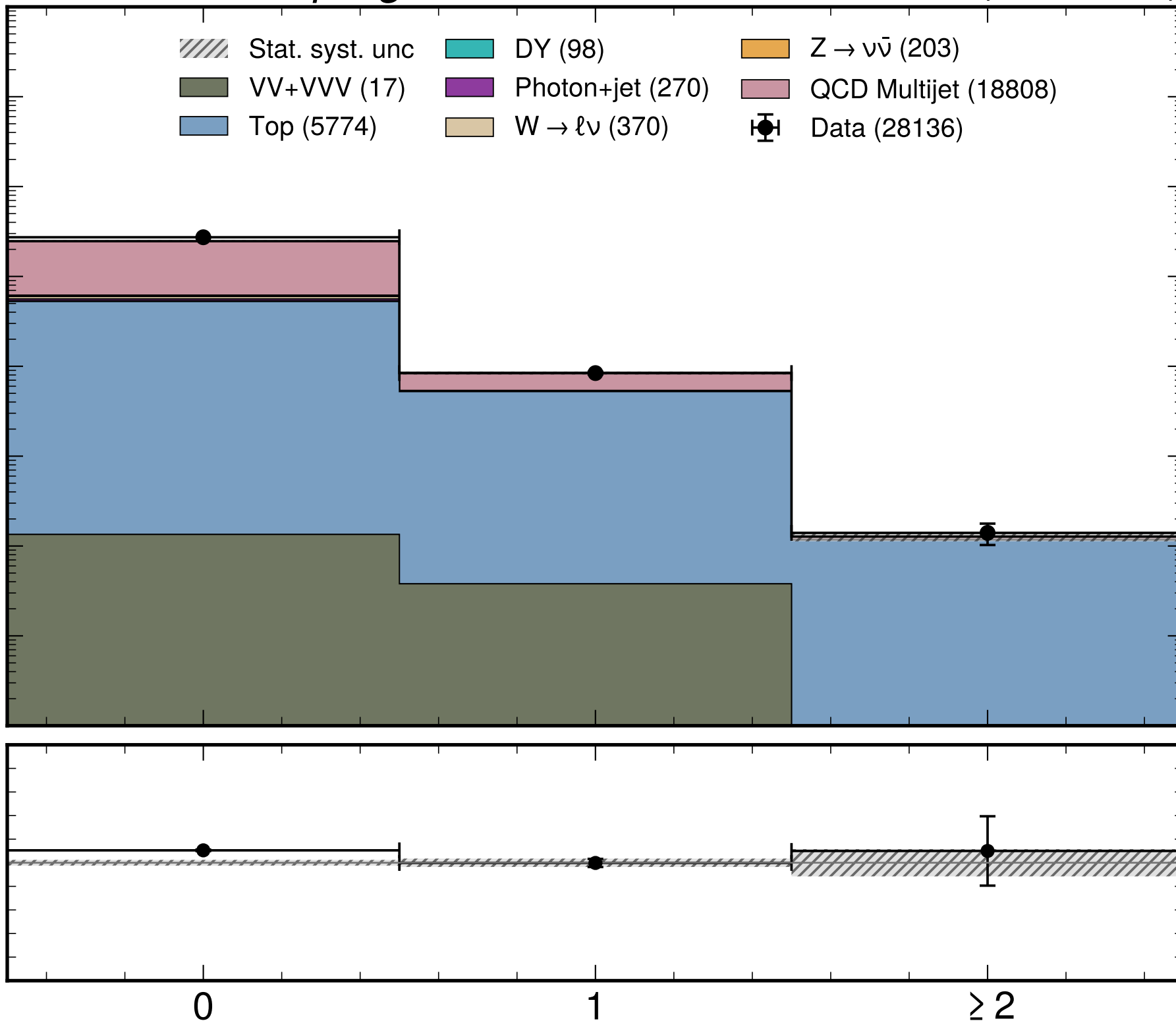
10⁻¹
2
1
0

0

1

≥ 2

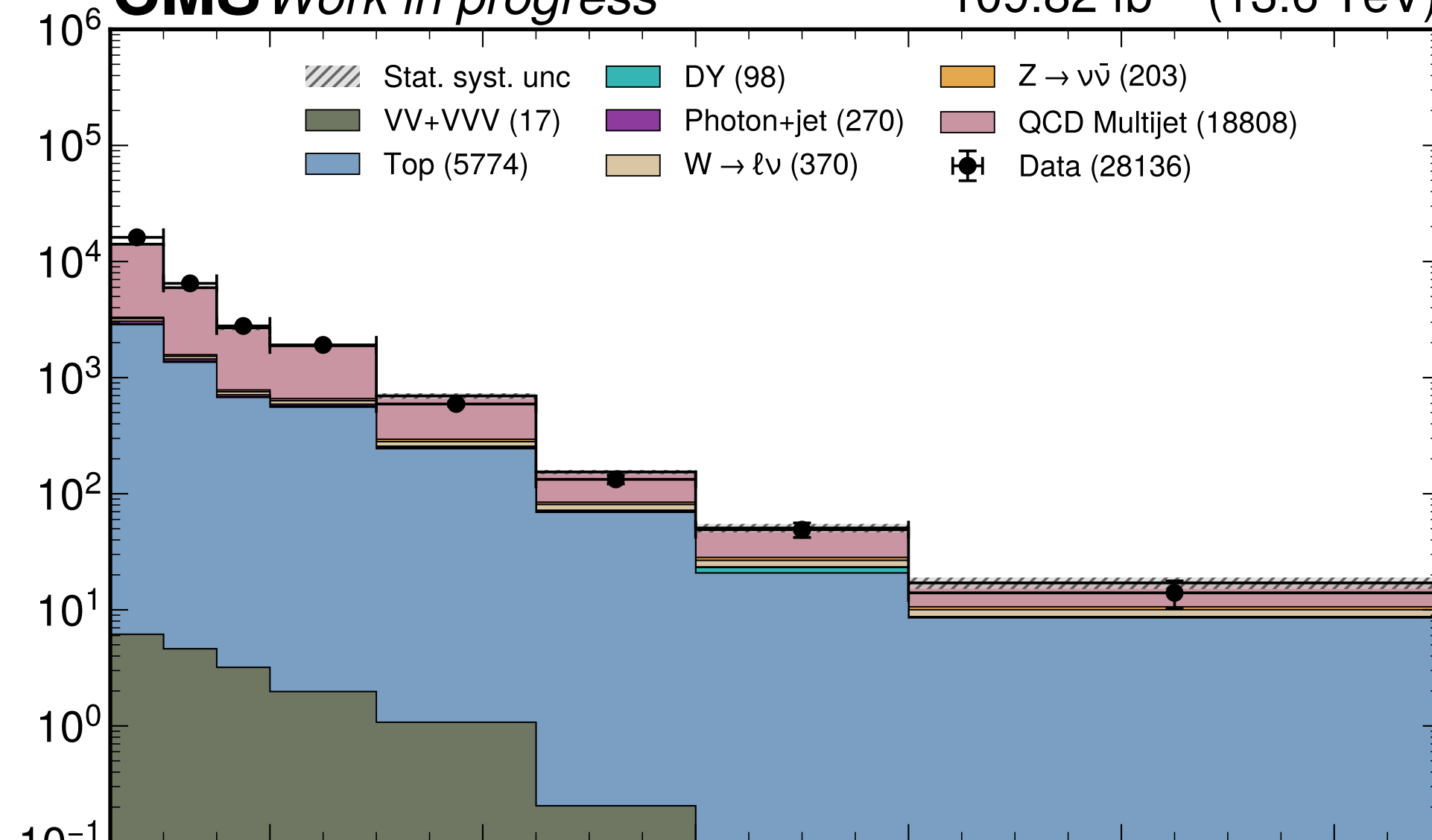
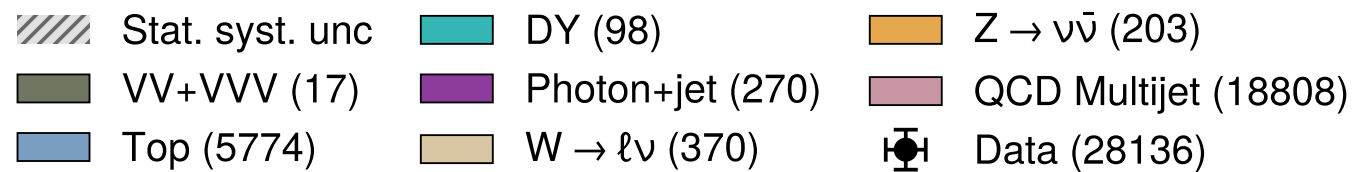
N_W



CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)

Events



Data/MC

400

600

800

1000

1200

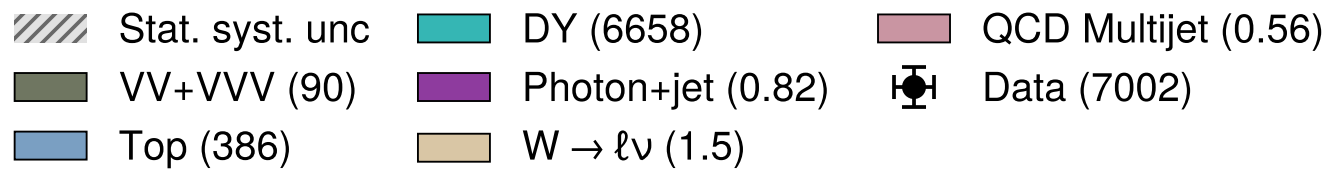
1400

U_T (GeV)

CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)

Events



Data/MC

10⁻¹

2

0

500

1000

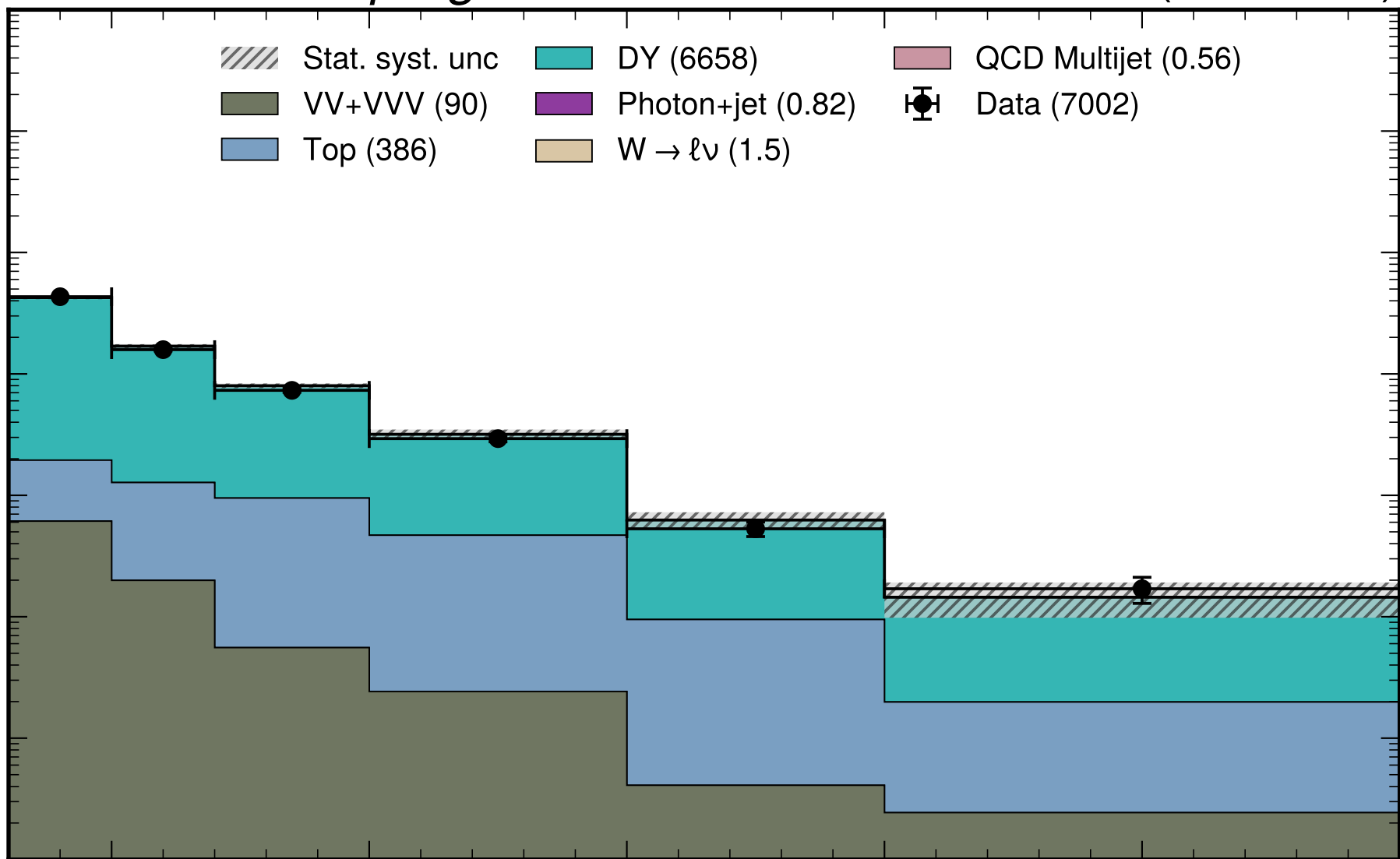
1500

2000

2500

3000

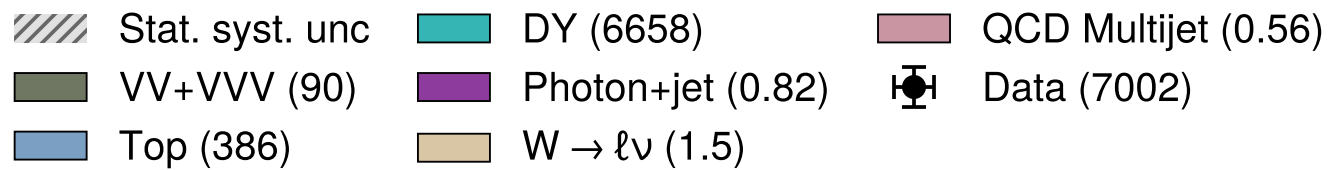
H_T (GeV)



CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)

Events



Data/MC

10⁻¹

2

0

500

1000

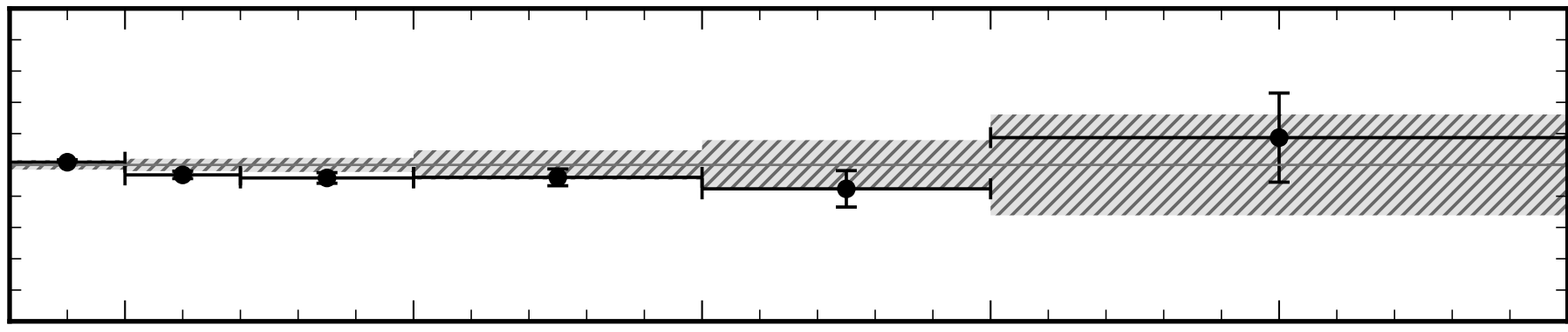
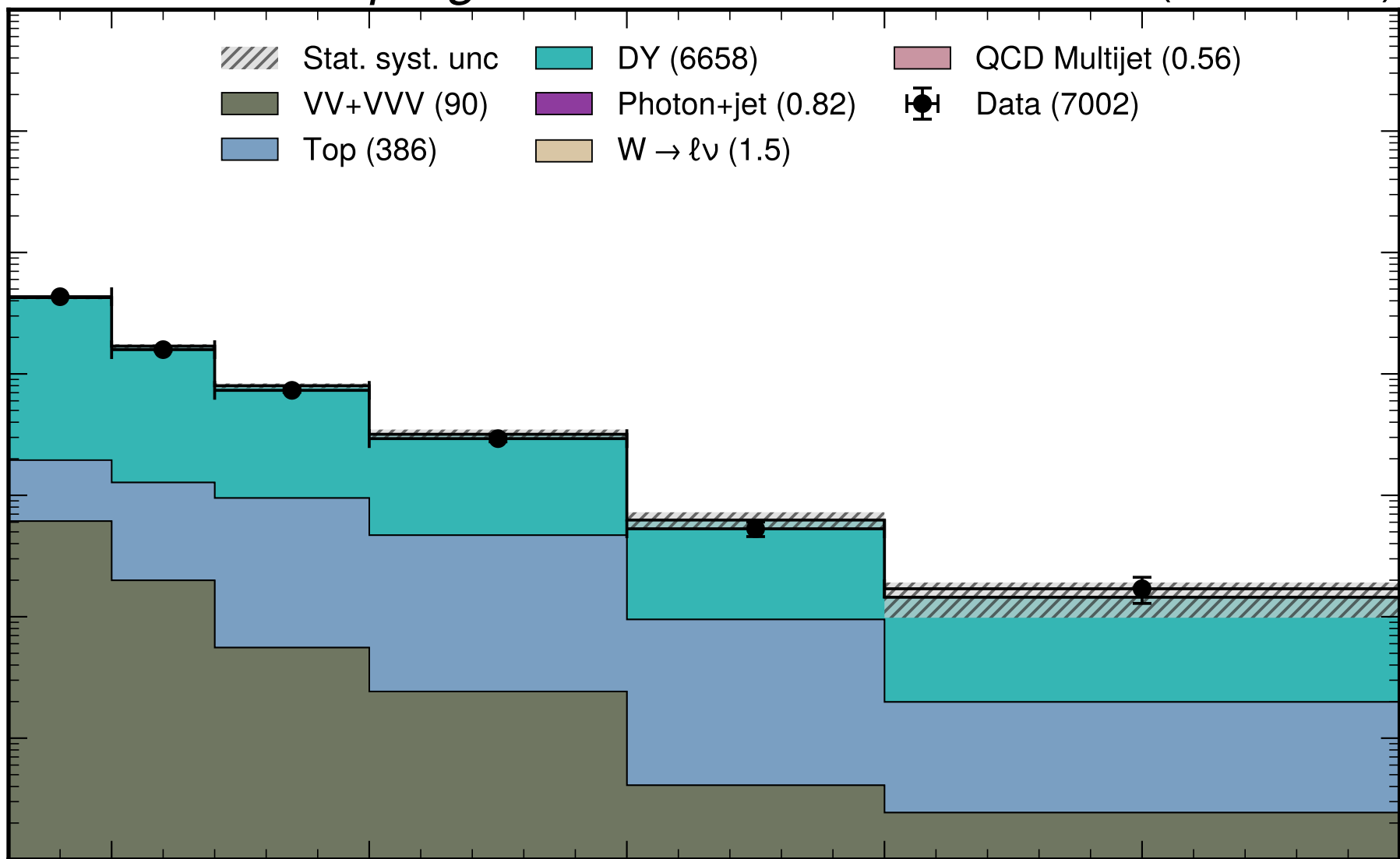
1500

2000

2500

3000

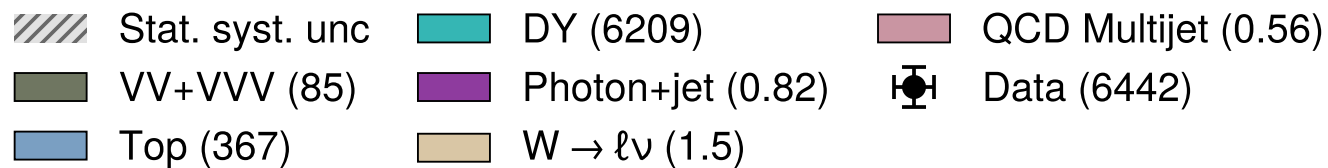
H_T (GeV)



CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)

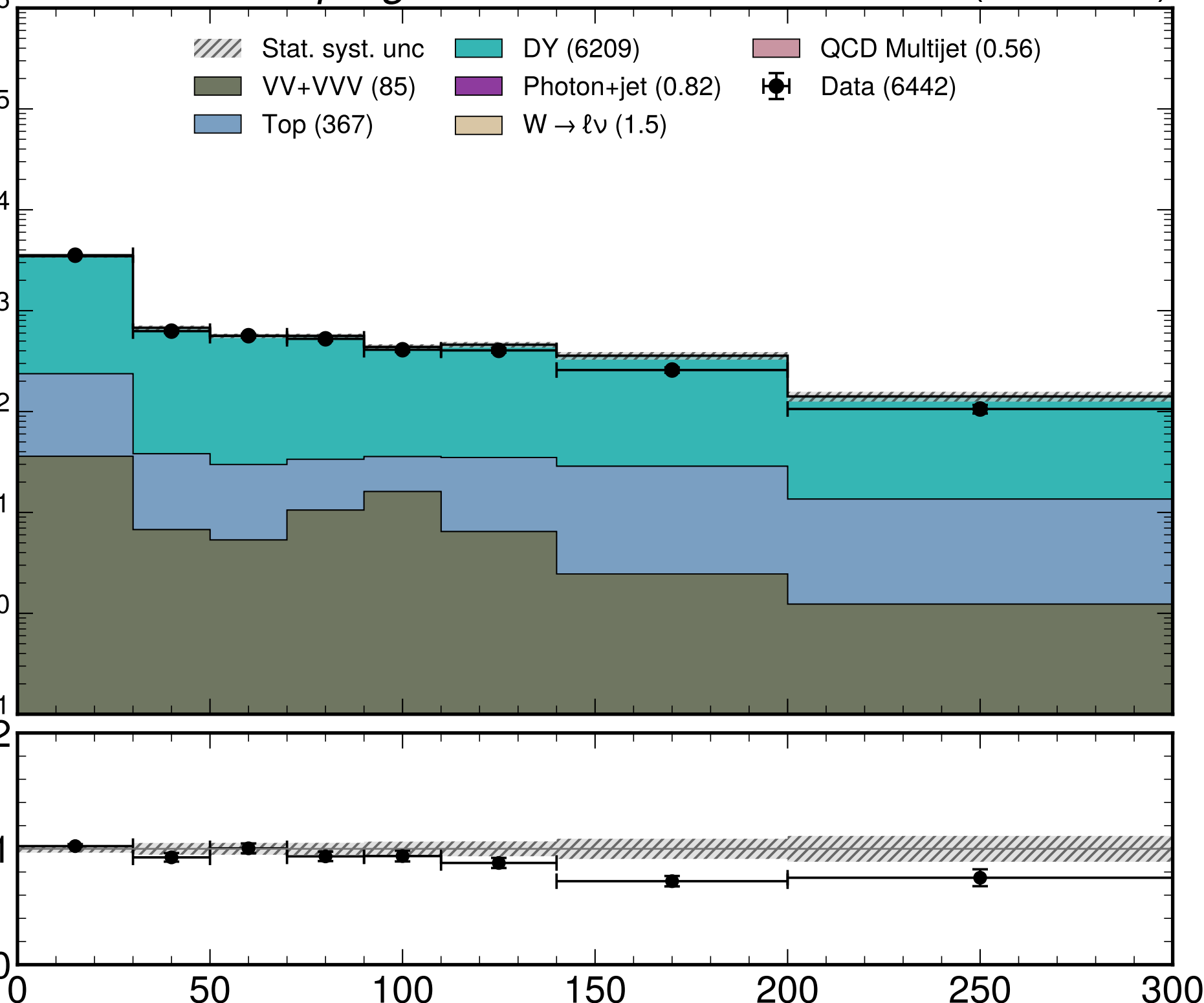
Events



Data/MC

10⁻¹
2
1
0

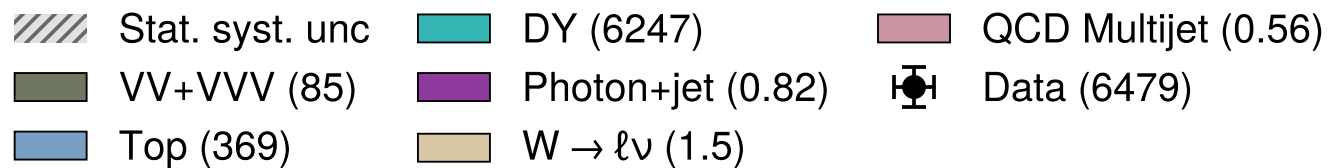
Leading AK8 jet m_{SD} (GeV)



CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)

Events



Data/MC

10⁻²

200

400

600

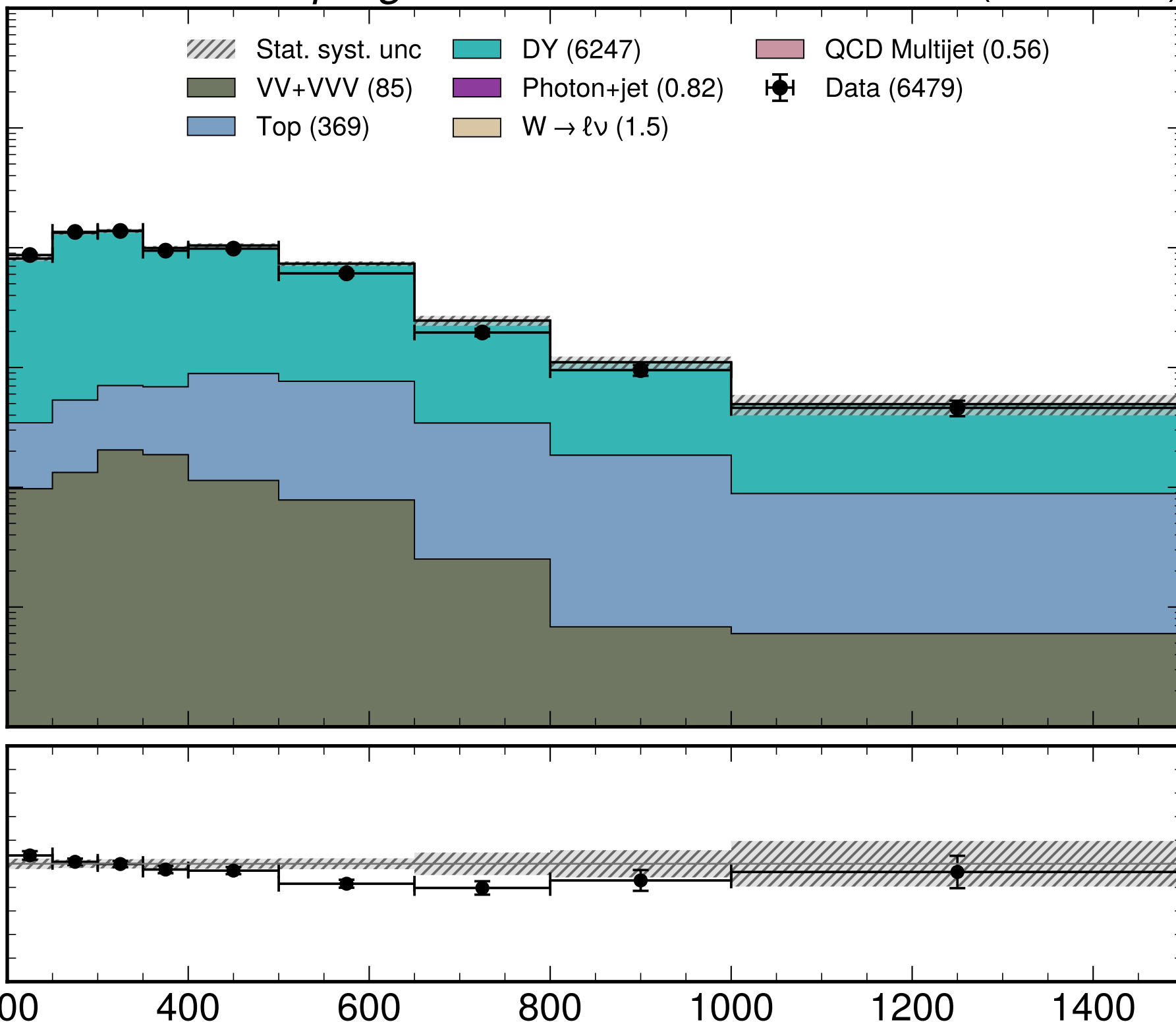
800

1000

1200

1400

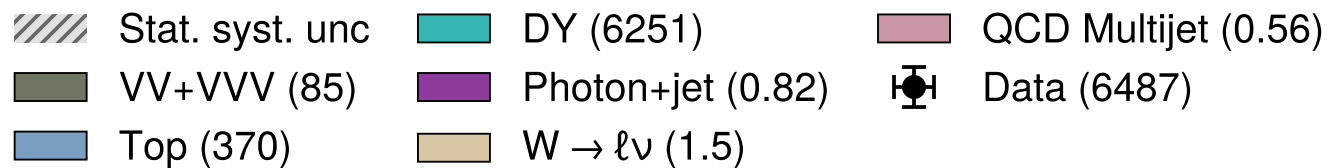
Leading AK8 jet p_T (GeV)



CMS *Work in progress*

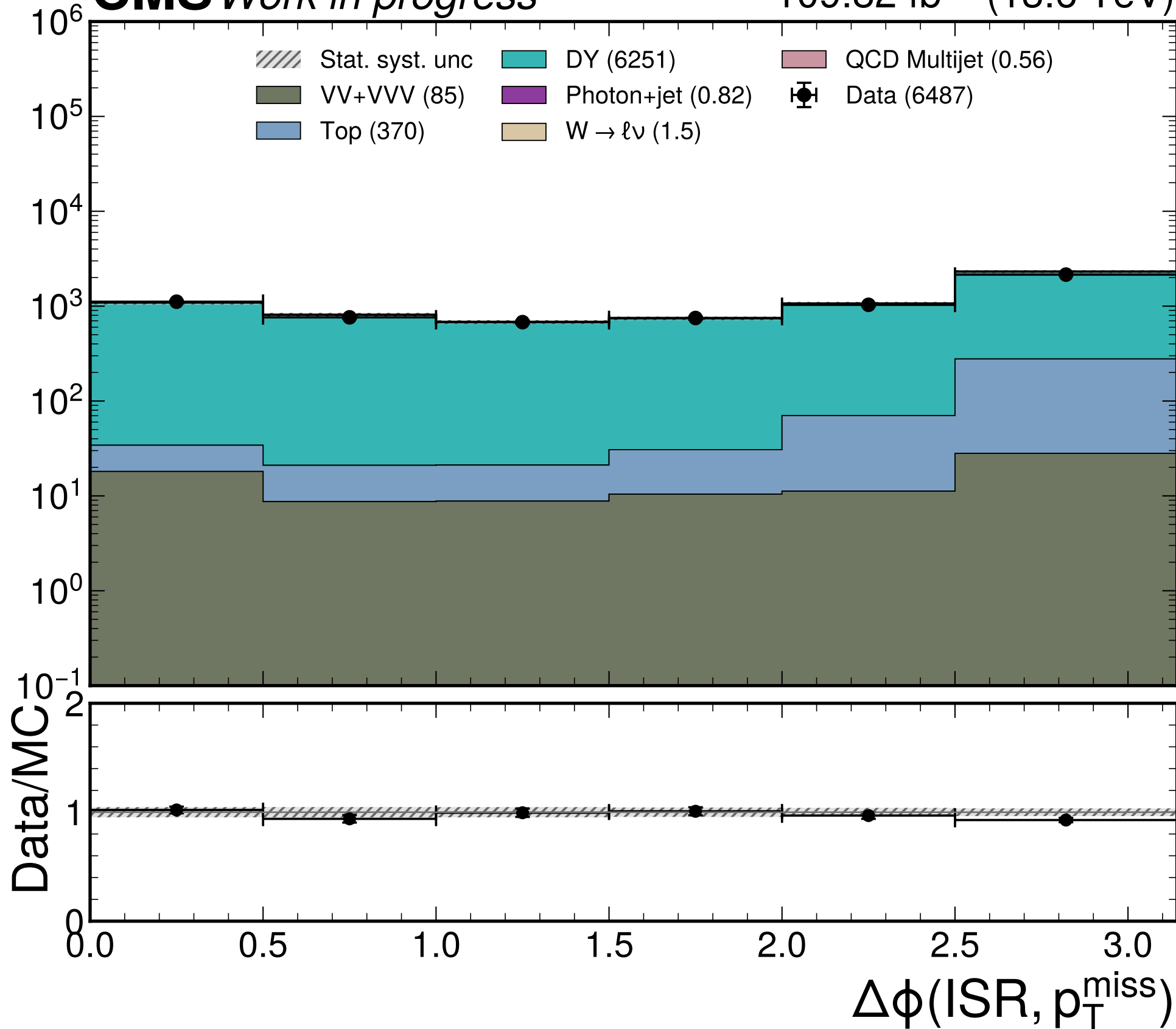
109.82 fb⁻¹ (13.6 TeV)

Events



Data/MC

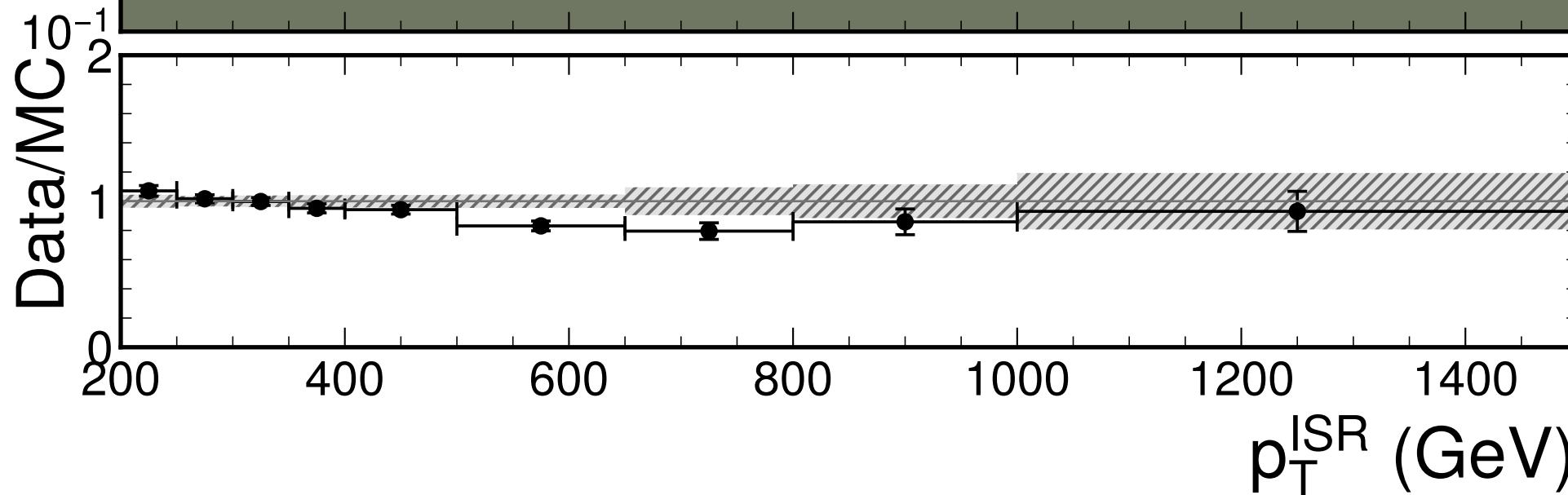
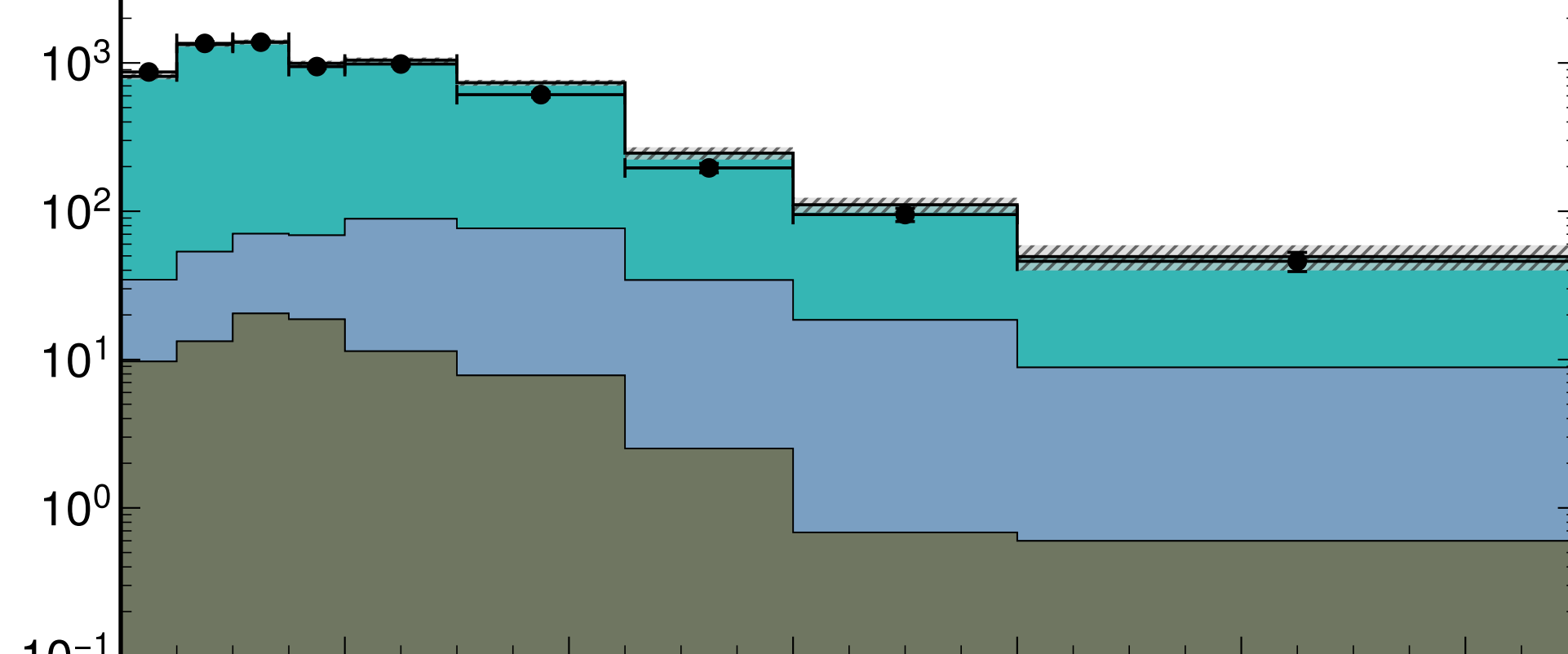
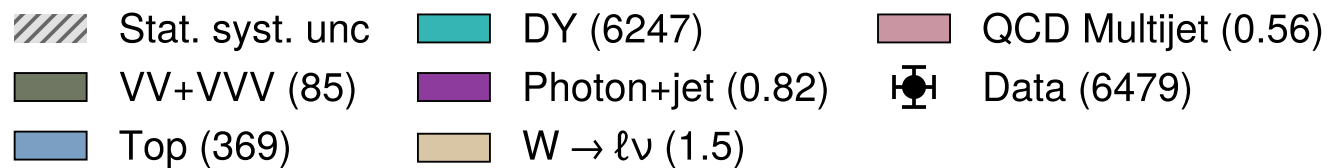
$\Delta\phi(\text{ISR}, p_{\text{T}}^{\text{miss}})$



CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)

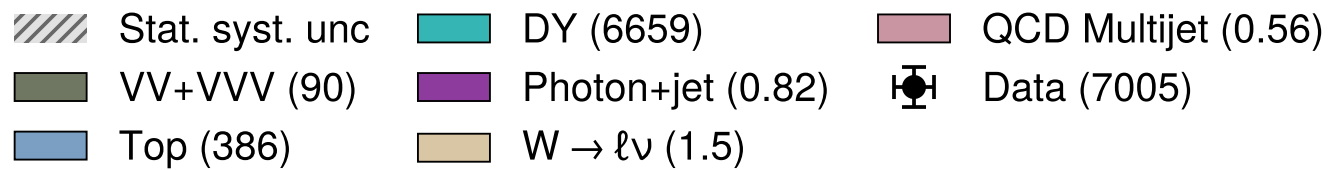
Events



CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)

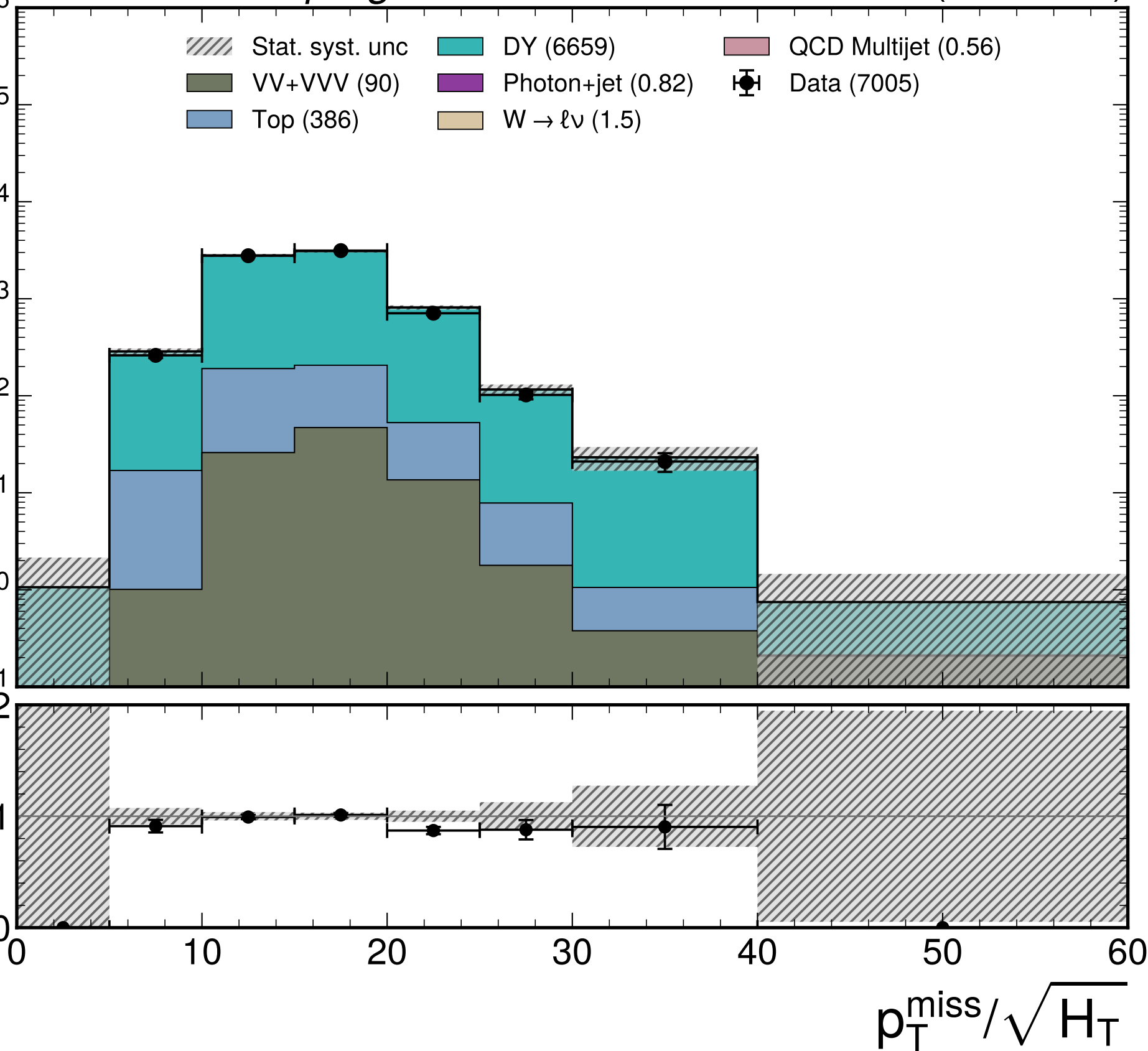
Events



Data/MC

10⁻¹

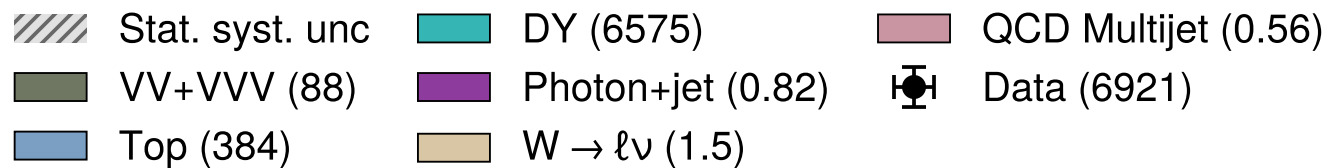
0



CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)

Events



Data/MC

10⁻¹
2
1
0

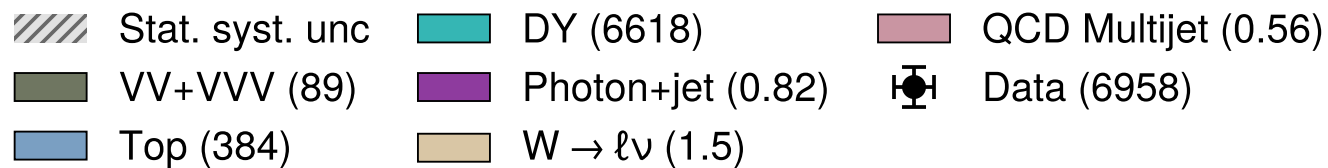
0 200 400 600 800 1000 1200

m_T^b (GeV)

CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)

Events



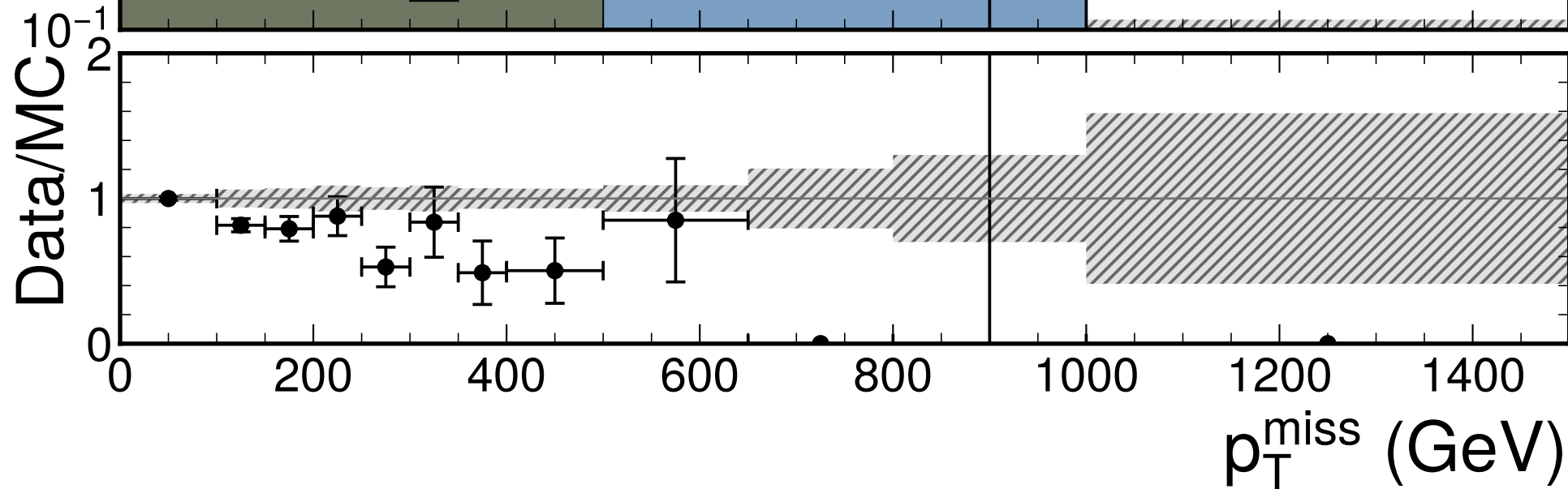
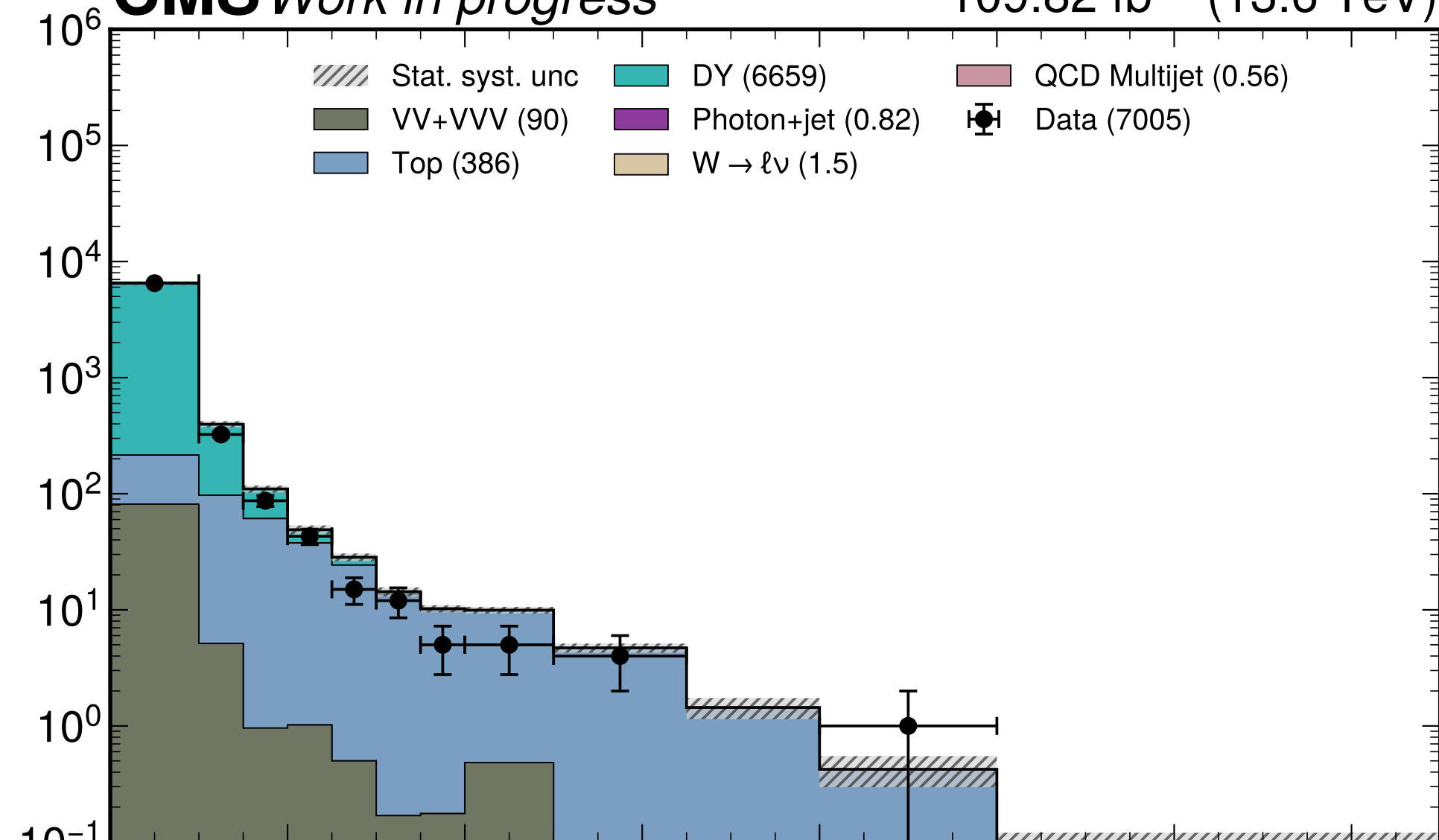
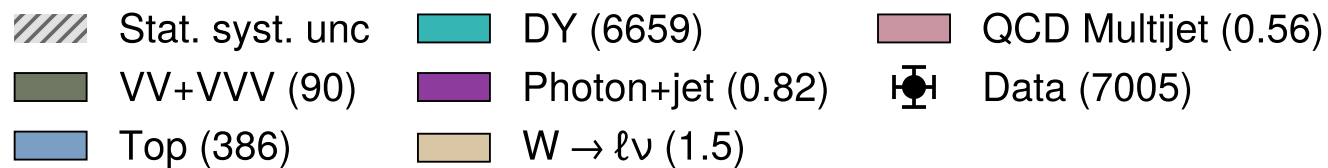
Data/MC

p_T^b (GeV)

CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)

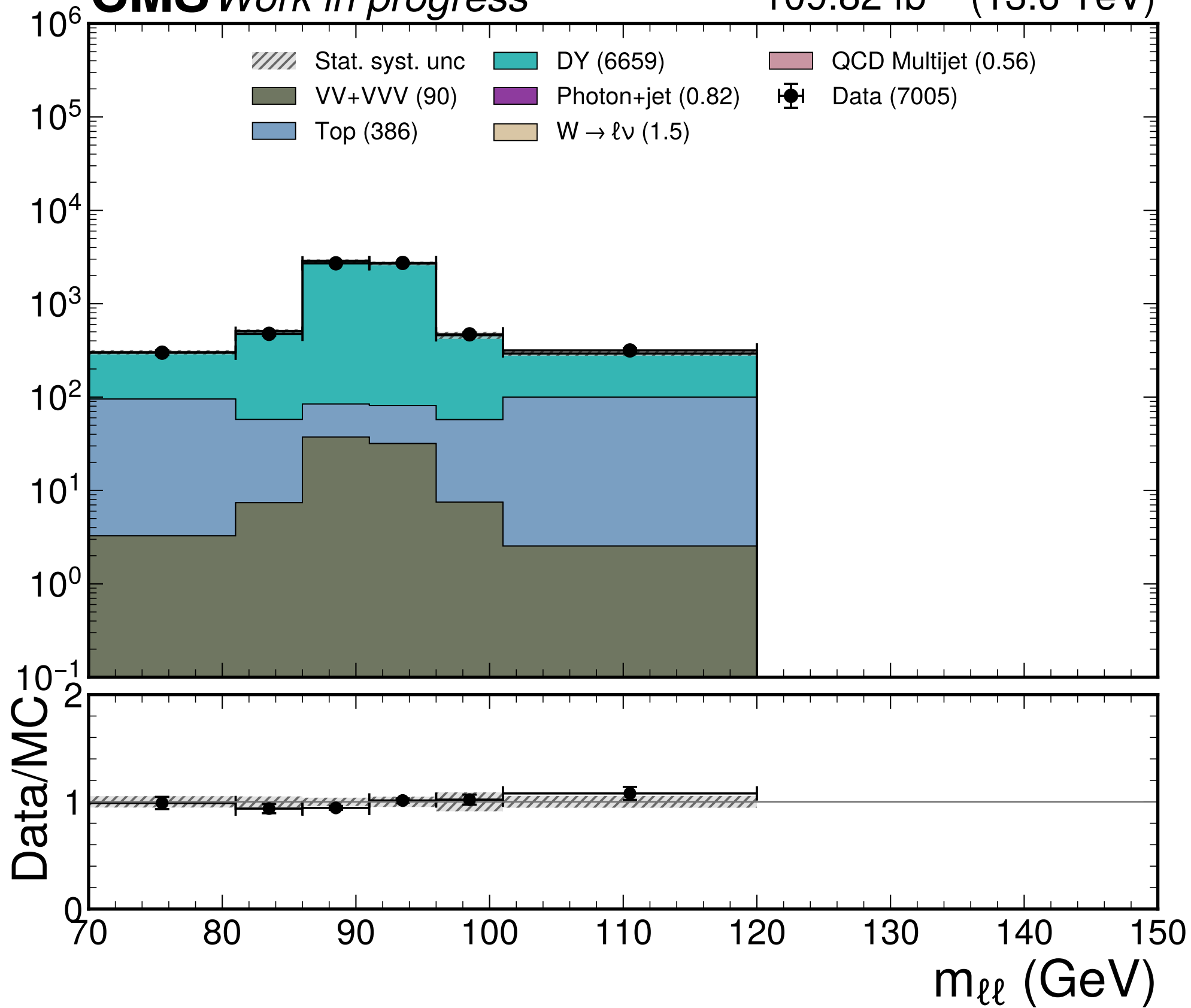
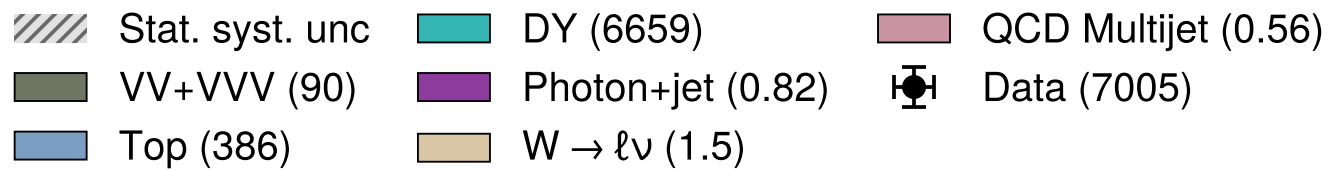
Events



CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)

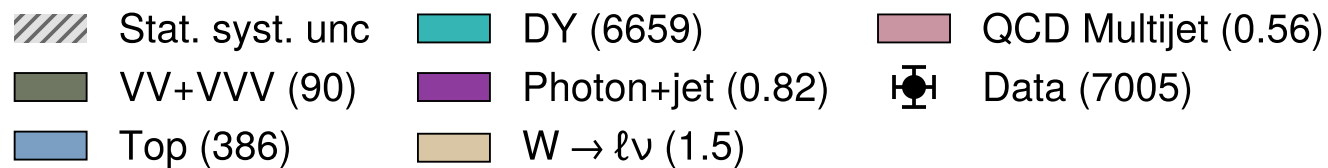
Events



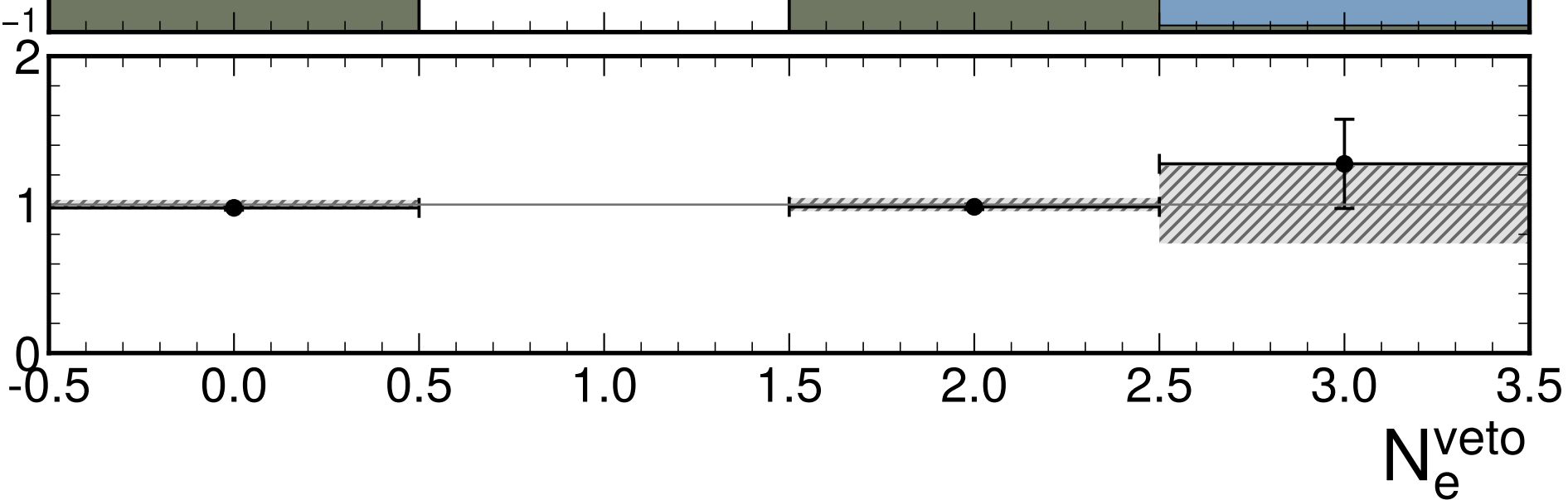
CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)

Events



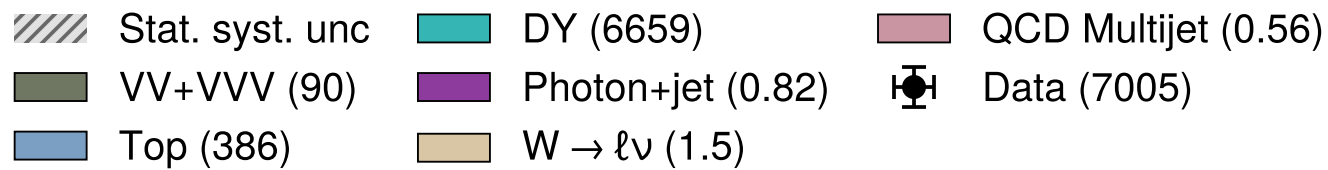
Data/MC



CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)

Events



Data/MC

10⁻¹

0

0

1

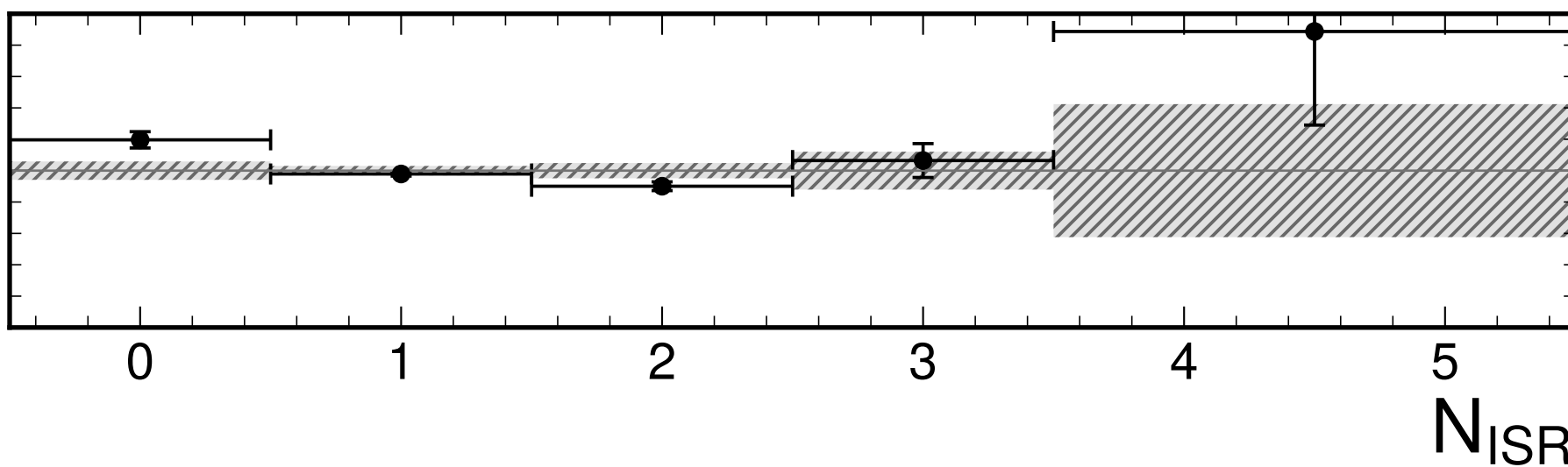
2

3

4

5

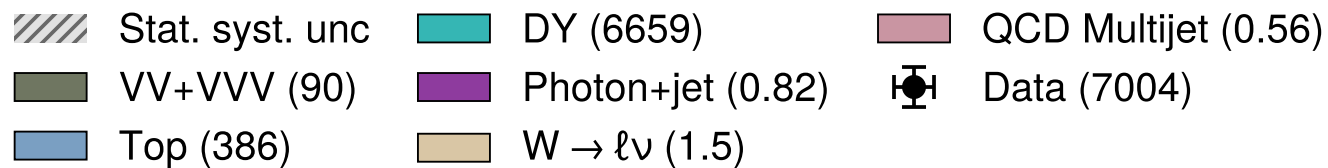
N_{ISR}



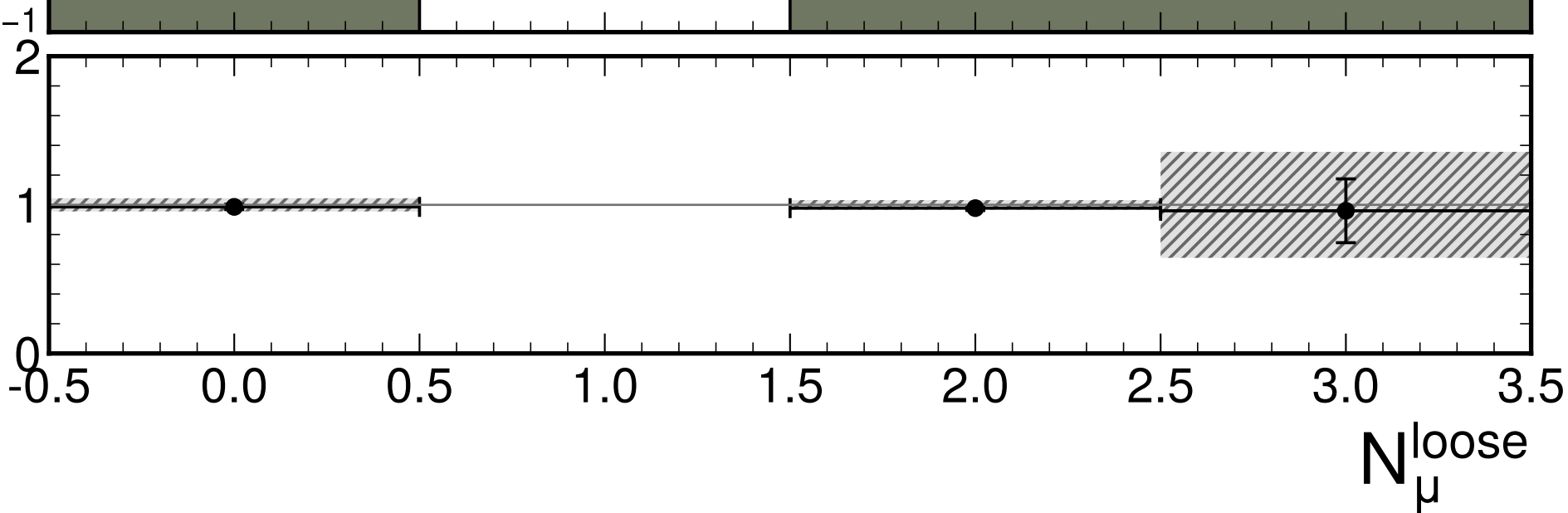
CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)

Events



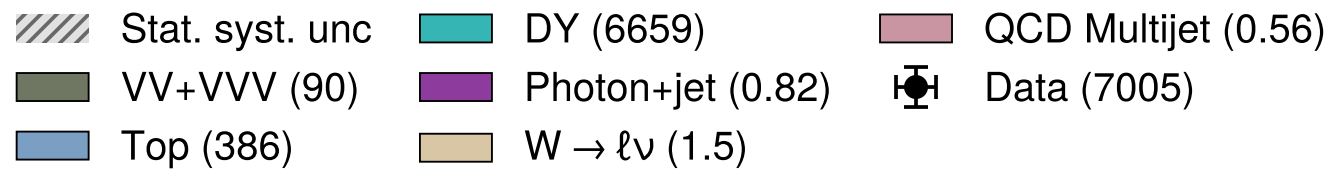
Data/MC



CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)

Events



Data/MC

10⁻¹

2

0

1

2

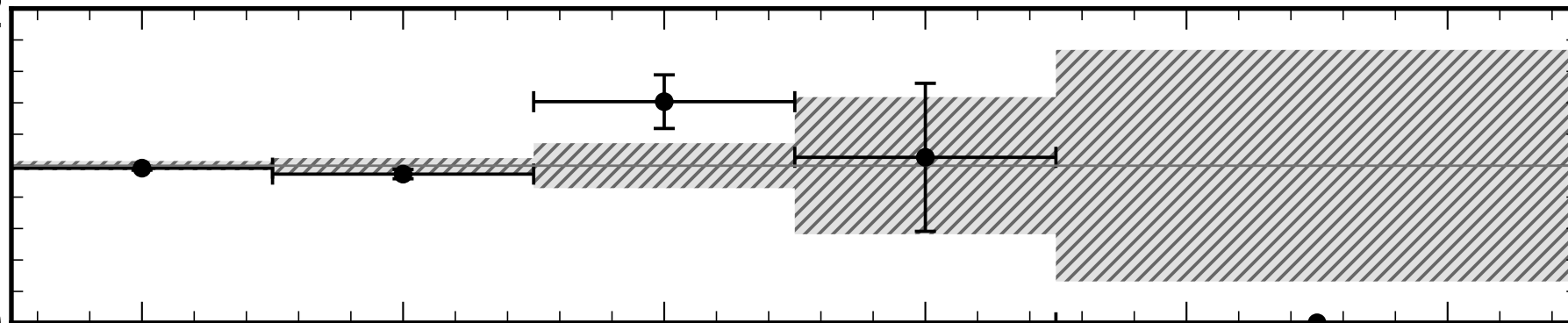
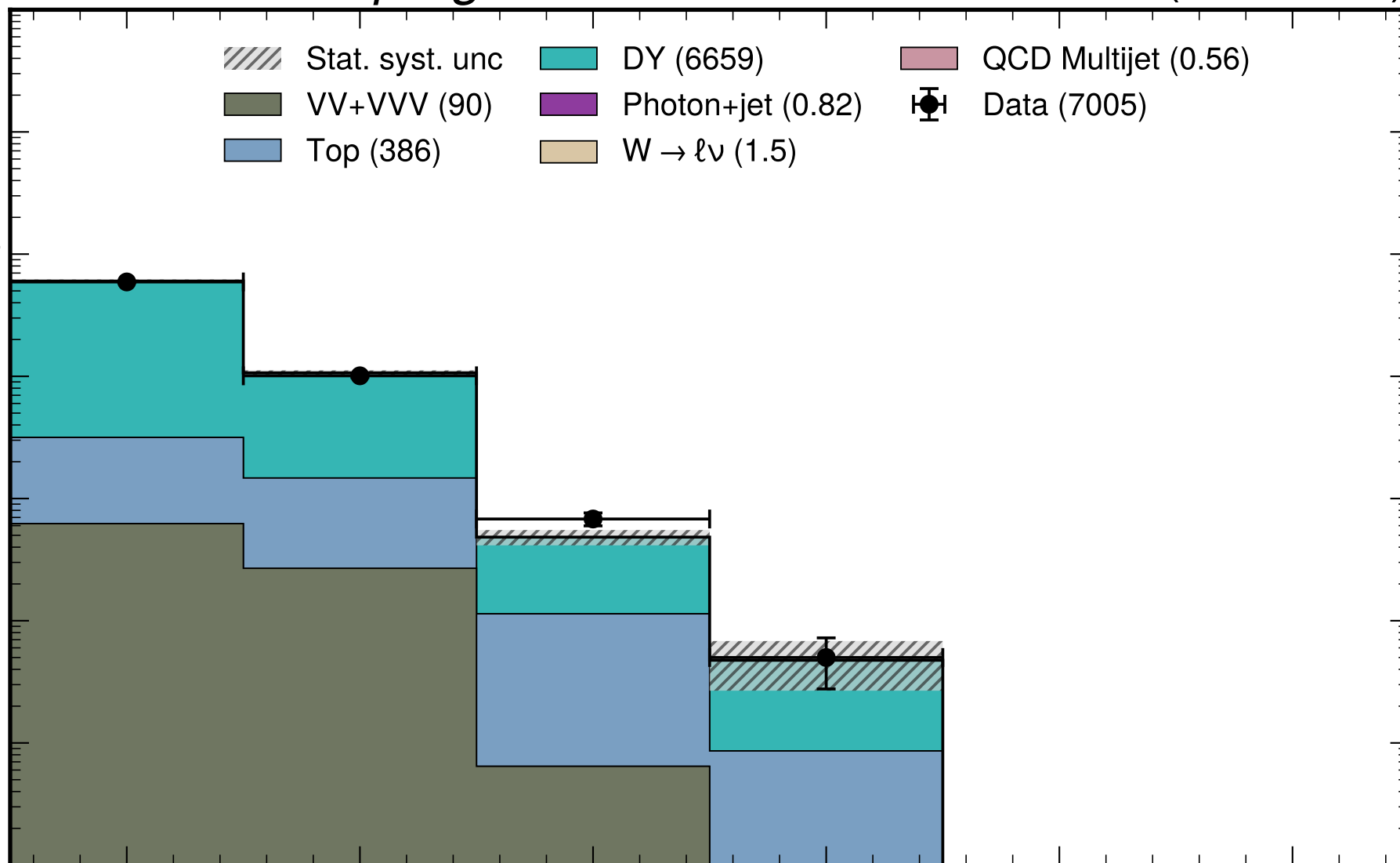
3

4

5

6

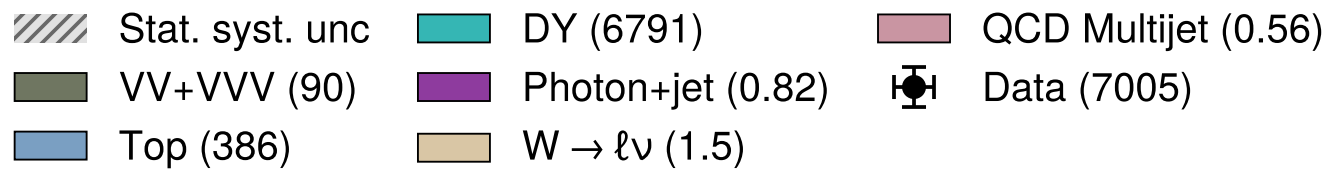
N_b



CMS *Work in progress*

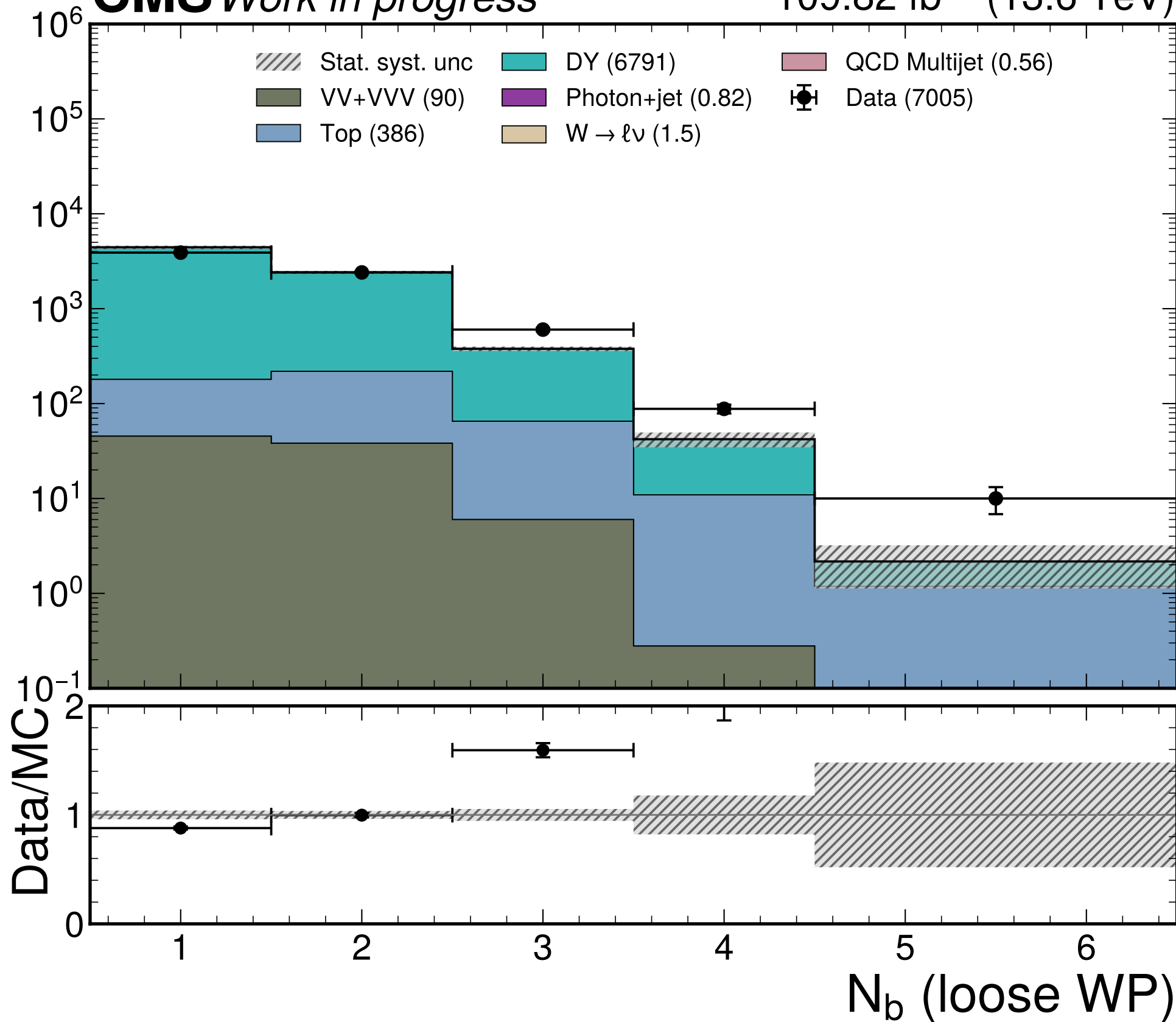
109.82 fb⁻¹ (13.6 TeV)

Events



Data/MC

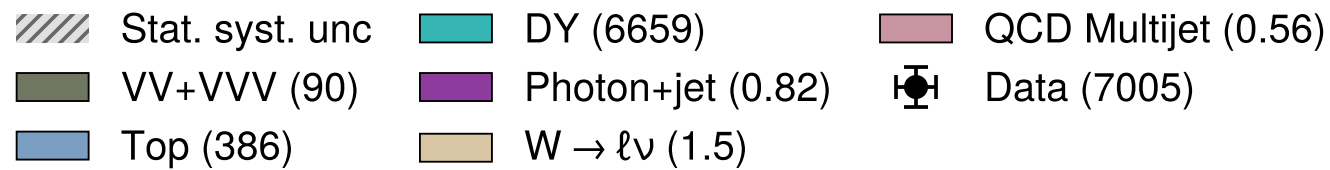
N_b (loose WP)



CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)

Events



Data/MC

10⁻¹
2
1
0

1

2

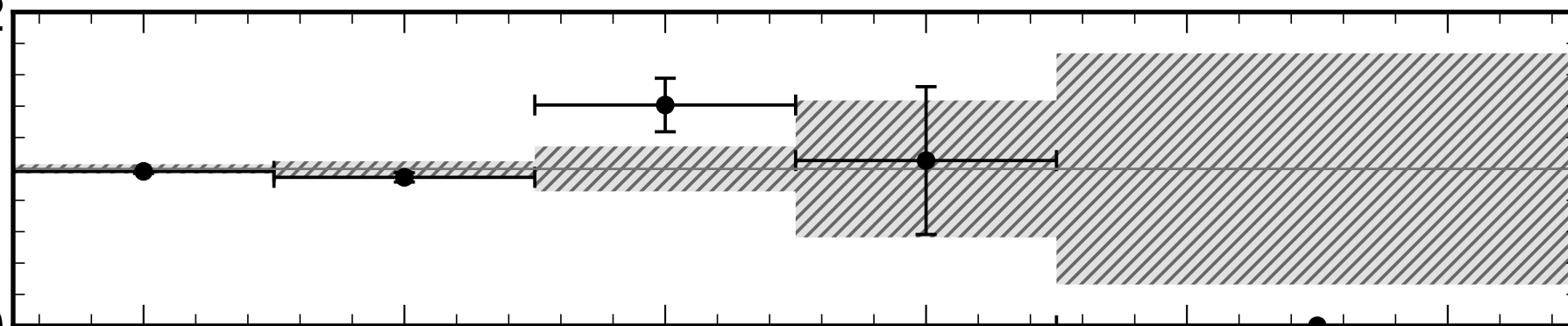
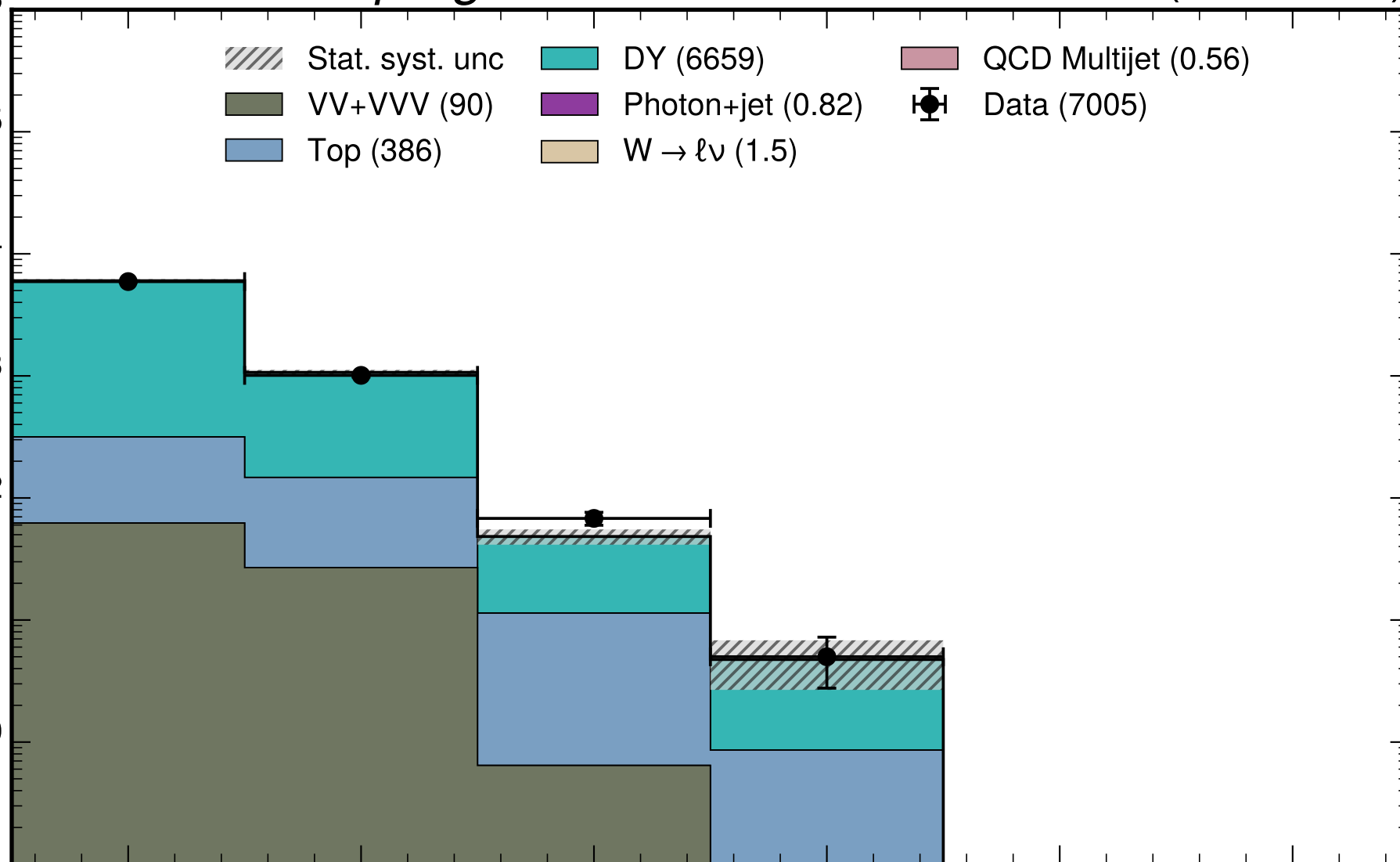
3

4

5

6

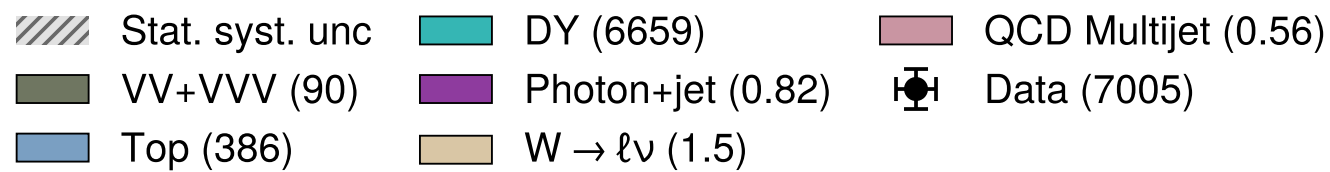
N_b



CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)

Events



Data/MC

10⁻¹

0

2

4

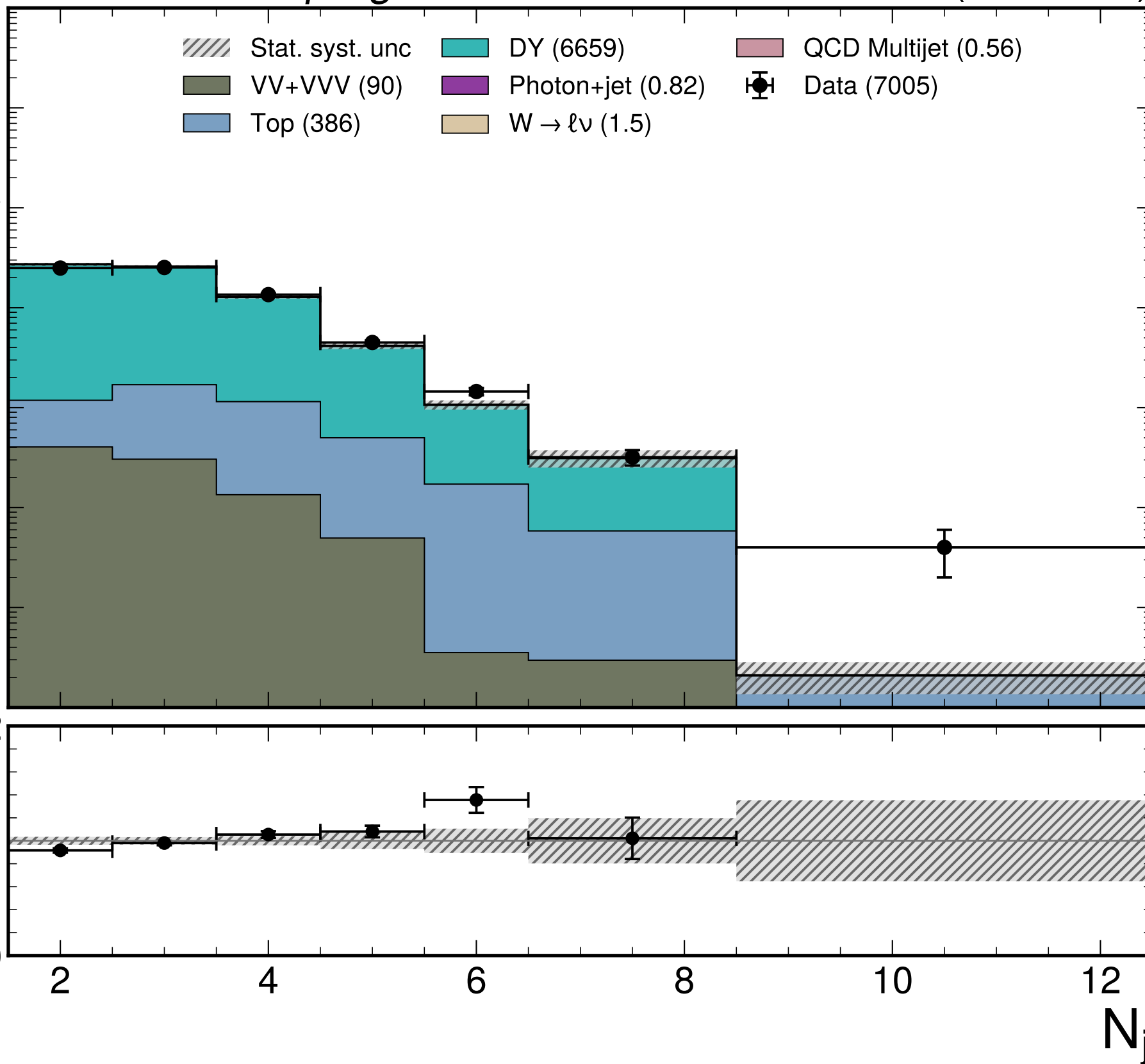
6

8

10

12

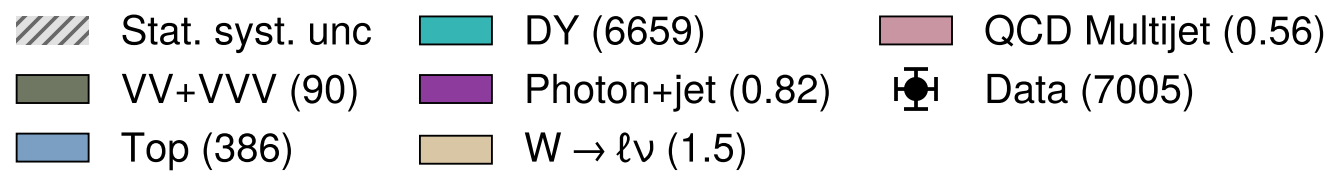
N_j



CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)

Events



Data/MC

10⁻¹

2

2

4

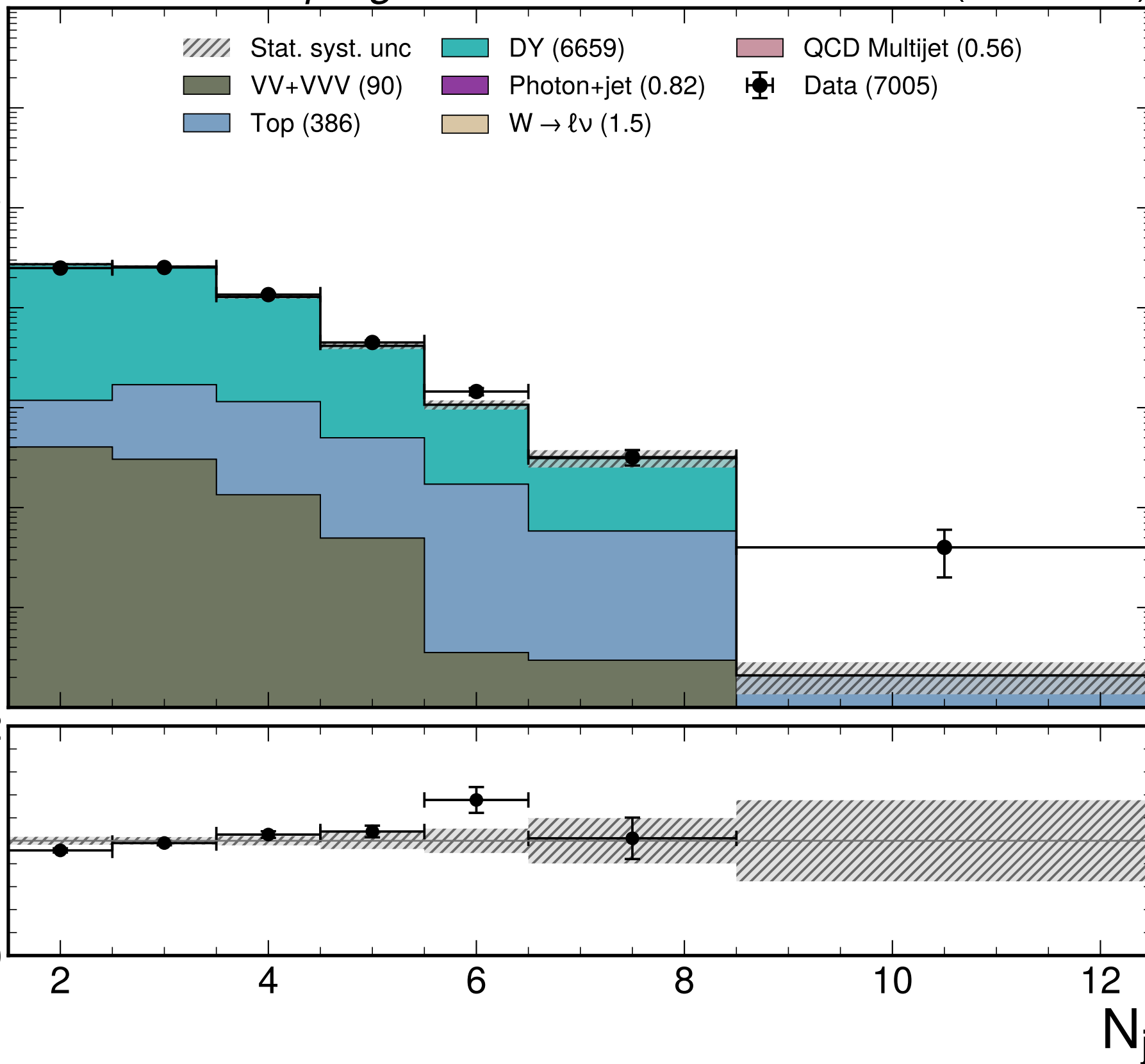
6

8

10

12

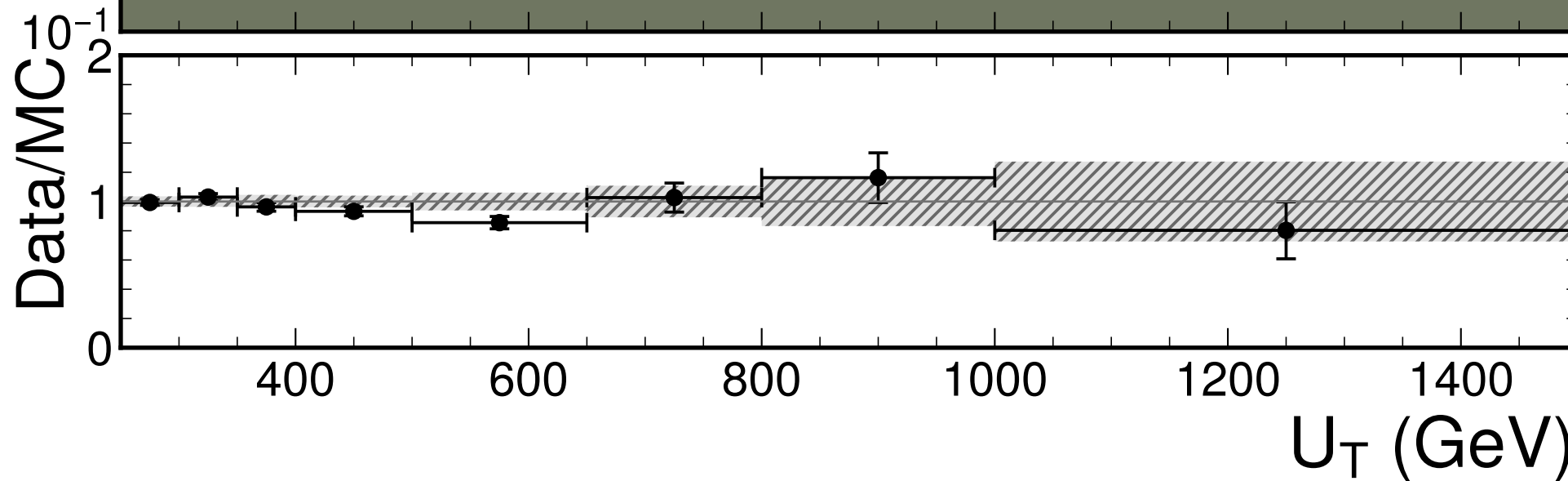
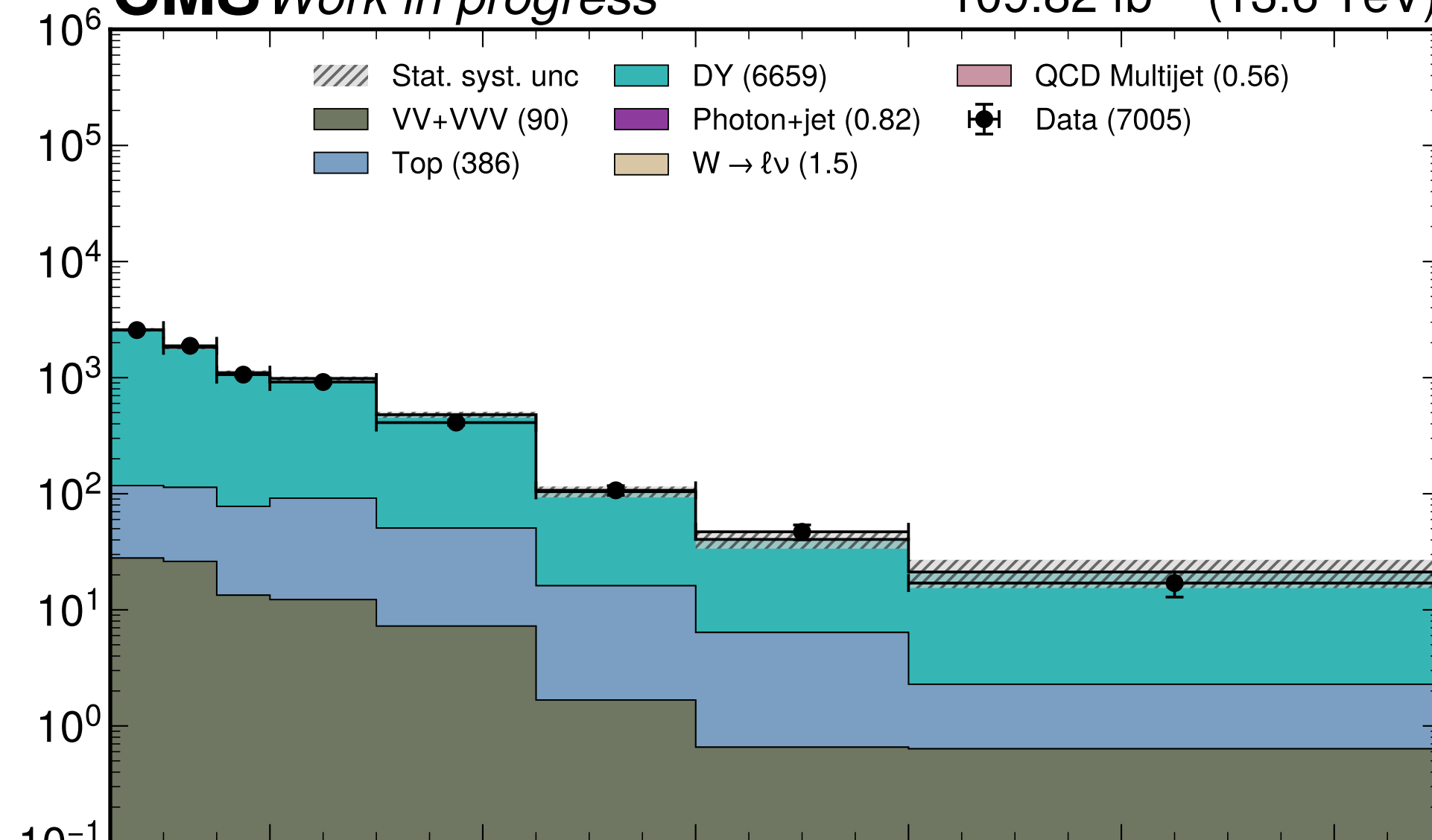
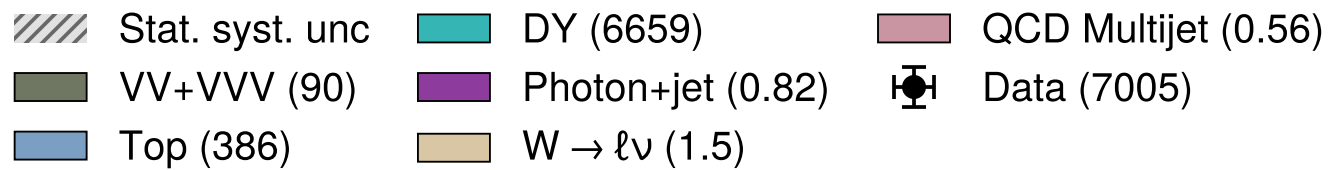
N_j



CMS *Work in progress*

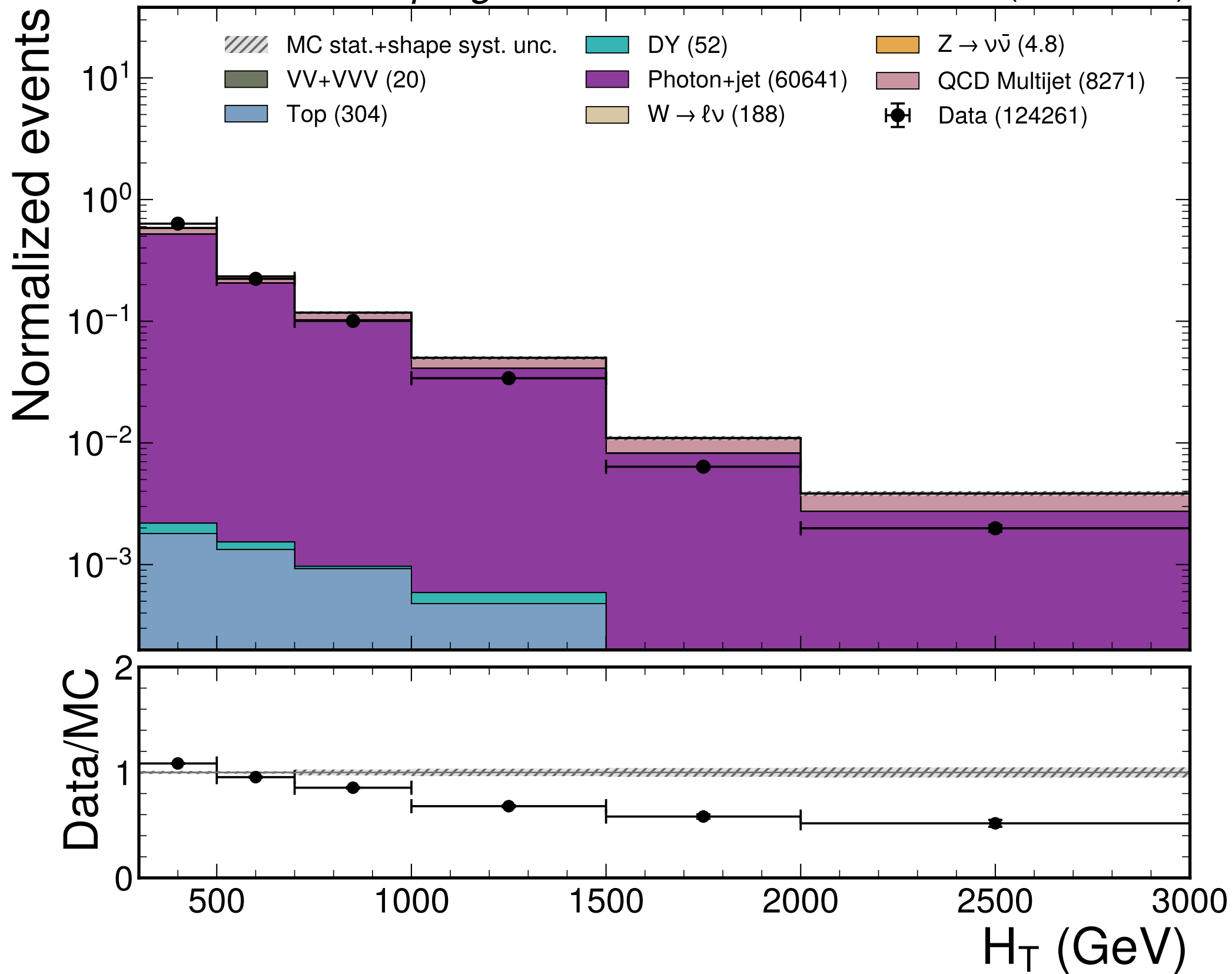
109.82 fb⁻¹ (13.6 TeV)

Events



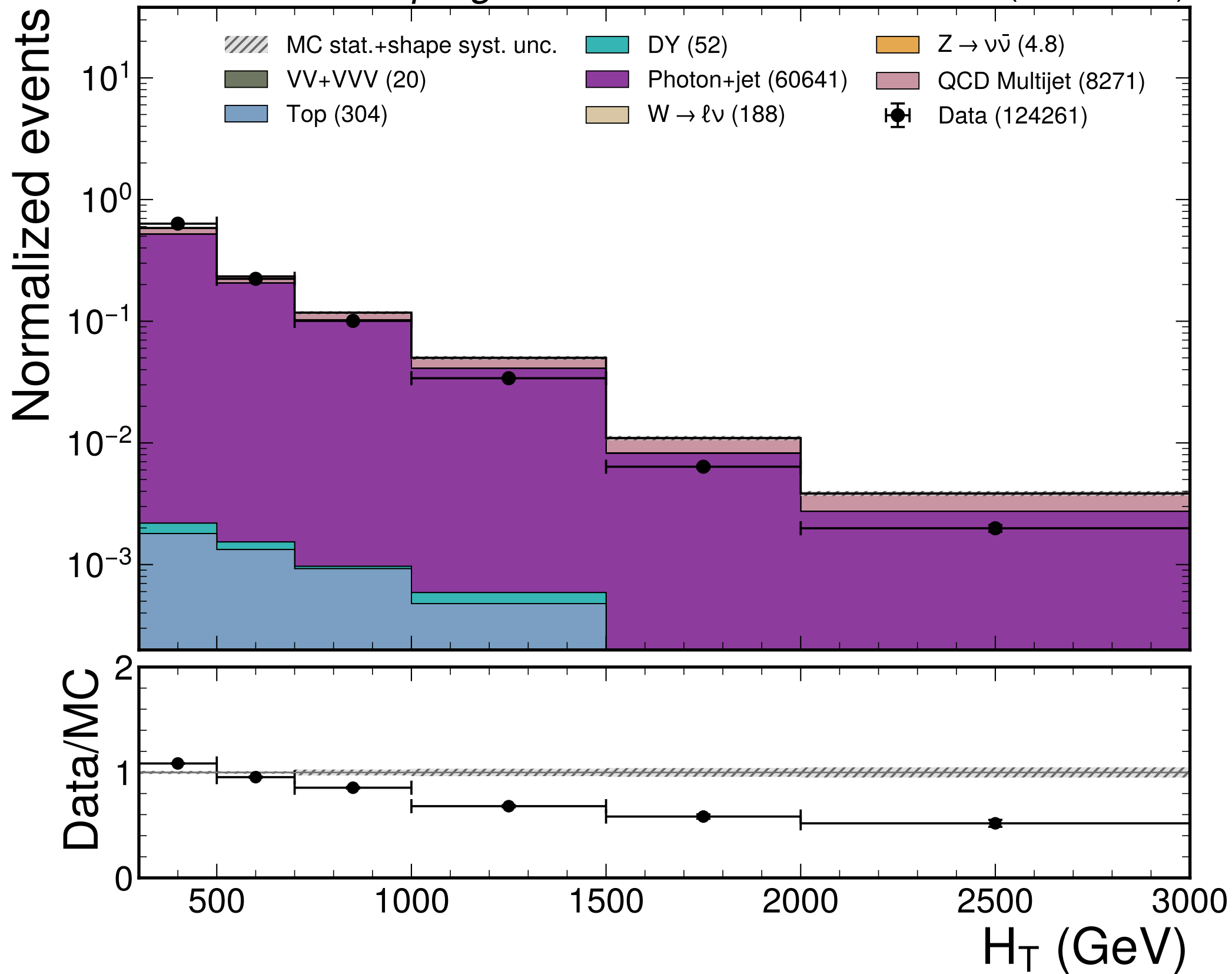
CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)



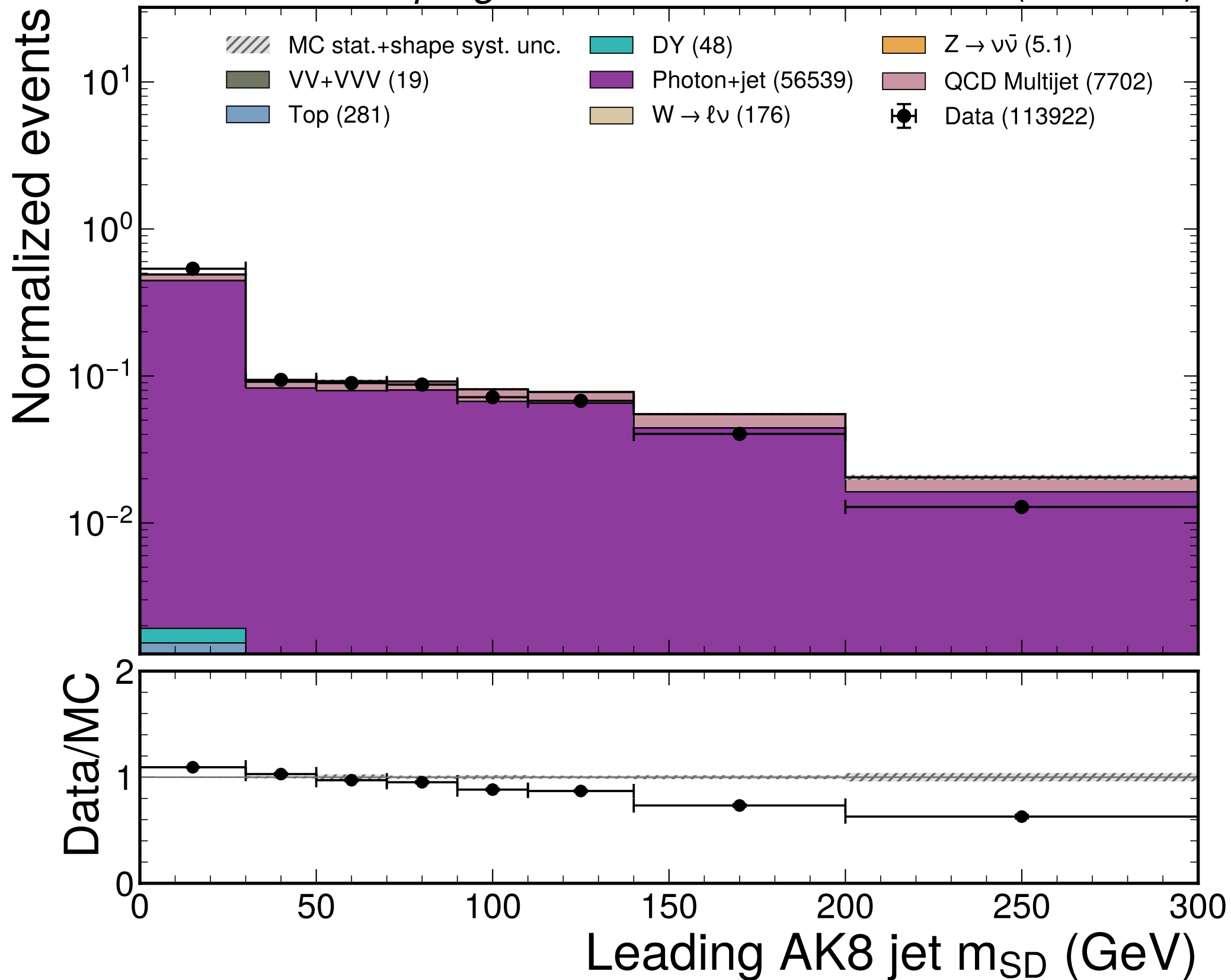
CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)



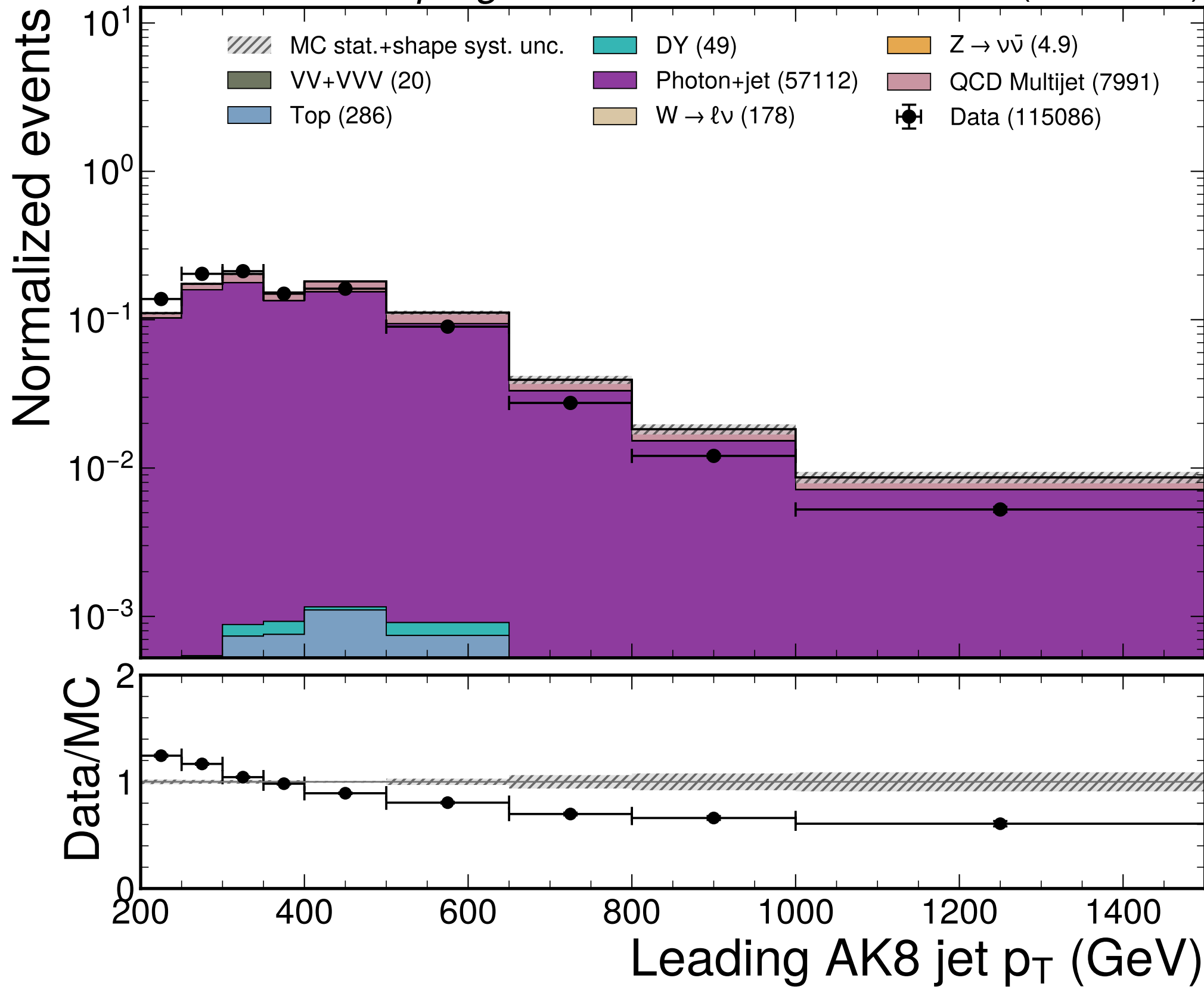
CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)



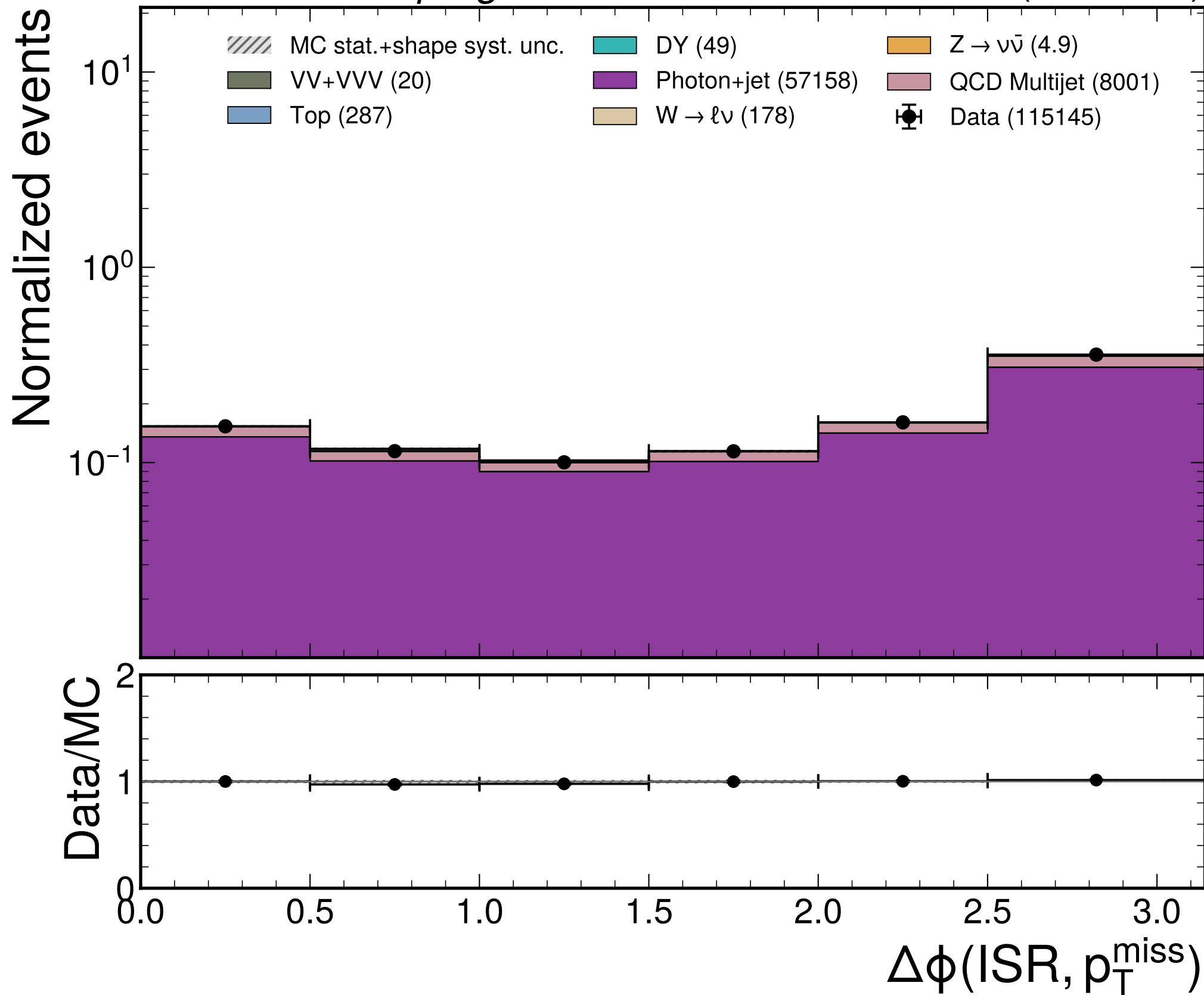
CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)



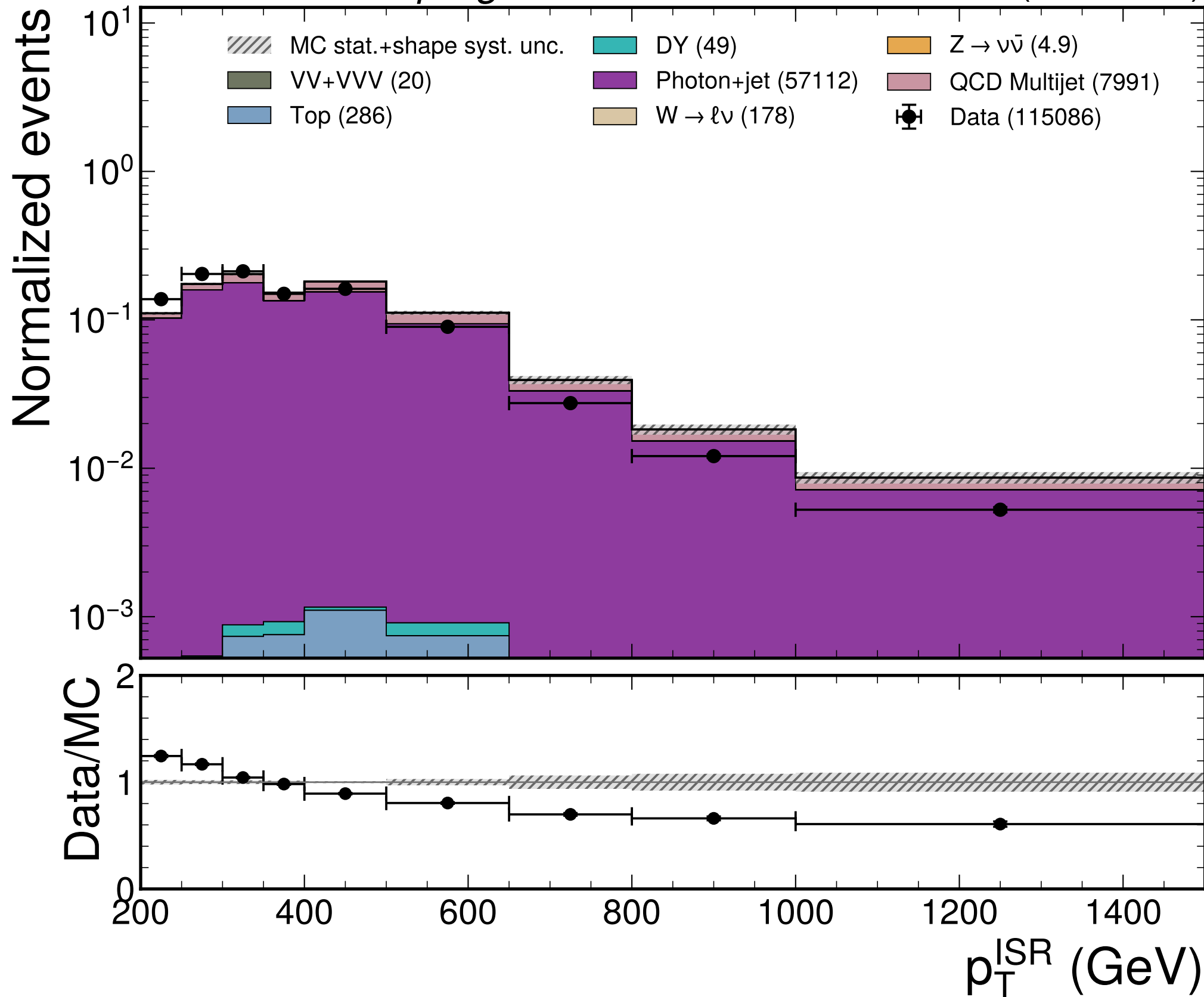
CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)



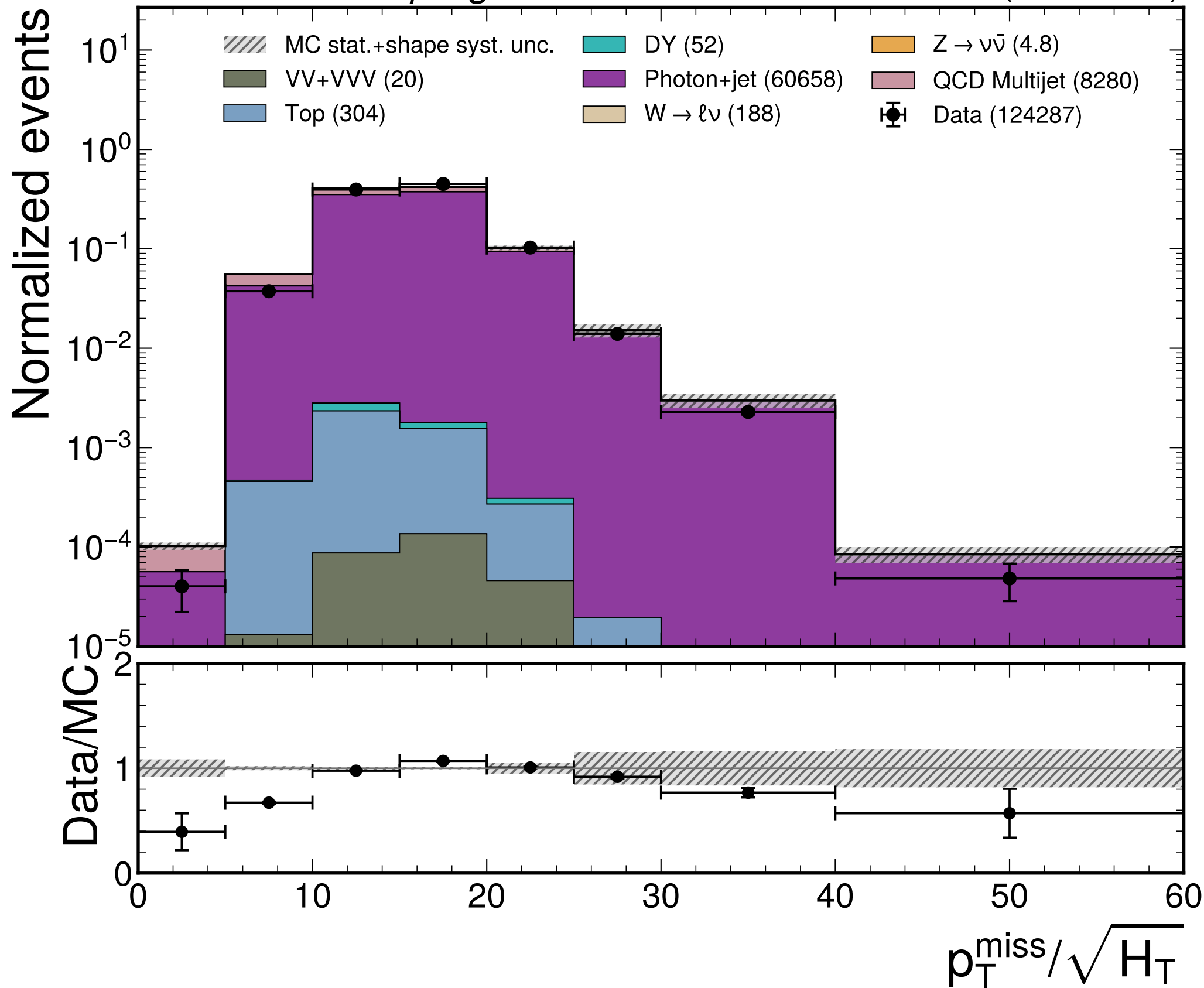
CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)



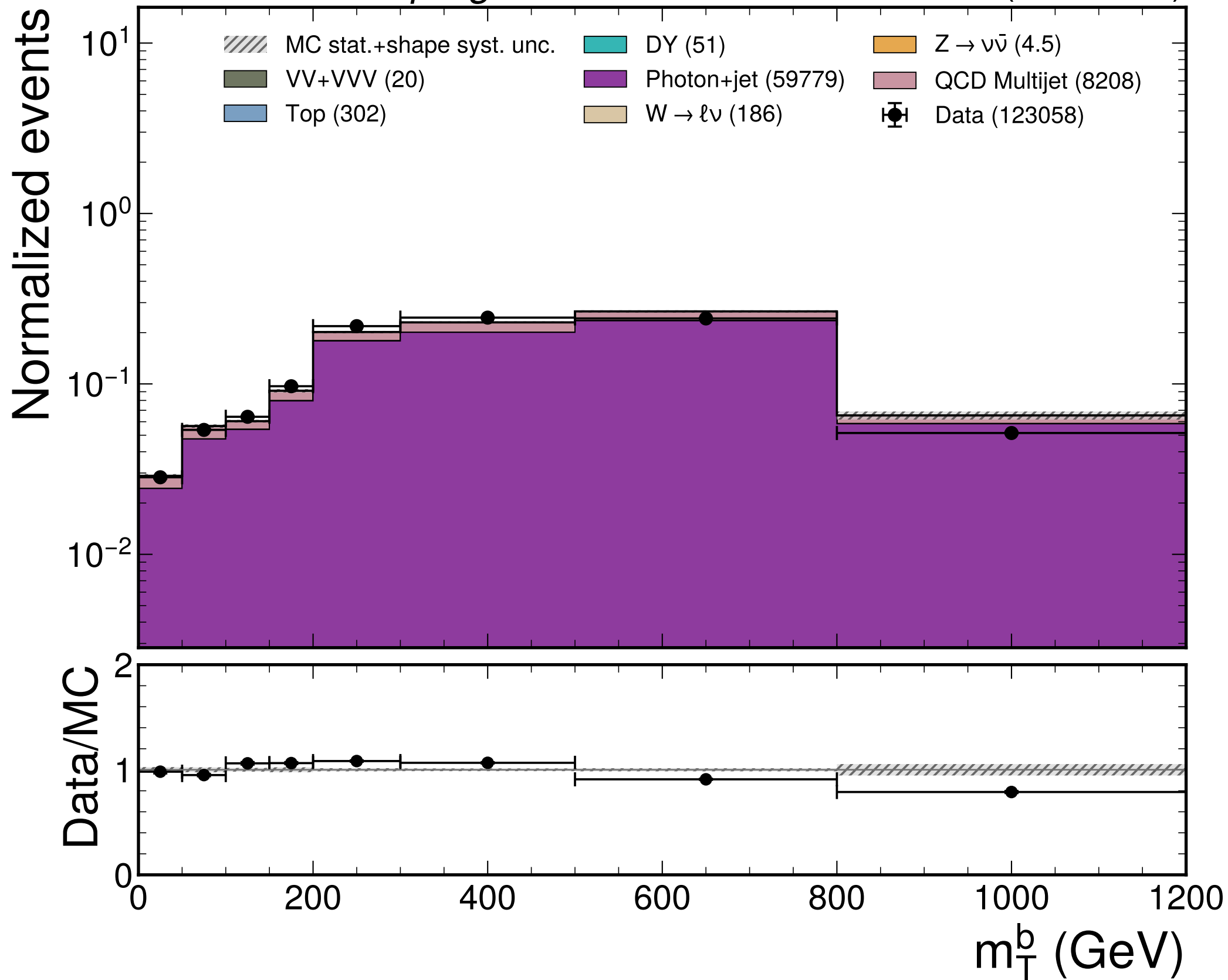
CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)



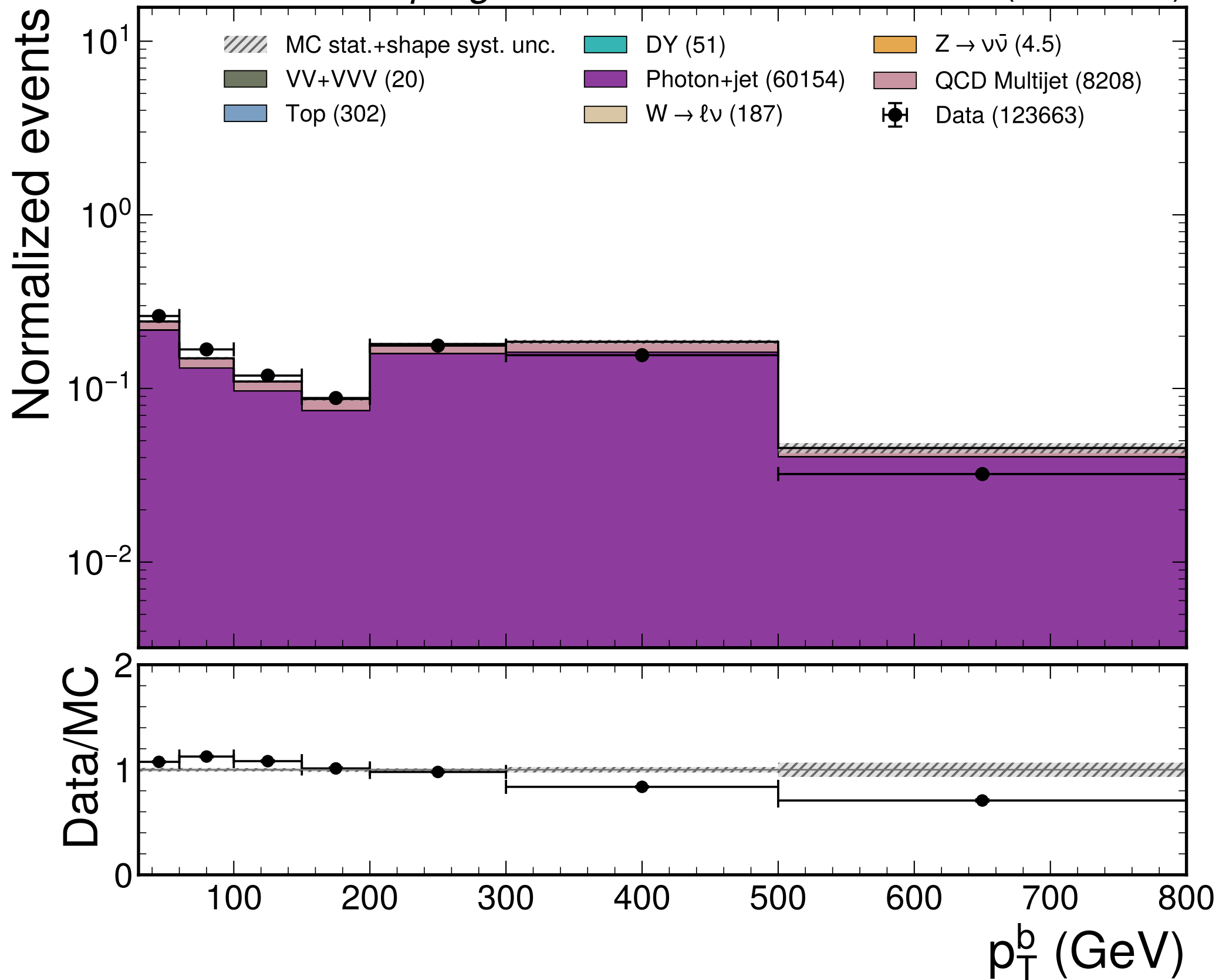
CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)



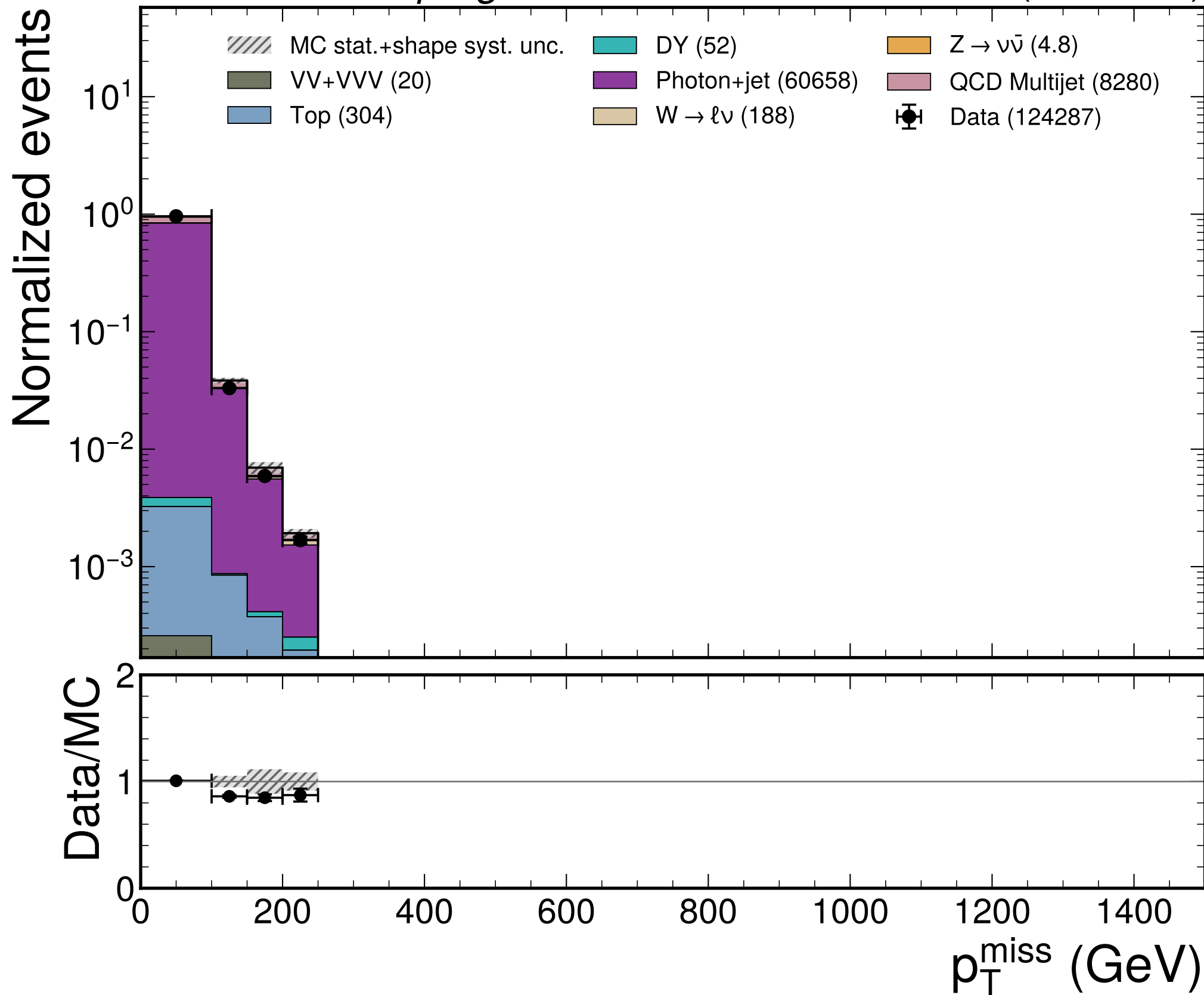
CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)



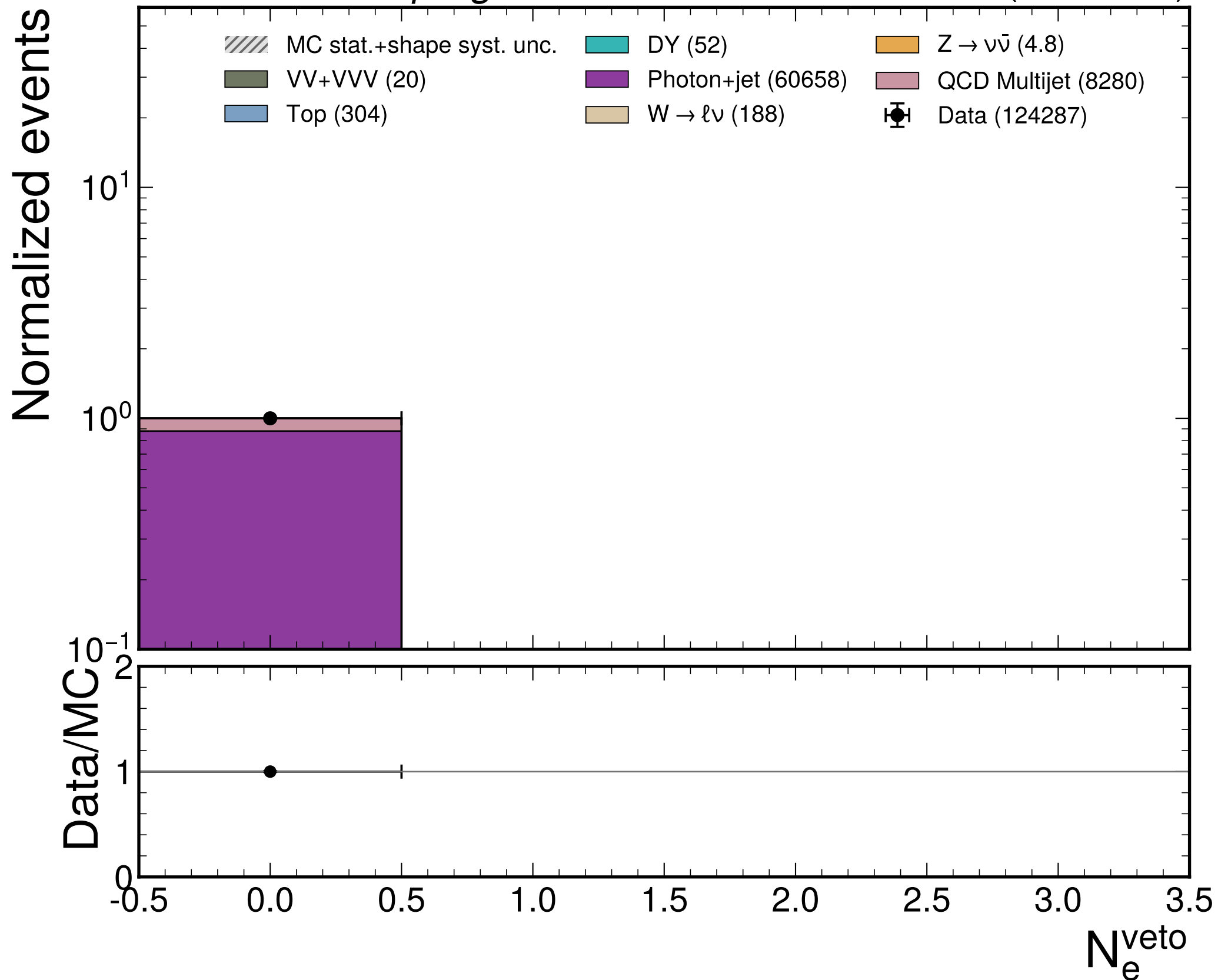
CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)



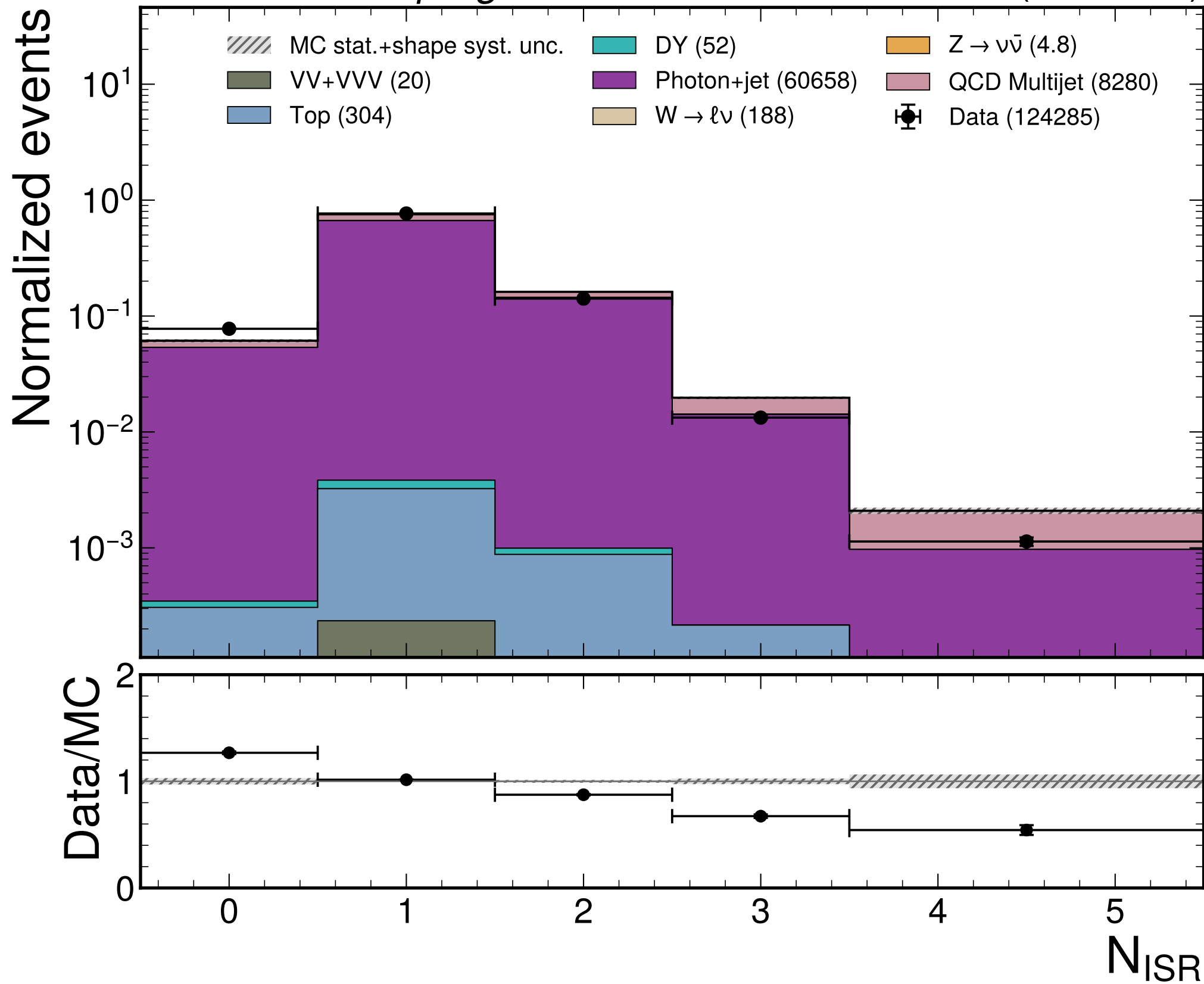
CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)



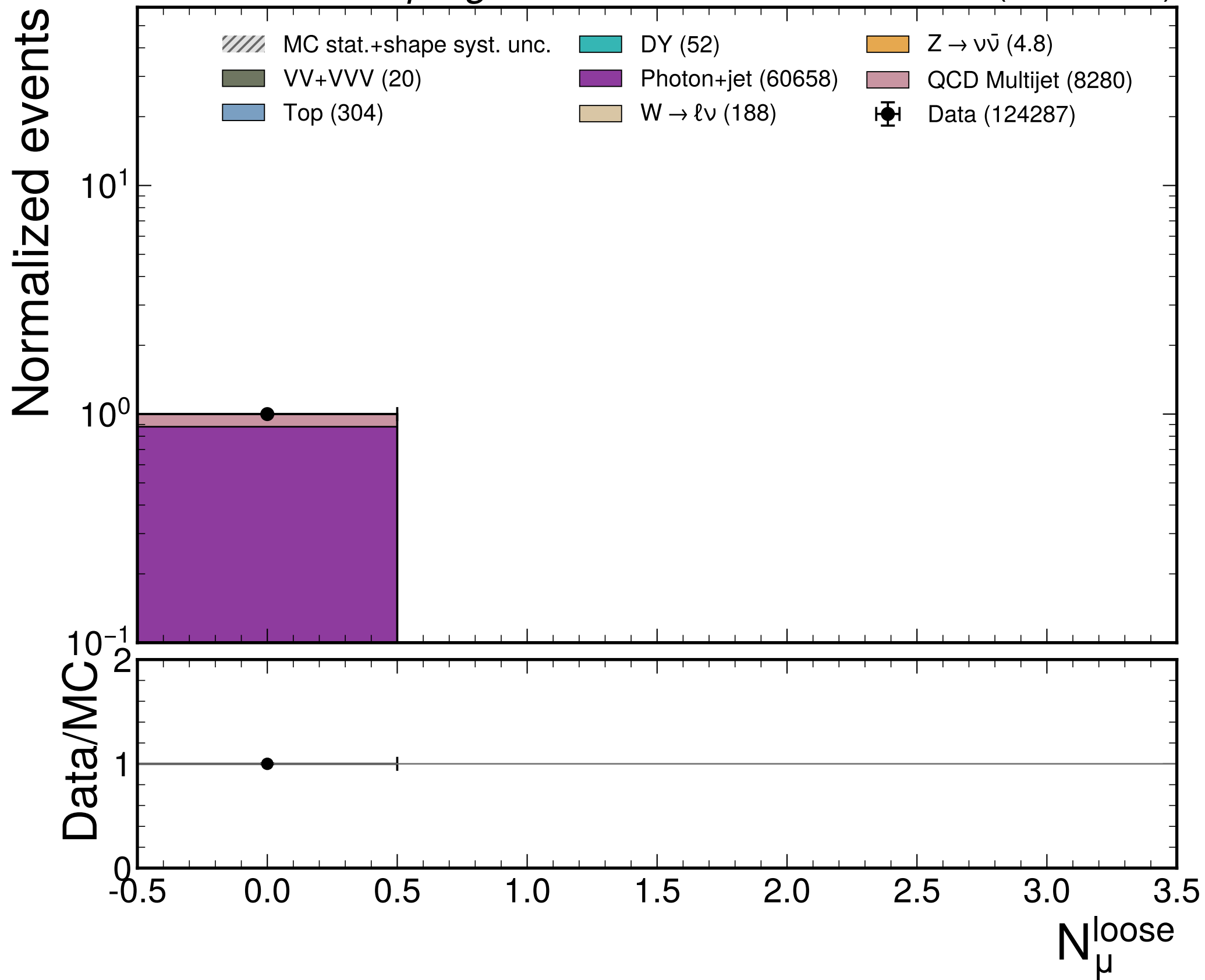
CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)



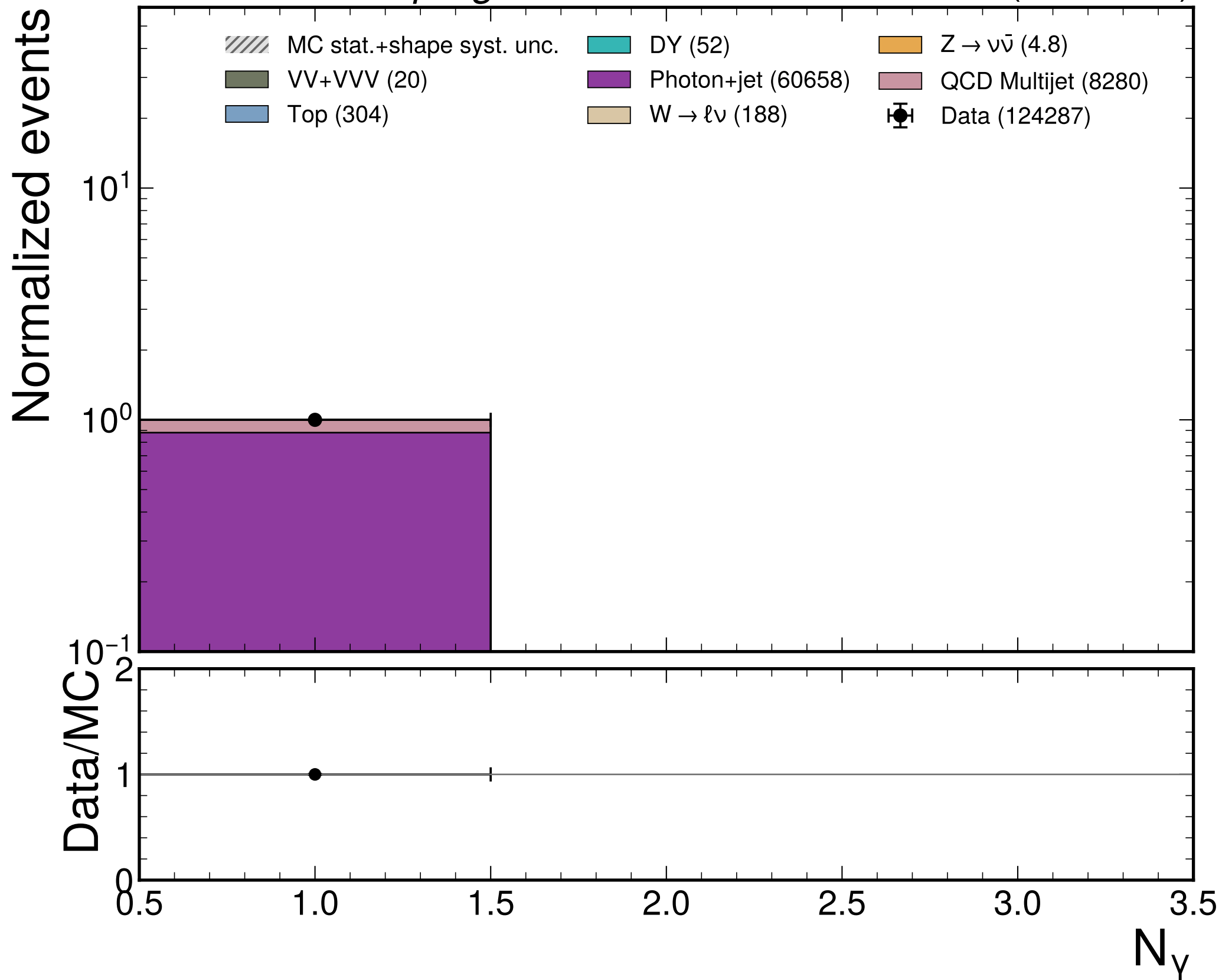
CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)



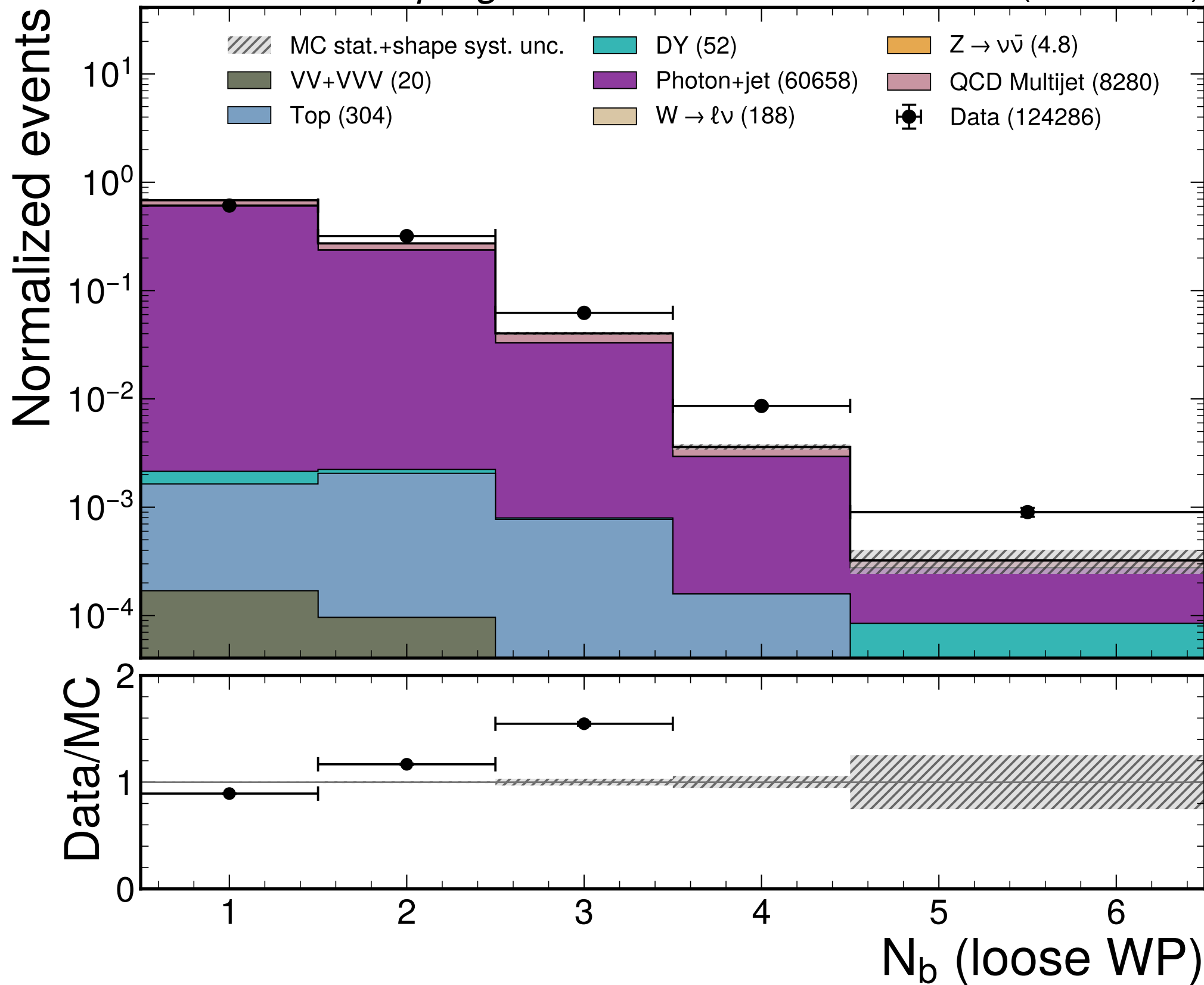
CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)



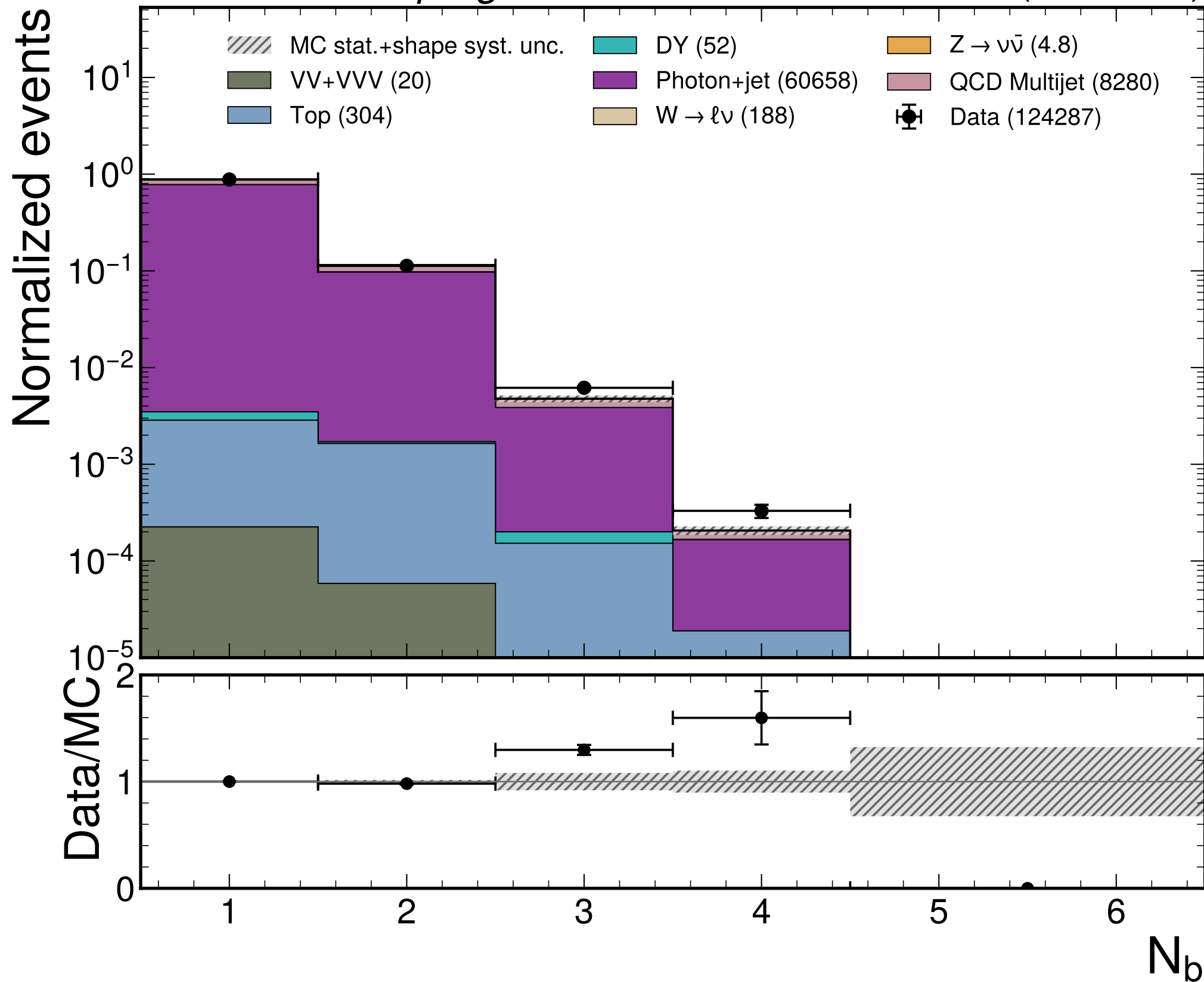
CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)



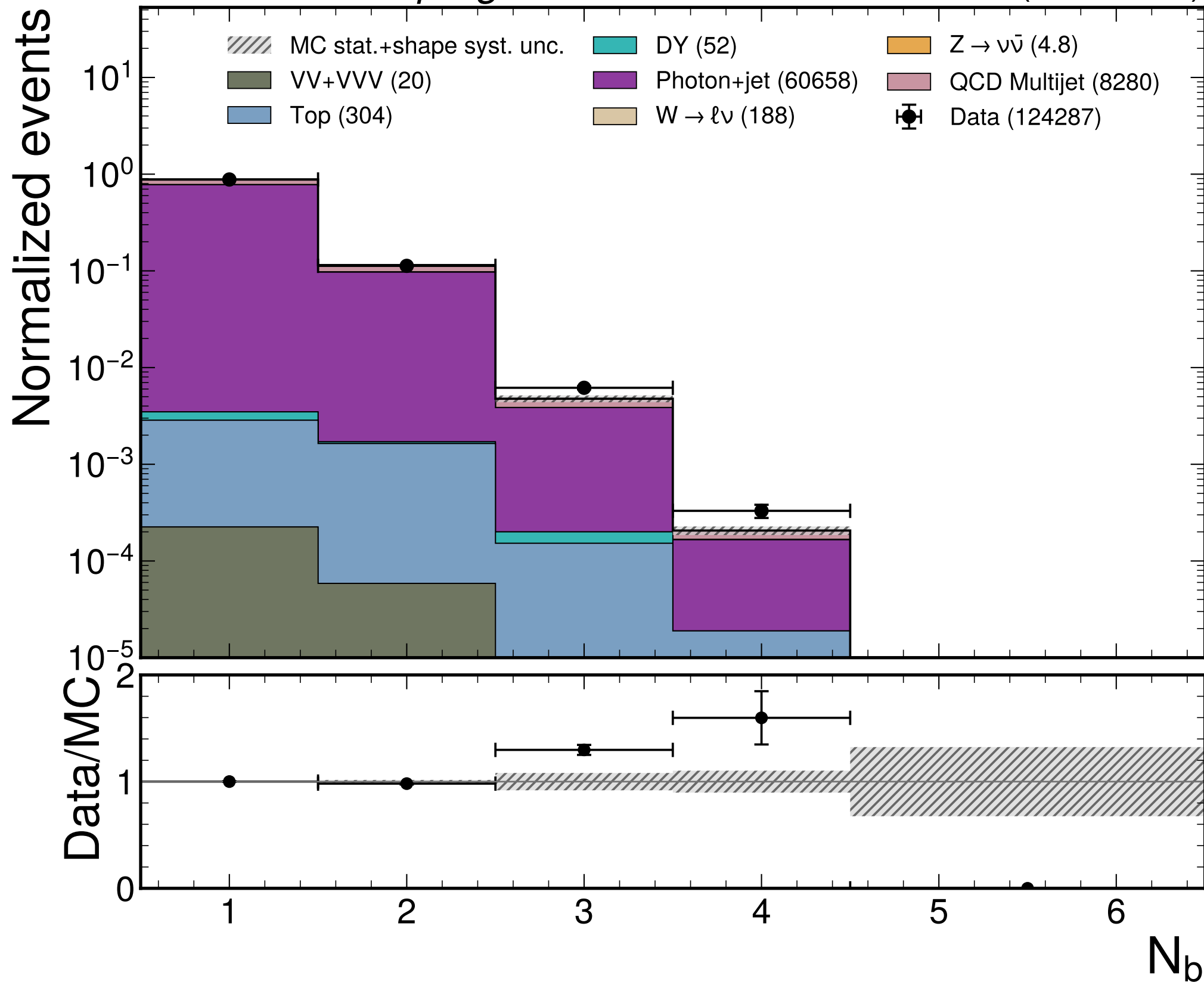
CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)



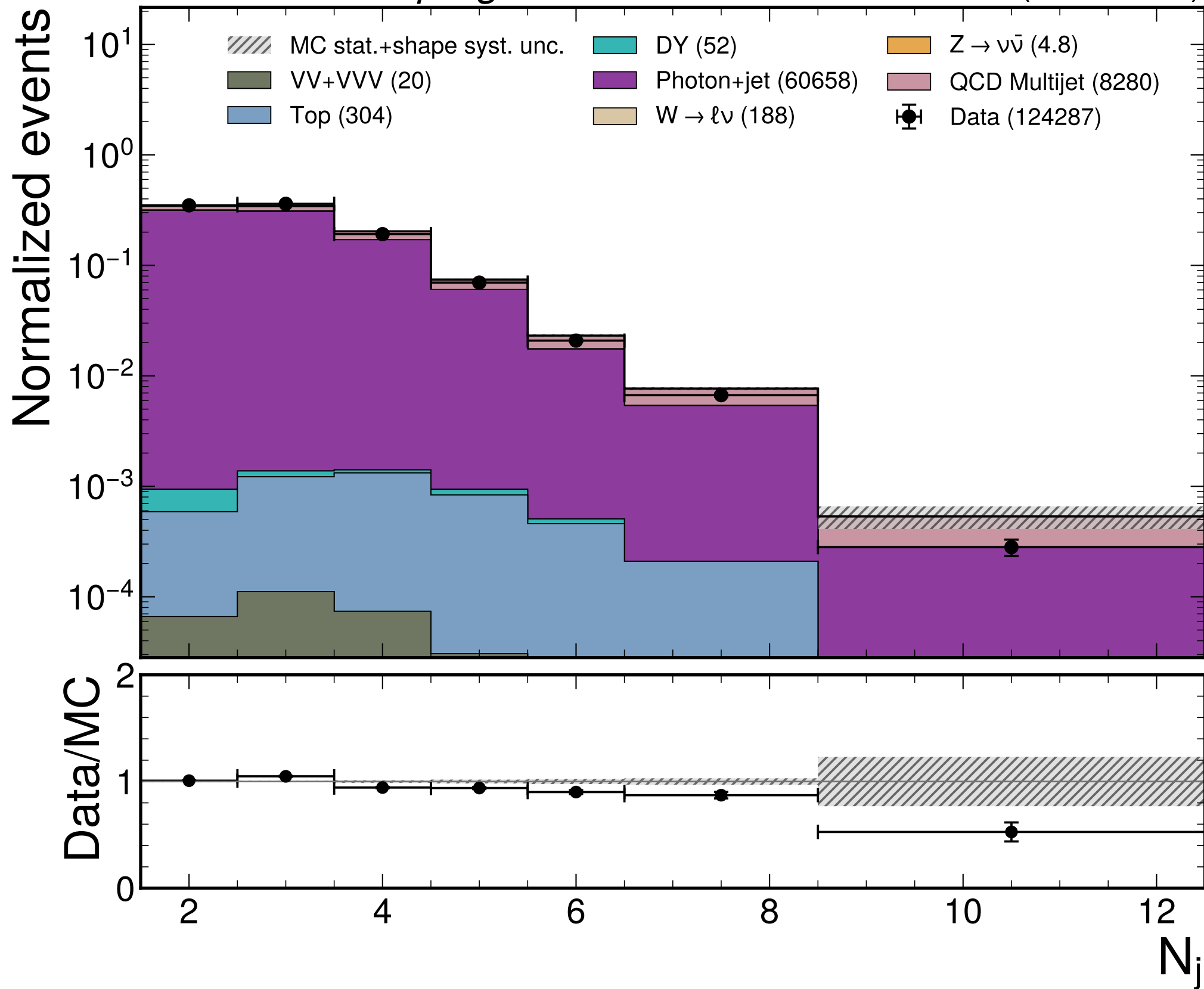
CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)



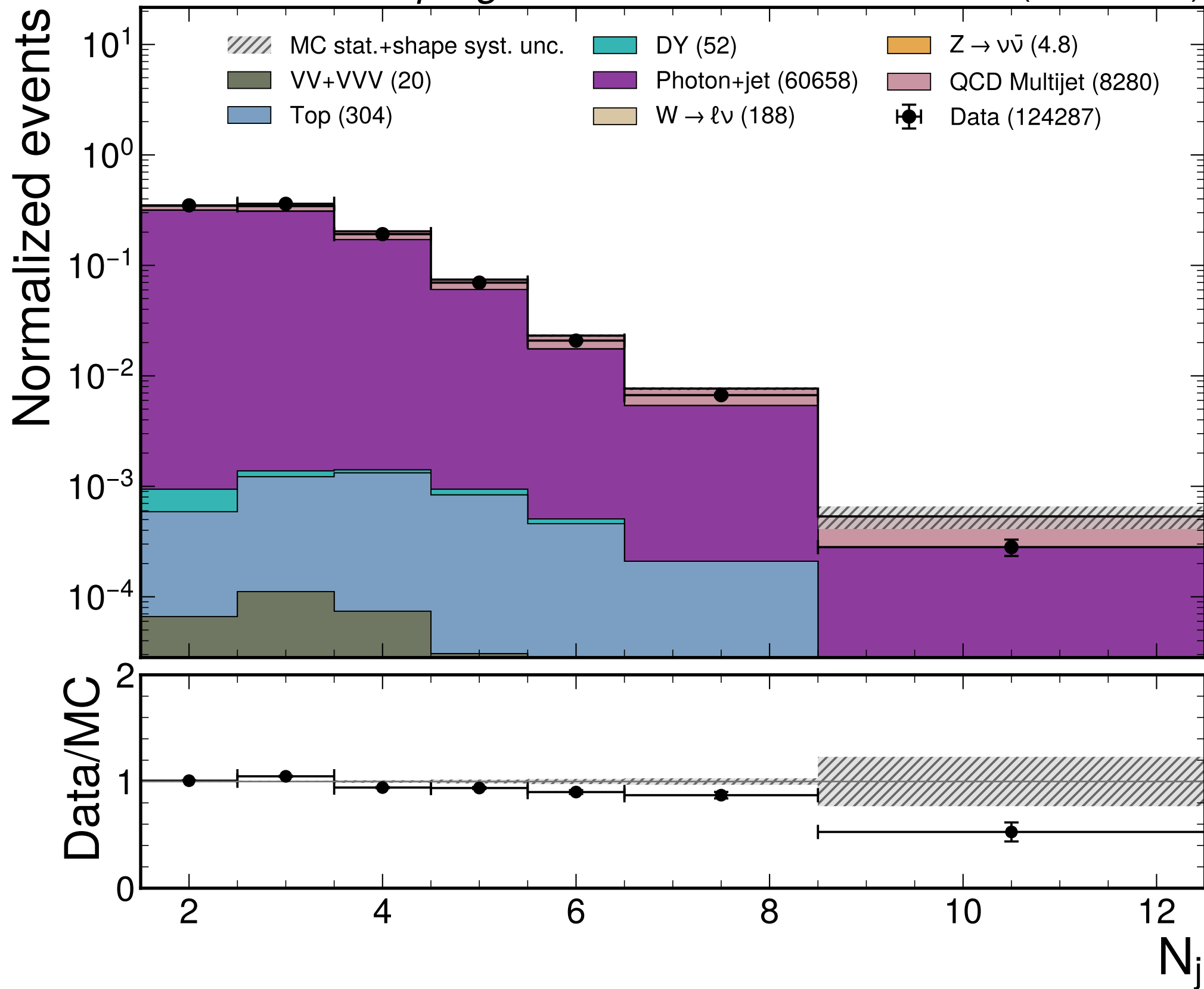
CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)



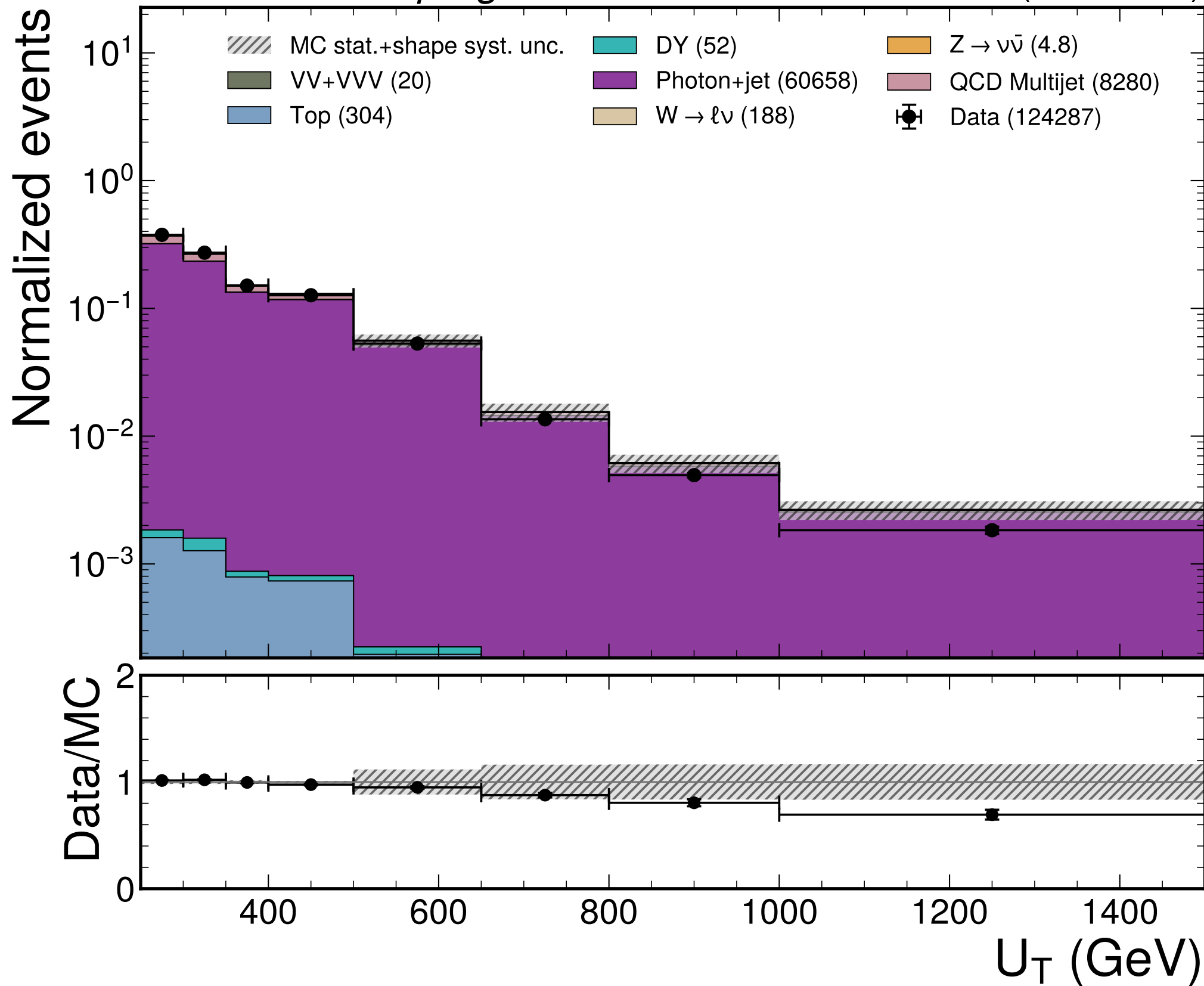
CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)



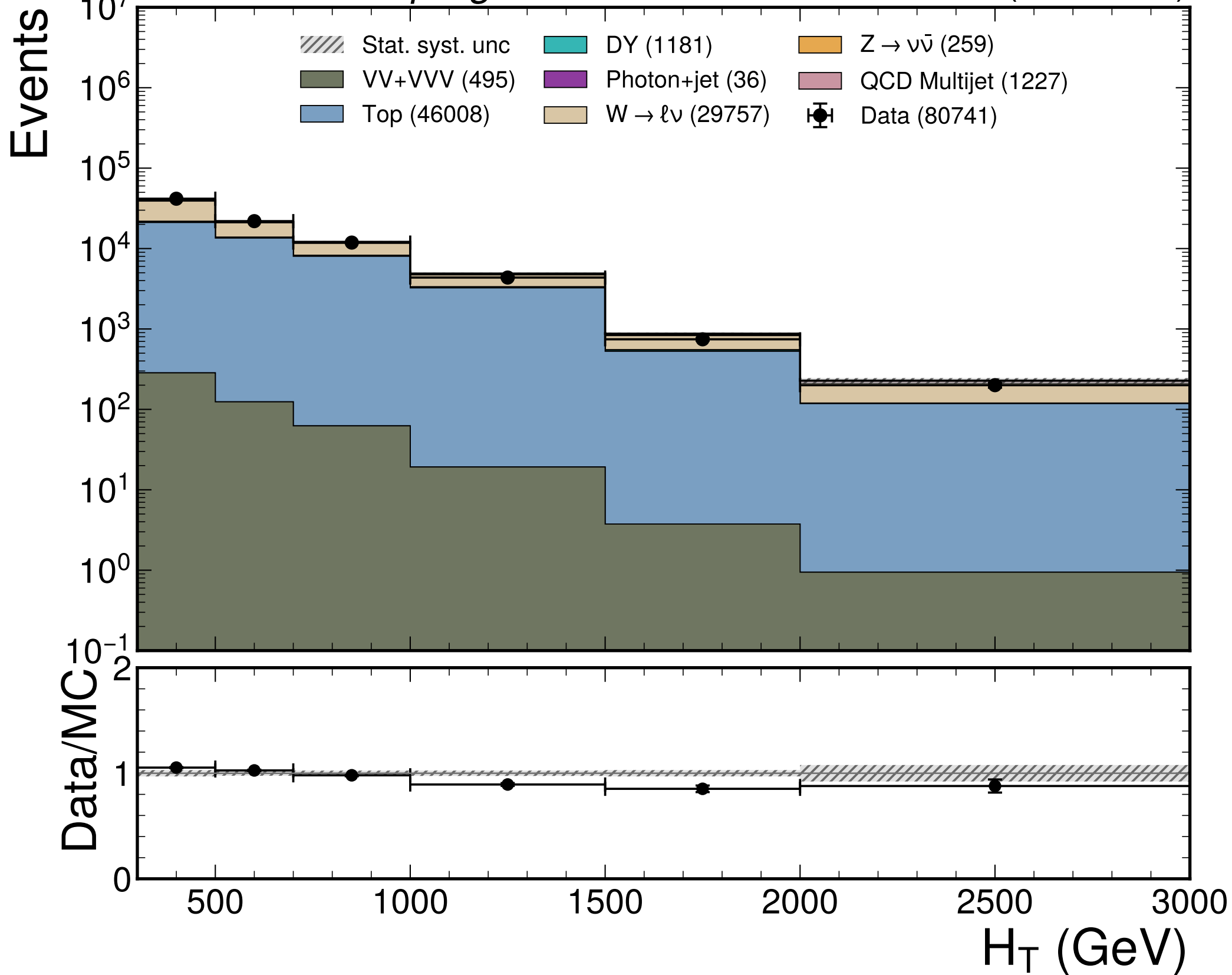
CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)



CMS *Work in progress*

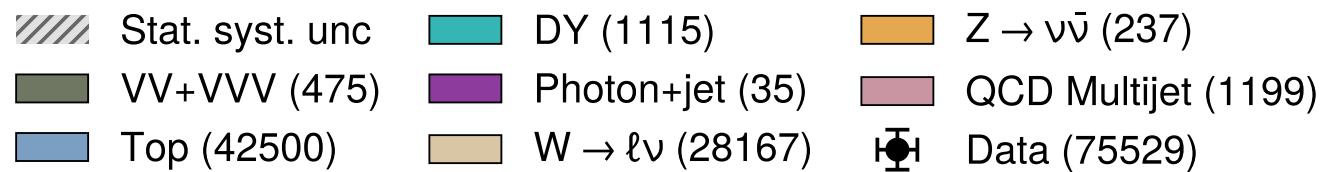
109.82 fb⁻¹ (13.6 TeV)



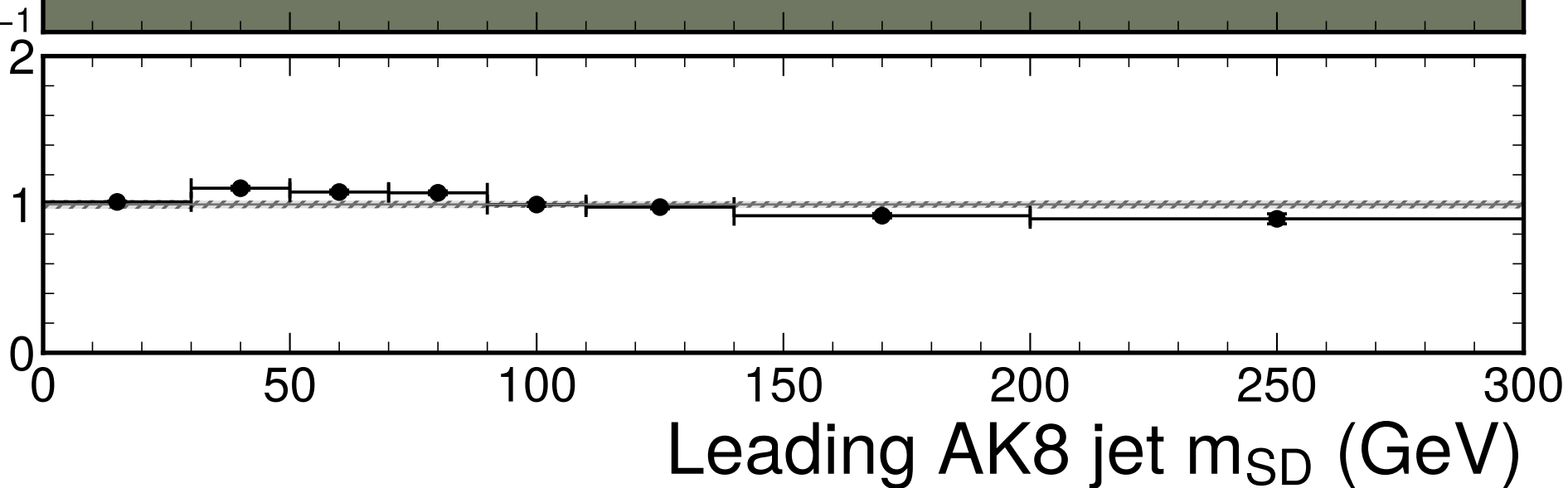
CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)

Events



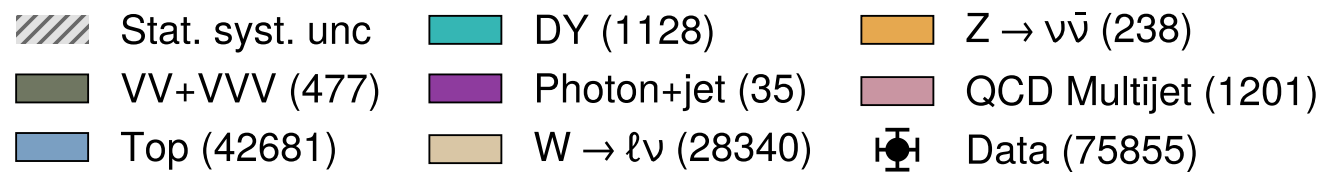
Data/MC



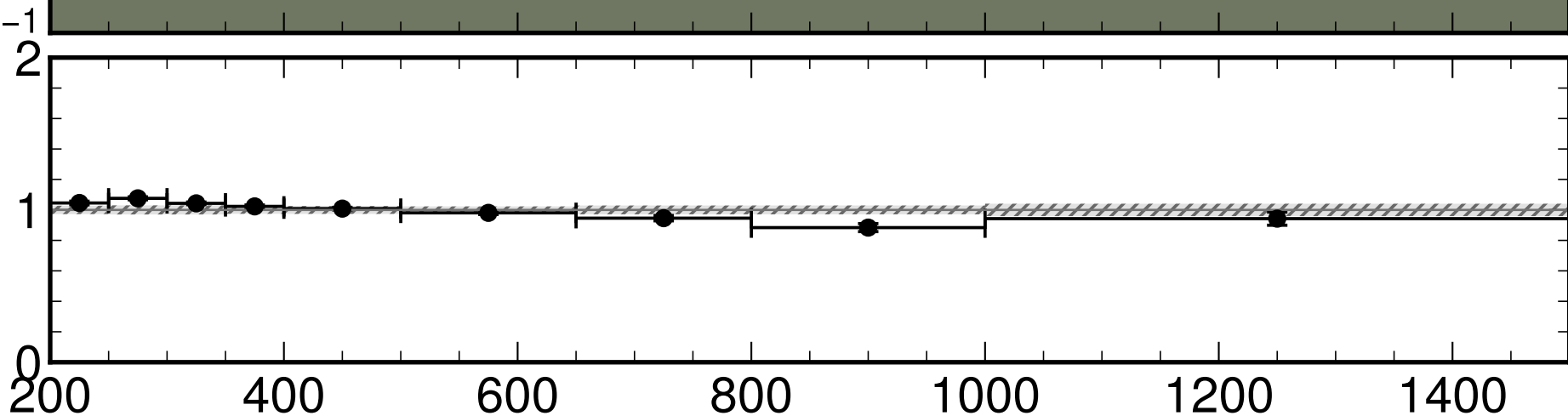
CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)

Events



Data/MC

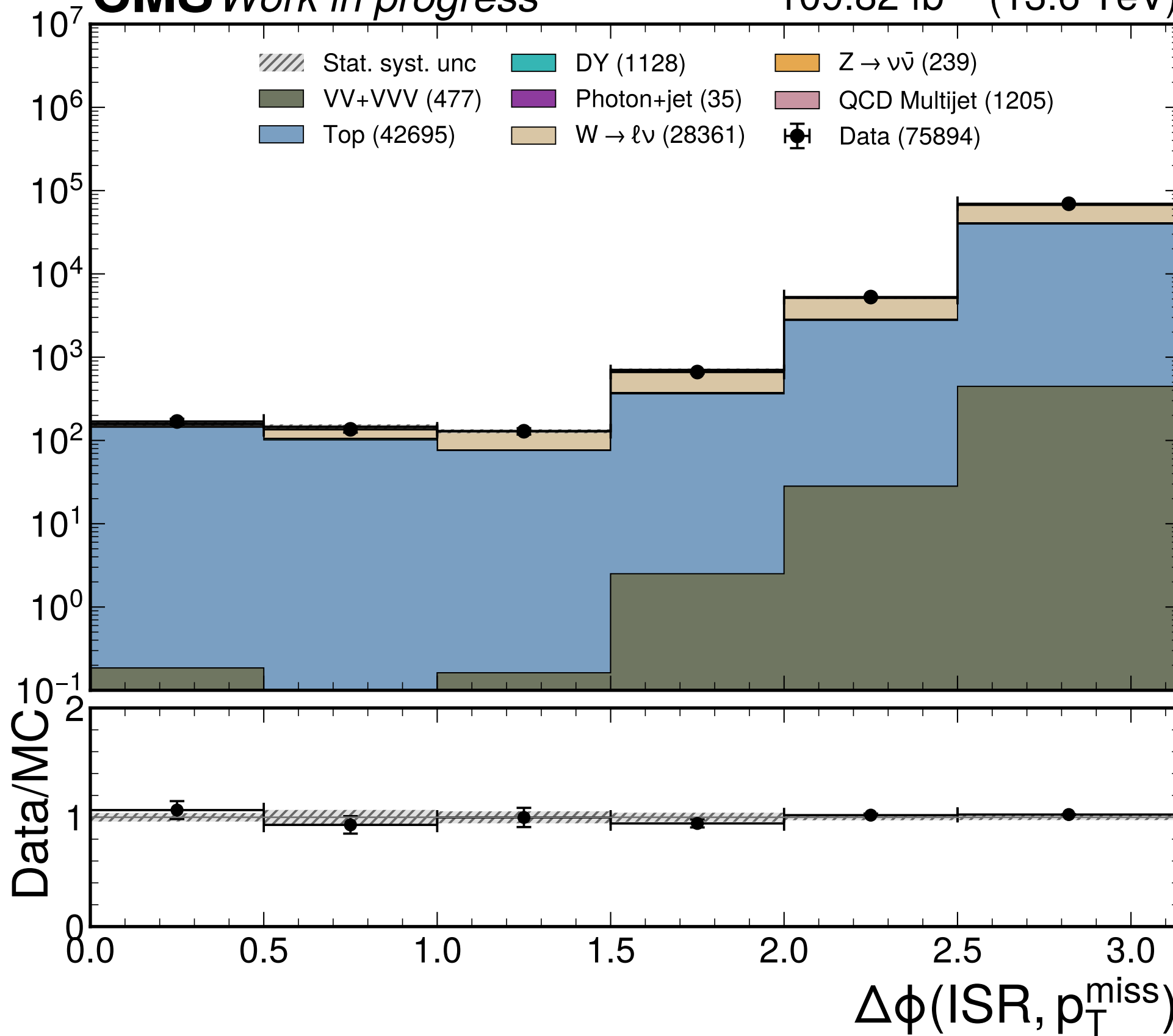
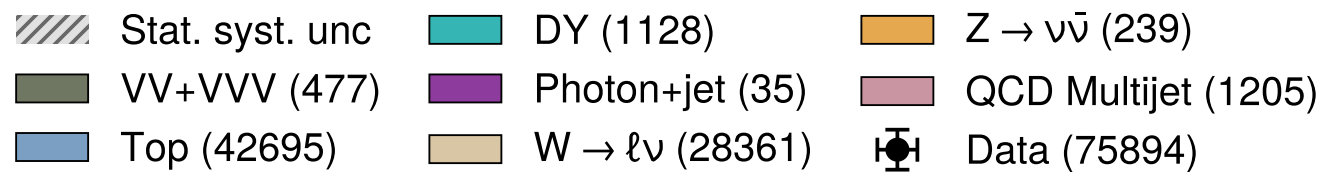


Leading AK8 jet p_T (GeV)

CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)

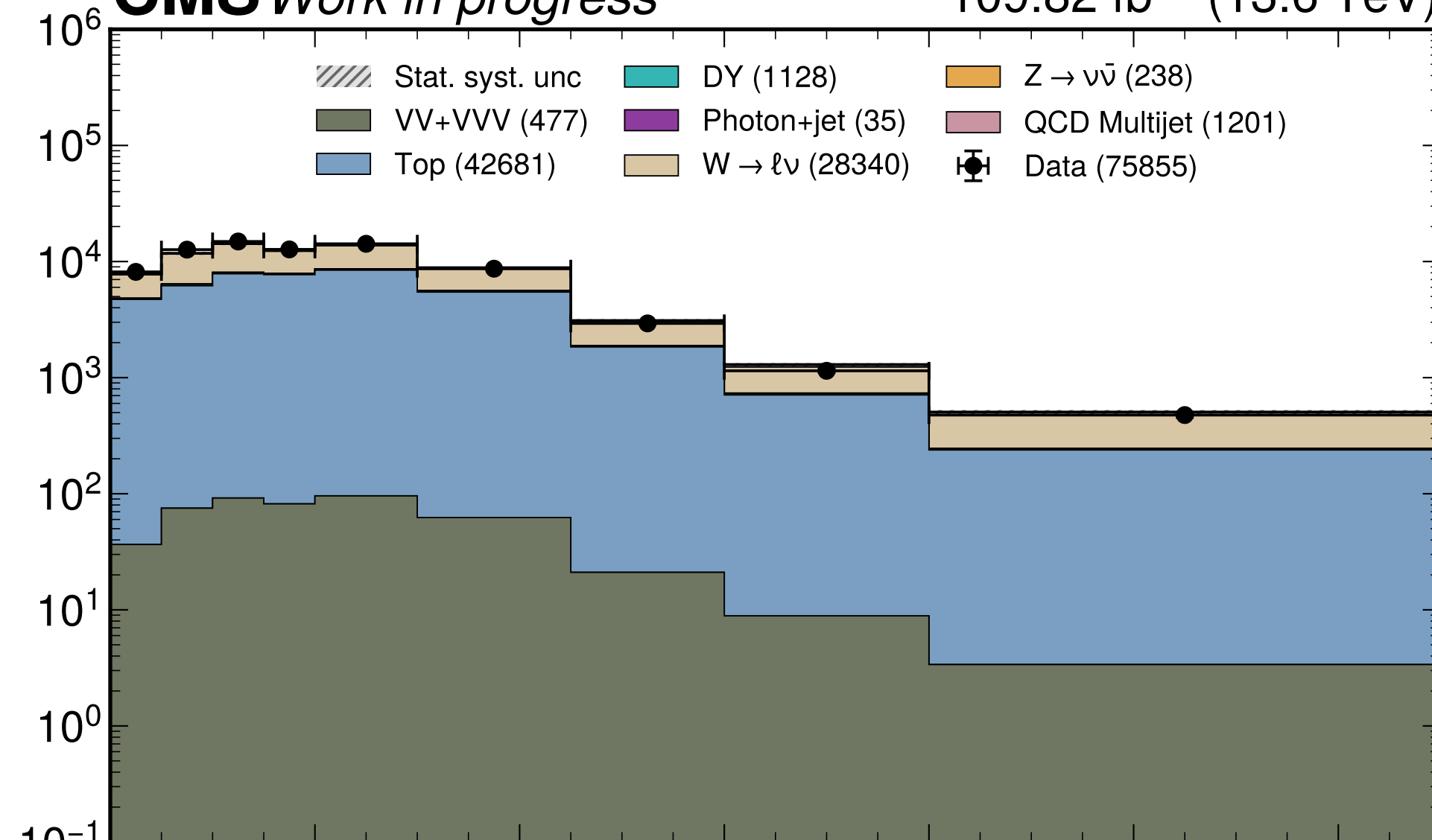
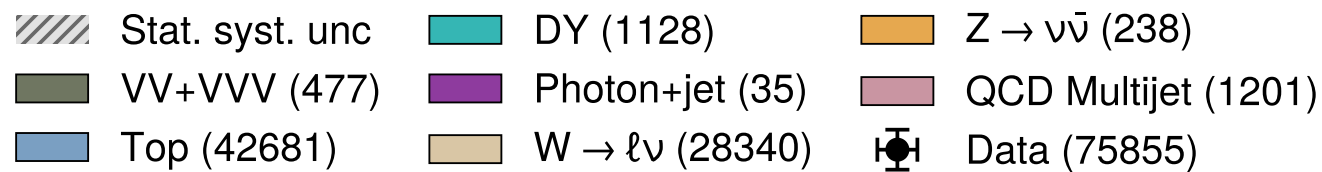
Events



CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)

Events



Data/MC

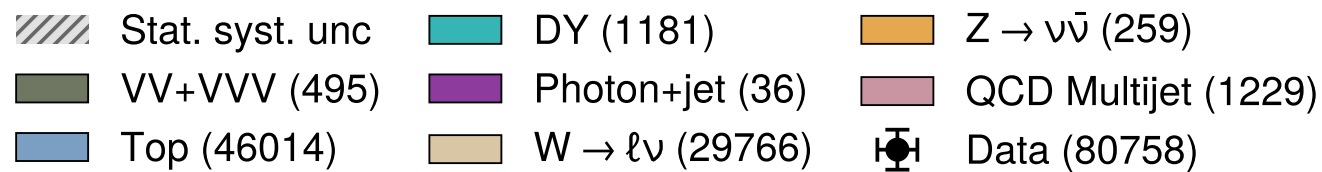
200 400 600 800 1000 1200 1400

p_T^{ISR} (GeV)

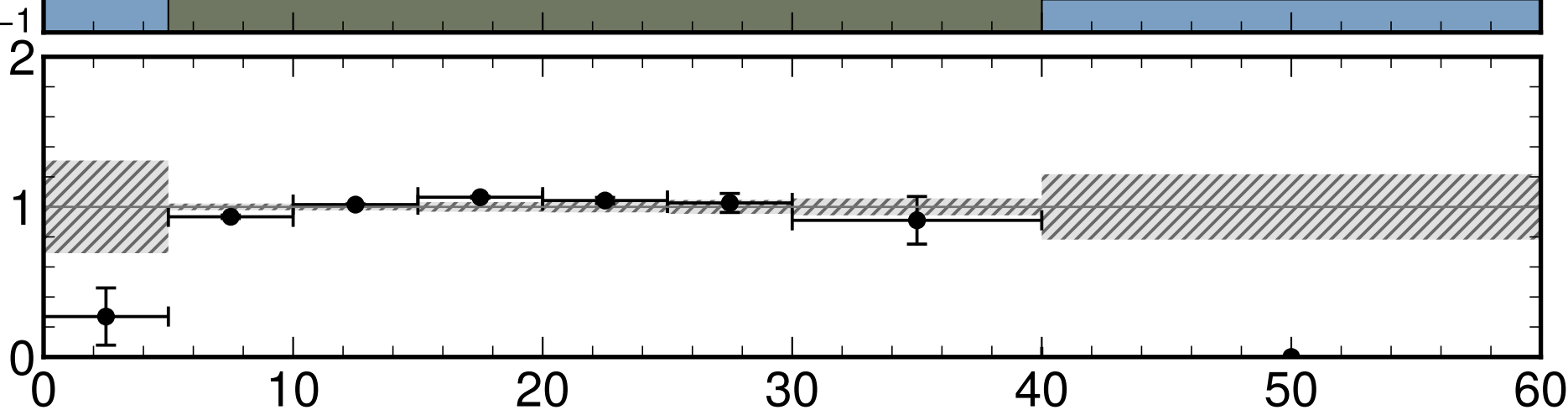
CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)

Events



Data/MC

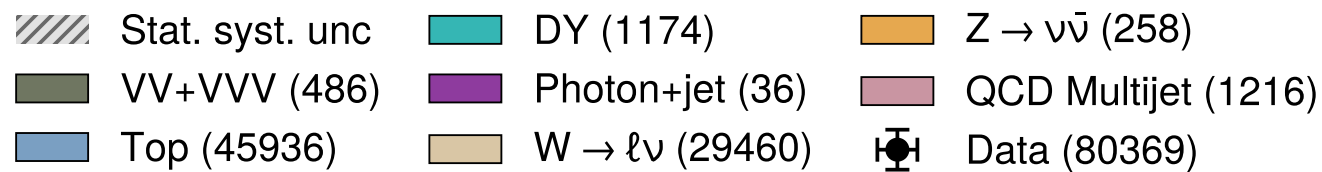


$p_T^{\text{miss}}/\sqrt{H_T}$

CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)

Events

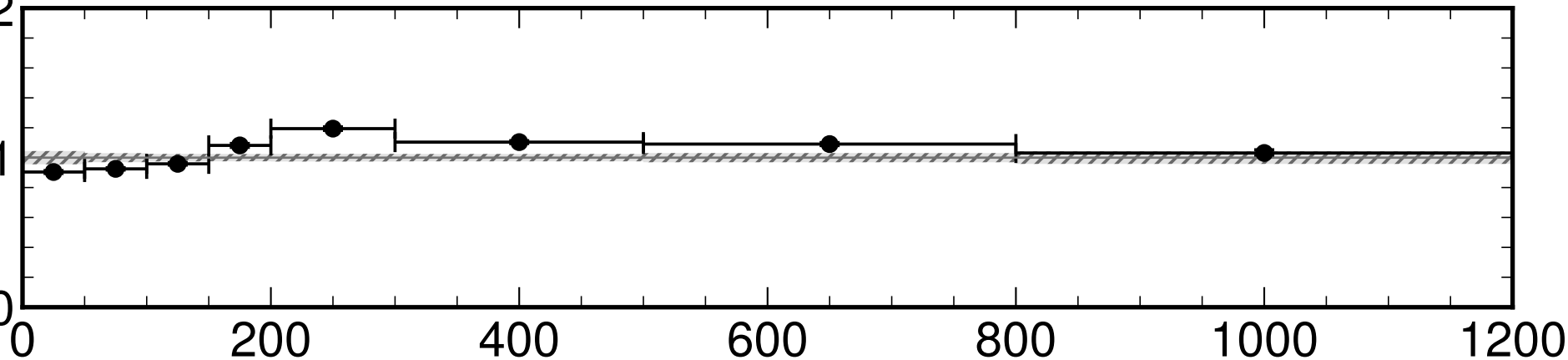


Data/MC

10⁻¹

2

0

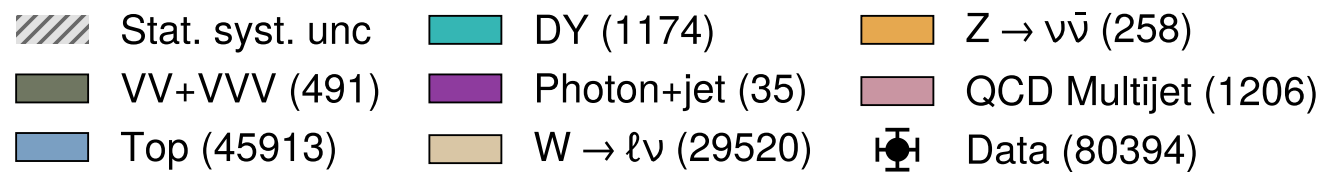


m_T^b (GeV)

CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)

Events



Data/MC

10⁻²

2

1

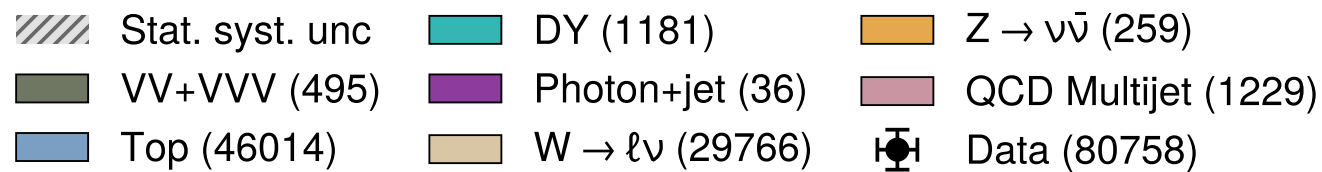
0

p_T^b (GeV)

CMS *Work in progress*

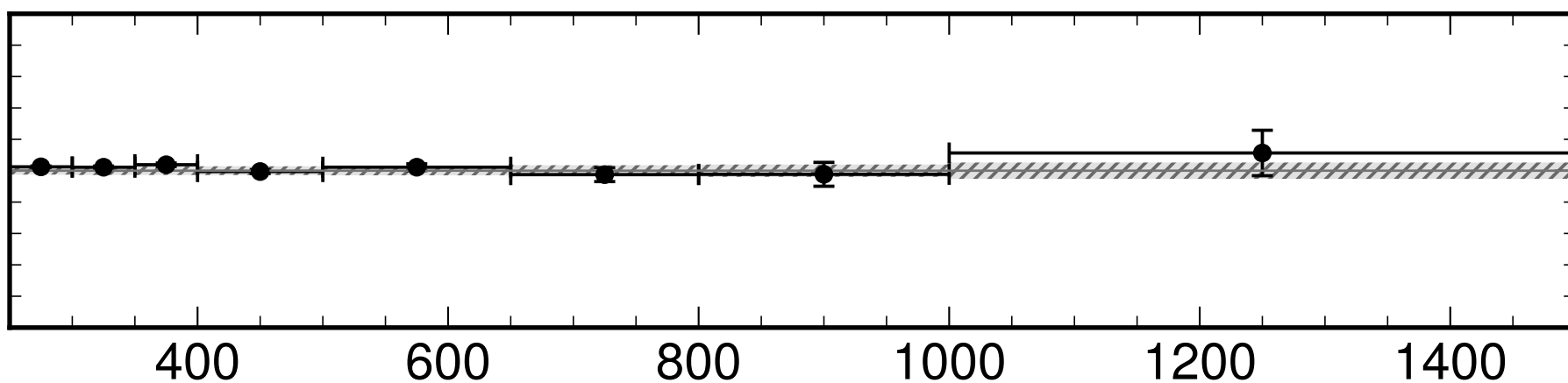
109.82 fb⁻¹ (13.6 TeV)

Events



Data/MC

10⁻²

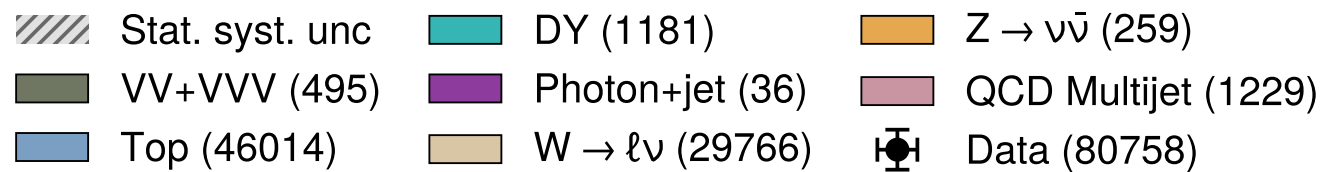


p_T^{miss} (GeV)

CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)

Events



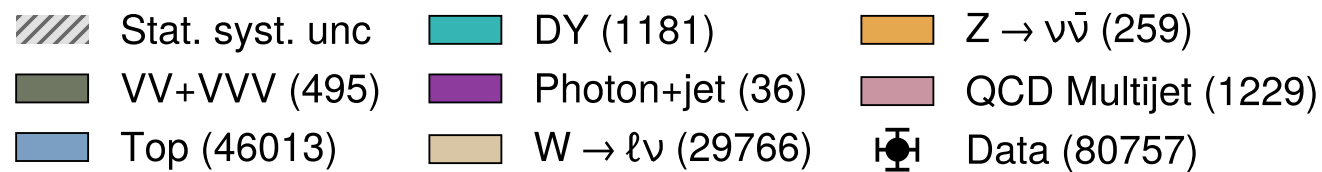
Data/MC

N_e^{veto}

CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)

Events



Data/MC

10⁻¹
2
1
0

0

1

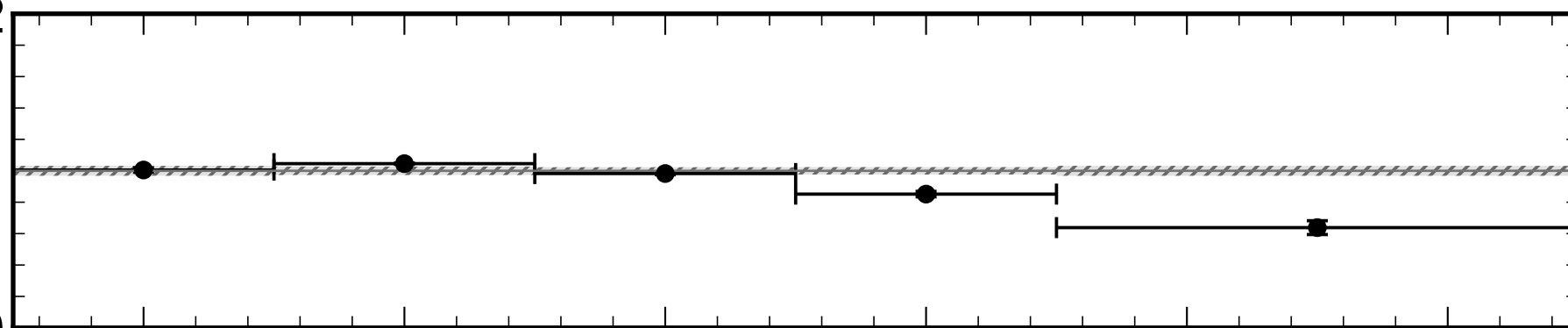
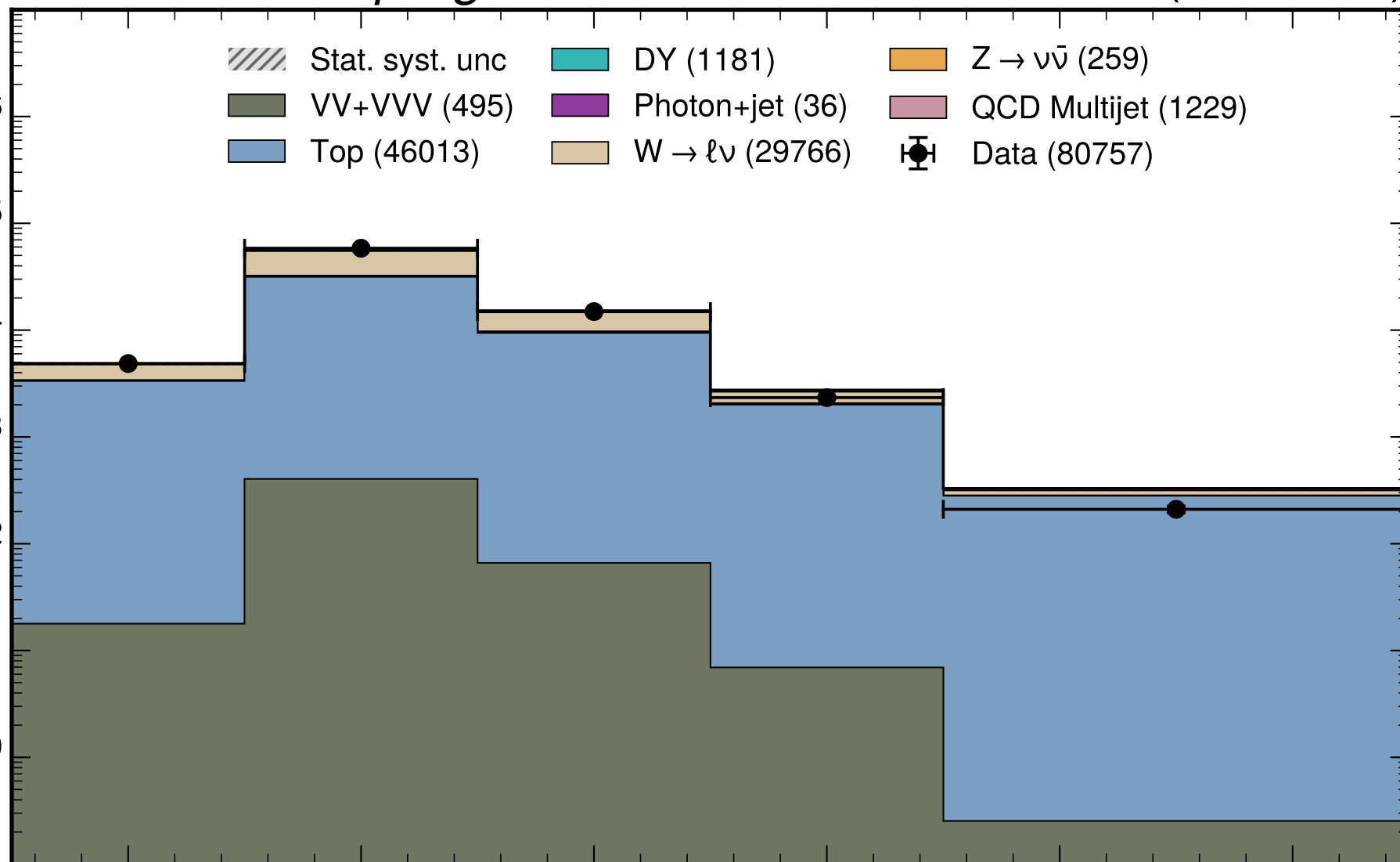
2

3

4

5

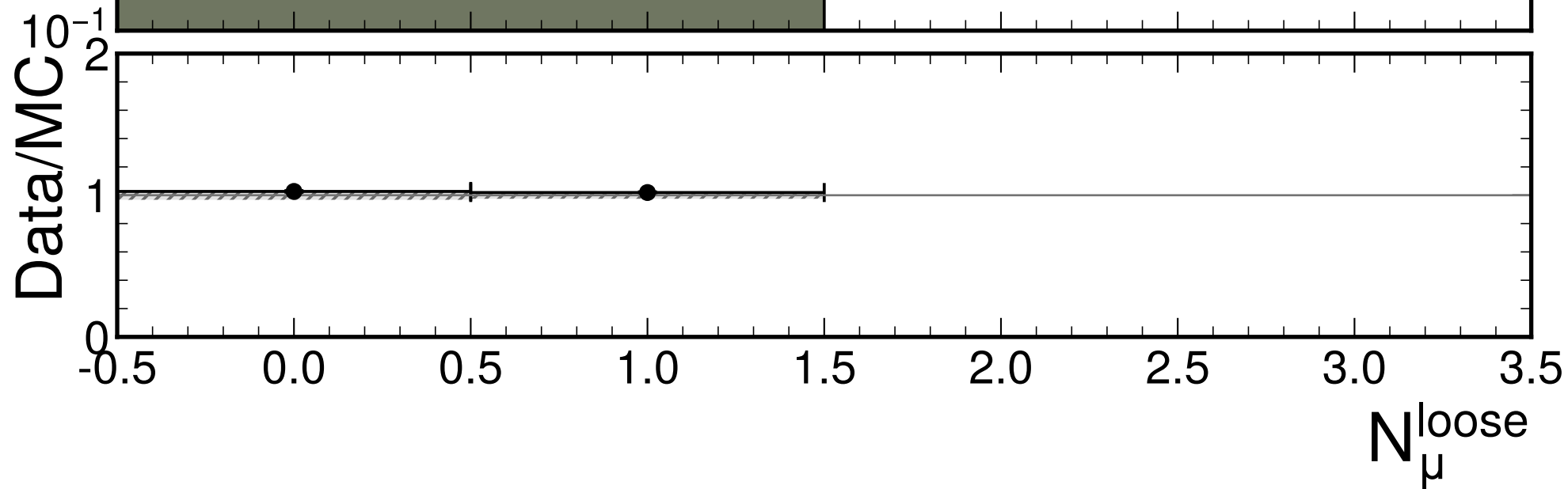
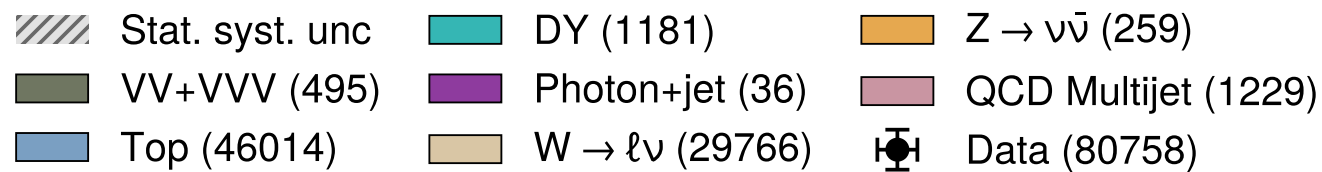
N_{ISR}



CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)

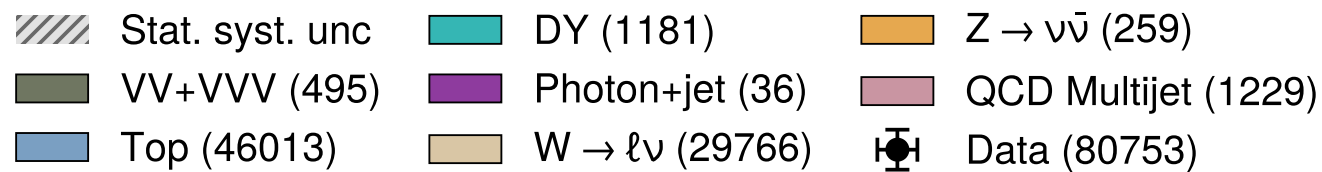
Events



CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)

Events



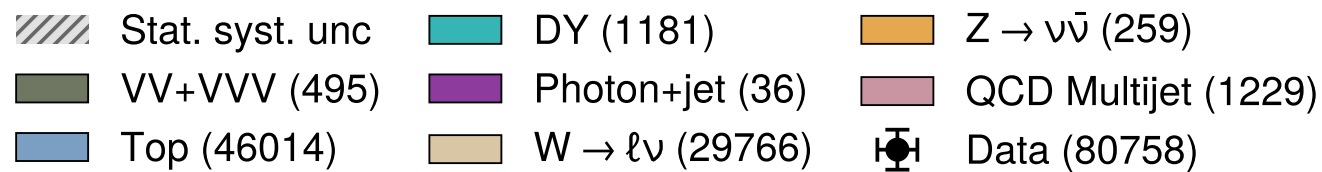
Data/MC

N_b (loose WP)

CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)

Events



Data/MC

10⁻¹
2
1
0

1

2

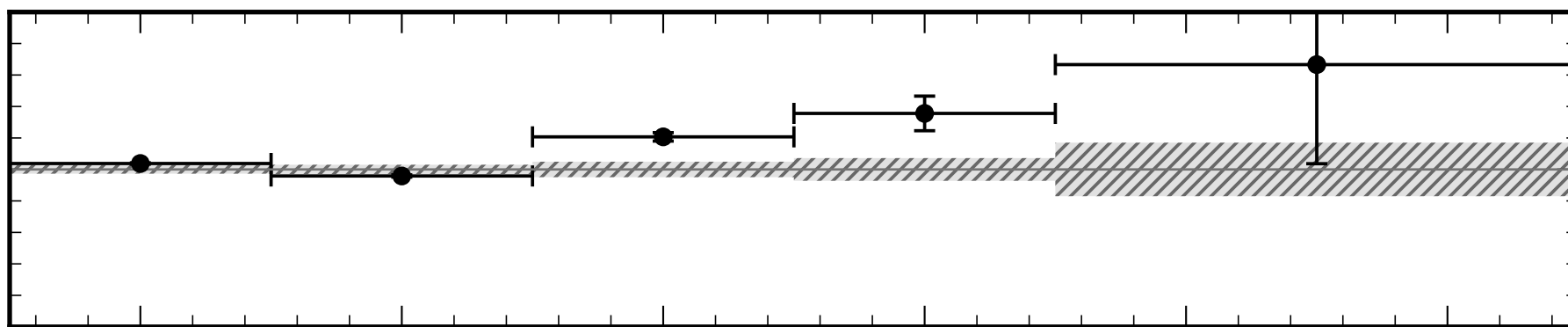
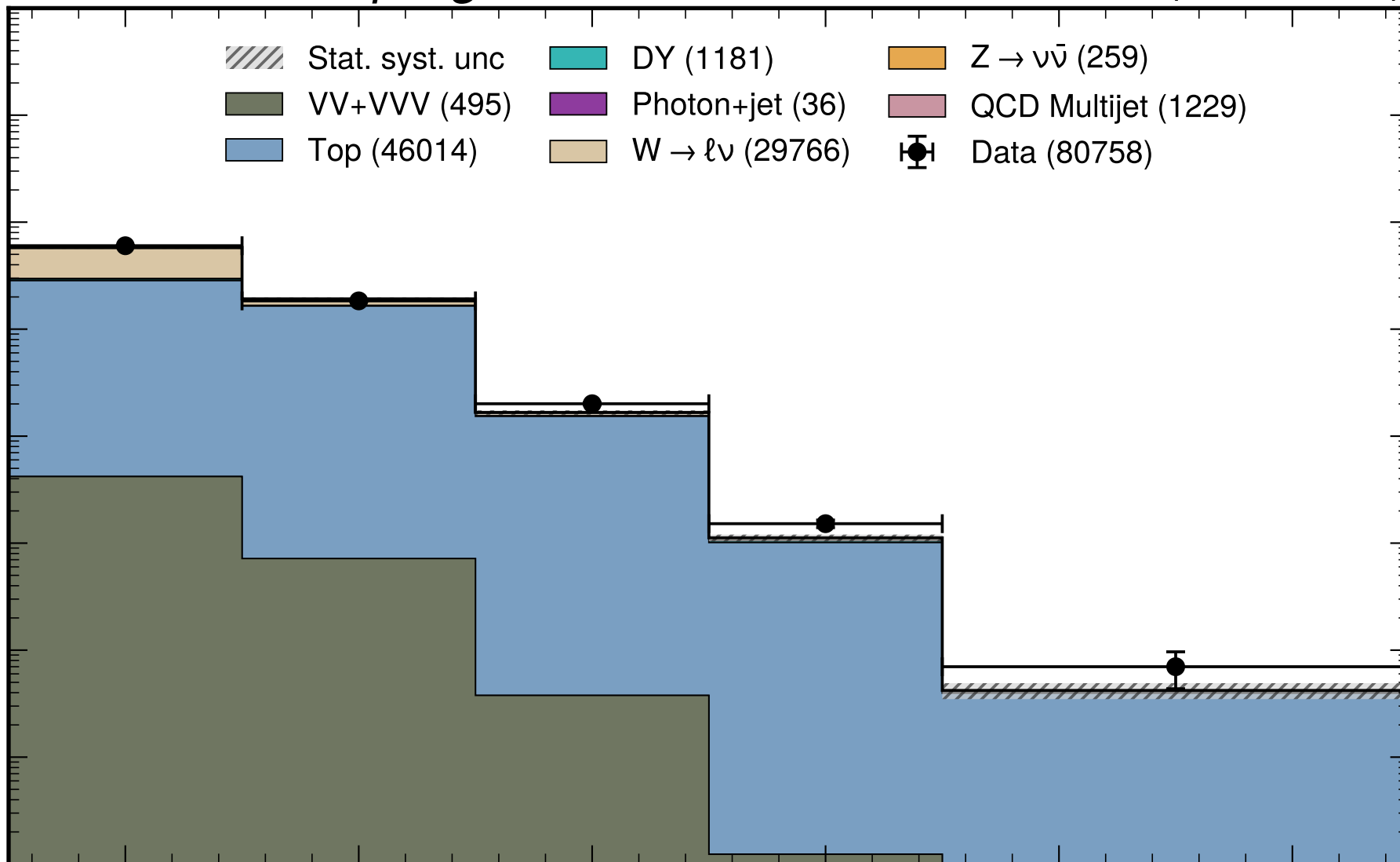
3

4

5

6

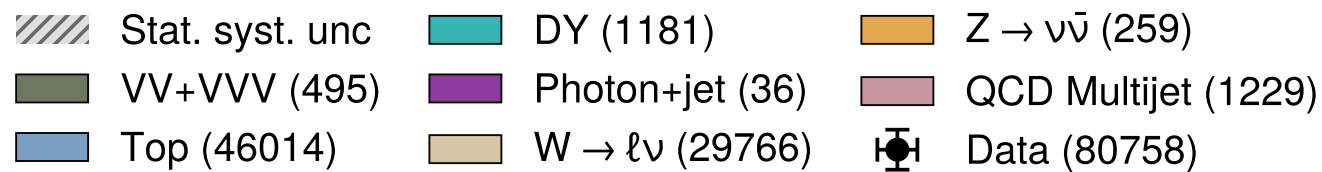
N_b



CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)

Events



Data/MC

10⁻¹

2

0

2

4

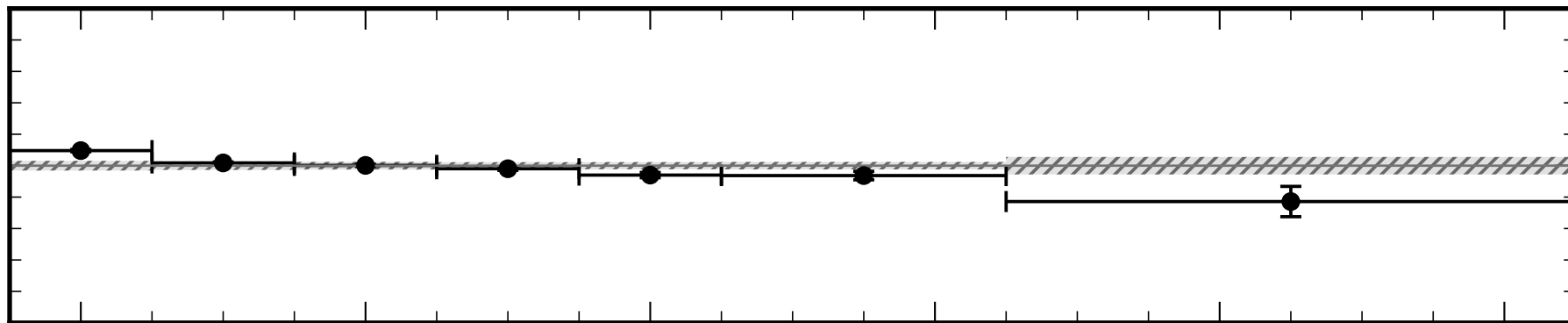
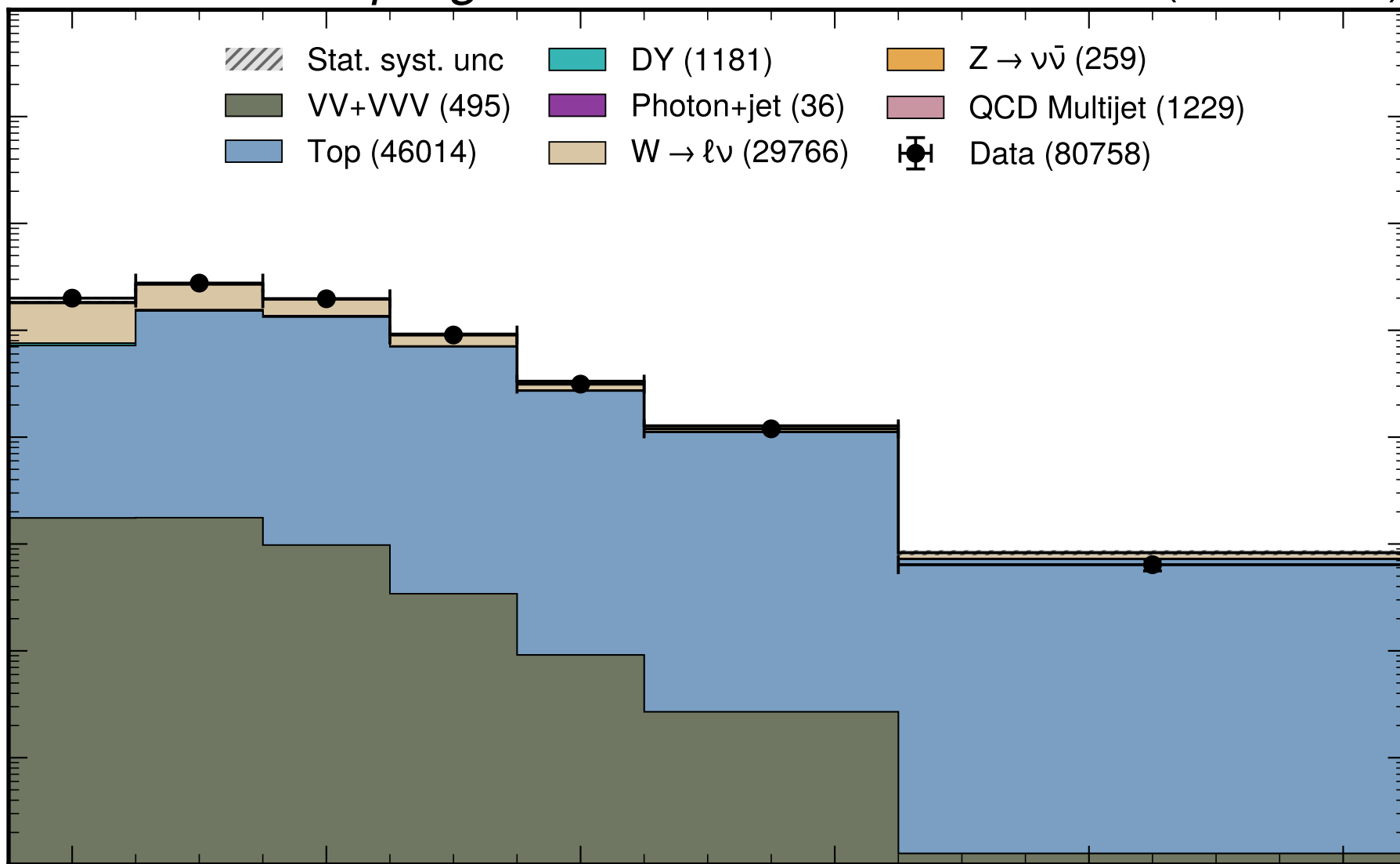
6

8

10

12

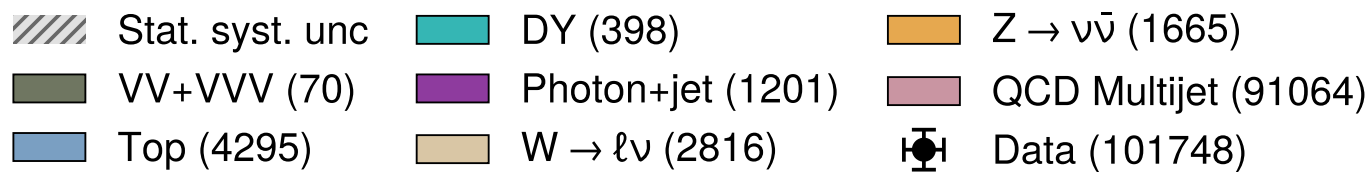
N_j



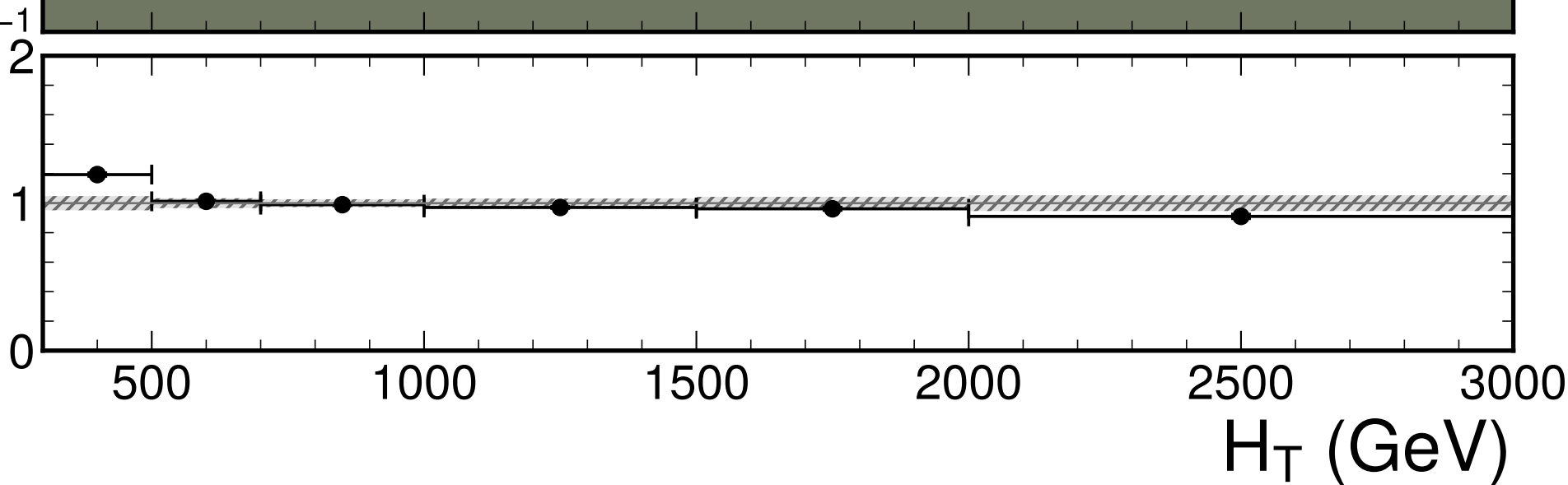
CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)

Events



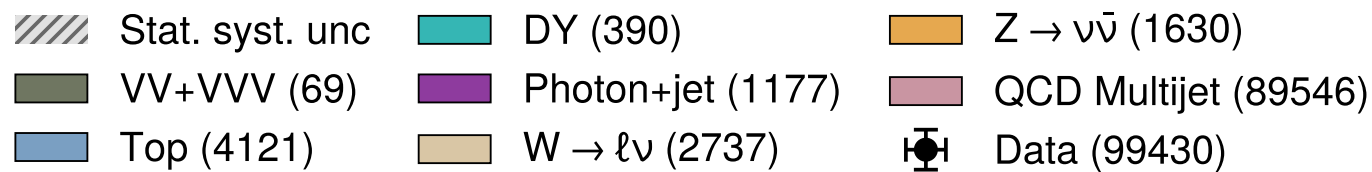
Data/MC



CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)

Events



Data/MC

0

1

10⁻²

2

Leading AK8 jet m_{SD} (GeV)

0

1

2

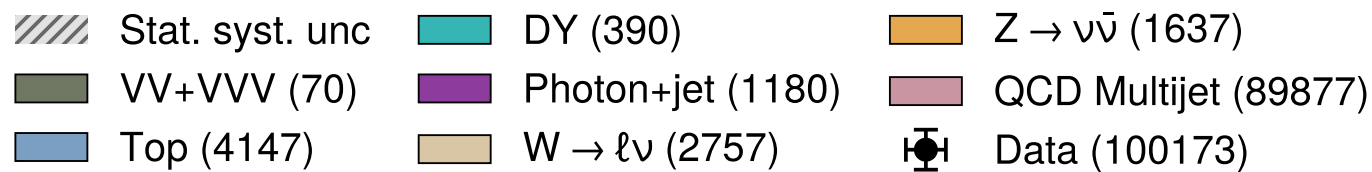
10⁻²

2

CMS *Work in progress*

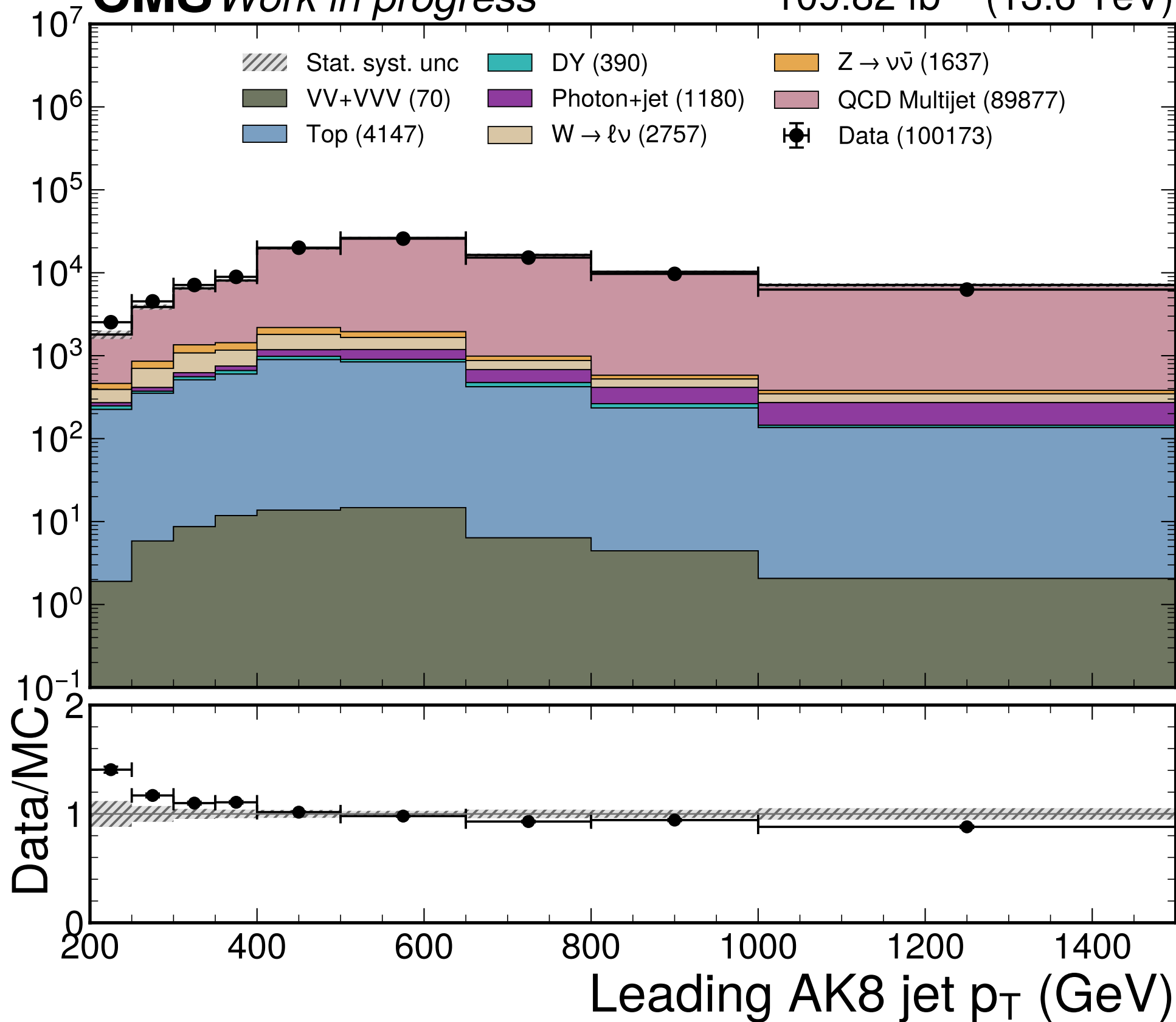
109.82 fb⁻¹ (13.6 TeV)

Events



Data/MC

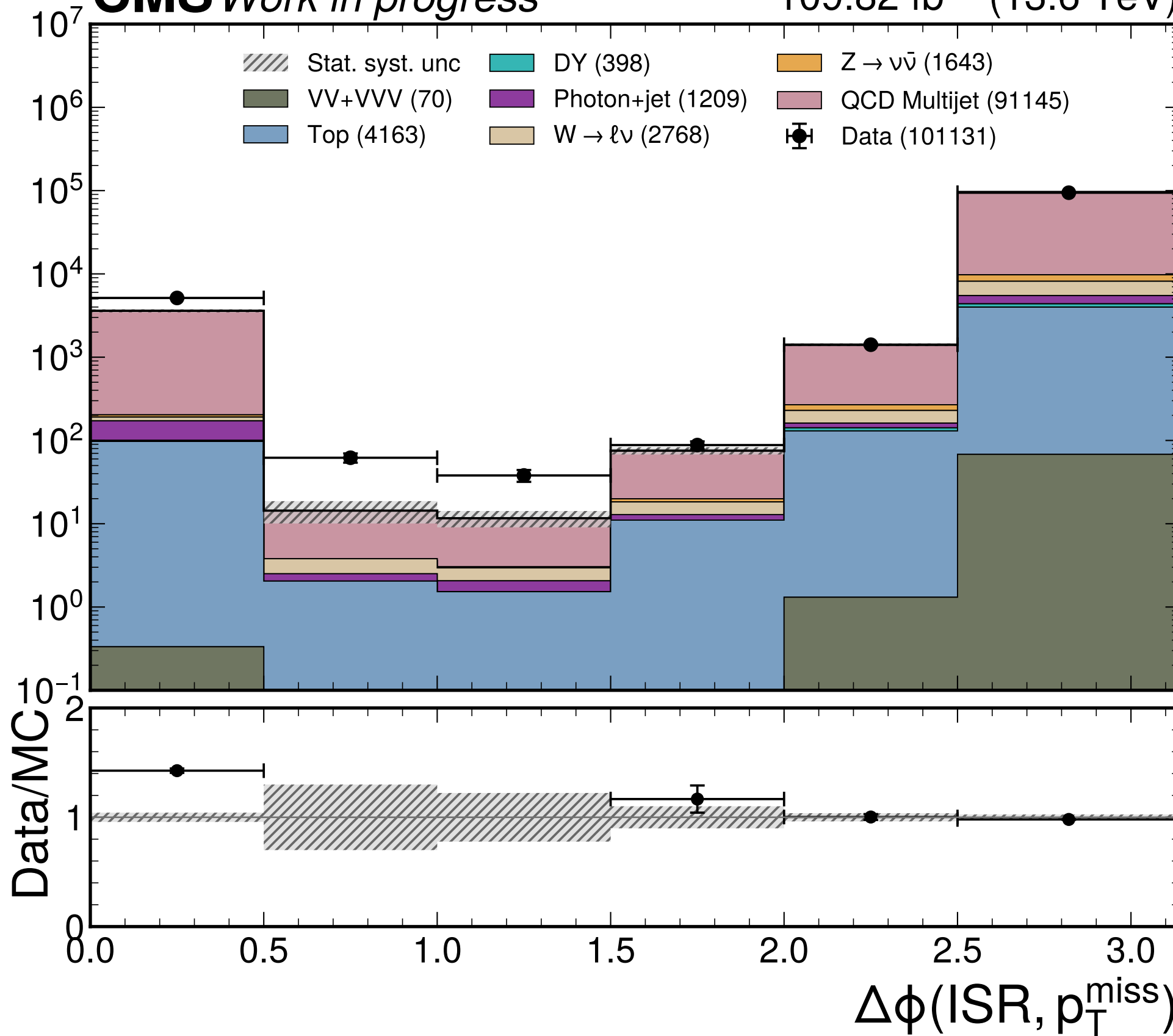
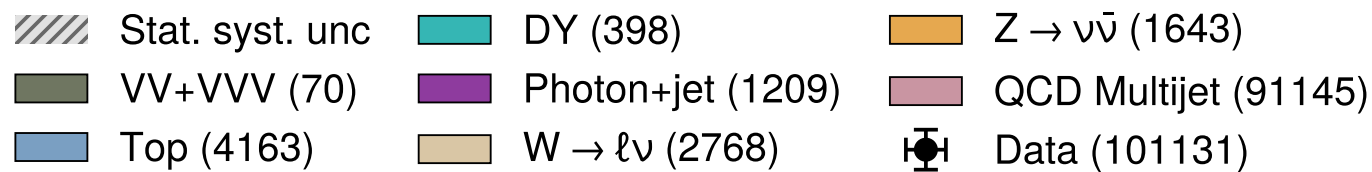
Leading AK8 jet p_T (GeV)



CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)

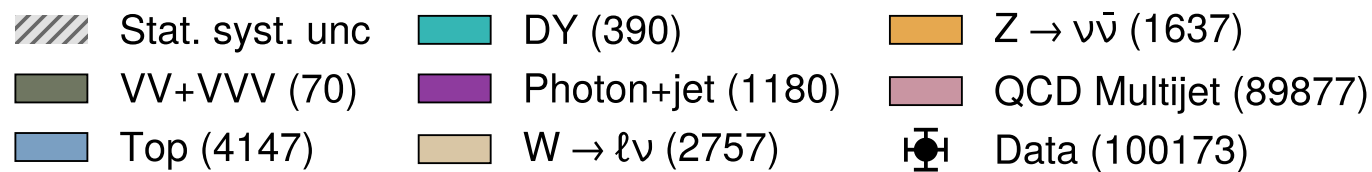
Events



CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)

Events



Data/MC

200

400

600

800

1000

1200

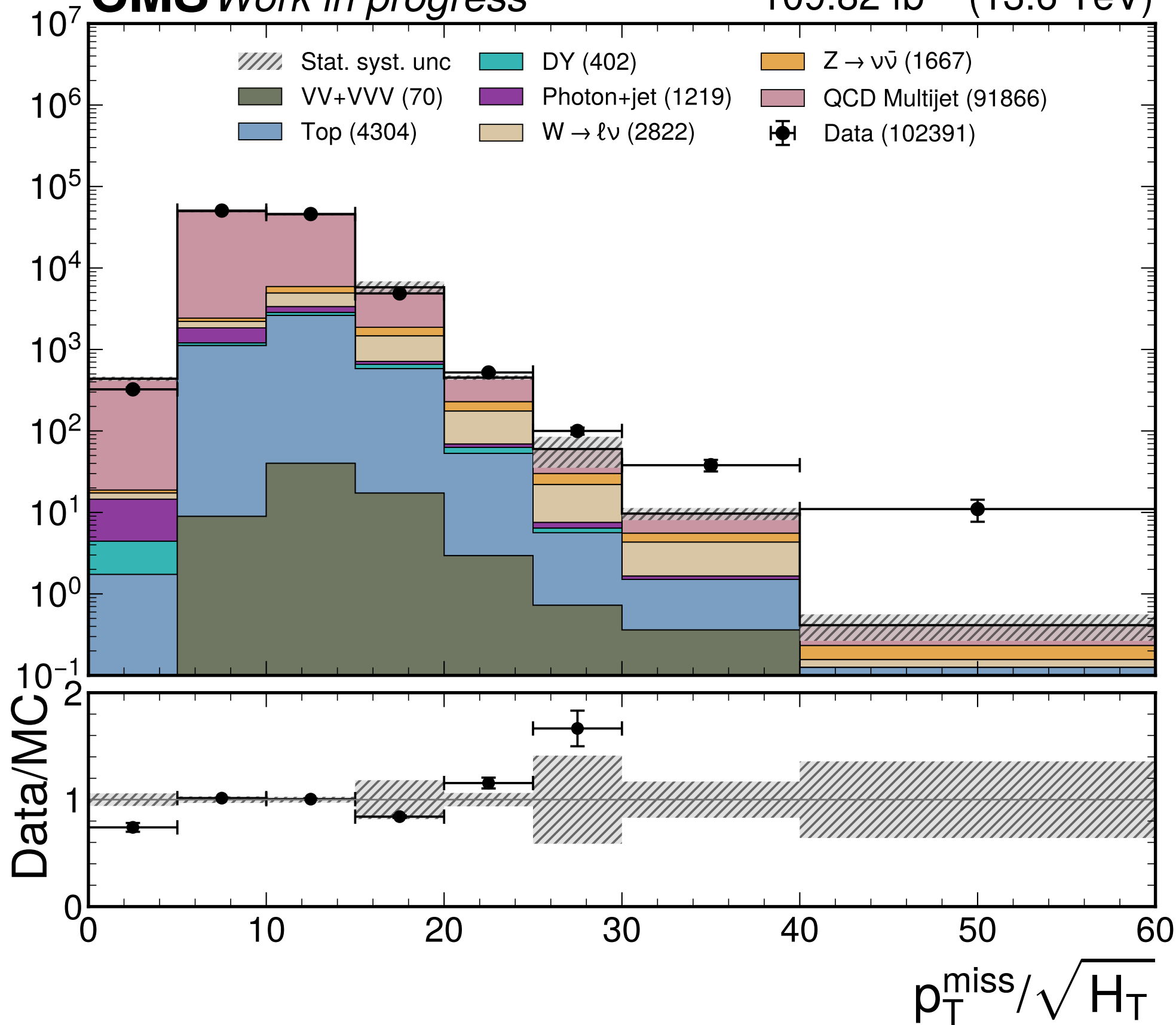
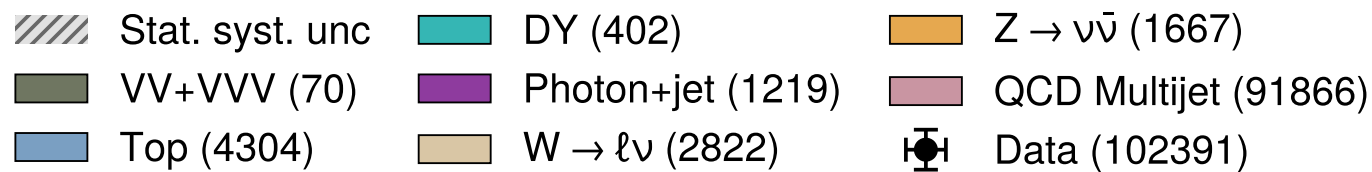
1400

p_T^{ISR} (GeV)

CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)

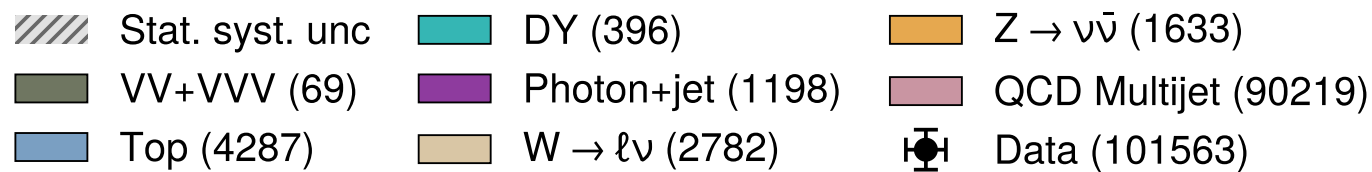
Events



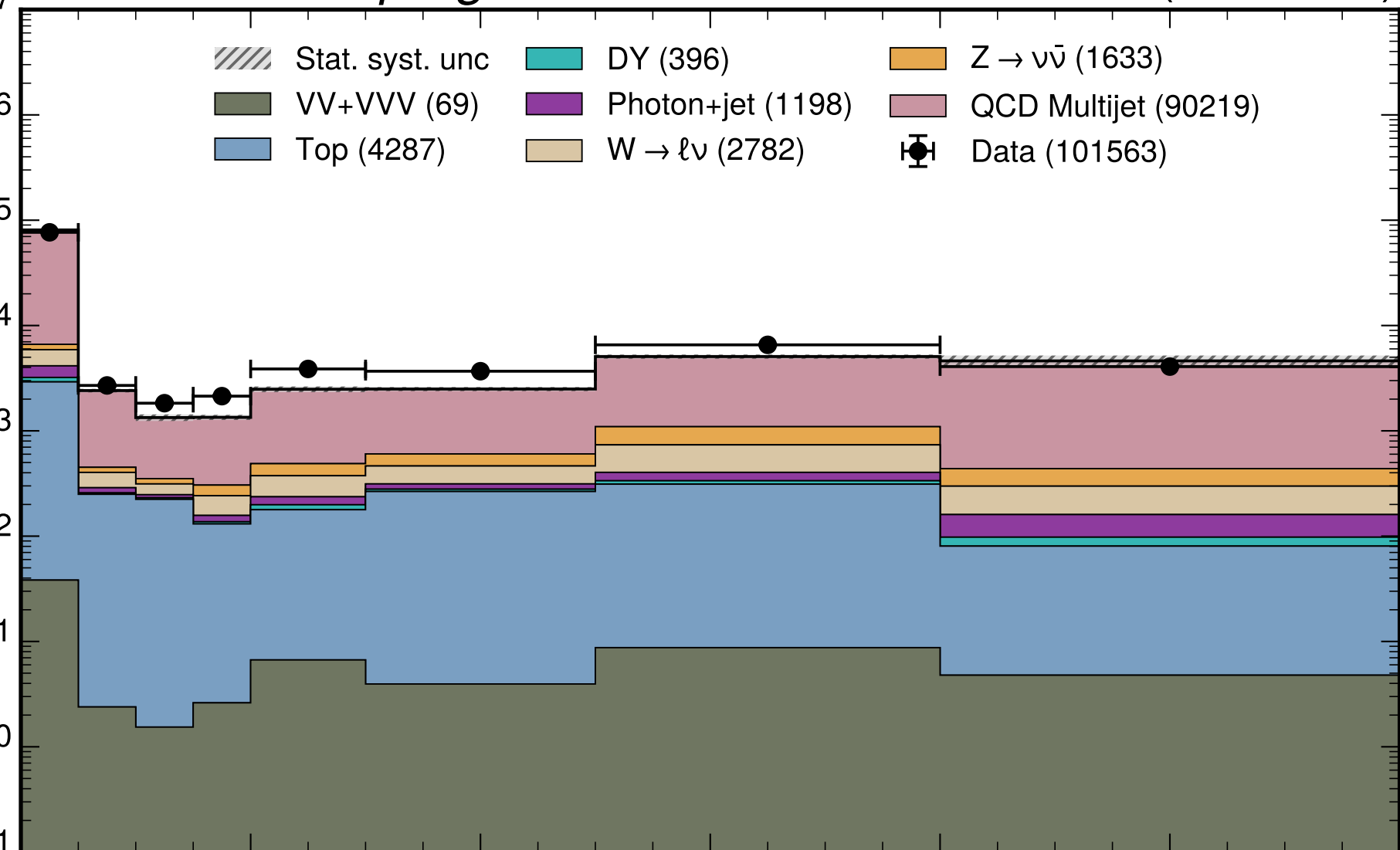
CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)

Events



Data/MC

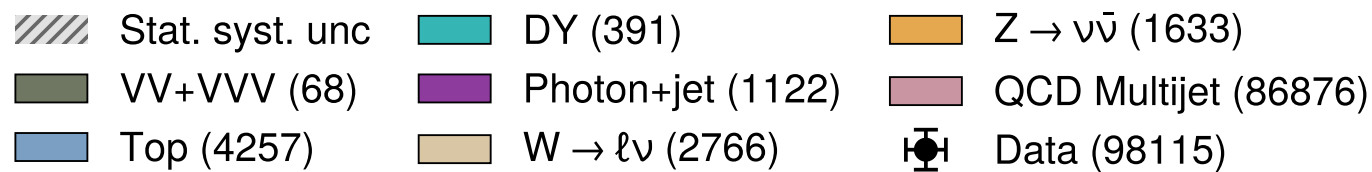


m_T^b (GeV)

CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)

Events



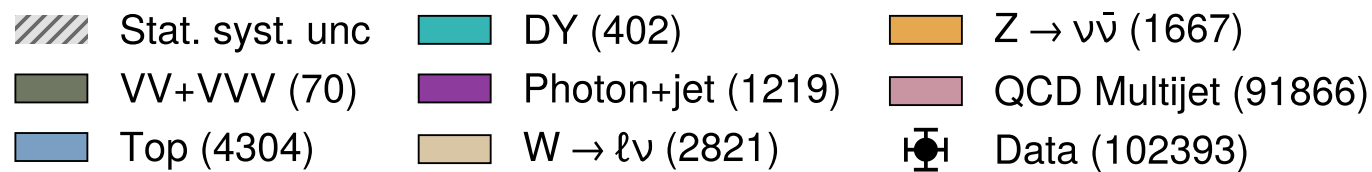
Data/MC

p_T^b (GeV)

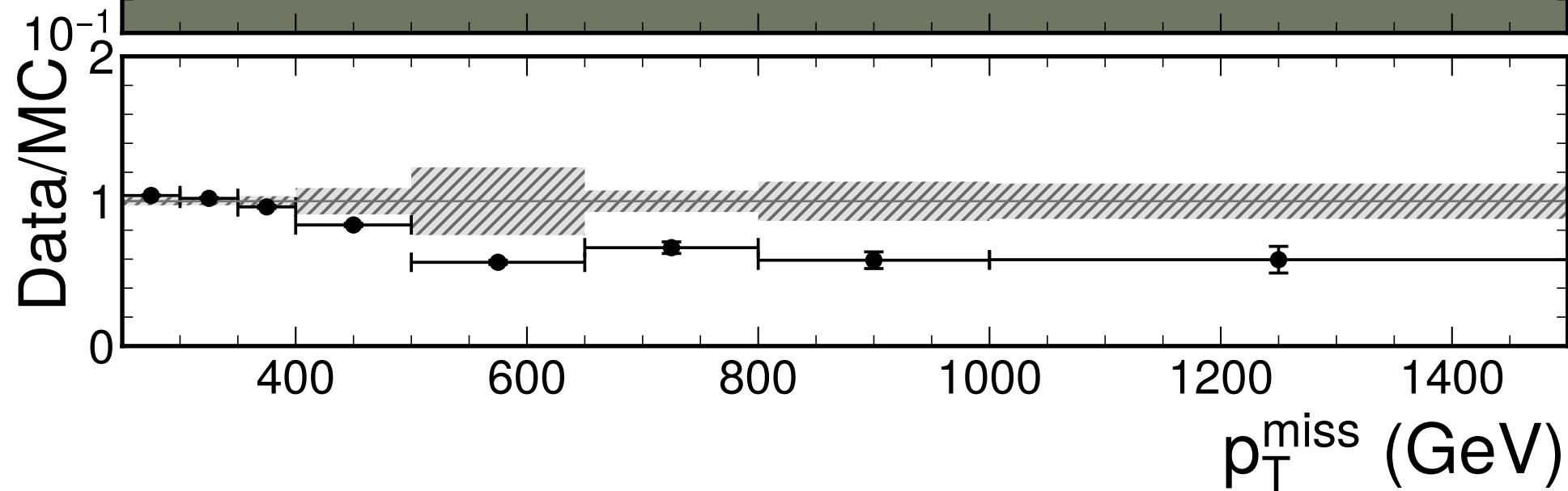
CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)

Events



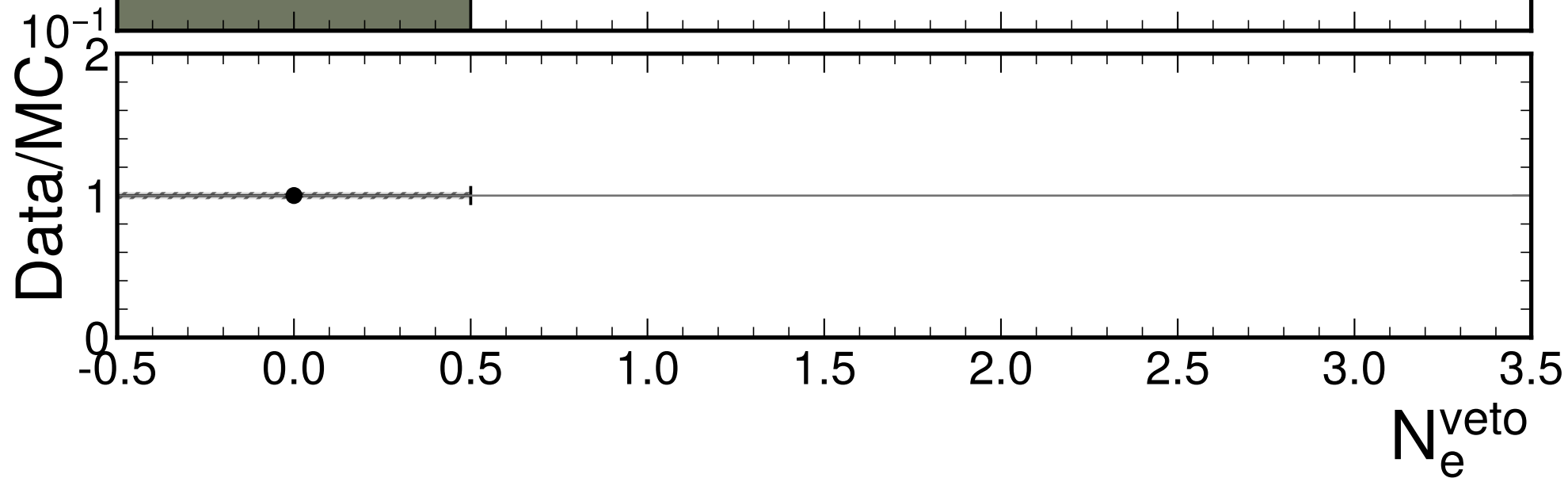
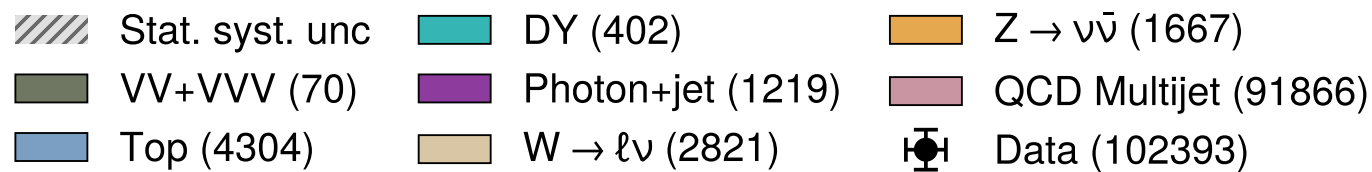
Data/MC



CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)

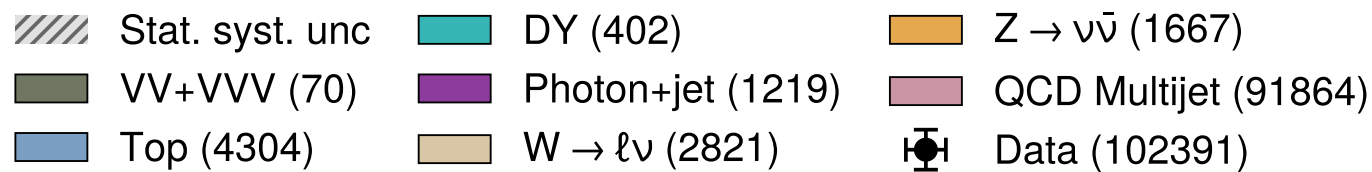
Events



CMS *Work in progress*

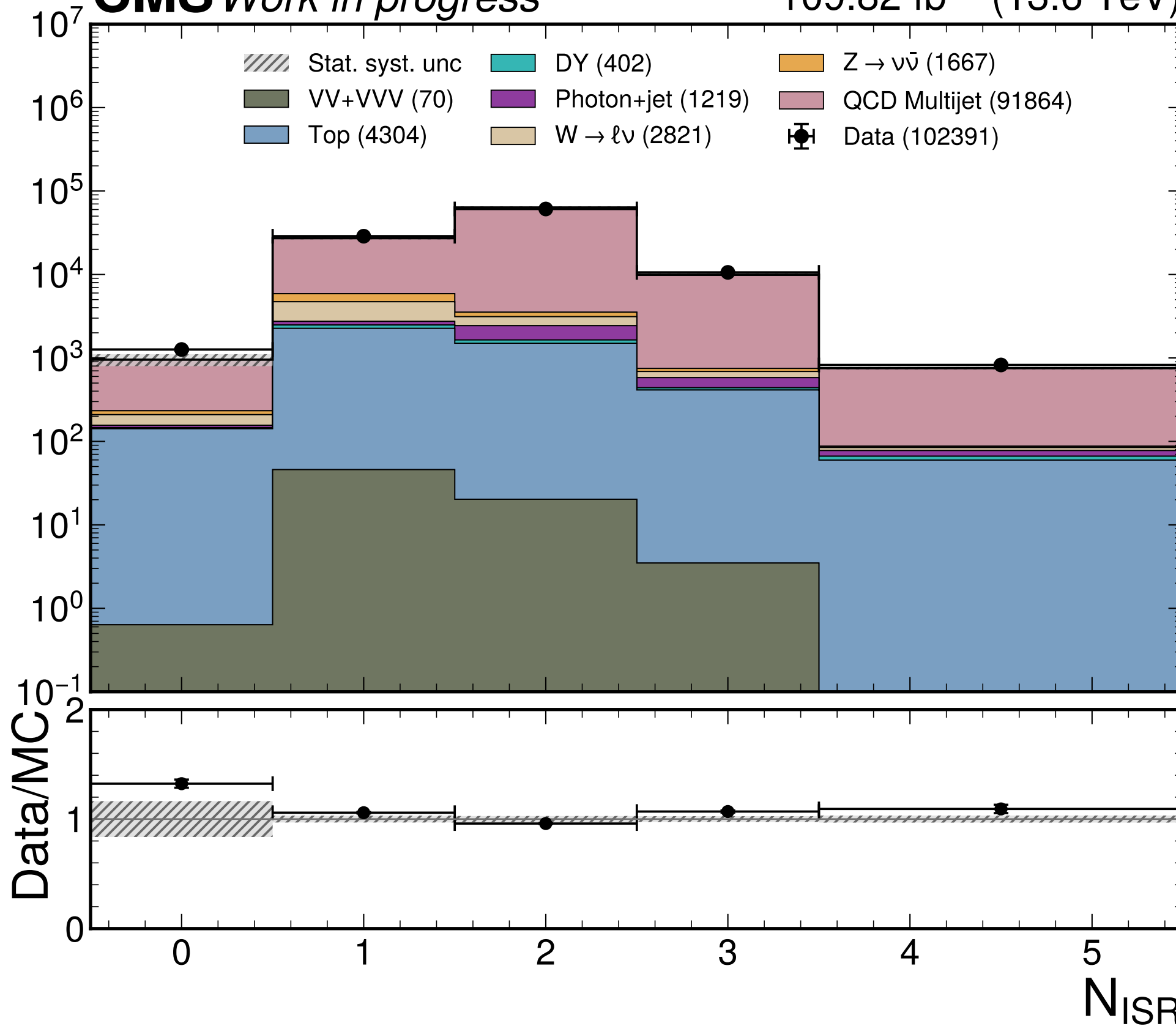
109.82 fb⁻¹ (13.6 TeV)

Events



Data/MC

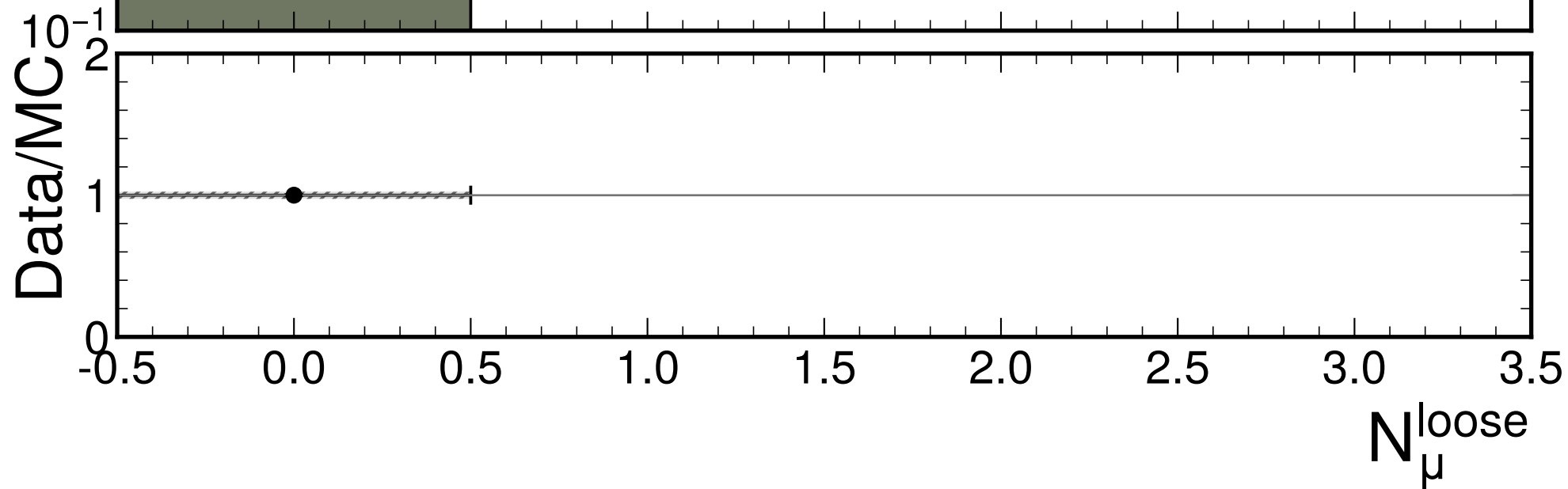
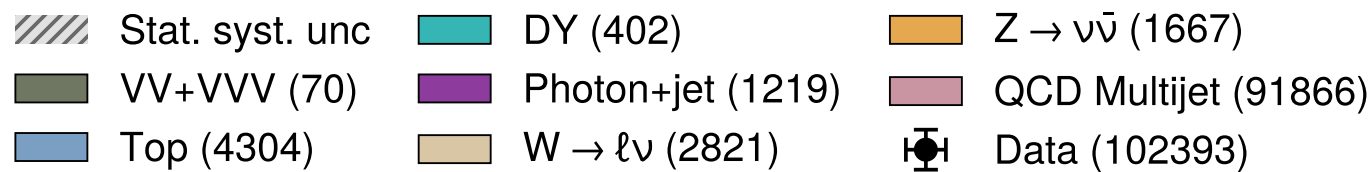
N_{ISR}



CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)

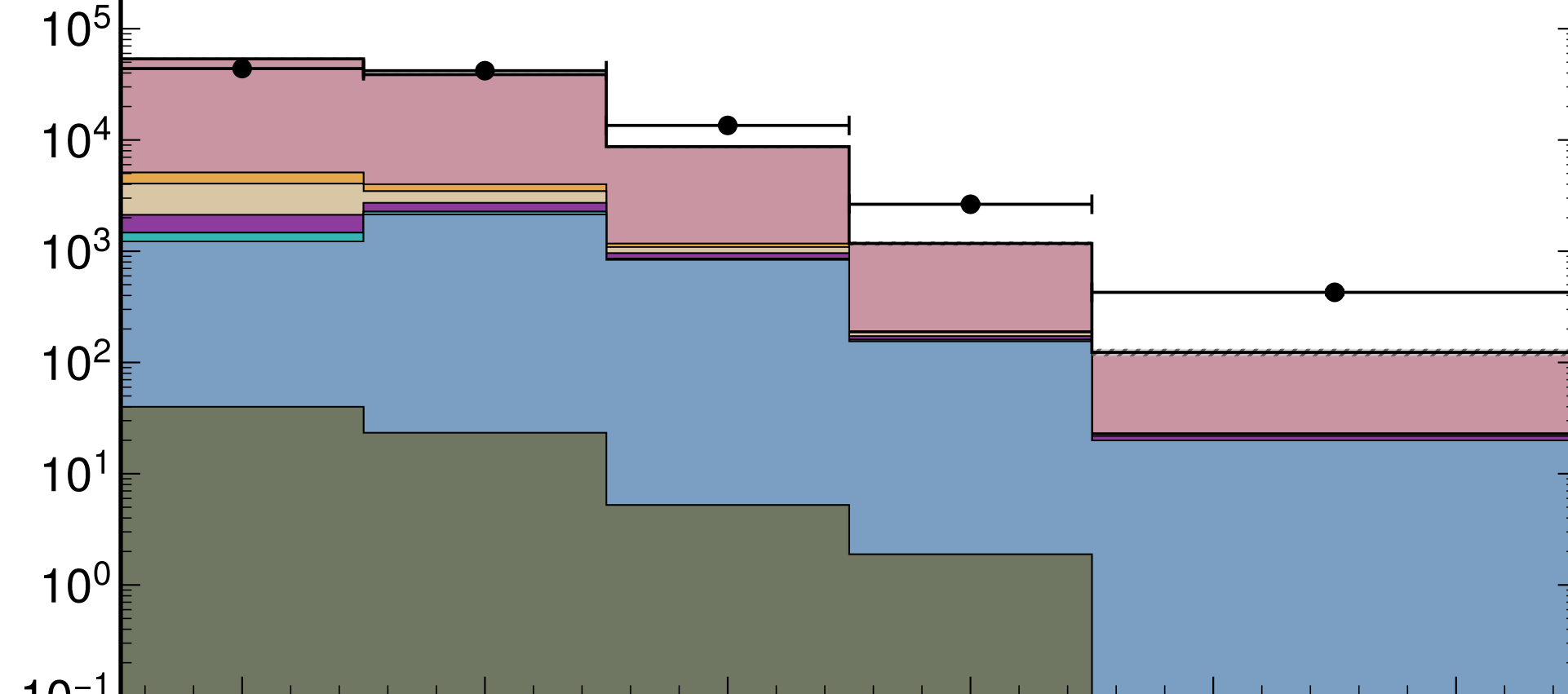
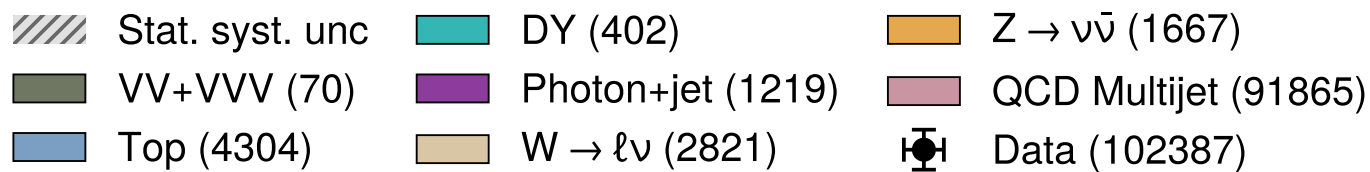
Events



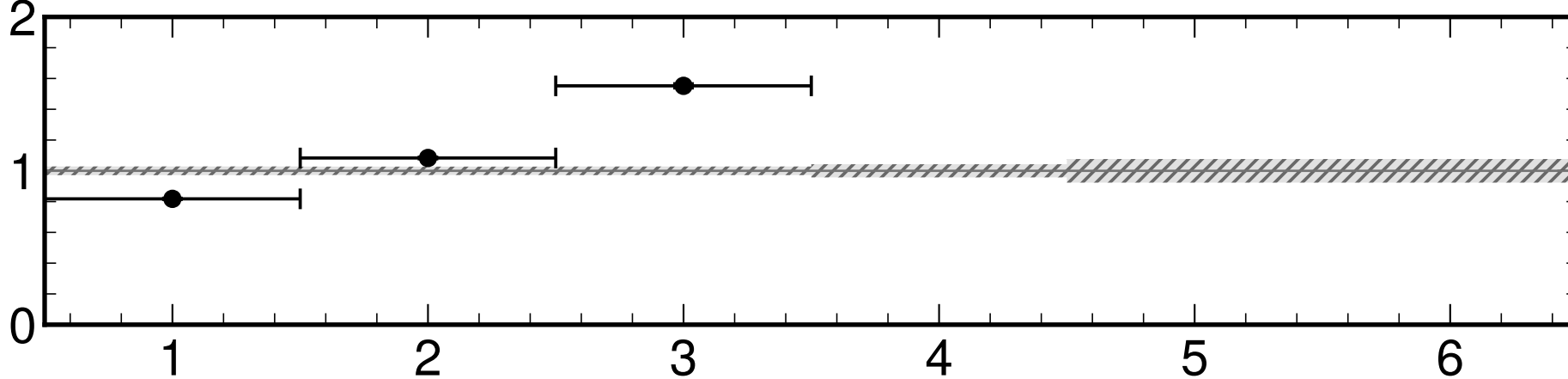
CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)

Events



Data/MC

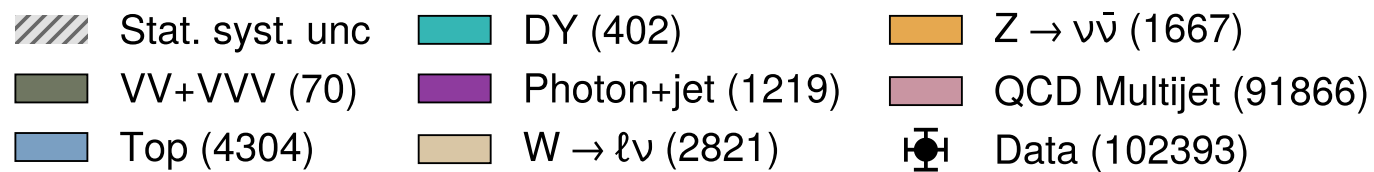


N_b (loose WP)

CMS *Work in progress*

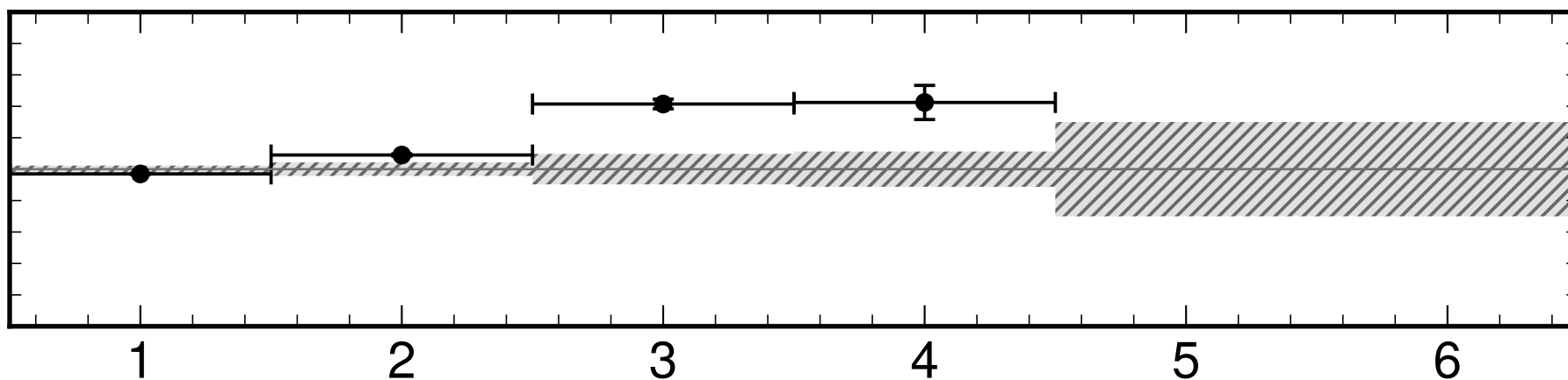
109.82 fb⁻¹ (13.6 TeV)

Events



Data/MC

10⁻¹
2
1
0

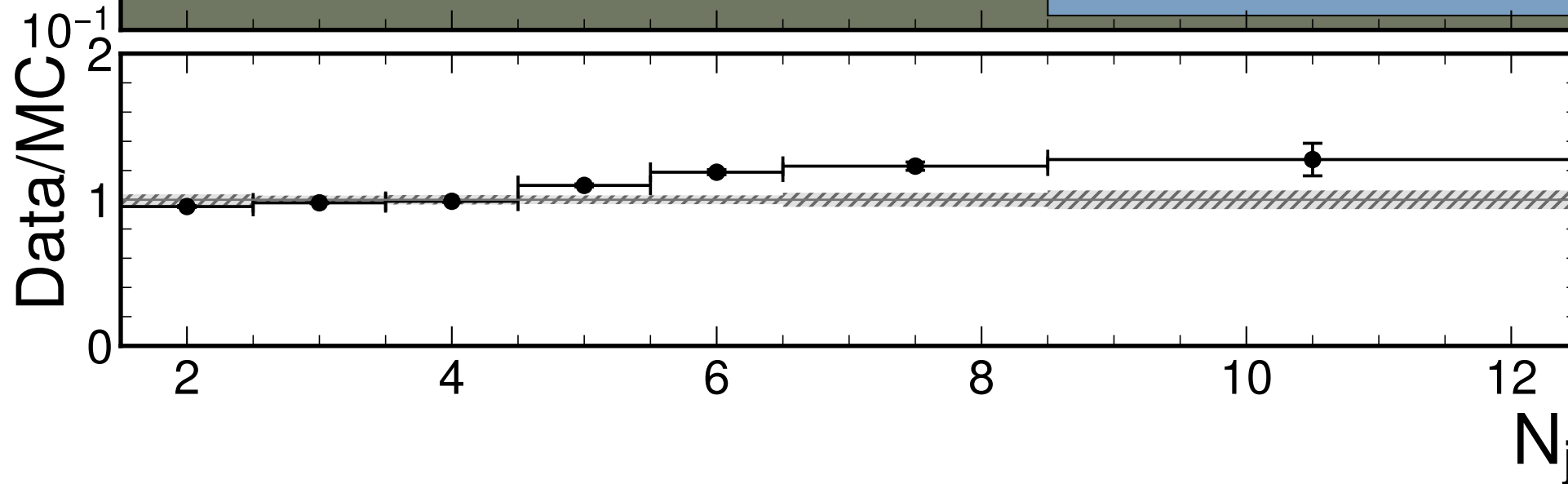
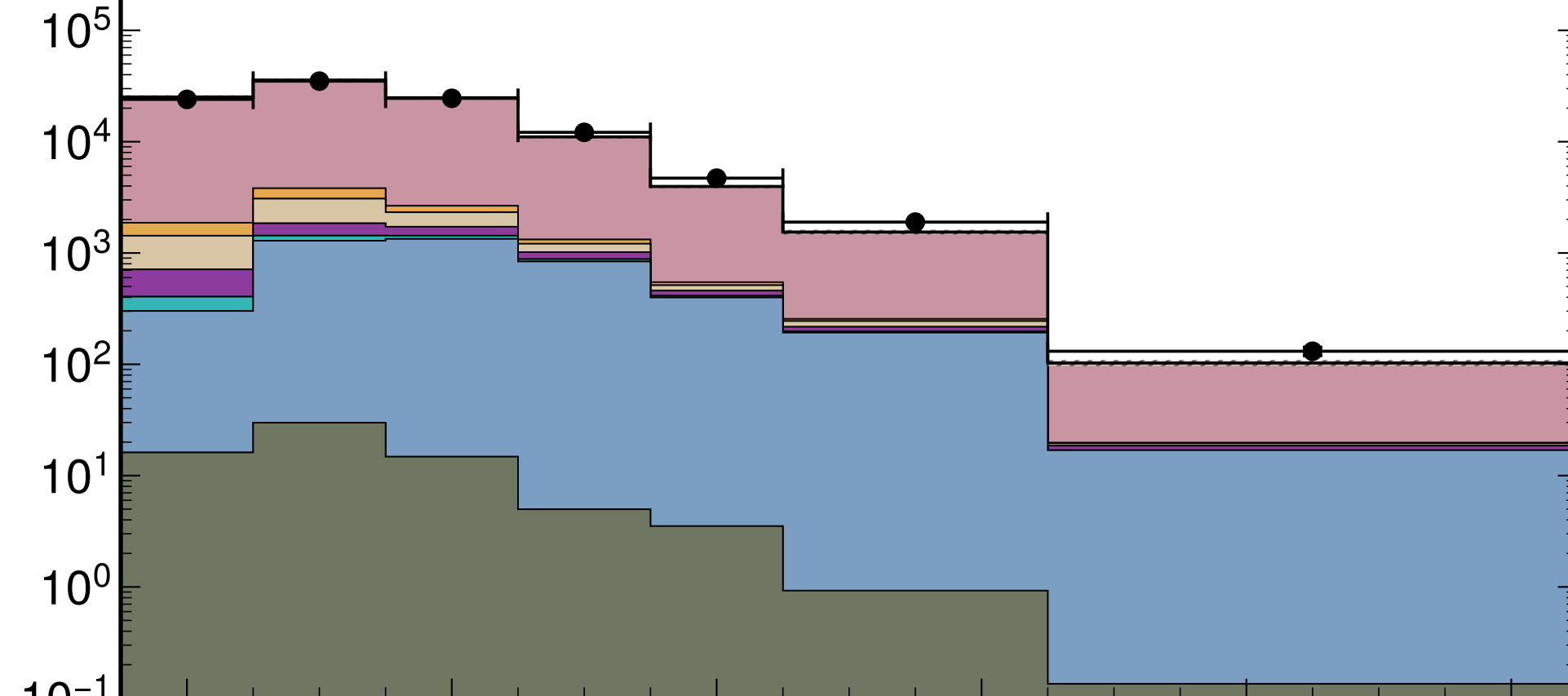
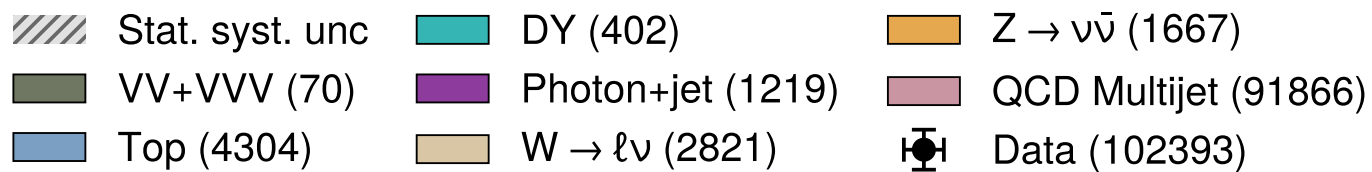


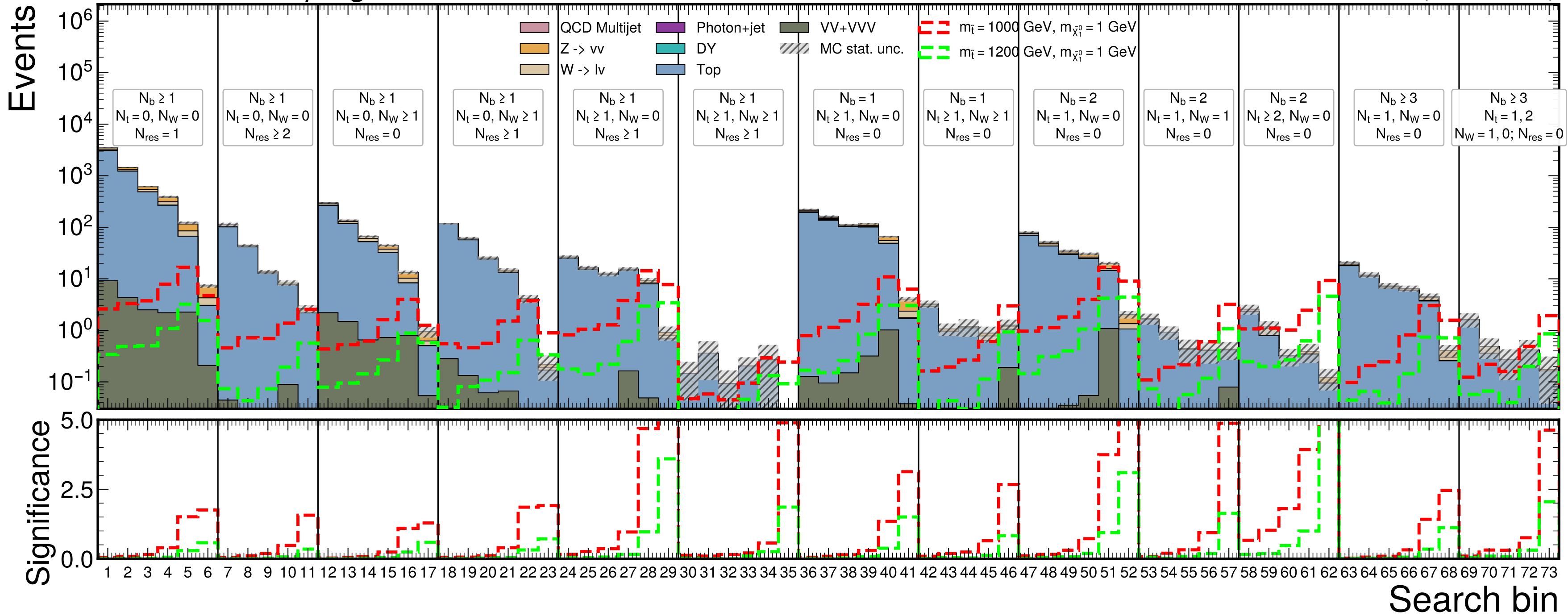
N_b

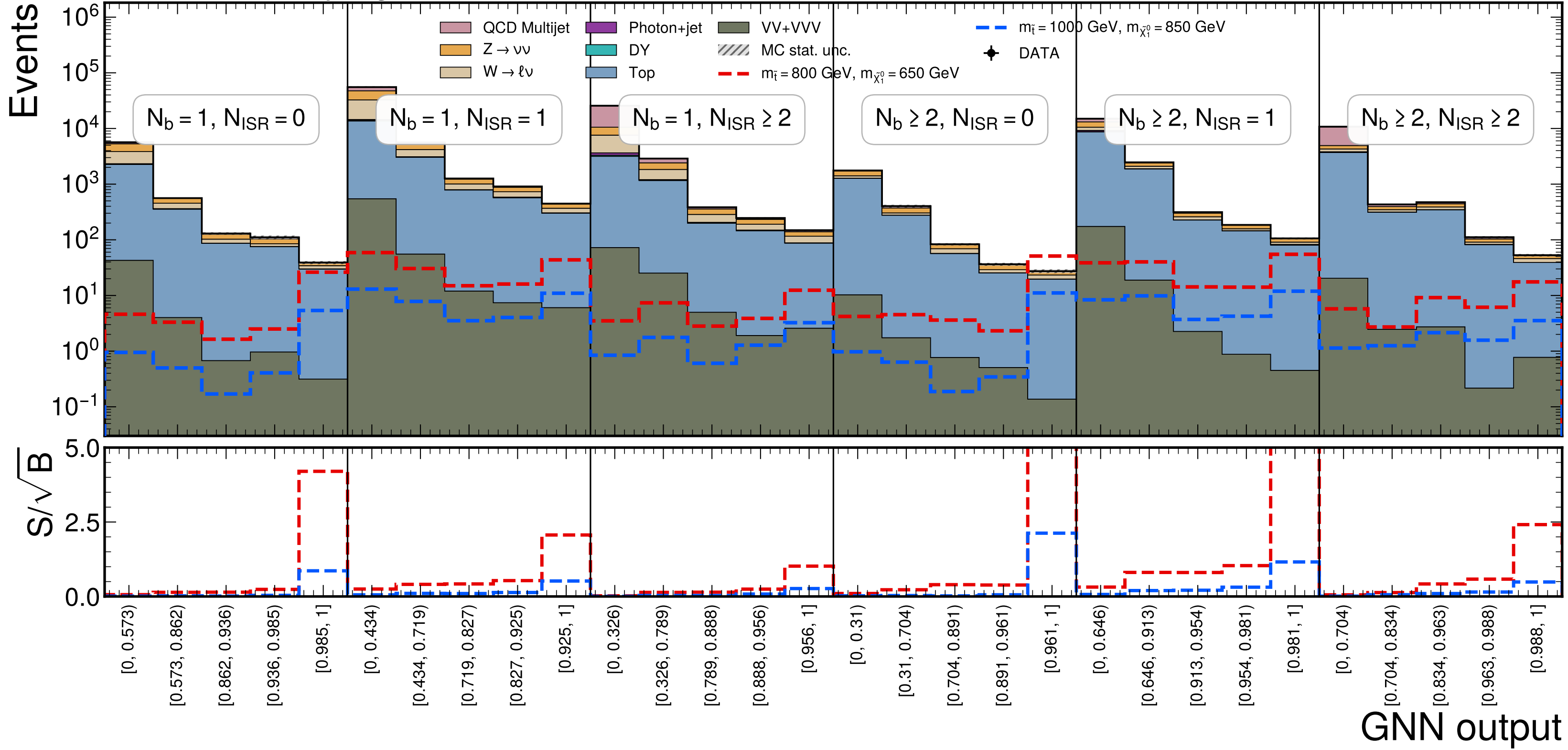
CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)

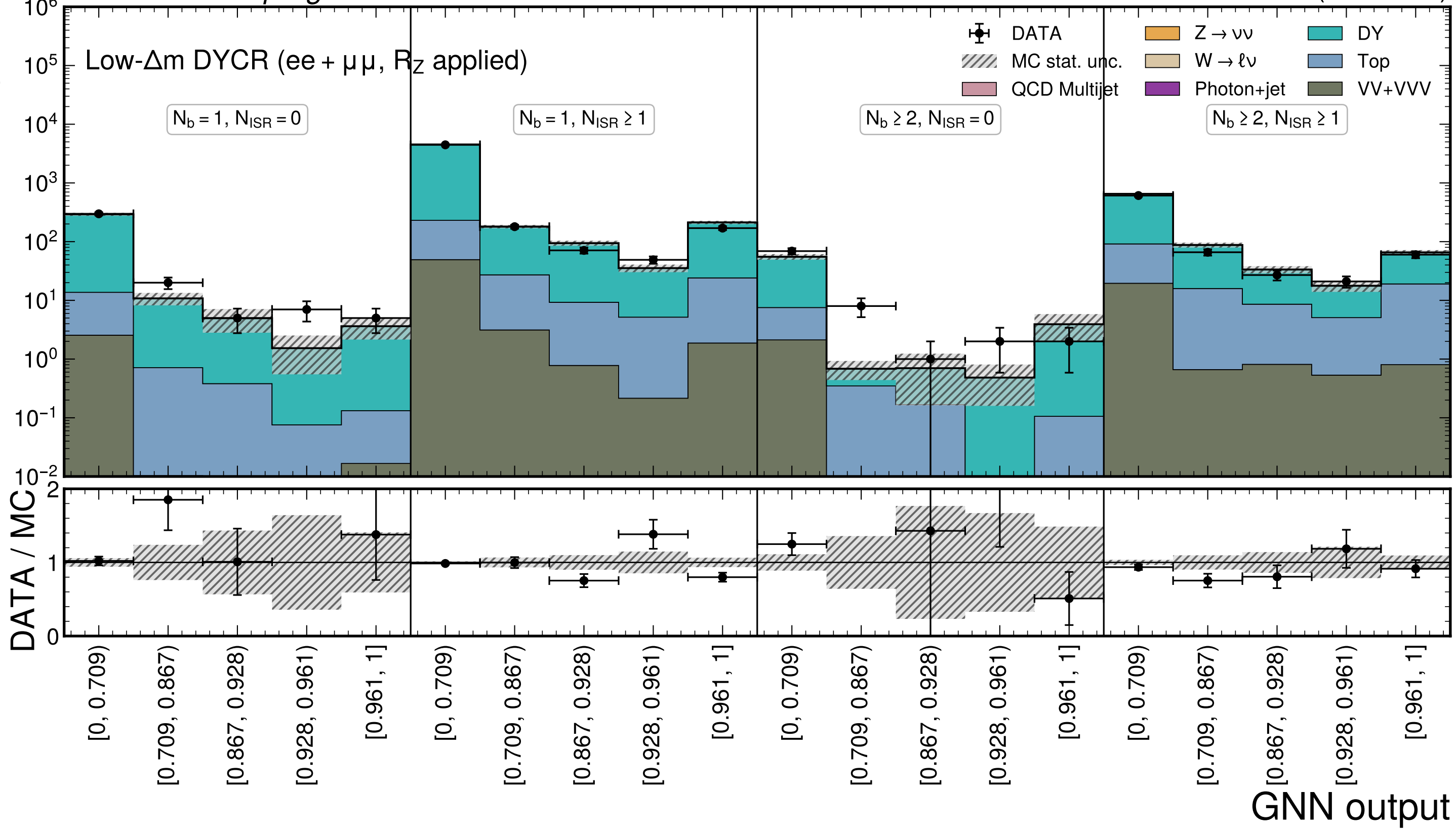
Events



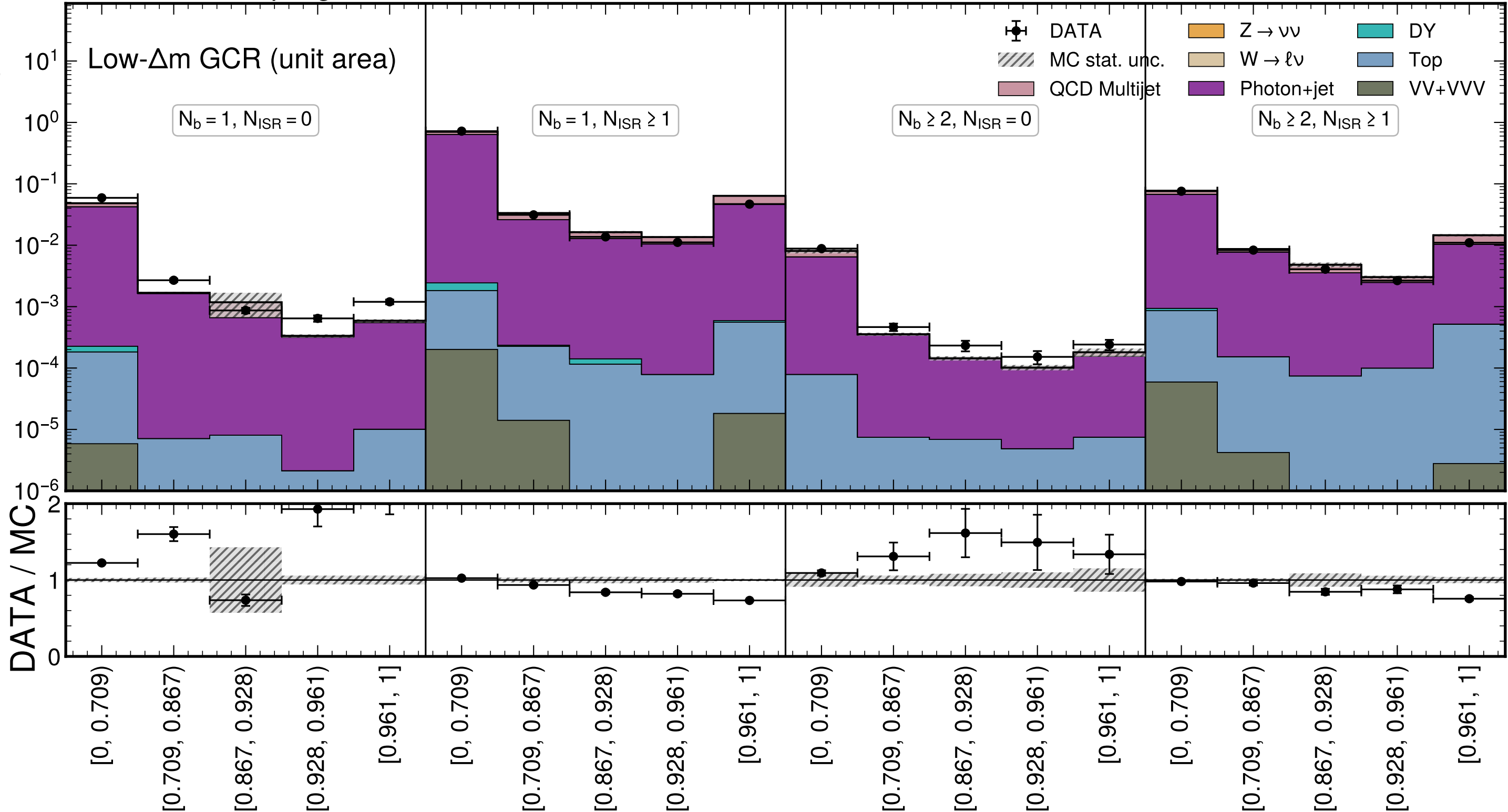




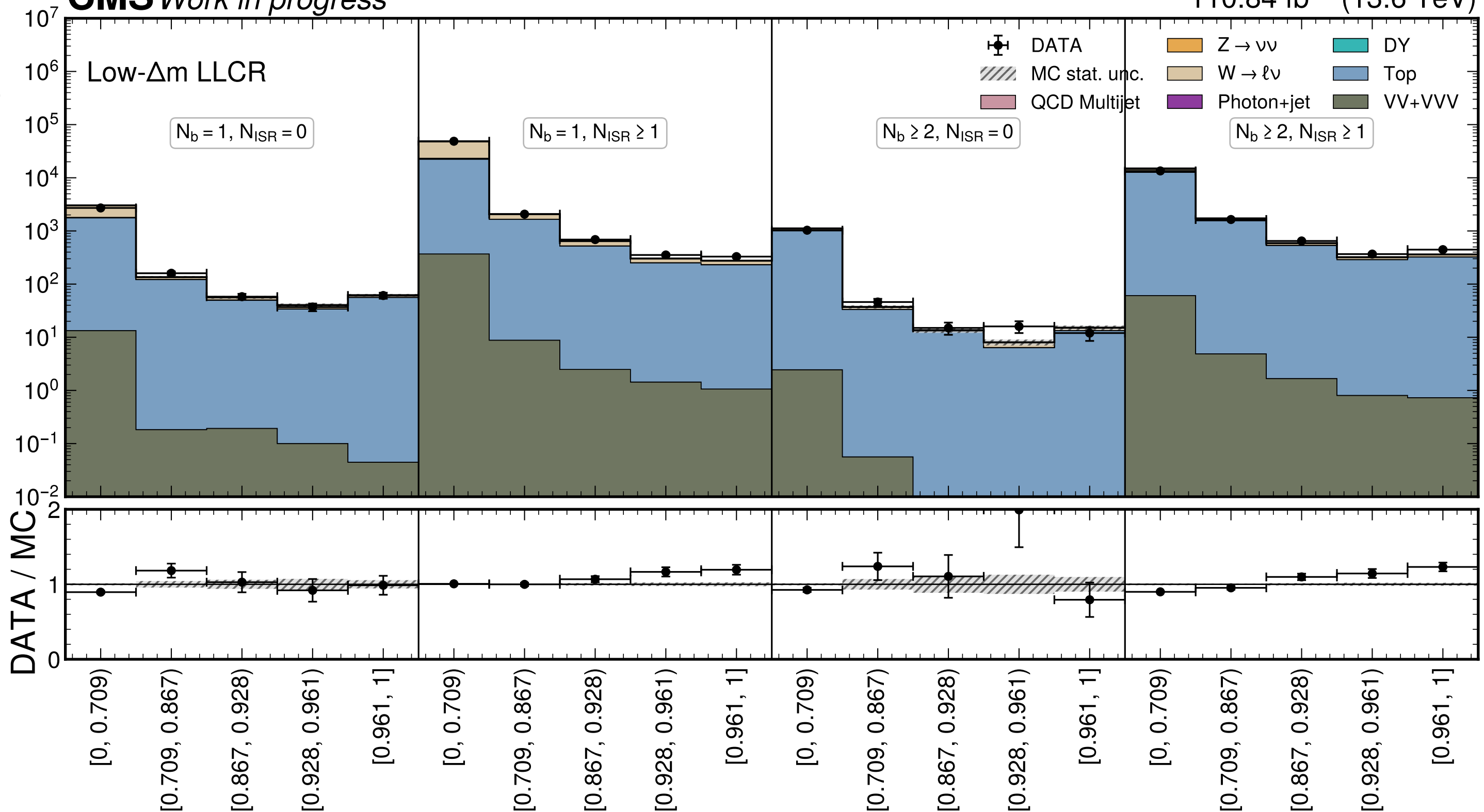
Events / bin



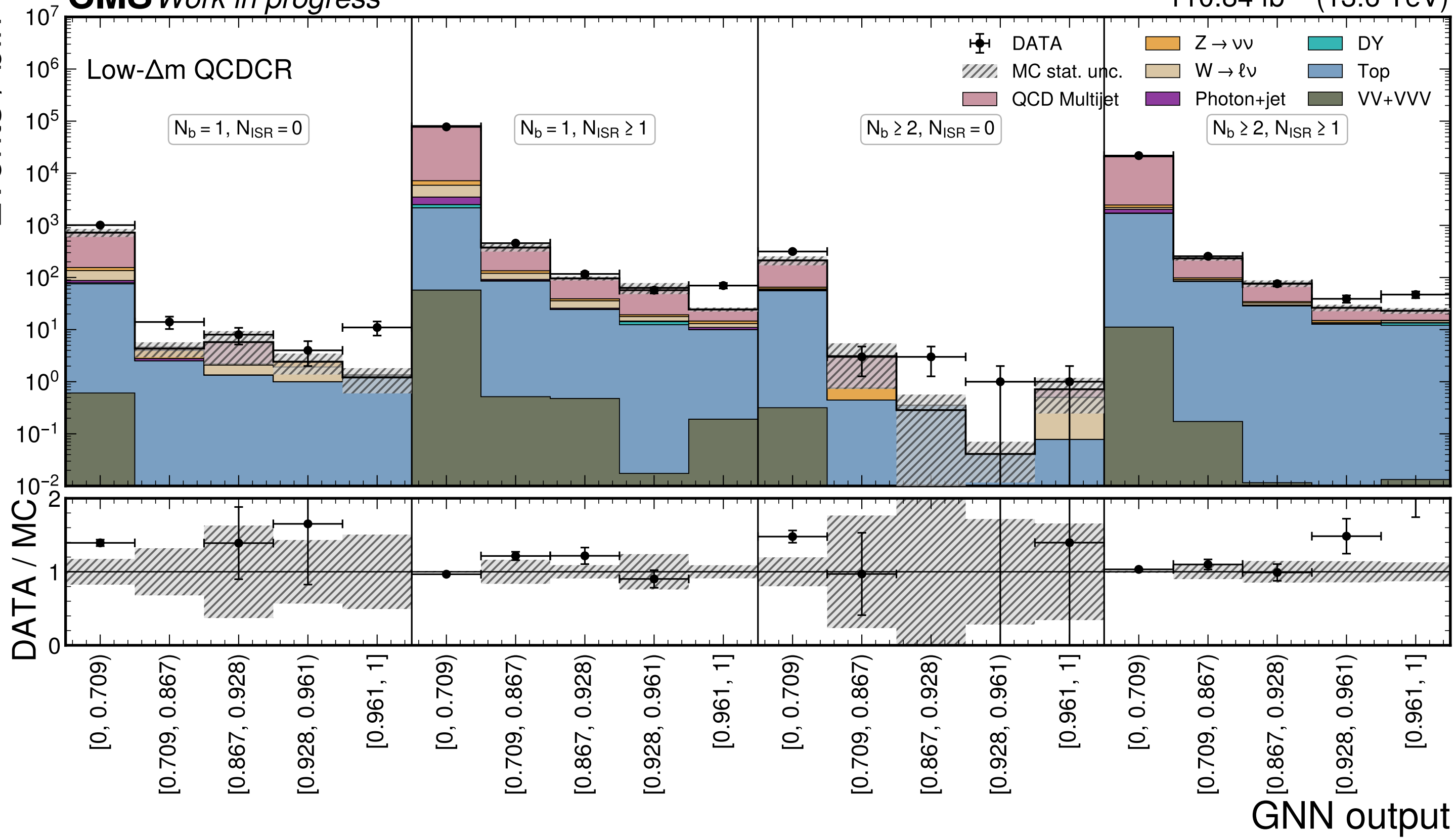
Normalized events / bin



Events / bin



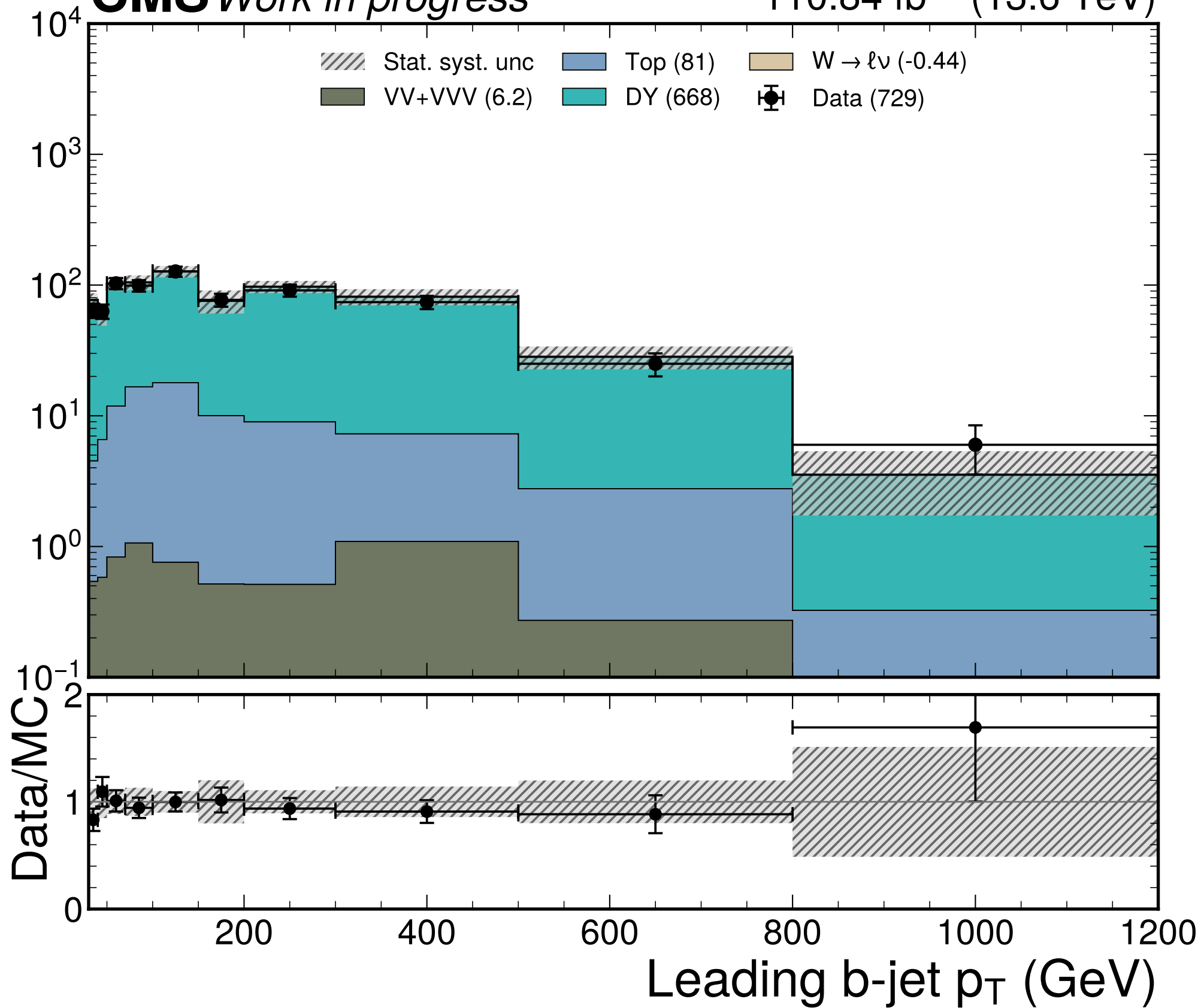
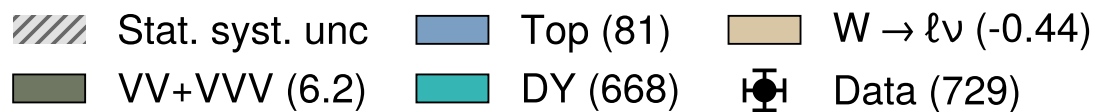
Events / bin



CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)

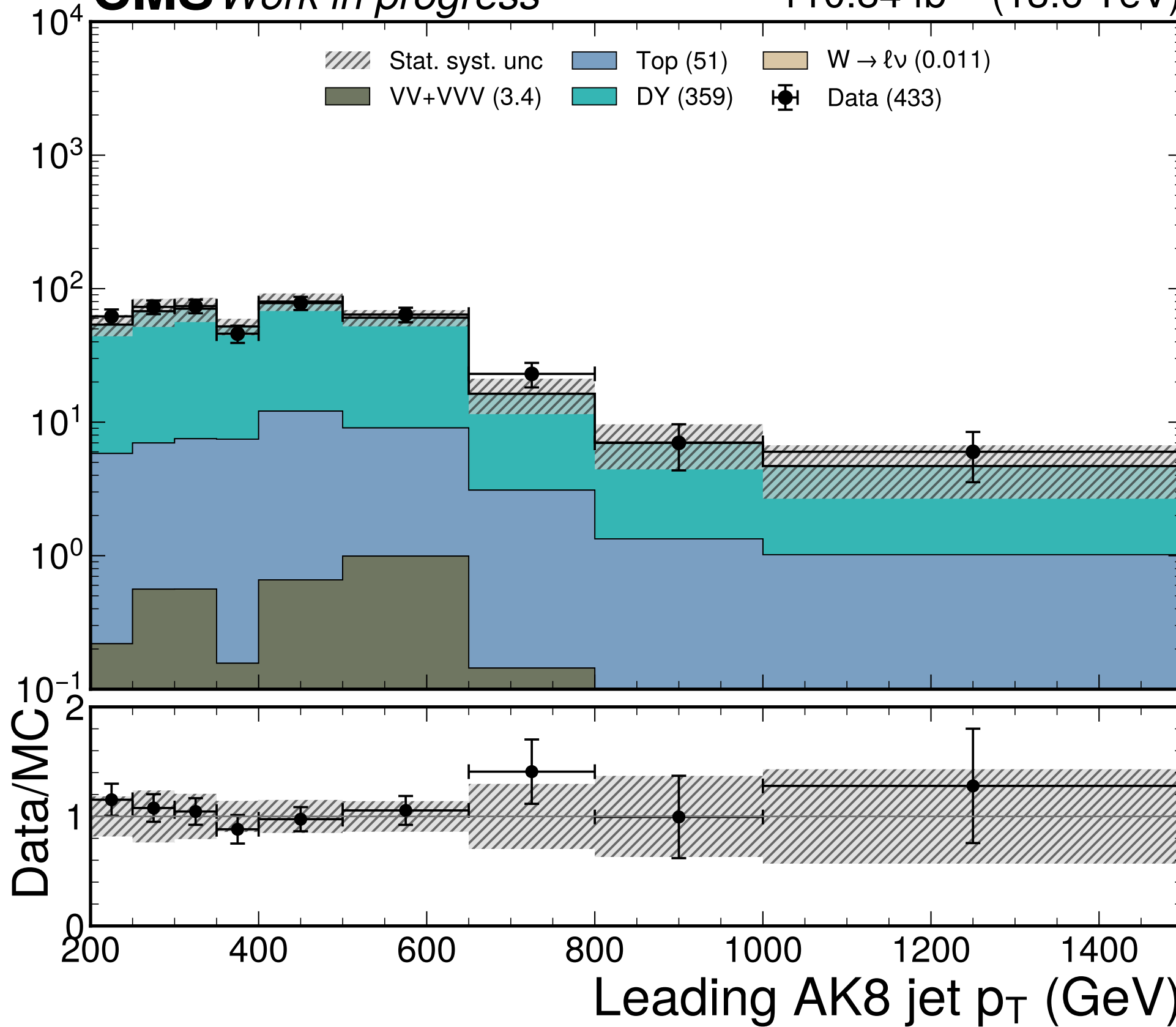
Events



CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)

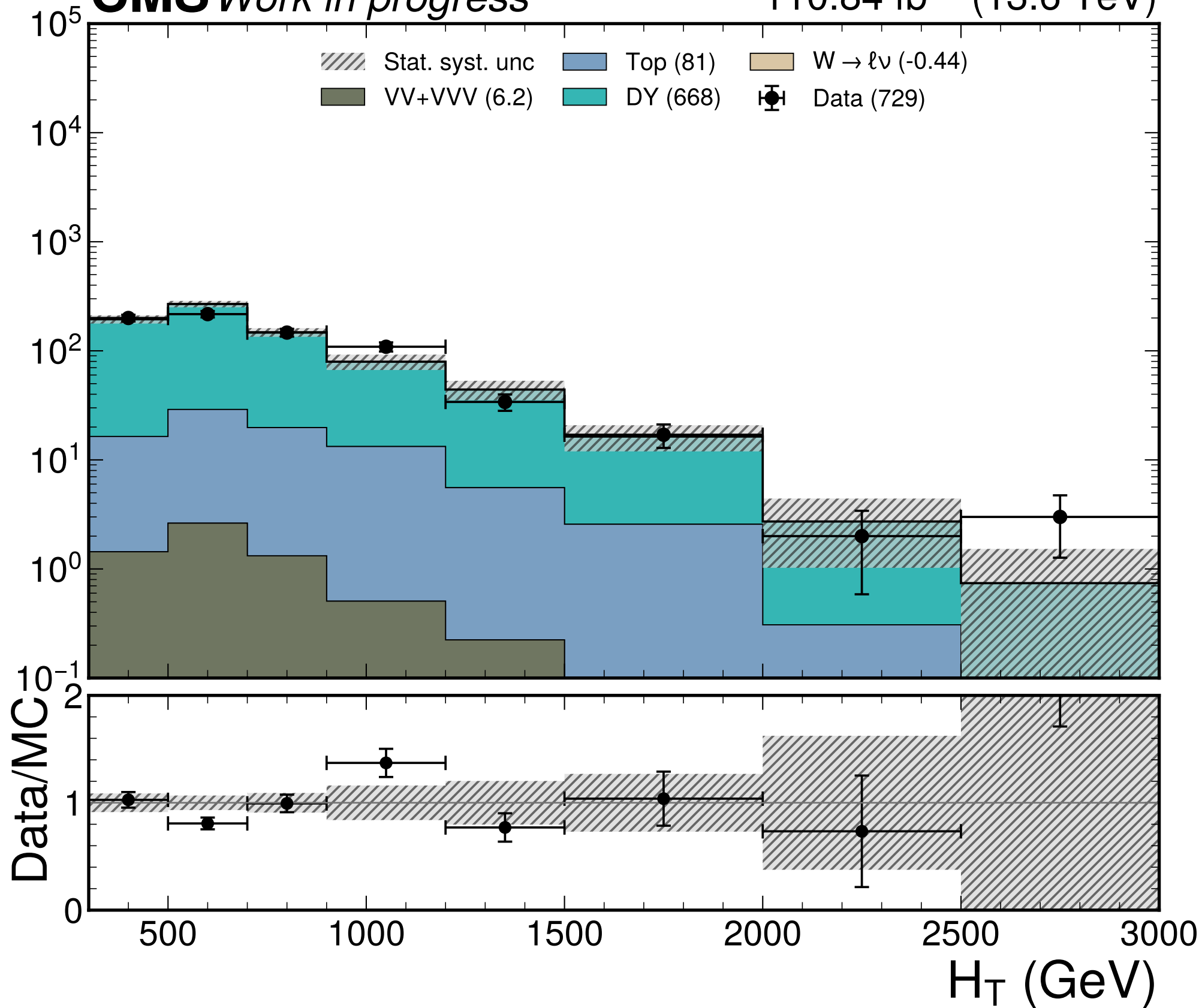
Events



CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)

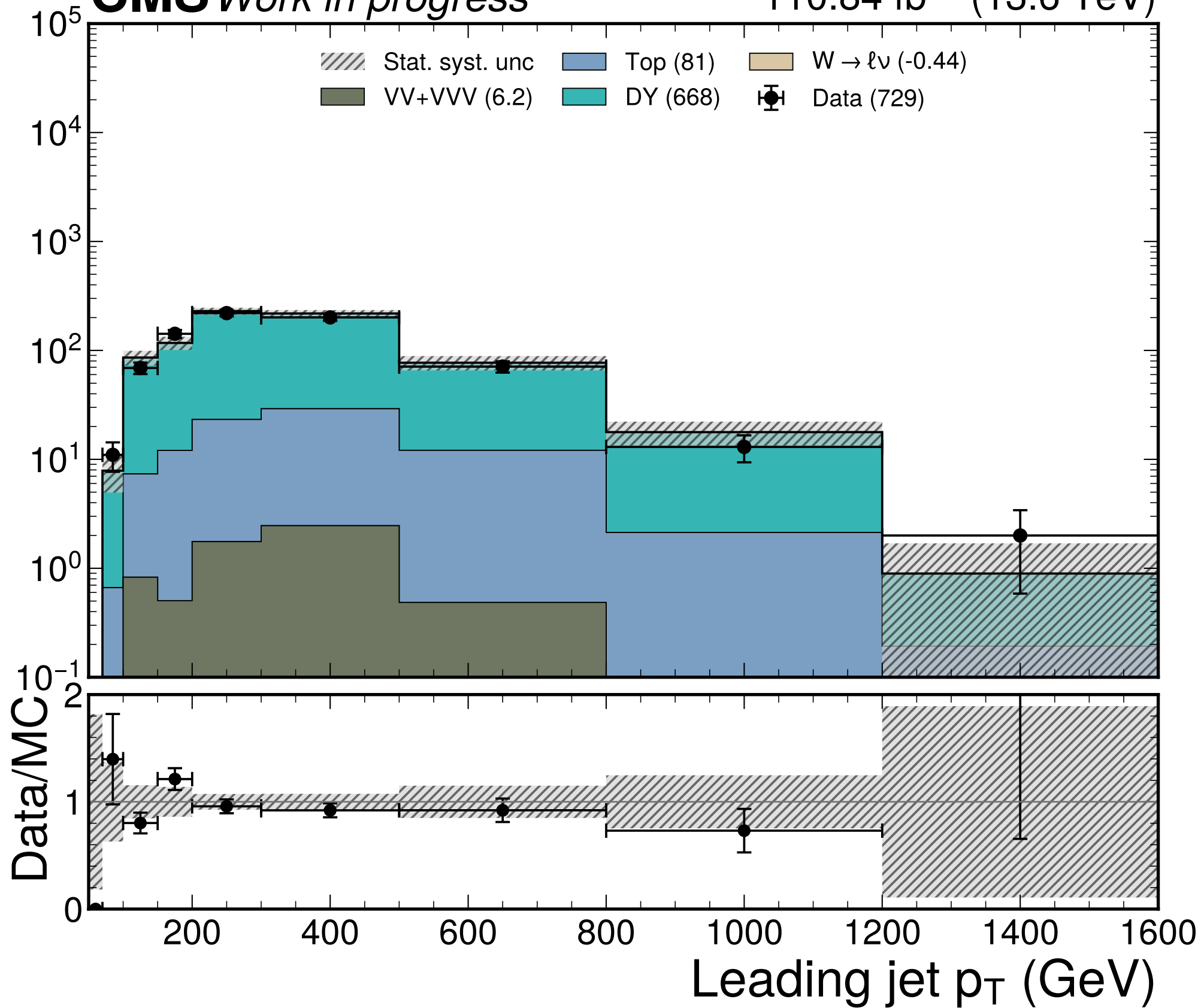
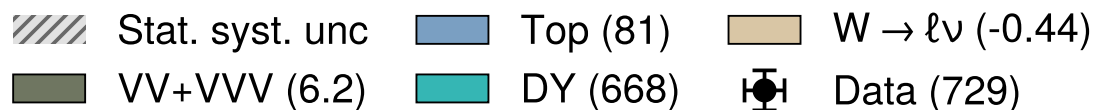
Events



CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)

Events



CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)

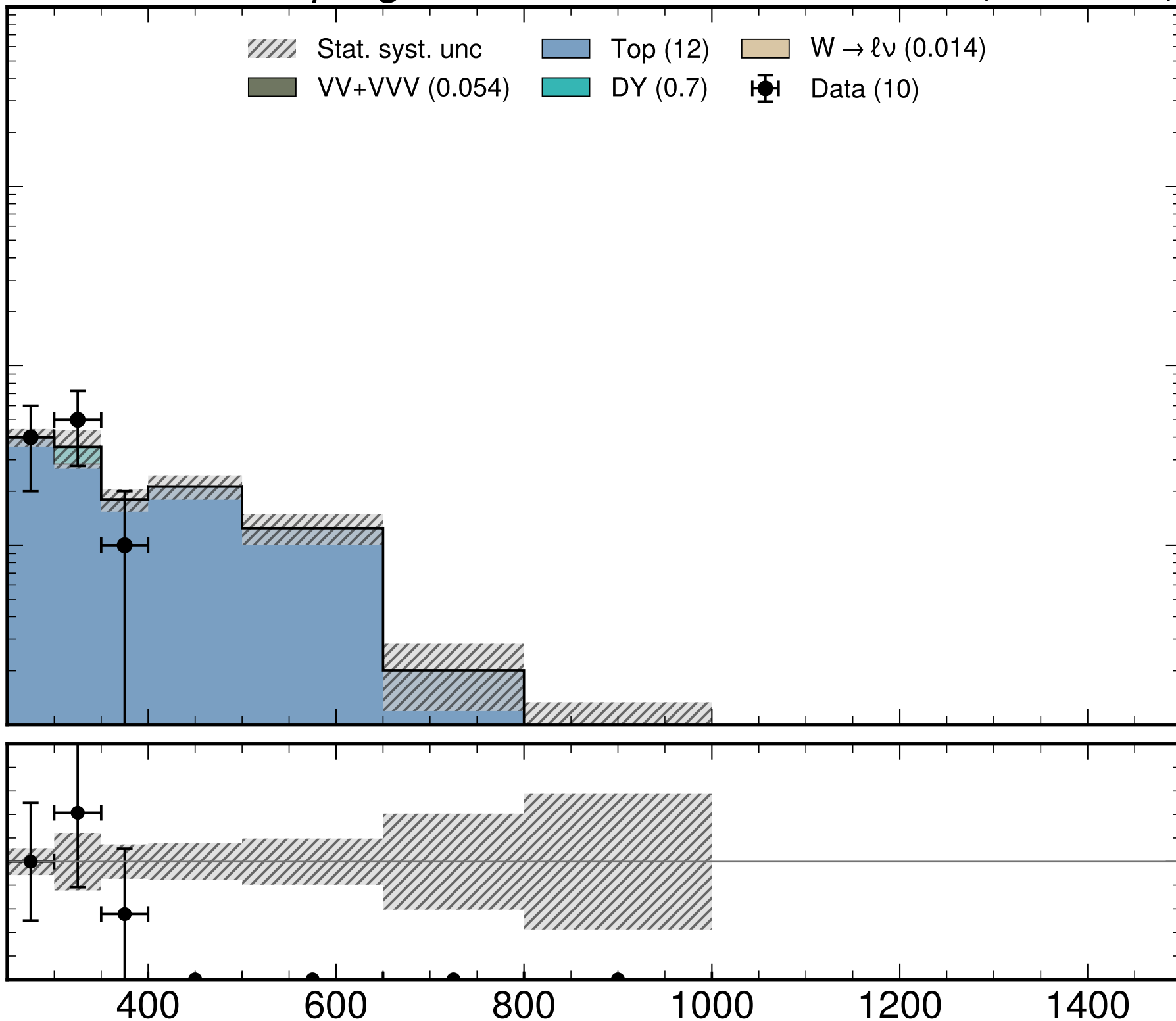
Events



Data/MC

10⁻¹
2
1
0

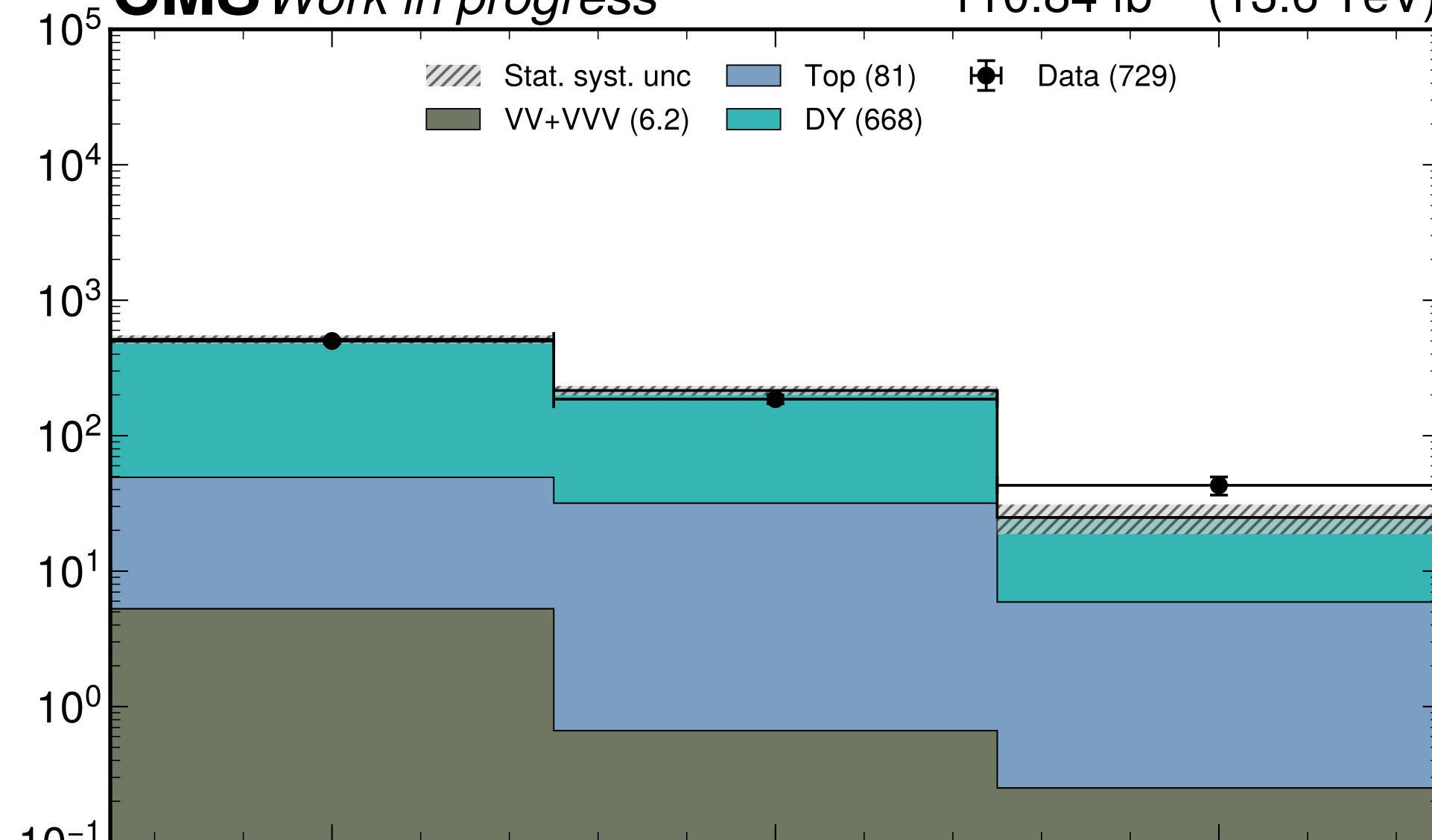
p_T^{miss} (GeV)



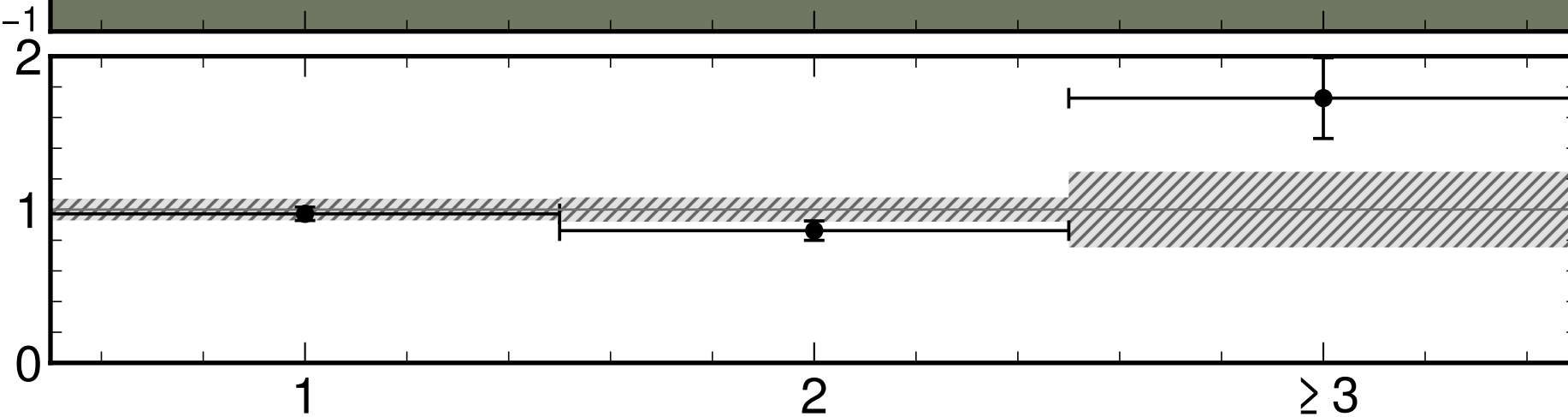
CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)

Events



Data/MC

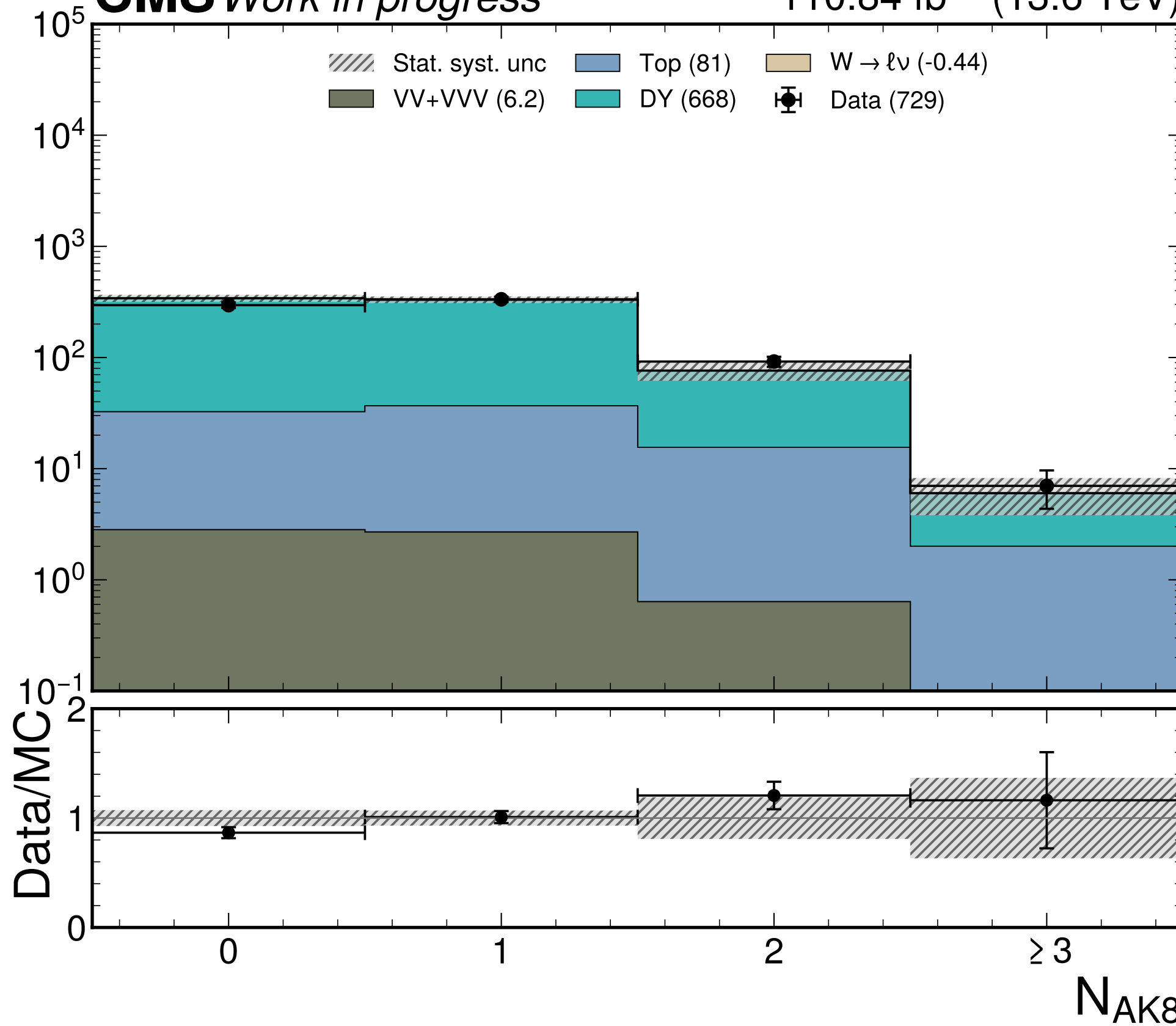
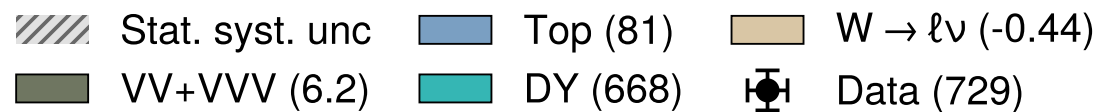


N_b

CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)

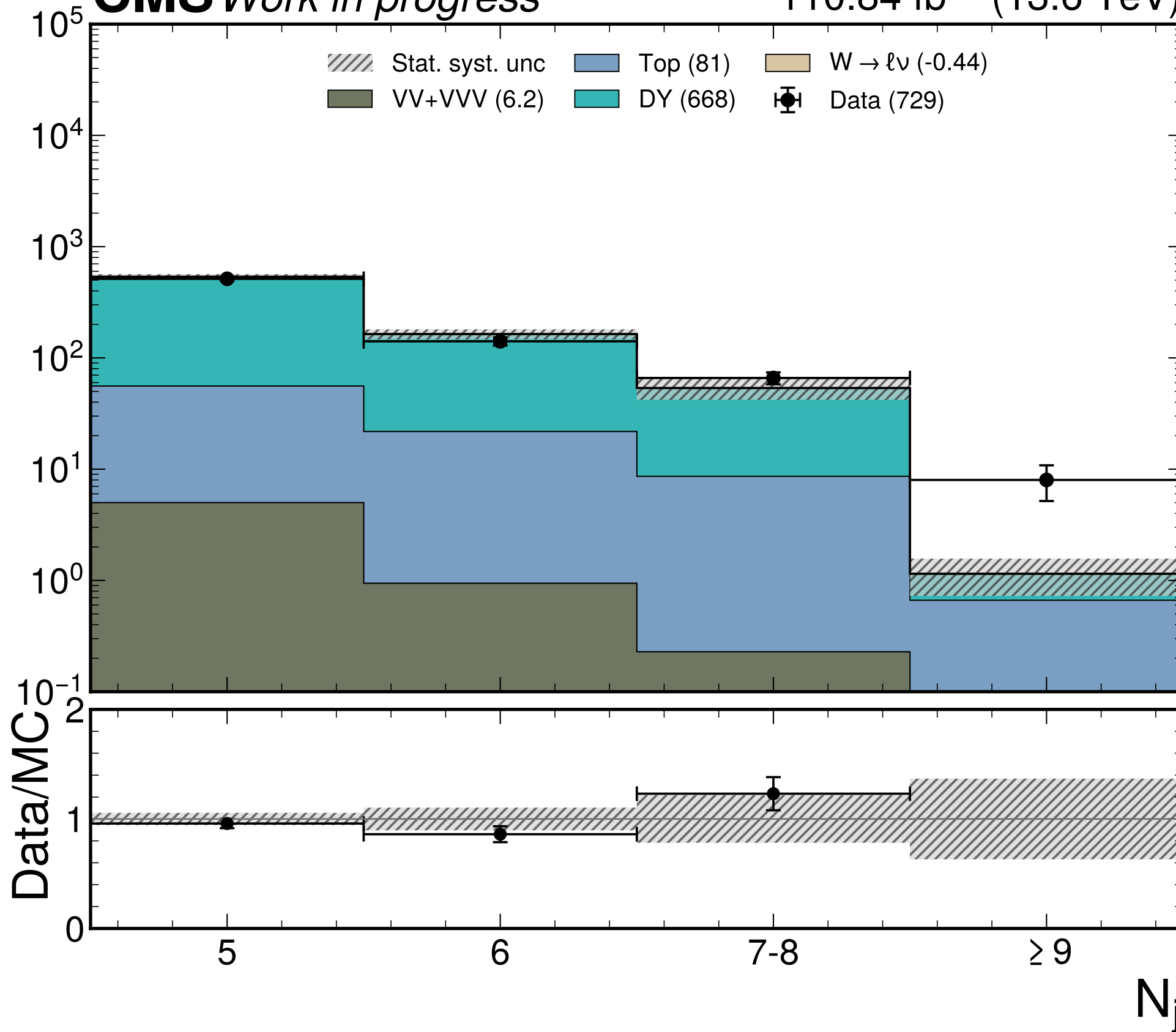
Events



CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)

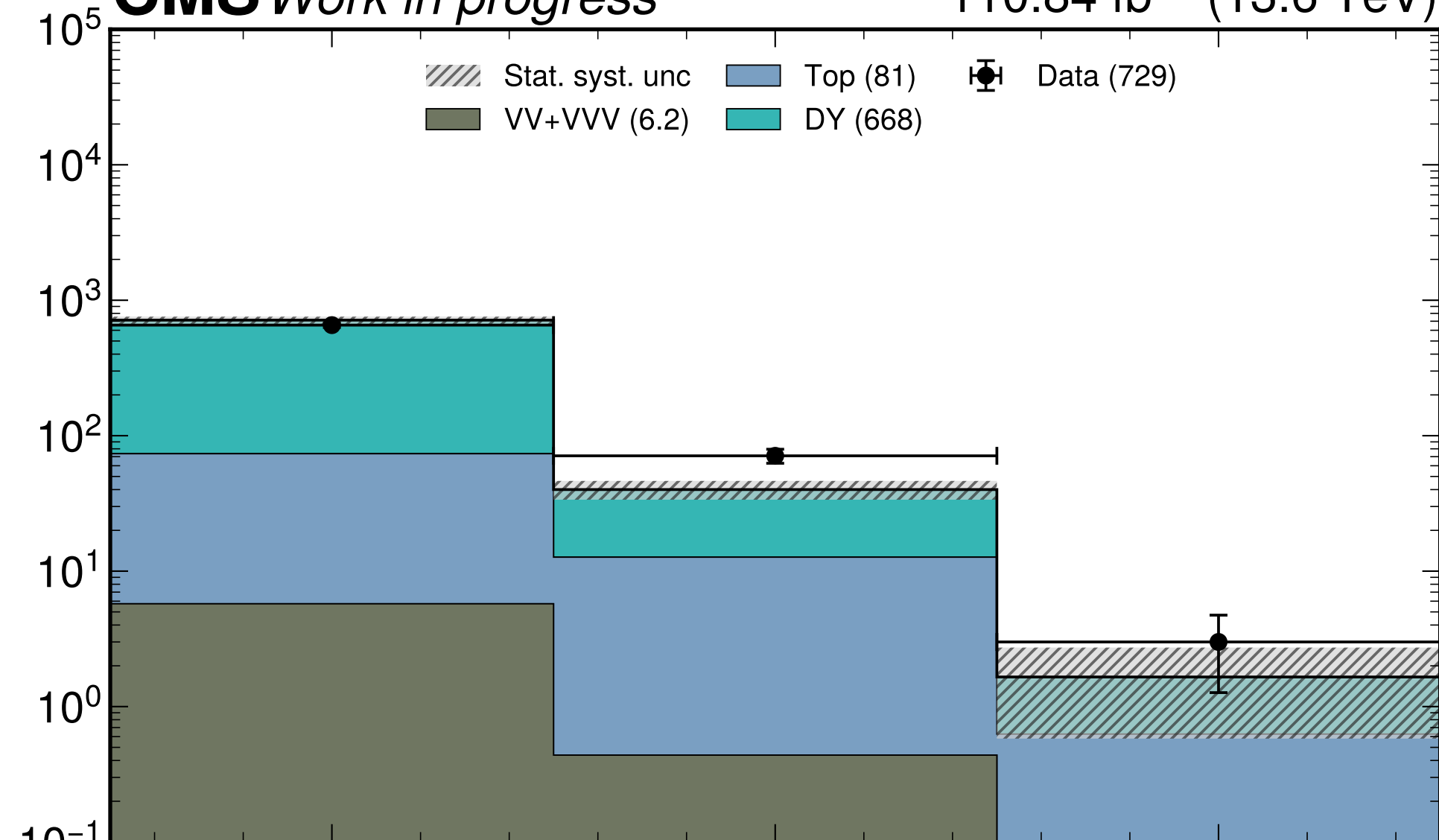
Events



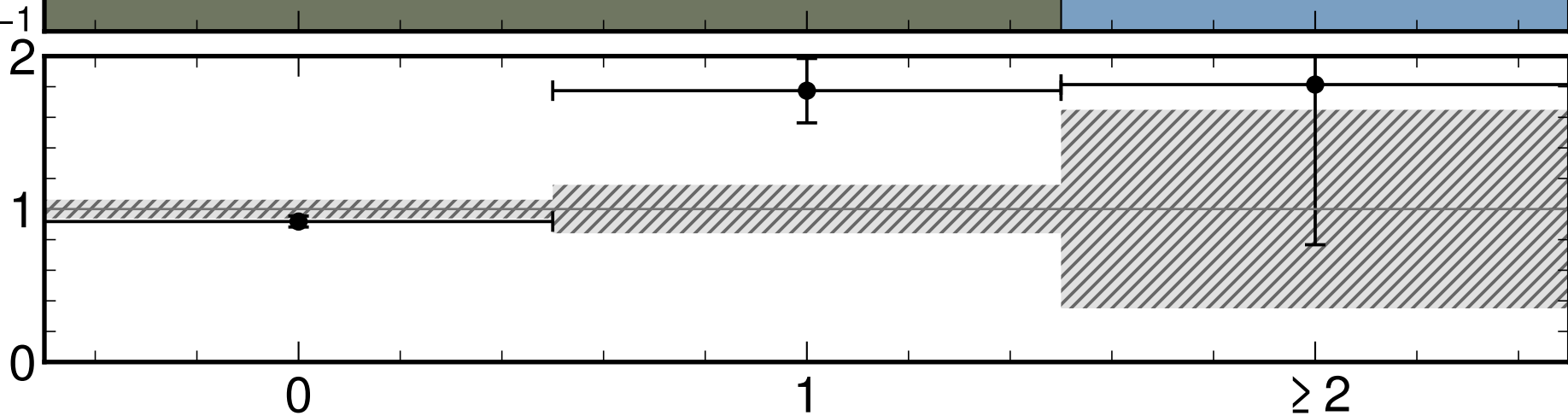
CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)

Events



Data/MC

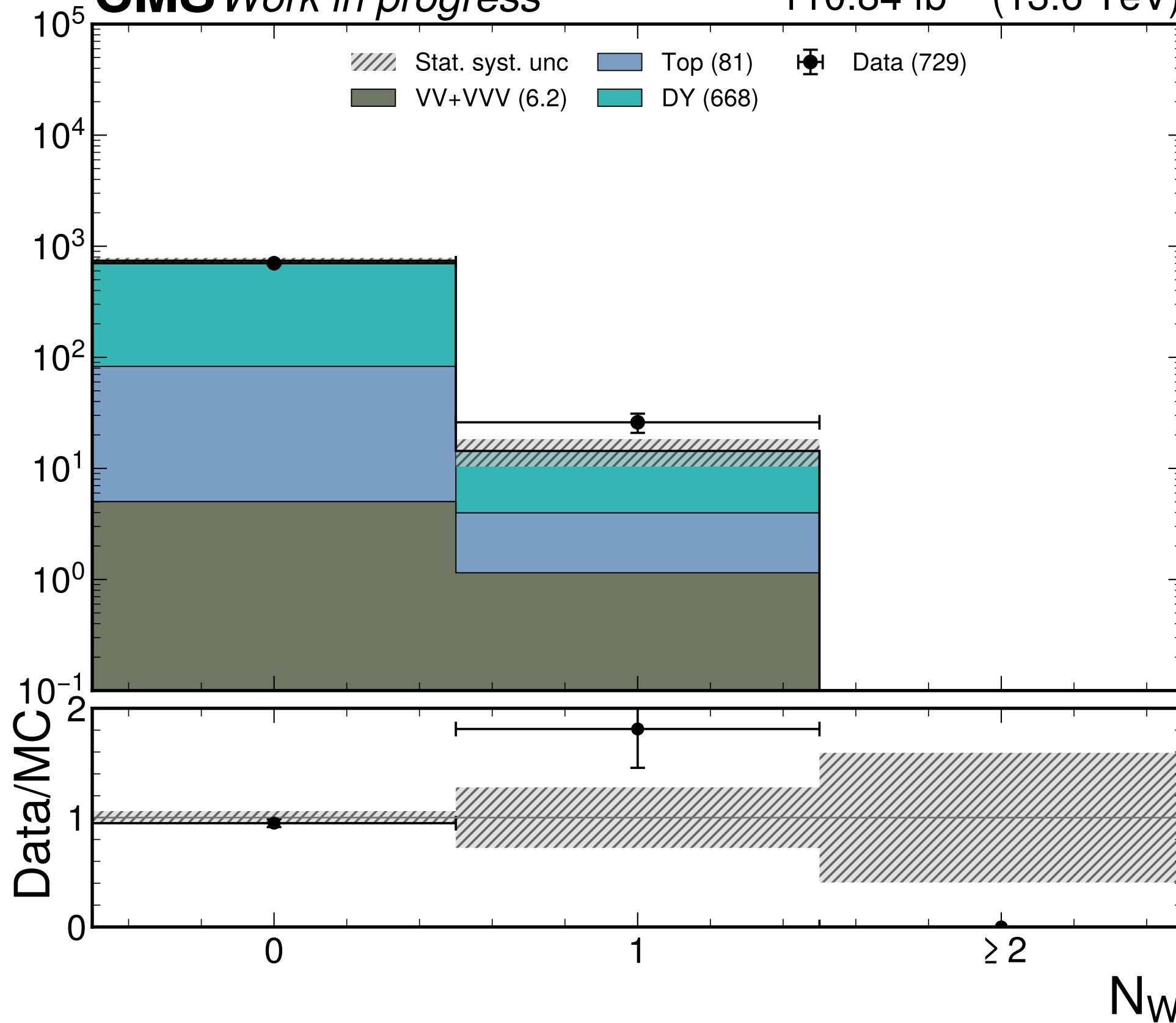


N_t

CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)

Events

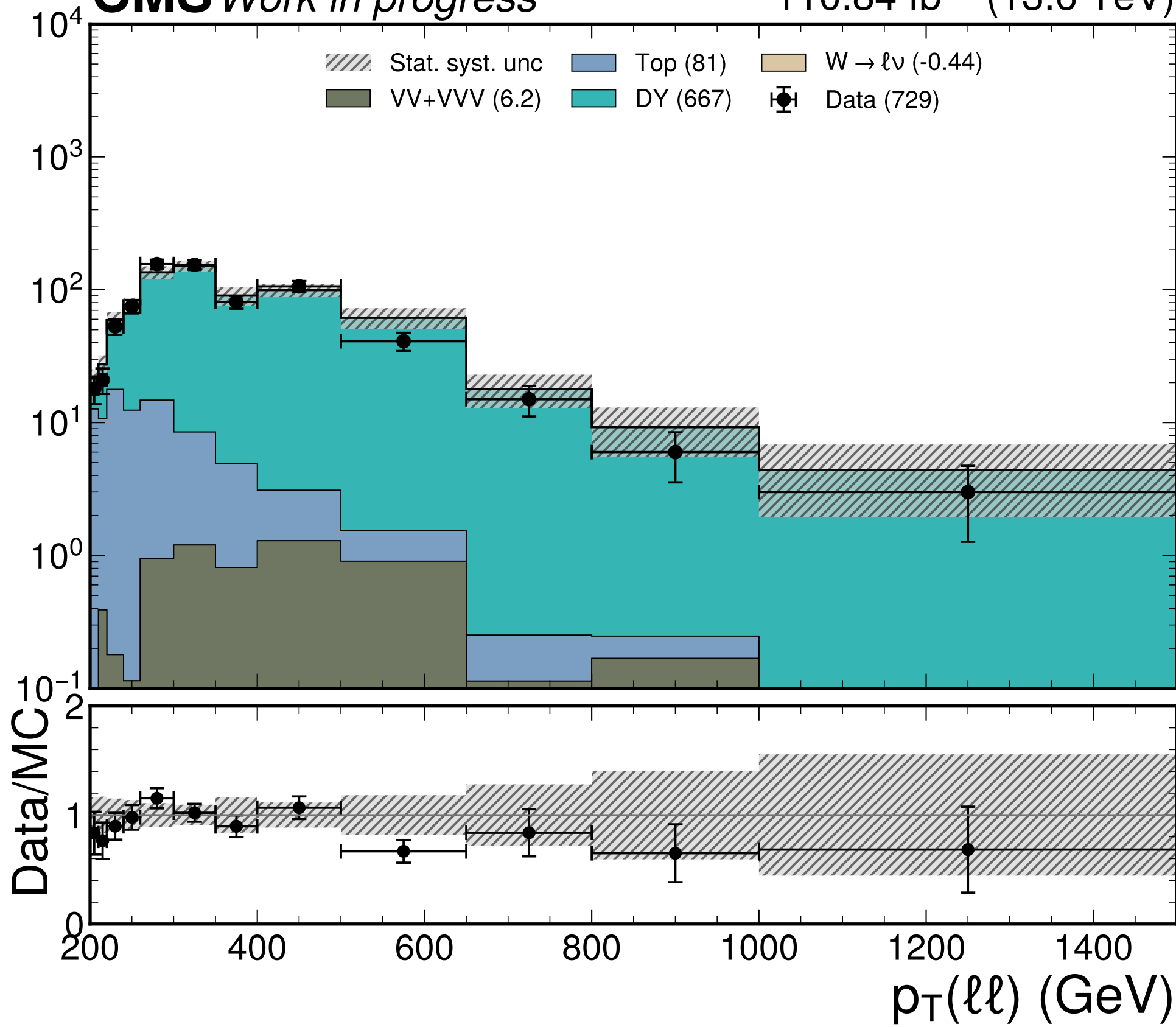


CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)

Events

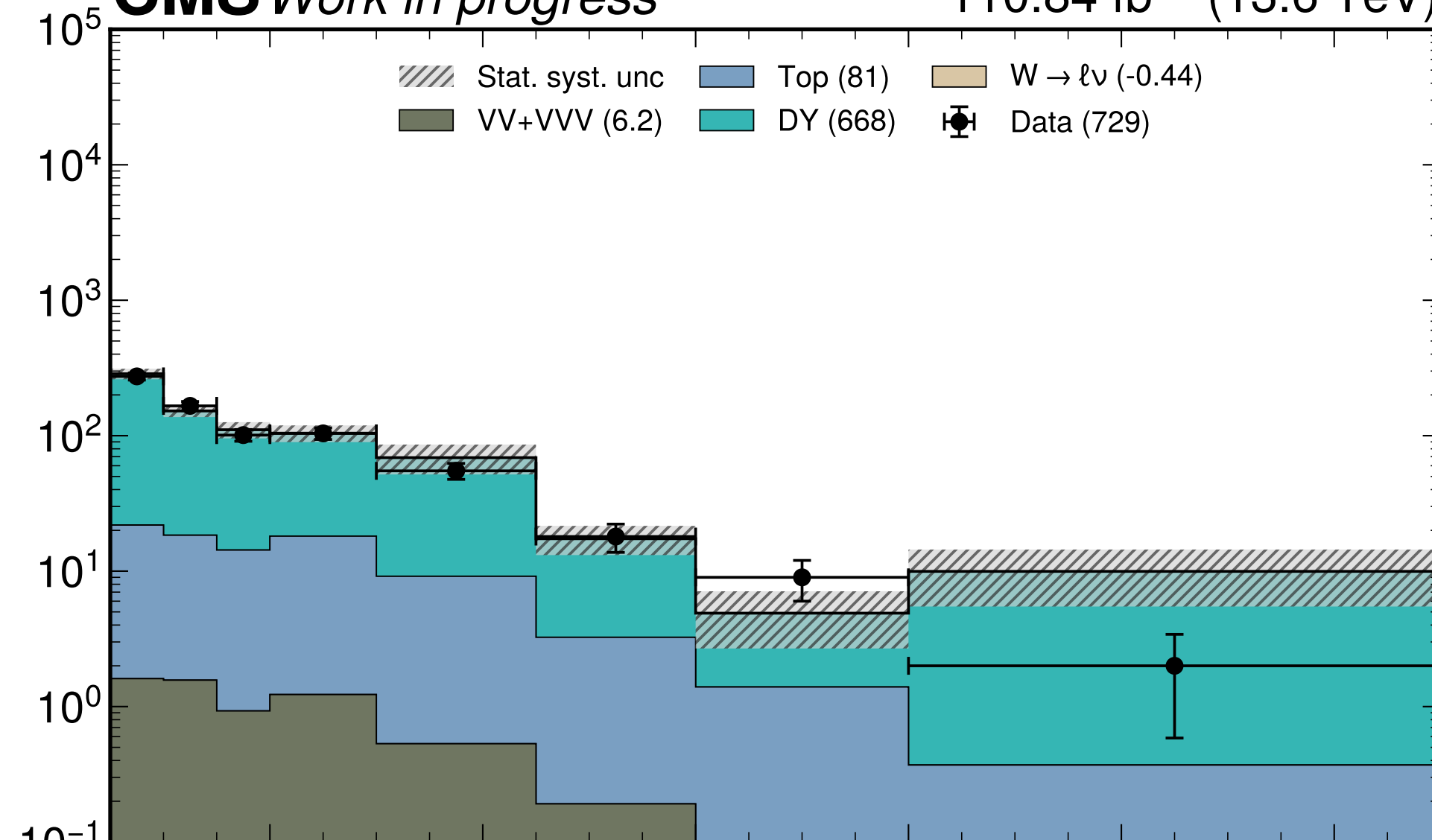
Stat. syst. unc Top (81) W → ℓν (-0.44)
VV+VVV (6.2) DY (667) Data (729)



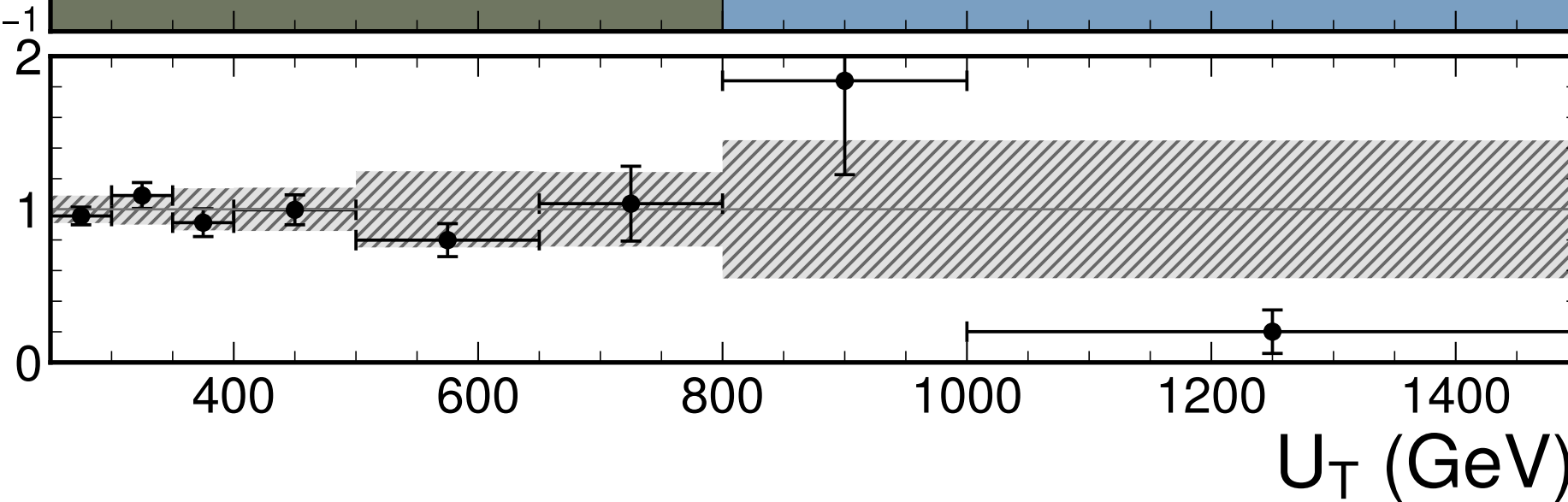
CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)

Events

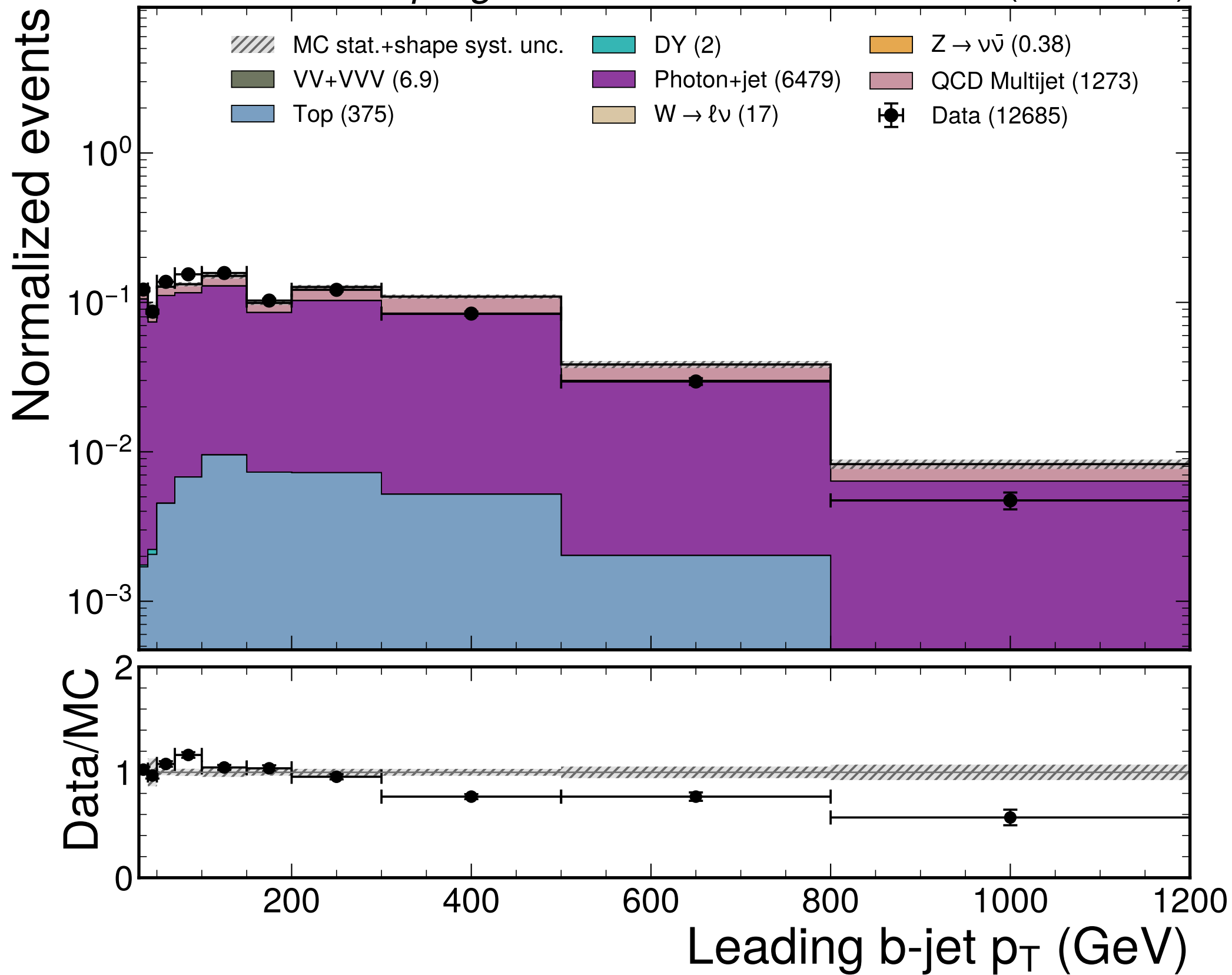


Data/MC



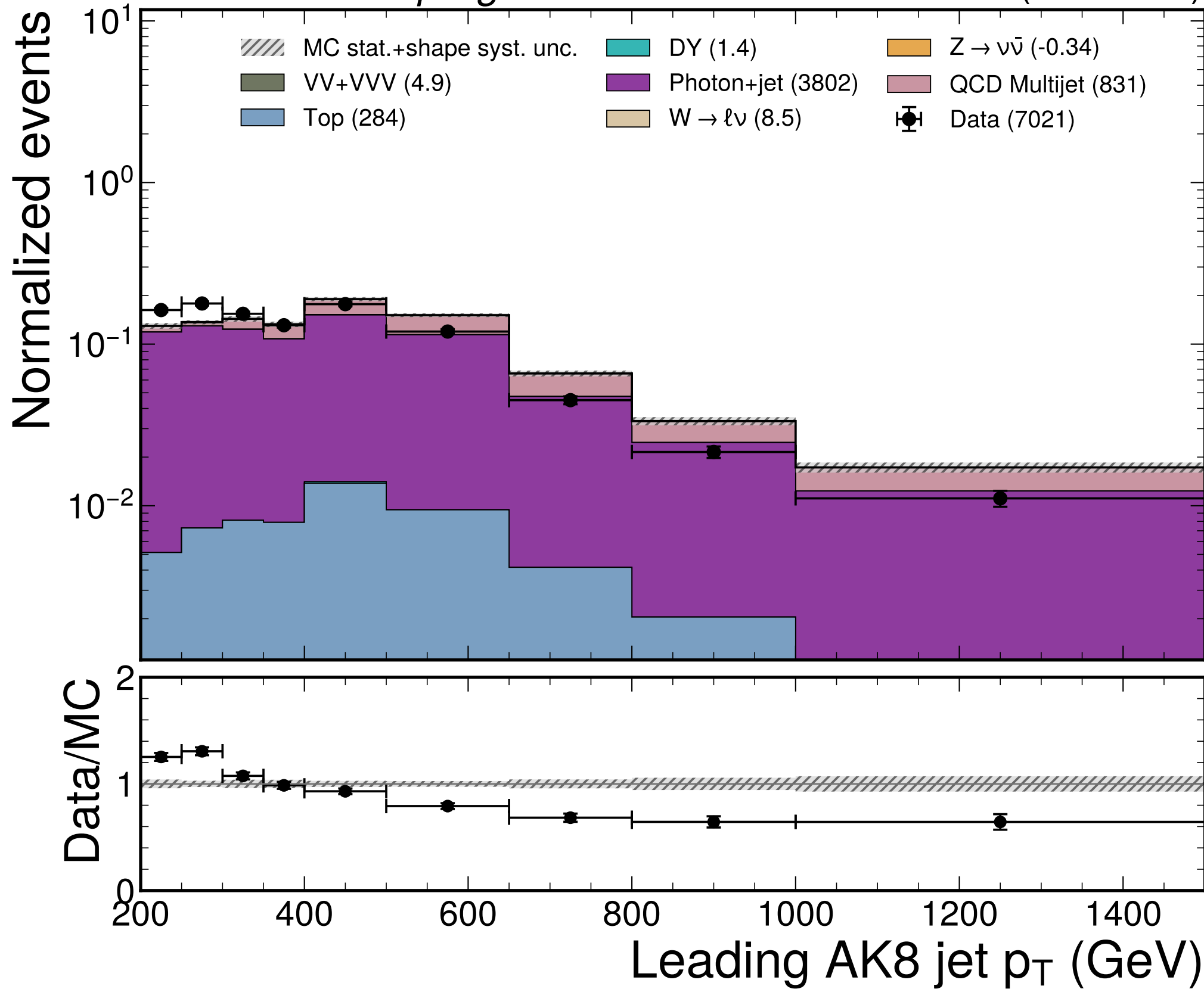
CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)



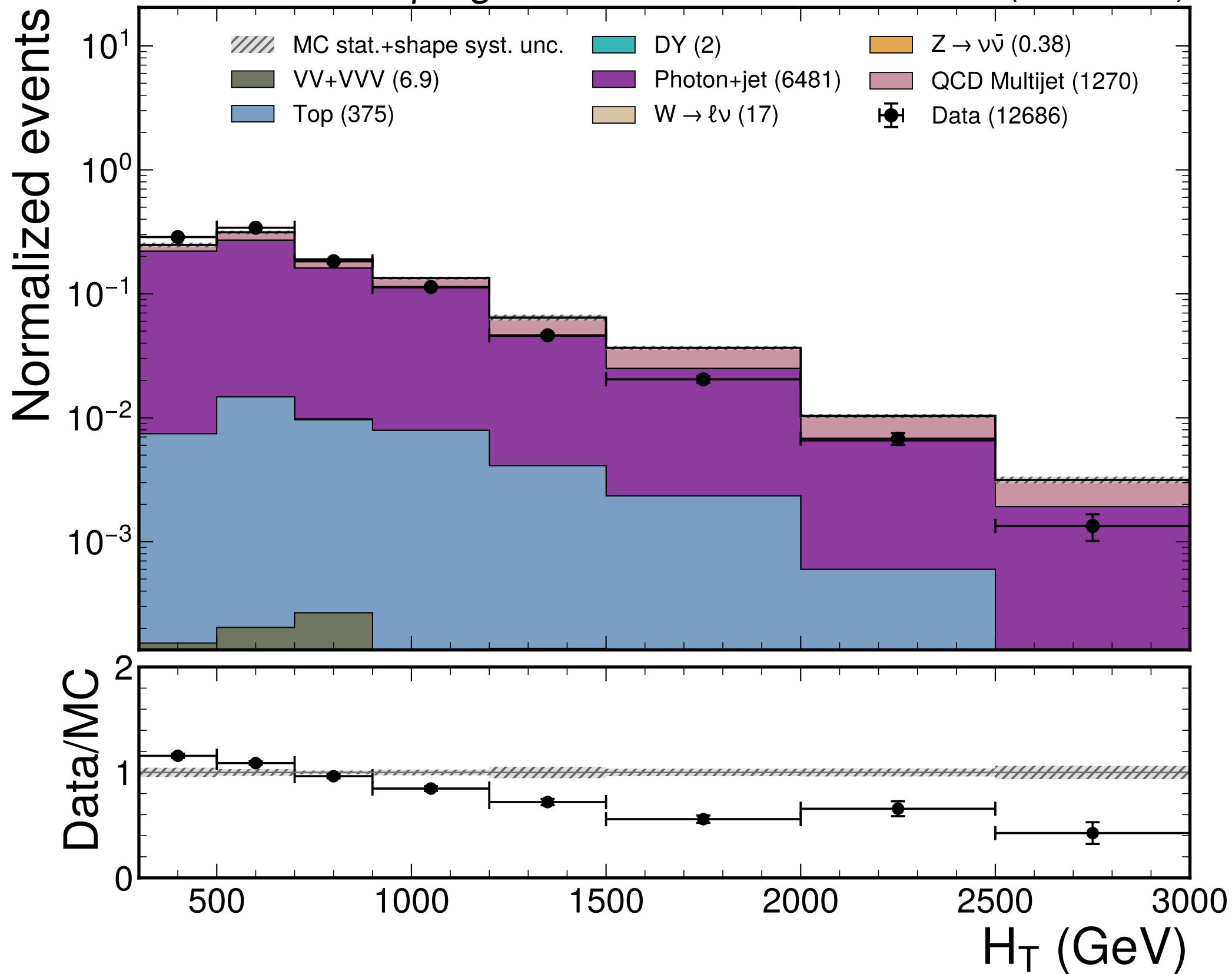
CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)



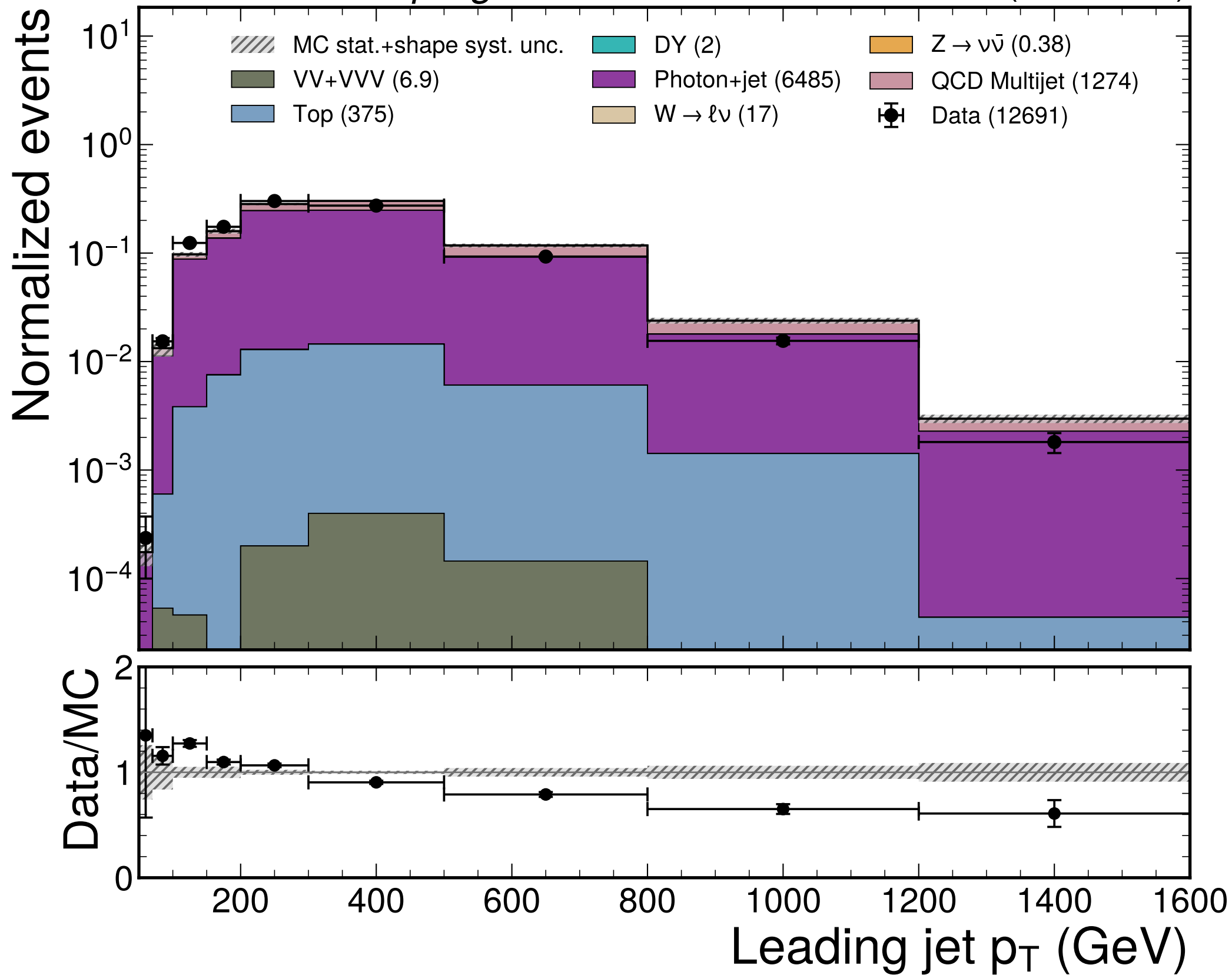
CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)



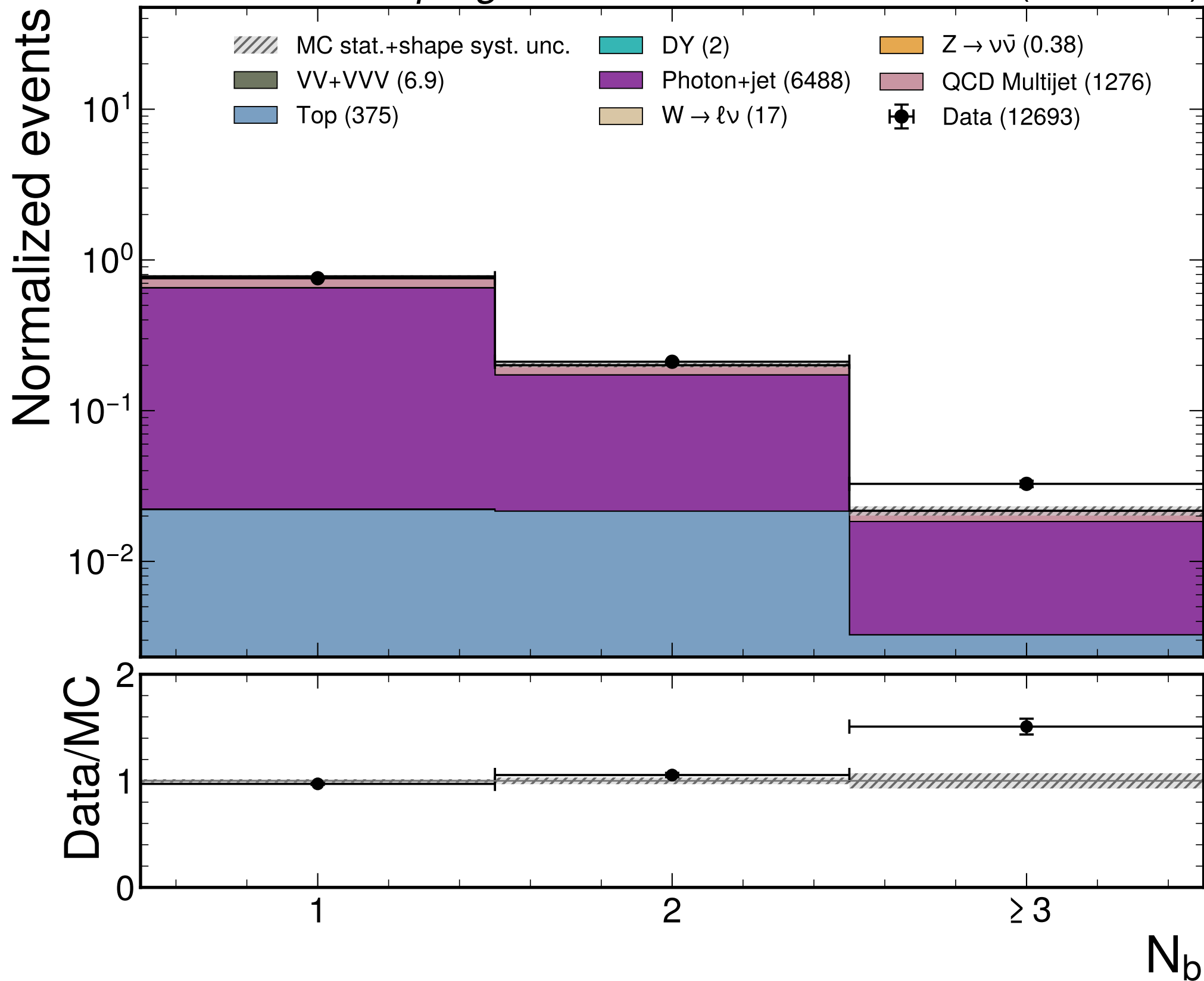
CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)



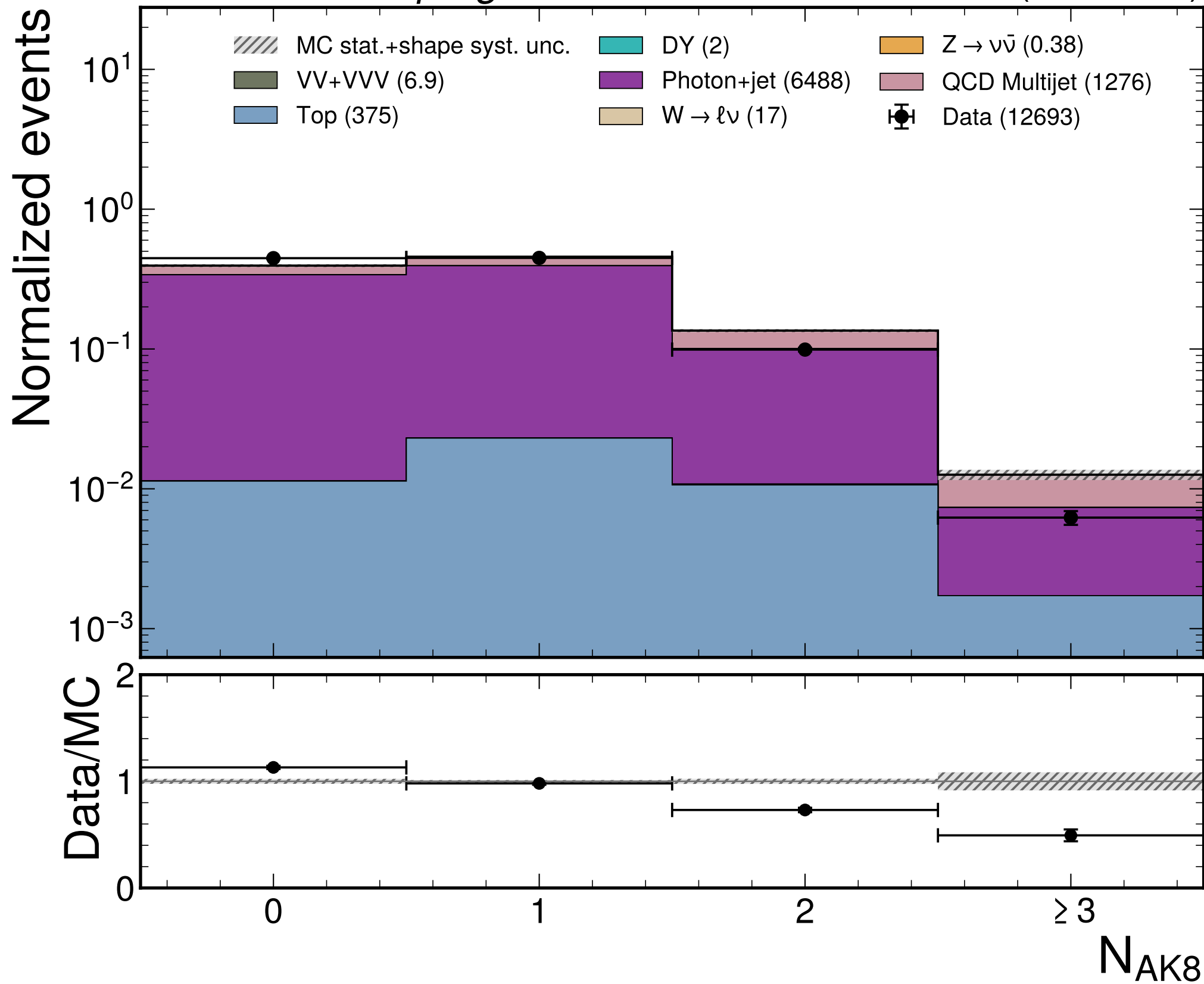
CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)



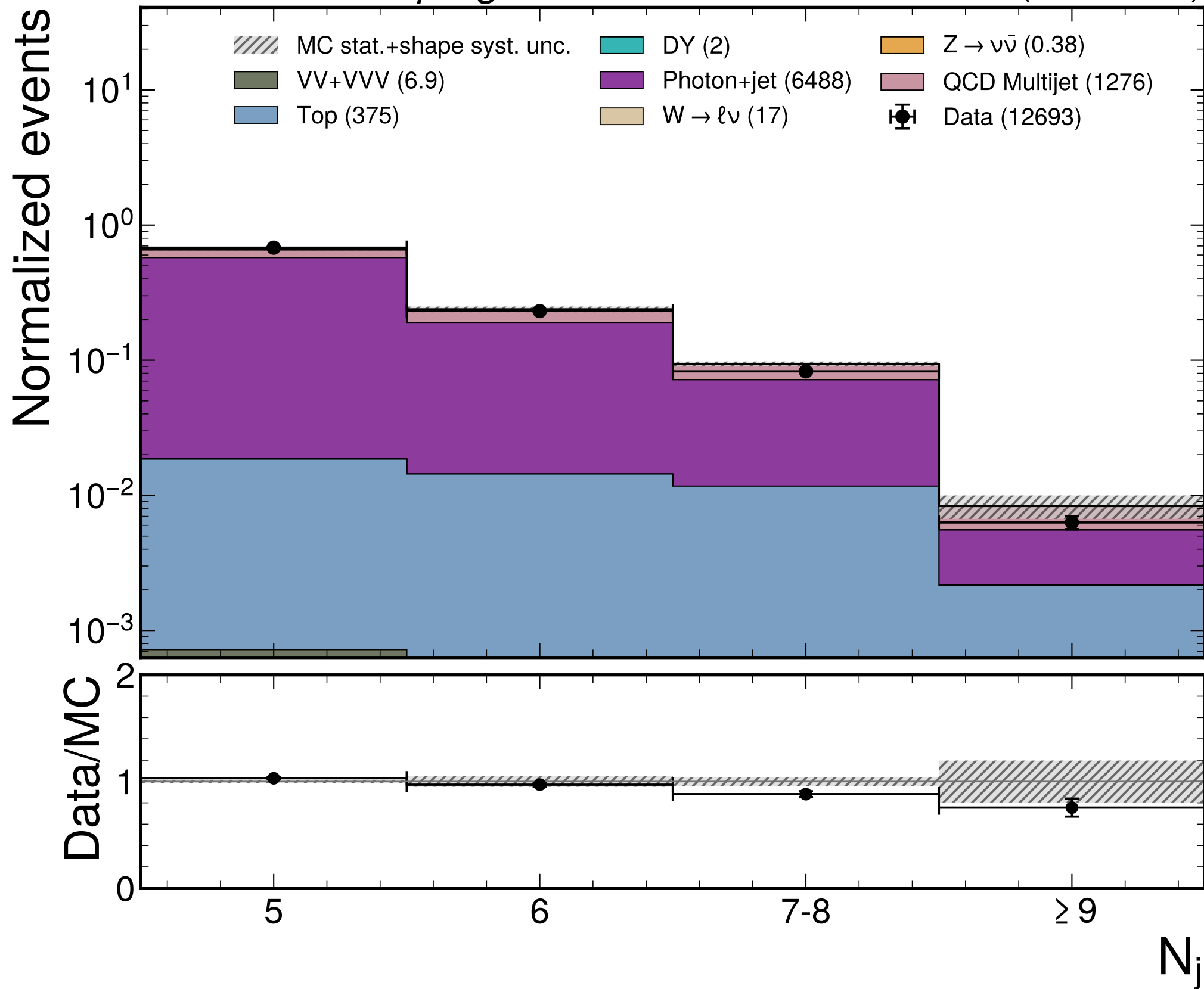
CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)



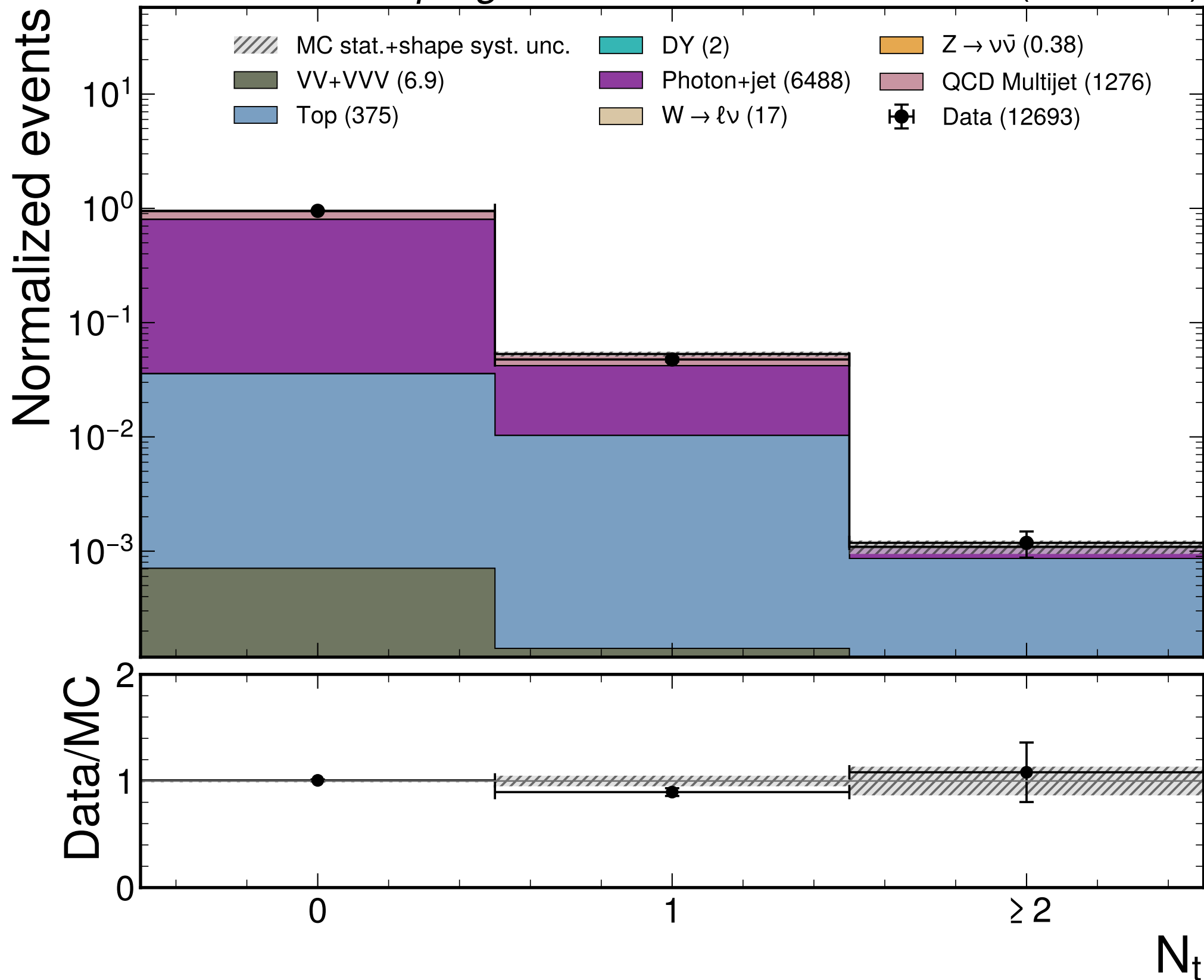
CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)



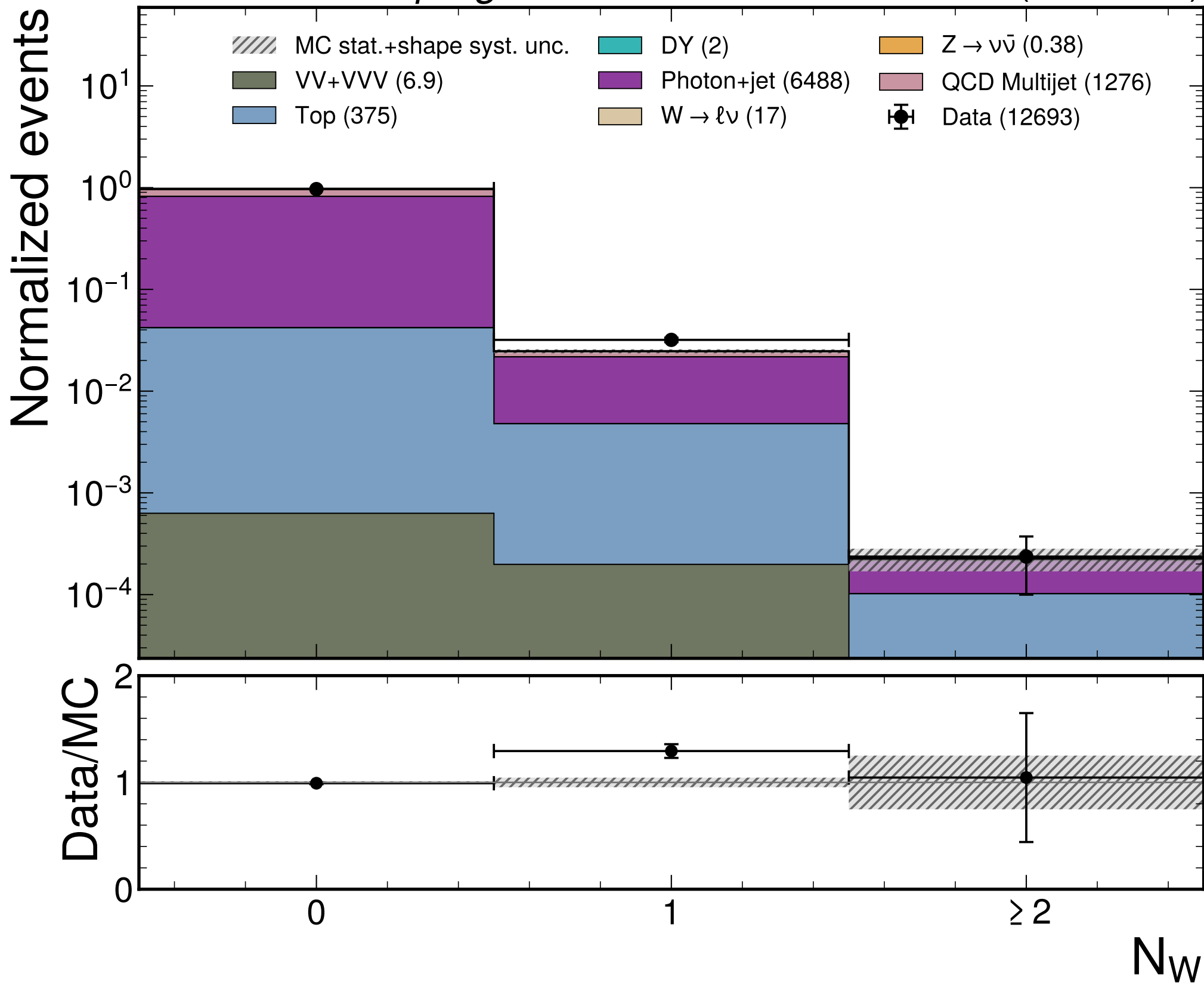
CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)



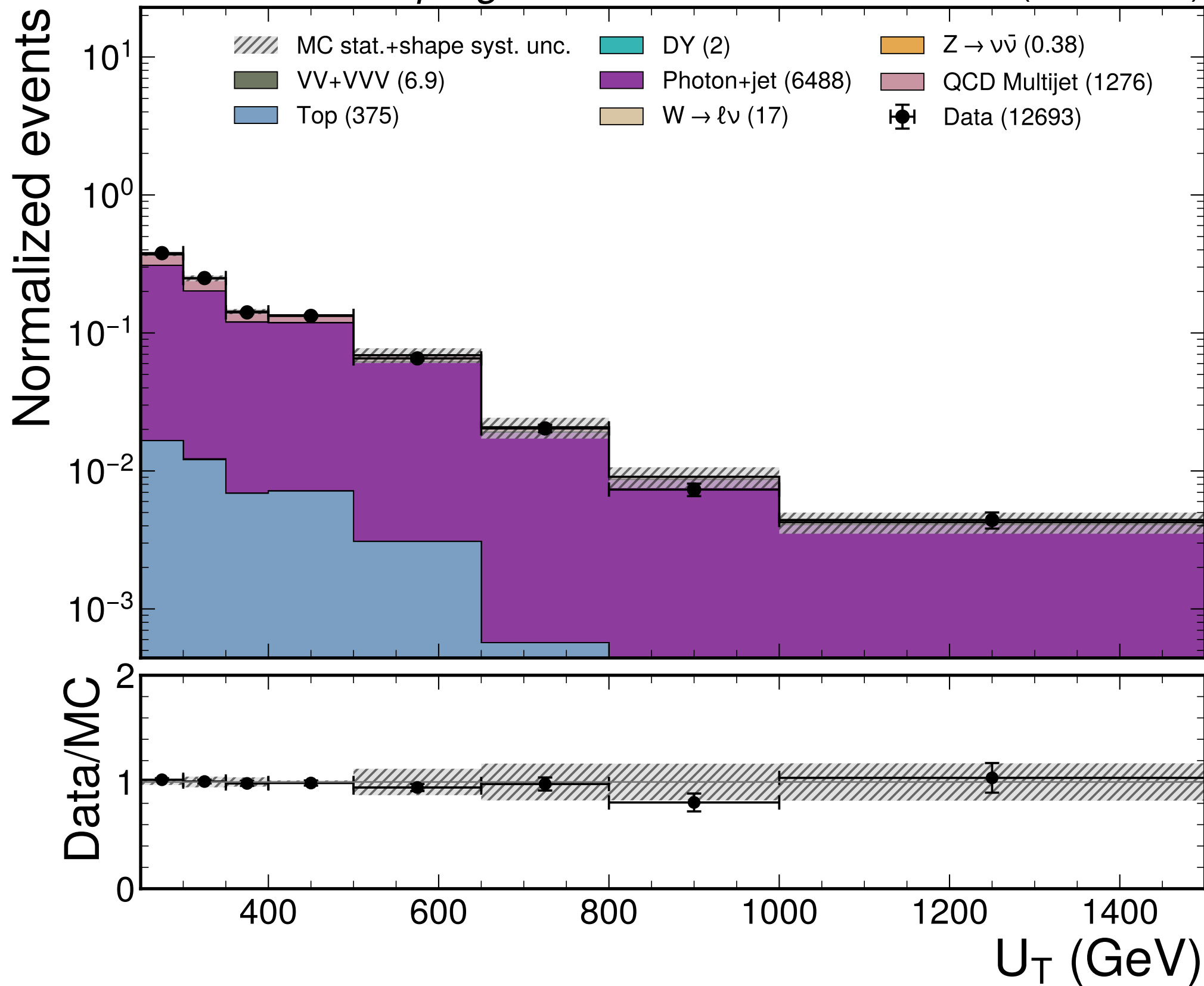
CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)



CMS *Work in progress*

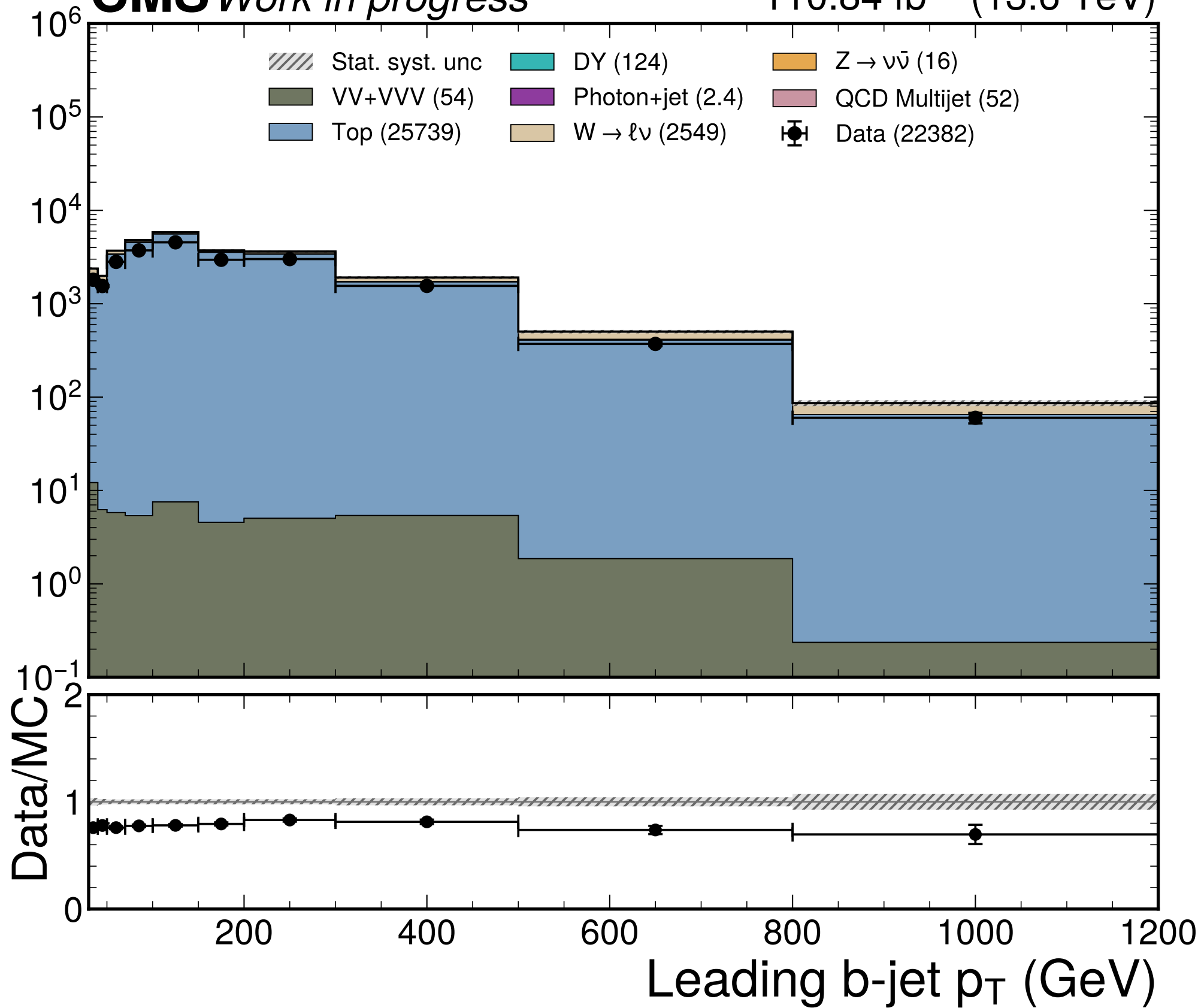
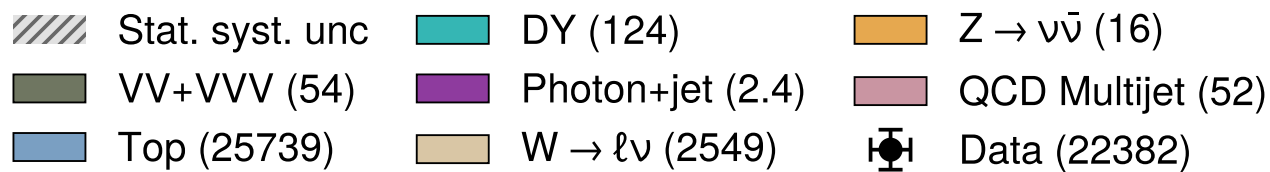
110.84 fb⁻¹ (13.6 TeV)



CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)

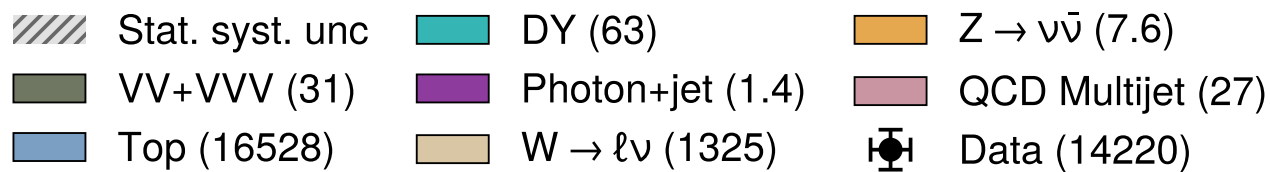
Events



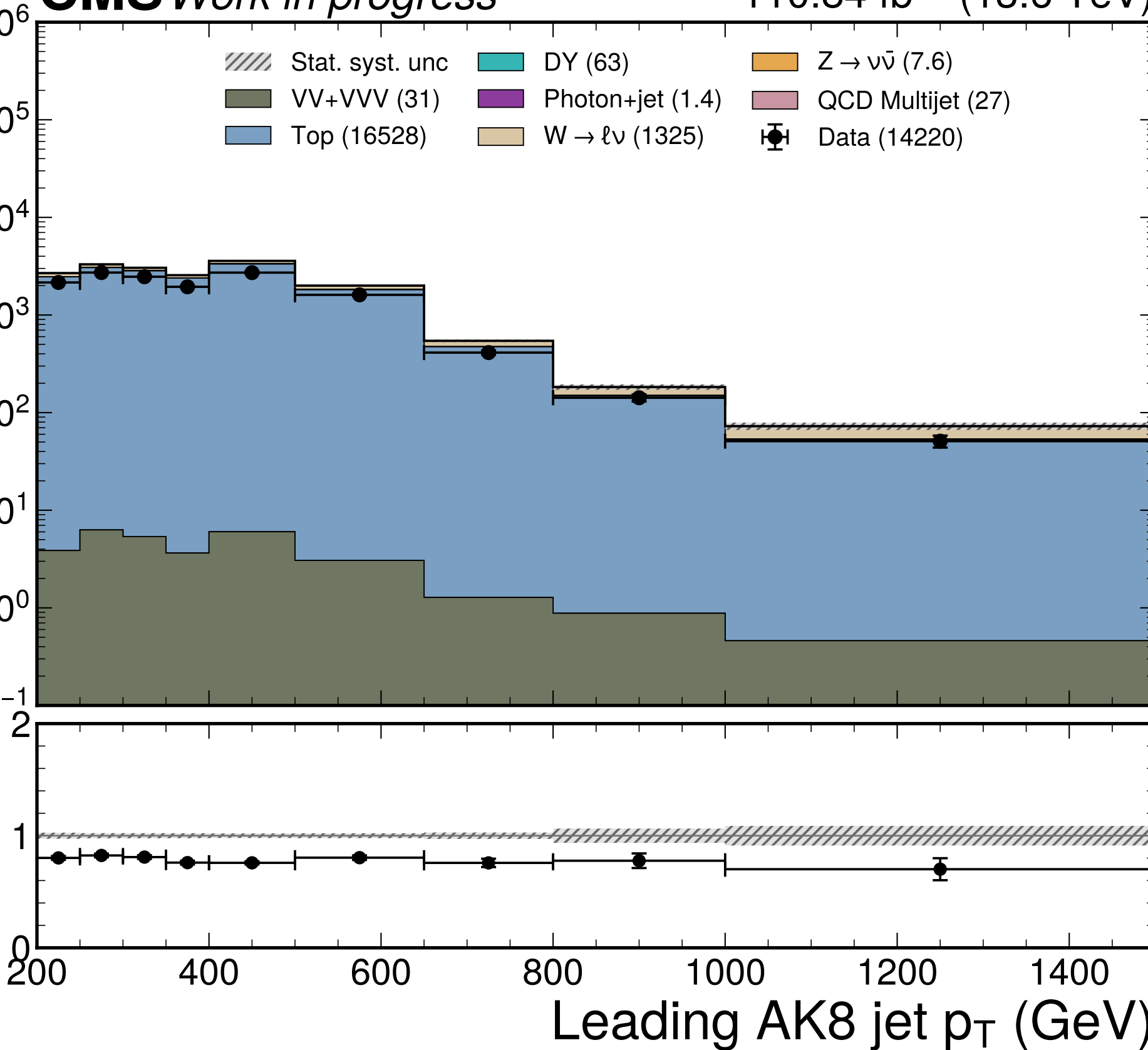
CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)

Events

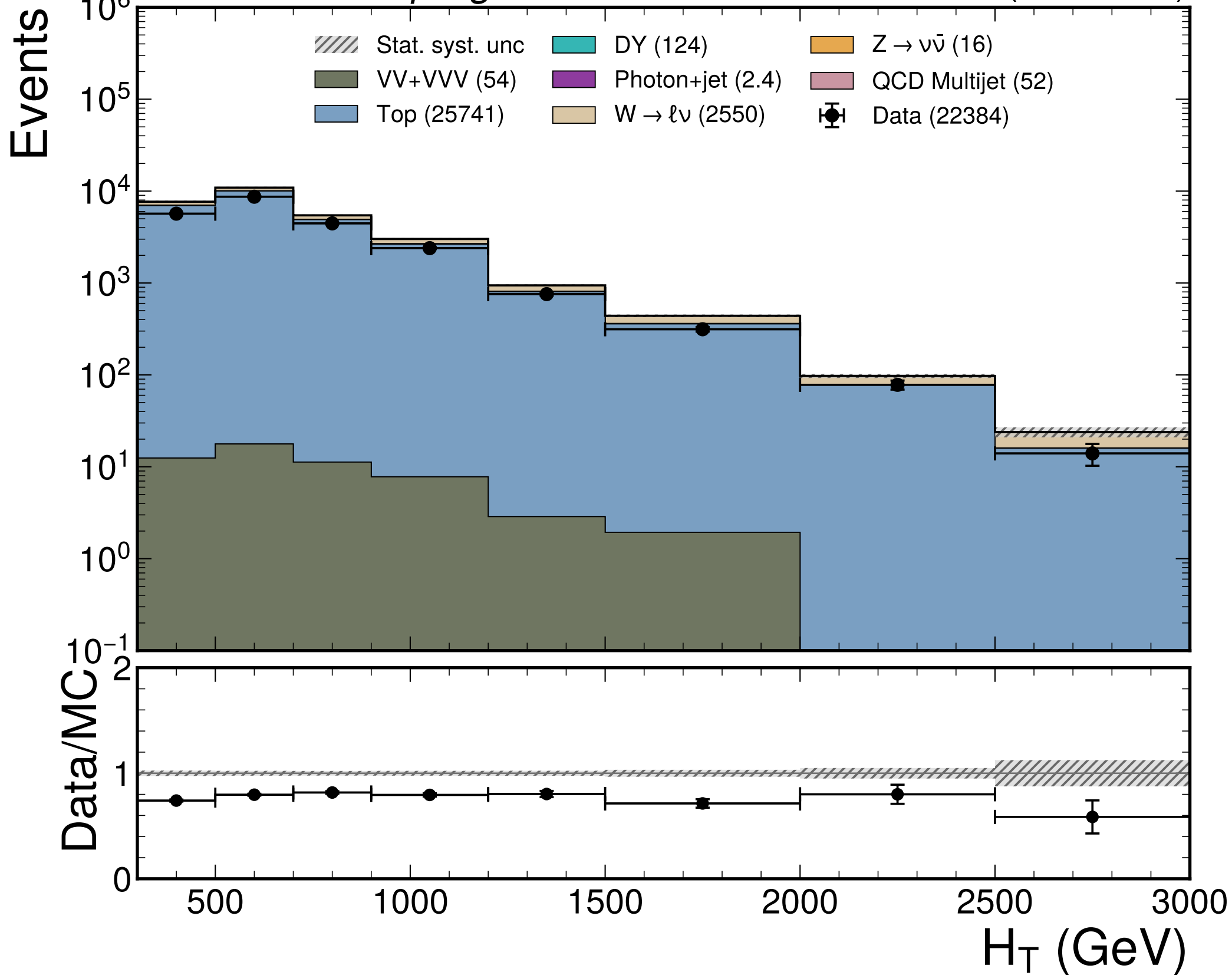


Data/MC



CMS *Work in progress*

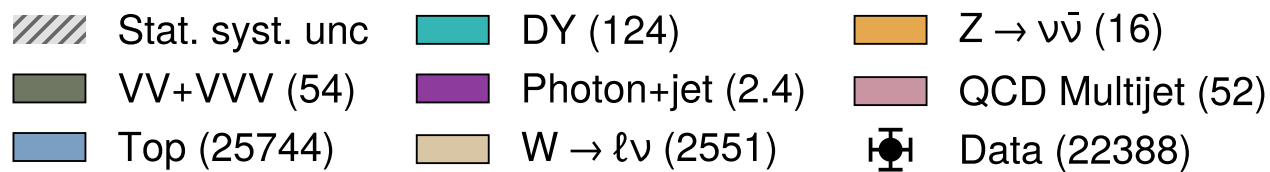
110.84 fb⁻¹ (13.6 TeV)



CMS *Work in progress*

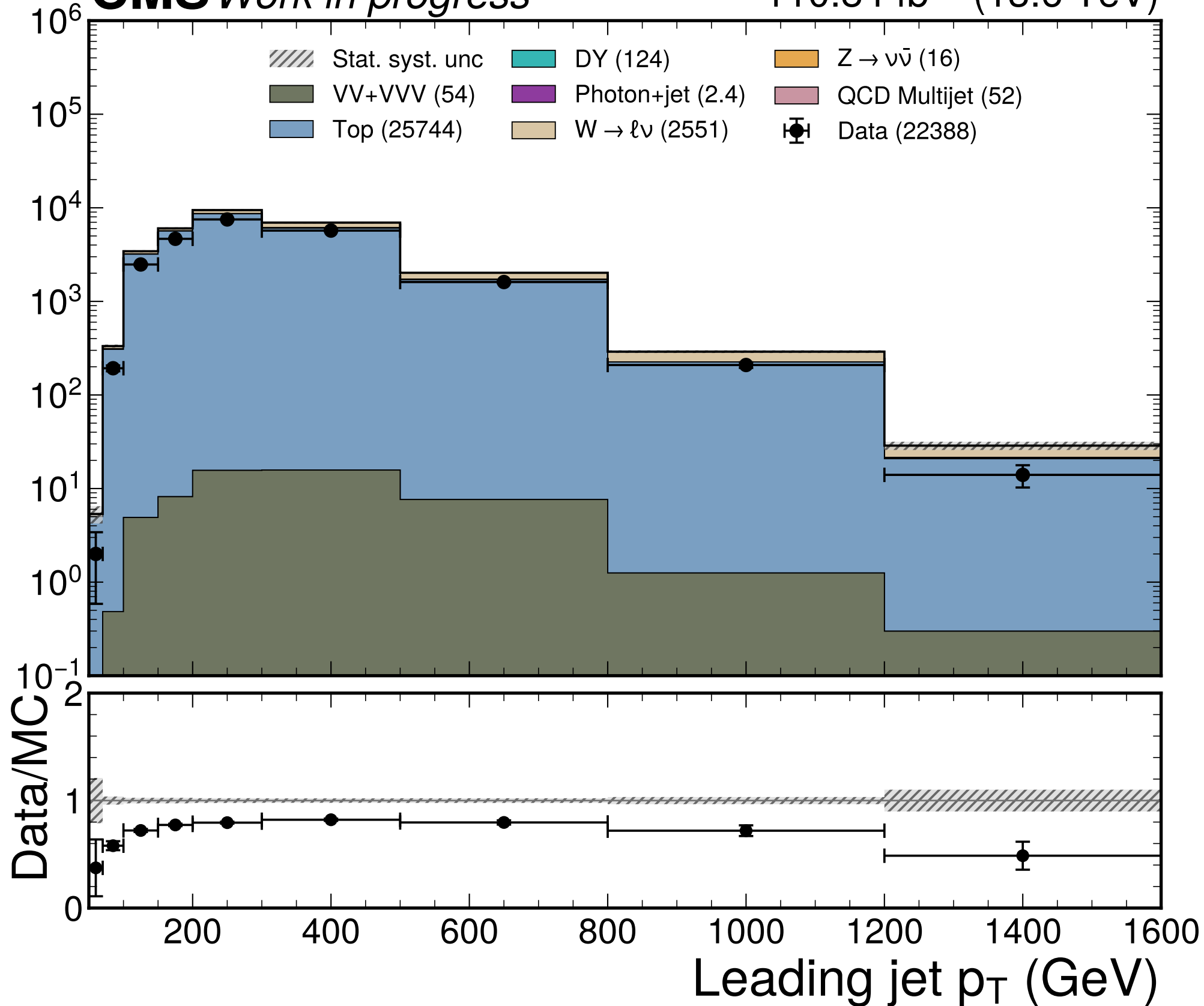
110.84 fb⁻¹ (13.6 TeV)

Events



Data/MC

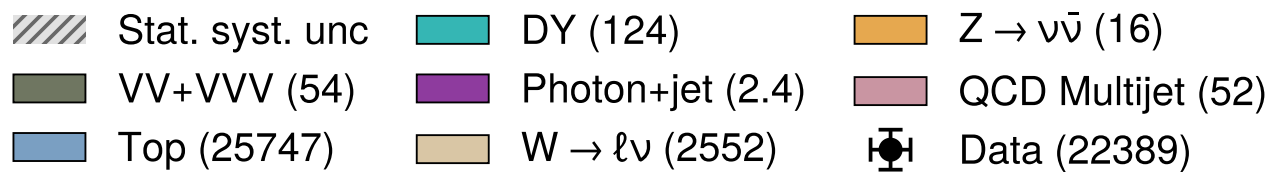
Leading jet p_T (GeV)



CMS *Work in progress*

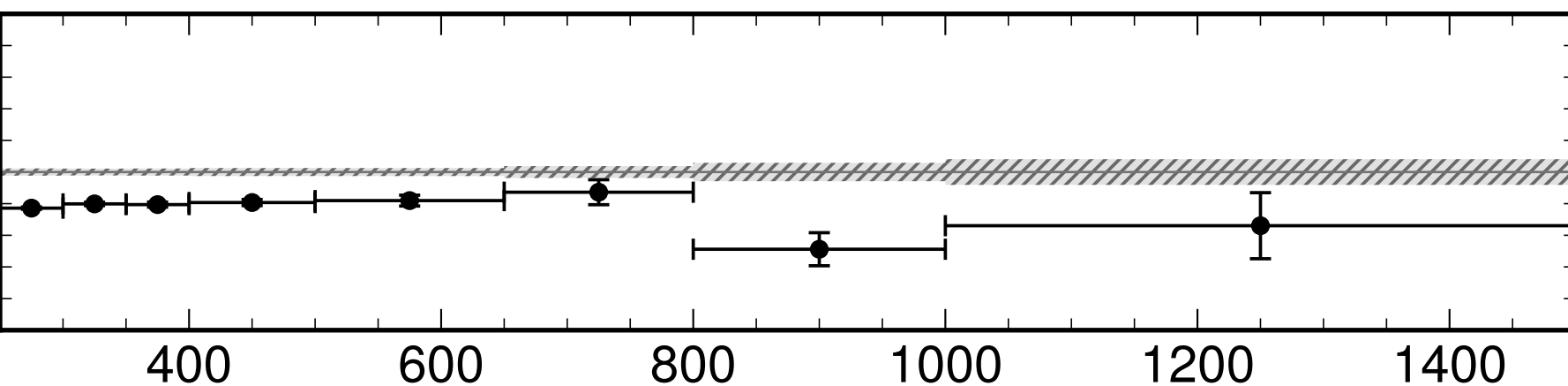
110.84 fb⁻¹ (13.6 TeV)

Events



Data/MC

10⁻²

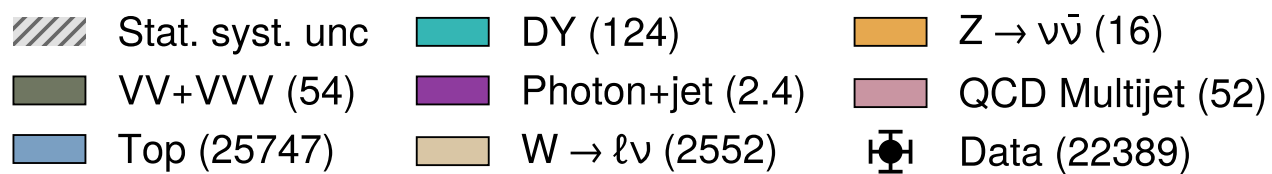


p_T^{miss} (GeV)

CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)

Events



Data/MC

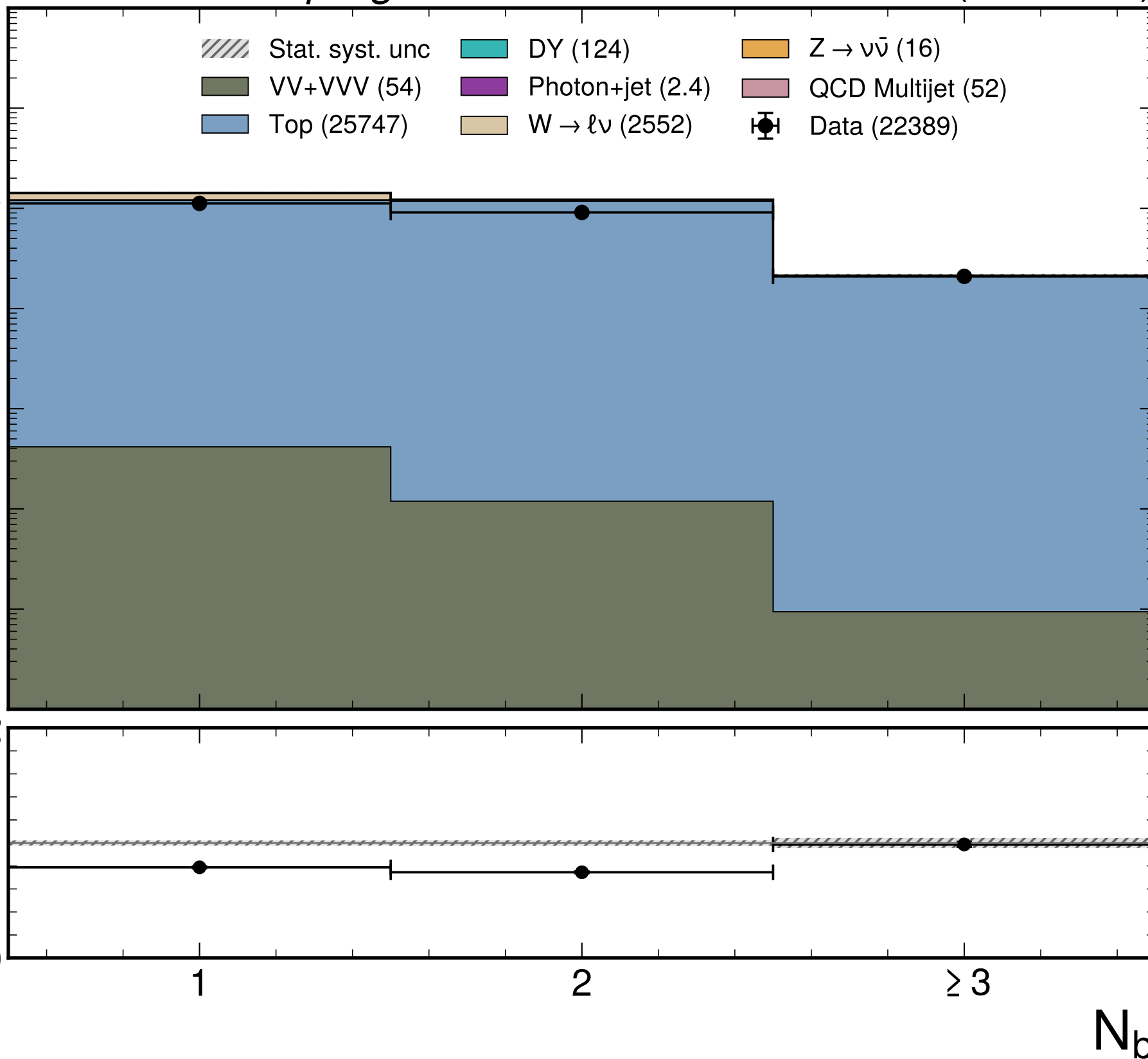
10⁻¹
2
1
0

1

2

≥ 3

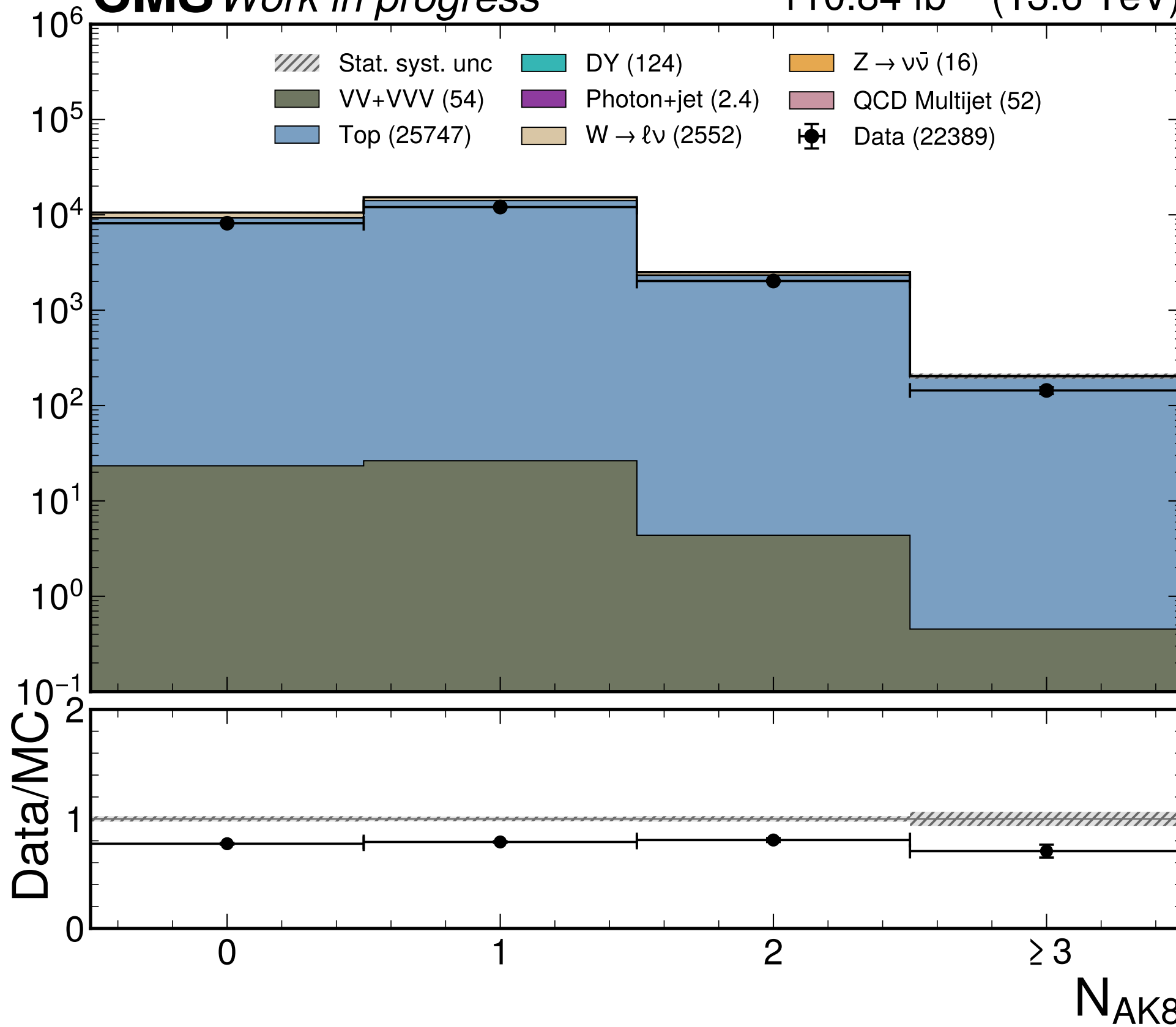
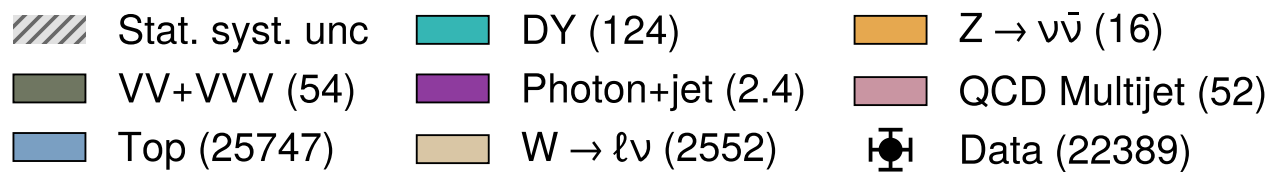
N_b



CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)

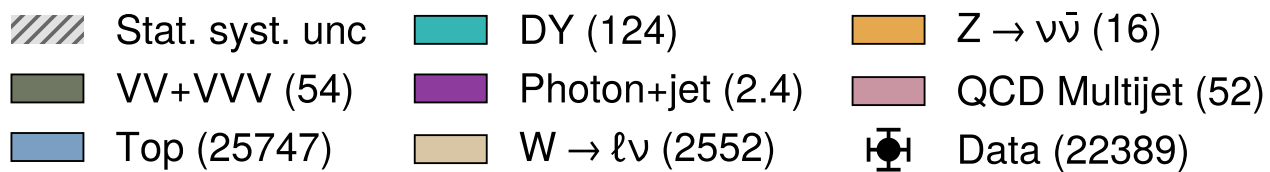
Events



CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)

Events



Data/MC

10⁻¹
2
1
0

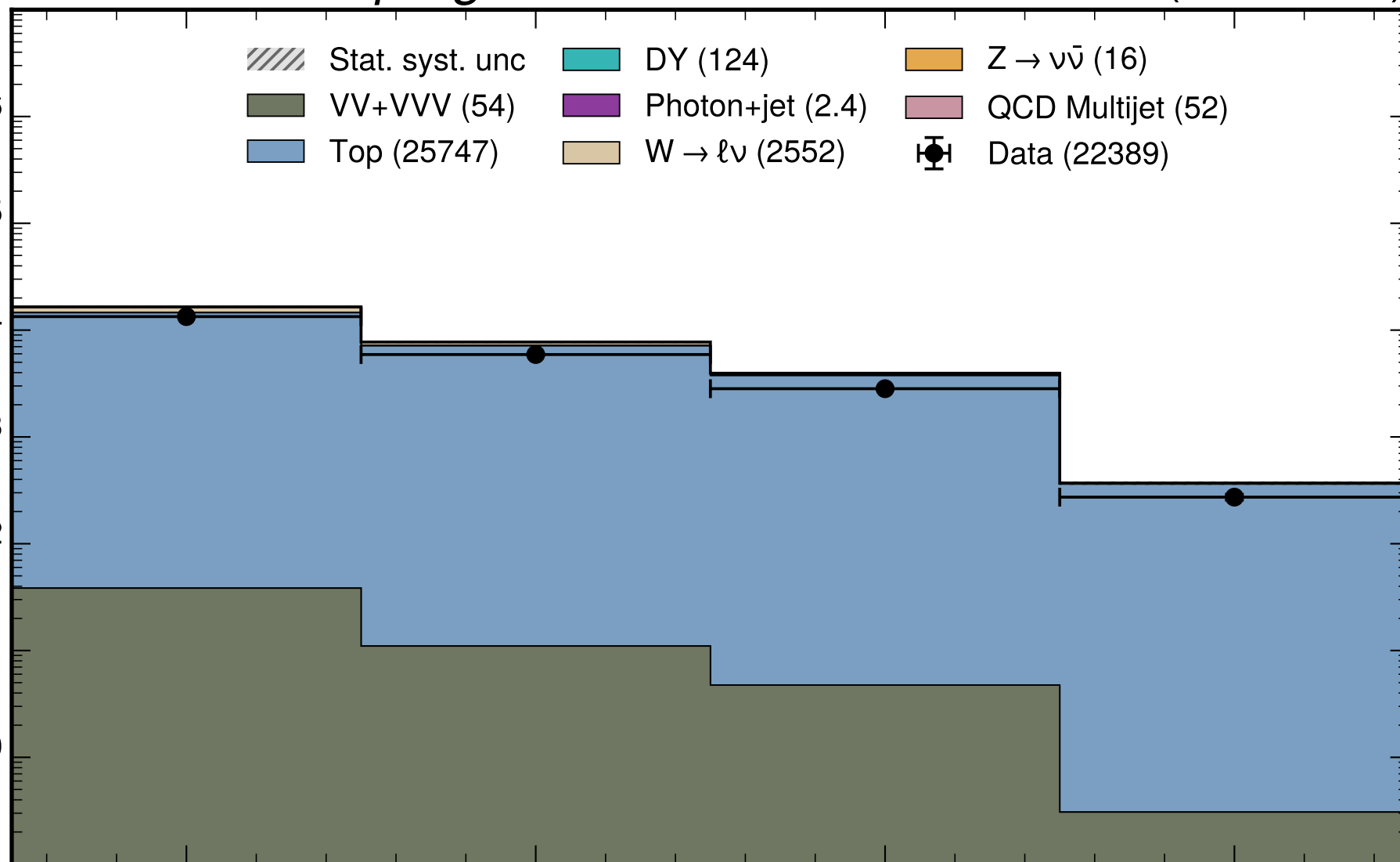
5

6

7-8

≥ 9

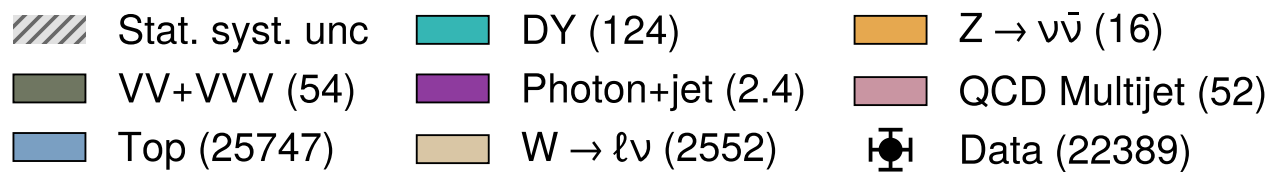
N_j



CMS *Work in progress*

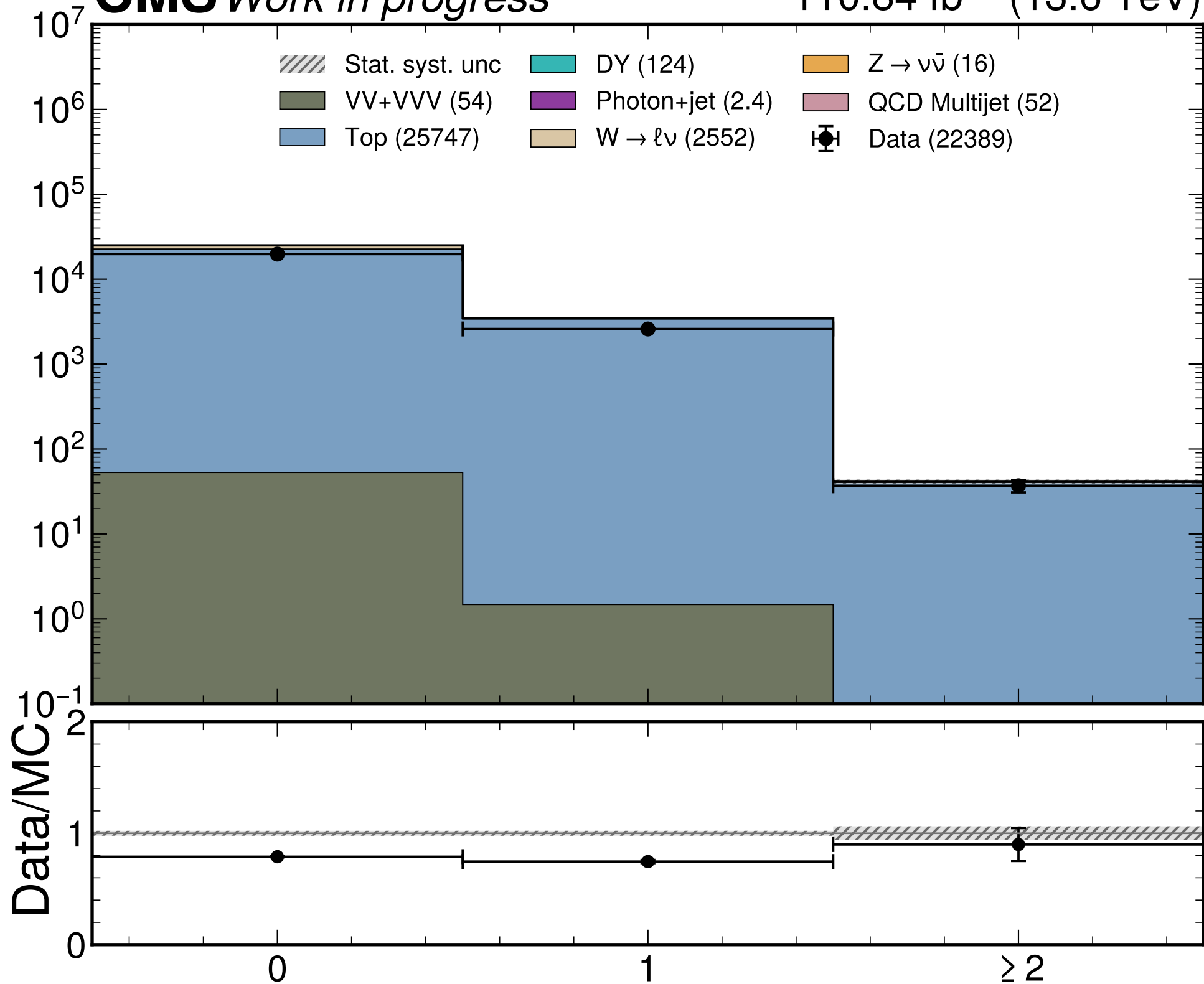
110.84 fb⁻¹ (13.6 TeV)

Events



Data/MC

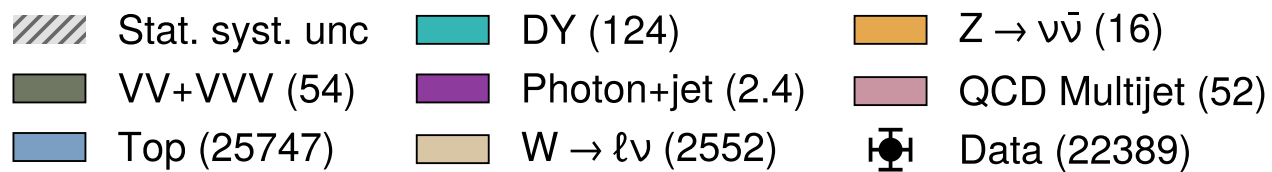
N_t



CMS *Work in progress*

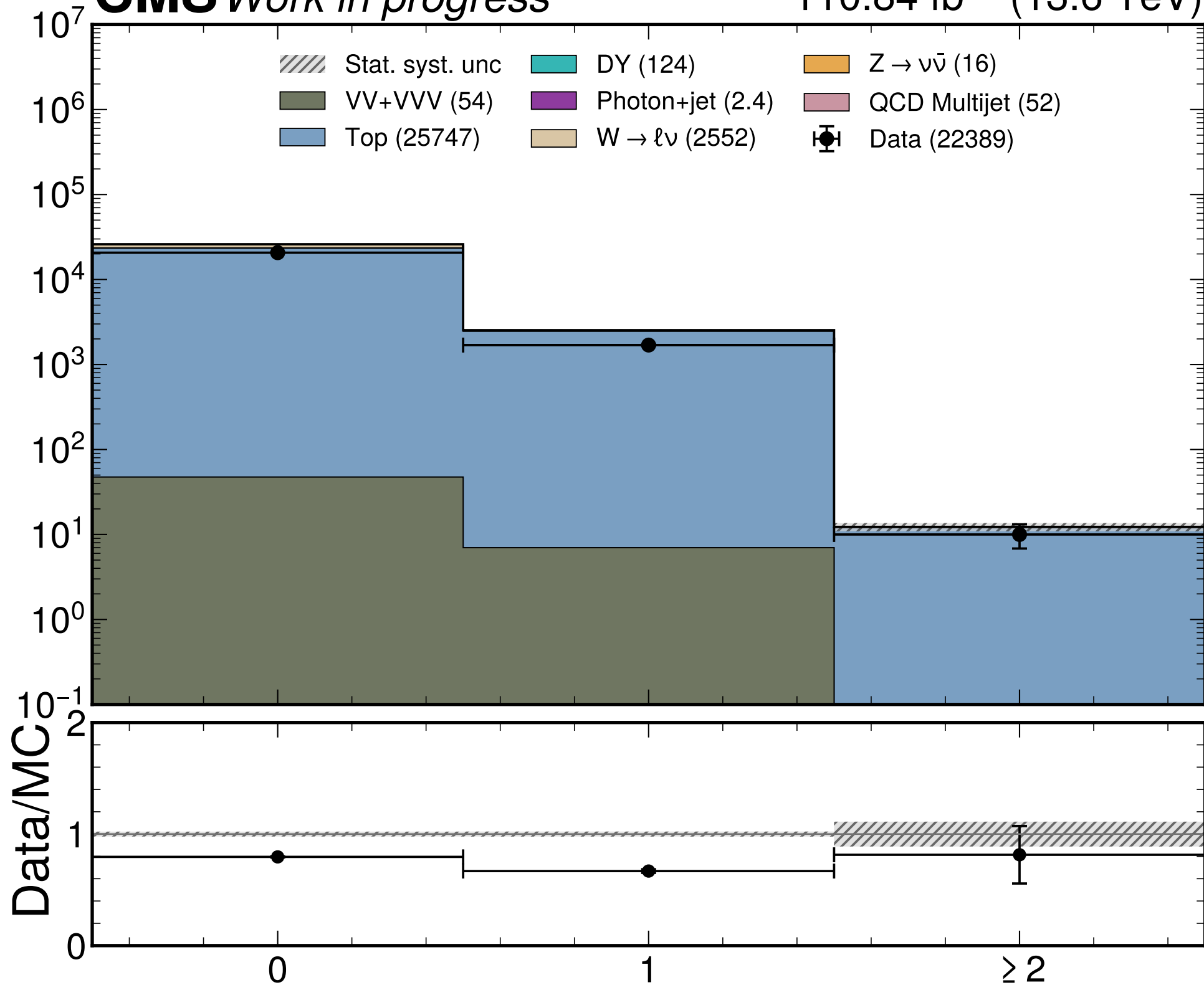
110.84 fb⁻¹ (13.6 TeV)

Events



Data/MC

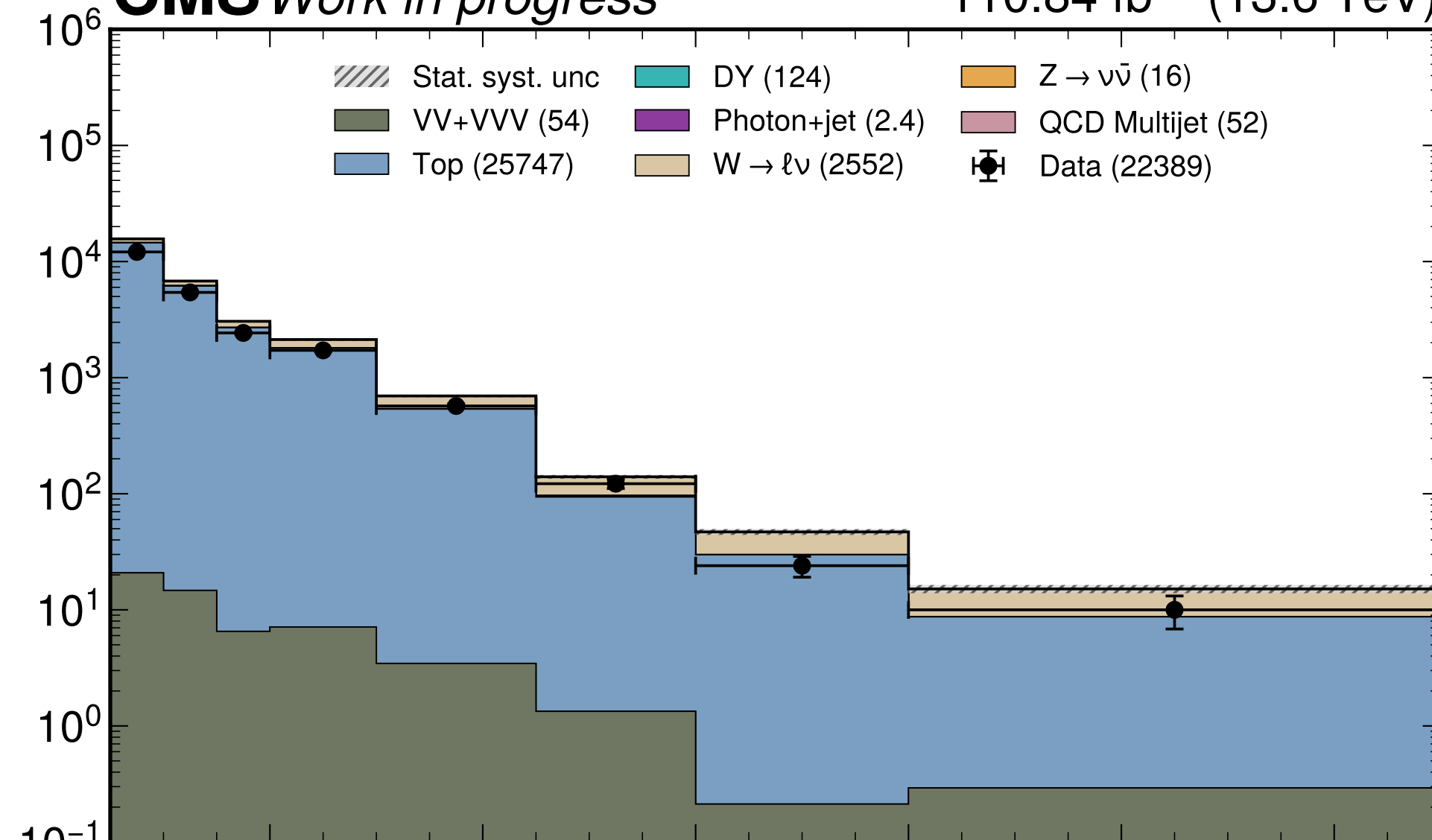
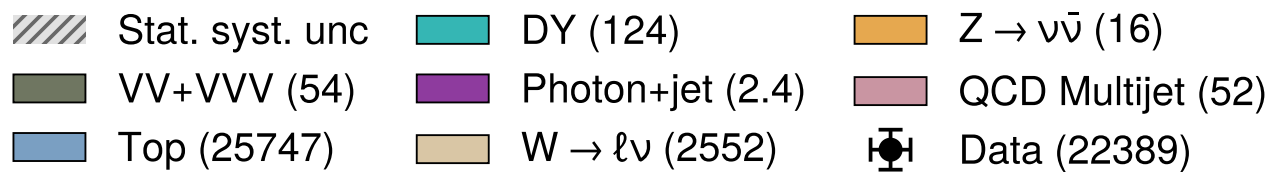
N_W



CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)

Events



Data/MC

400

600

800

1000

1200

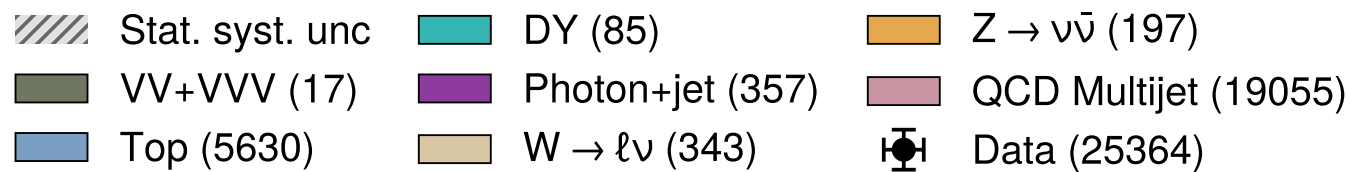
1400

U_T (GeV)

CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)

Events



Data/MC

10⁻¹
2
1
0

200

400

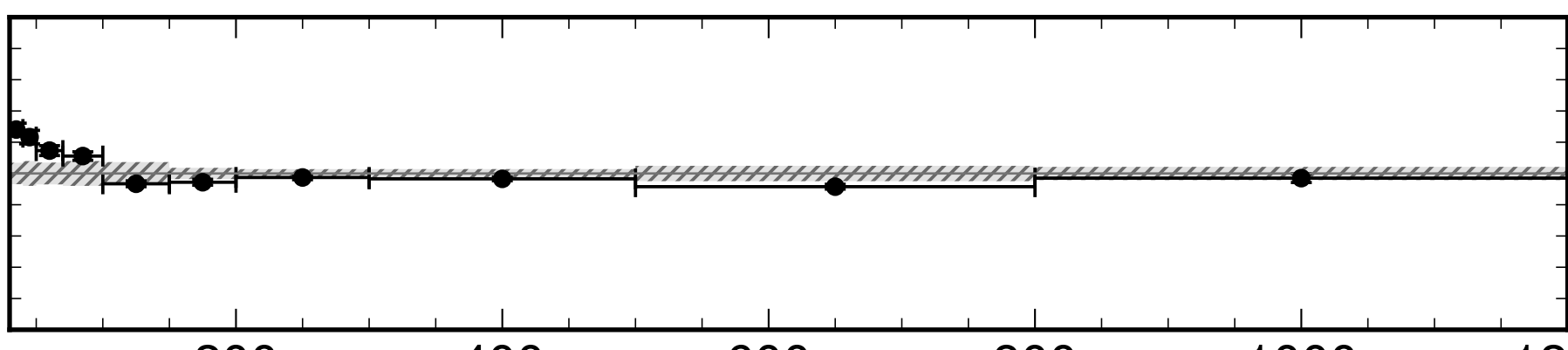
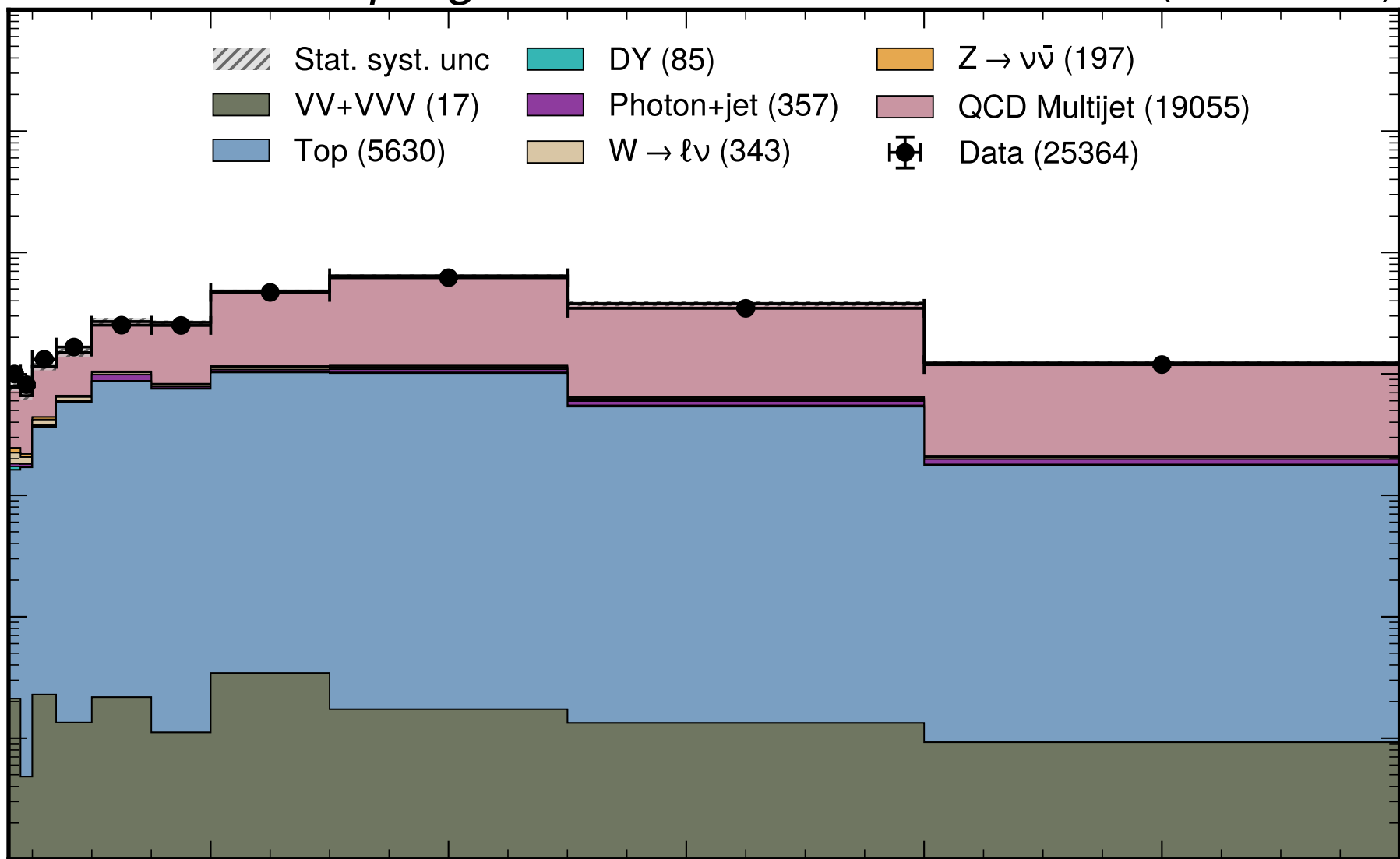
600

800

1000

1200

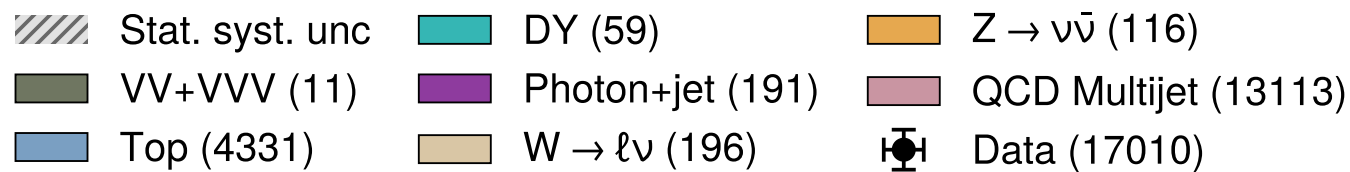
Leading b-jet p_T (GeV)



CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)

Events



Data/MC

10⁻²

200

400

600

800

1000

1200

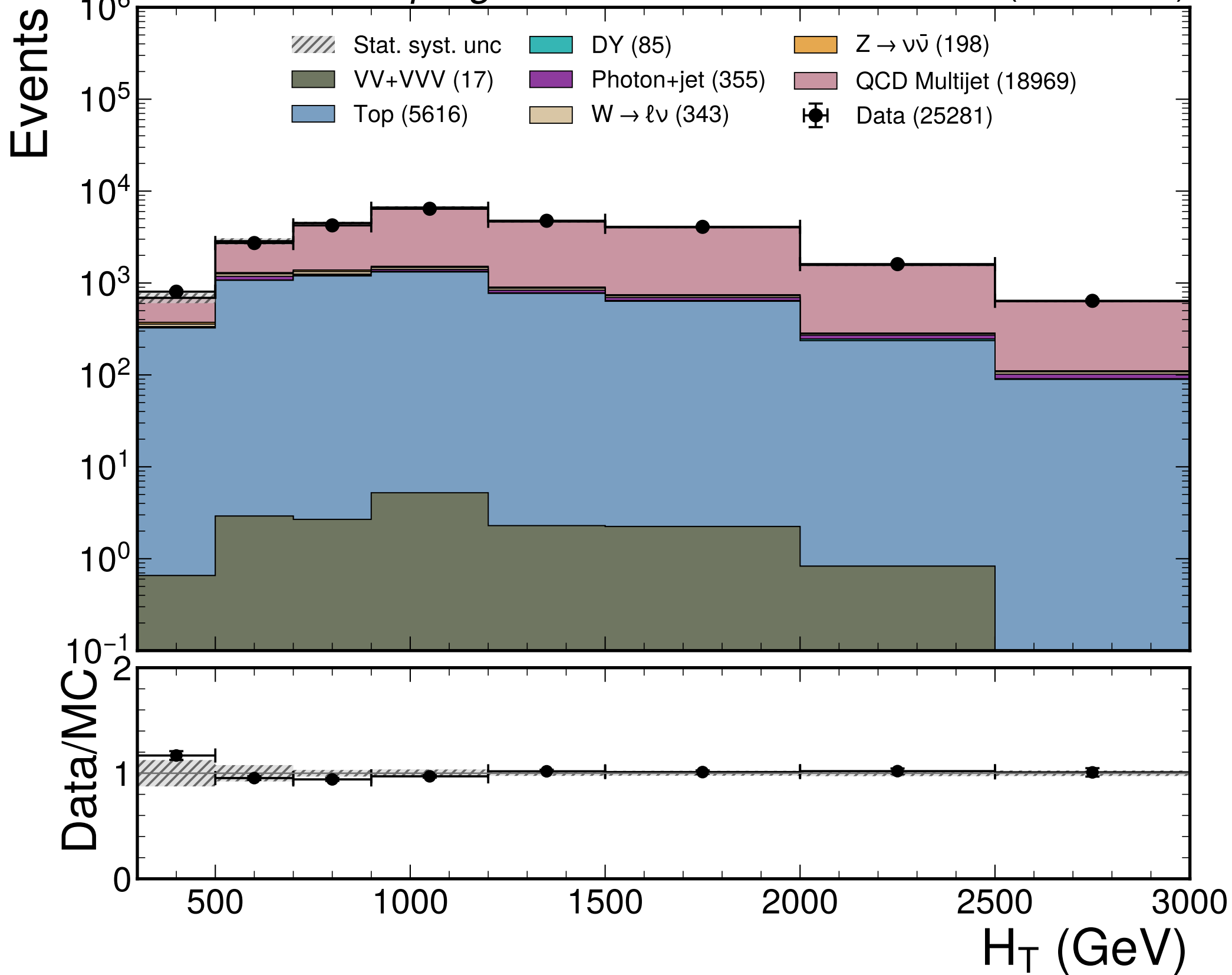
1400

Leading AK8 jet p_T (GeV)

Detailed description: The figure consists of two panels sharing a common x-axis labeled 'Leading AK8 jet p_T (GeV)' ranging from 200 to 1500. The top panel is a stacked histogram showing event counts on a logarithmic y-axis from 10⁻¹ to 10⁶. The background is dominated by 'Top (4331)' in blue. Other visible components include 'QCD Multijet (13113)' in pink, 'DY (59)' in teal, 'Photon+jet (191)' in purple, 'W → ℓν (196)' in tan, 'Z → νν̄ (116)' in orange, and 'VV+VVV (11)' in dark olive green. Data points are shown as black circles with error bars. The bottom panel shows the 'Data/MC' ratio on a linear y-axis from 0 to 2. Data points are black circles with error bars, and a horizontal line at 1.0 represents the expected ratio. A hatched band around 1.0 indicates the statistical and systematic uncertainties.

CMS *Work in progress*

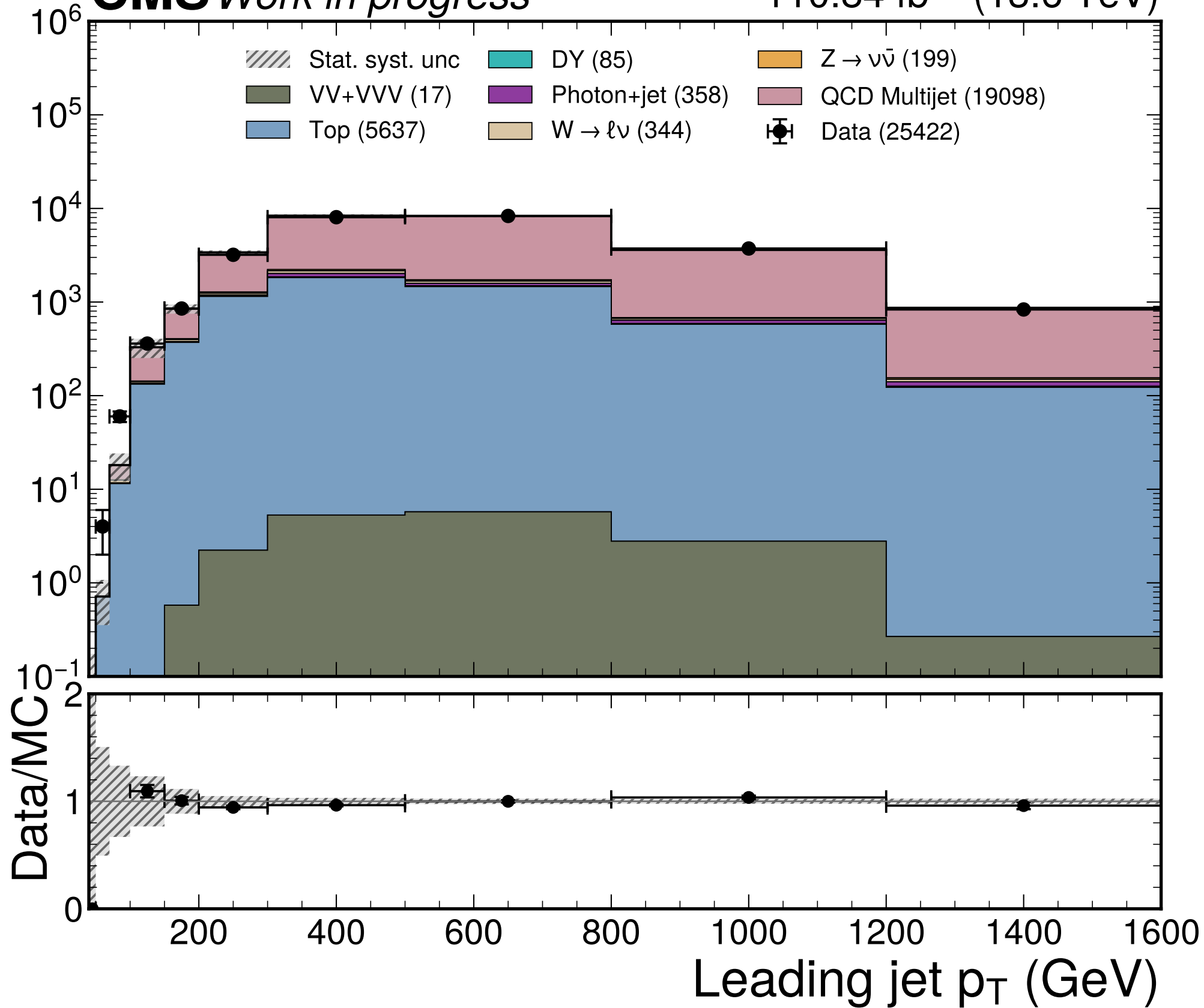
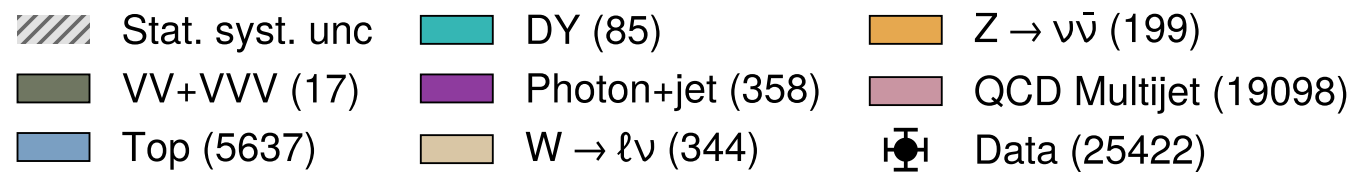
110.84 fb⁻¹ (13.6 TeV)



CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)

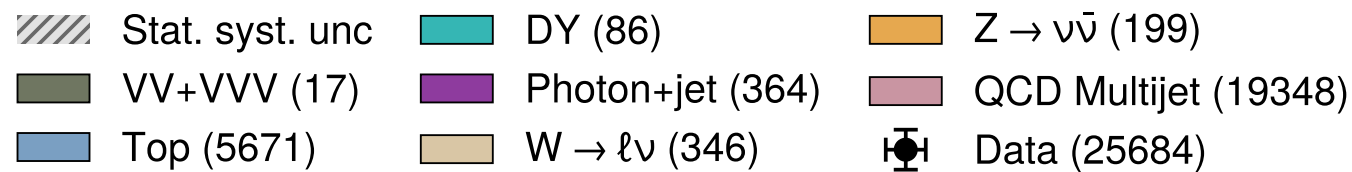
Events



CMS *Work in progress*

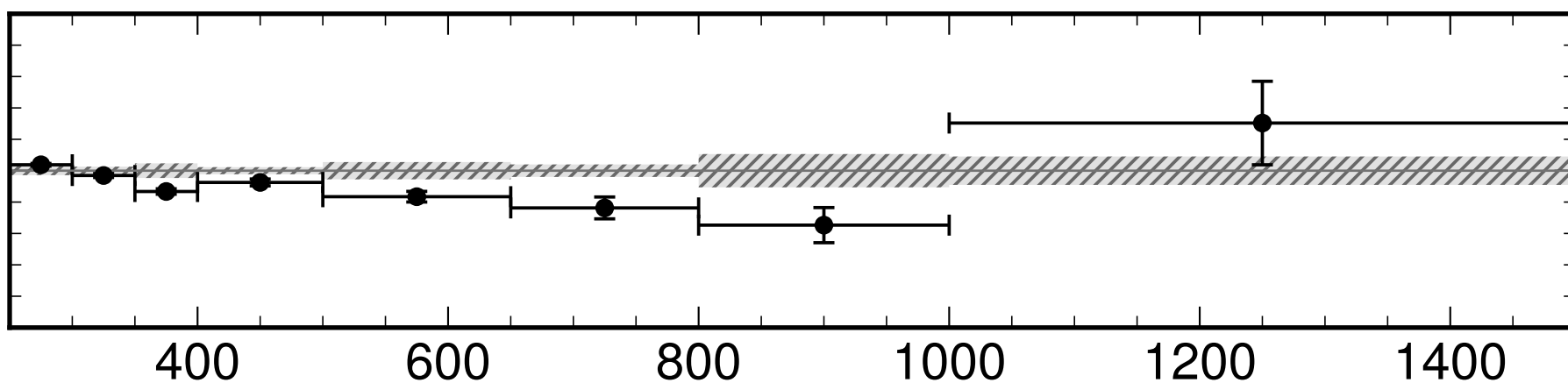
110.84 fb⁻¹ (13.6 TeV)

Events



Data/MC

10⁻¹
2

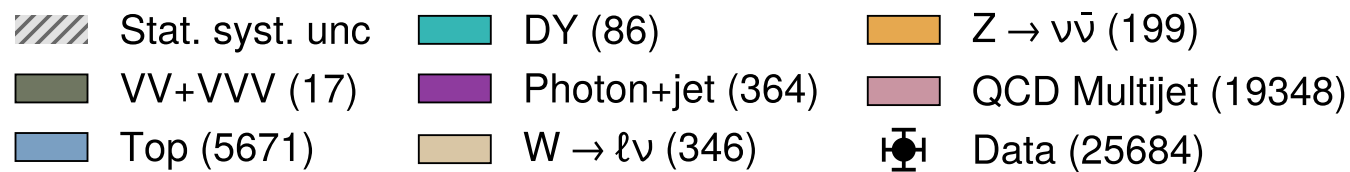


p_T^{miss} (GeV)

CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)

Events



Data/MC

1

2

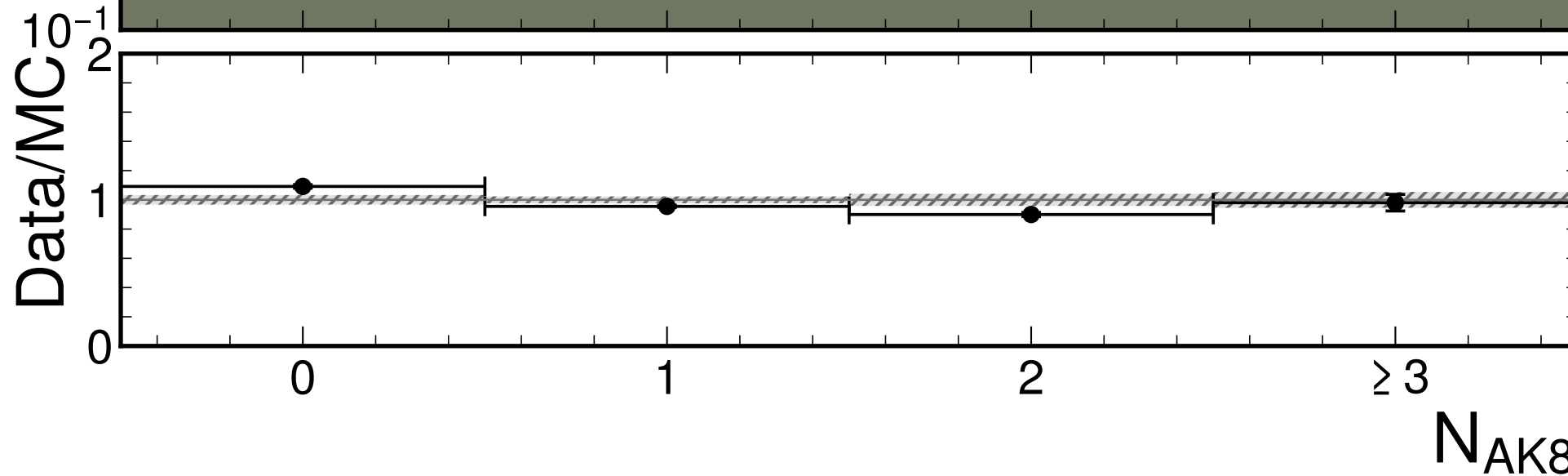
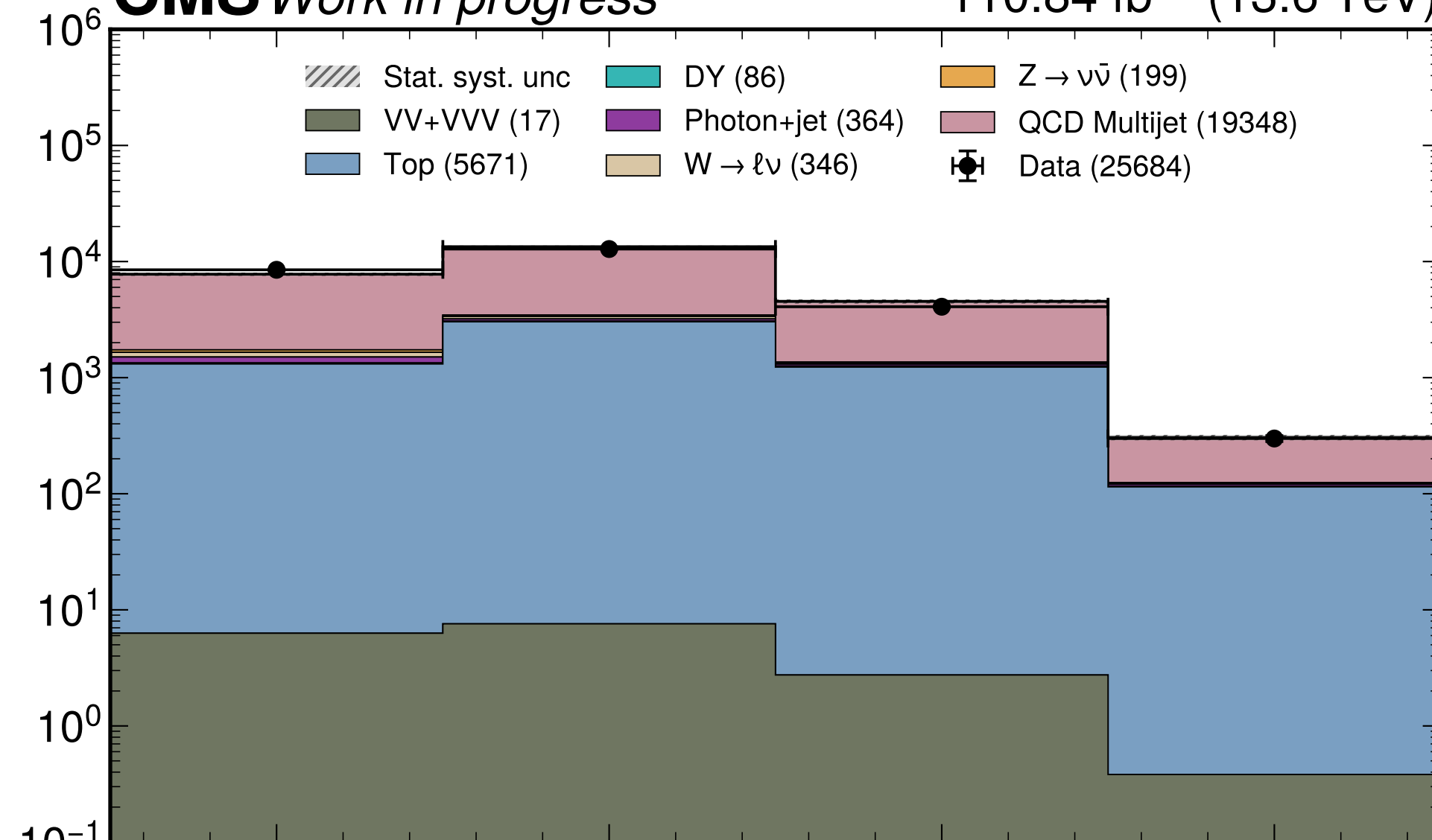
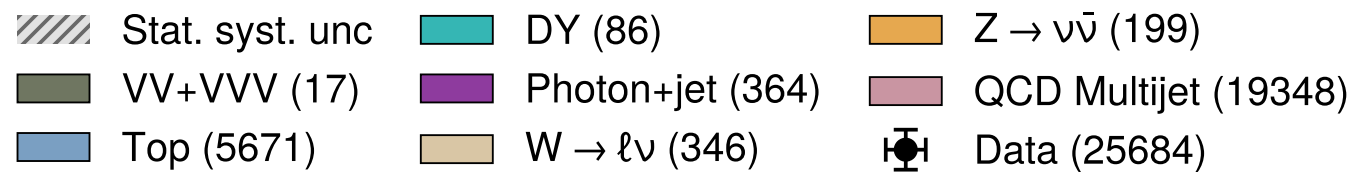
≥ 3

N_b

CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)

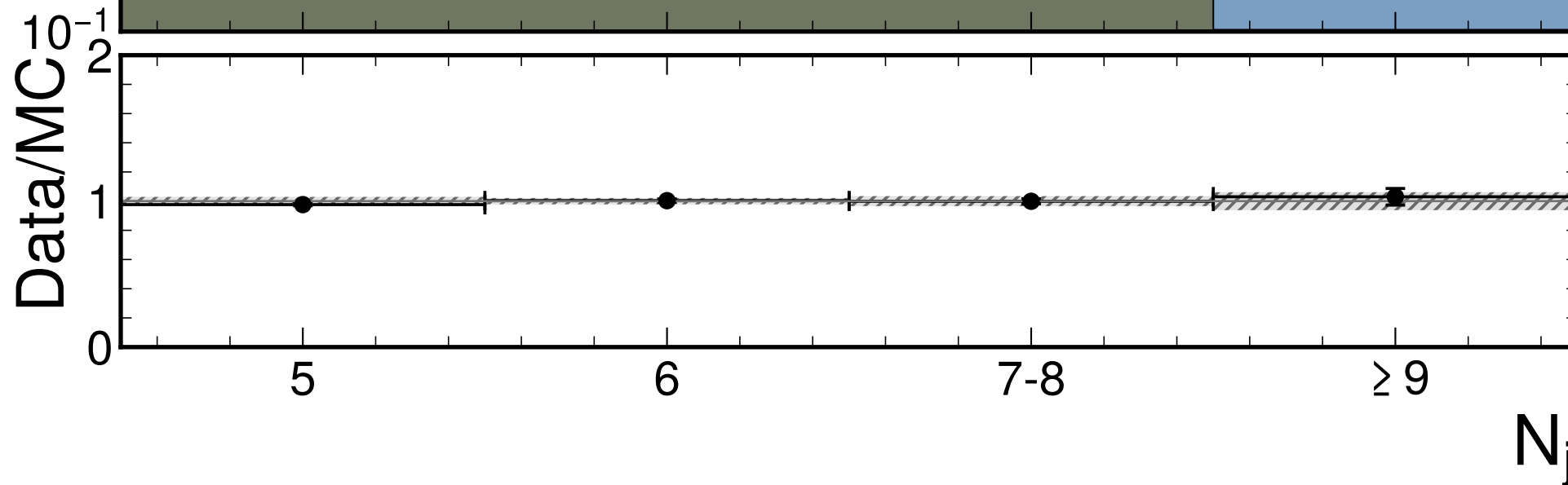
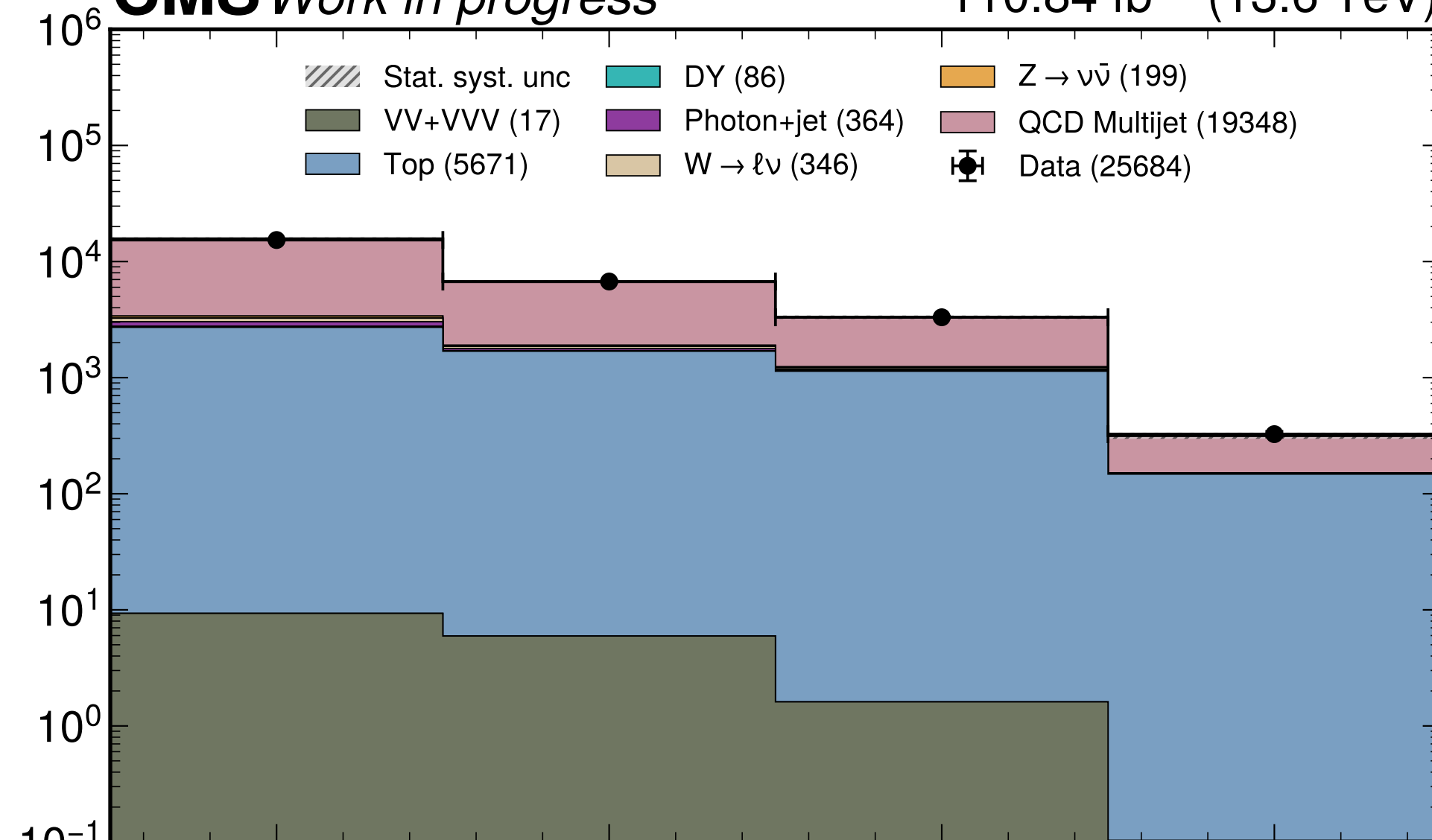
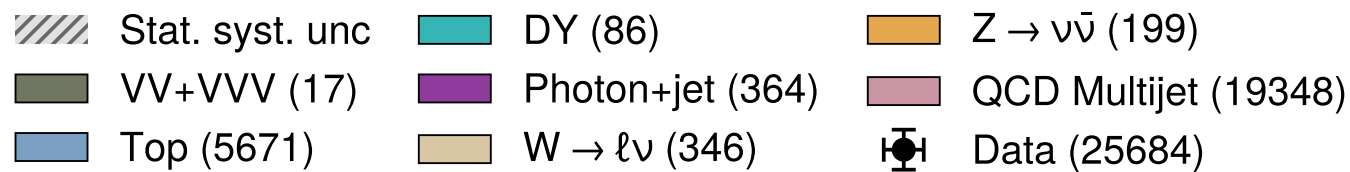
Events



CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)

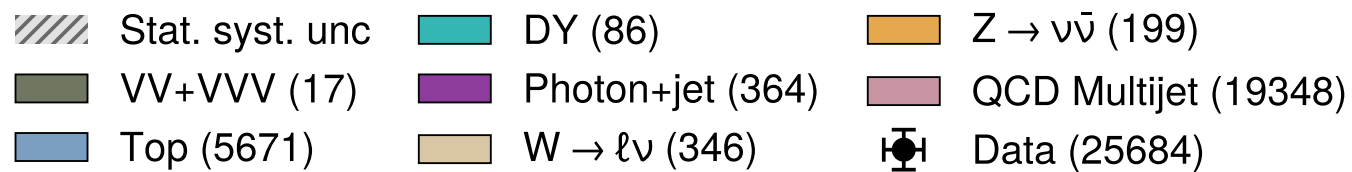
Events



CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)

Events



Data/MC

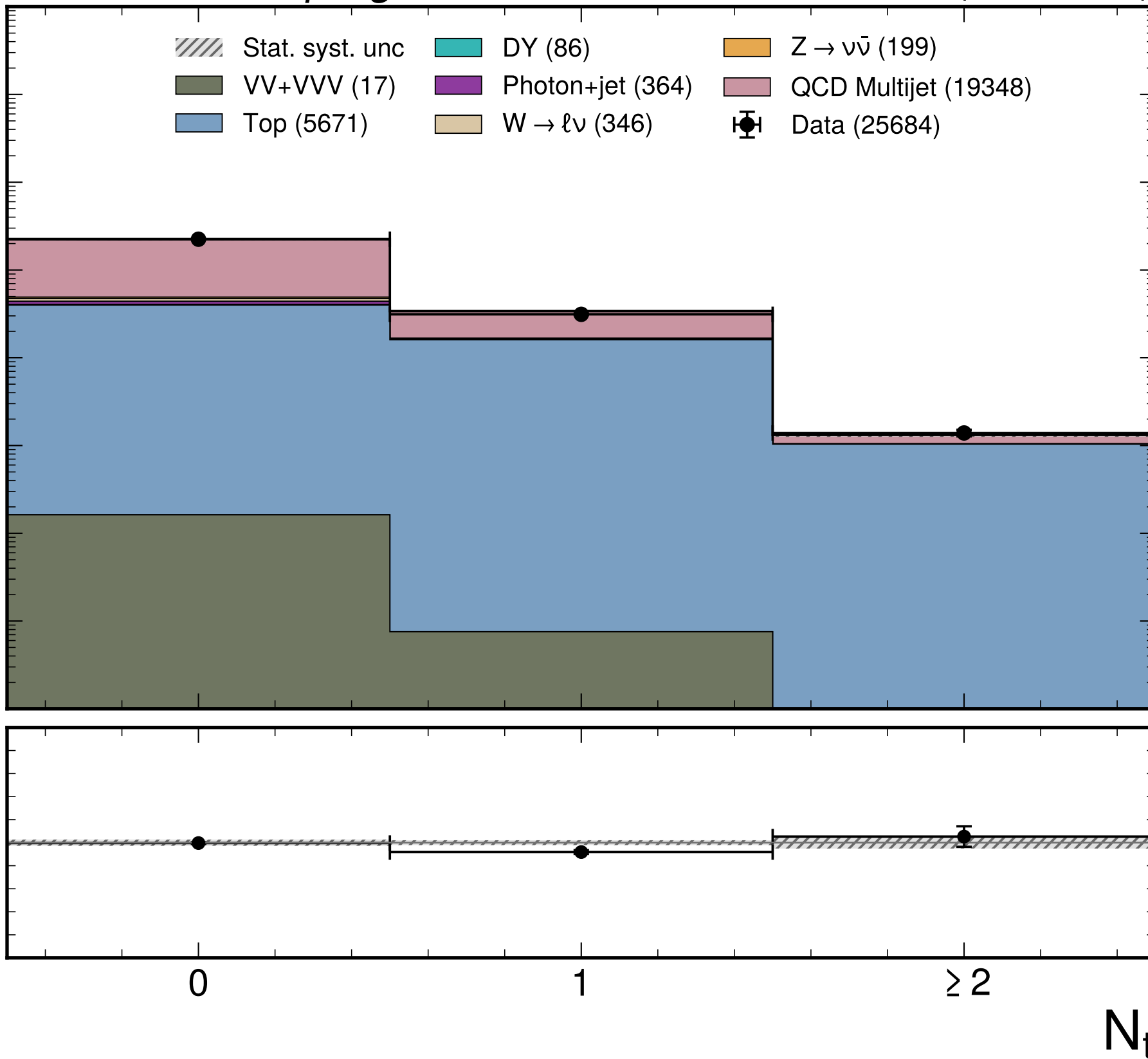
10⁻¹
2
1
0

0

1

≥ 2

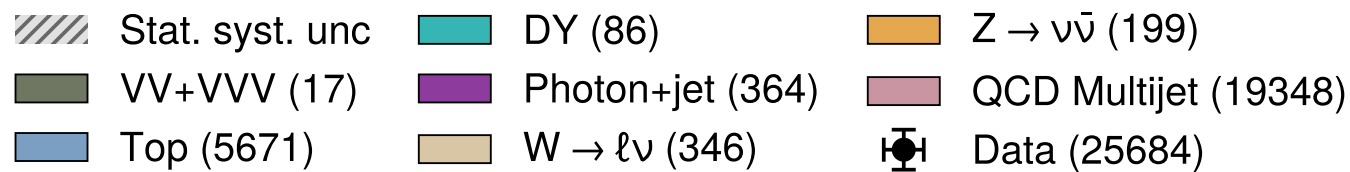
N_t



CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)

Events



Data/MC

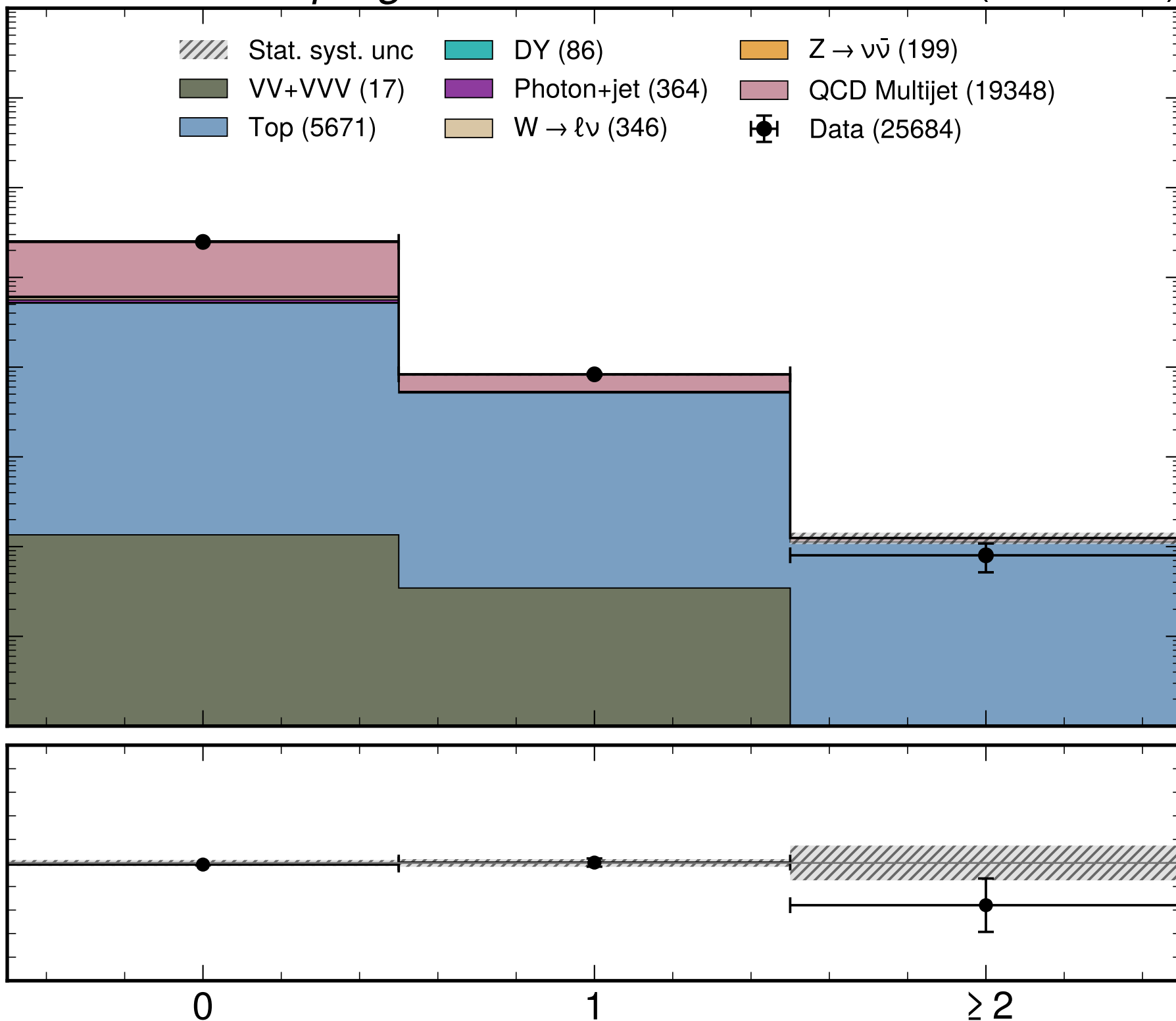
10⁻¹
2
1
0

0

1

≥ 2

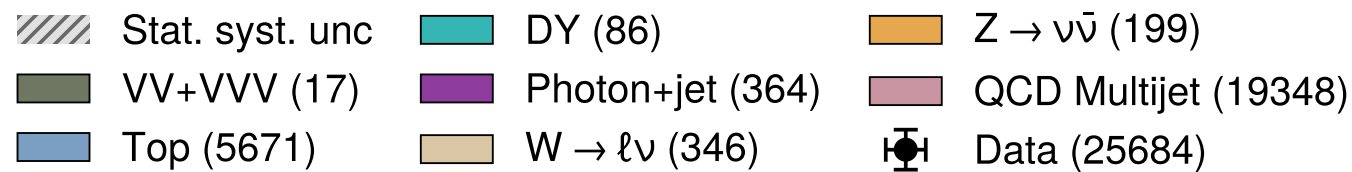
N_W



CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)

Events



Data/MC

10⁻¹
2
1
0

400

600

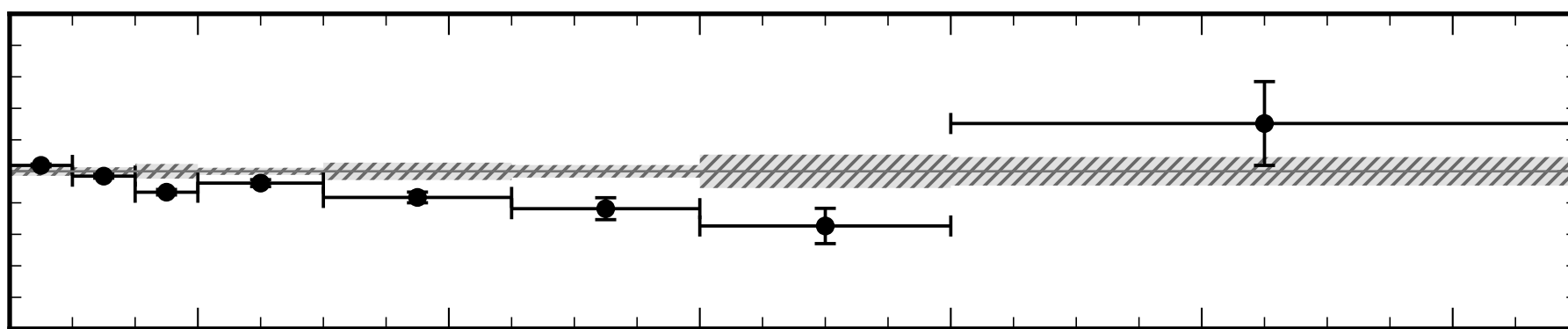
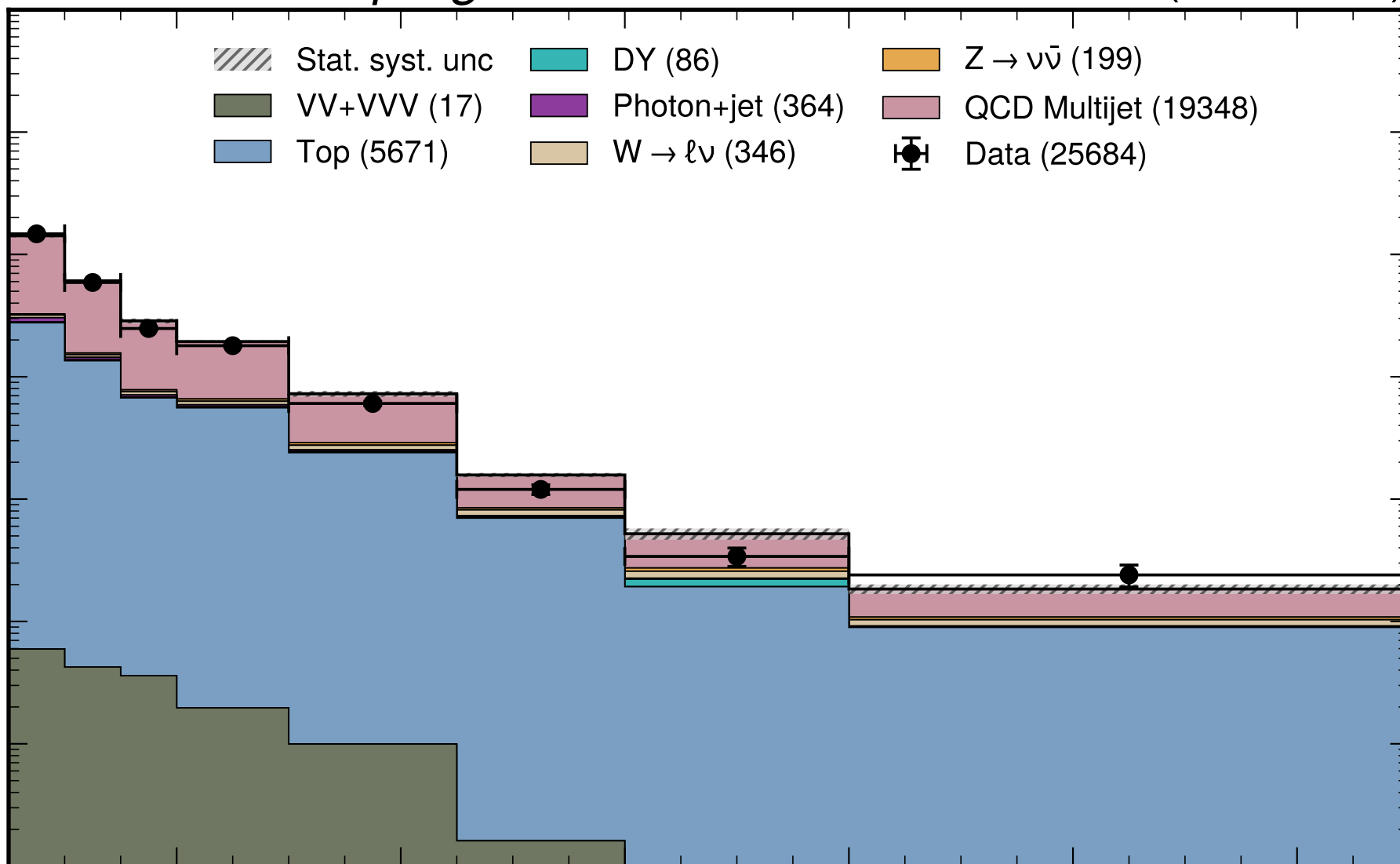
800

1000

1200

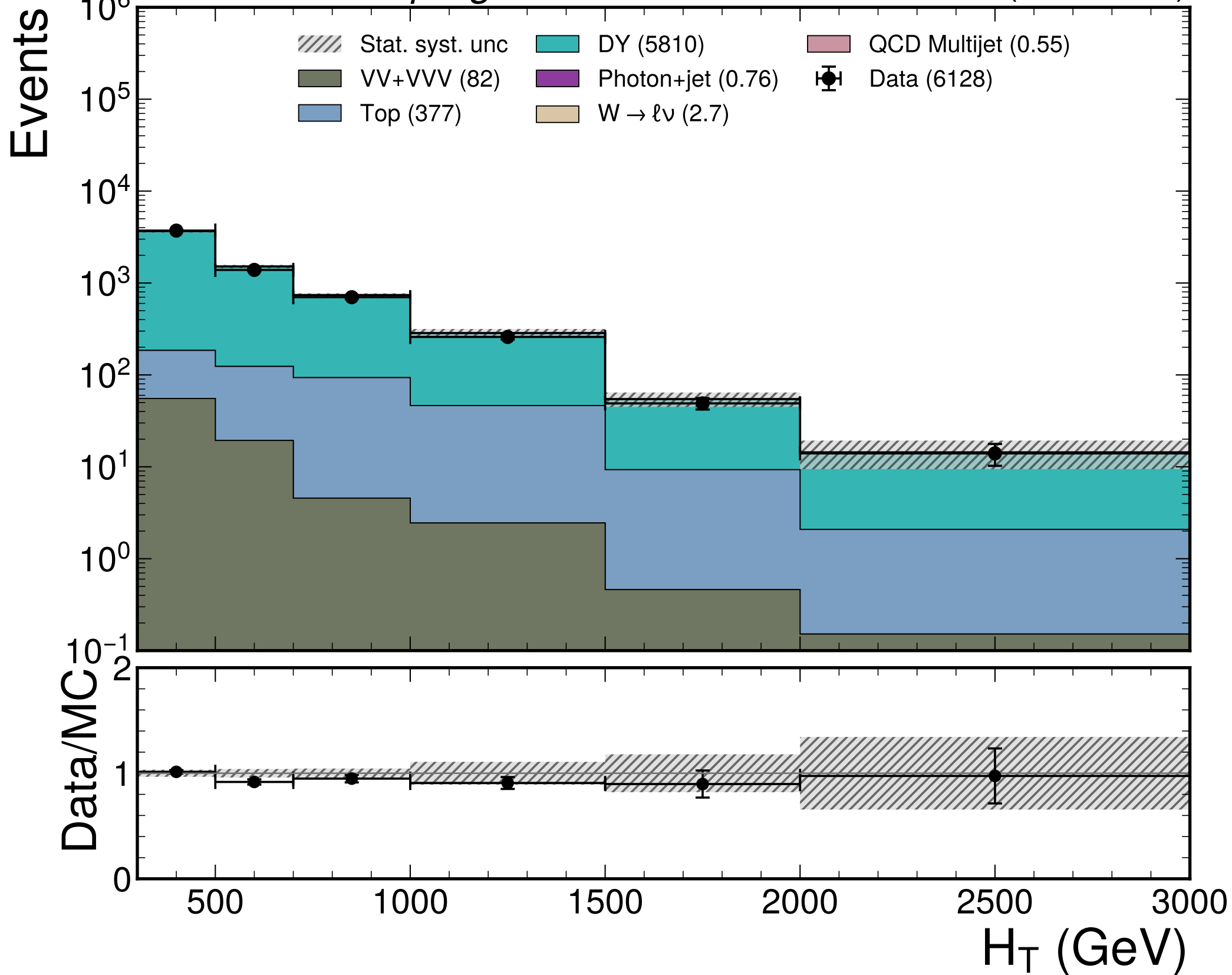
1400

U_T (GeV)



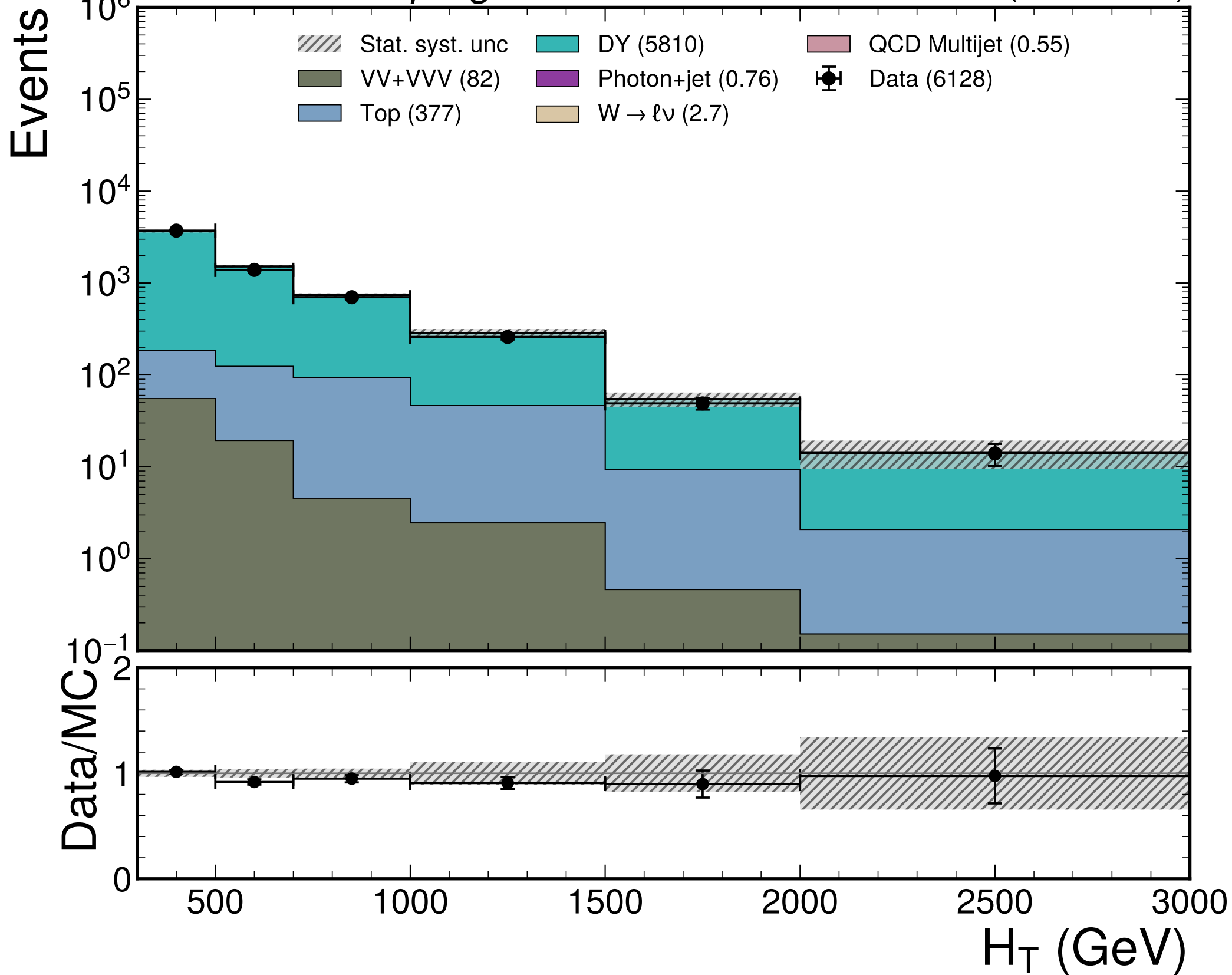
CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)



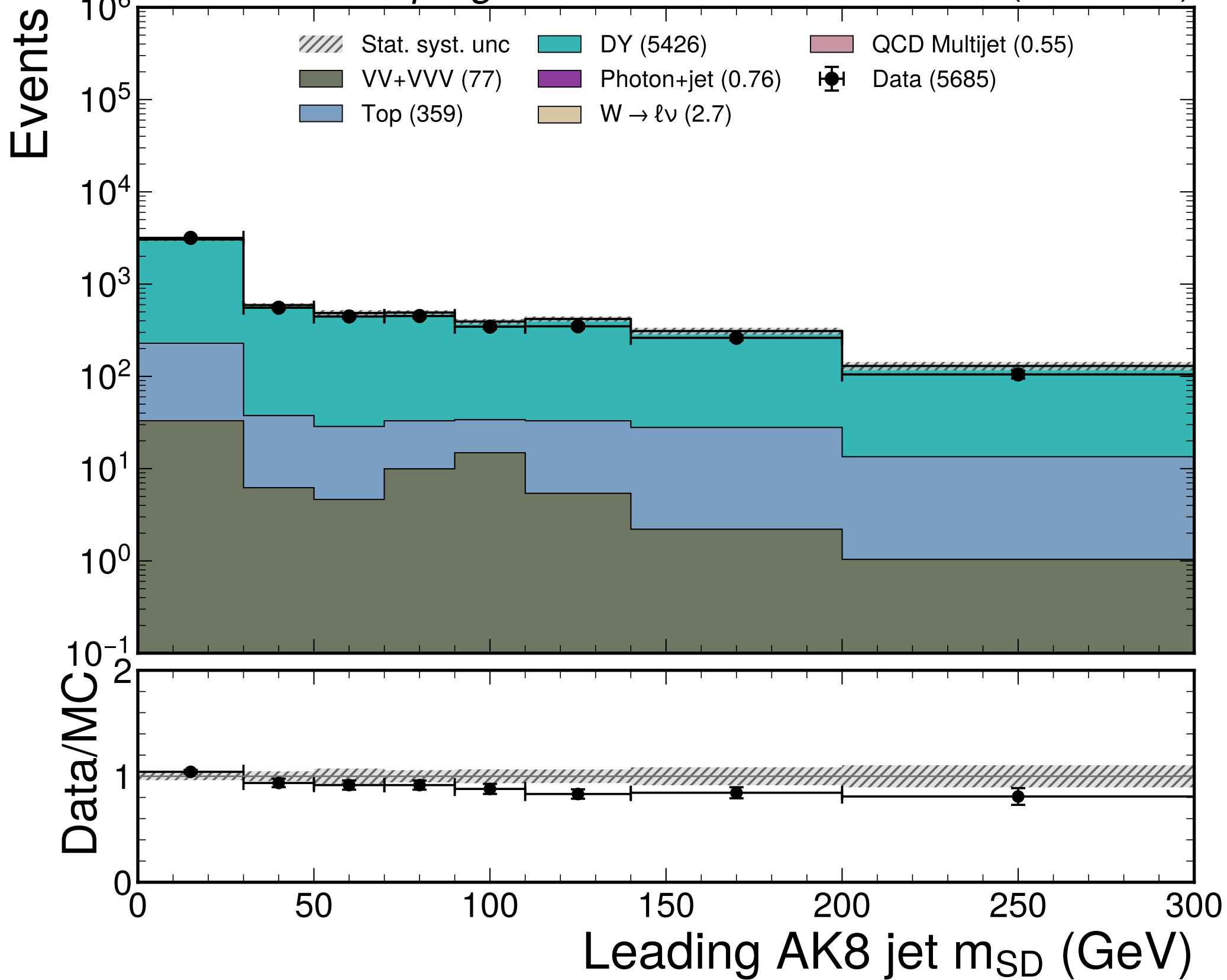
CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)



CMS *Work in progress*

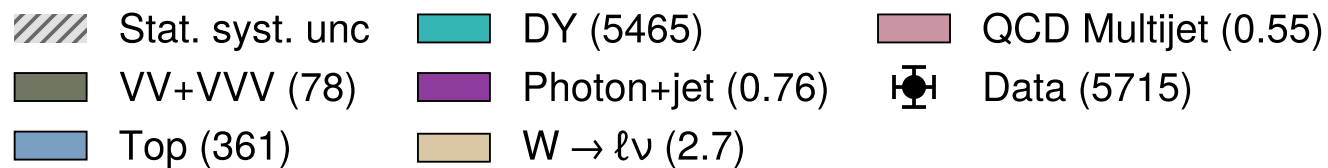
110.84 fb⁻¹ (13.6 TeV)



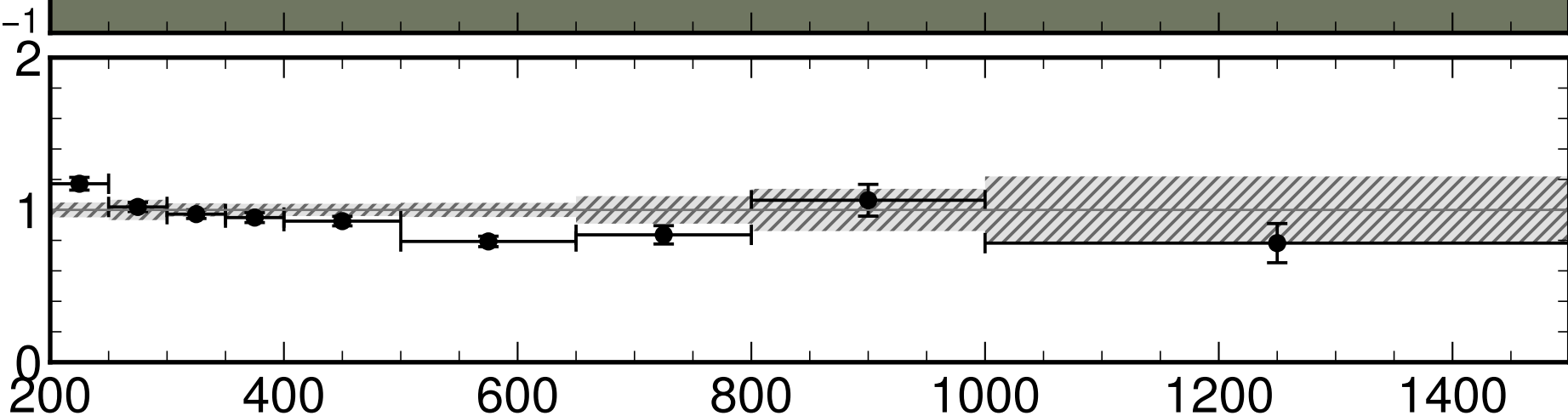
CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)

Events



Data/MC

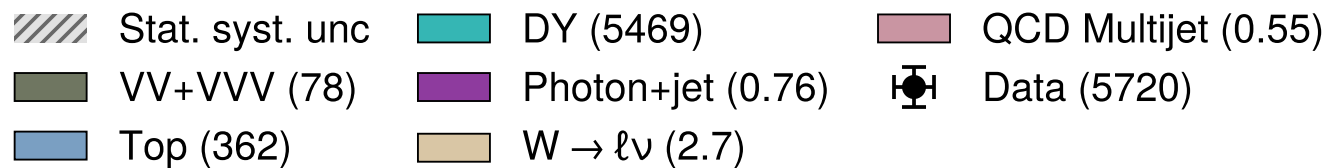


Leading AK8 jet p_T (GeV)

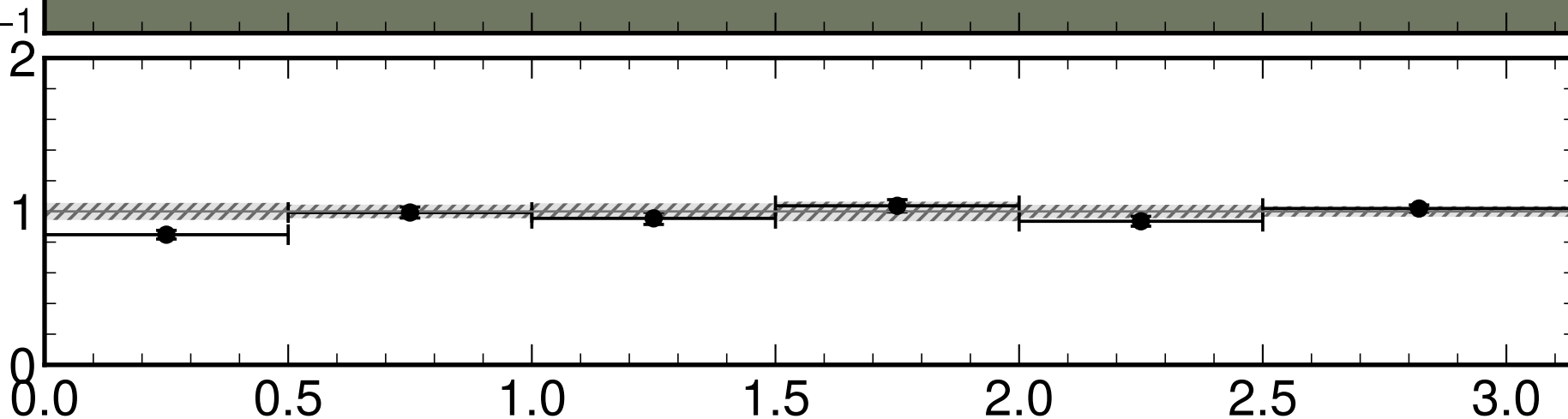
CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)

Events



Data/MC

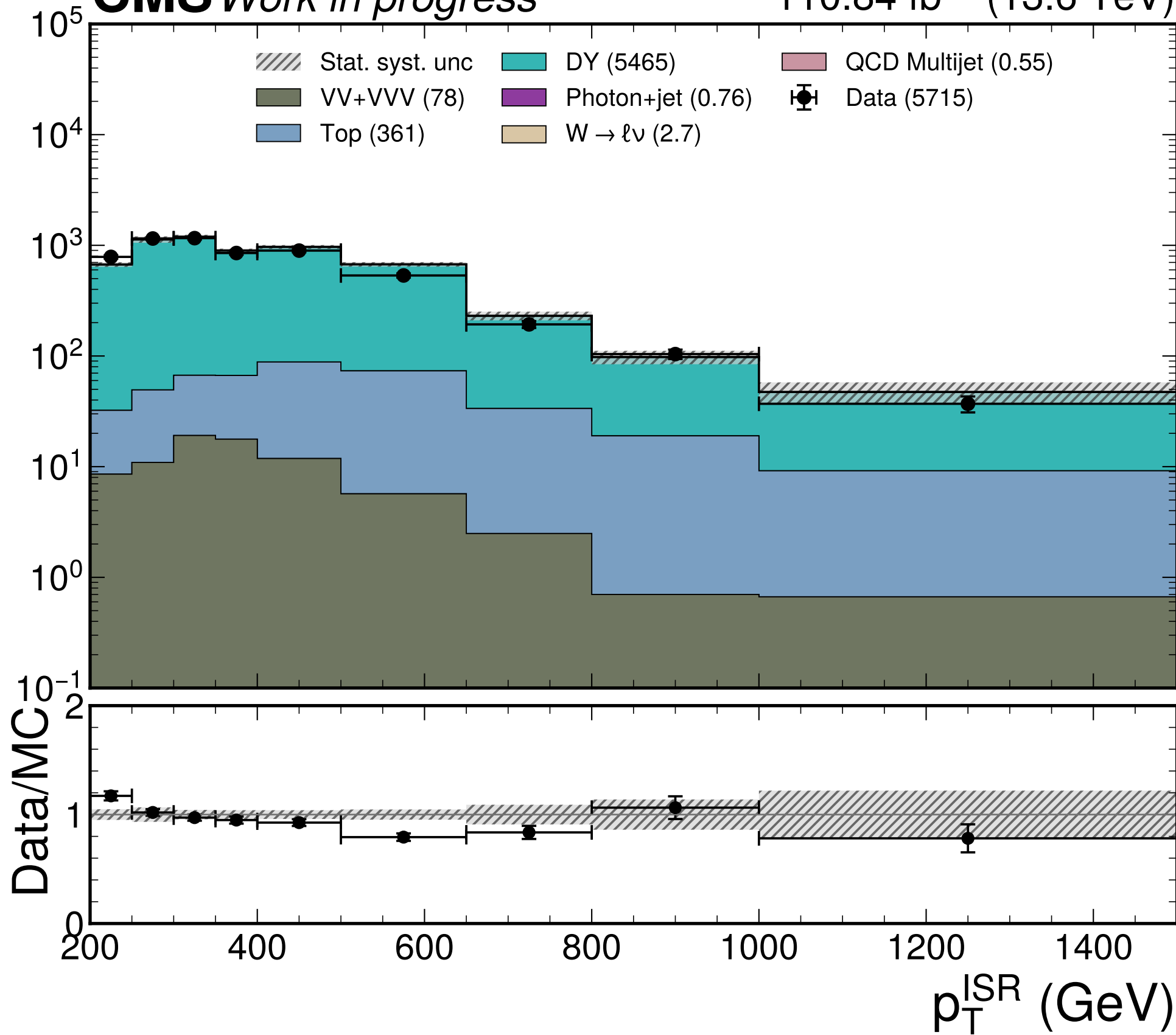
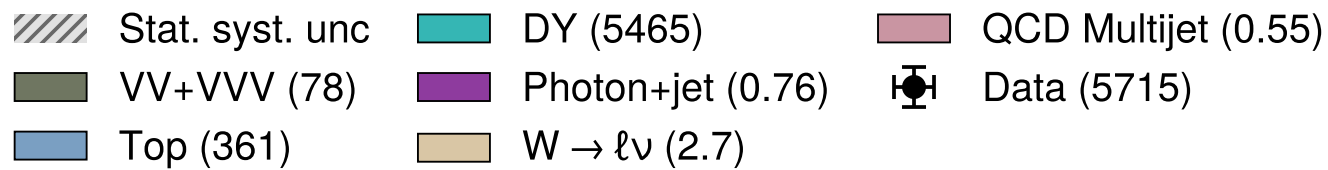


$\Delta\phi(\text{ISR}, p_{\text{T}}^{\text{miss}})$

CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)

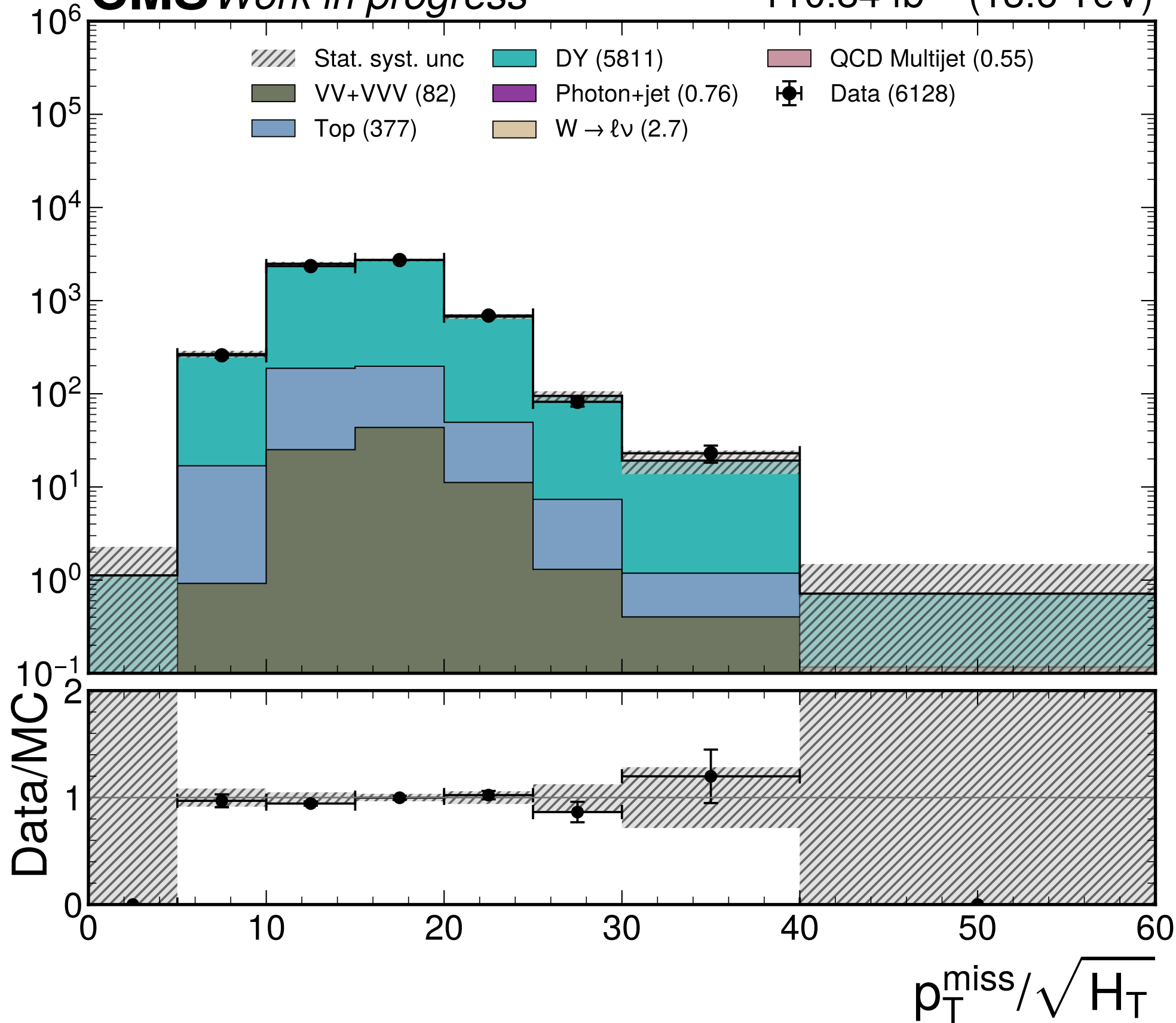
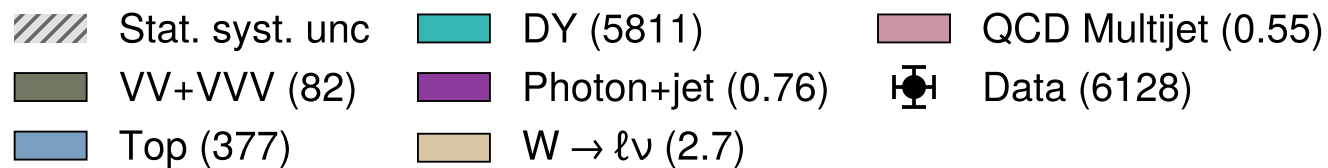
Events



CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)

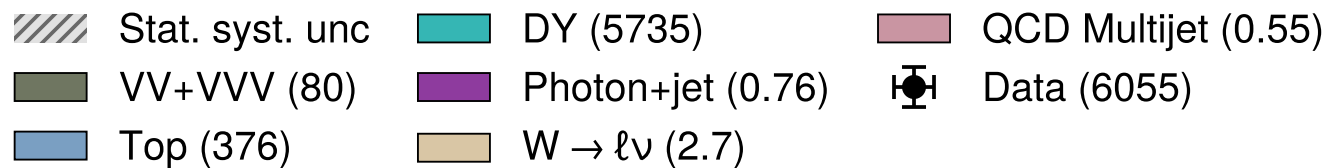
Events



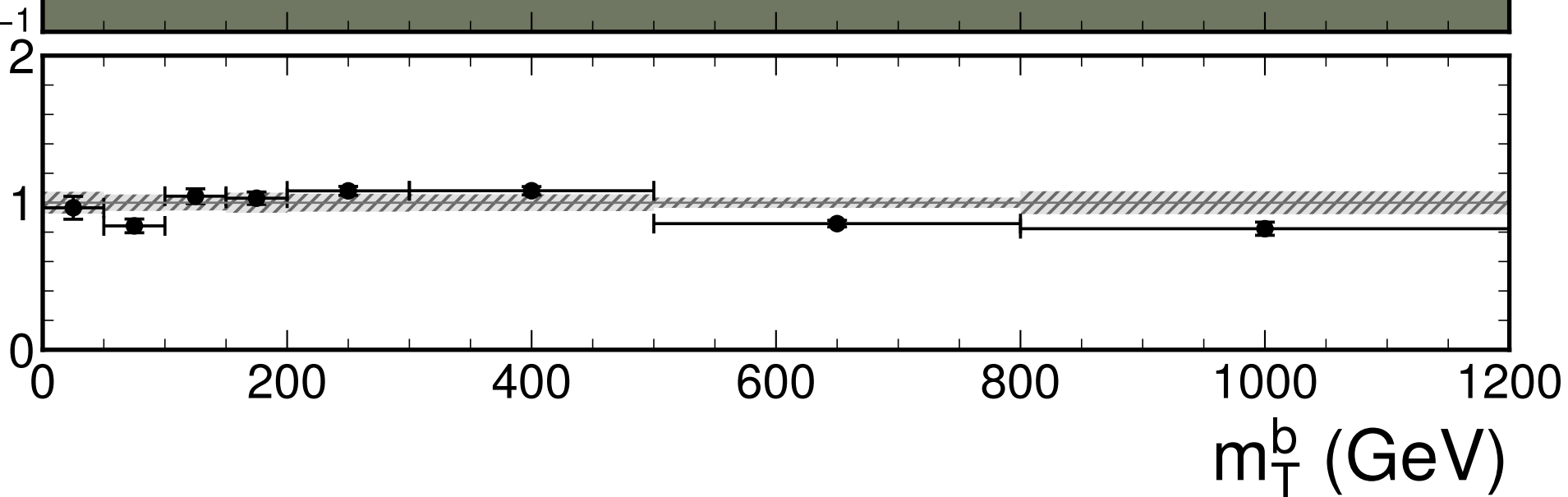
CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)

Events



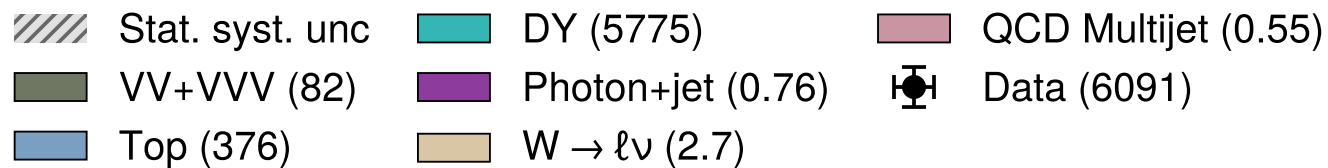
Data/MC



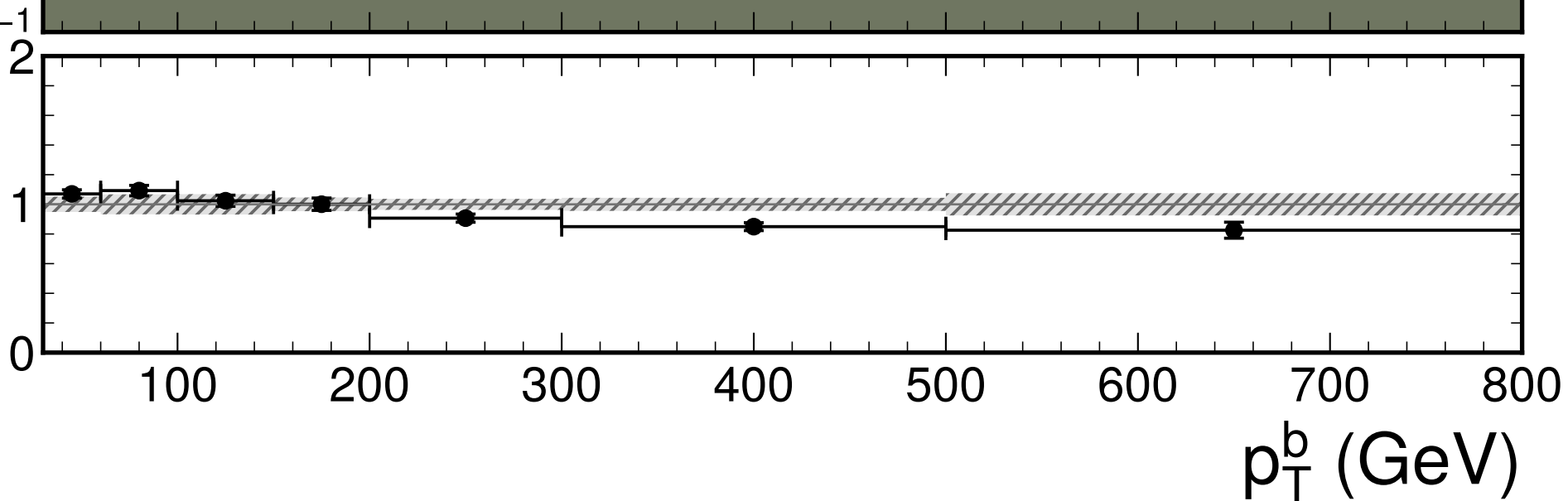
CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)

Events



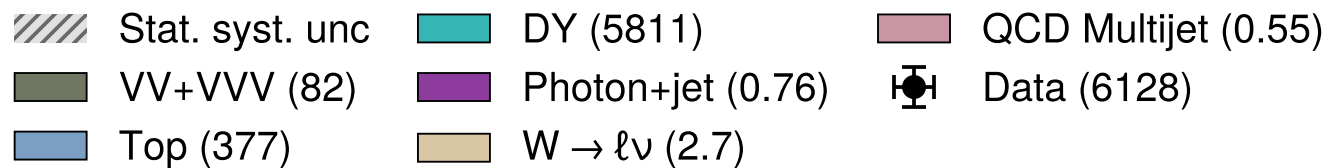
Data/MC



CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)

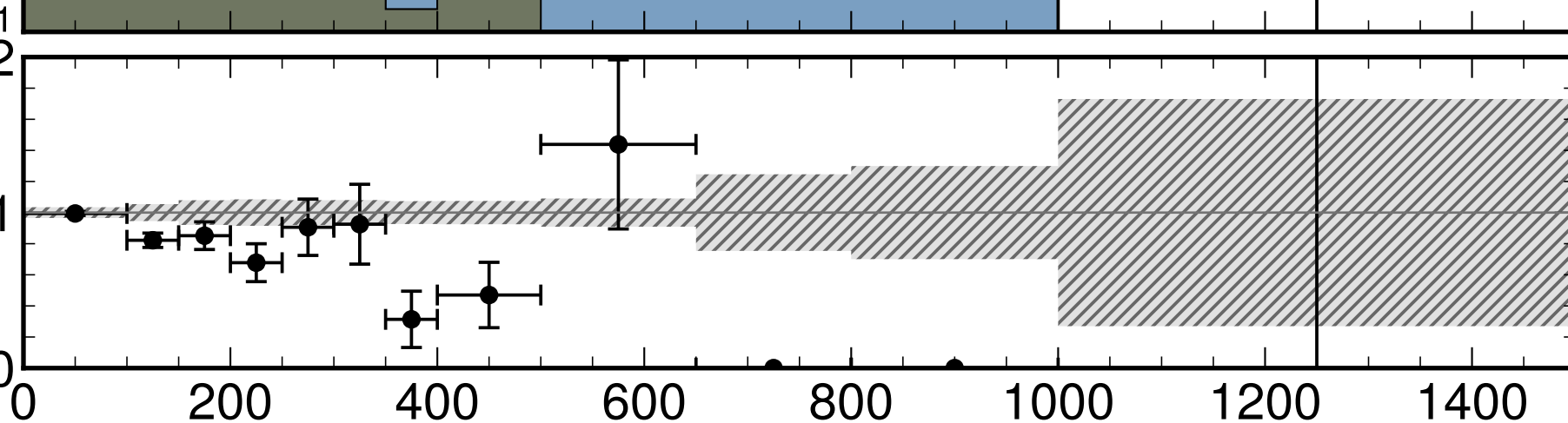
Events



Data/MC

10⁻²

0

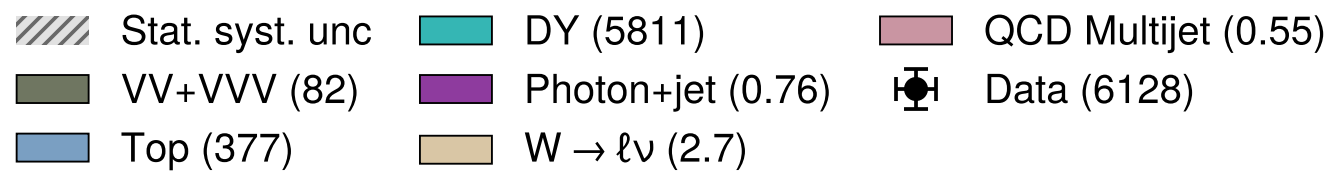


p_T^{miss} (GeV)

CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)

Events



Data/MC

10⁻¹

2

70

80

90

100

110

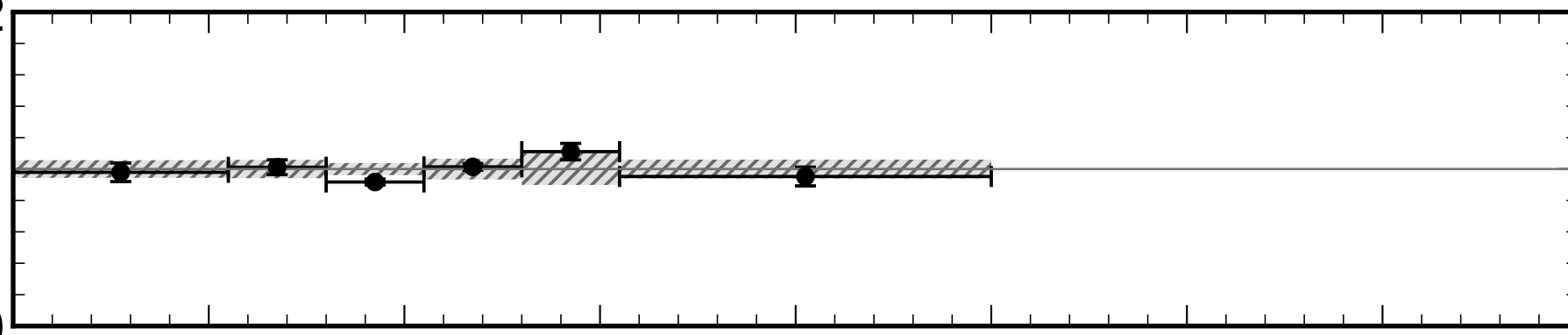
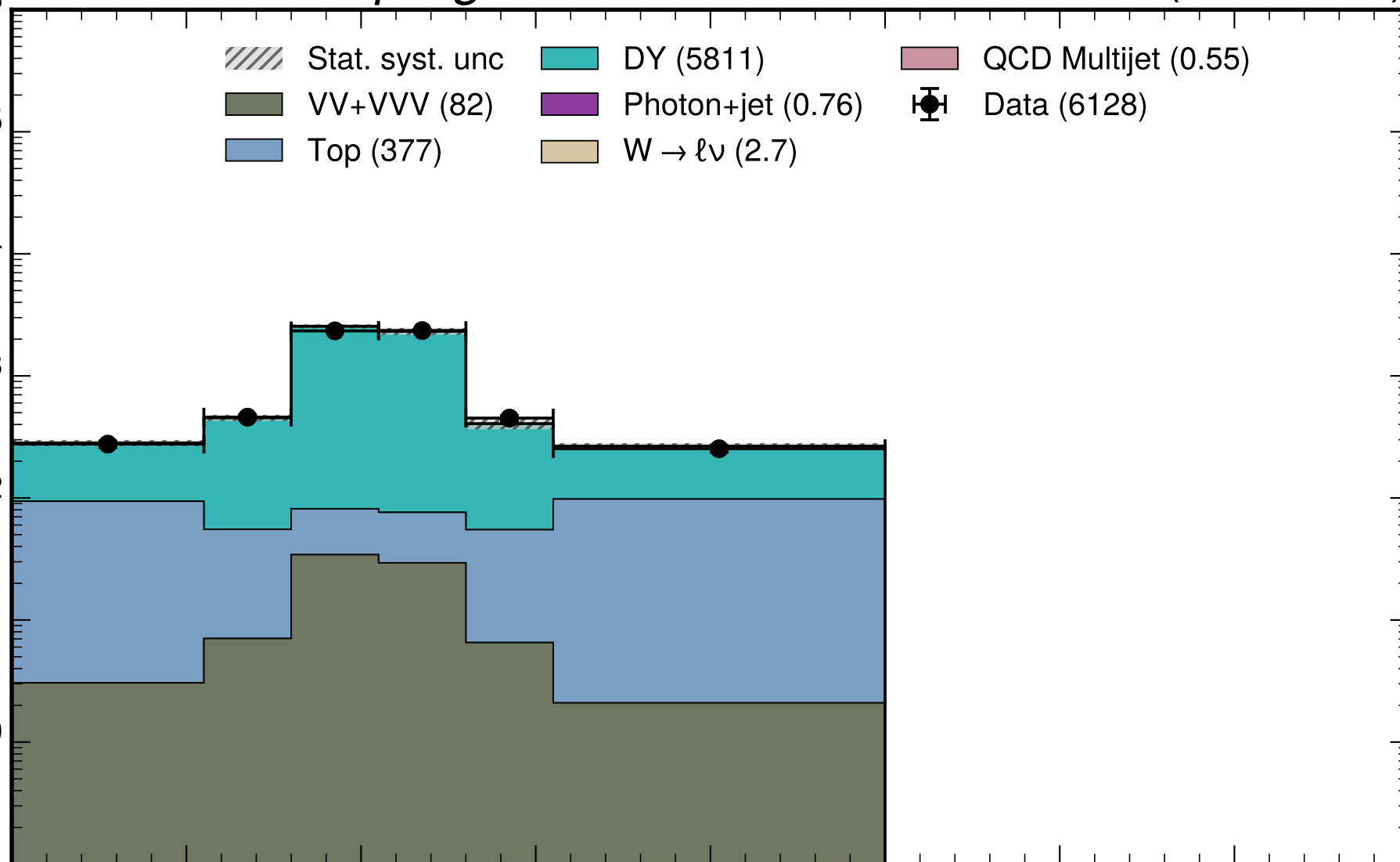
120

130

140

150

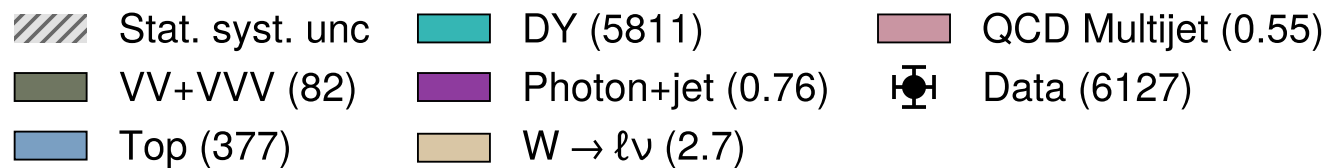
$m_{\ell\ell}$ (GeV)



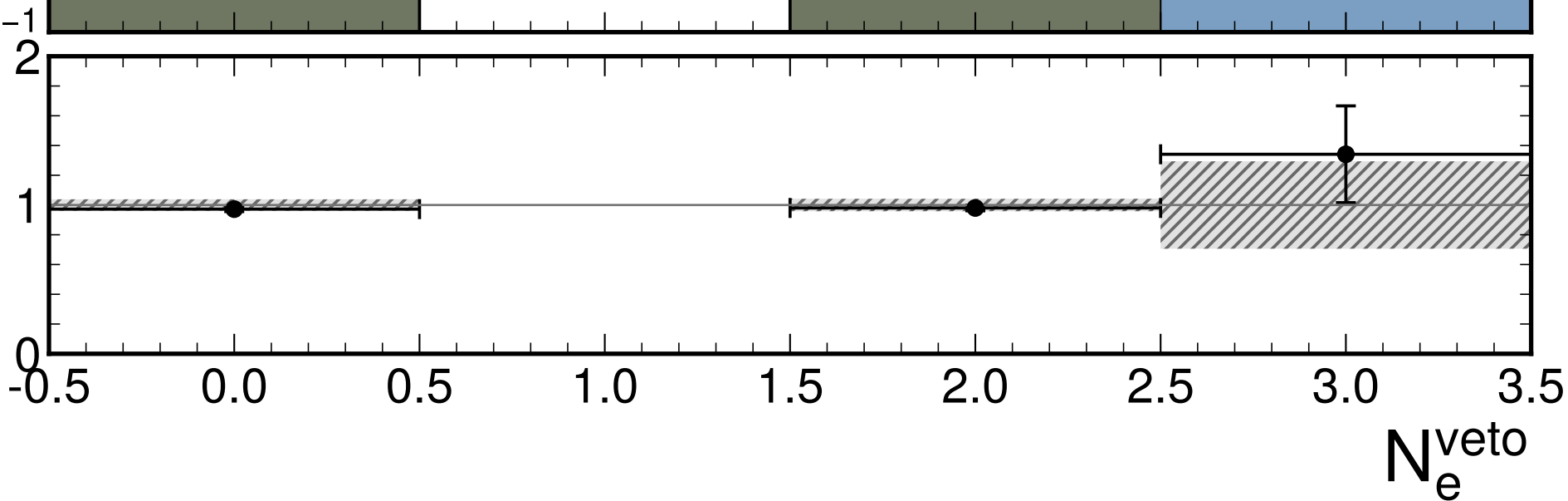
CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)

Events



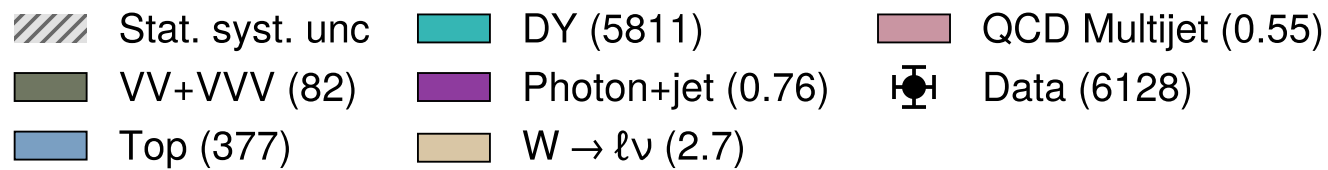
Data/MC



CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)

Events



Data/MC

10⁻¹

2

0

0

1

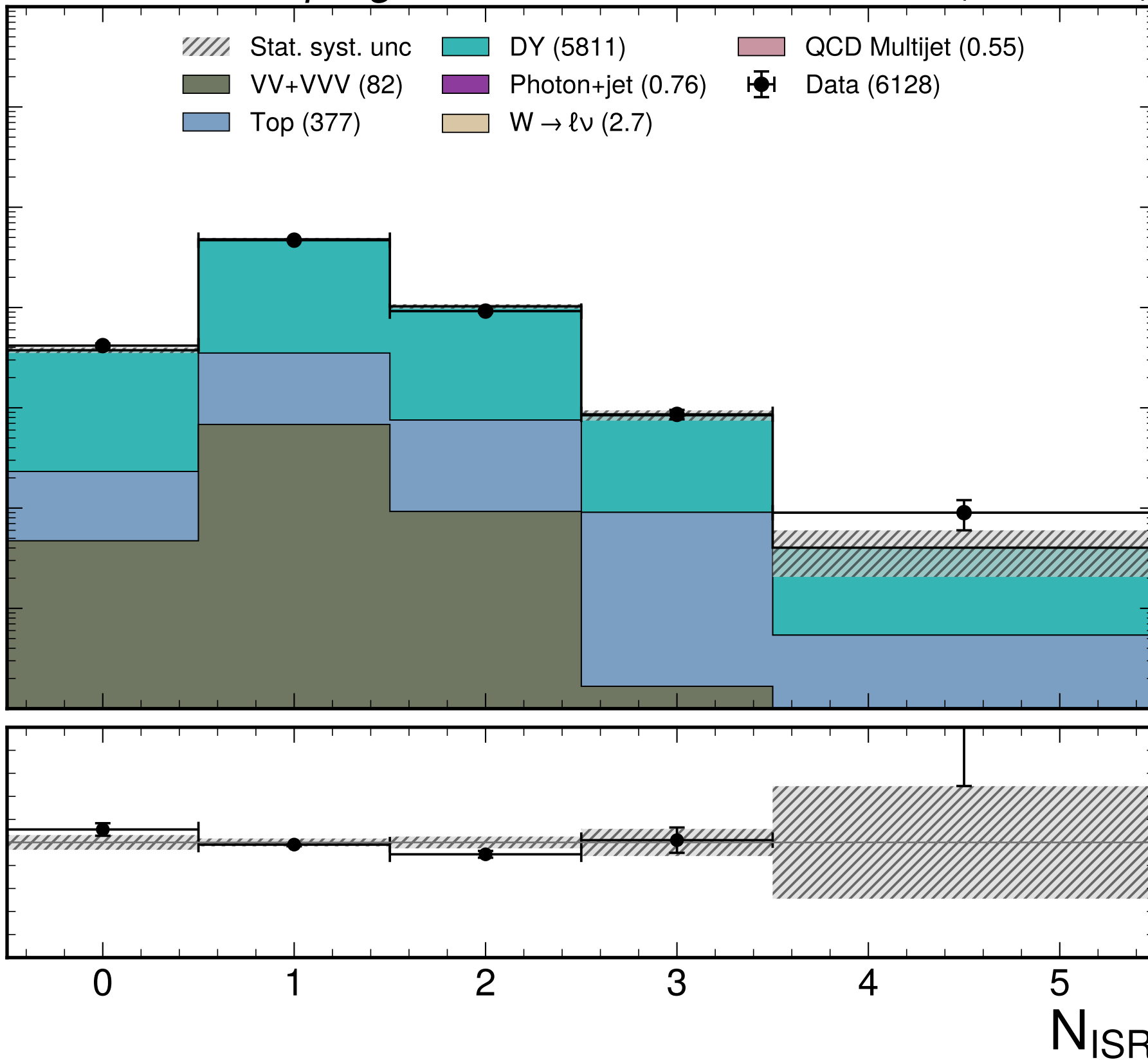
2

3

4

5

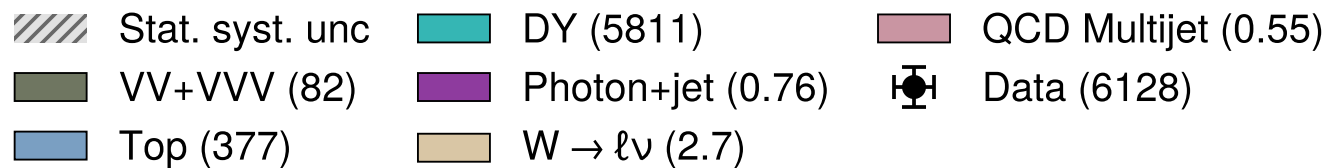
N_{ISR}



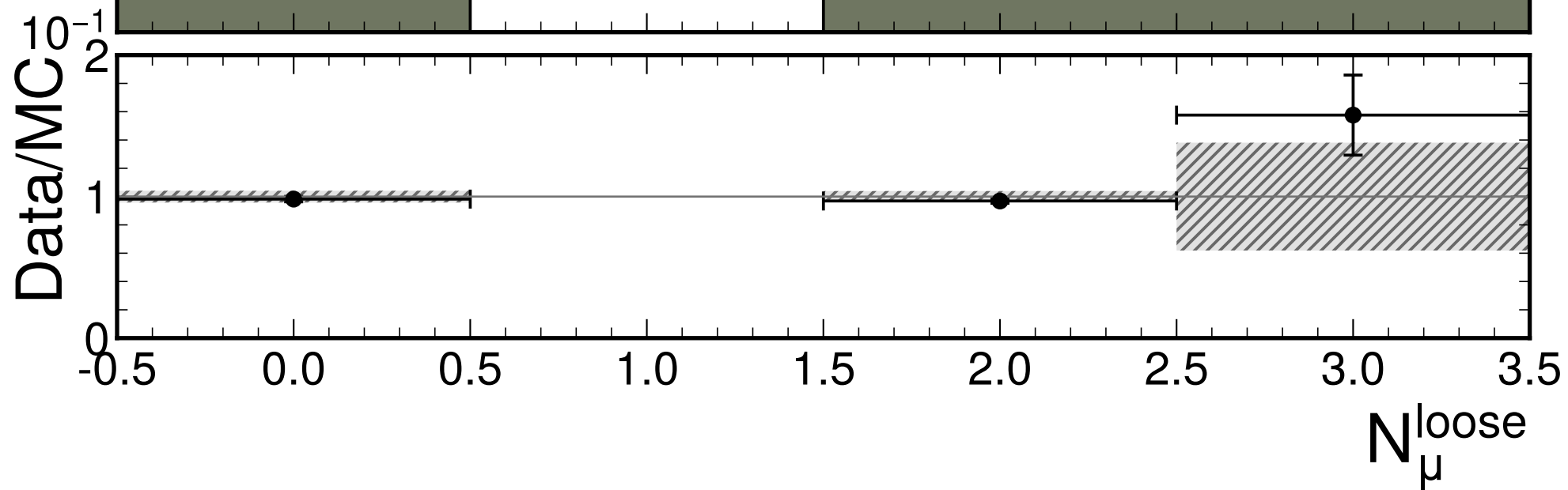
CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)

Events



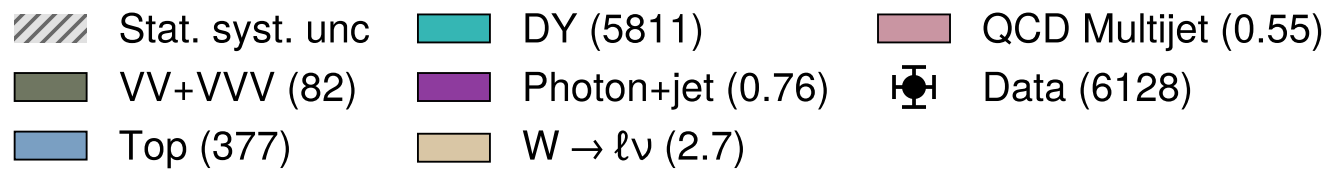
Data/MC



CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)

Events



Data/MC

10⁻¹
2
1
0

1

2

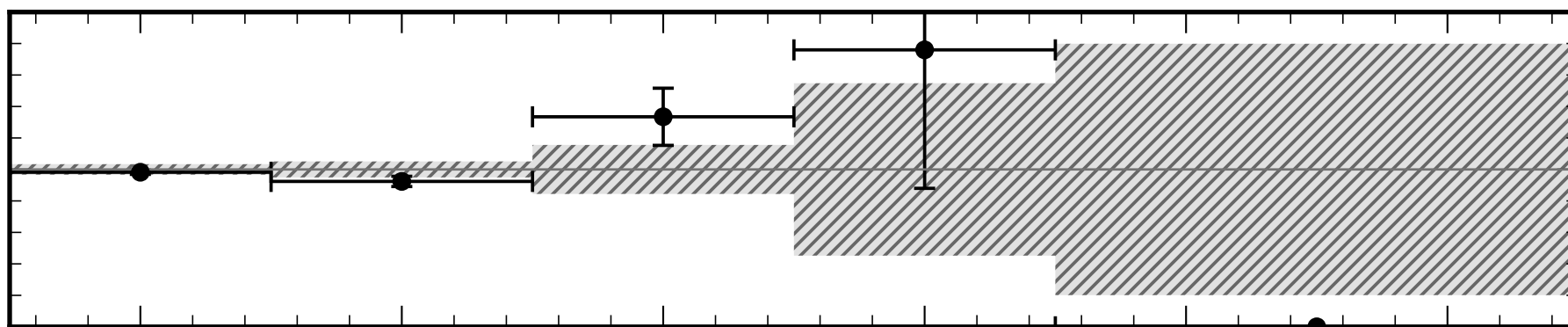
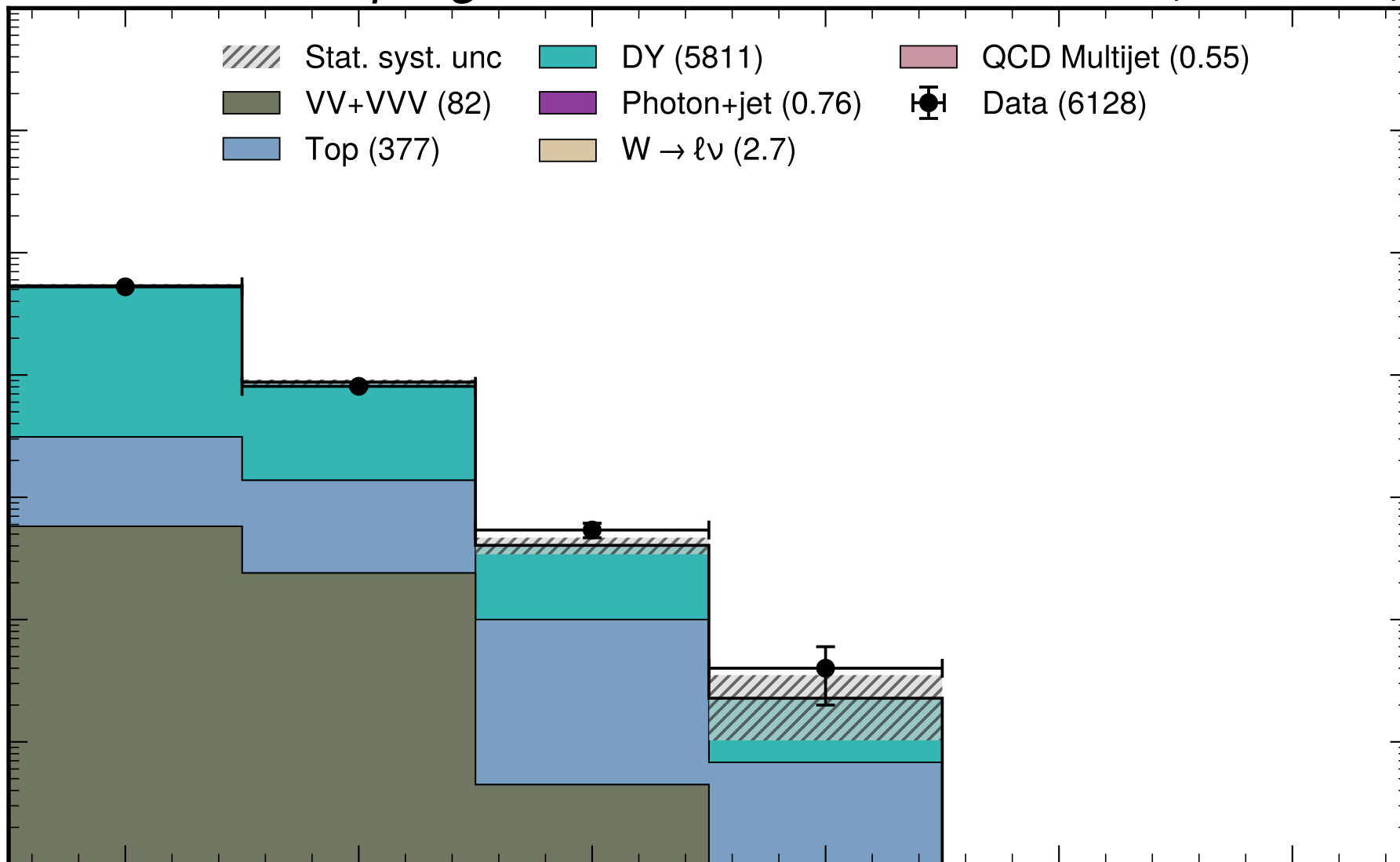
3

4

5

6

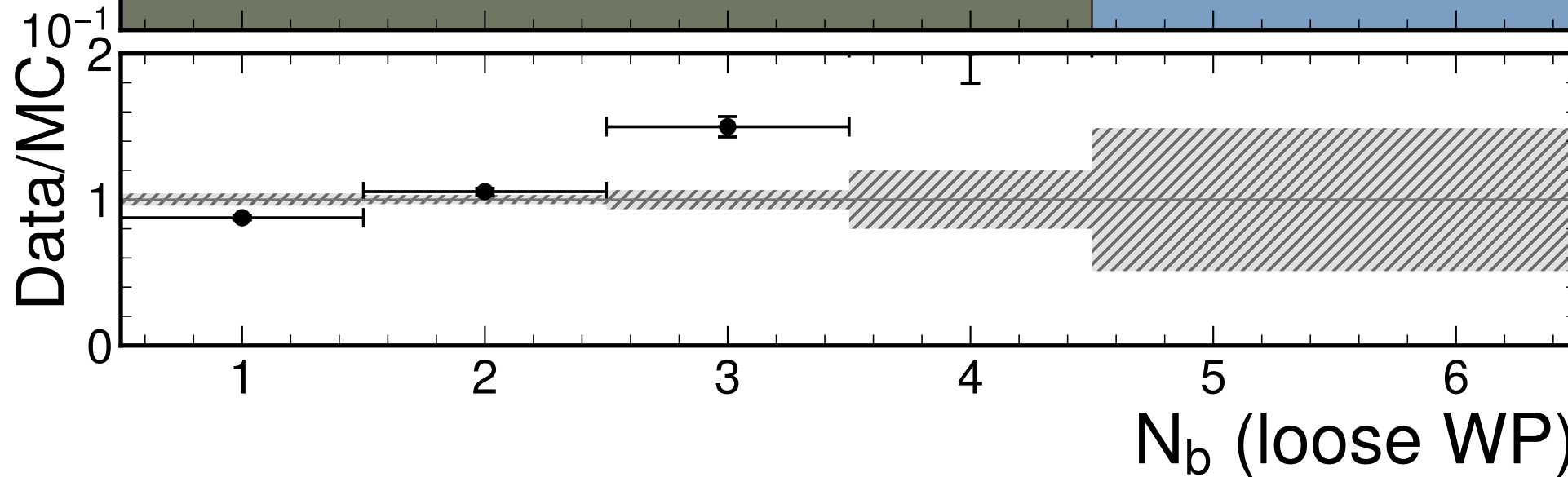
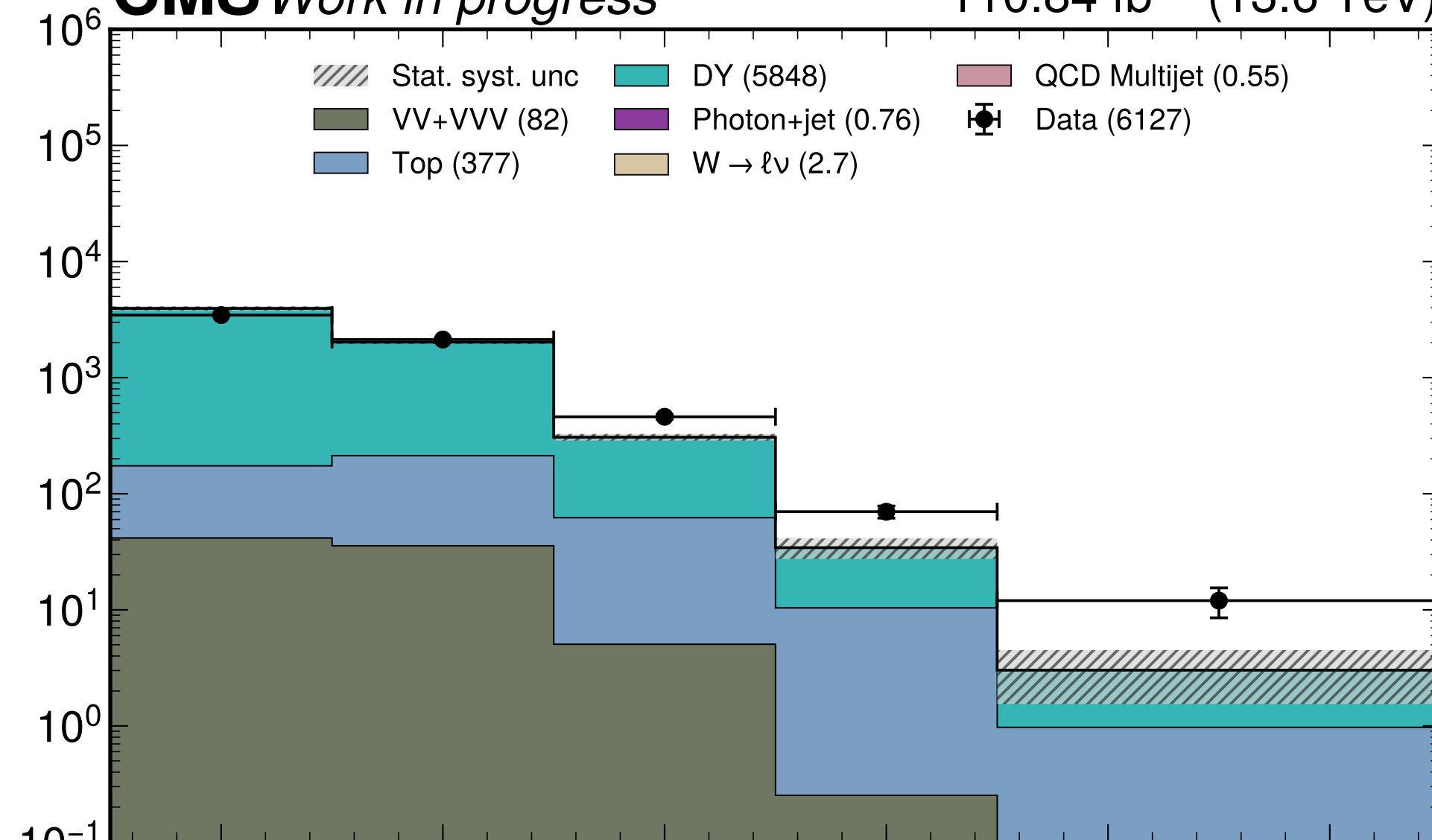
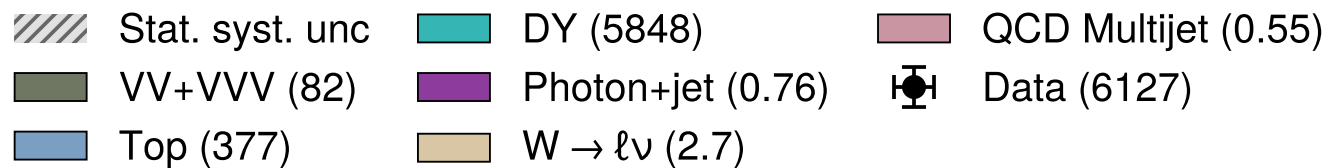
N_b



CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)

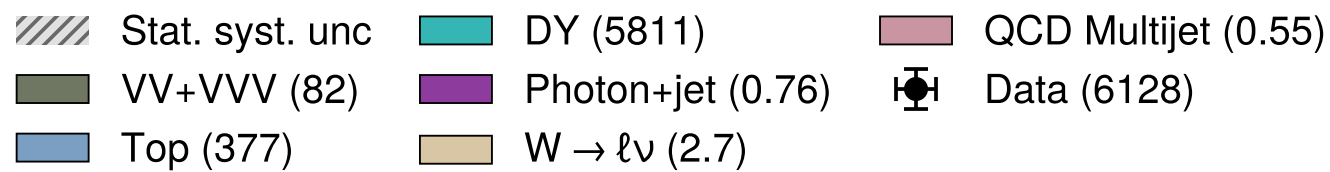
Events



CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)

Events



Data/MC

10⁻¹
2
1
0

1

2

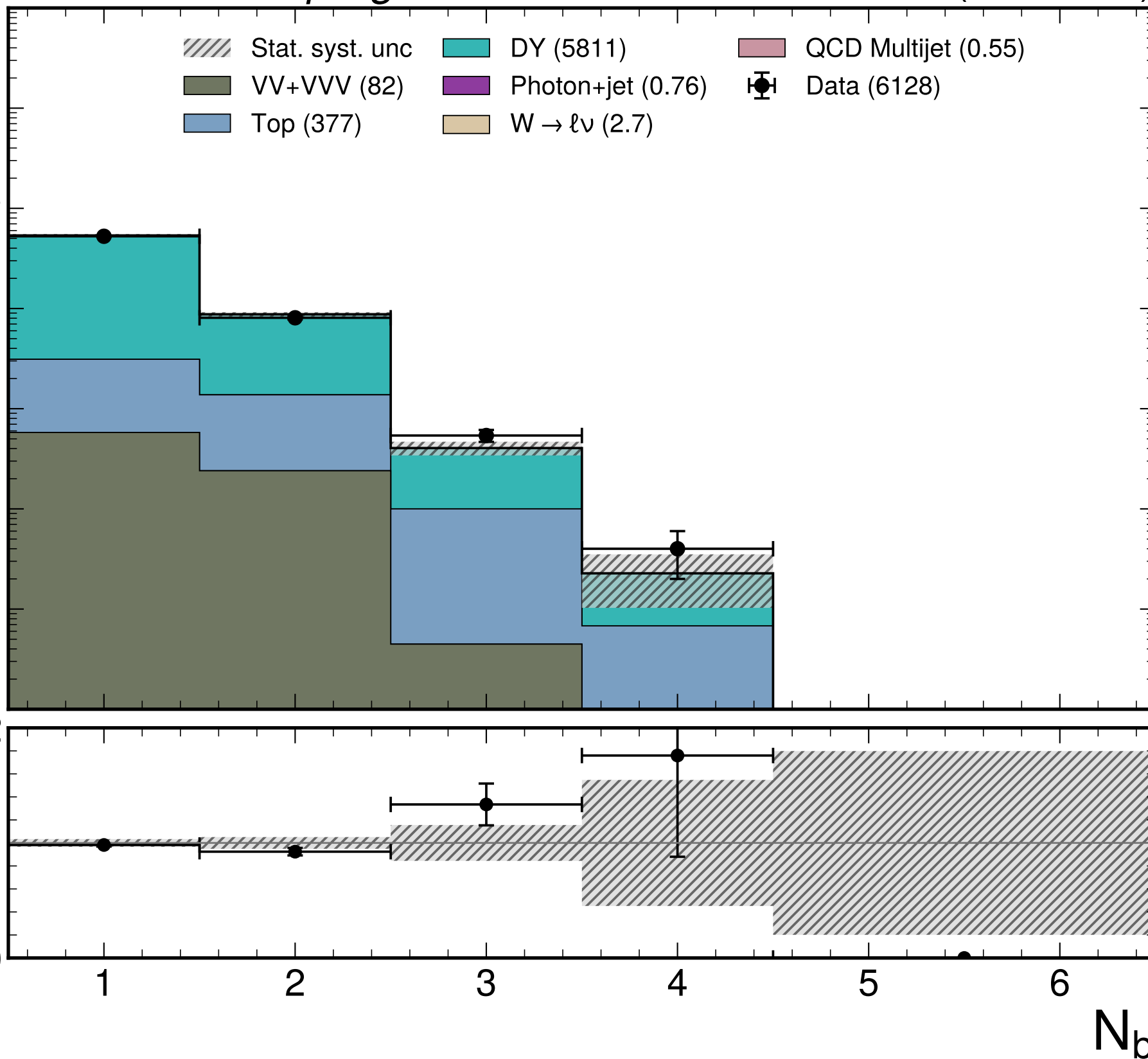
3

4

5

6

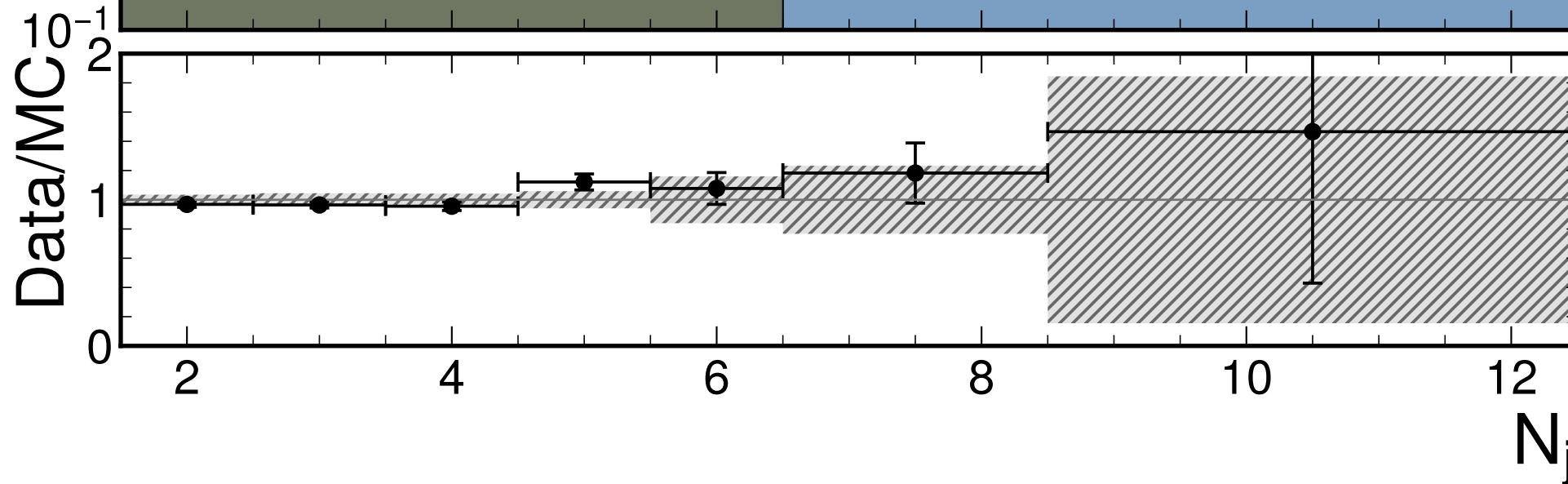
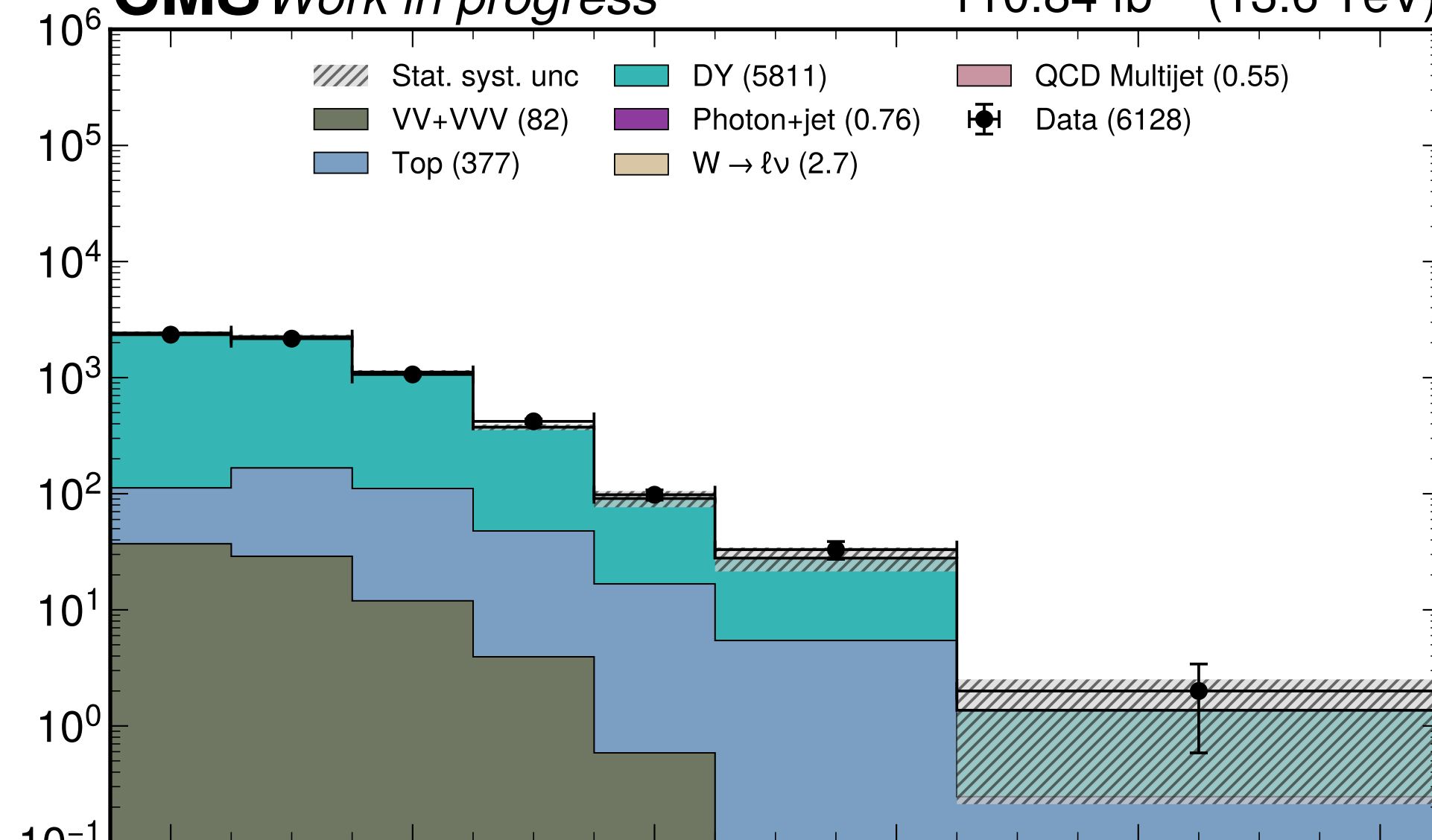
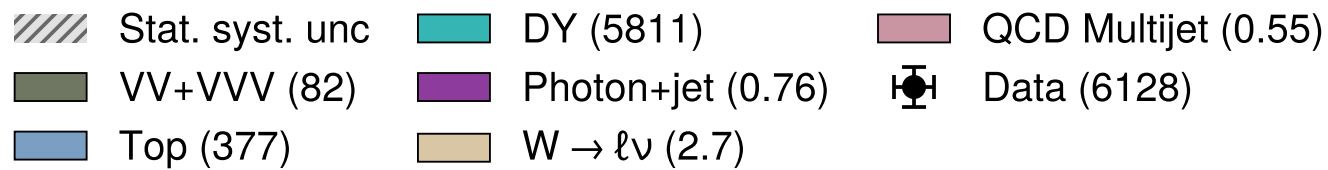
N_b



CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)

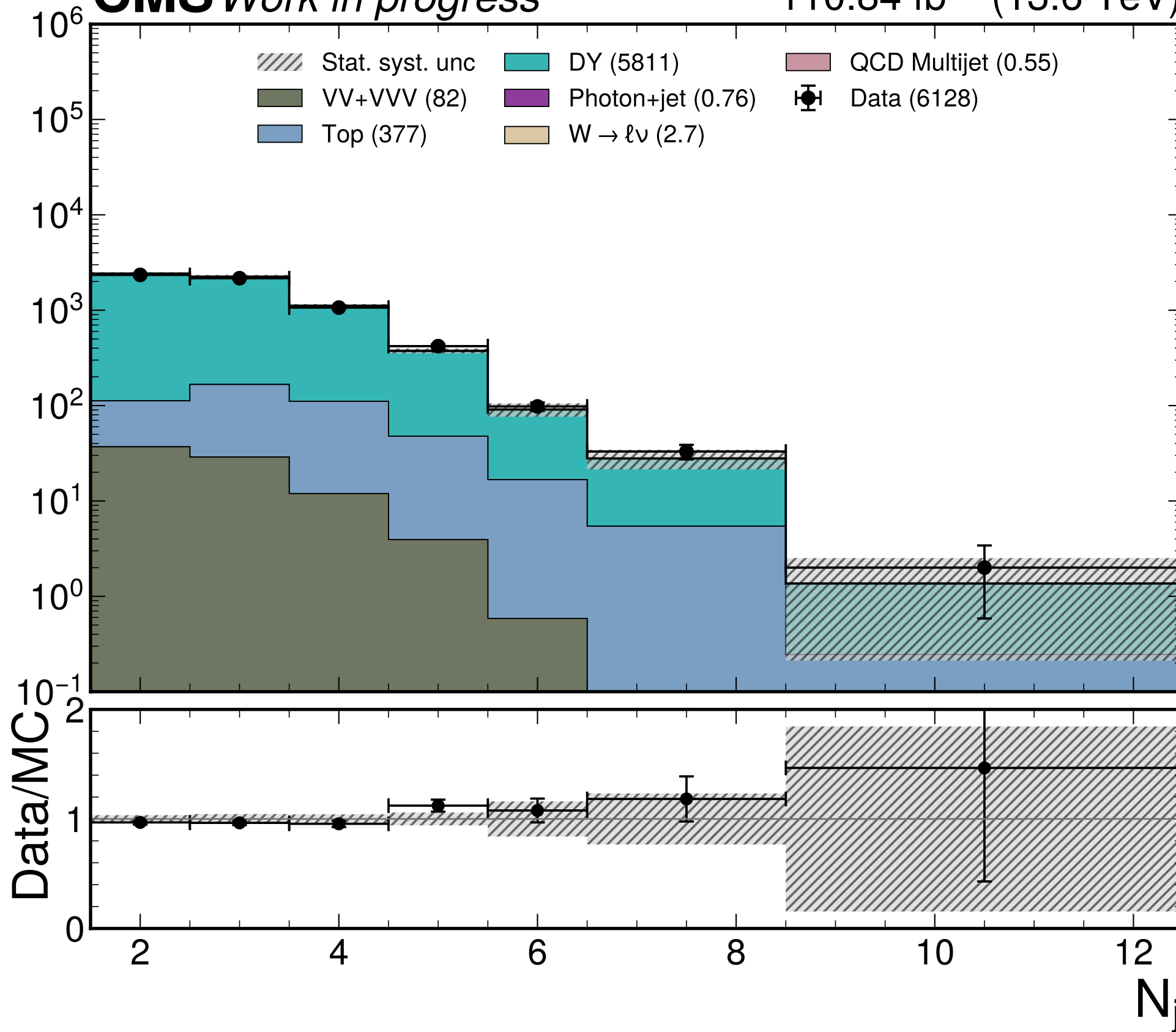
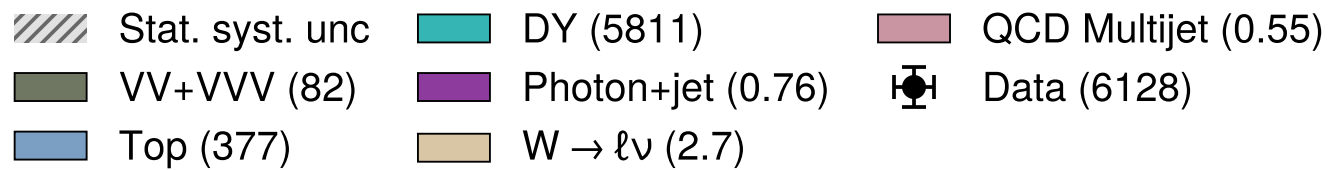
Events



CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)

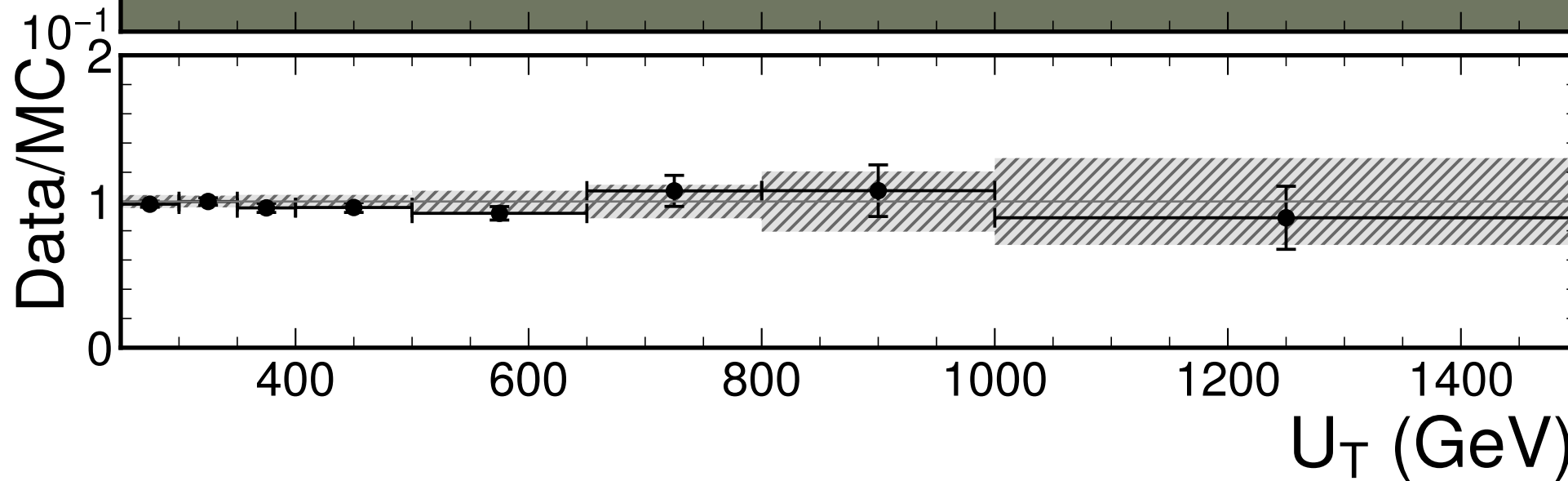
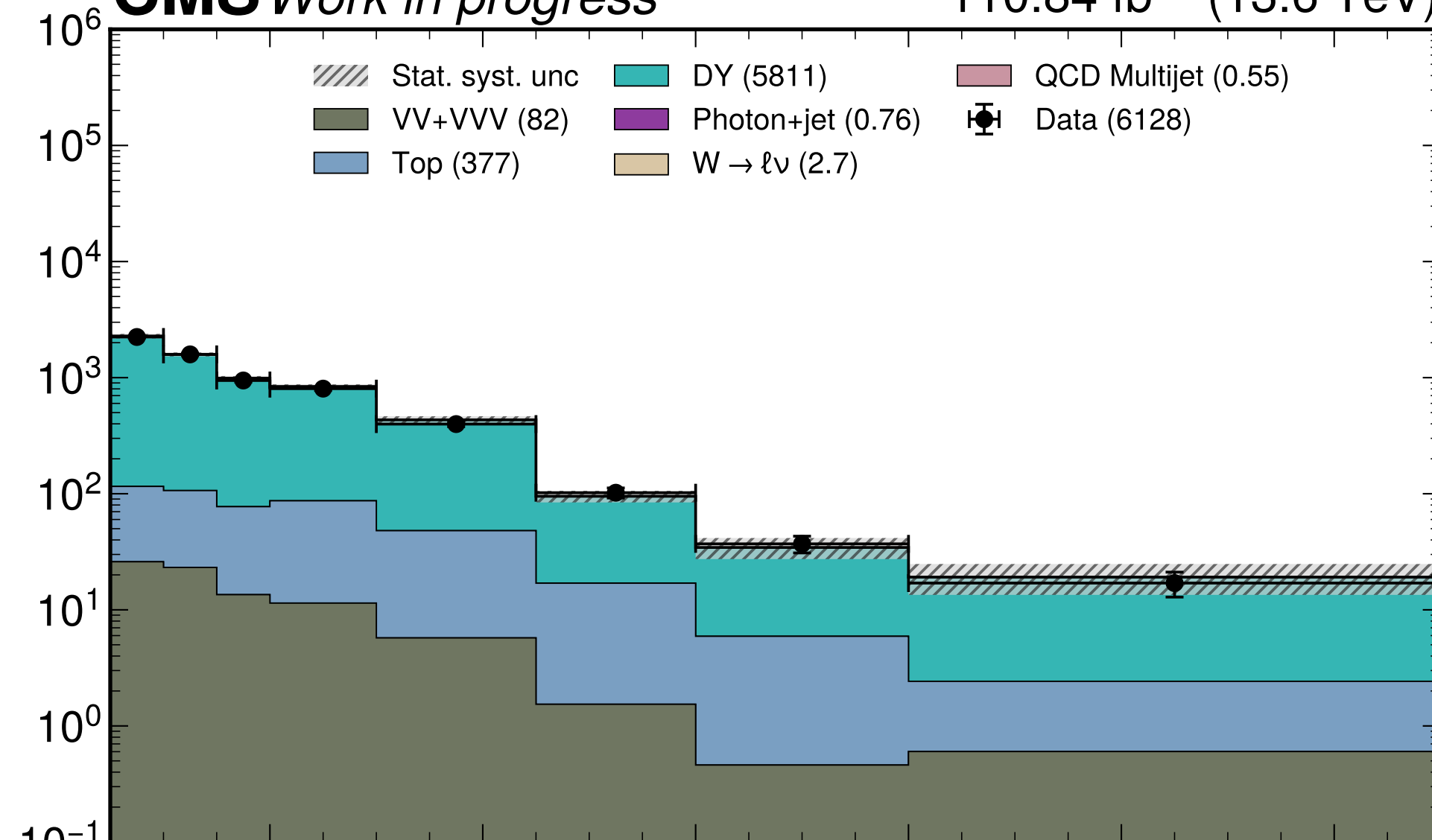
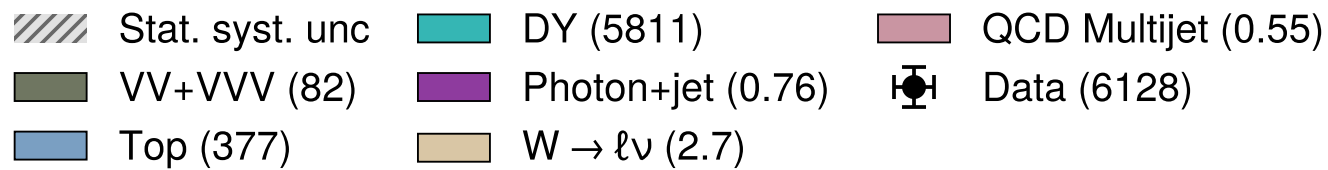
Events



CMS *Work in progress*

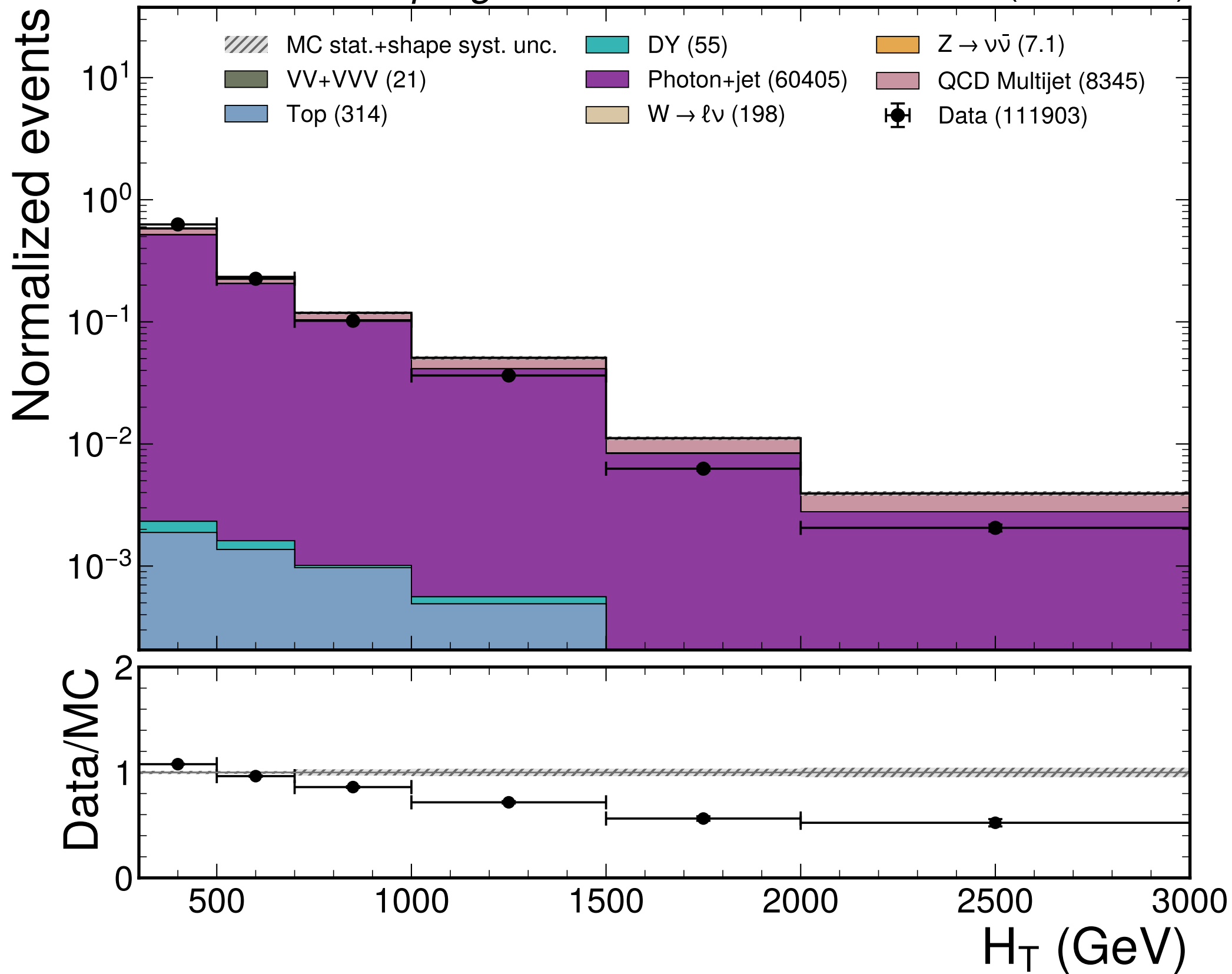
110.84 fb⁻¹ (13.6 TeV)

Events



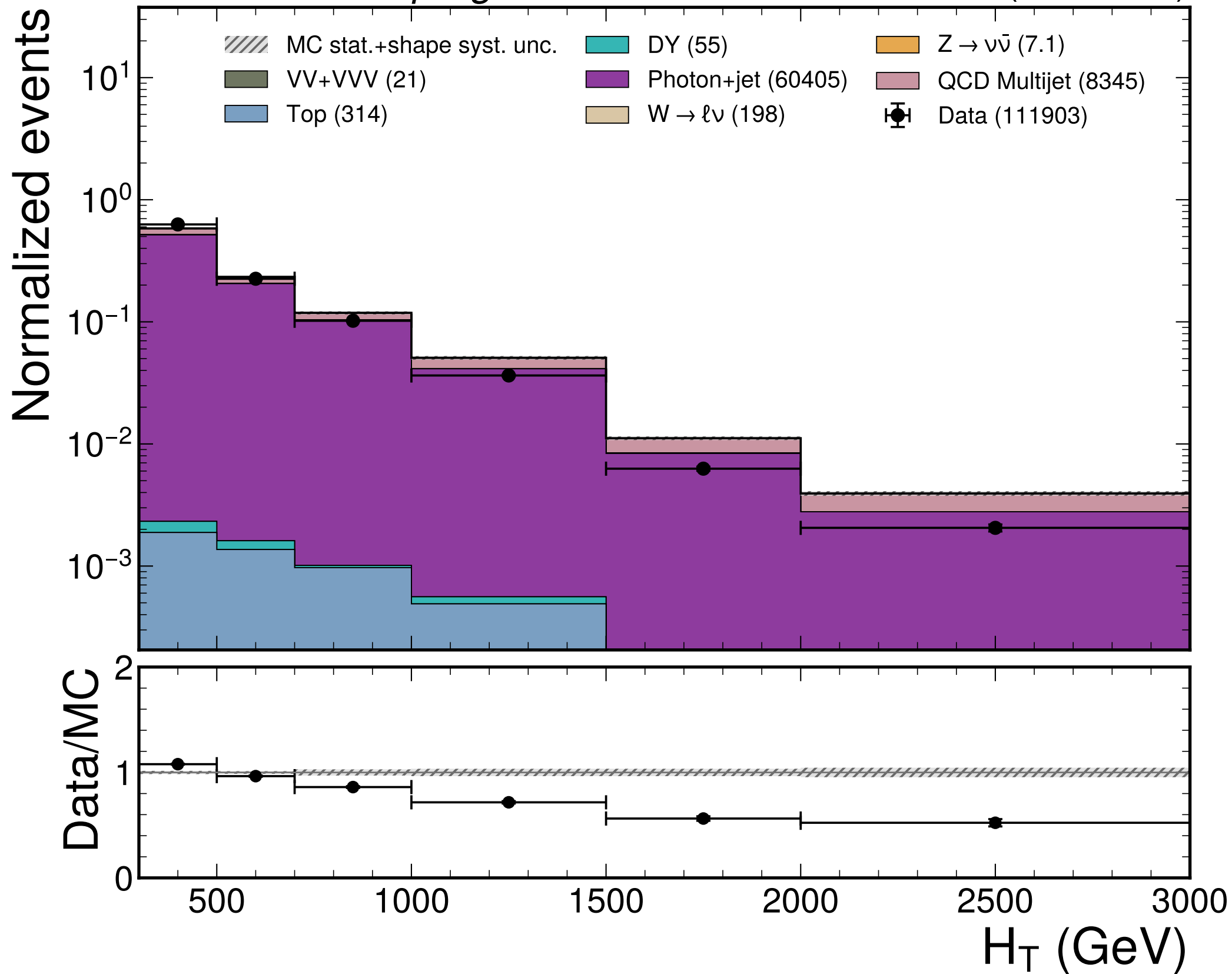
CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)



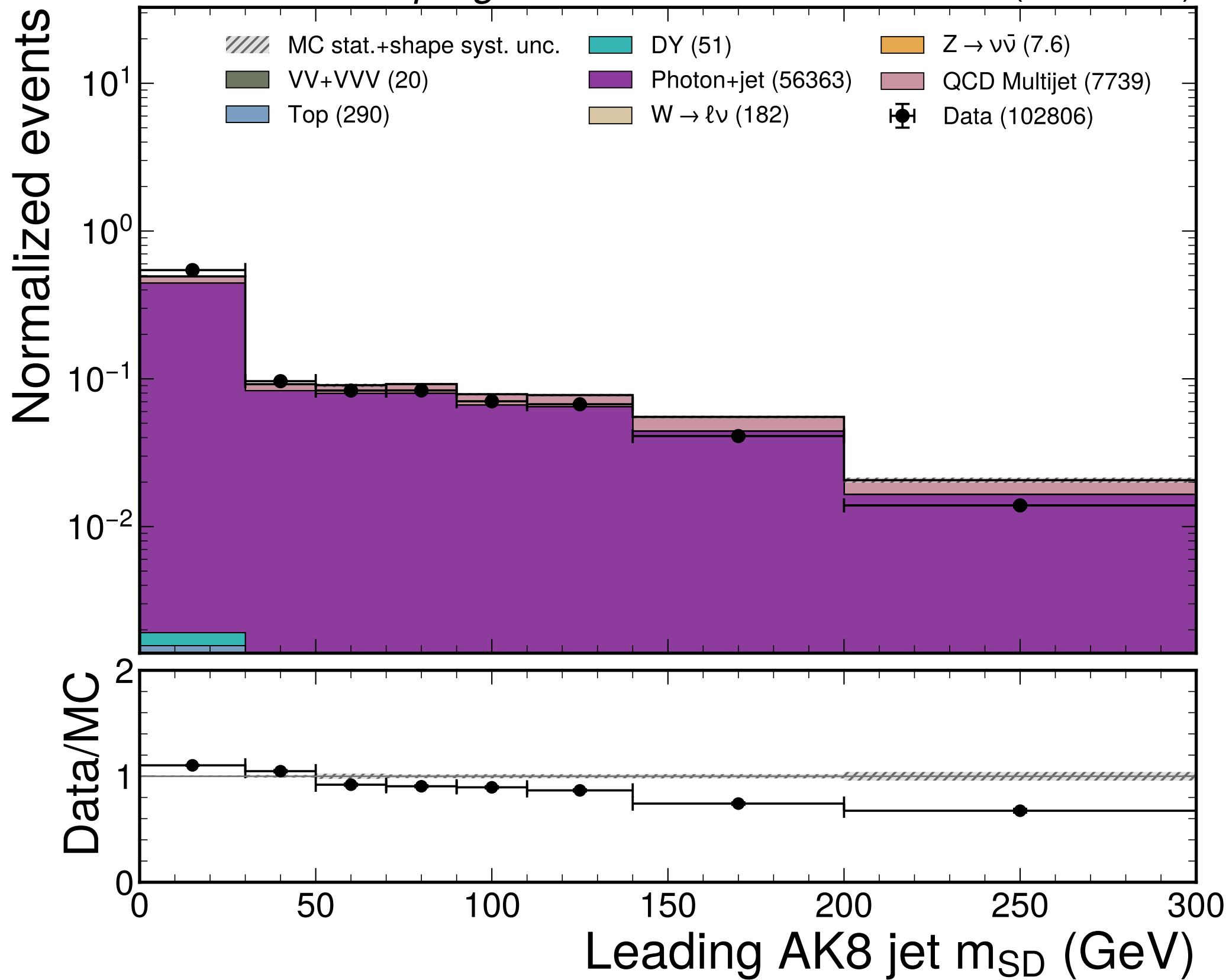
CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)



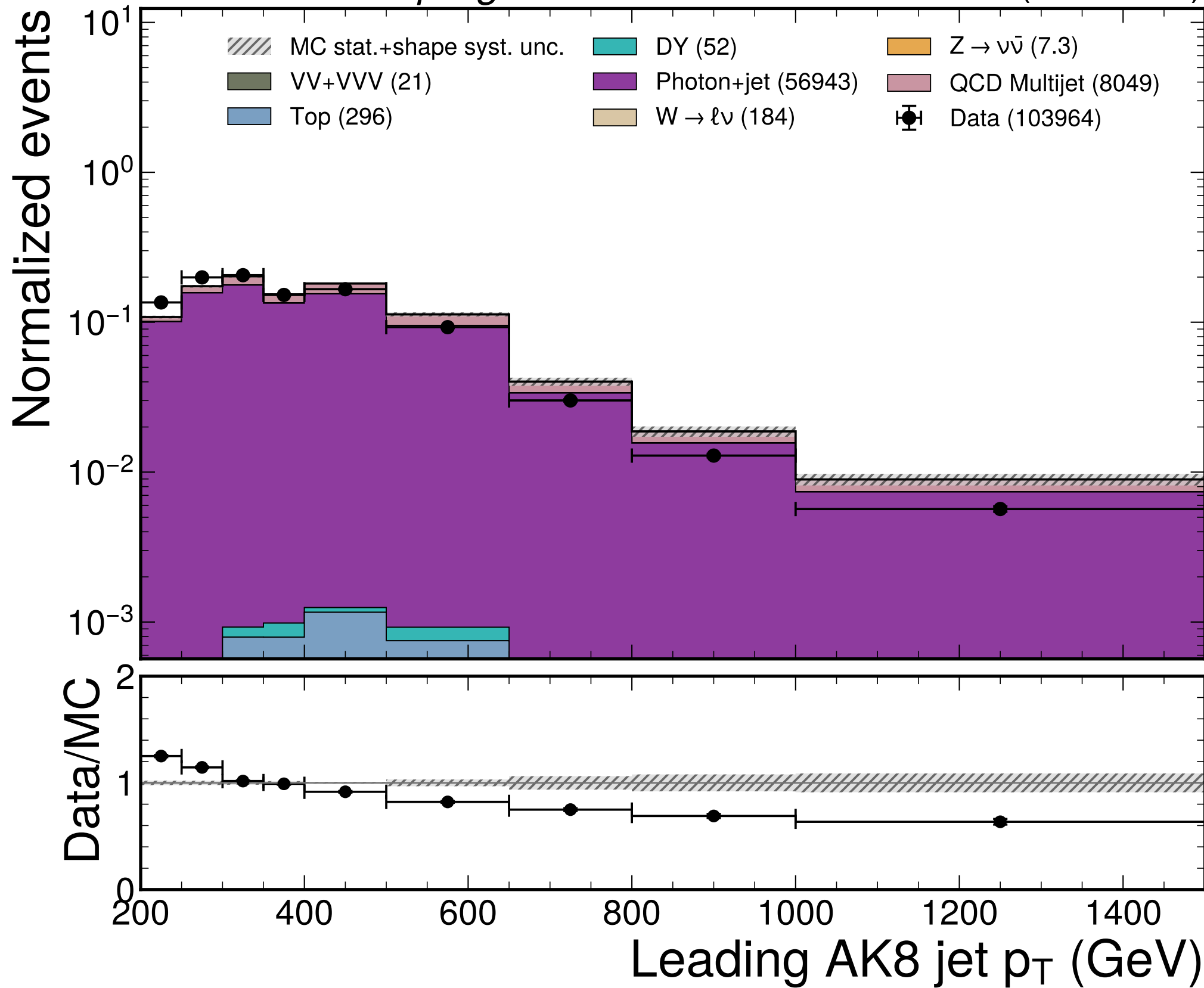
CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)



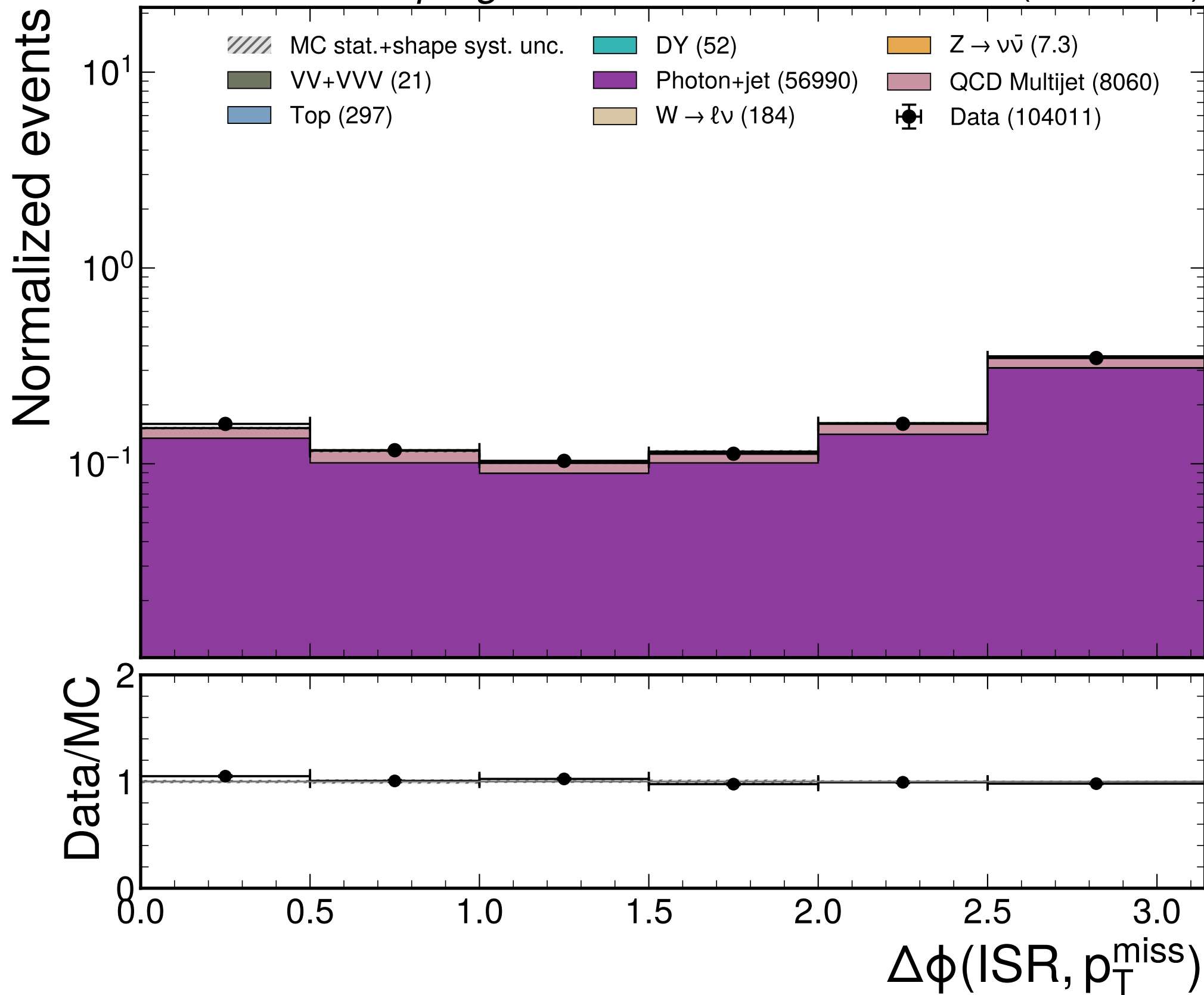
CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)



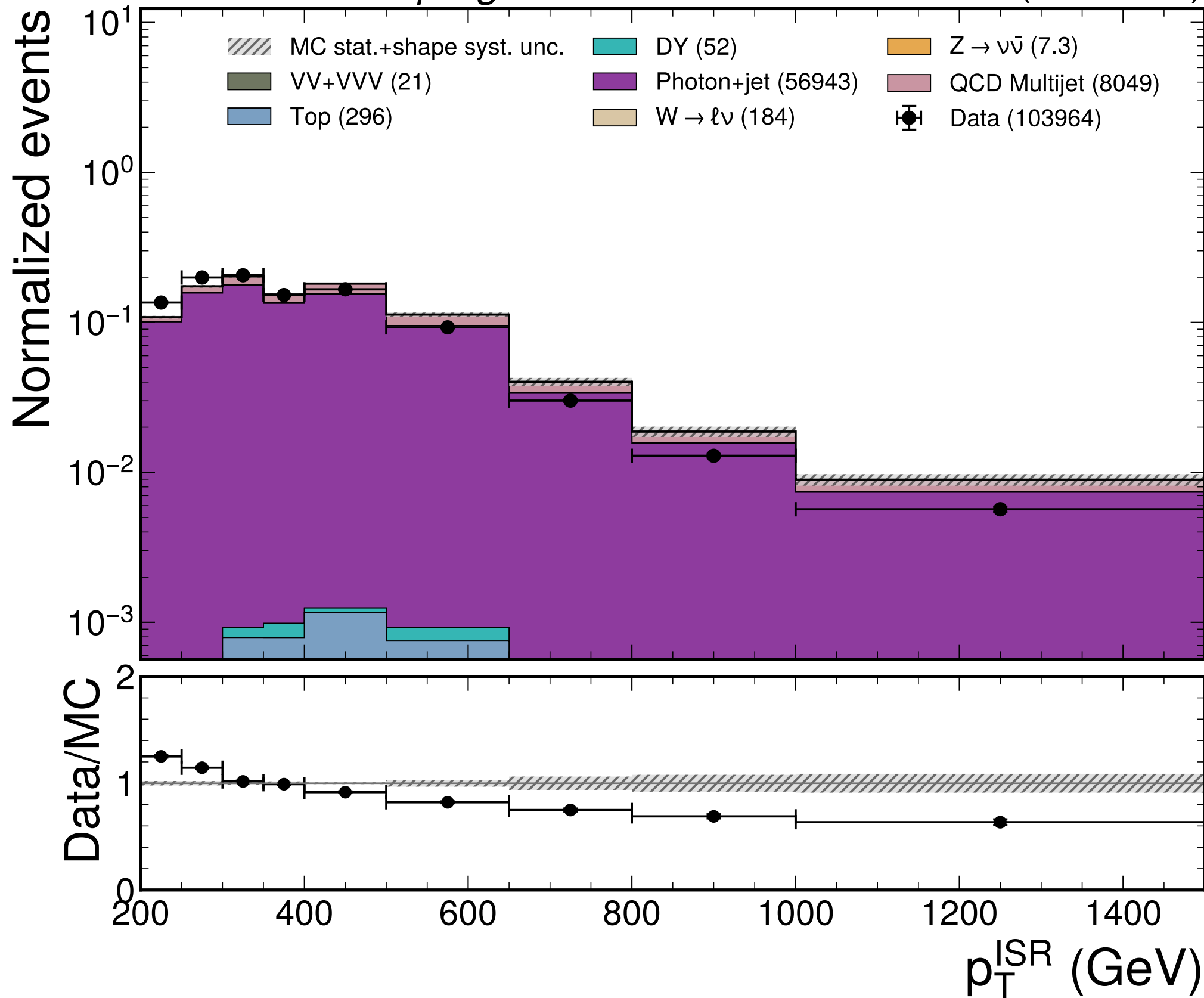
CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)



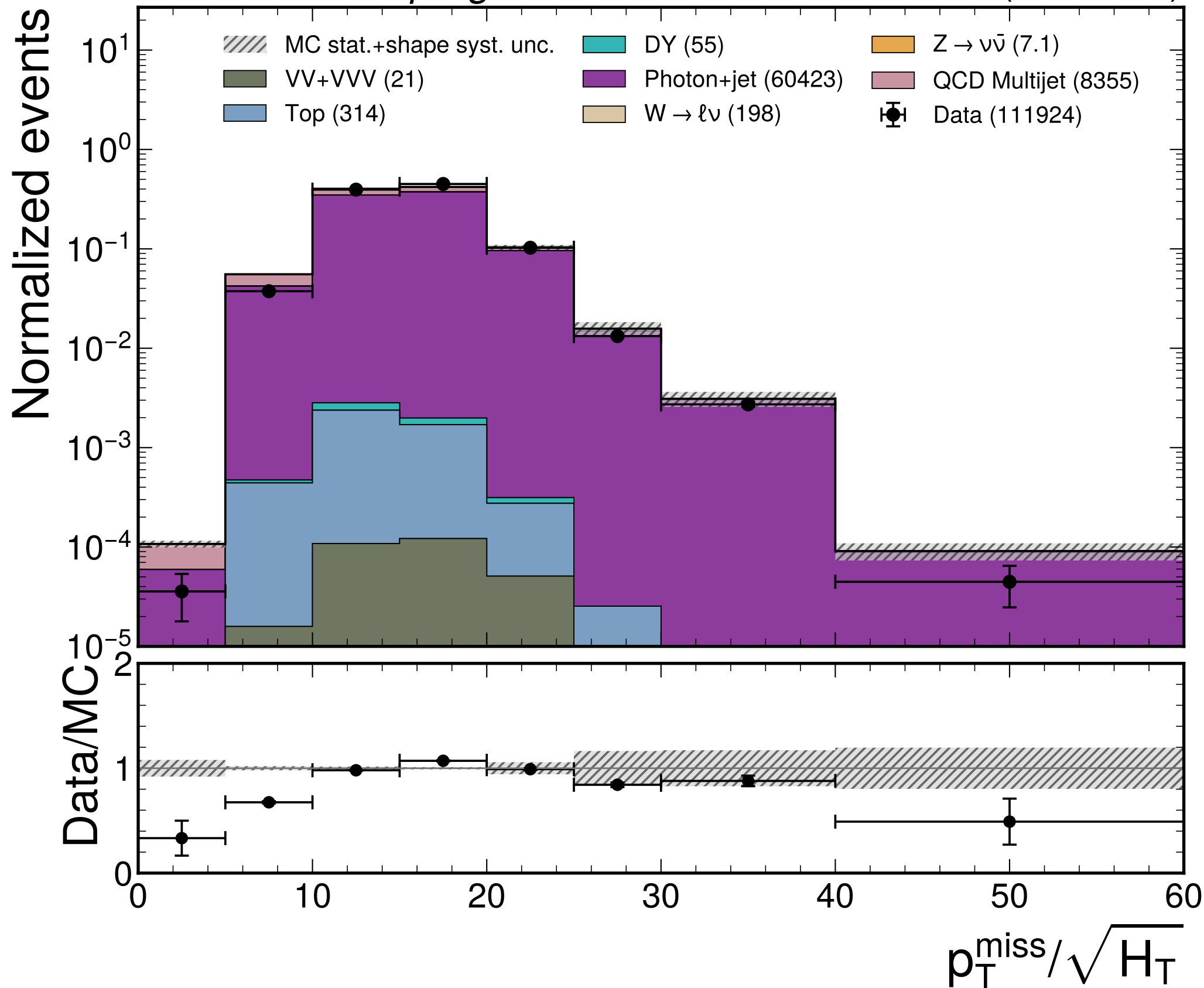
CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)



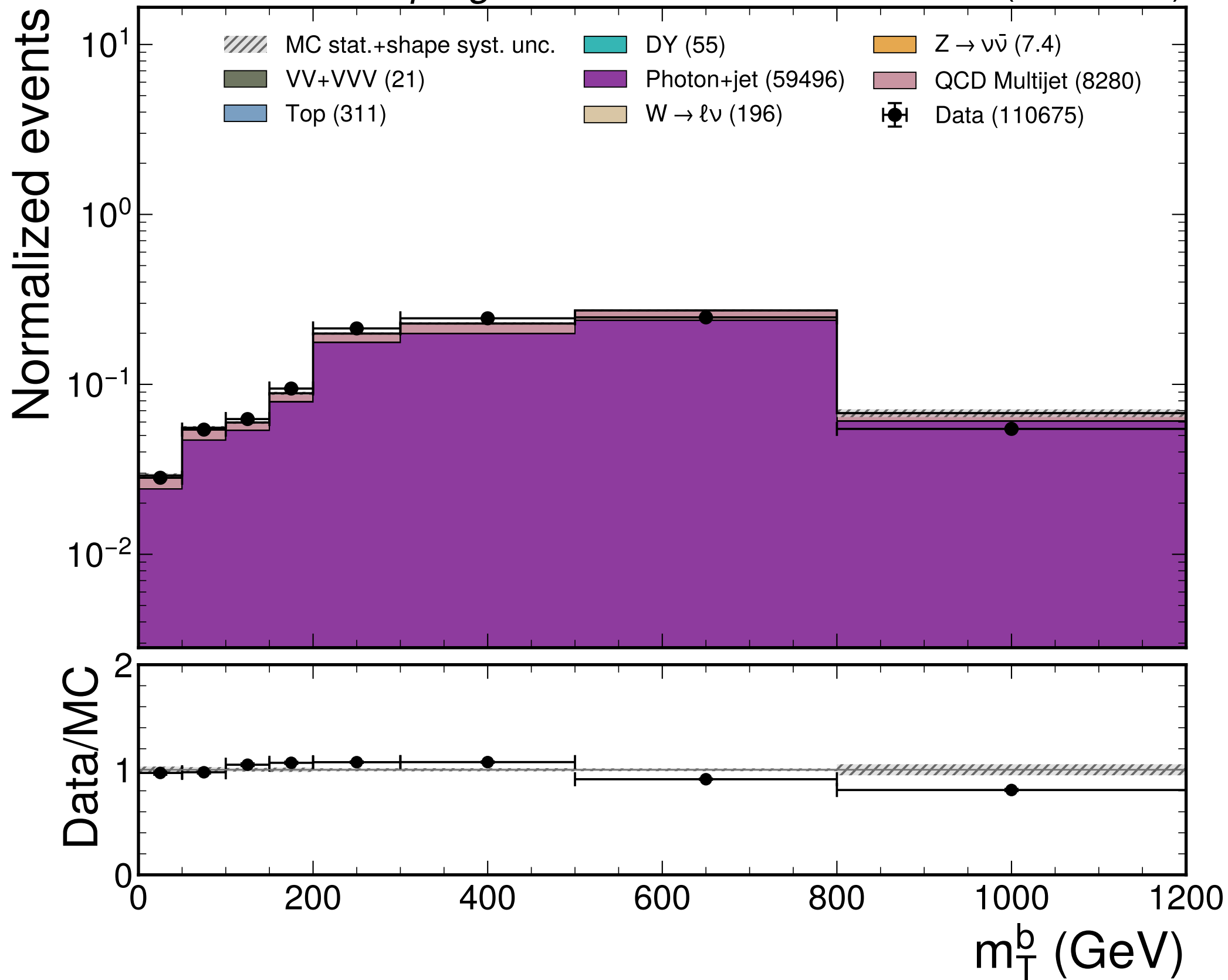
CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)



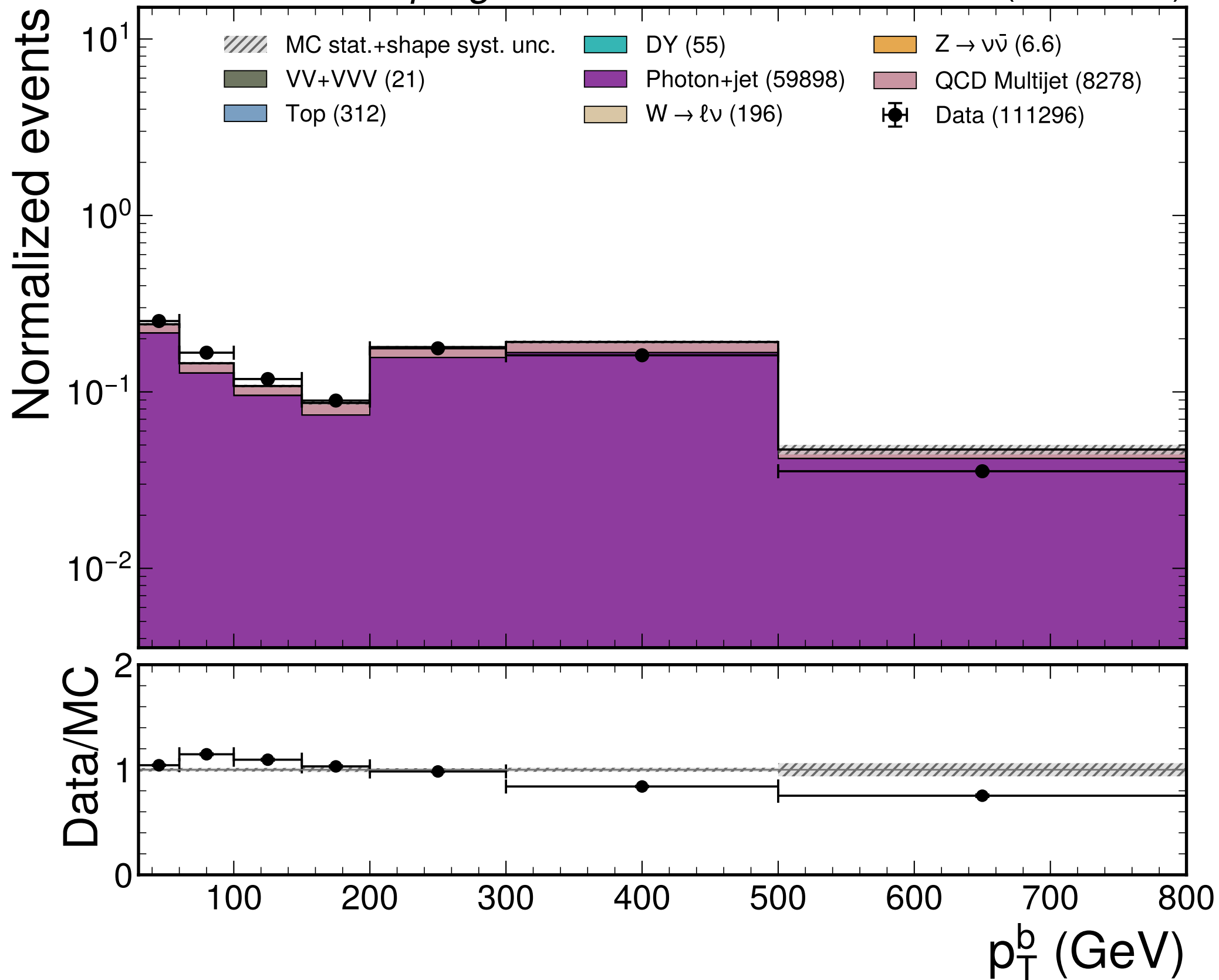
CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)



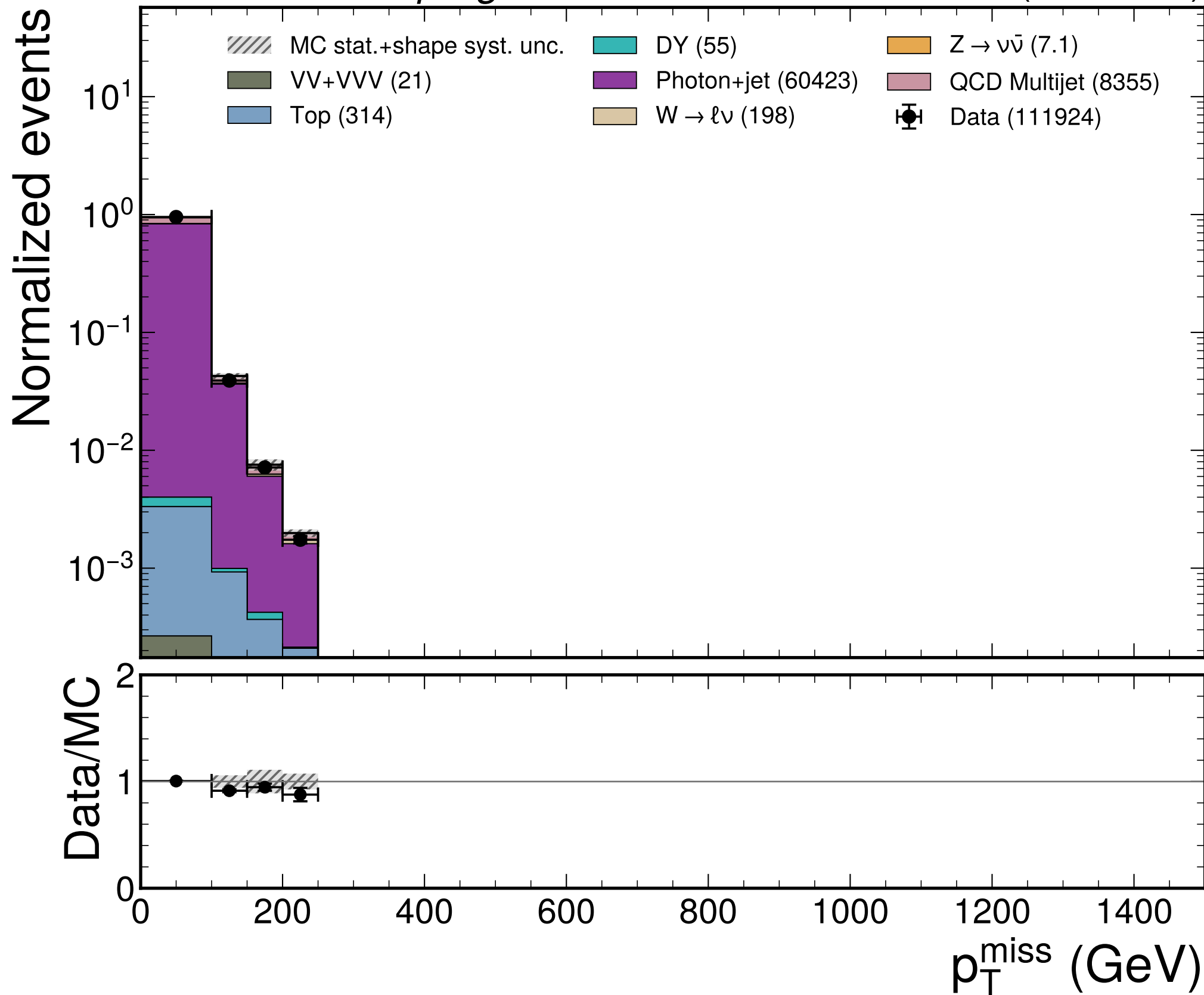
CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)



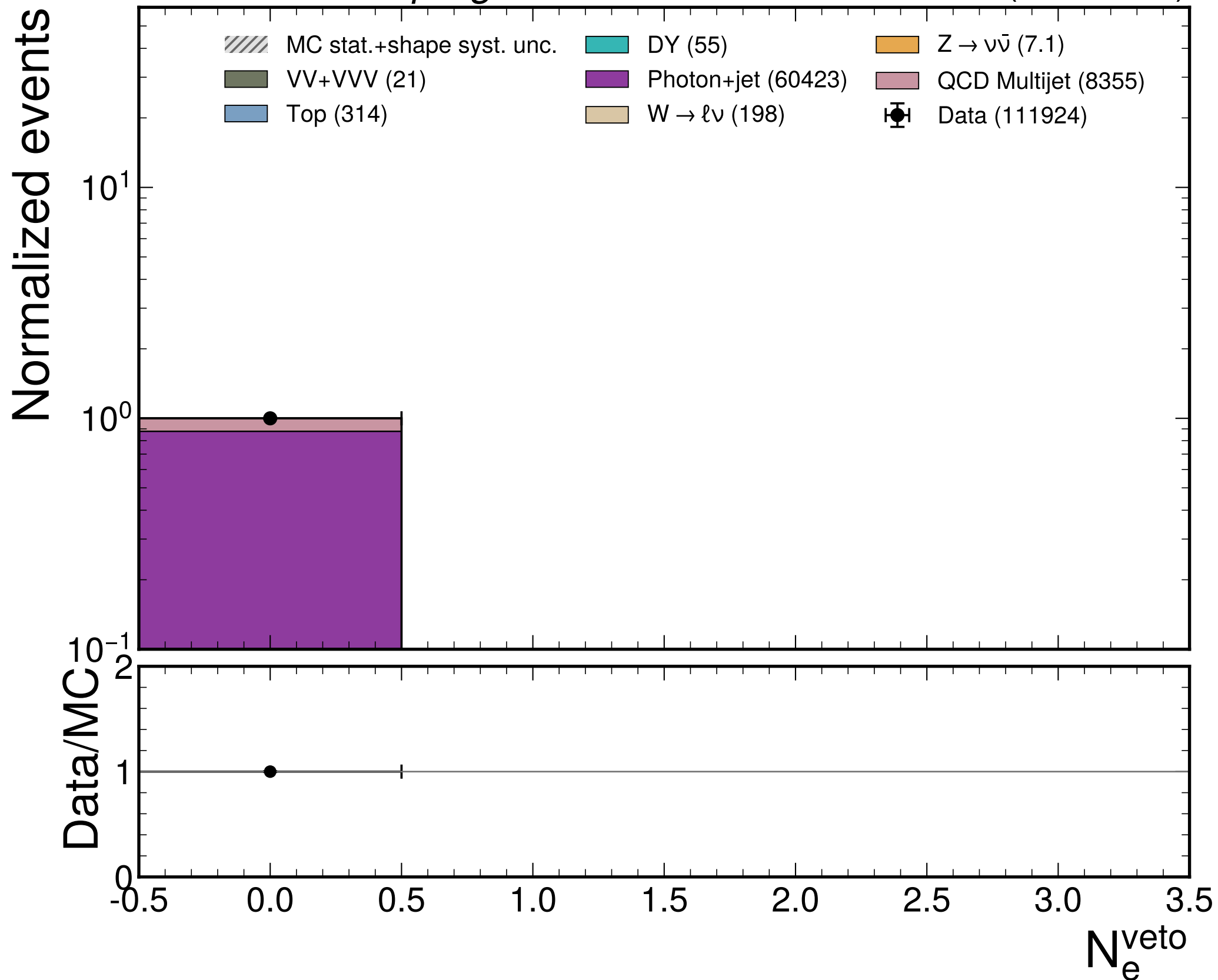
CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)



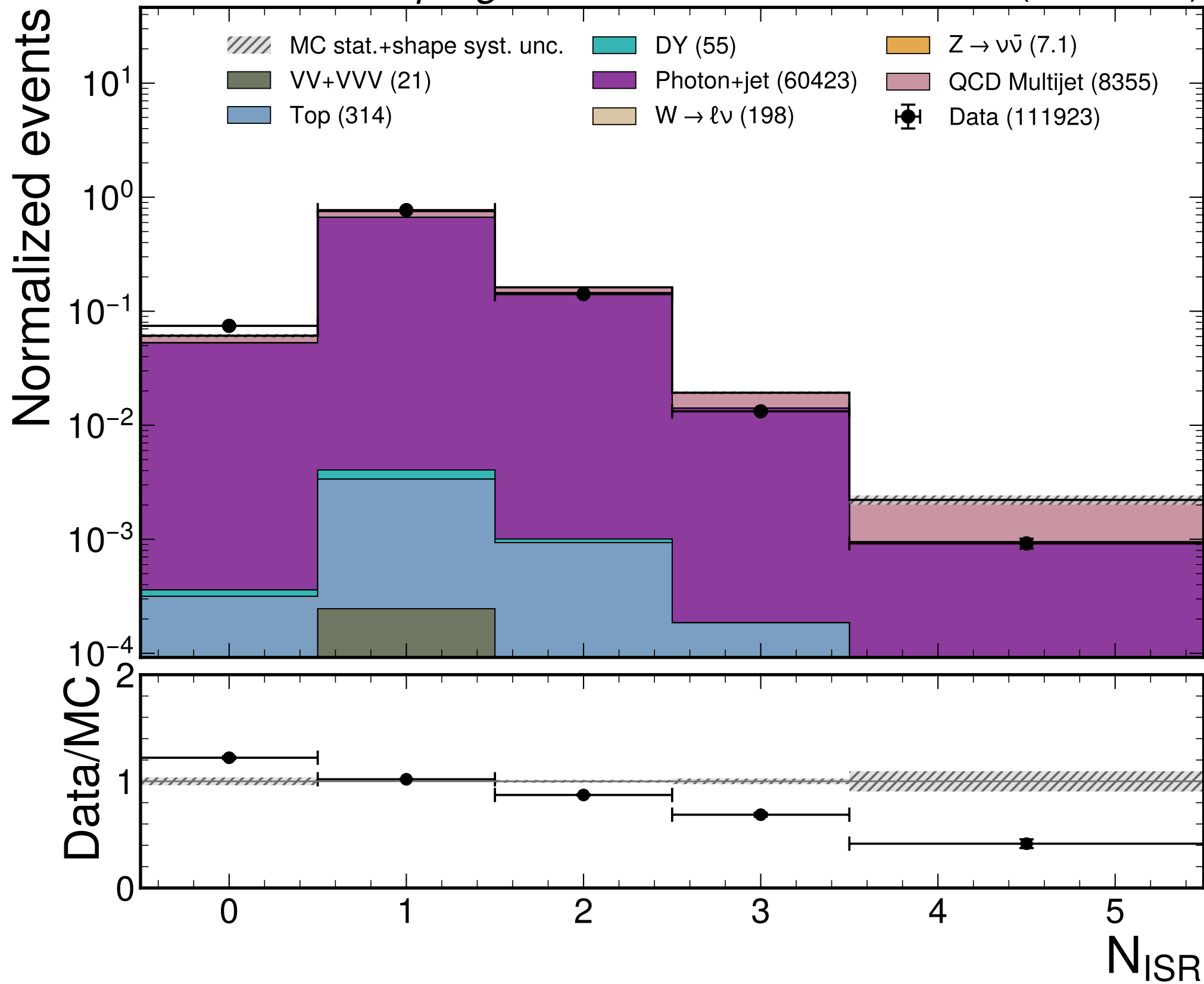
CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)



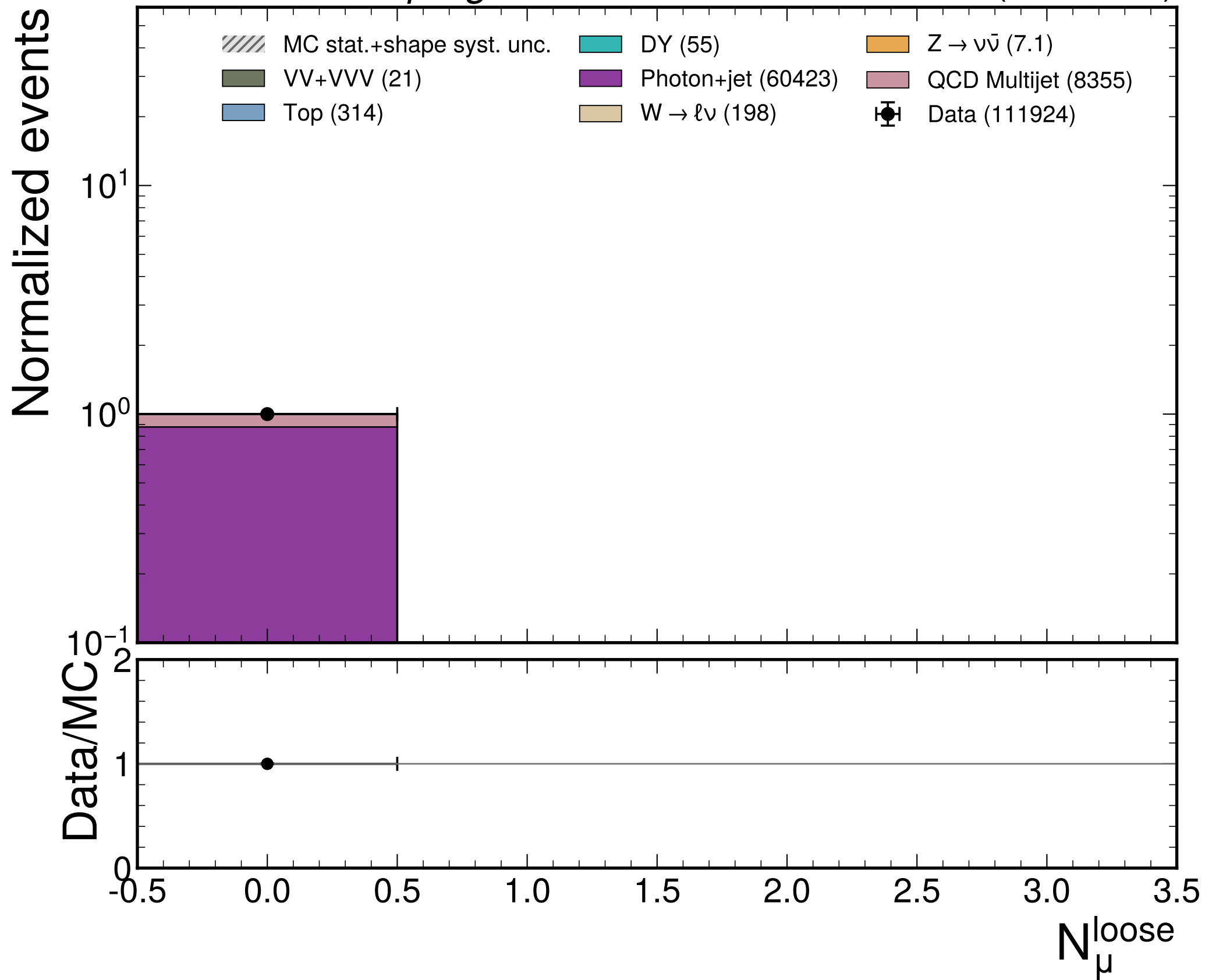
CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)



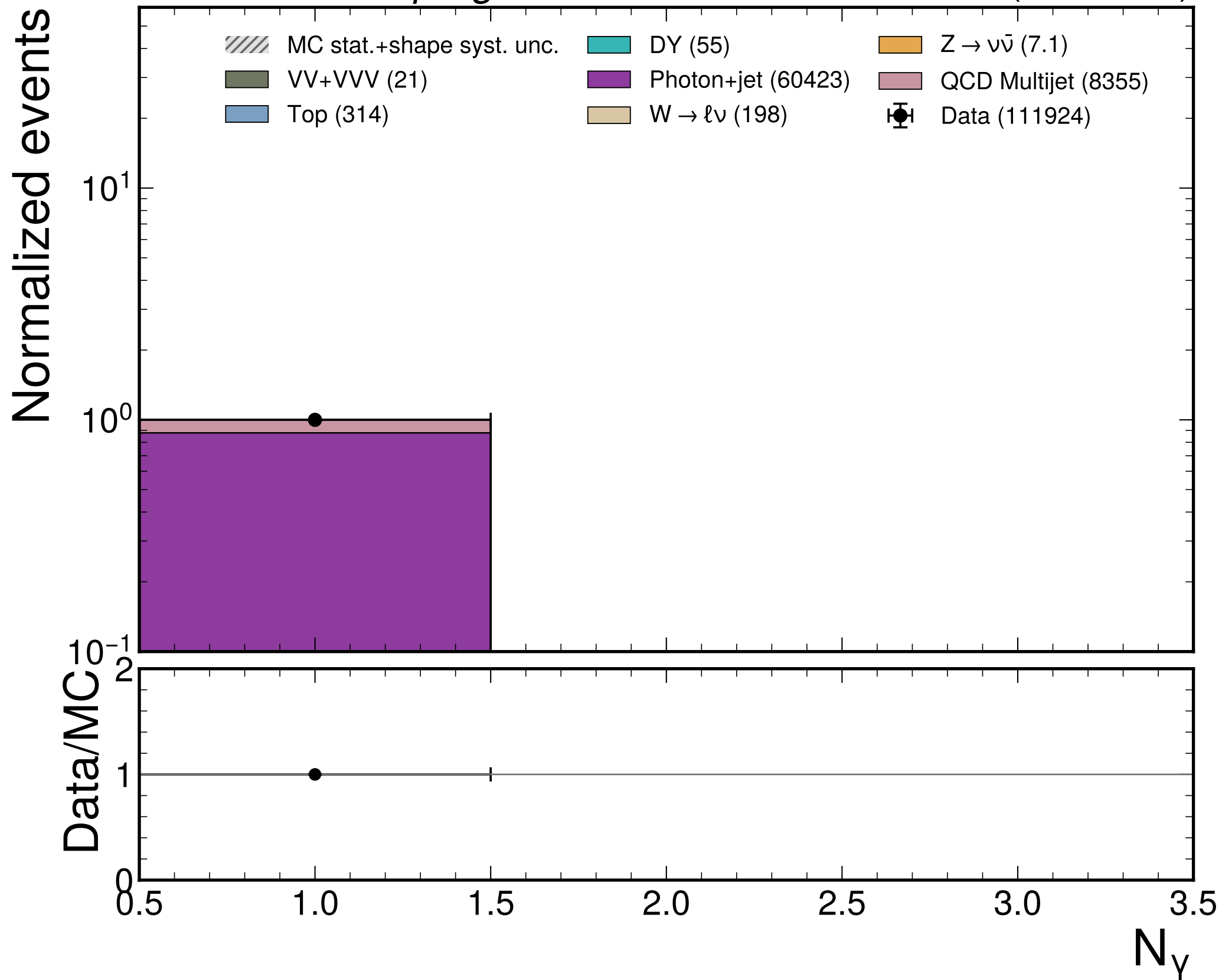
CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)



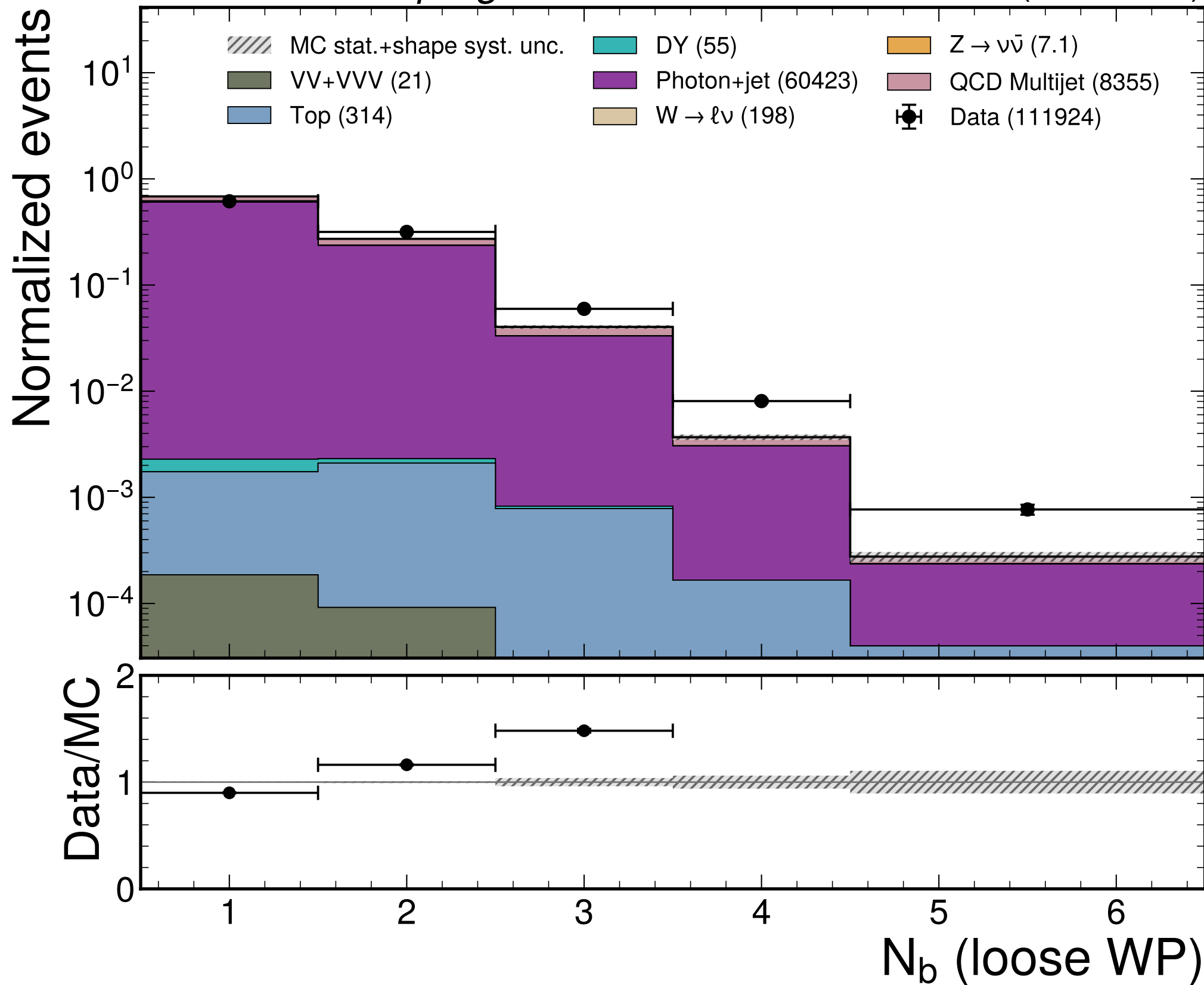
CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)



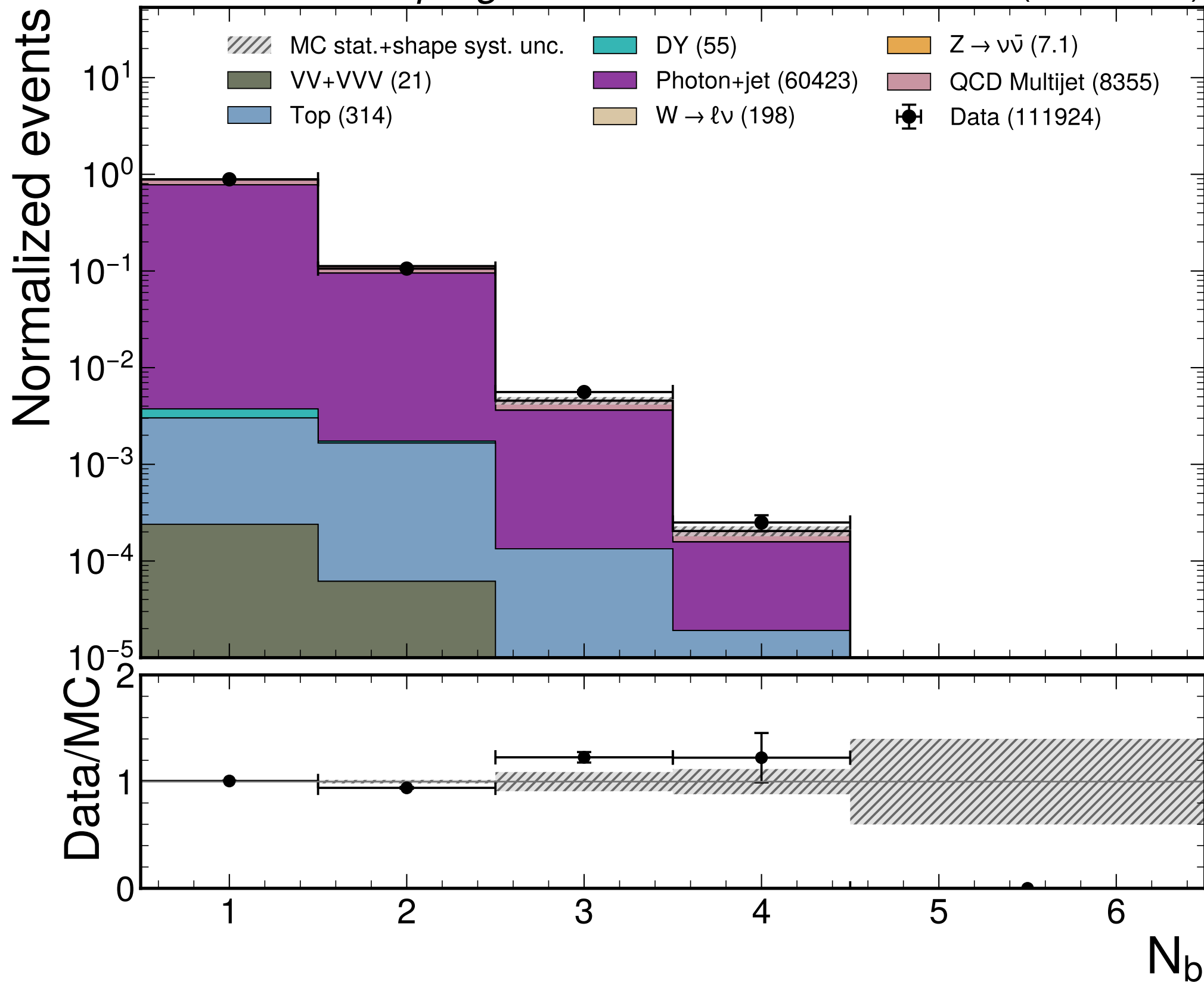
CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)



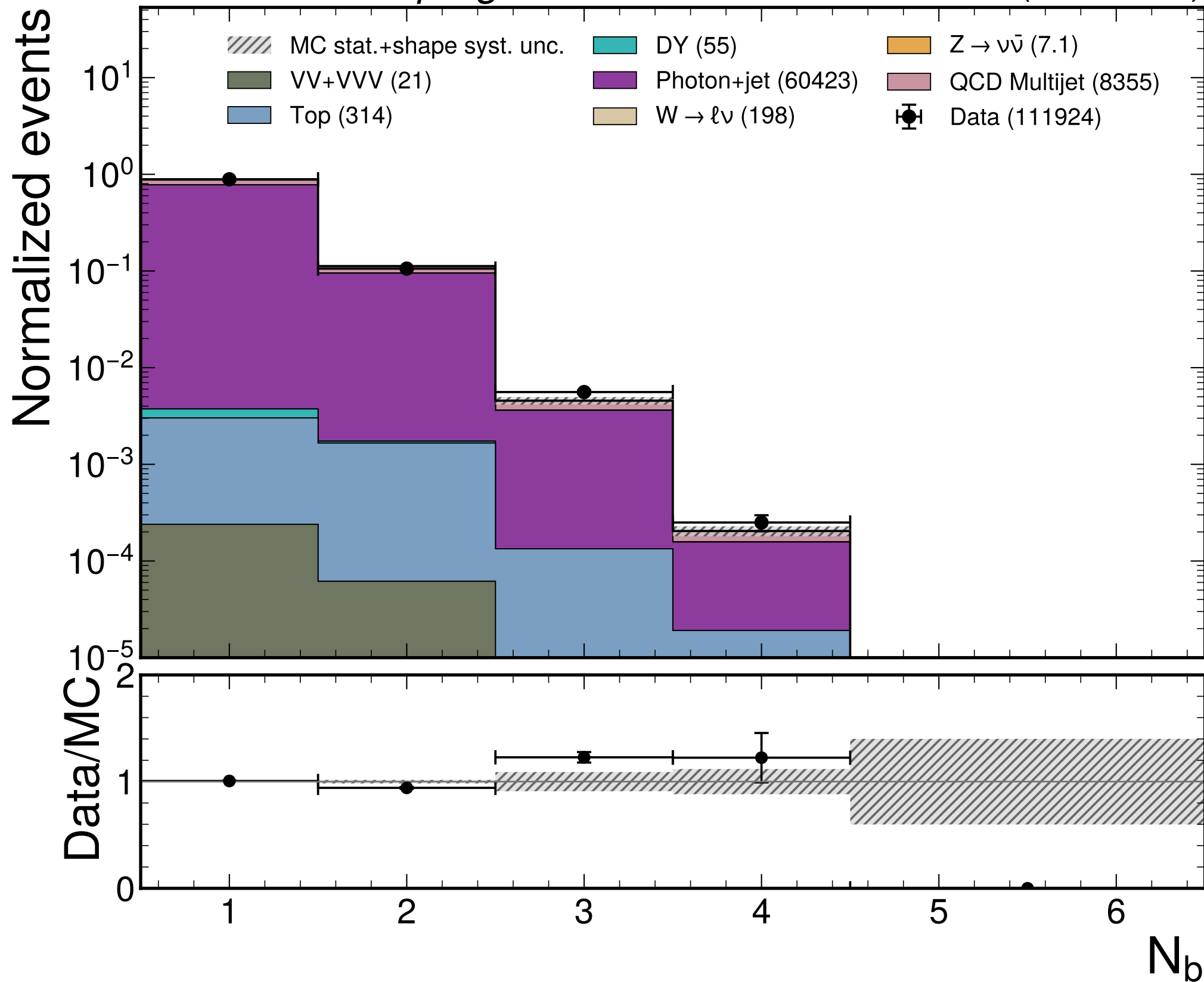
CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)



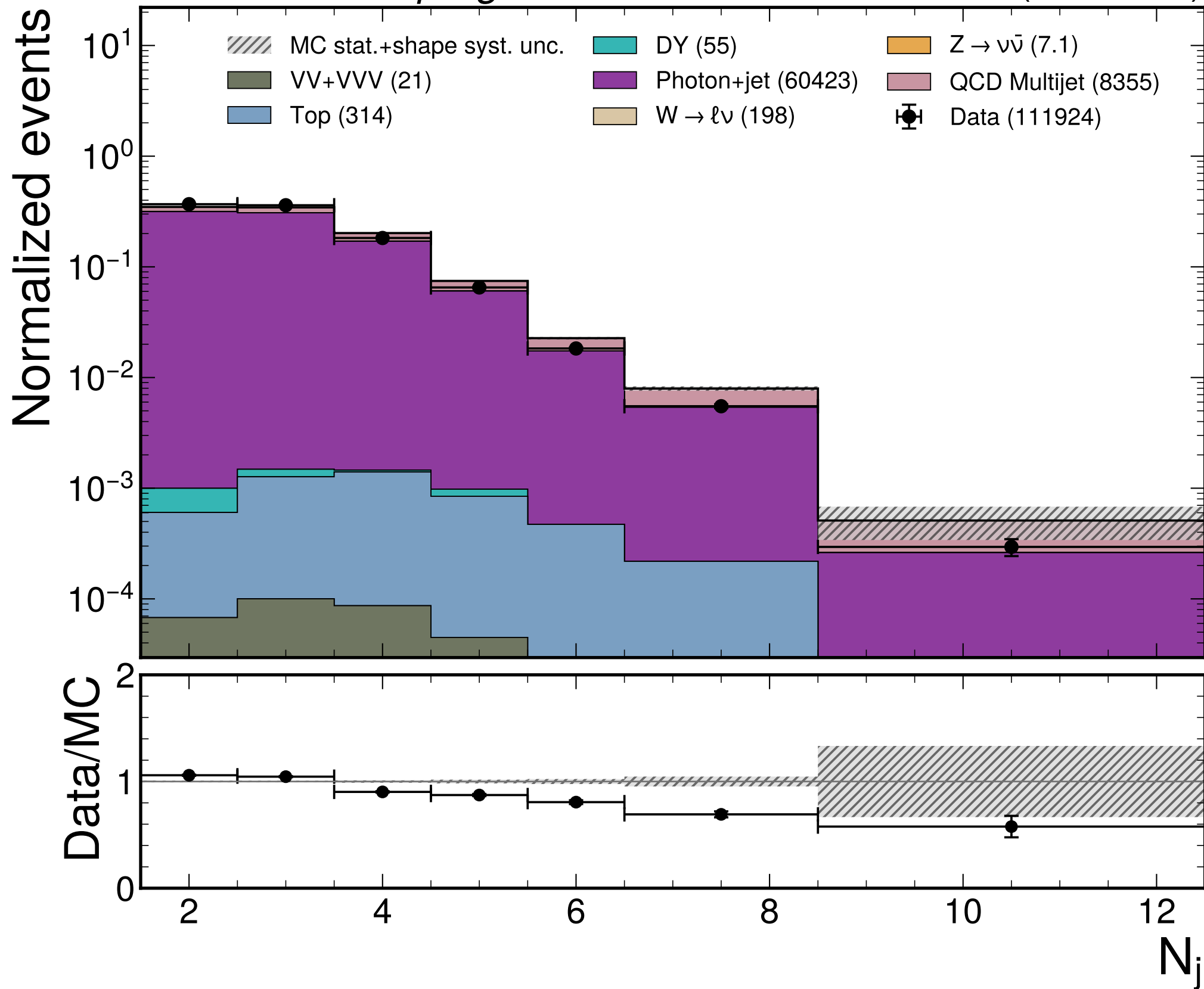
CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)



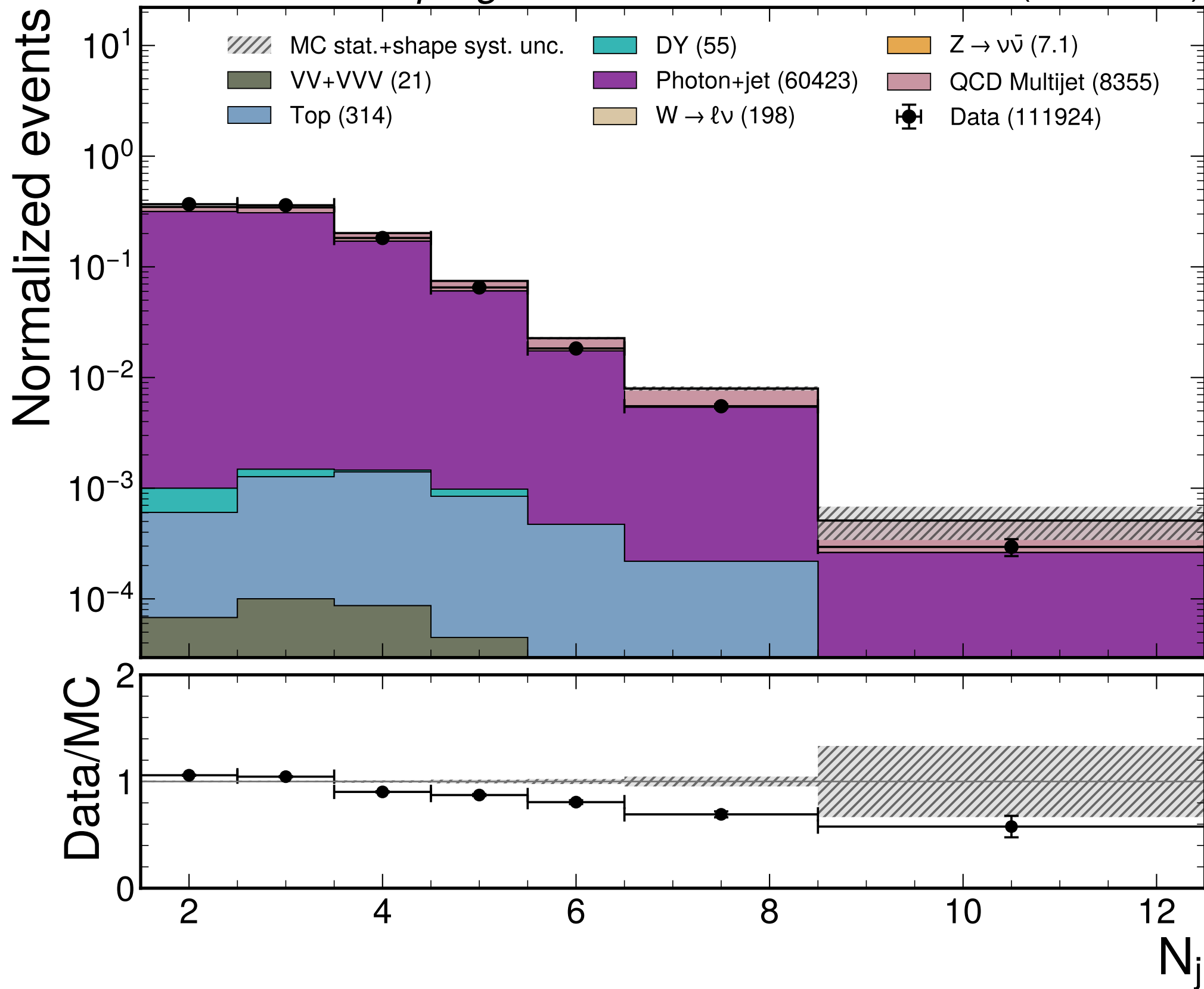
CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)



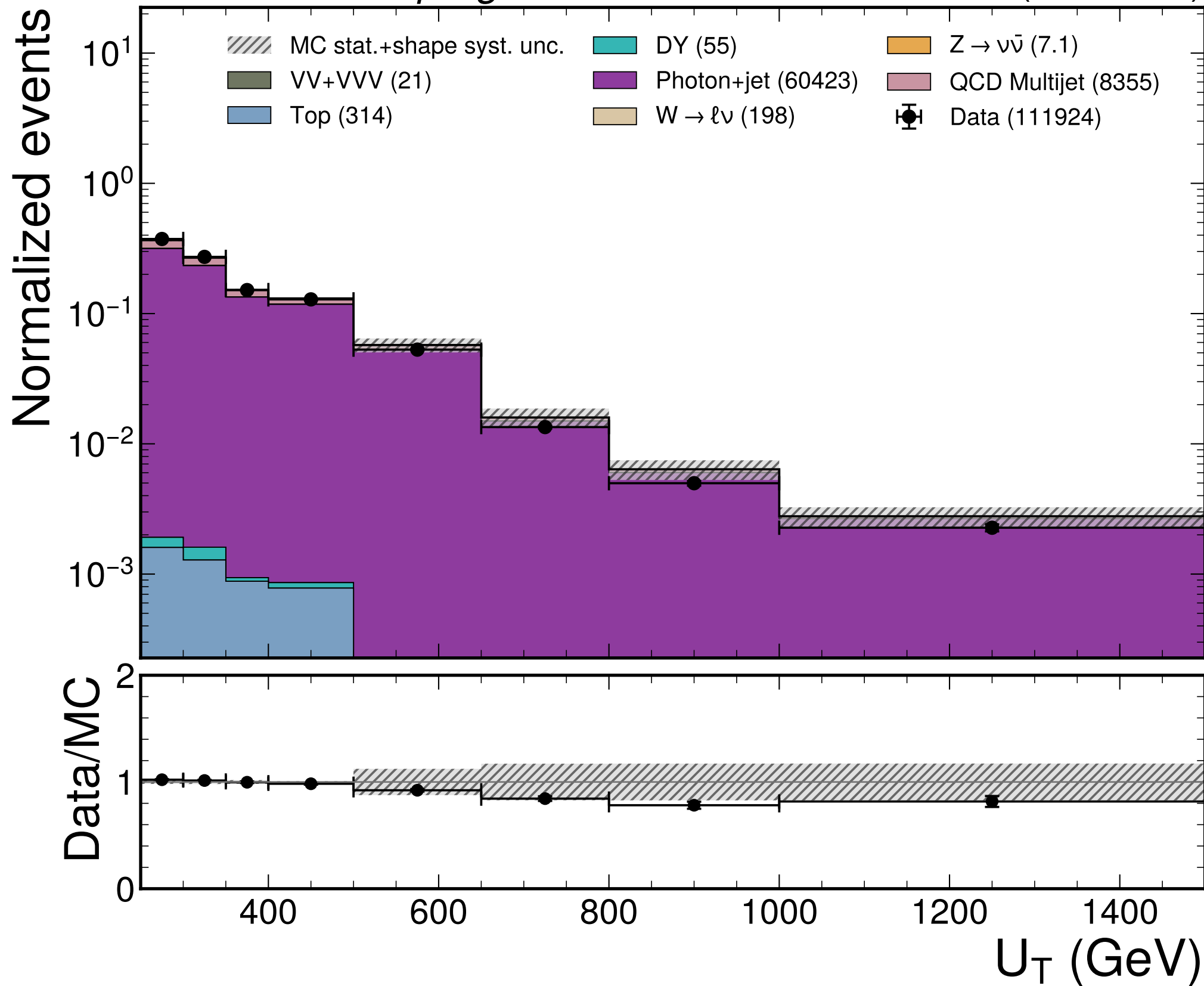
CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)



CMS *Work in progress*

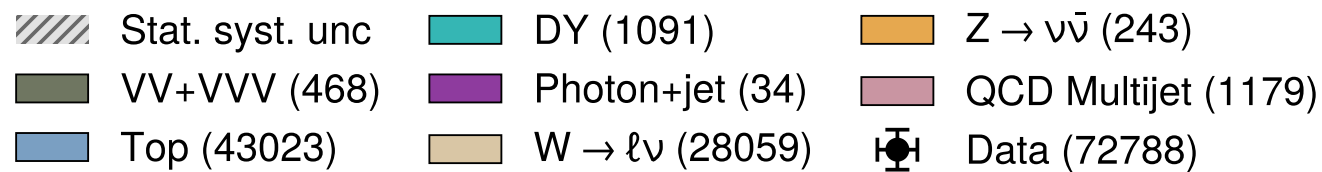
110.84 fb⁻¹ (13.6 TeV)



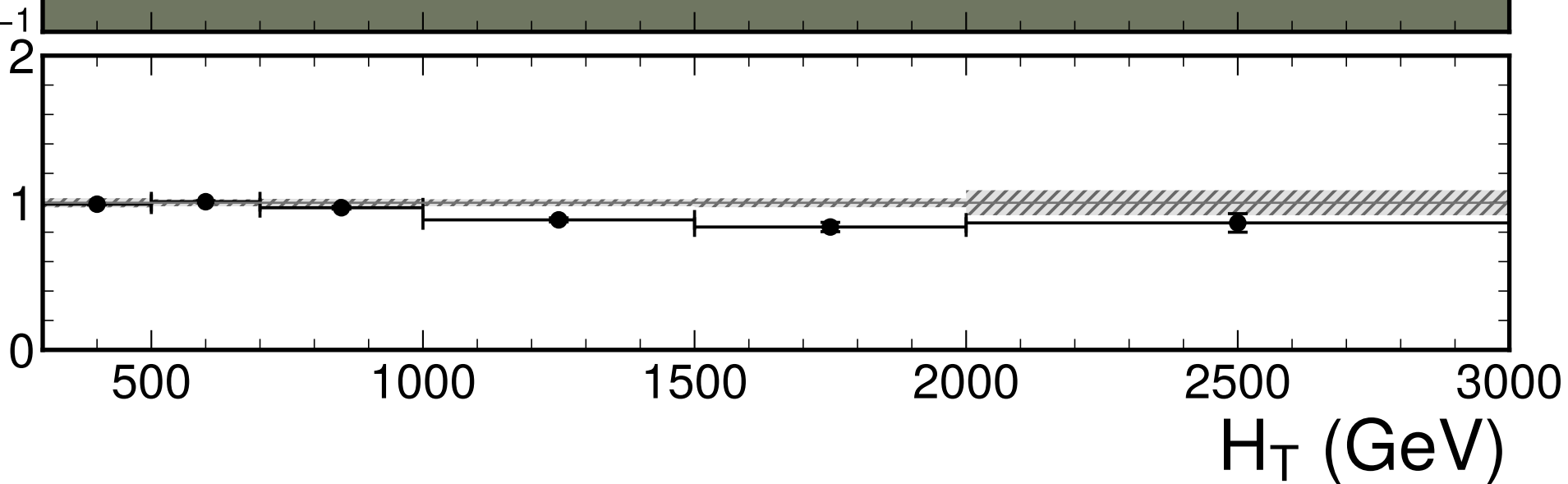
CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)

Events



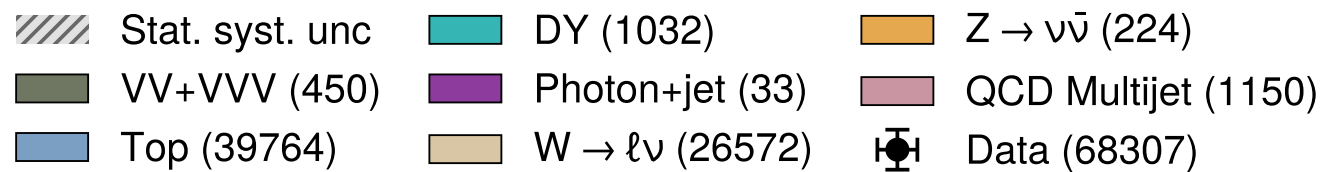
Data/MC



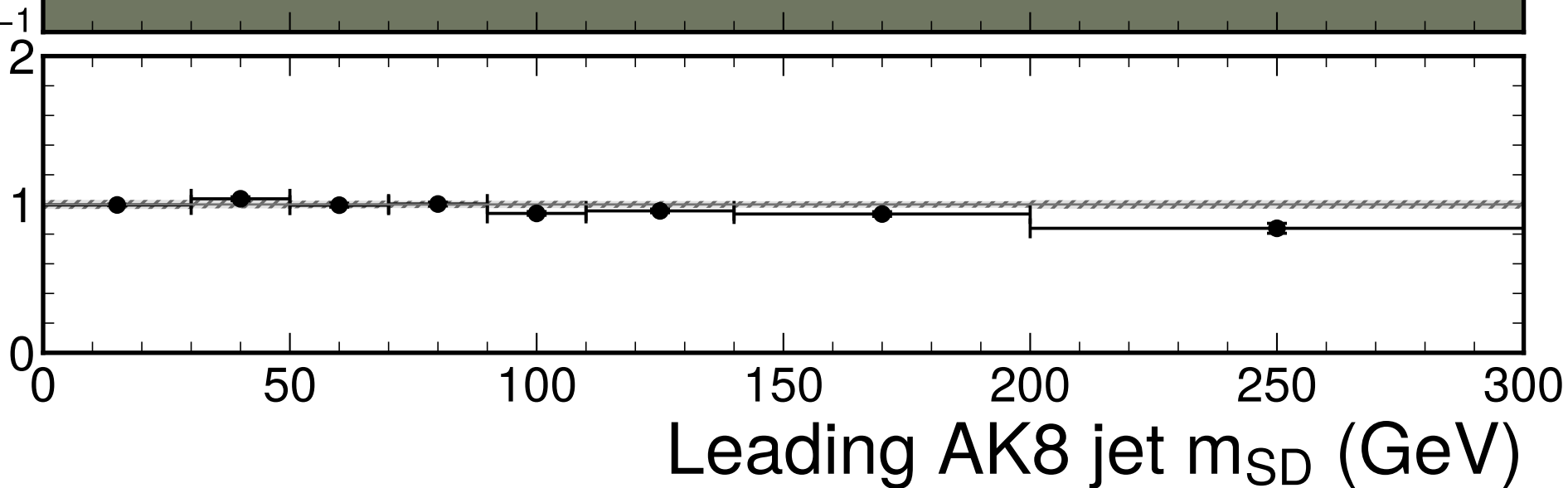
CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)

Events



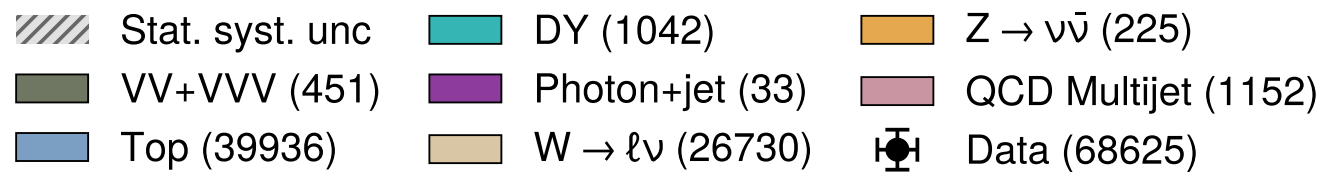
Data/MC



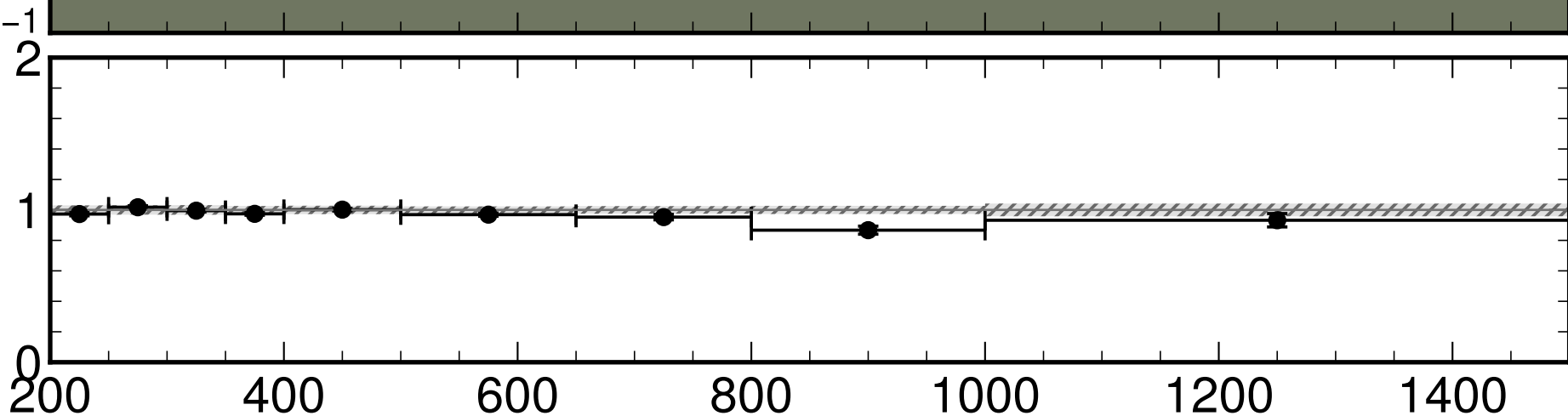
CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)

Events



Data/MC

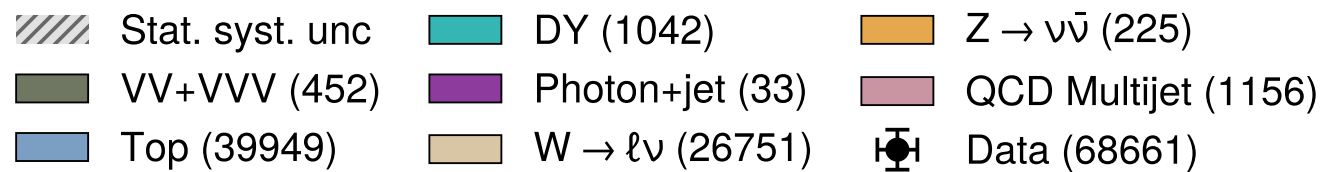


Leading AK8 jet p_T (GeV)

CMS *Work in progress*

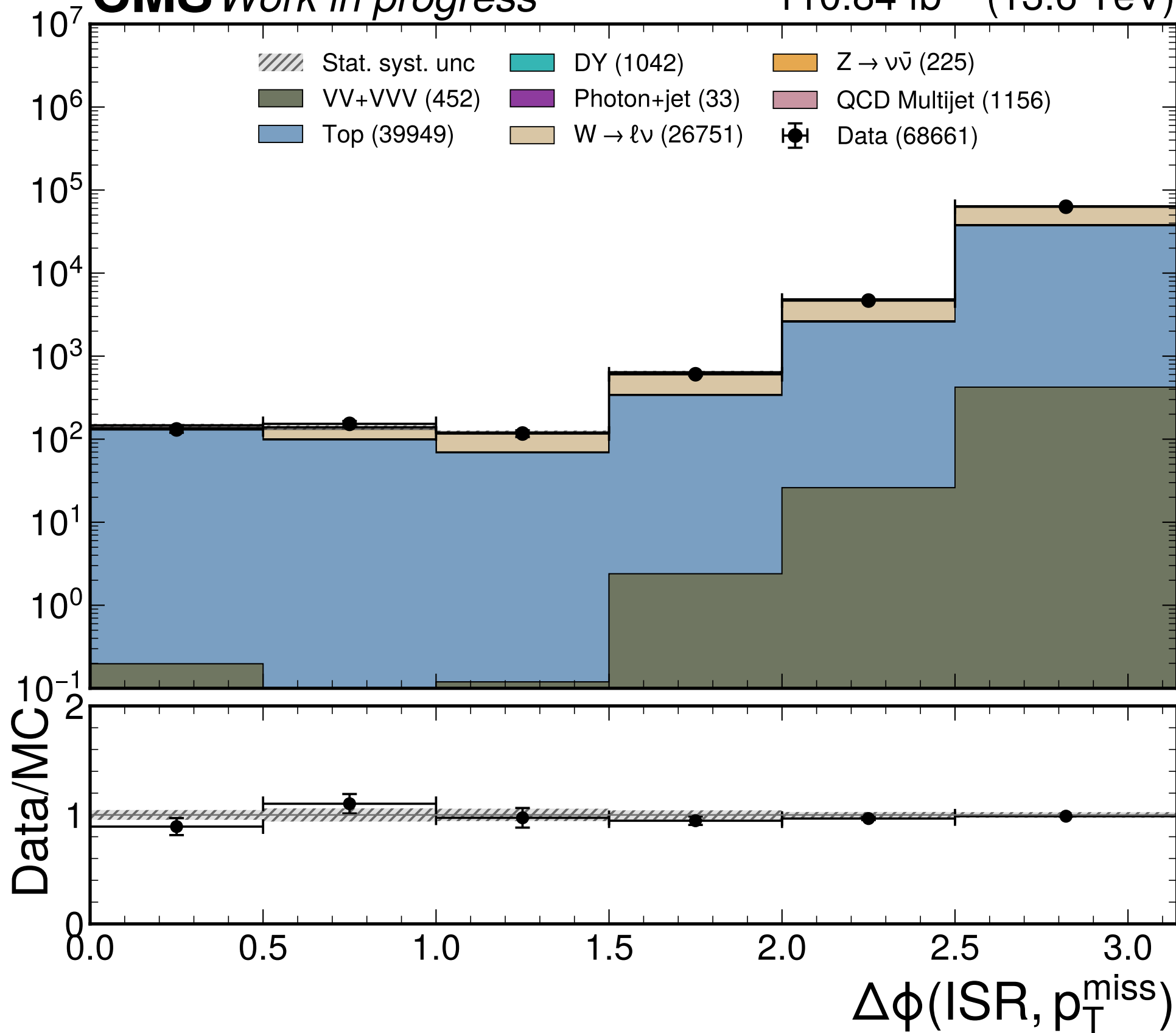
110.84 fb⁻¹ (13.6 TeV)

Events



Data/MC

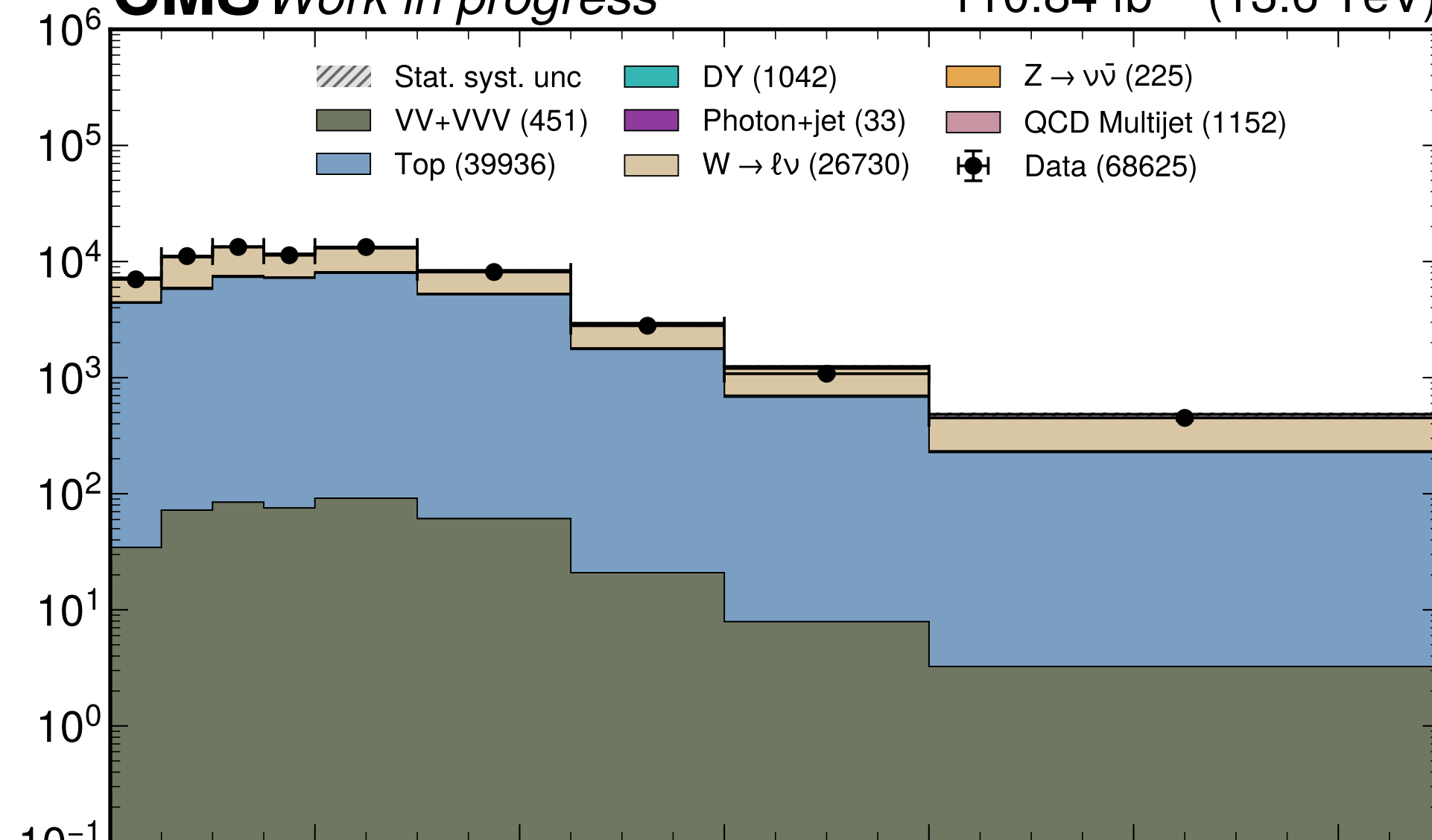
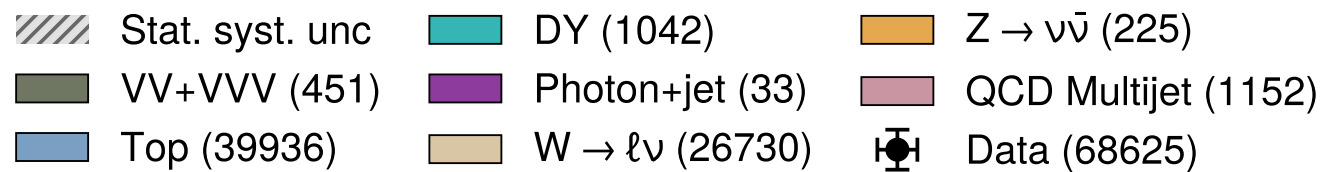
$\Delta\phi(\text{ISR}, p_{\text{T}}^{\text{miss}})$



CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)

Events



Data/MC

200

400

600

800

1000

1200

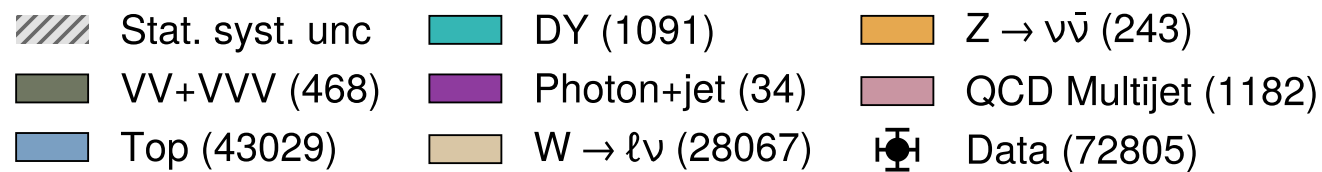
1400

p_T^{ISR} (GeV)

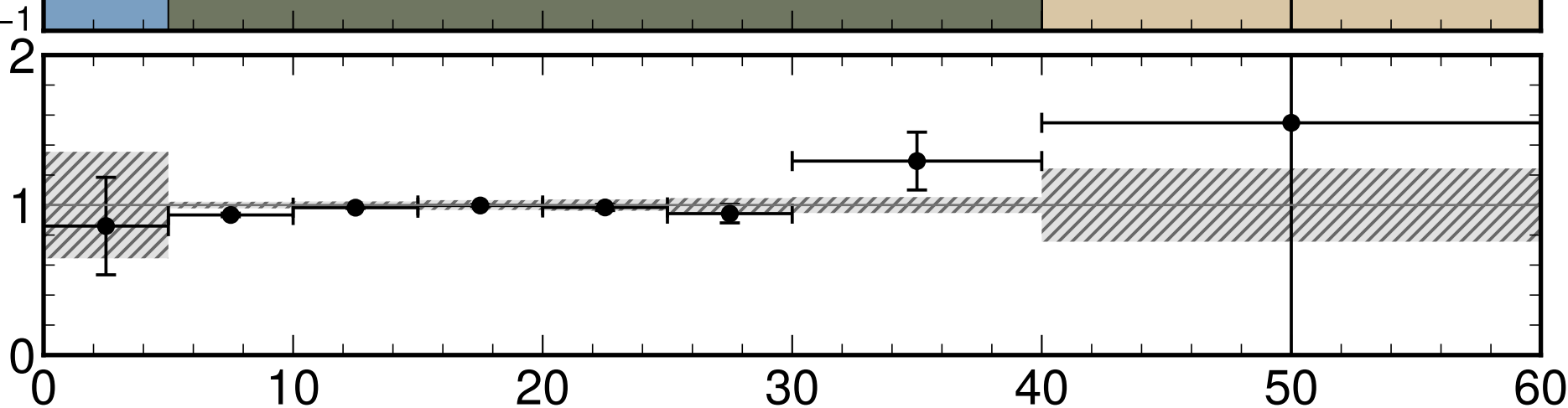
CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)

Events



Data/MC

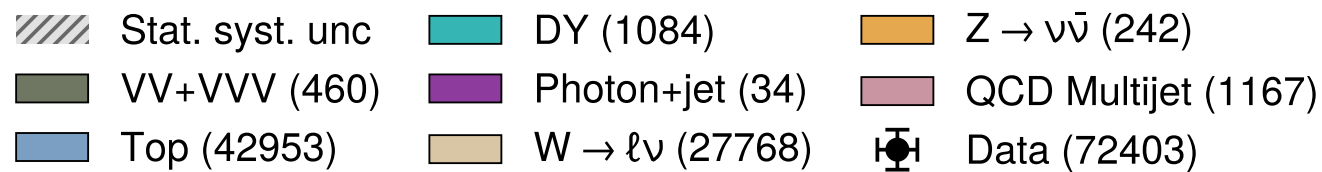


$p_T^{\text{miss}}/\sqrt{H_T}$

CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)

Events

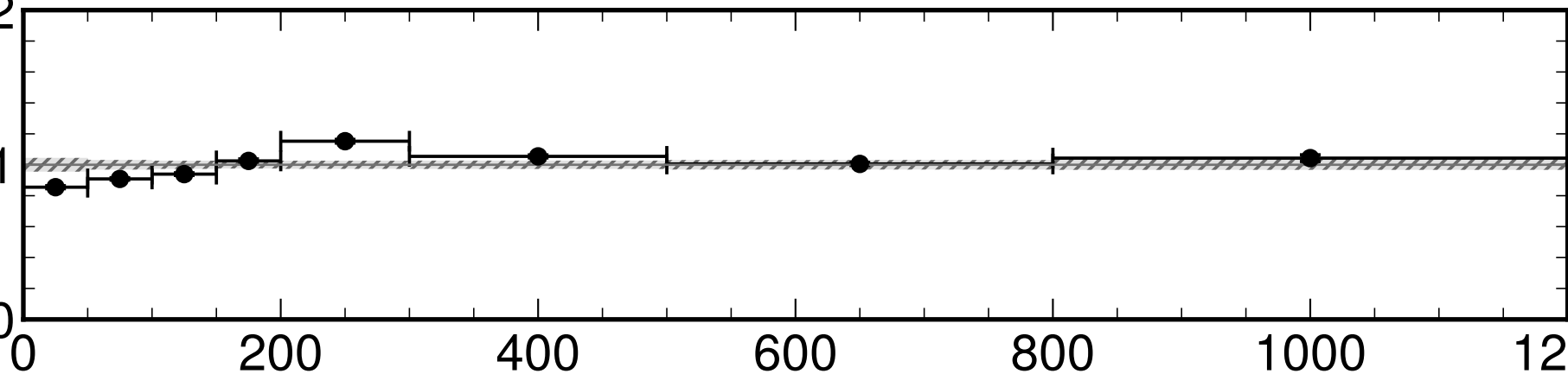


Data/MC

10⁻¹

2

0

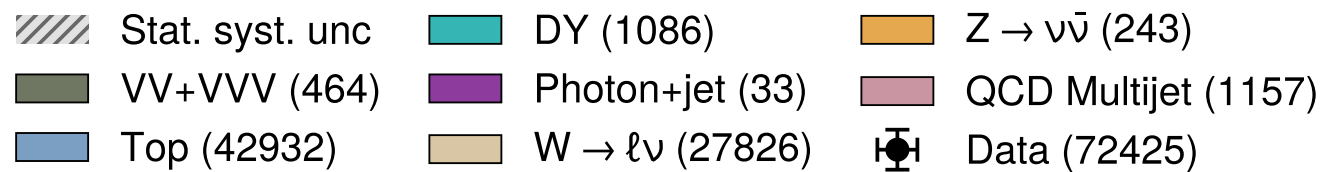


m_T^b (GeV)

CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)

Events



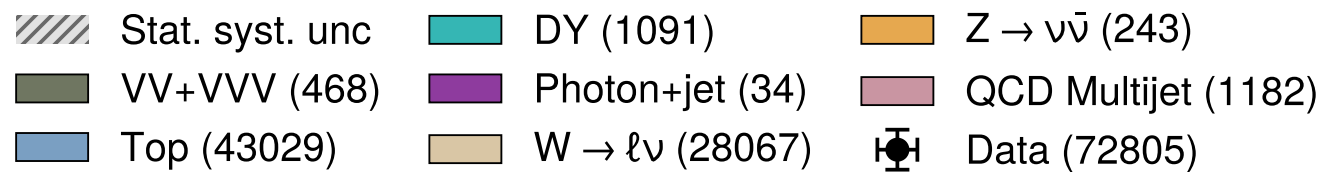
Data/MC

p_T^b (GeV)

CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)

Events



Data/MC

10⁻²

0

400

600

800

1000

1200

1400

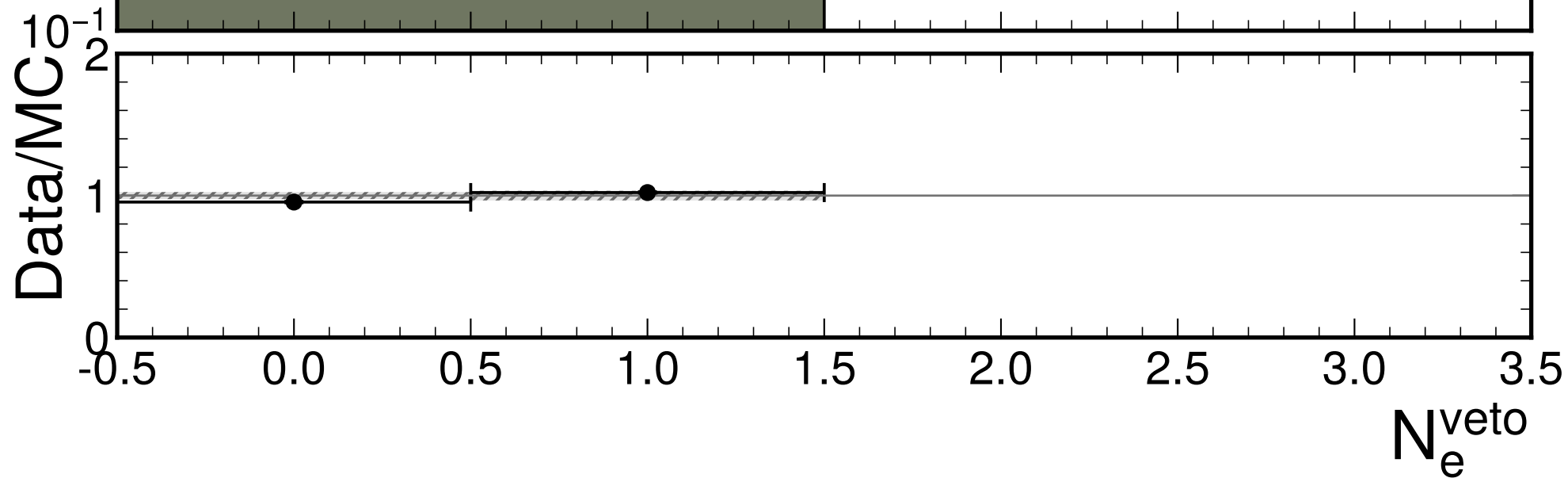
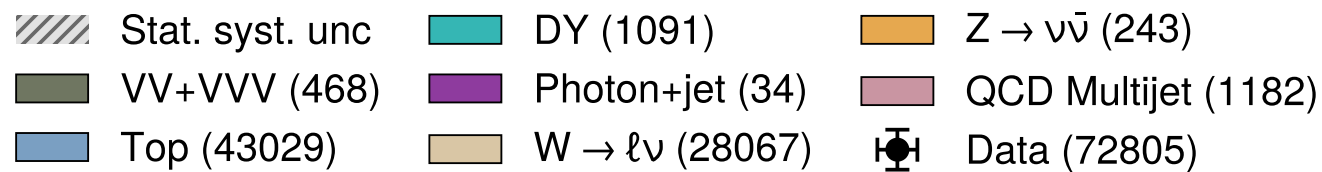
p_T^{miss} (GeV)

Detailed description: The main plot is a stacked histogram showing event counts on a logarithmic y-axis (from 10⁰ to 10⁷) versus p_T^{miss} (GeV) on a linear x-axis (from 0 to 1500). The background components are stacked: VV+VVV (dark olive green), Top (blue), W → ℓν (tan), Z → νν̄ (orange), and QCD Multijet (pink). Data points are shown as black circles with error bars. The bottom plot shows the ratio Data/MC on a linear y-axis (from 0 to 2) versus p_T^{miss} (GeV) on a linear x-axis. A horizontal line at 1.0 represents the expected ratio, with a hatched band indicating the statistical and systematic uncertainty. Data points with error bars are plotted, showing a slight deviation from 1.0 at higher p_T^{miss}.

CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)

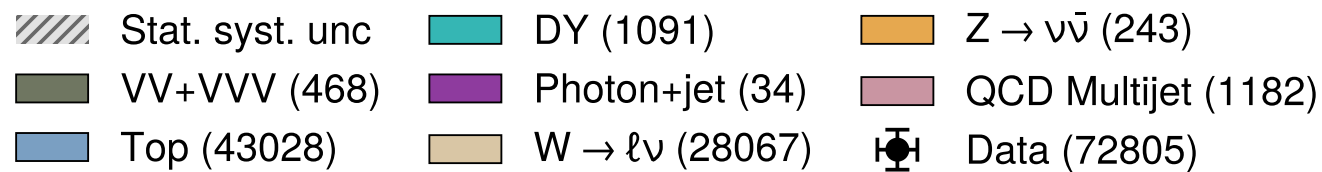
Events



CMS *Work in progress*

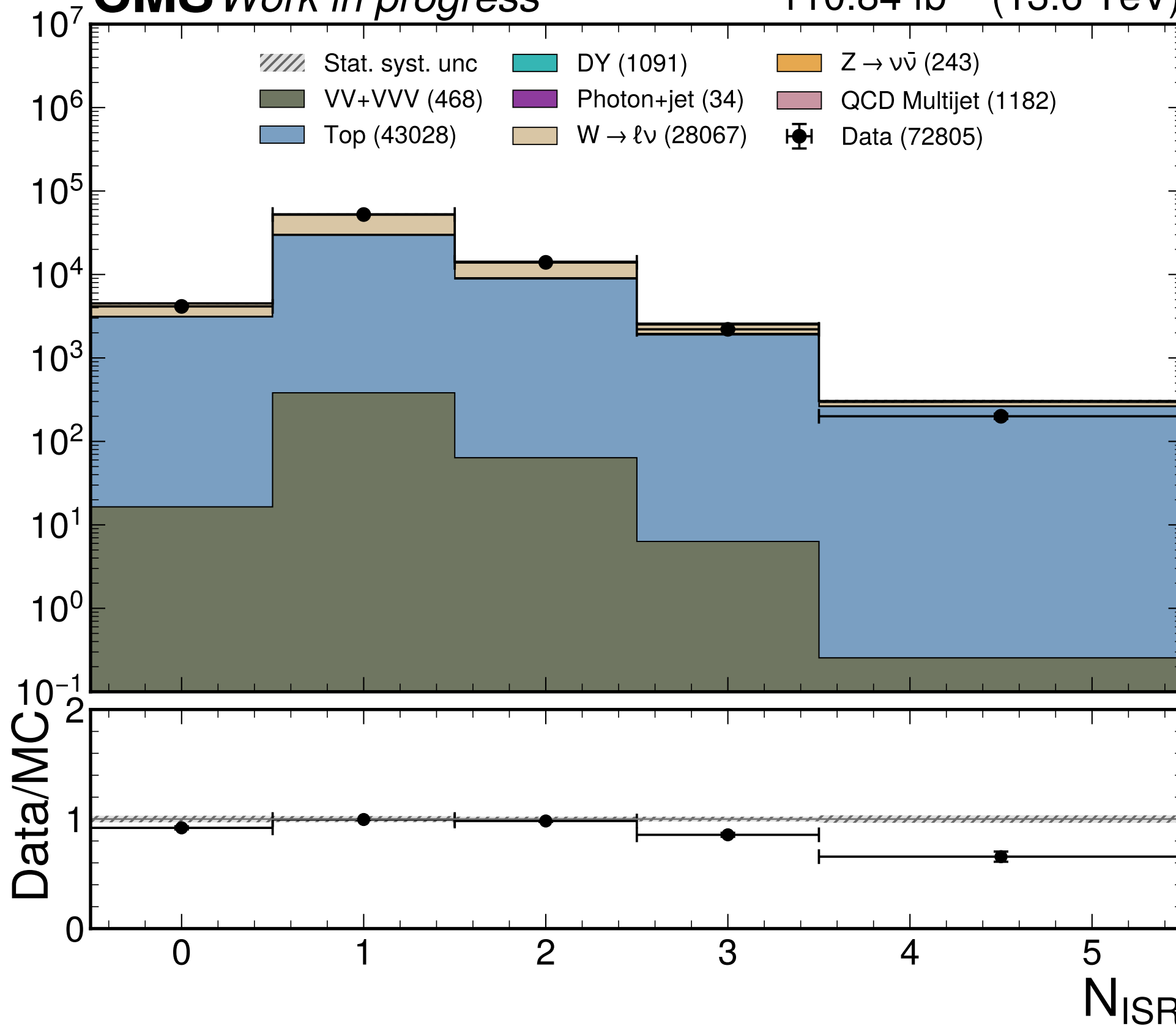
110.84 fb⁻¹ (13.6 TeV)

Events



Data/MC

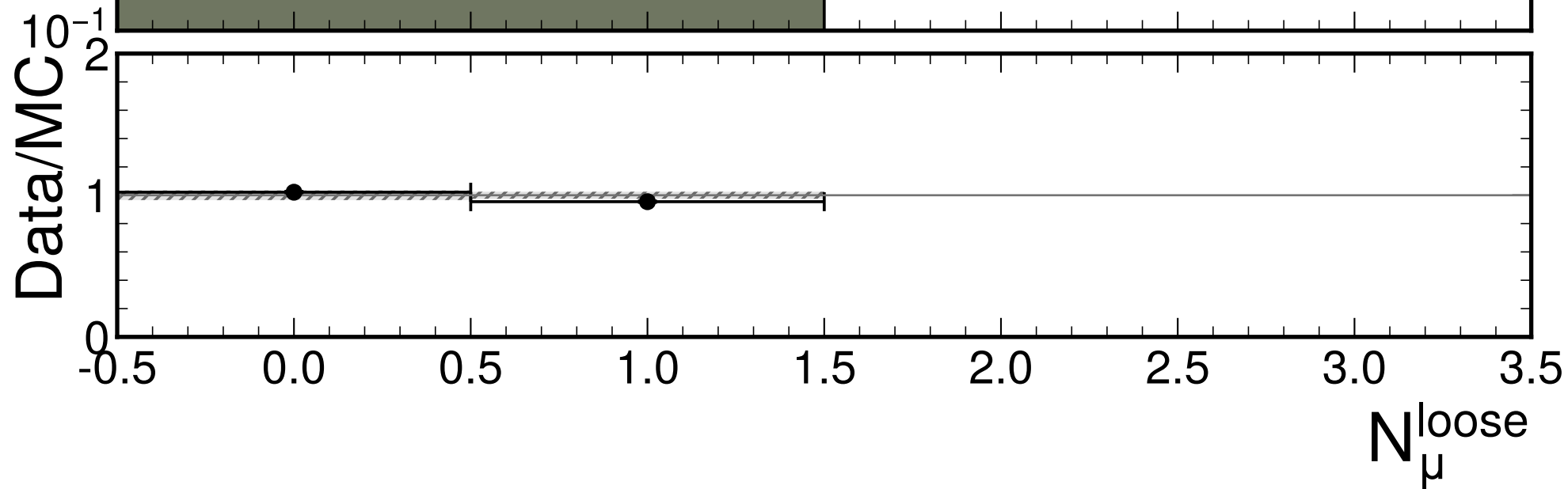
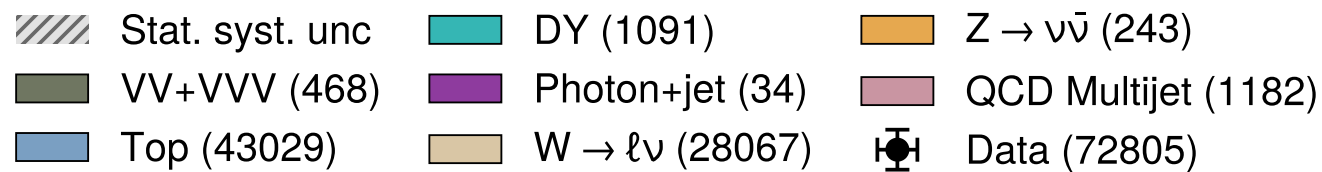
N_{ISR}



CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)

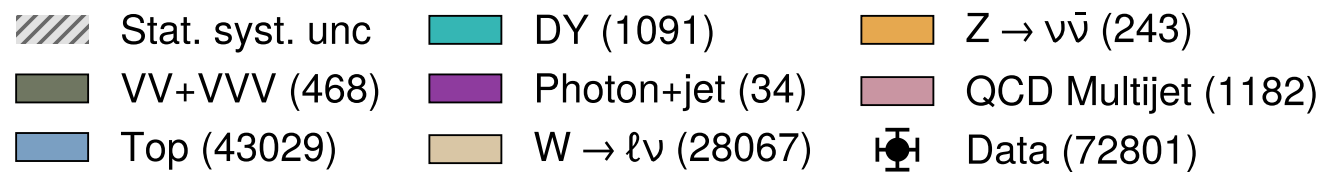
Events



CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)

Events



Data/MC

10⁻¹
2
1
0

1

2

3

4

5

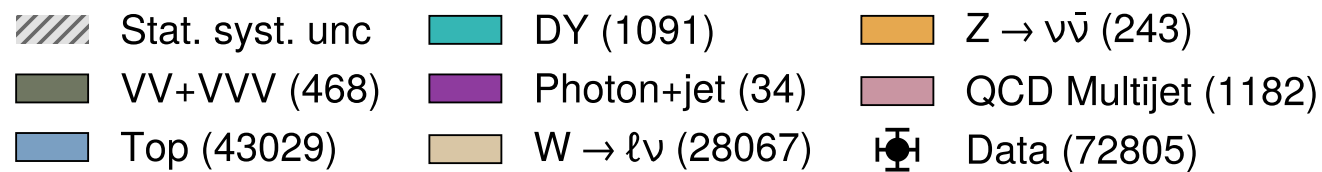
6

N_b (loose WP)

CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)

Events



Data/MC

10⁻¹
2
1
0

1

2

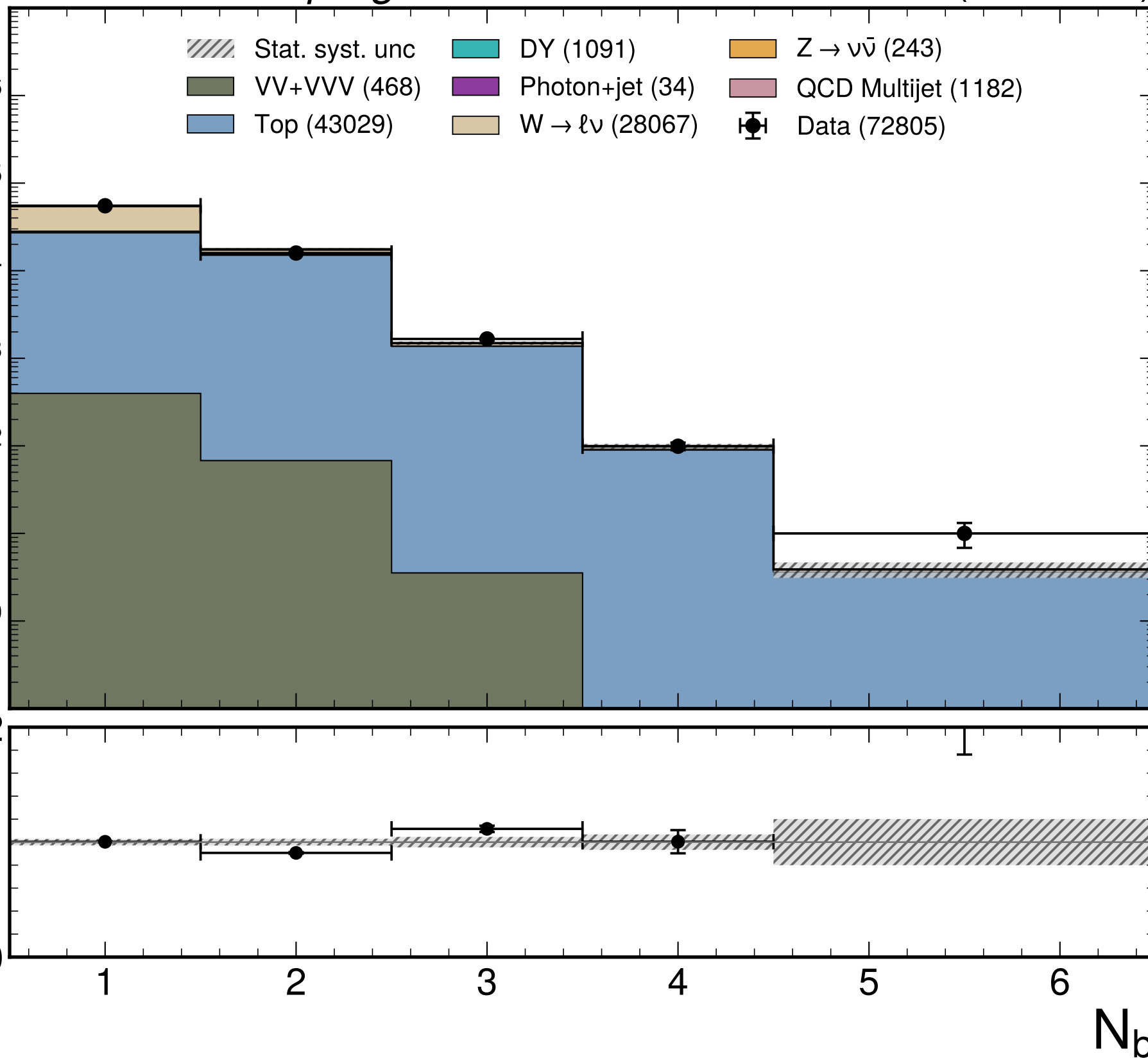
3

4

5

6

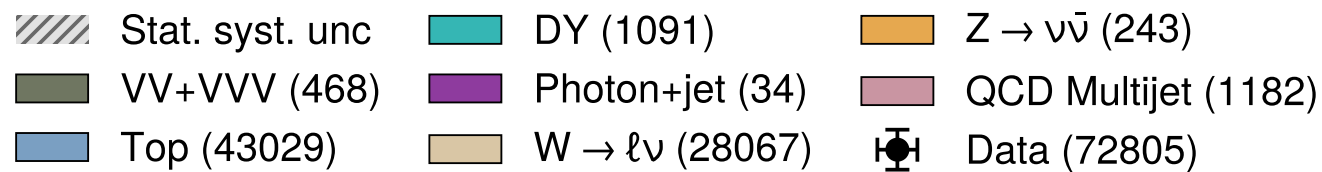
N_b



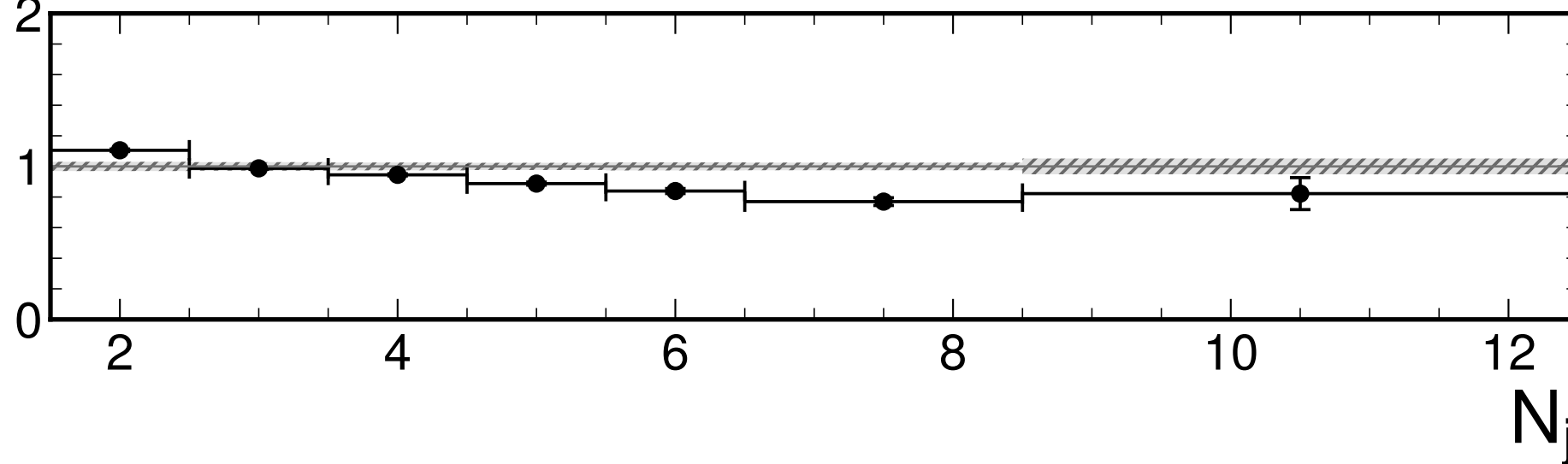
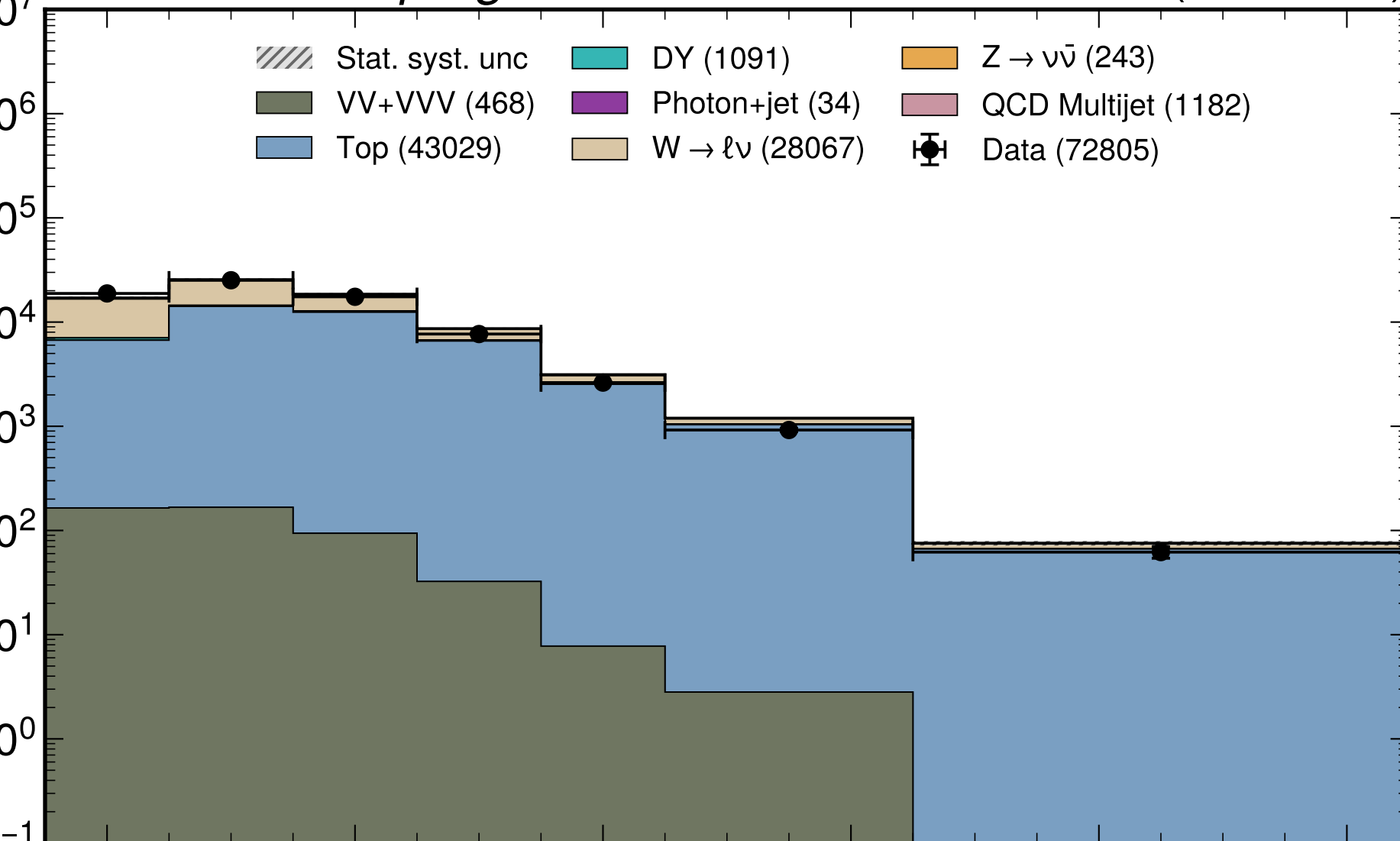
CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)

Events



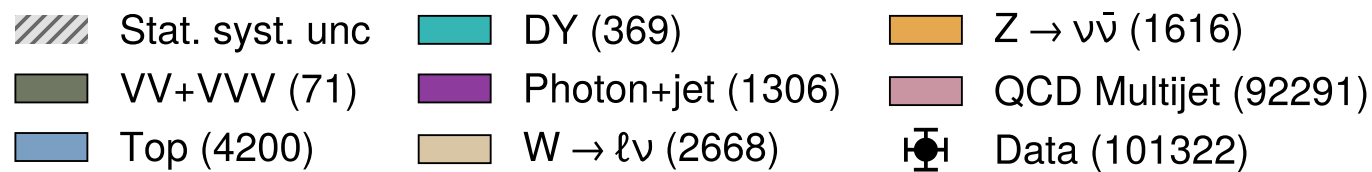
Data/MC



CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)

Events



Data/MC

10⁻¹

2

0

500

1000

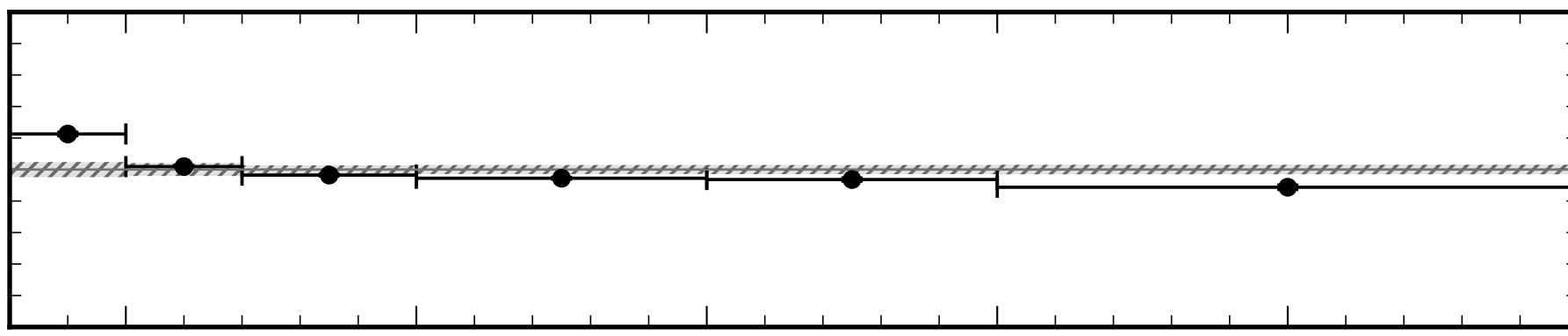
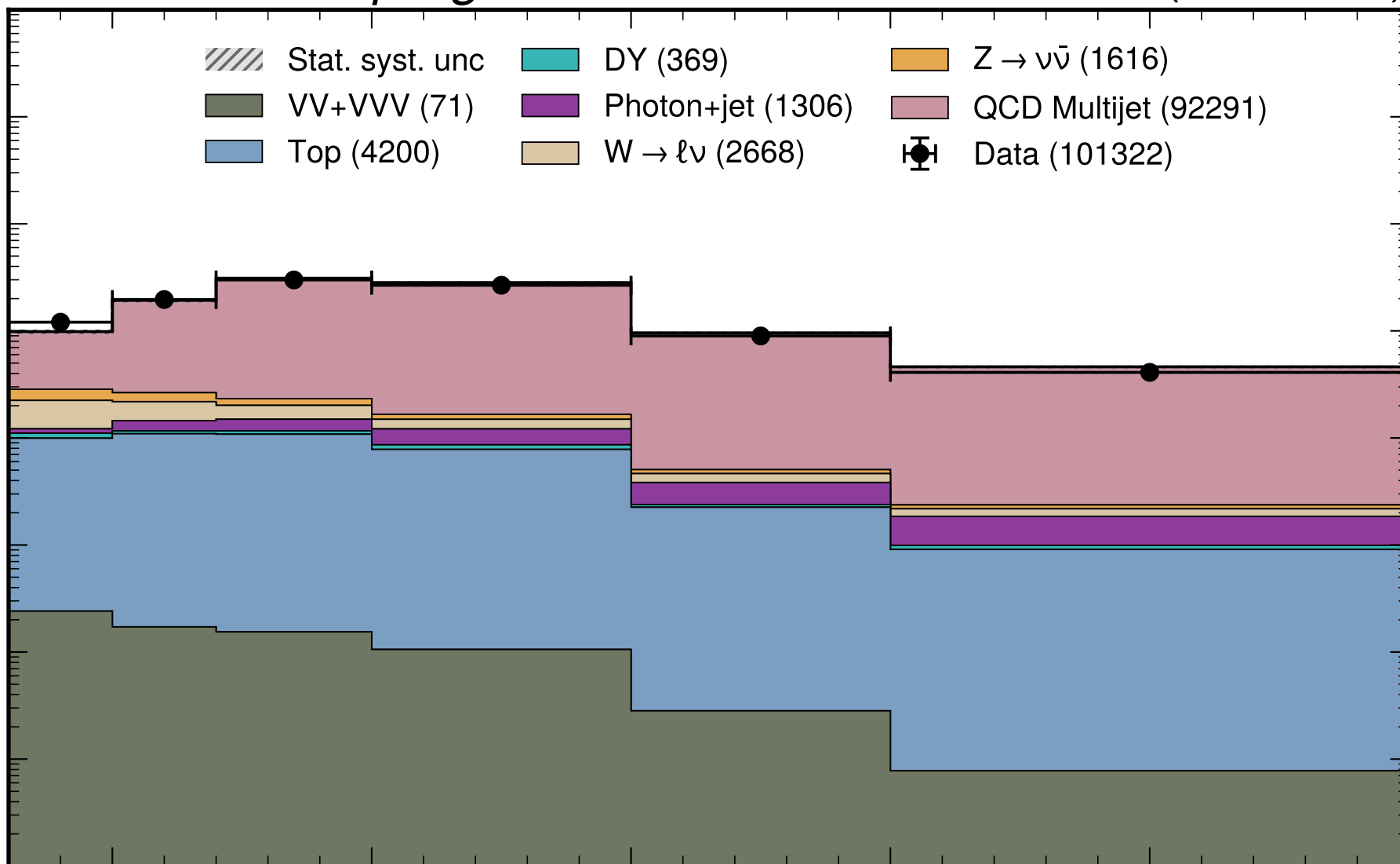
1500

2000

2500

3000

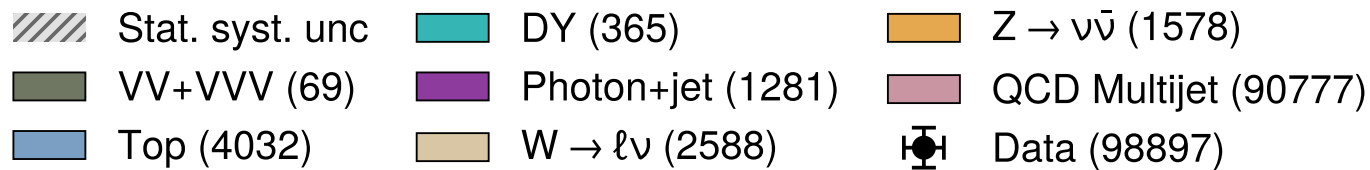
H_T (GeV)



CMS *Work in progress*

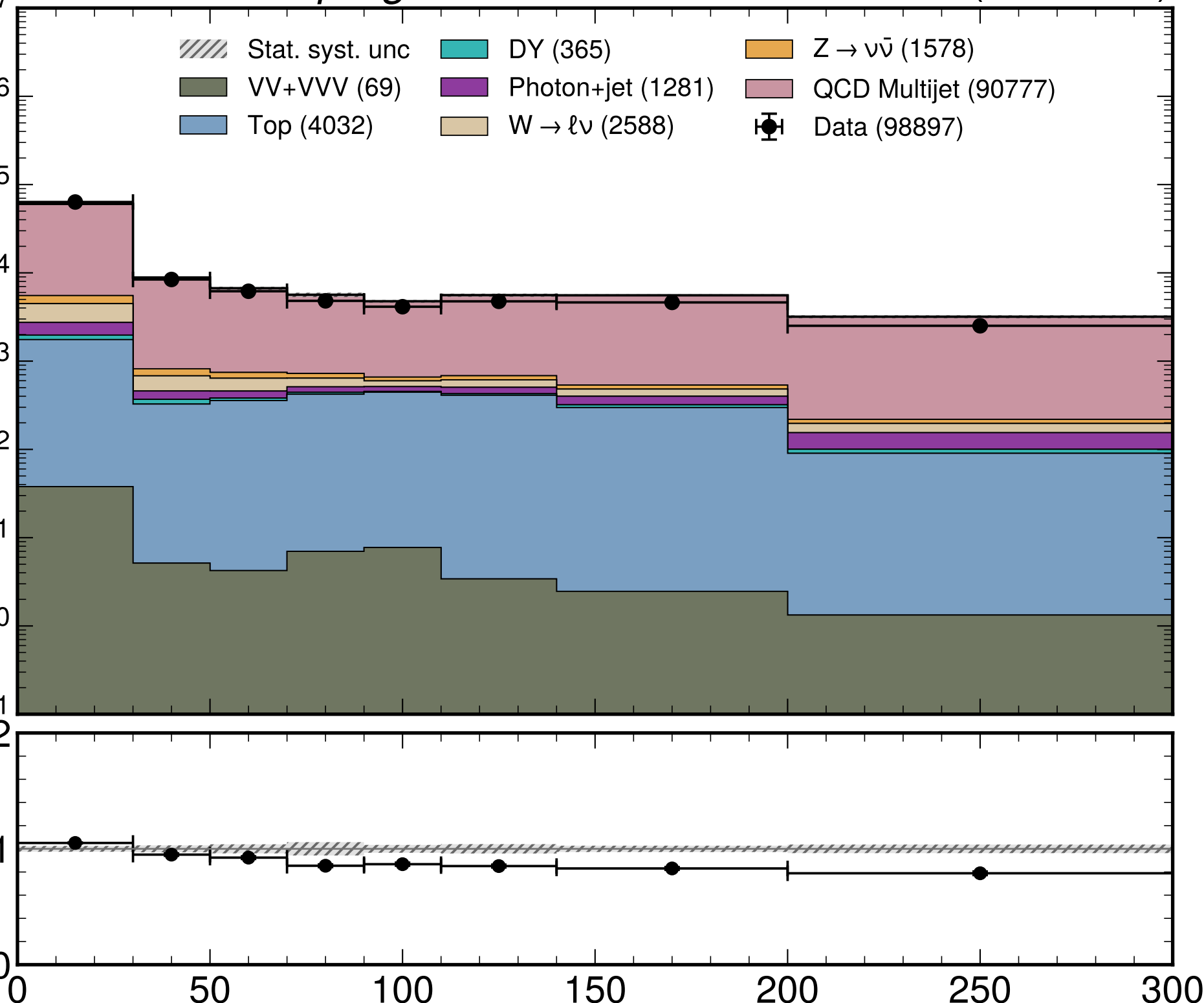
110.84 fb⁻¹ (13.6 TeV)

Events



Data/MC

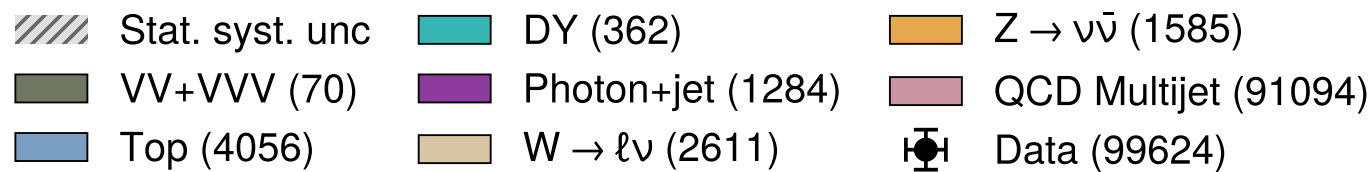
Leading AK8 jet m_{SD} (GeV)



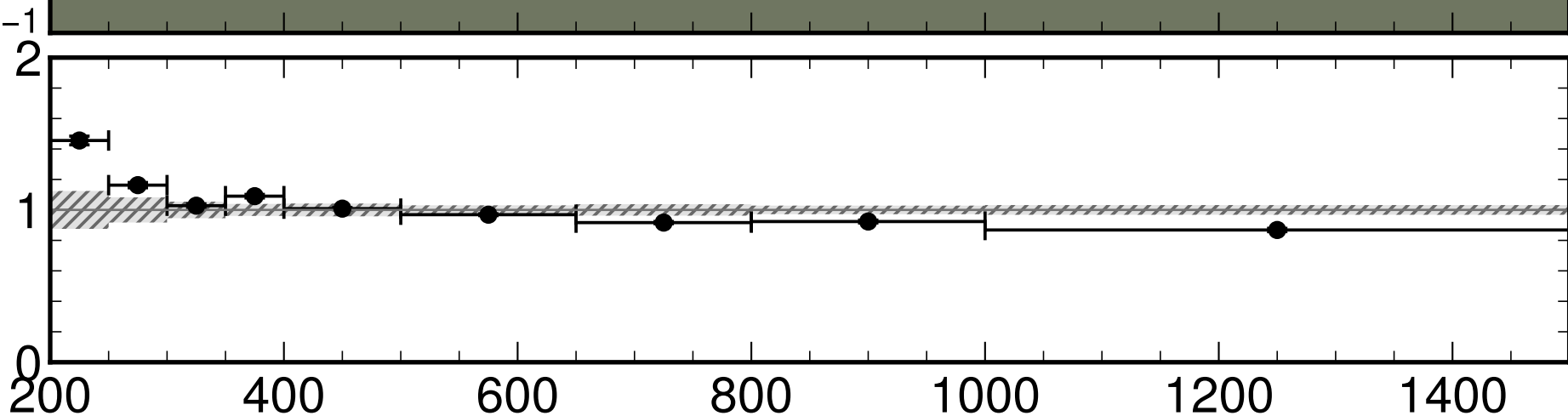
CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)

Events



Data/MC

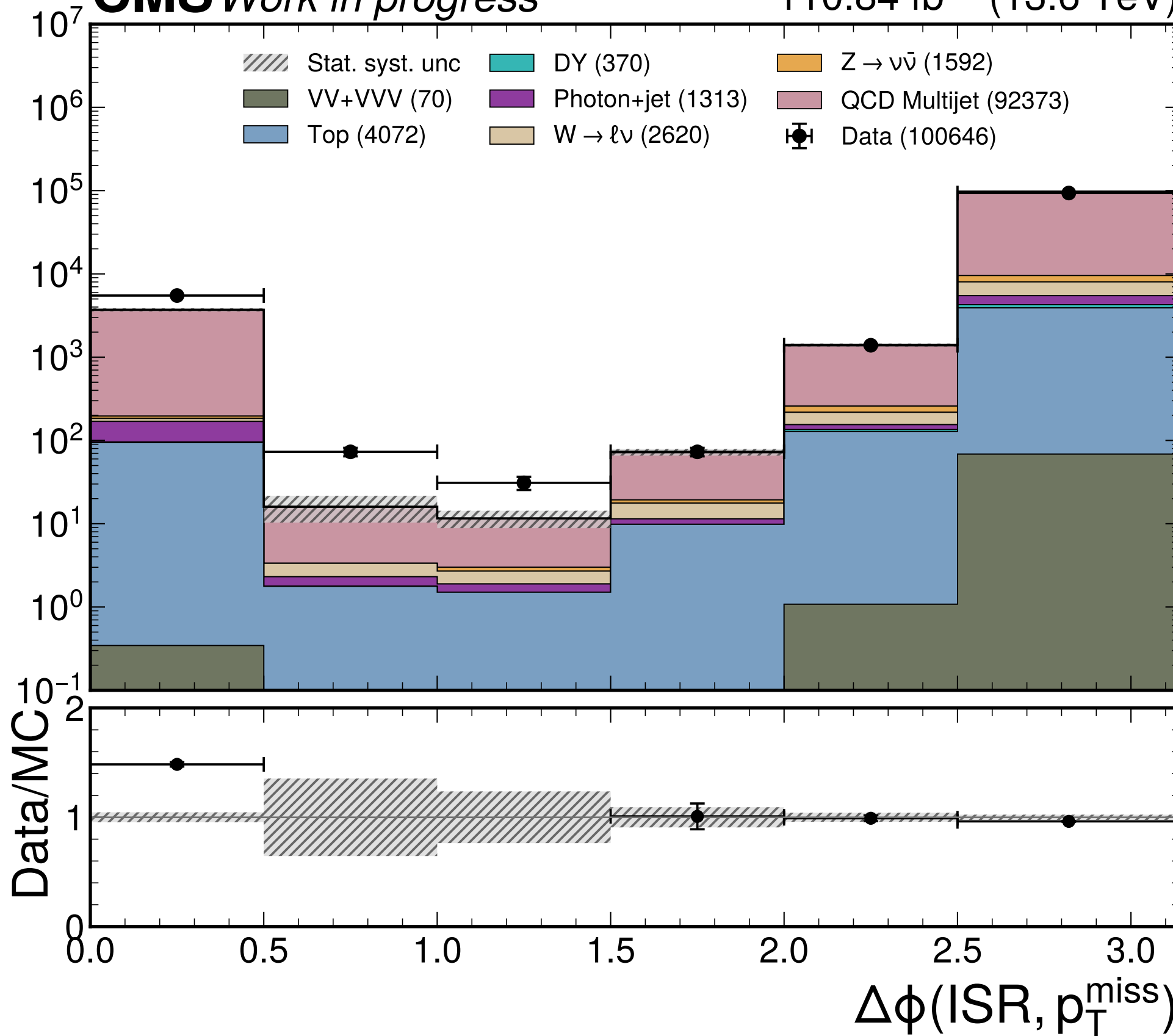
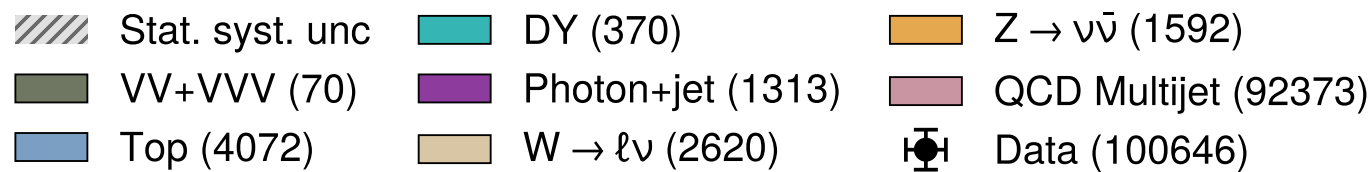


Leading AK8 jet p_T (GeV)

CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)

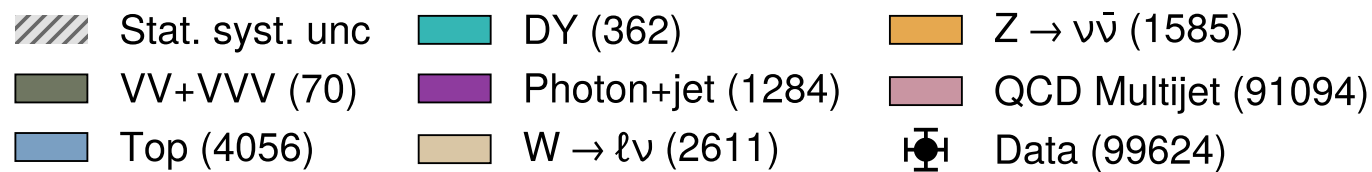
Events



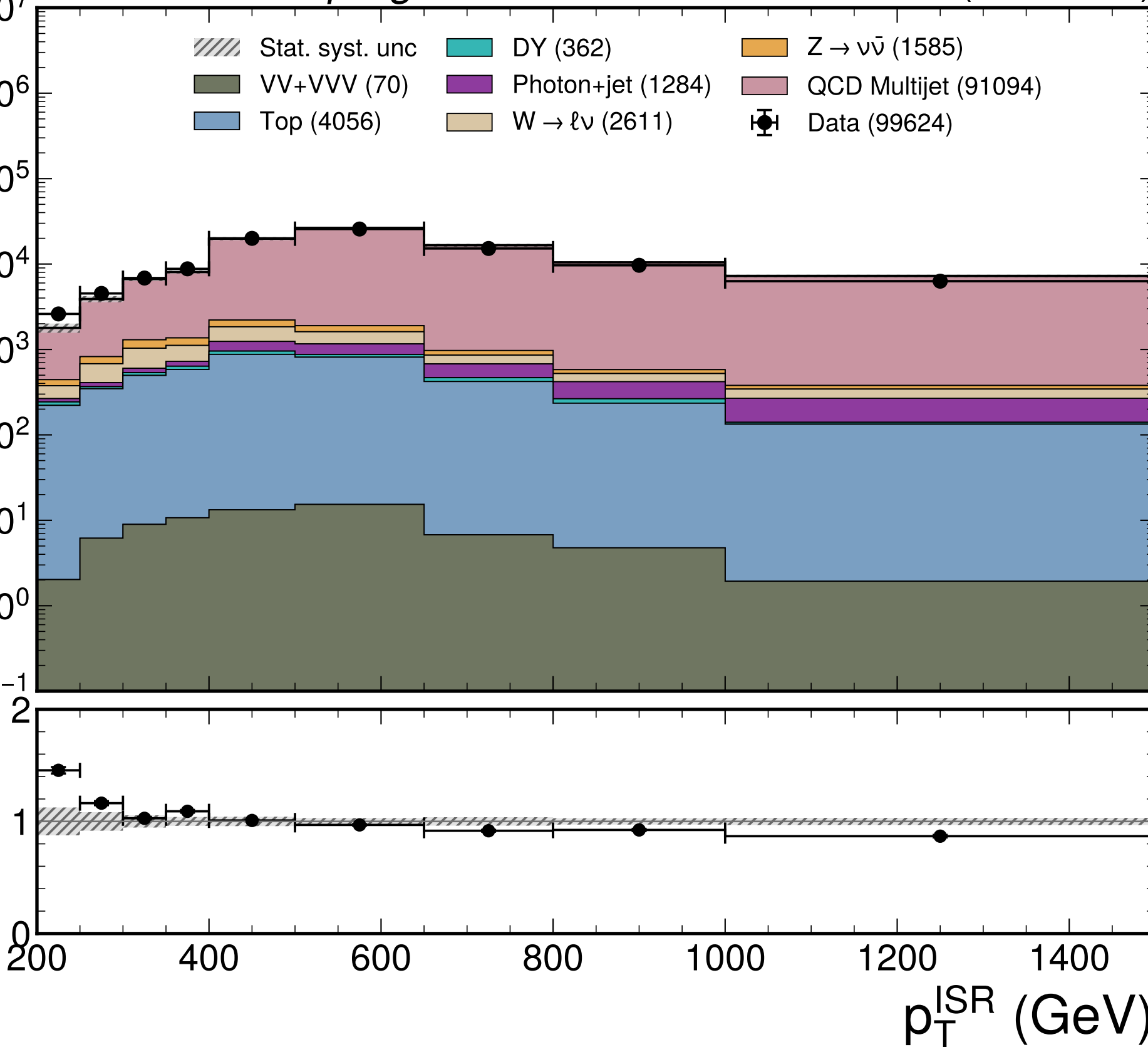
CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)

Events



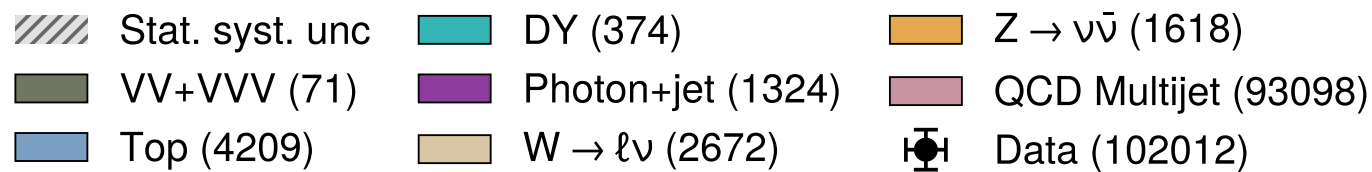
Data/MC



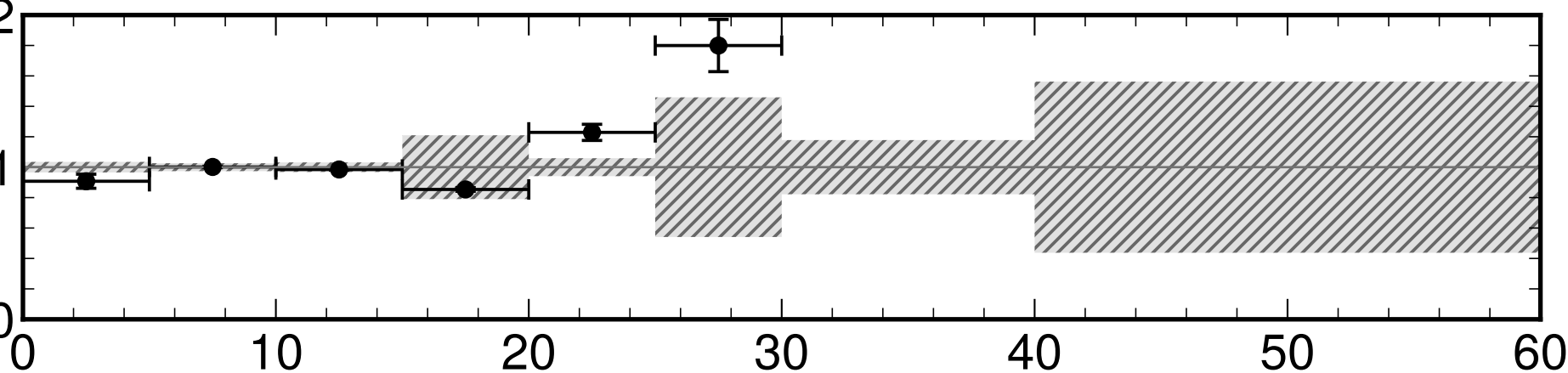
CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)

Events



Data/MC

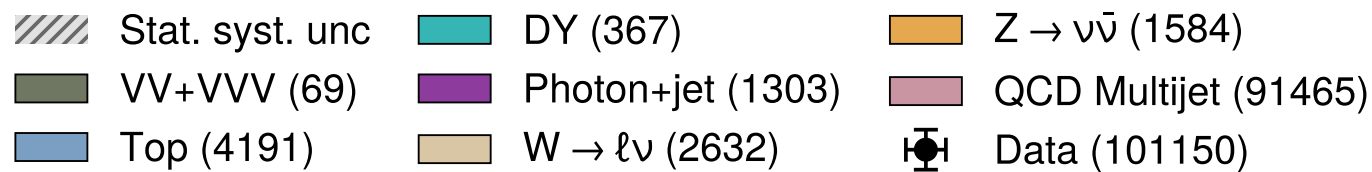


$p_T^{\text{miss}}/\sqrt{H_T}$

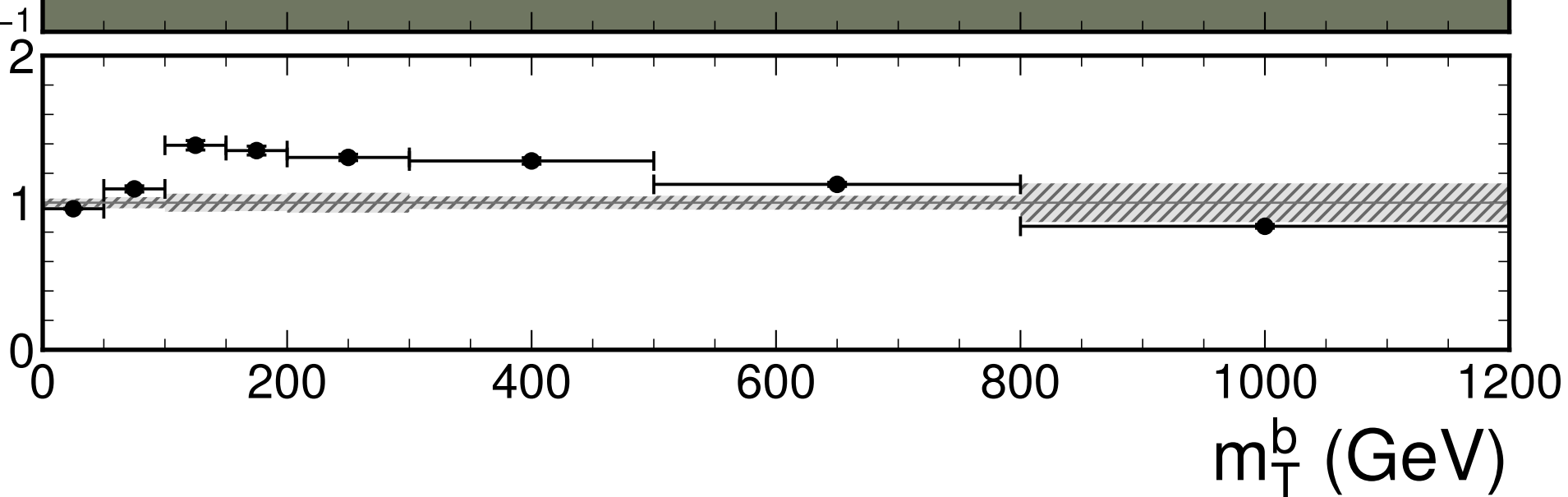
CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)

Events



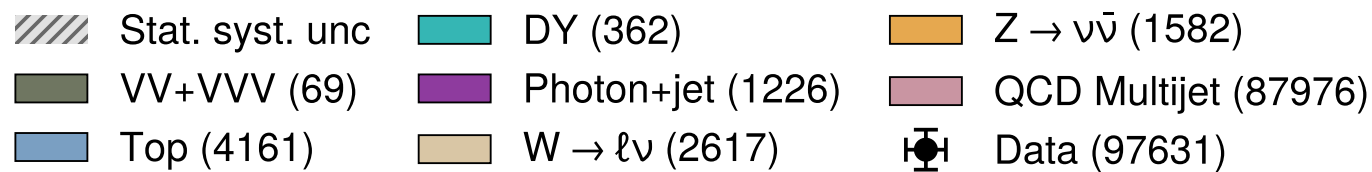
Data/MC



CMS *Work in progress*

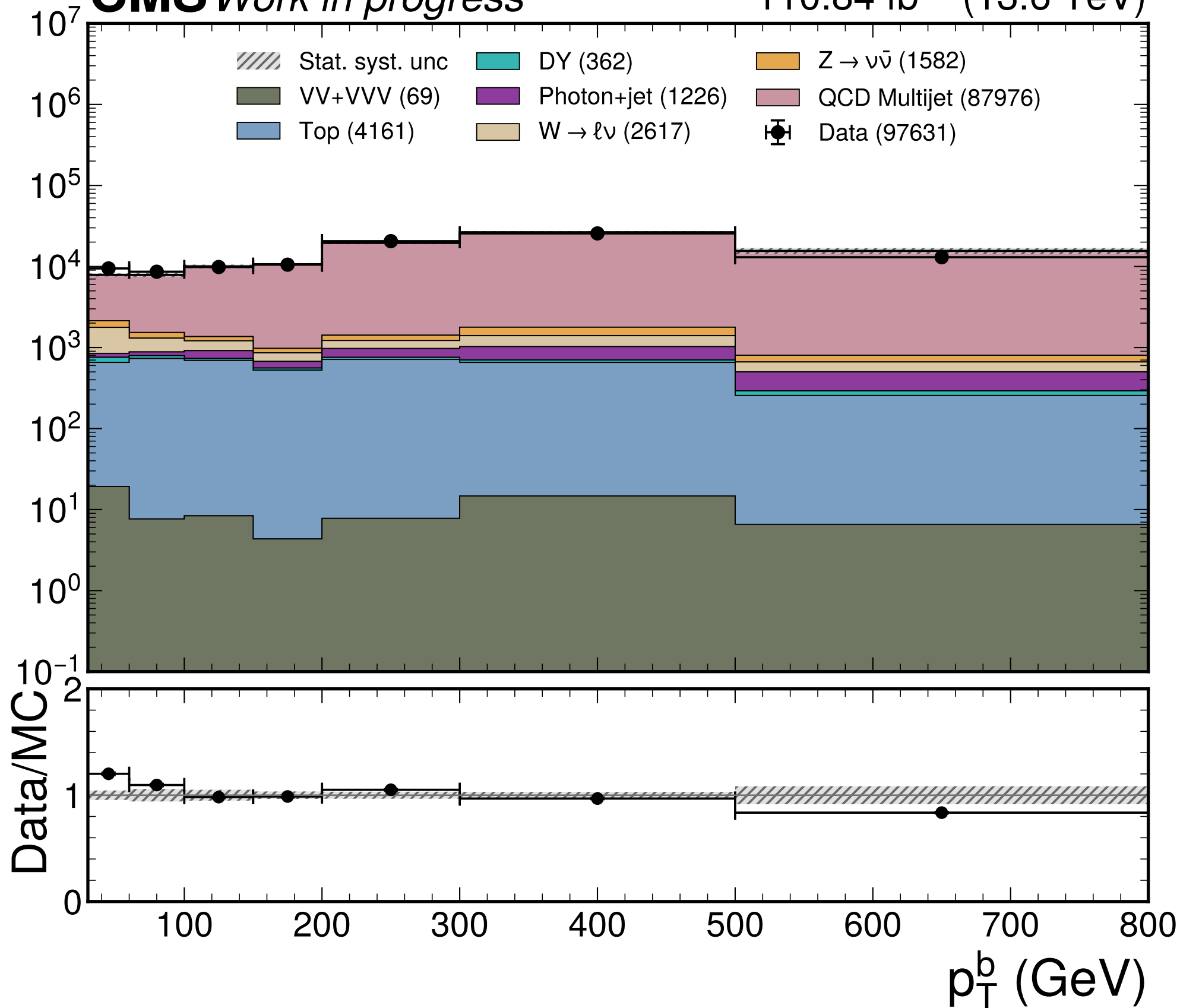
110.84 fb⁻¹ (13.6 TeV)

Events



Data/MC

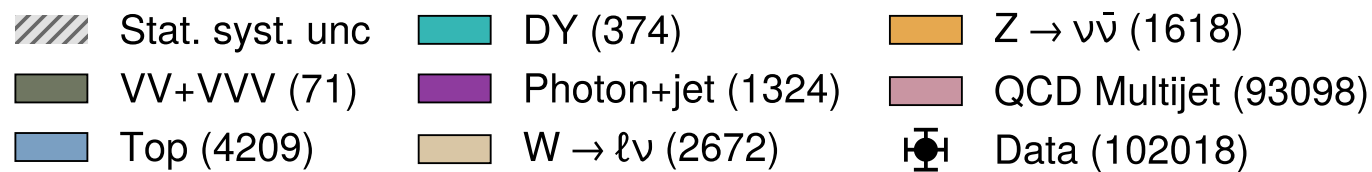
p_T^b (GeV)



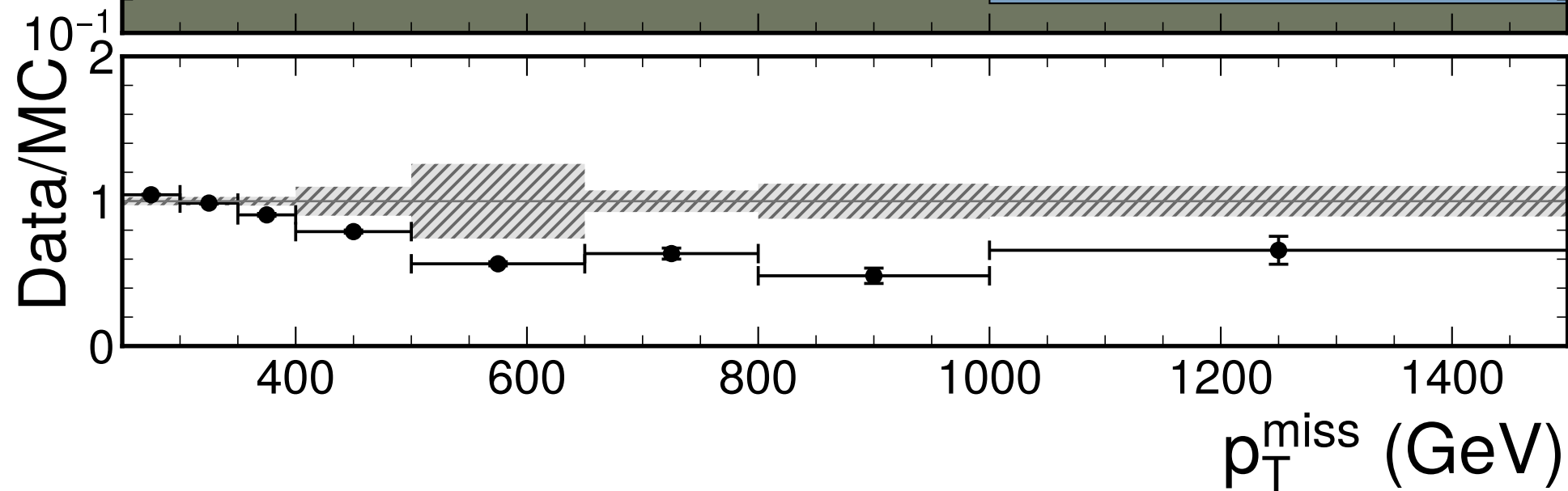
CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)

Events



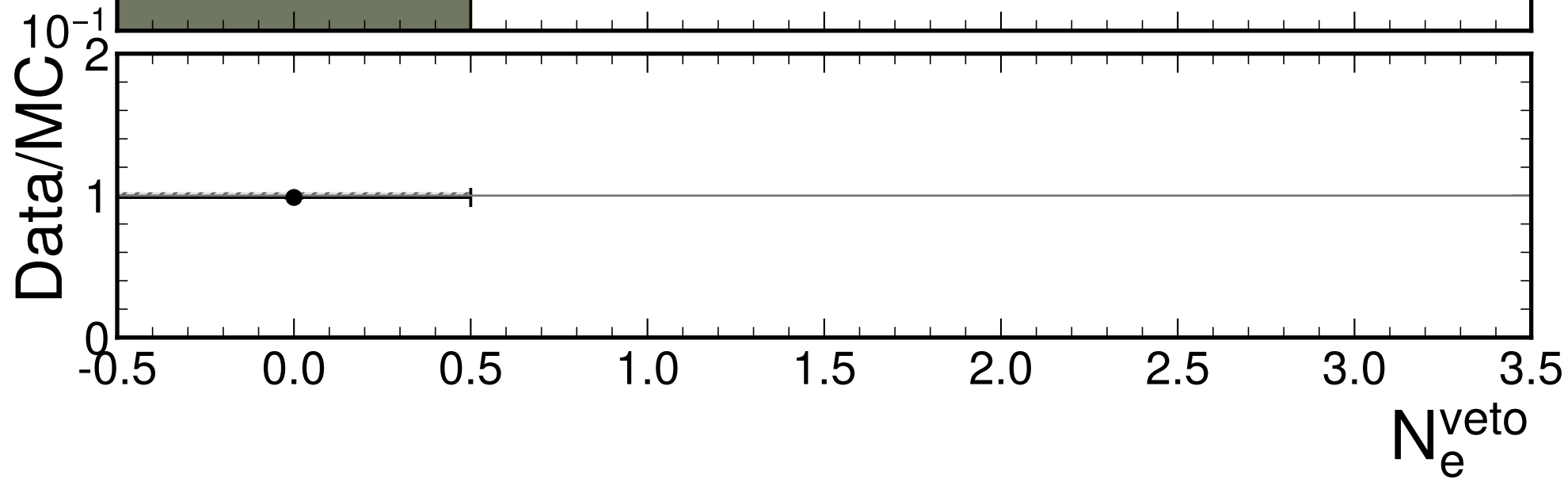
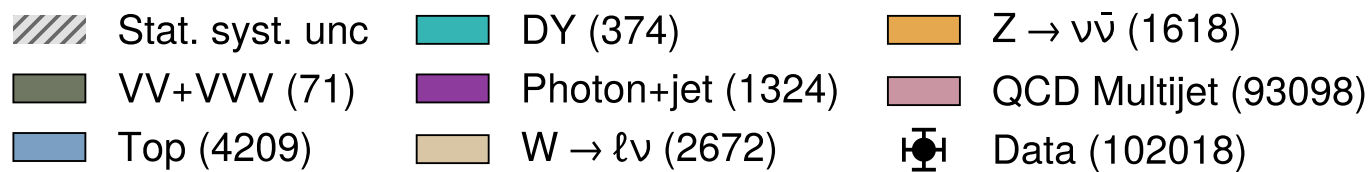
Data/MC



CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)

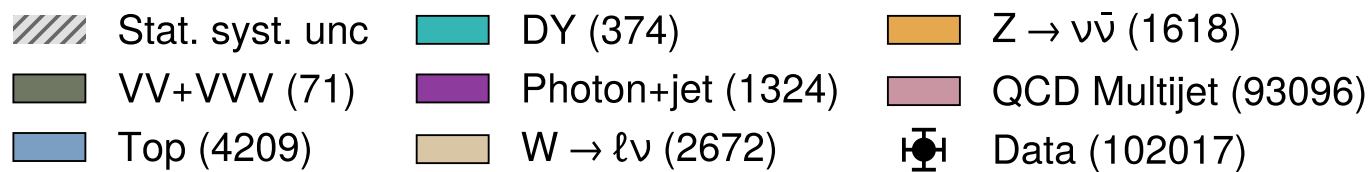
Events



CMS *Work in progress*

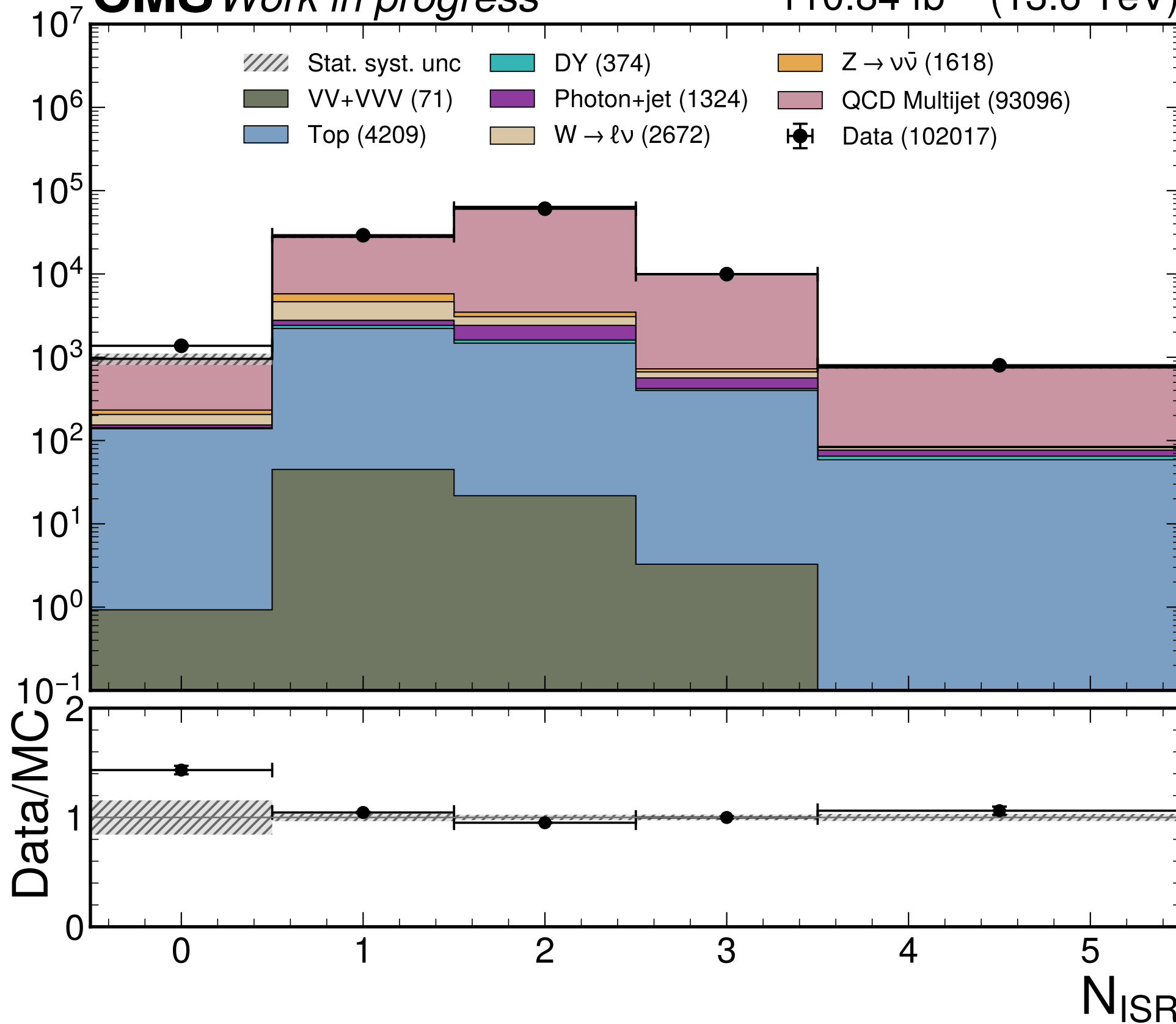
110.84 fb⁻¹ (13.6 TeV)

Events



Data/MC

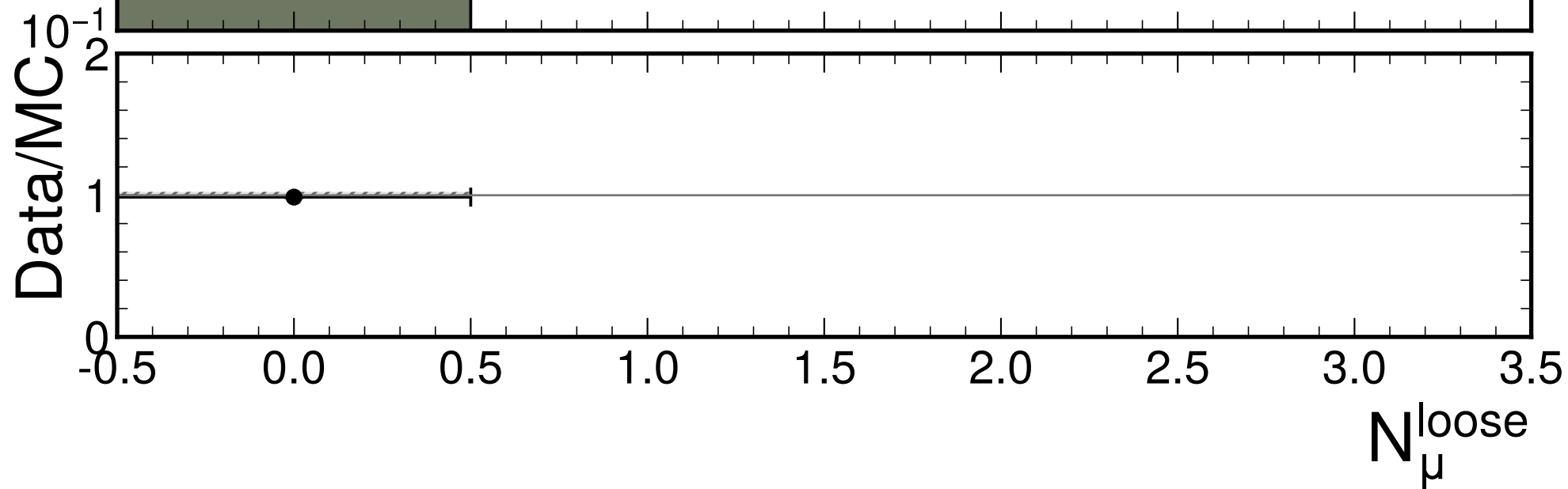
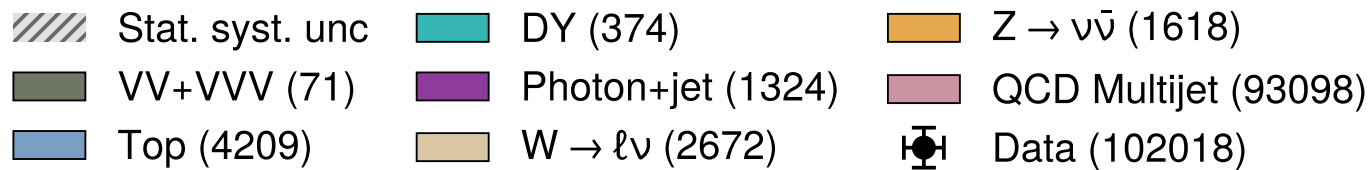
N_{ISR}



CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)

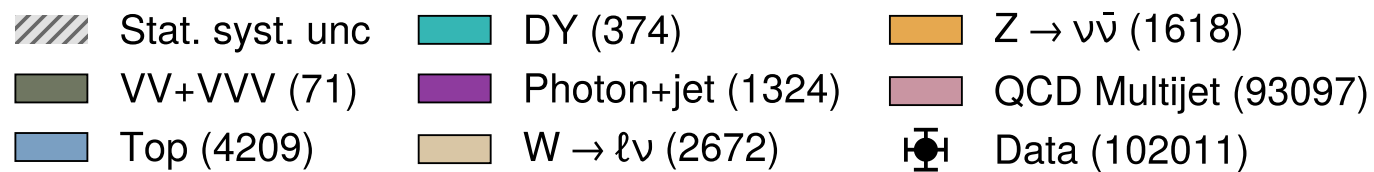
Events



CMS *Work in progress*

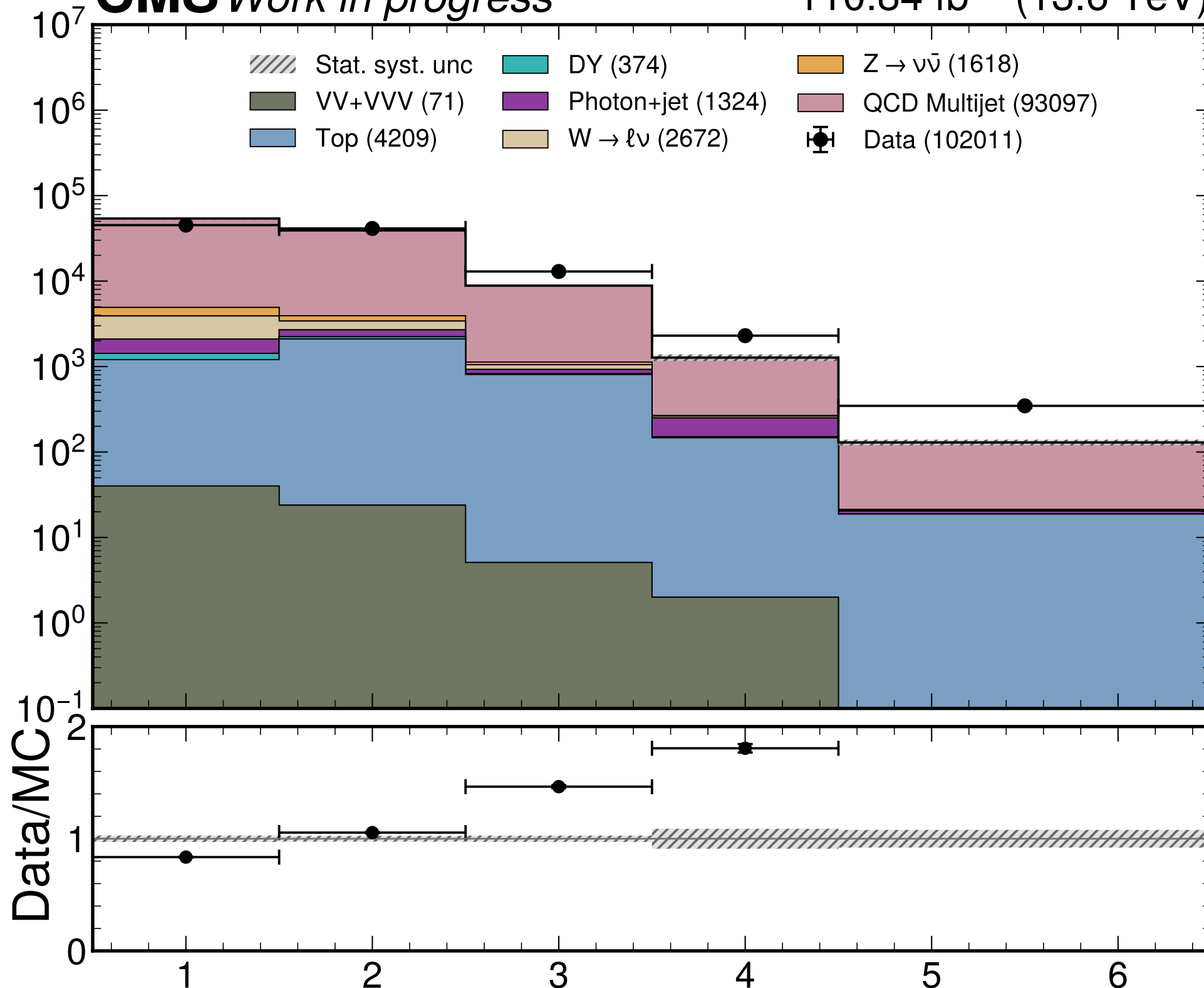
110.84 fb⁻¹ (13.6 TeV)

Events



Data/MC

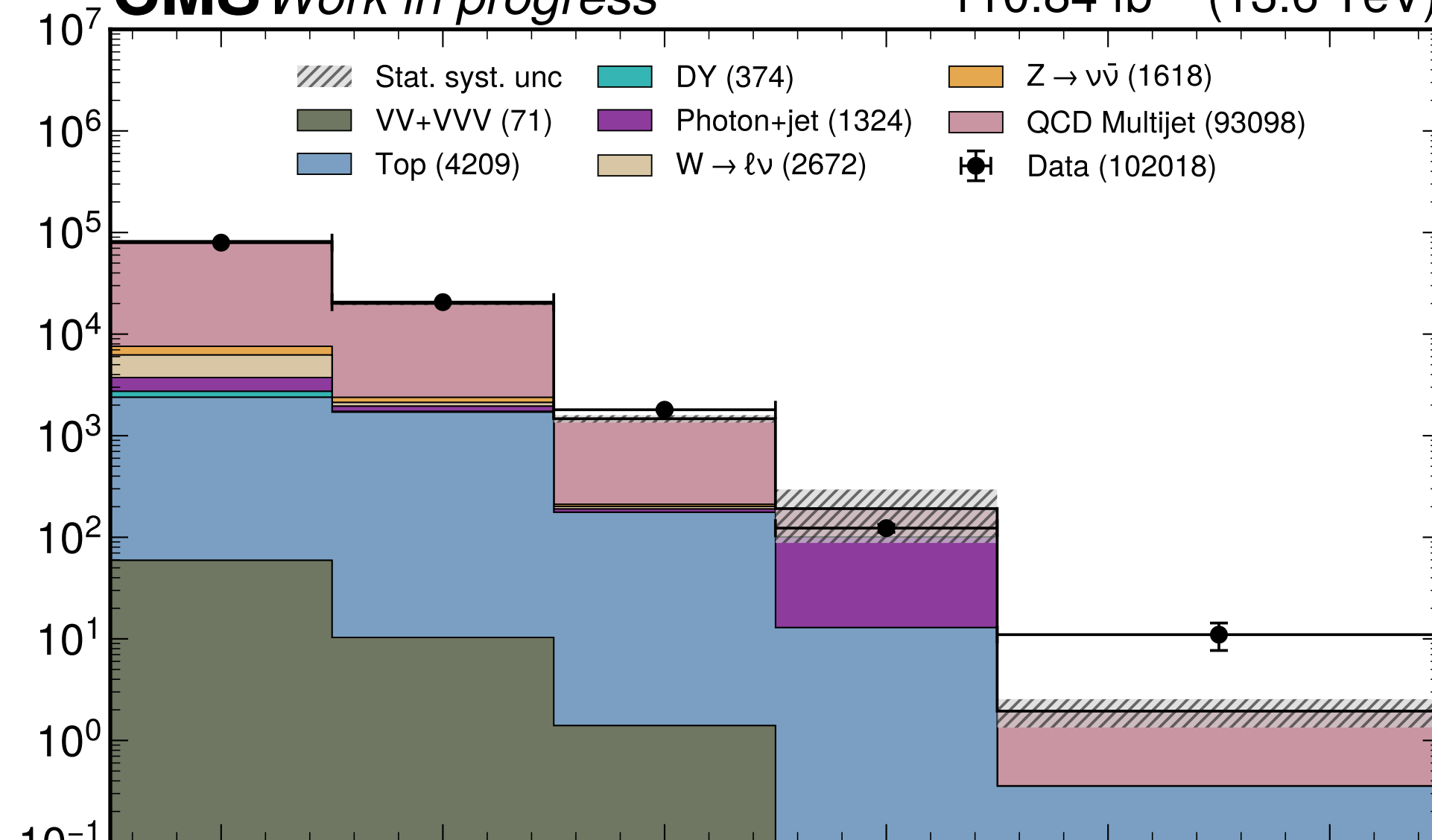
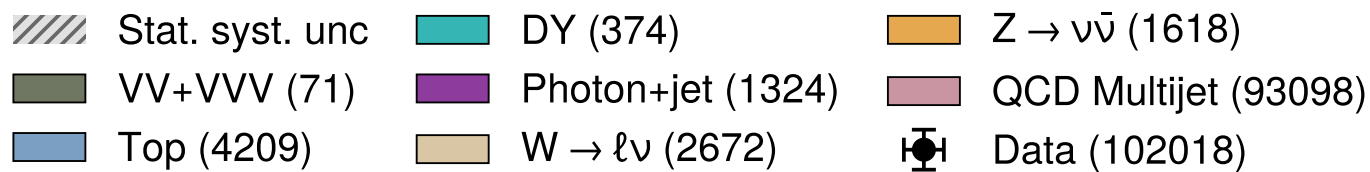
N_b (loose WP)



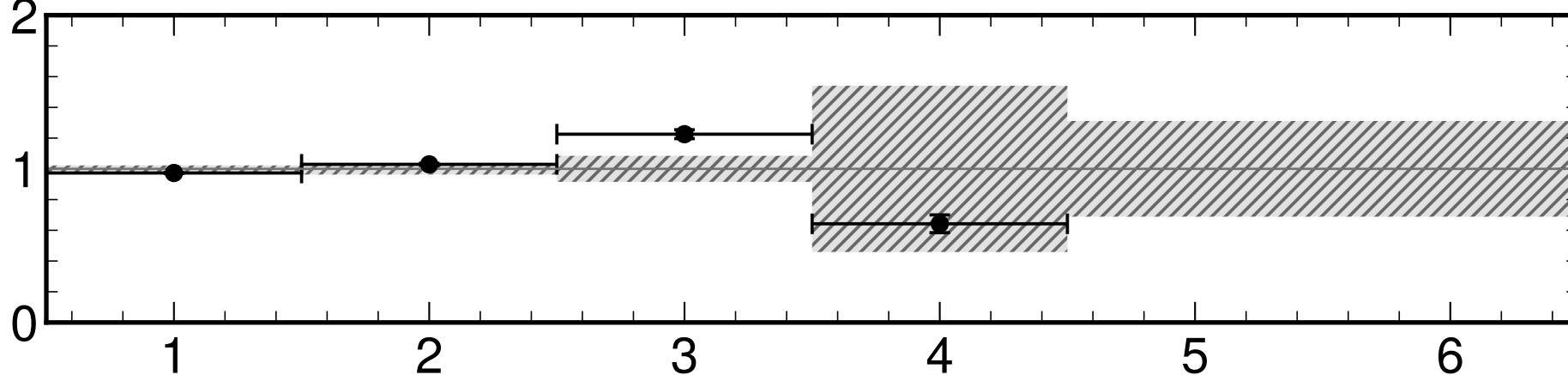
CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)

Events



Data/MC

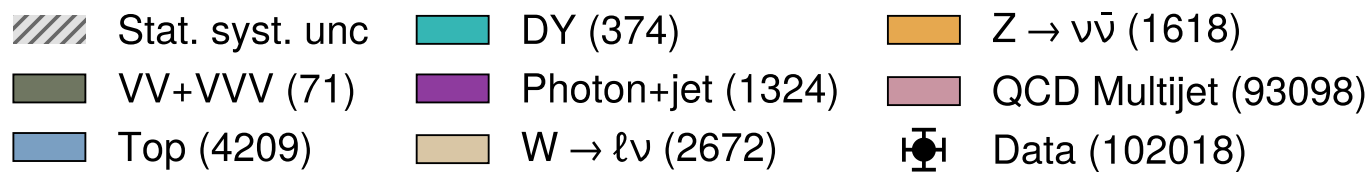


N_b

CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)

Events



Data/MC

N_j

